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# Auto CAD 2008 2D TQC電腦輔助設計類
# 檢定題庫解題秘笈


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# 目錄

**目錄** ..... **I**

**圖目錄** ..... **VII**

**表目錄** ..... **IX**

**成果資料說明** ..... **1**

**第 1 章 基本概念** ..... **3**

1.1 **測驗內容與評分標準** ..... 3

1.2 **座標系統－右手定則** ..... 3

1.3 **基本環境設定－十字游標、弧與圓平滑度、精確值、點型式** ..... 4

1.4 **指令繪圖－上圖下文、左鍵右鼠** ..... 6

1.5 **使用慣例說明** ..... 7

1.6 **常用指令簡介** ..... 8

*1.6.1 Line，**線*** ..... *8*

*1.6.2 Circle，**圓*** ..... *9*

*1.6.3 Arc，**弧*** ..... *9*

*1.6.4 RECtangle，**矩形*** ..... *10*

*1.6.5 POLygon，**多邊形*** ..... *10*

*1.6.6 eXplode，**炸開*** ..... *11*

*1.6.7 Fillet，**切圓弧*** ..... *11*

*1.6.8 CHAmfer，**倒角*** ..... *12*

*1.6.9 ARray，**環形陣列*** ..... *12*

*1.6.10 ARray，**矩形陣列*** ..... *13*

*1.6.11 Offset，**偏移複製*** ..... *13*

*1.6.12 COpy，**複製*** ..... *14*

*1.6.13 Move，搬移*.....14

*1.6.14 EXtend，延伸*.....15

*1.6.15 TRim，修剪*.....15

*1.6.16 ROtate，轉動*.....16

*1.6.17 MIrror，鏡射*.....16

*1.6.18 LENgthen，變更長度*.....17

*1.6.19 DIVide，等分*.....17

*1.6.20 SCale，放大*.....18

*1.6.21 物件鎖點1*.....18

*1.6.22 物件鎖點2*.....19

*1.6.23 相切圓性質*.....19

*1.6.24 BOundary，邊界*.....20

*1.6.25 Hatch，剖面線*.....20

*1.6.26 DIst，距離*.....20

*1.6.27 ID，座標值*.....21

*1.6.28 PEdit，編輯聚合線*.....21

1.7 AutoCAD 2D **常用指令集**.....22

**第 2 章 檢定題庫**.....**25**

2.1 **第一類：綜合應用一**.....25

*2.1.1 AutoCAD2D2008-101*.....25

*2.1.2 AutoCAD2D2008-102*.....28

*2.1.3 AutoCAD2D2008-103*.....31

*2.1.4 AutoCAD2D2008-104*.....34

*2.1.5 AutoCAD2D2008-105*.....36

*2.1.6 AutoCAD2D2008-106*.....38

*2.1.7 AutoCAD2D2008-107*.....40

*2.1.8 AutoCAD2D2008-108*.....42

*2.1.9 AutoCAD2D2008-109*.....45

*2.1.10 AutoCAD2D2008-110*.....47

*2.1.11 AutoCAD2D2008-111*.....49

*2.1.12 AutoCAD2D2008-112*.....51

*2.1.13 AutoCAD2D2008-113*.....53

*2.1.14 AutoCAD2D2008-114*.....55

*2.1.15 AutoCAD2D2008-115*.....57

2.2 **第二類：綜合應用二** ..... 61

*2.2.1 AutoCAD2D2008-201*.....61

*2.2.2 AutoCAD2D2008-202*.....64

*2.2.3 AutoCAD2D2008-203*.....66

*2.2.4 AutoCAD2D2008-204*.....69

*2.2.5 AutoCAD2D2008-205*.....71

*2.2.6 AutoCAD2D2008-206*.....74

*2.2.7 AutoCAD2D2008-207*.....77

*2.2.8 AutoCAD2D2008-208*.....79

*2.2.9 AutoCAD2D2008-209*.....82

*2.2.10 AutoCAD2D2008-210*.....84

*2.2.11 AutoCAD2D2008-211* .....87

*2.2.12 AutoCAD2D2008-212*.....89

*2.2.13 AutoCAD2D2008-213*.....91

*2.2.14 AutoCAD2D2008-214*.....94

*2.2.15 AutoCAD2D2008-215*.....96

2.3 **第三類：綜合應用三** ..... 99

*2.3.1 AutoCAD2D2008-301*.....99

*2.3.2 AutoCAD2D2008-302* ..... *101*

*2.3.3 AutoCAD2D2008-303* ..... *103*

*2.3.4 AutoCAD2D2008-304* ..... *106*

*2.3.5 AutoCAD2D2008-305* ..... *108*

*2.3.6 AutoCAD2D2008-306* ..... *112*

*2.3.7 AutoCAD2D2008-307* ..... *115*

*2.3.8 AutoCAD2D2008-308* ..... *118*

*2.3.9 AutoCAD2D2008-309* ..... *120*

*2.3.10 AutoCAD2D2008-310* ..... *123*

*2.3.11 AutoCAD2D2008-311* ..... *126*

*2.3.12 AutoCAD2D2008-312* ..... *130*

*2.3.13 AutoCAD2D2008-313* ..... *133*

*2.3.14 AutoCAD2D2008-314* ..... *136*

*2.3.15 AutoCAD2D2008-315* ..... *138*

2.4 **第四類：玩具&生活用品應用** ..... 143

*2.4.1 AutoCAD2D2008-401* ..... *143*

*2.4.2 AutoCAD2D2008-402* ..... *145*

*2.4.3 AutoCAD2D2008-403* ..... *147*

*2.4.4 AutoCAD2D2008-404* ..... *149*

*2.4.5 AutoCAD2D2008-405* ..... *151*

*2.4.6 AutoCAD2D2008-406* ..... *154*

*2.4.7 AutoCAD2D2008-407* ..... *157*

*2.4.8 AutoCAD2D2008-408* ..... *160*

*2.4.9 AutoCAD2D2008-409* ..... *163*

*2.4.10 AutoCAD2D2008-410* ..... *166*

*2.4.11 AutoCAD2D2008-411* ..... *170*

*2.4.12 AutoCAD2D2008-412*.....172

*2.4.13 AutoCAD2D2008-413*.....174

*2.4.14 AutoCAD2D2008-414*.....177

*2.4.15 AutoCAD2D2008-415*.....180

2.5 **第五類：機械設計應用**.....183

*2.5.1 AutoCAD2D2008-501*.....183

*2.5.2 AutoCAD2D2008-502*.....187

*2.5.3 AutoCAD2D2008-503*.....190

*2.5.4 AutoCAD2D2008-504*.....194

*2.5.5 AutoCAD2D2008-505*.....196

*2.5.6 AutoCAD2D2008-506*.....200

*2.5.7 AutoCAD2D2008-507*.....203

*2.5.8 AutoCAD2D2008-508*.....206

*2.5.9 AutoCAD2D2008-509*.....208

*2.5.10 AutoCAD2D2008-510*.....211

*2.5.11 AutoCAD2D2008-511*.....213

*2.5.12 AutoCAD2D2008-512*.....216

*2.5.13 AutoCAD2D2008-513*.....218

*2.5.14 AutoCAD2D2008-514*.....220

*2.5.15 AutoCAD2D2008-515*.....222

2.6 **第六類：建築&室內設計應用**.....225

*2.6.1 AutoCAD2D2008-601*.....225

*2.6.2 AutoCAD2D2008-602*.....229

*2.6.3 AutoCAD2D2008-603*.....233

*2.6.4 AutoCAD2D2008-604*.....236

*2.6.5 AutoCAD2D2008-605*.....241

*2.6.6 AutoCAD2D2008-606* ..... *245*

*2.6.7 AutoCAD2D2008-607* ..... *248*

*2.6.8 AutoCAD2D2008-608* ..... *252*

*2.6.9 AutoCAD2D2008-609* ..... *254*

*2.6.10 AutoCAD2D2008-610* ..... *257*

*2.6.11 AutoCAD2D2008-611* ..... *260*

*2.6.12 AutoCAD2D2008-612* ..... *263*

*2.6.13 AutoCAD2D2008-613* ..... *266*

*2.6.14 AutoCAD2D2008-614* ..... *269*

*2.6.15 AutoCAD2D2008-615* ..... *273*

**参考文献** ..... **277**

# 圖目錄

圖 1、右手定則 1 ..... 4
圖 2、右手定則 2 ..... 4
圖 3、選項設定視窗 ..... 5
圖 4、精確值設定視窗 ..... 5
圖 5、點型式設定視窗 ..... 6
圖 6、範例：Line，線 ..... 8
圖 7、範例：Circle，圓 ..... 9
圖 8、範例：Arc，弧 ..... 9
圖 9、範例：RECtangle，矩形 ..... 10
圖 10、範例：POLygon，多邊形 ..... 10
圖 11、範例：eXplode，炸開 ..... 11
圖 12、範例：Fillet，切圓弧 ..... 11
圖 13、範例：CHAmfer，倒角 ..... 12
圖 14、範例：ARray，環形陣列 ..... 12
圖 15、範例：ARray，矩形陣列 ..... 13
圖 16、範例：Offset，偏移複製 ..... 13
圖 17、範例：COpy，複製 ..... 14
圖 18、範例：Move，搬移 ..... 14
圖 19、範例：EXtend，延伸 ..... 15
圖 20、範例：TRim，修剪 ..... 15
圖 21、範例：ROtate，轉動 ..... 16
圖 22、範例：MIrror，鏡射 ..... 16
圖 23、範例：LENgthen，變更長度 ..... 17
圖 24、範例：DIVide，等分 ..... 17

圖 25、範例：SCale，放大 ..... 18

圖 26、範例：物件鎖點 1 ..... 18

圖 27、範例：物件鎖點 2 ..... 19

圖 28、範例：相切圓性質 ..... 19

圖 29、範例：BOundary，邊界 ..... 20

圖 30、範例：Hatch，剖面線 ..... 20

圖 31、範例：DIst，距離 ..... 20

圖 32、範例：ID，座標值 ..... 21

圖 32、範例： PEdit，編輯聚合線 ..... 21

# 表目錄

**表** 1、TQC Auto CAD 2D **檢定測驗內容與評分標準對應說明表**.....3

**表** 2、**字頭碼與物件類別對應說明表**.....7

**表** 3、**常用之鎖點與符號說明表**.....8

**表** 4、AutoCAD 2D **常用指令集**.....22

**表** 5、AutoCAD 2D **常用物件鎖點表**.....23

# 成果資料說明

## 成果資料說明

### 一、教材摘要說明

本教材為配合 Auto CAD 2008 實力養成暨評量[1]乙書編撰而成，主要收納 TQC Auto CAD 2008 術科 2D 檢定題庫，計 6 大類共 90 題。本教材同時以 PowerCam5[2] 錄製教學影片放置於本校「LMS 數位學習系統」教學網頁，以供學生隨時上網瀏覽觀看學習。另外，以重點提示方式整理列出 Auto CAD 2D 之基本繪圖概念與指令。

### 二、教材宗旨、架構說明

目前坊間 TQC Auto CAD 檢定相關資料，只有財團法人中華民國電腦技能基金會出版 Auto CAD 術科檢定題庫，該會至今尚未有該類術科檢定題庫之詳細解題步驟之相關書籍資料。因此，為配合課程教學需要及幫助讀者能順利輕鬆地考取 TQC Auto CAD 2D 專業證照，特編撰本教材。

本教材主要將 Auto CAD 2D 2008 術科檢定題庫每一題之解題步驟詳細列出，並以圖形配合說明，光碟資料片內亦有實際之圖形檔案，本教材在架構上同於 Auto CAD 2008 實力養成暨評量乙書之編排。

### 三、學習者將獲得的知識、技能或態度的教材目標

藉由每道題庫的解說，讀者能學習 Auto CAD 相關指令及電腦繪圖操作技能，在不斷練習與熟悉下，可順利考取 TQC Auto CAD 2D 專業證照。

### 四、教材實際運用績效

96 學年度營建工系(所)共計 35 位學生考取 TQC Auto CAD 2D 專業證照，其中研究生 6 位、大四生 12 位、大三生 14 位、大二生 2 位、及大一生計 1 位(資料統計至 97/06/27)。

# 第1章 基本概念

Auto CAD 2008 2D 解題秘笈為配合 Auto CAD 2008 實力養成暨評量[1]乙書編撰而成，收納 TQC Auto CAD 2008 術科 2D 檢定題庫，計 6 大類共 90 題，每題皆有詳細之解題步驟與圖形說明，在圖形與指令解題步驟相互配合說明下，相信能讓讀者輕鬆掌握每一題，經不斷練習與熟悉，即可順利考取 TQC 製圖類之 CAD 2D 專業證照。

## 1.1 測驗內容與評分標準

TQC 製圖類之 Auto CAD 2D 檢定測驗項目，計有實用級與專業級兩項，其測驗內容與評分標準如表 1 所示。

表 1、TQC Auto CAD 2D 檢定測驗內容與評分標準對應說明表

| | 學科測驗 | 術科測驗 | 及格成績 | 測驗時間 |
|---|---|---|---|---|
| 實用級 | 無 | 第一至六類各由題庫 15 題中抽出 1 題，第一至二題每題 10 分，第三題 15 分，第四至五題每題 20 分，第六題 25 分，共計 100 分。 | 70 分 | 80 分鐘 |
| 專業級 | 單、複選混合，由題庫 598 題中抽出 20 題，每題 1 分，答錯不倒扣，共計 20 分。 | 第一至六類各由題庫 15 題中抽出 1 題，第一至三題每題 10 分，第四至五題每題 15 分，第六題 20 分，共計 80 分。 | 70 分 | 100 分鐘 |

## 1.2 座標系統－右手定則

Auto CAD 採用之座標系統同於數學上之座標系統，而其在立體上之表現是以右手定則加以規範之，不論在 2D 或 3D 此定則皆常用之，讀者應加以了解與熟悉以利繪圖工作。其包含有下列 2 項：

1、右手定則 1：用以決定三軸，右手拇指為 X 方向、右手食指為 Y 方向、右手中指為 Z 方向，且三指之間互相垂直，如圖 1 所示。

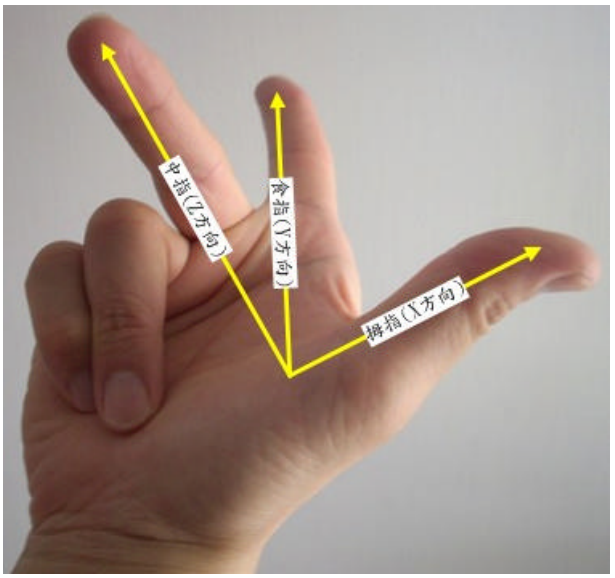



**圖** 1、**右手定則** 1

2、**右手定則** 2：**用以決定旋轉角方向，右手拇指指向旋轉基準軸方向(註：當旋轉基準軸為線段時，方向為線段起點指向線段終點)，其餘** 4 **指即為轉動正方向，如圖** 2 **所示。**

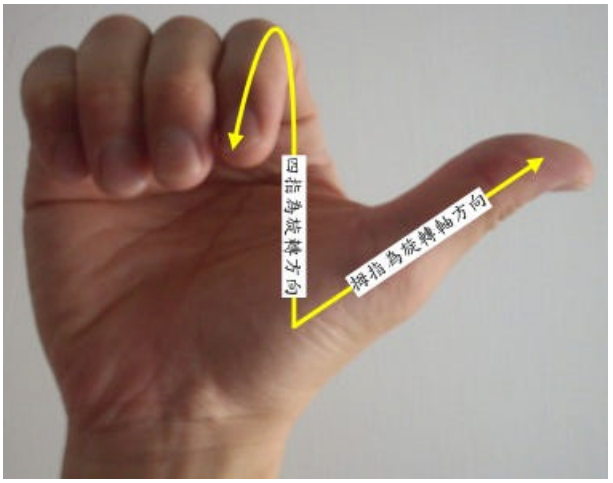



**圖** 2、**右手定則** 2

## 1.3 基本環境設定－十字游標、弧與圓平滑度、精確值、點型式

**在開始繪圖時應先設定** Auto CAD **作業環境，其最基本的設定有：十字游標、弧與圓平滑度、精確值、點型式等，其設定如下所述：**

1、**十字游標、弧與圓平滑度：點選「工具」→「選項」或在指令視窗中輸入** Options **即**

出現如圖 3 之視窗。十字游標大小位於左下角，弧與圓平滑度於右上角，通用其設定值分別為 100 與 10000(讀者可自行設定)。

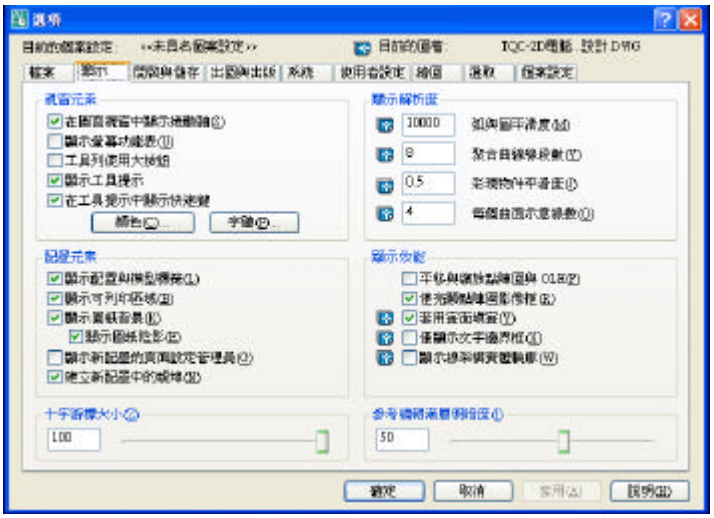


圖 3、選項設定視窗

2、精確值：此設定值主要用於輸出，目的在於查詢圖面上相關資料，例如：兩點間距離、圓直徑大小、或弧的角度等等數據資料之小數位數。在 TQC Auto CAD 2D 檢定測驗中，其距離與角度精確值皆規定為 4 位小數值。設定方式為點選「格式」→「單位」或在指令視窗中輸入 UNits 即出現如圖 4 之視窗。

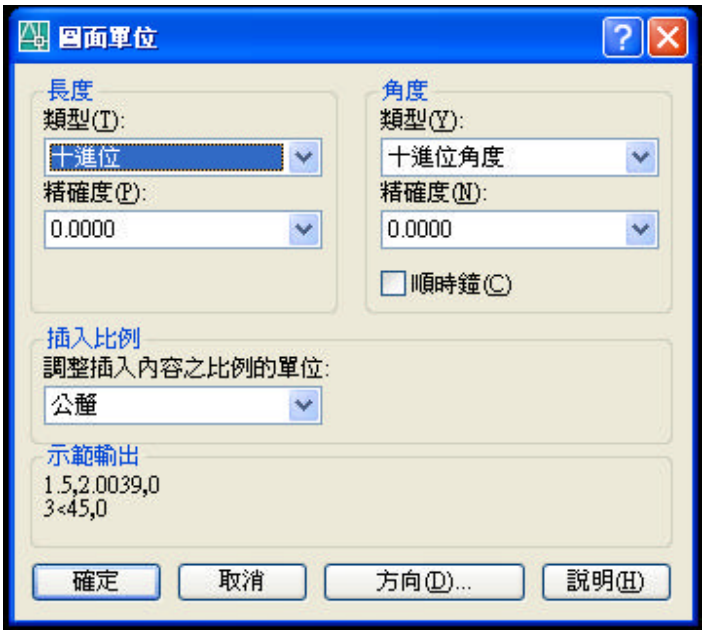



圖 4、精確值設定視窗

3、點型式：在 Auto CAD 中，點物件沒有任何性質可言(即沒有大小問題)，其只有代表

某一點之位置。在表現上 Auto CAD 允許使用者自行設定其表現型式，設定方式為點選「格式」→「點型式」或在指令視窗中輸入 DDPTYPE 即出現如圖 4 之視窗、使用者可自行選取設定之。

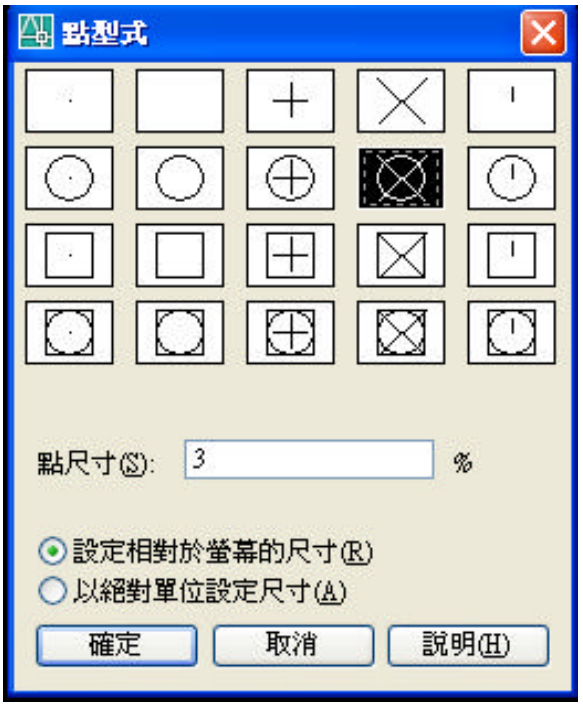



圖 5、點型式設定視窗

## 1.4 指令繪圖－上圖下文、左鍵右鼠

Auto CAD 畫面基本上主要區分為：1)位於左右兩側與上方之圖示(Icon)指令點選區、2)位於正中間的圖形顯示區、及 3)位於正上方的文字輸出入區等 3 大部份 (以上畫面配置使用者可自行設定)。筆者個人習慣，繪圖時注視圖形顯示區與文字輸出入區，左手不離鍵盤，右手不離滑鼠。如此即為「上圖下文、左鍵右鼠」，習慣養成，自然能以指令繪圖。

基於上述，本解題秘笈之繪圖方式完全以鍵盤輸入指令方式編撰而成，筆者個人認為如能克服英文單字指令的障礙且熟悉與熟記指令，在繪圖時即能將心理所想的繪圖動作自然而然地轉換成英文單字指令。舉例而言：「現在要畫出長度等於半徑且端點分別在圓心與圓周上水平直線」。轉換成指令時即為：「Line(畫線), cen(鎖定圓心), 點圓物件之圓周, qua(鎖定圓之四分點), 點圓物件之水平方向四分點」，如此即可輕鬆完成上述繪圖動作。在上述之指令中

利用畫線指令「Line」、圓心鎖點指令「cen」、及圓四分點鎖點指令「qua」等。而其實這些英文指令即為其真正的英文含義，例如：Line 即為線、cen 即為 center(中心)之縮寫、qua 即為 quarter(四分之一、季)之縮寫。雖然 Auto CAD 指令多且複雜，但是常用的指令並不多，約莫 20 而已, 且在繪圖過程中 Auto CAD 亦會在文字視窗中出現每一步驟的選項供提示使用者輸入。因此，讀者如能養成「上圖下文、左鍵右鼠」的繪圖習性，相信在讀者用心之下必能輕鬆駕馭 Auto CAD。

## 1.5 使用慣例說明

本解題秘笈以圖形配合指令文字說明，包含有下列 2 種格式：

1、左圖右文格式：左方框內為圖形說明，右方框內為指令文字說明。

2、上圖下文格式：上方框內為圖形說明，下方框內為指令文字說明。

上述 2 種格式，在圖形框內如出現 1 個以上的圖形，則指令文字框內會有直線以區隔上下或左右圖形所對應之指令文字說明。圖形框內之 P01(或 P02，或 P03)為任意位置，讀者可自行決定之位置，即每題的起畫位置。另外，圖形框內有 50、51、52…等之數字順序編號，其代表意義為解題之真正步驟次序，50 為第 1 步，51 為第 2 步，依此類推開始。而其數字編號前面之英文字母為物件(Entity、或稱圖素，即在 CAD 中畫出之線、圓等實物)類別之字頭碼，常用之字頭碼與物件類別對應說明，如表 2 所示。指令文字框內常用之鎖點與符號，如表 3 所示。

表 2、字頭碼與物件類別對應說明表

| 項次 | 字頭碼 | 中文物件類別名稱 | 原文物件類別名稱 |
|---|---|---|---|
| 01 | A | 弧 | Arc |
| 02 | C | 圓 | Circle |
| 03 | L | 線 | Line |
| 04 | P | 點 | Point |
| 05 | R | 矩形(含圓角與斜角)/多邊形 | RECtangle/Polygon |
| 06 | S | 物件組合 | Group(Set) |

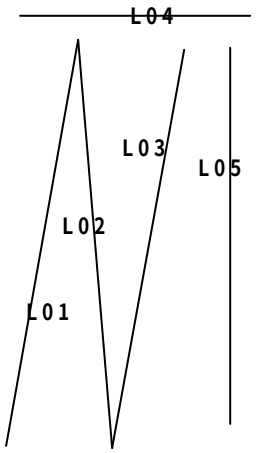
表 3、常用之鎖點與符號說明表

| 項次 | 文字(鎖點)、或符號 | 意義或使用說明 |
|---|---|---|
| 01 | → | 產生 |
| 02 | , | 空白鍵或 Enter 鍵或滑鼠按鍵 |
| 03 | ; | 多重選取 |
| 04 | & | 2 物件交點(配合 int 鎖點) |
| 05 | @ | 鎖定上次點/相對座標值 |
| 06 | < | 角度座標值 |
| 07 | (正交模式) | 以「F8」鍵設定為正交模式 |
| 08 | 向上 xx | 將滑鼠向上移，輸入 xx 數值 |
| 09 | int, L50 & A51 | 鎖定 L50 與 A51 之交點 |
| 10 | cen, A50 | 鎖定 A50 之中心點 |
| 11 | end, L50 左 | 鎖定 L50 之左端點(可為右、上、下端等) |
| 12 | mid, L50 | 鎖定 L50 之中間點 |
| 13 | mtp/m2p, 點 1, 點 2 | 鎖定點 1、點 2 直線間之中間點<br>(點 1、點 2 可能實際點，或物件端中、交、點等) |
| 14 | qua, C50 上 | 鎖定 C50 之下方四分點(可為左右、上方等) |
| 15 | nod, P50 | 鎖定 P50 點位 |
| 16 | CTRL+1 | 點選物件，按「CTRL」鍵與「1」鍵，查物件性質 |
| 17 | per, L50 | 鎖定垂直於 L50 之點 |

## 1.6 常用指令簡介

以下以簡單圖形配合指令格式以說明本解題秘笈使用之指令格式意義，讀者如果需要查詢詳細的指令格式請自行參閱 Auto CAD 2008 特訓教材基礎篇[3]乙書。

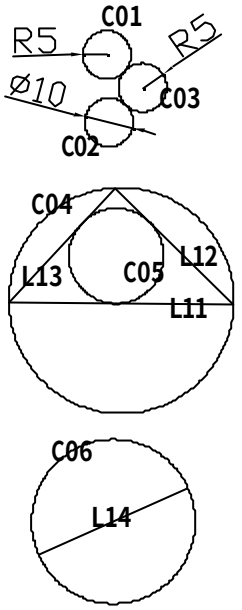
### 1.6.1 Line，線




1、Line, L01 下端, L01 上端, →L01,　L02 下端, →L02, L03 上端, →L03
2、Line, L04 左端, (正交模式), L04 右端, →L04
3、Line, L05 上端, (正交模式), L05 下端, →L05

圖 6、範例：Line，線

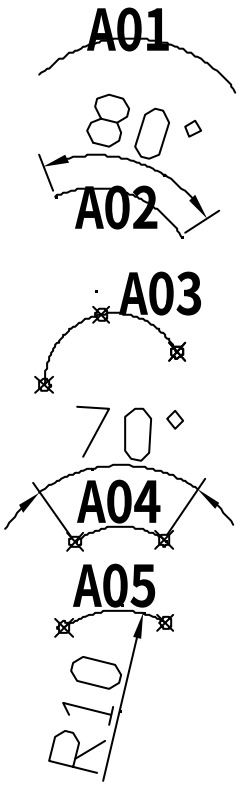
### 1.6.2 Circle，圓




1、Circle, **圓心**, 5, →C01
2、Circle, **圓心**, D, 5, →C02
3、Circle, T, tan, C01, tan, C02, 5, →C03
4、Circle, 3P, int, L11 & L12, int, L12 & L13, int, L11 & L13, →C04
5、Circle, 3P, tan, L11, tan, L12, tan, L13, →C05
6、Circle, 2P, end, L14 **左**, end, L14 **右**, →C06

**圖** 7、**範例：**Circle，**圓**

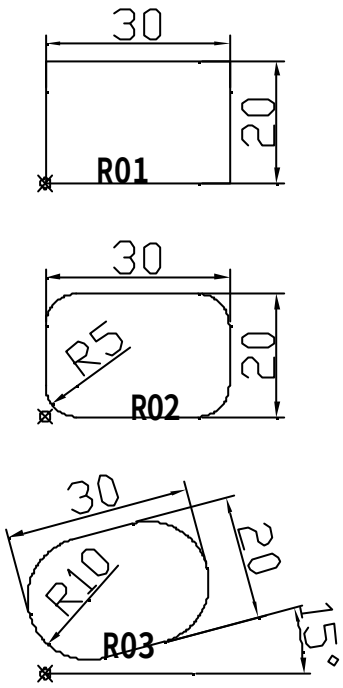
### 1.6.3 Arc，弧




1、Arc, C, A01 **中心點**, A01 **右端**, A01 **左端**, →A01
2、Arc, C, A02 **中心點**, A02 **右端**, A, 80, →A02
3、Arc, A03 **右端**, A03 **弧上**, A03 **左端**, →A03
4、Arc, A04 **右端**, E, A04 **左端**, A, 70, →A04
5、Arc, A05 **右端**, E, A05 **左端**, R, 10, →A05

**圖** 8、**範例：**Arc，**弧**

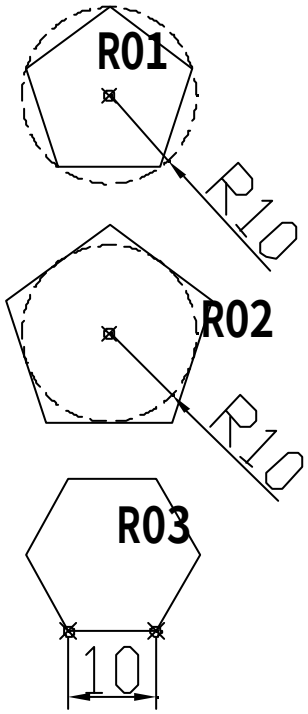
### 1.6.4 RECtangle，矩形




1、RECtangle, F, 0, **起點**, D, 30, 20, **向右上點**, →R01
2、RECtangle, F, 5, **起點**, D, 30, 20, **向右上點**, →R02
3、RECtangle, F, 10, **起點**, R, 15, D, 30, 20, **向右上點**, →R03

**圖** 9、**範例：**RECtangle，**矩形**

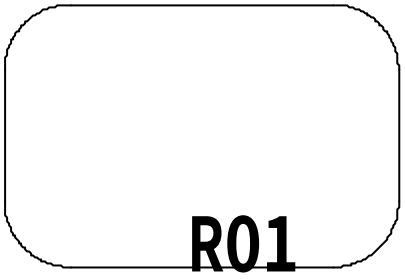
### 1.6.5 POLygon，多邊形




1、POLygon, 5, **中心點**, I, 10, **向上點**, →R01
2、POLygon, 5, **中心點**, C, 10, **向上點**, →R02
3、POLygon, 6, E, **左側點**, **右側點**, →R03

**圖** 10、**範例：**POLygon，**多邊形**

### 1.6.6 eXplode，炸開




1、eXplode, R01, →4 個線段與 4 個弧
2、eXplode, R02, →6 個線段

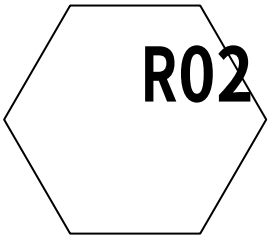
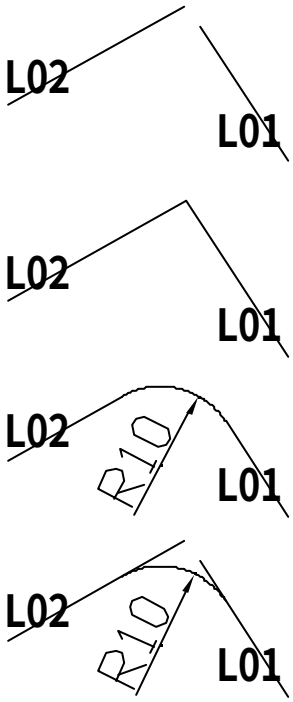



圖 11、範例：eXplode，炸開

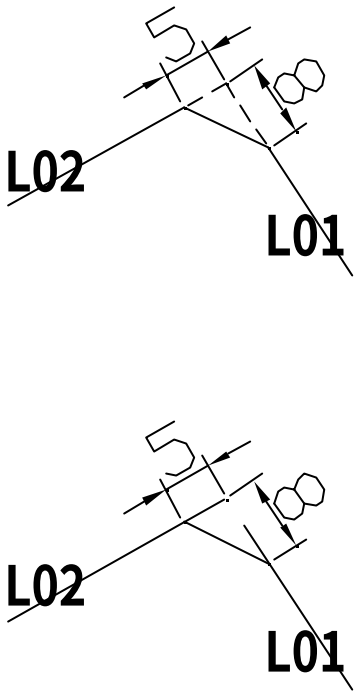
### 1.6.7 Fillet，切圓弧




1、Fillet, T, T, R, 0, L01, L02, →L01 與 L02 相交
2、Fillet, T, T, R, 10, L01, L02, →產生 R10 之弧，並修剪
3、Fillet, T, N, R, 10, L01, L02, →產生 R10 之弧，但不修剪

圖 12、範例：Fillet，切圓弧

### 1.6.8 CHAmfer，倒角

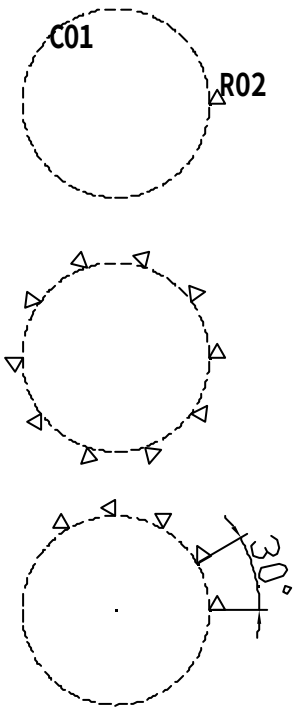



1、CHAmfer, T, T, D, 8, 5, L01, L02, →**產生倒角，並修剪**

2、CHAmfer, T, N, D, 8, 5, L01, L02, →**產生倒角, 但不修剪**

**圖** 13、**範例：**CHAmfer，**倒角**

### 1.6.9 ARray，環形陣列

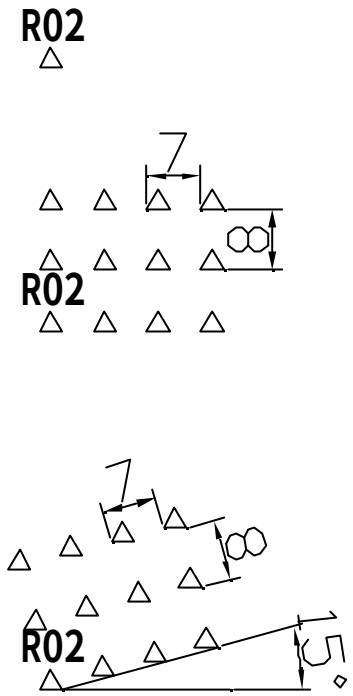



1、ARray, **環形陣列**, **中心點**, cen, C01, **項目總數**, 10, **佈滿角度**, 360, **物件**, R02, →**產生** 10 **個** R02

2、ARray, **環形陣列**, **中心點**, cen, C01, **項目總數**, 5, **項目間角度**, 30, **物件**, R02, →**產生** 5 **個** R02

**圖** 14、**範例：**ARray，**環形陣列**

## 1.6.10 ARray，矩形陣列

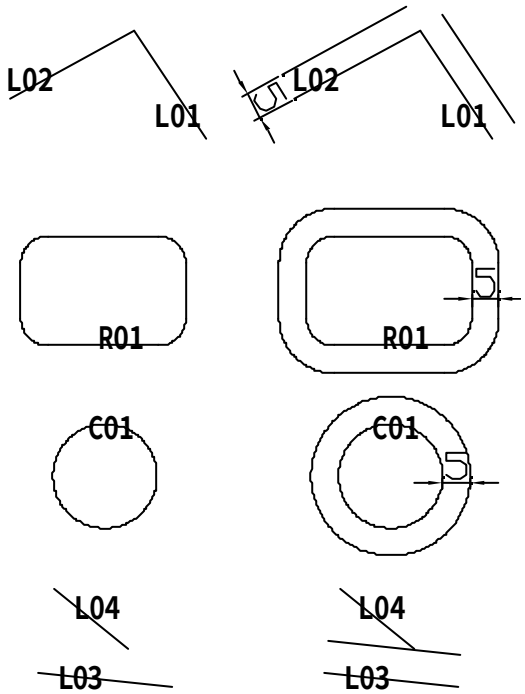



1、ARray, **矩形陣列**, **列**, 3, **行**, 4, **列偏移**, 8, **行偏移**, 7, **物件**, R02, →**產生** 12 **個** R02

2、ARray, **矩形陣列**, **列**, 3, **行**, 4, **列偏移**, 8, **行偏移**, 7, **陣列角度**, 15,**物件**, R02, →**產生** 12 **個** R02 (**角度** 15)

**圖** 15、**範例：**ARray，**矩形陣列**

## 1.6.11 Offset，偏移複製




1、Offset, 5, L01, **向上點**,

2、Offset, 5, L02, **向上點**,

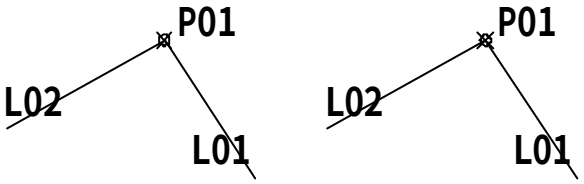
3、Offset, 5, R01, **向外點**,

4、Offset, 5, C01, **向外點**,

5、Offset, T, L03, end, L04 **下**,

**圖** 16、**範例：**Offset，**偏移複製**

### 1.6.12 COpy，複製




1、COpy, L01;L02, nod, P01, nod, P02,

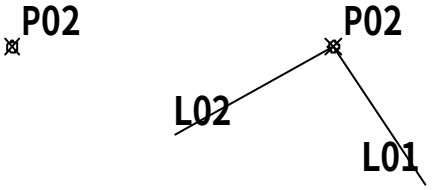
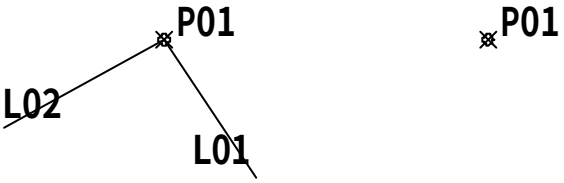



**圖** 17、**範例：**COpy，**複製**

### 1.6.13 Move，搬移




1、Move, L01;L02, nod, P01, nod, P02,

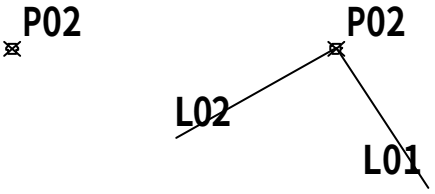
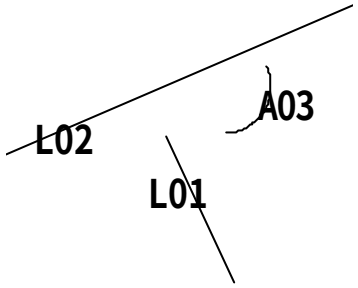



**圖** 18、**範例：**Move，**搬移**

### 1.6.14 EXtend，延伸




1、EXtend, L02, L01;A03, →L01;A03 皆與 L02 相交

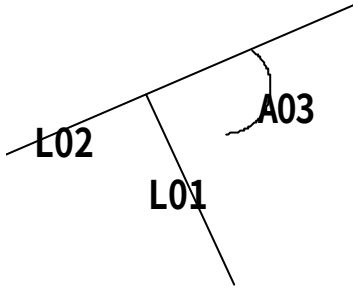
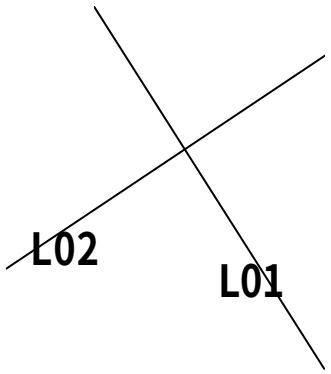



圖 19、範例：EXtend，延伸

### 1.6.15 TRim，修剪




1、TRim, L02;L01, L02 上;L01 上, →L01 與 L02 相交，且修剪

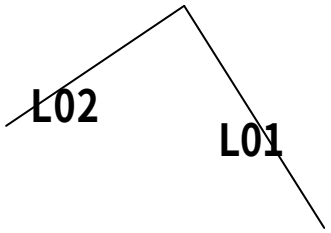
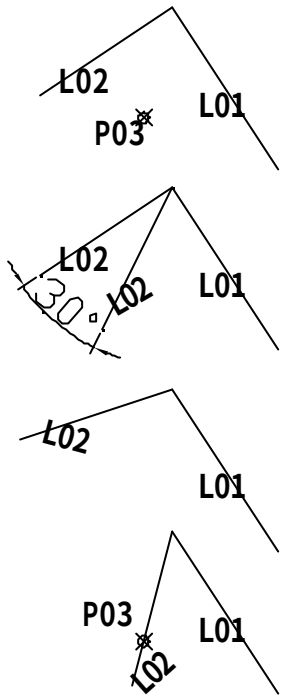



圖 20、範例：TRim，修剪

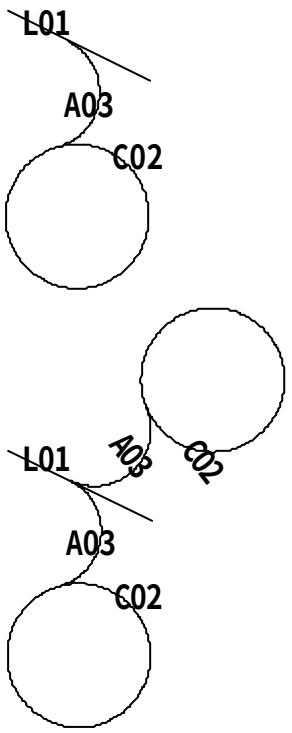
## 1.6.16 ROtate，轉動




1、ROtate, L02, int, L01 & L02, C, 30, →新 L02 與原物件呈 30 度夾角
2、ROtate, L02, int, L01 & L02, -15, →L02 轉動-15 度
3、ROtate, L02, int, L01 & L02, R, int, L01 & L02, end, L02 左, nod, P03, →L02 轉動通過 P03

圖 21、範例：ROtate，轉動

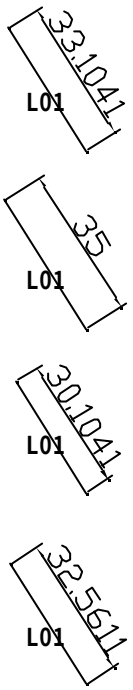
## 1.6.17 MIrror，鏡射




1、MIrror, C02;A03, end, L01 上, end, L01 下, →新 C02 與 A03 在 L01 對向

圖 22、範例：MIrror，鏡射

## 1.6.18 LENgthen，變更長度




1、LENgthen, T, 35, L01 上, →L01 總長 35
2、LENgthen, DE, -3, L01 上, →L01 縮短 3
3、LENgthen, DY, L01 上, 任意位置, →L01 長度任意變更

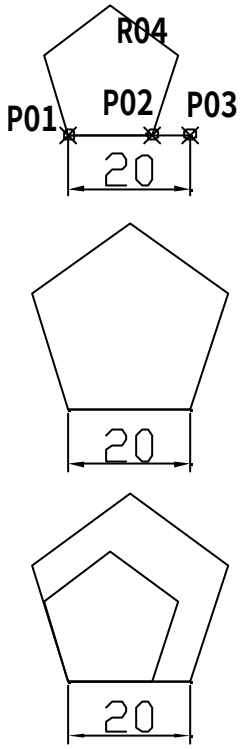
圖 23、範例：LENgthen，變更長度

## 1.6.19 DIVide，等分

1、DIVide, L01, 5, →L01 上產生 4 個點，將 L01 等分成 5 等份

圖 24、範例：DIVide，等分

## 1.6.20 SCale，放大

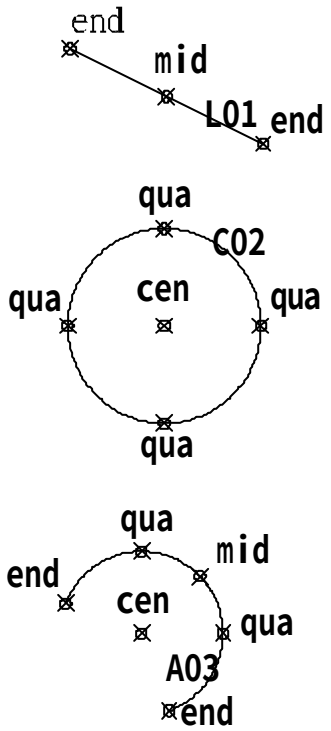



1、SCale, R04, P01, R, P01, P02, P03, →R04 **變大**

2、SCale, R04, P01, C, R, P01, P02, P03, →**原尺寸** R04 **保留，另產生新** R04 **且變大**

**圖** 25、**範例：**SCale**，放大**

## 1.6.21 物件鎖點 1



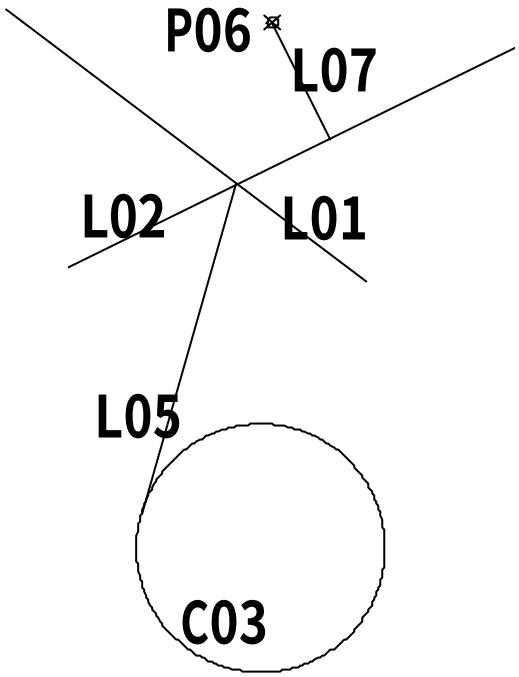

1、Line：**端點**, end;　**中點**, mid

2、Circle：**四分點**, qua;　**中心點**, cen(**四分點一定在** X **軸與** Y **軸上**)

3、Arc：**四分點**, qua; (**四分點一定在** X **軸與** Y **軸上**); **中心點**, cen　**端點**, end;　**中點**, mid; (**中點在弧之正中間**)

**圖** 26、**範例：物件鎖點** 1

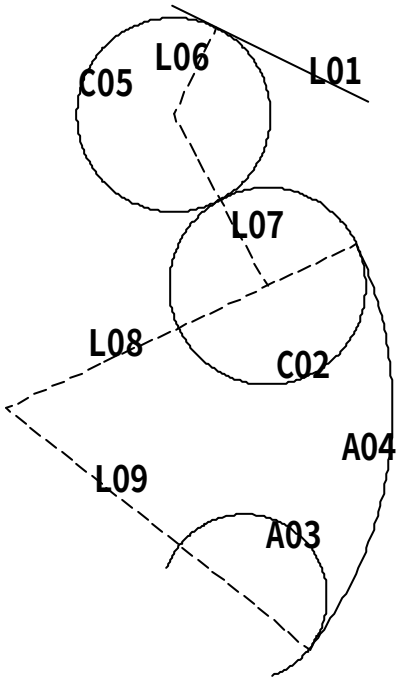
### 1.6.22 物件鎖點 2




1、Line, int, L01 & L02, tan, C03, →L05
2、POint, m2p, end, L01, end, L02, →P06
3、Line, nod, P06, per, L02, →L07

圖 27、範例：物件鎖點 2

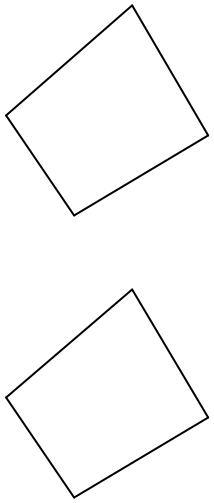
### 1.6.23 相切圓性質




1、C05 分別與 L01 及 C02 相切，則 L06 垂直於 L01,且通過 C05 圓心及 L01 與 C05 之切點。L07 則為 C02 圓心與 C05 圓心之連線且通過 C02 與 C05 之切點
2、A04 分別與 A03 及 C02 相切，則 L08 為 C02 圓心與 A04 圓心之連線且其延伸一定通過 C02 與 A04 之切點。L08 亦同。
3、Circle, T, C02, A03, 半徑值, →A04(再 TRim 之)
4、Circle, T, C02, L01, 半徑值, →C05
5、Fillet, T, N, R, 半徑值, →C05(產生右側弧段)

圖 28、範例：相切圓性質

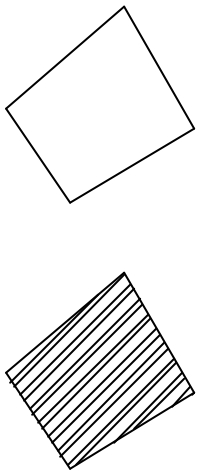
## 1.6.24 BOundary，邊界



1、BOundary, **點閉合區域**, →**封閉聚合線**
2、**點封閉聚合線**, CTRL+1, →**查幾何長度值**

**圖** 29、**範例：**BOundary，**邊界**

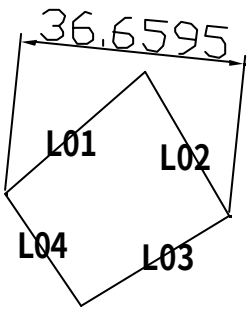
## 1.6.25 Hatch，剖面線



1、Hatch, **點閉合區域**, →**斜線區域**
2、**點斜線區域**, CTRL+1, →**查幾何面積值**

**圖** 30、**範例：**Hatch，**剖面線**

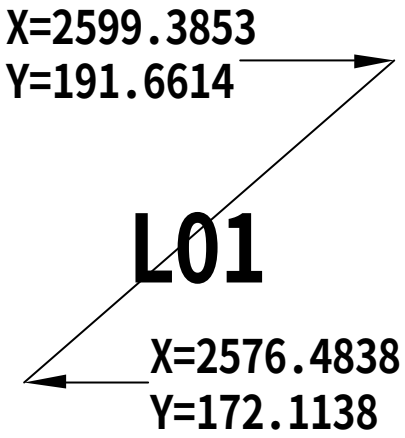
## 1.6.26 DIst，距離




1、DIst, **位置點** 1, **位置點** 2, →**兩點間距離**
2、DIst, int, L01 & L04, int, L02 & L03, →36.6595

**圖** 31、**範例：**DIst，**距離**

### 1.6.27 ID，座標值



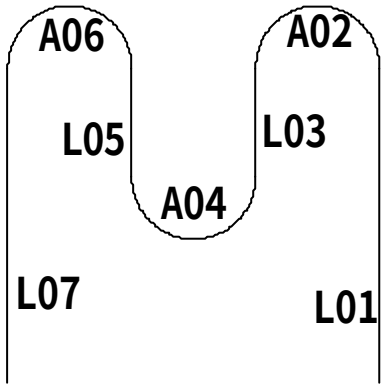

1、ID, **位置點**, →**座標值**

2、ID, end, L01 **上**, →X=2599.3853, Y=191.6614

3、ID, end, L01 **下**, →X=2576.4838, Y=172.1138

**圖** 32、**範例：**ID，**座標值**

### 1.6.28 PEdit，編輯聚合線




1、PEdit, L01, Y, J, A02;L03;A04;L05:A06;L07, →S01

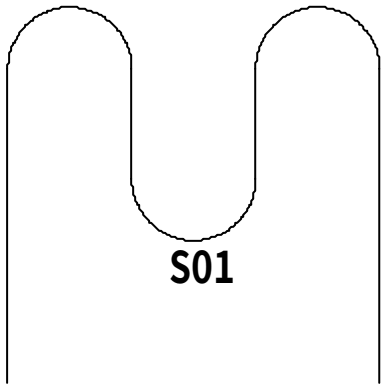



**圖** 32、**範例：** PEdit，**編輯聚合線**

## 1.7 AutoCAD 2D 常用指令集

本解題秘笈將 AutoCAD 2D 常用之指令與物件鎖點分別整理成表 4 與表 5, 供讀者參考。讀者如能熟諳，即能達成以「指令繪圖－上圖下文、左鍵右鼠」之目標。

表 4、AutoCAD 2D 常用指令集

| 項次 | 指令功能 | 一般指令 | 快捷指令 |
|---|---|---|---|
| 01 | 畫弧 | Arc | A |
| 02 | 矩/環形陣列複製 | ARray | AR |
| 03 | 建立封閉邊界 | BOundary | BO |
| 04 | 畫圓 | Circle | C |
| 05 | 畫倒角 | CHAmfer | CHA |
| 06 | 複製物件 | COpy | CO |
| 07 | 查兩點距離值與平面角度值 | DIst | DI |
| 08 | 將物件等分(產生等分點) | DIVide | DIV |
| 09 | 刪除物件 | Erase | E |
| 10 | 將物件延伸 | EXtend | EX |
| 11 | 畫圓角 | Fillet | F |
| 12 | 群組物件 | Group | G |
| 13 | 畫剖面線 | Hatch | H |
| 14 | 查點座標值 | ID | ID |
| 15 | 畫線 | Line | L |
| 16 | 調整物件長度 | LENgthen | LEN |
| 17 | 搬移物件 | Move | M |
| 18 | 鏡射物件 | MIrror | MI |
| 19 | 偏移複製物件 | Offset | O |
| 20 | 編輯聚合線 | PEdit | PE |
| 21 | 畫點 | POint | PO |
| 22 | 畫多邊形 | POLygon | POL |
| 23 | 畫矩形 | RECtangle | REC |
| 24 | 旋轉物件 | ROtate | RO |
| 25 | 比例縮放物件 | SCale | SC |
| 26 | 修剪物件 | TRim | TR |
| 27 | 炸開物件 | eXplode | X |
| 28 | 無限延伸建構線 | XLine | XL |
| 29 | 縮放 | Zoom | Z |

**表 5、AutoCAD 2D 常用物件鎖點表**

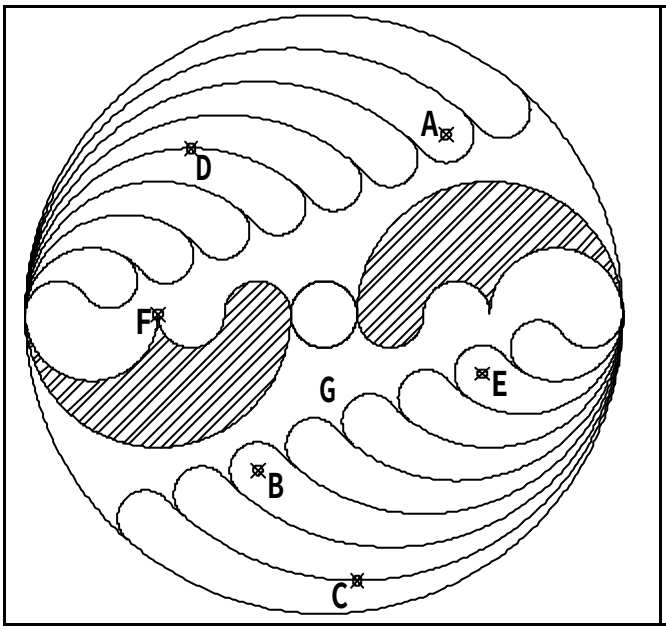
| 項次 | 指令功能 | 一般指令 |
|---|---|---|
| 01 | 指令中計算式 | 'cal |
| 02 | 選取全部物件 | all |
| 03 | 鎖定物件之端點 | end |
| 04 | 鎖定物件之中點 | mid |
| 05 | 鎖定物件之實際交點 | int |
| 06 | 鎖定兩物件之延伸交點 | app |
| 07 | 鎖定物件之中心點 | cen |
| 08 | 鎖定物件之四分點 | qua |
| 09 | 鎖定物件之切點 | tan |
| 10 | 鎖定與物件垂直 | per |
| 11 | 鎖定點物件 | nod |
| 12 | 鎖定與物件平行 | par |
| 13 | 鎖定兩點間之中點 | mtp |
| 14 | 鎖定兩點間之中點 | m2p |
| 15 | 鎖定上次點 | @ |
| 16 | 執行上次指令 | 空白鍵 |
| 17 | 執行上次指令 | ENTER 鍵 |
| 18 | 取消指令 | ESC 鍵 |
| 19 | 物件鎖點開關 | F3 鍵 |
| 20 | 物件鎖點開關 | CTRL+F |
| 21 | 正交模式開關 | F8 鍵 |
| 22 | 正交模式開關 | ORTHO |
| 23 | 鎖點模式開關 | F9 鍵 |
| 24 | 動態輸入開關 | F12 鍵 |
| 25 | 查詢物件性質 | CTRL+1 |
| 26 | 選取全部物件 | CTRL+A |

# 第2章 檢定題庫

Auto CAD 2008 2D 之 TQC 術科 2D 檢定題庫，計 6 大類，每類計 15 題，共 90 題，為遵重智慧財產權，每題之圖形與尺寸資料請參閱 Auto CAD 2008 實力養成暨評量乙書[1]，本解題秘笈僅就解題步驟說明。

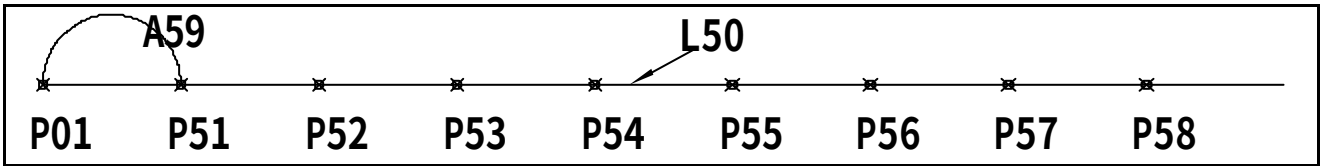
## 2.1 第一類：綜合應用一

### 2.1.1 AutoCAD2D2008-101

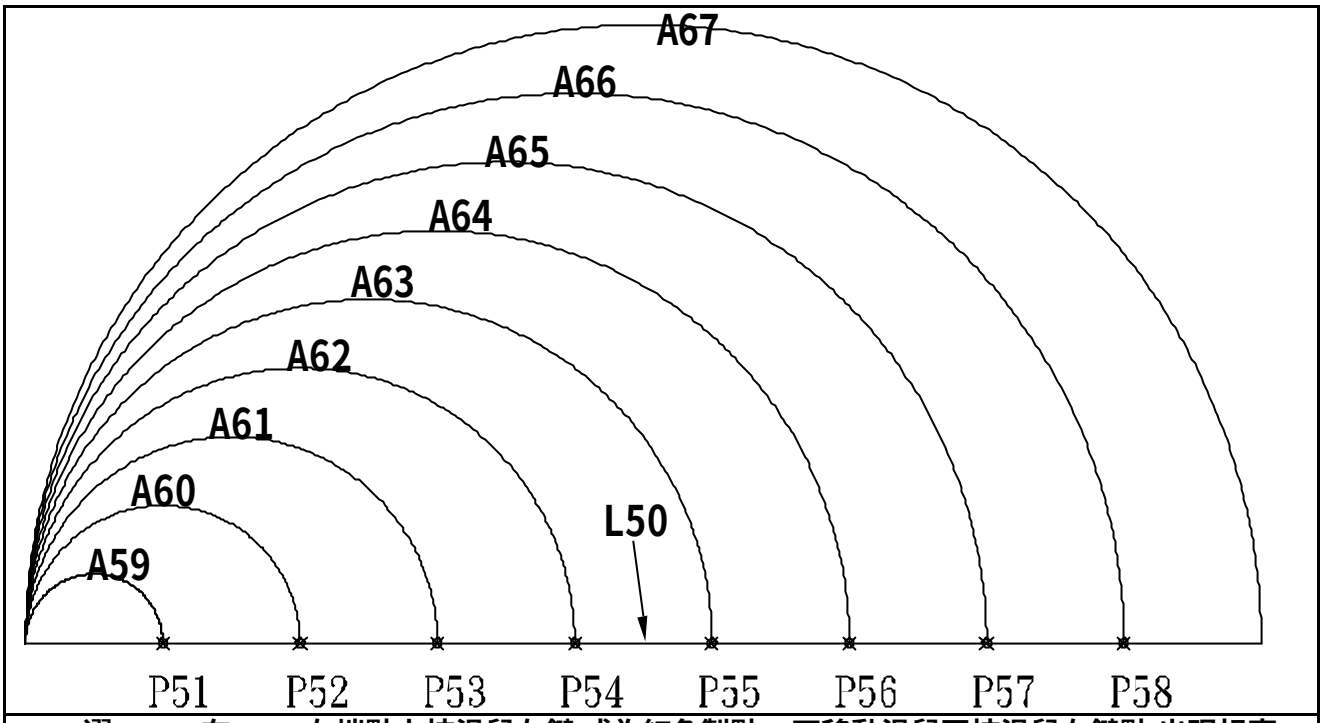



1、 中心點 A 至中心點 B 距離為何？ 79.4578
2、 四分點 C 至四分點 D 高度為何？ 89.1944
3、 中心點 E 至交點 F 距離為何？ 68.1480
4、 斜線區域總面積為何？ 1774.6799
5、 G 區域周長為何？ 455.3037

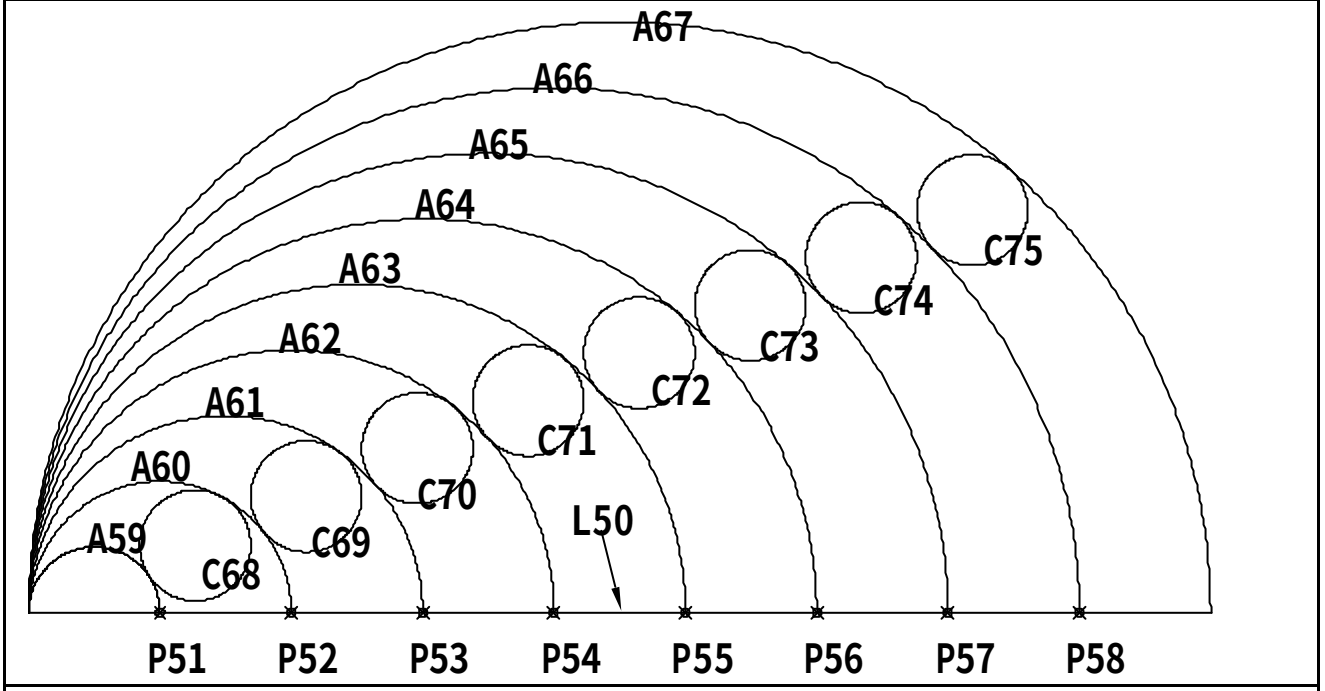
**解題步驟：**



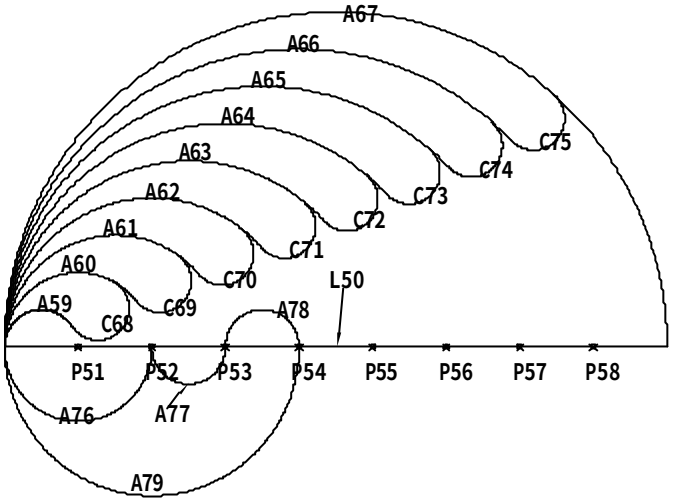

01、 Line, 任一點 P01, (正交模式), 向右, 123.5, →L50
02、 DIVide, L50, 9, →P51~P58
03、 Arc, C, m2p(mtp), nod, P51, end, L50 左, nod, P51, end, L50 左, →A59



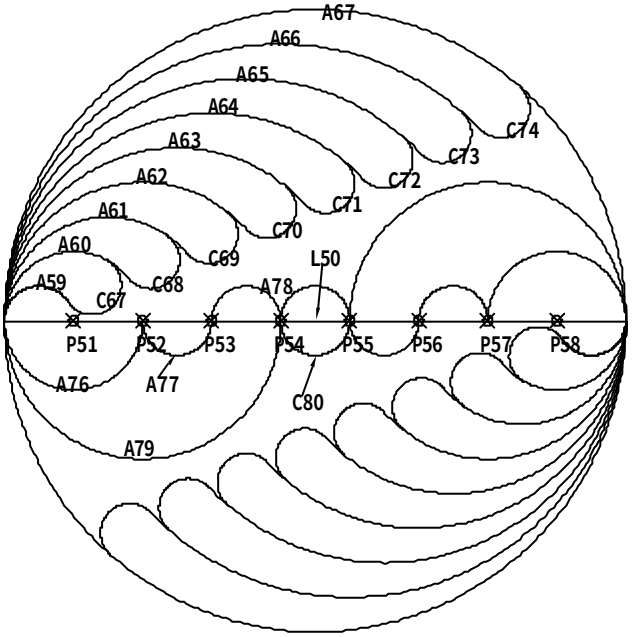

04、 **選** A59, **在** L50 **左端點上按滑鼠左鍵(成為紅色掣點)**, **不移動滑鼠再按滑鼠右鍵點(出現視窗)**, **點比例**, **鍵盤輸入** C, **鍵盤輸入** R, end, L50 **左**, nod, P51, nod, P52, →A60, nod, P53, →A61, nod, P54, →A62, nod, P55, →A63 nod, P56, →A64, nod, P57, →A65, nod, P58, →A66, end, L50 **右**, →A67



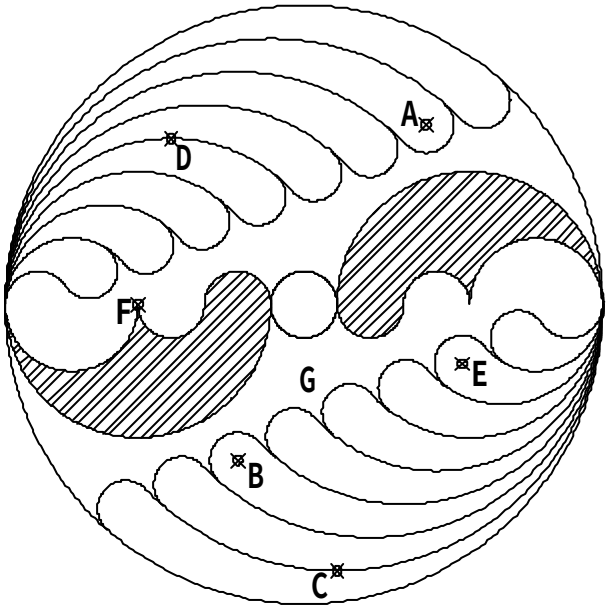

05、 Circle, T, A59, A60, 5.8, →C68
06、 Circle, T, A60, A61, 5.8, →C69
07、 Circle, T, A61, A62, 5.8, →C70
08、 Circle, T, A62, A63, 5.8, →C71
09、 Circle, T, A63, A64, 5.8, →C72
10、 Circle, T, A64, A65, 5.8, →C73
11、 Circle, T, A65, A66, 5.8, →C74
12、 Circle, T, A66, A67, 5.8, →C75




13、 TRim,
14、 Arc, C, nod, P51, end, L50 **左**, nod, P52, →A76
15、 Arc, C, m2p(mtp), nod, P52, nod, P53 nod, P52, nod, P53, →A77
16、 Arc, C, m2p(mtp), nod, P54, nod, P53, nod, P54, nod, P53, →A78
17、 Arc, C, nod, P52, end, L50 **左**, nod, P54, →A79

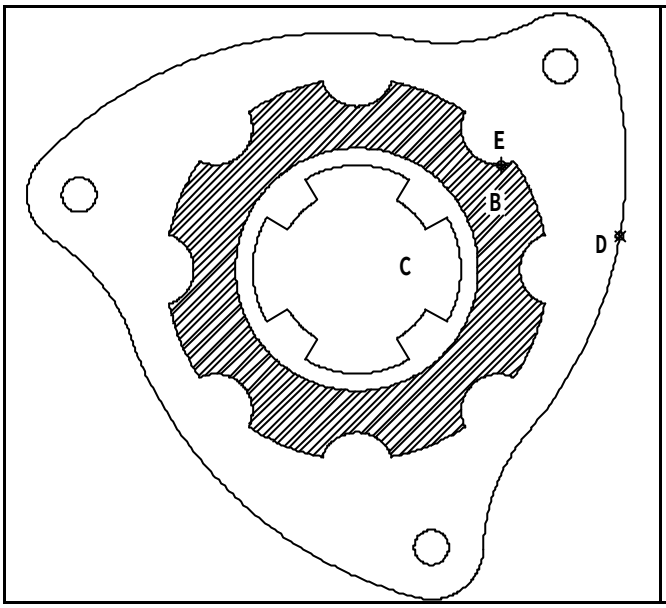



18、 ROtate, **除** L50 **外之所有物件**, cen, A67, C, 180, →**上下**, **左右對稱物件**
19、 Circle, 2P, nod, P55, nod, P54, →C80



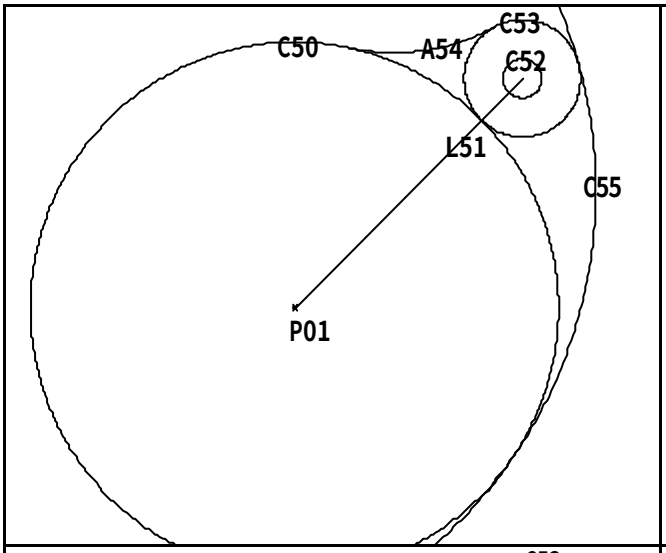

20、 Erase,
21、 Hatch, →**產生斜線區域**
22、 DIst, cen, A, cen, B, →**距離**=79.4578
23、 DIst, qua,, C, qua, D, →Y **差值**=89.1944
24、 DIist, cen, E, int, F, →**距離**=68.1480,
25、 **點斜線區域**, CTRL+1, →**查幾何面積值**=1774.6799
26、 BOundary, **點** G **區域**,
27、 **點** G **區域邊緣**, CTRL+1, →**查幾何長度值**=455.3037

## 2.1.2 AutoCAD2D2008-102

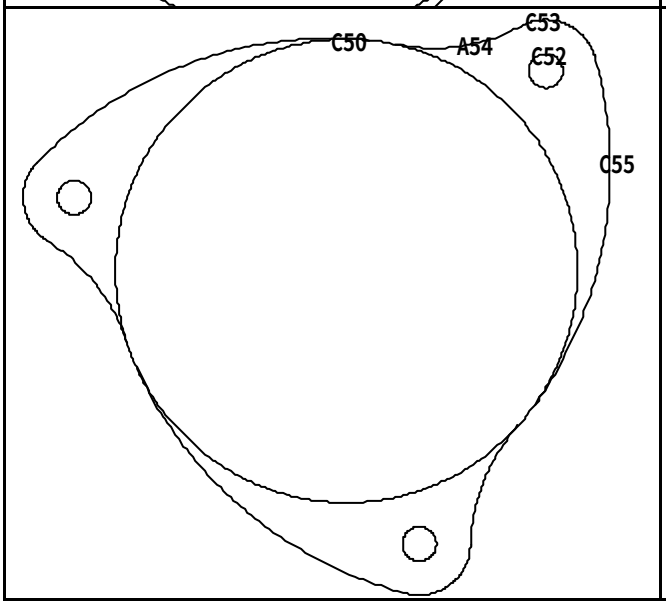



1、 **圖形最外圍周長為何？** 541.4946
2、 **斜線** B **區域面積為何？** 4730.2428
3、 C **區域周長為何？** 229.8384
4、 **弧** D **夾角為何？** 51.2169
5、 **弧** E **半徑為何？** 10.6005

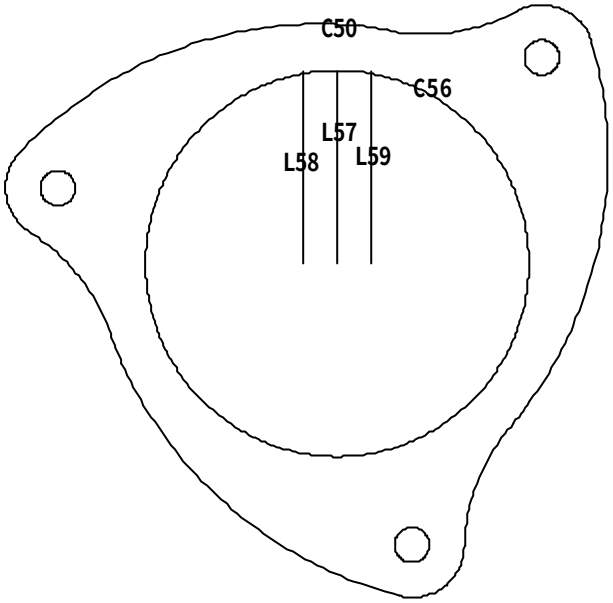
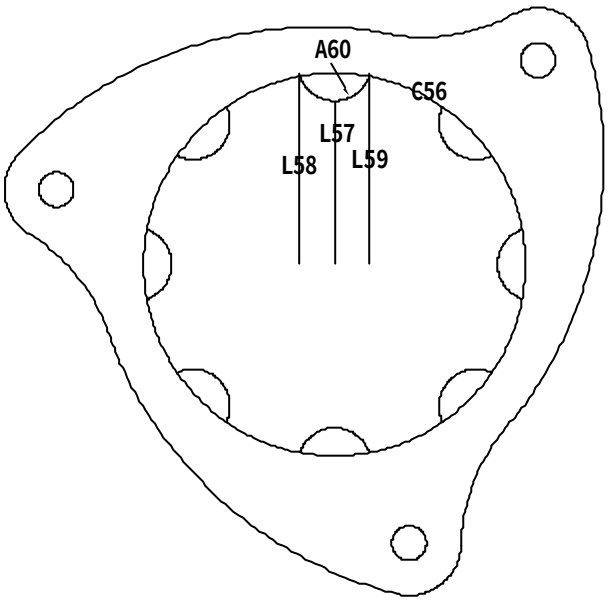
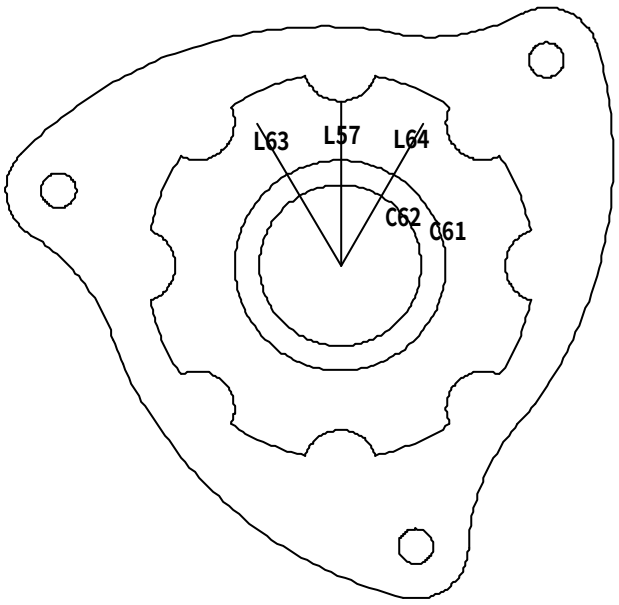
**解題步驟：**

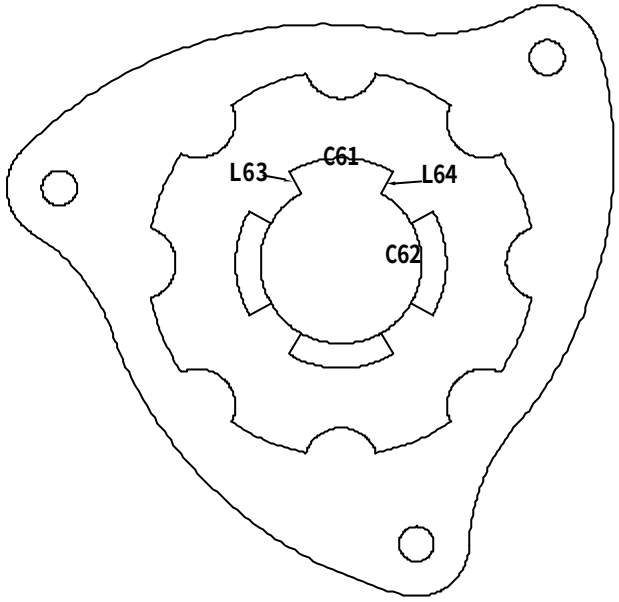
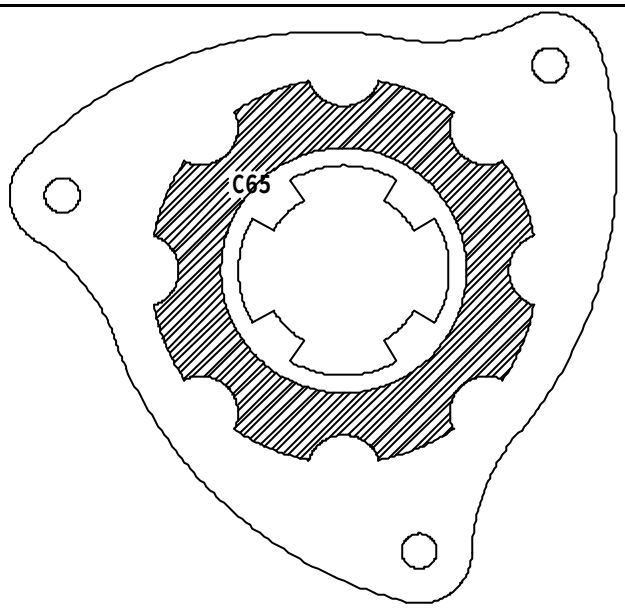
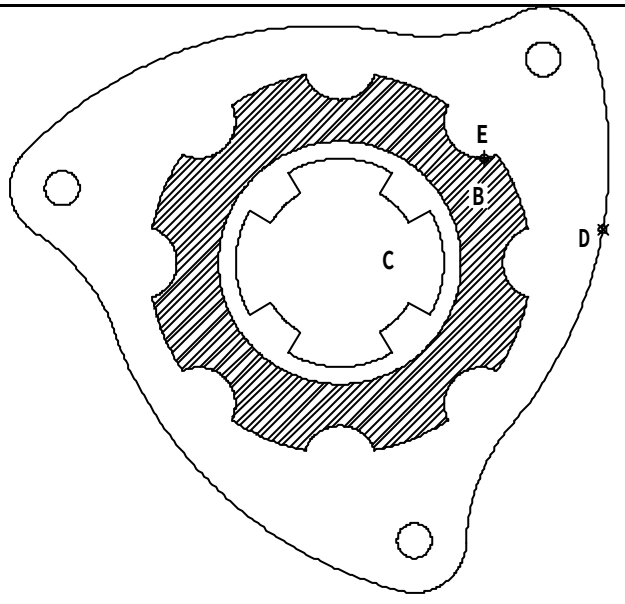



01、 Circle, **任一點** P01, D, 136, →C50
02、 Line, cen, C50, @83<45, →L51
03、 Circle, end, L51 **上**, D, 10, →C52
04、 Circle, @(**鎖定上次點**), 15, →C53
05、 Fillet, T, N, R, 55, C50 **上**, C53 **上**, →A54
06、 Circle, T, C50 **右下**, C53 **右上**, 120,→C55

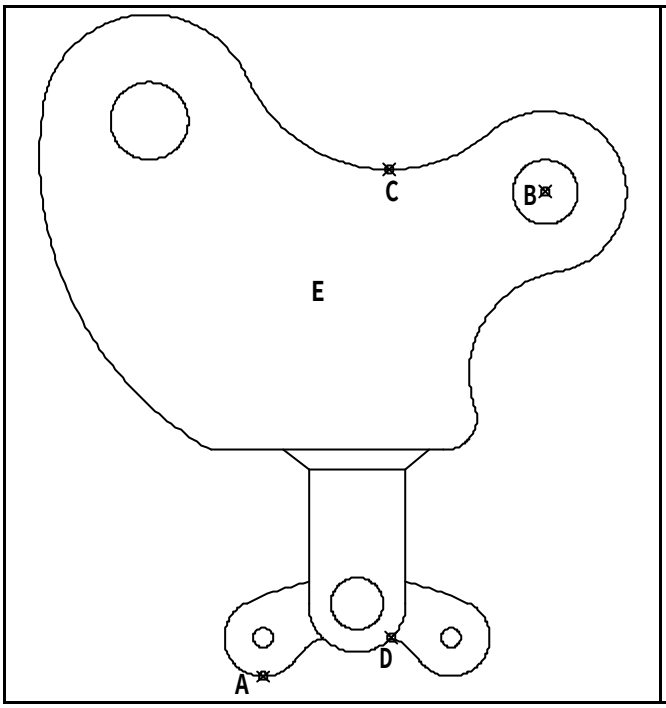



07、 TRim,
08、 Erase,
09、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 3, **佈滿角度**, 360, **物件**, C52;C3;C54;C55, →**相同物件** 3 **組**

| | |
|---|---|
|  | 10、 Circle, cen, C50, D, 110, →C56<br>11、 TRim,<br>12、 Line, cen, C56, qua, C56 **上**, →L57<br>13、 Offset, 10, L57, **向左點**, →L58<br>14、 Offset, 10, L57, **向右點**, →L59 |
|  | 15、 LENgthen, DE, -8, L57 **上**, →L57 **縮短**<br>16、 Arc, int, L58 & C56, end, L57 **上**, int, L59 & C56, →A60<br>17、 ARray, **環形陣列**, **中心點**, cen, C56, **項目總數**, 8, **佈滿角度**, 360, **物件**, A60, **→相同物件** 8 **個** |
|  | 18、 TRim,<br>19、 Erase,<br>20、 Circle, end, L57 **下**, D, 60, →C61<br>21、 Circle, @(**鎖定上次點**), D, 46, →C62<br>22、 ROtate, L57, cen, C62, C, 30, →L63<br>23、 ROtate, L57, cen, C62, C, -30, →L64 |

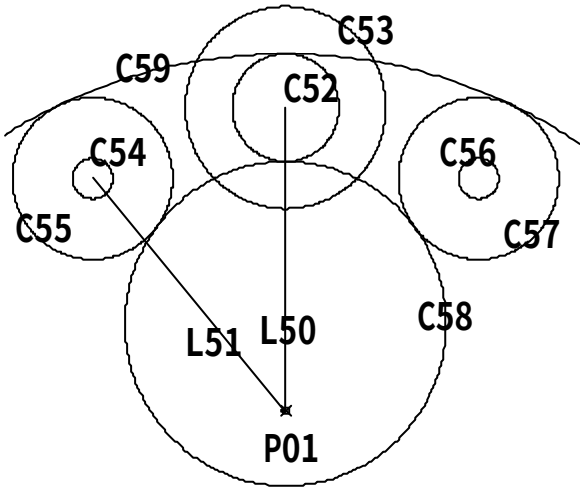
| | |
|---|---|
|  | 24、 TRim,<br>25、 Erase,<br>26、 ARray, **環形陣列**, **中心點**, cen, C62, **項目總數**, 4, **佈滿角度**, 360, **物件**, C61;L63;L64, **→相同物件** 4 **組** |
|  | 27、 TRim,<br>28、 Circle, cen, C 61, D, 70, →C65<br>29、 Hatch, B, **移除最內圈**, **→產生斜線區域** |
|  | 30、 BOundary, **點** D **區**, **點最外圍邊緣**, CTRL+1, **→查幾何長度值**=541.4946<br>31、 **點斜線區域**, CTRL+1, **→查幾何面積值** =4730.2428<br>32、 BOundary, **點** C **區**,<br>33、 **點** C **區邊緣**, CTRL+1, **→查幾何長度值** =229.8348<br>34、 **點弧** D, CTRL+1, **→查幾何總角度值** =51.2169<br>35、 **點弧** E, CTRL+1, **→查幾何半徑值**=10.6005 |

## 2.1.3 AutoCAD2D2008-103

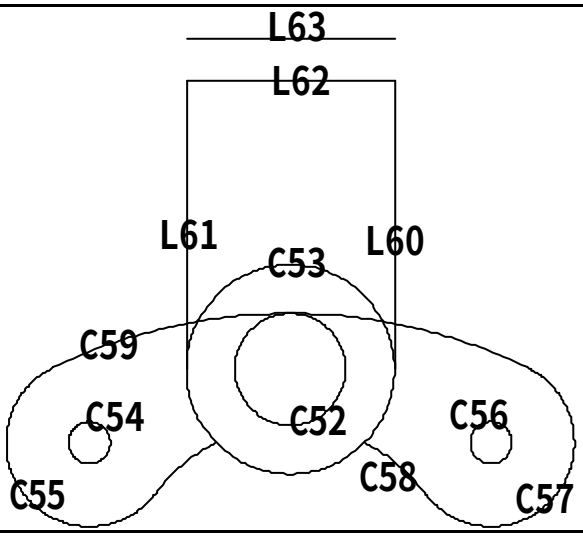



1、 **圖形最大高度為何？** 205.0280
2、 **弧** A **中心點至中心點** B **距離為何？** 163.1189
3、 **區域** E **扣除內孔面積為何？**14672.9551
4、 **弧** C **中心點至交點** D **距離為何？** 193.0122
5、 **圖形最外圍的周長為何？** 758.2307

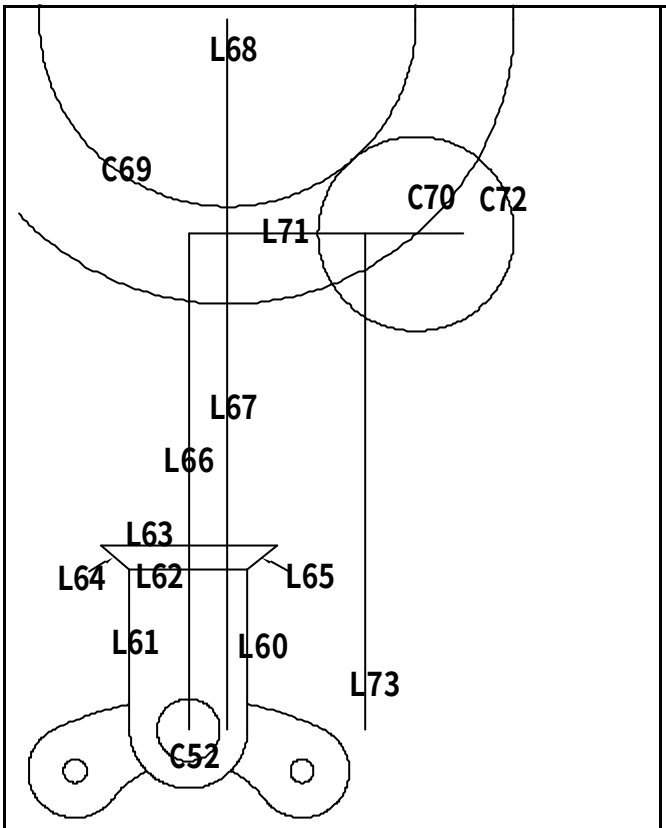
**解題步驟：**



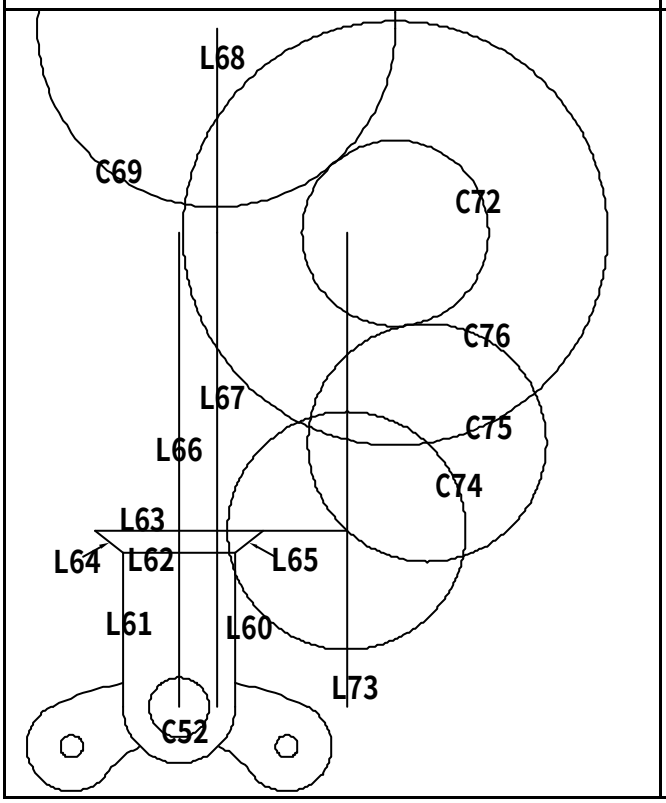

01、 Line, **任一點** P01, (**正交模式**), **向上** 45, →L50
02、 ROtate, L50, end, L50 **下**, C, 40, →L51
03、 Circle, end, L50 **上**, D, 16, →C52
04、 Circle, @(**鎖定上次點**), 15, →C53
05、 Circle, end, L51 **上**, D, 6, →C54
06、 Circle, @(**鎖定上次點**), D, 24, →C55
07、 MIrror, C54;C55, end, L50 **上**, end, L50 **下**, →C56;C57
08、 Circle, 3P, tan, C57 **左下**, tan, C52 **下**, tan, C55 **右下**, →C58
09、 Circle, 3P, tan, C57 **右上**, tan, C52 **上**, tan, C55 **左上**, →C59



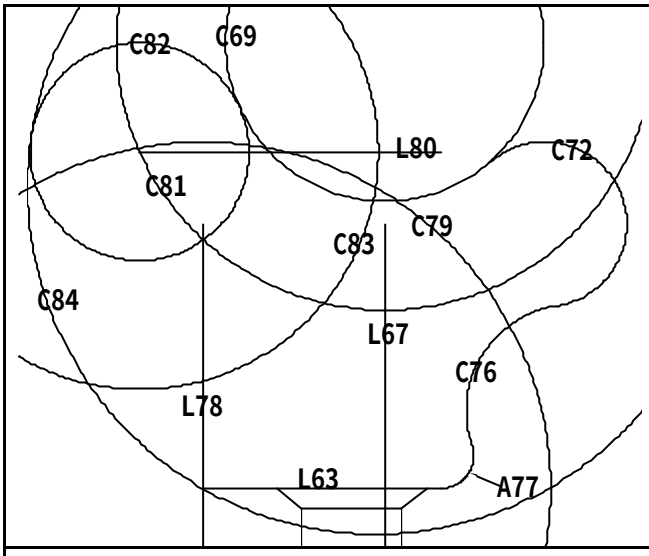
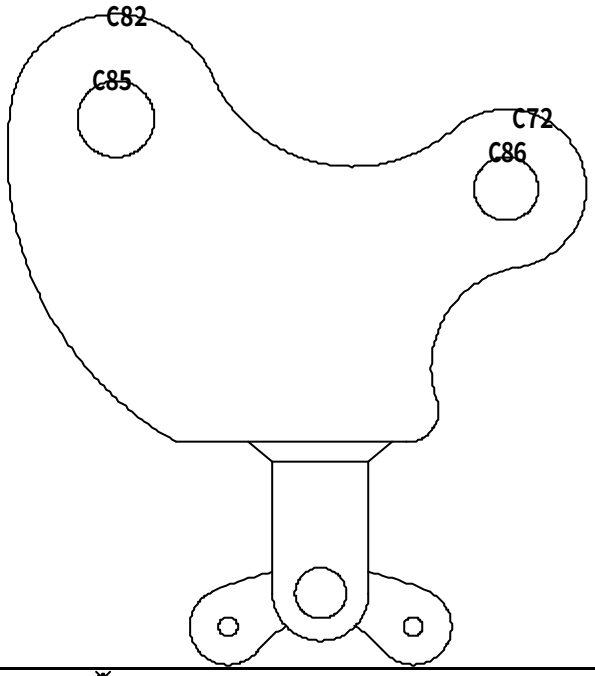
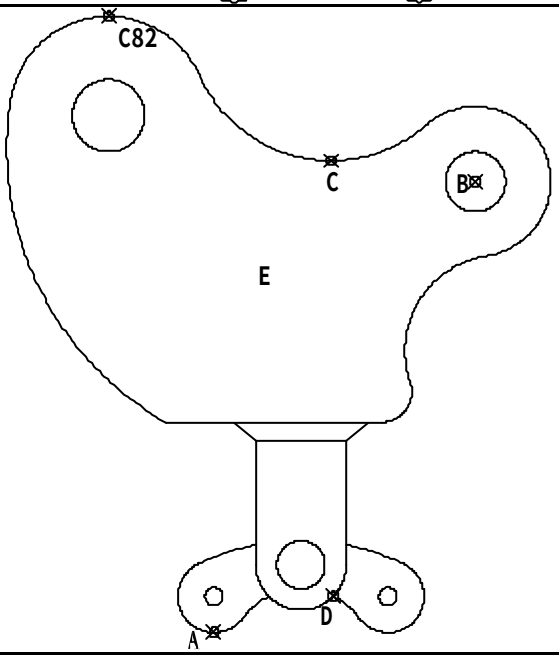

10、 Erase,
11、 TRim,
12、 Line, qua, C53 **右**, (**正交模式**), **向上** 41.5(=47.5-6), →L60
13、 Line, qua, C53 **左**, (**正交模式**), **向上** 41.5(=47.5-6), →L61
14、 Line, end, L60 **上**, end, L61 **上**, →L62
15、 Offset, 6, L62, **向上點**, →L63



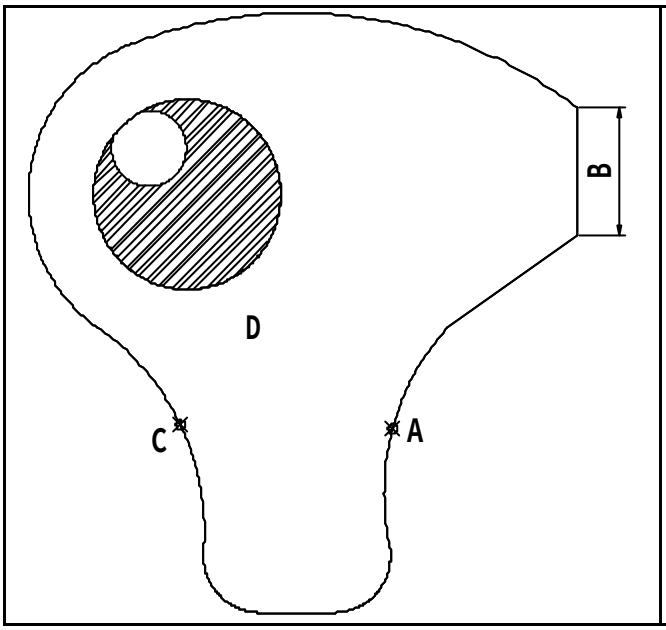

16、 LENgthen, DE, 7.5, L63 **左**, →L63 **向左延伸**, L63 **右**, →L63 **向右延伸**
17、 Line, end, L63 **左**, int, L61 & L62, →L64
18、 Line, end, L63 **右**, int, L60 & L62, →L65
19、 Line, cen, C52, (**正交模式**), **向上** 127.5, →L66
20、 Offset, 10, L66, **向右點**, →L67
21、 Line, end, L67 **上**, (**正交模式**), **向上** 55, →L68
22、 Circle, end, L68 **上**, 48, →C69
23、 Offset, 25, C69, **向外點**, →C70
24、 Line, end, L66 **上**, (**正交模式**) **向右** 70, →L71
25、 Circle, int, L71 & C70, 25, →C72
26、 Offset, 35, L67, **向右點**, →L73




27、 Erase,
28、 EXtend, L73, L63 **右**, →L63 **向右延伸**
29、 Circle, int, L63 & L73, 32, →C74
30、 Offset, 32, C72, **向外點**, →C75
31、 Circle, int, C74 & C75, 32, →C76

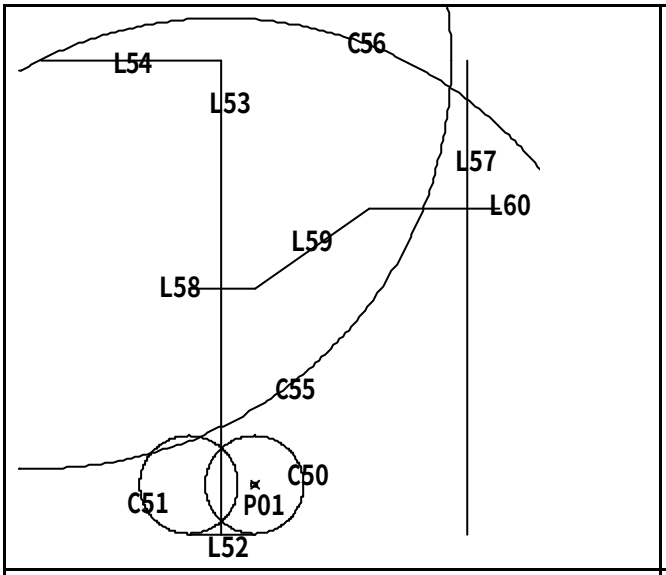
| | |
|---|---|
|  | 32、 Erase,<br>33、 TRim,<br>34、 Fillet, T, T, R, 10, C76 **左下**, L63 **右**, →A77<br>35、 Offset, 55, L67, **向左點**, →L78<br>36、 EXtend, L78, L63 **左**, →L63 **左延伸**<br>37、 Circle, int, L63 & L78, 105, →C79<br>38、 Offset, 102, L63, **向上點**, →L80<br>39、 Offset, 33, C69, **向外點**, →C81<br>40、 EXtend, C81, L80 **左**, →L80 **左延伸**<br>41、 Circle, int, L80 & C81, 33, →C82<br>42、 Circle, cen, C82, 72(=105-33), →C83<br>43、 Circle, int, C79 & C83, 105, →C84 |
|  | 44、 TRim,<br>45、 Erase,<br>46、 Circle, cen, C82, D, 24, →C85<br>47、 Circle, cen, C72, D, 20, →C86 |
|  | 48、 DIst, qua, C82, qua, A, →Y **差值**=205.0280<br>49、 DIst, cen, A, cen, B, →**距離**=163.1189<br>50、 Hatch, **點** E **區域**, →**產生斜線**<br>51、 **點** E **區域斜線**, CTRL+1, →**查幾何面積值** =14672.9551<br>52、 DIst, cen, C, int, D, →**距離**=193.0122<br>53、 **另產生一只有最外圍之物件**,<br>54、 BOundary, **點選內面**, **點選最外圍邊緣**, CTRL+1, →**查幾何長度值**=758.2307 |

## 2.1.4 AutoCAD2D2008-104

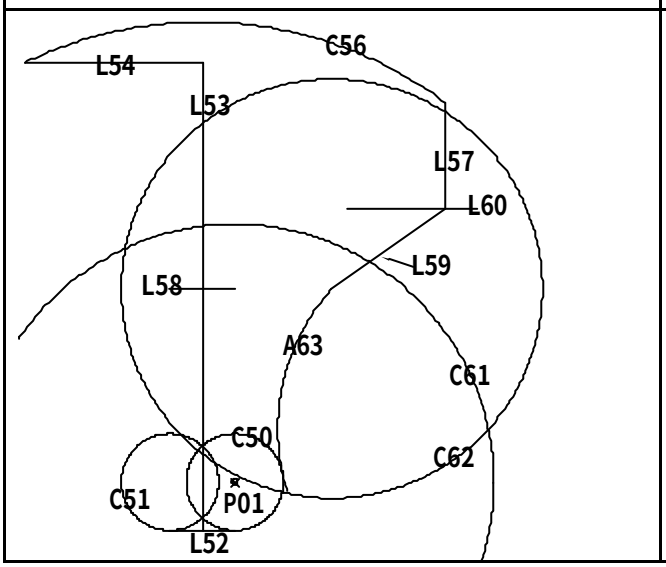



1、 **弧** A **長度為何？** 61.1784
2、 **尺寸** B **長度為何？** 33.2586
3、 **弧** A **至弧** C **中心距離為何？** 178.3374
4、 **斜線區域面積為何？**1655.0205
5、 **圖形最外圍的周長為何？** 495.1087

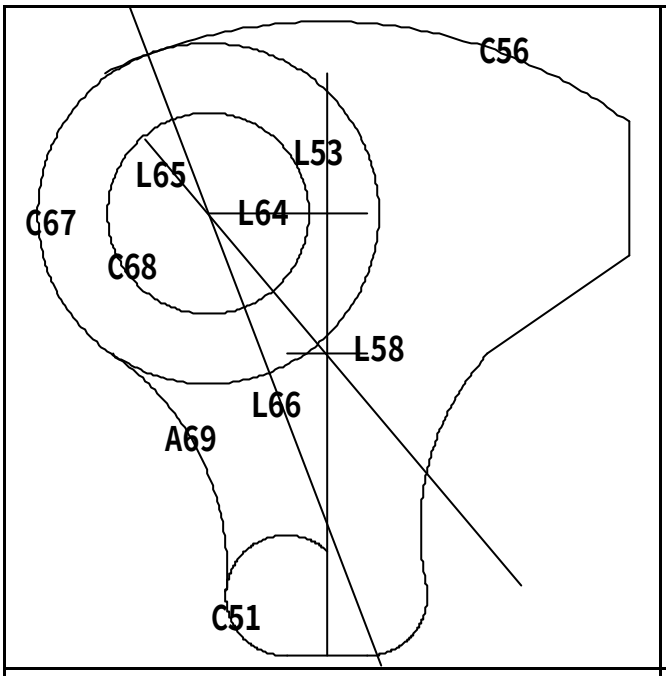
**解題步驟：**



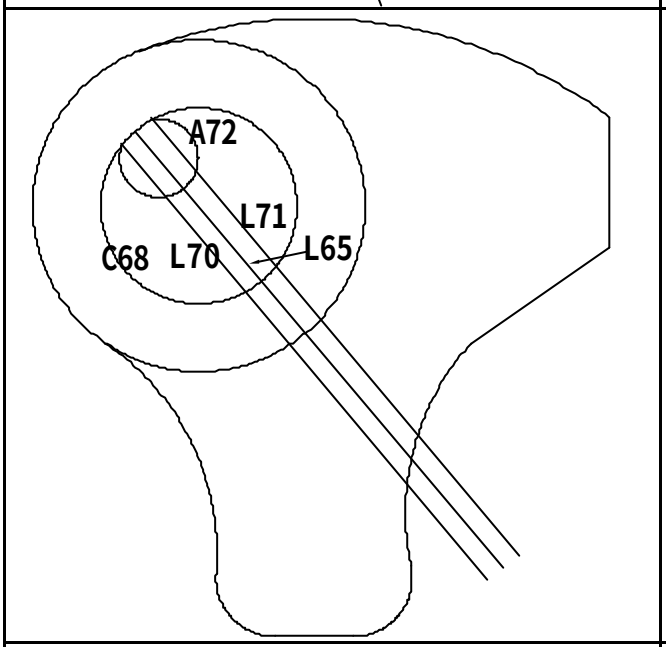

01、 Circle, **任一點** P01, 15, →C50
02、 COpy, C50, cen, C50, (**正交模式**), **向左** 20, →C51
03、 Line, qua, C50 **下**, qua, C51 **下**, →L52
04、 Line, mid, L52, (**正交模式**), **向上** 145(=15+60+35+35), →L53
05、 Line, end, L53 **上**, (**正交模式**), **向左** 55, →L54
06、 Circle, end, L54 **左**, 125, →C55
07、 Circle, int, L53 & C55, 125, →C56
08、 Offset, 75, L53, **向右點**, →L57
09、 Offset, 75(=15+60), L52, **向上點**, →L58
10、 Line, end, L58 **右**, @42.7<35, →L59
11、 Line, end, L59 **上**, (**正交模式**), **向右** 40, →L60



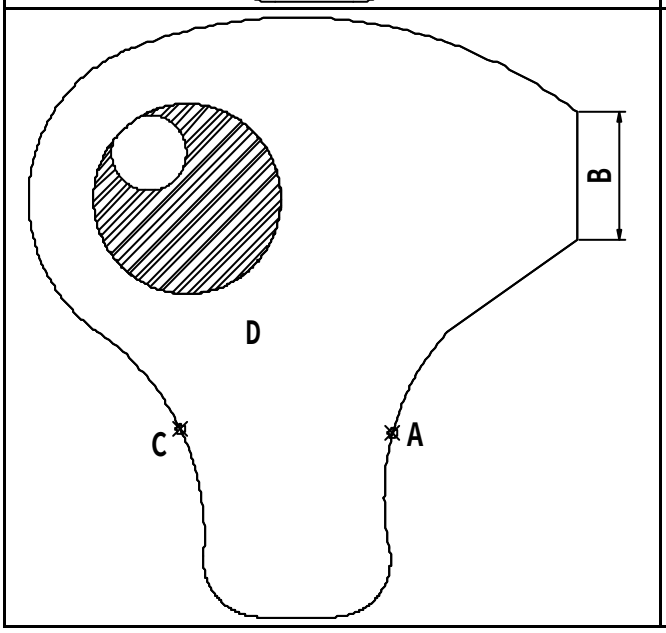

12、 TRim,
13、 Erase,
14、 Move, L59, end, L60 **左**, int, L57 & L60, →L59 **向左移**
15、 Offset, 65, C50, **向外點**, →C61
16、 Circle, end, L59 **下**, 65, →C62
17、 Arc, C, int, C61 & C62, end, L59 **下**, int, C50 & C62, →A63




18、 TRim,
19、 Erase,
20、 Offset, 35, L58, **向上點**, →L64
21、 ROtate, L53, int, L53 & L58, C, 40, →L65
22、 EXtend, L65, L64 **左**, →L64 **向左延伸**
23、 XLine, cen, C56, int, L64 & L65, →L66
24、 Circle, int, L64 & L65, int, C56 & L66, →C67
25、 Circle, @(**鎖定上次點**), D, 50, →C68
26、 Fillet, T, N, R, 65, C51 **左**, C67 **右下**, →A69

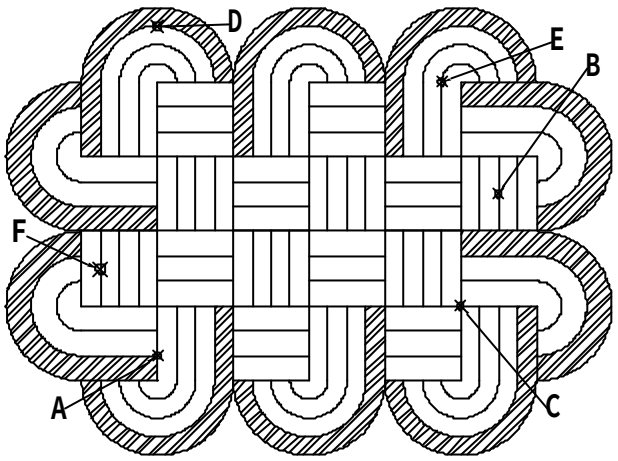



27、 TRim,
28、 Erase,
29、 EXtend, C68, L65, →L65 **與** C68 **相交**
30、 Offset, 5, L65, **向左下點**, →L70, L65, **向右上點**, →L71
31、 Arc, int, C68 & L70, E, int, C68 & L71, R, -10, →A72



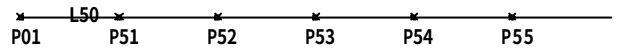

32、 Erase,
33、 Hatch, →**產生斜線區域**
34、 **點選弧** A, CTRL+1, →**查幾何弧長值** =61.1784
35、 Dist, int, B **上**, int, B **下**, →**距離**=33.2586
36、 Dist, cen, A, cen, C, →**距離**=178.3374
37、 **點選斜線區域**, CTRL+1, →**查幾何面積值** =1655.0205
38、 BOundary, **點選** D **區域**,
39、 **點選最外圍邊緣**, CTRL+1, →**查幾何長度值** =495.1087

## 2.1.5 AutoCAD2D2008-105



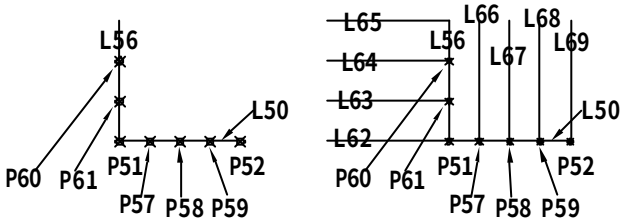

1、 **交點** A **至中點** B **距離為何？** 102.8023
2、 **交點** C **至四分點** D **角度為何？** 136.8476
3、 **交點** E **至中點** F **距離為何？** 105.9592
4、 **斜線區域總面積為何？** 4426.2590
5、 **圖形所圍成的面積為何？** 16823.2162

**解題步驟：**

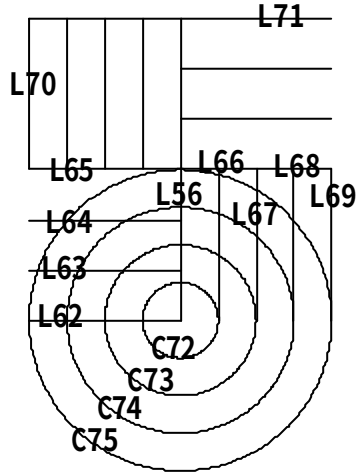



01、 Line, **任一點** P01, (**正交模式**), **向右**, 123.5, →L50
02、 DIVide, L50, 6, →P51~P55

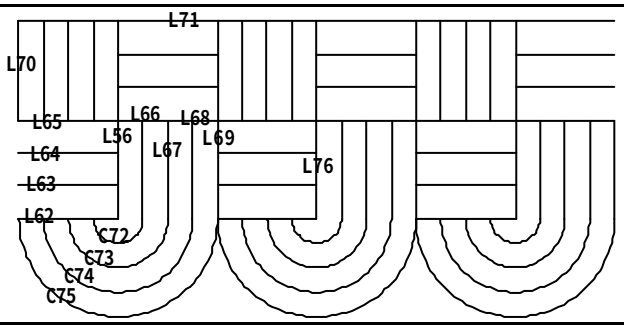
03、 LENgthen, DY, L50 **右**, nod, P52, →L50 **向左縮短至** P52, L50 **左**, nod, P51, →L50 **向右縮短至** P51



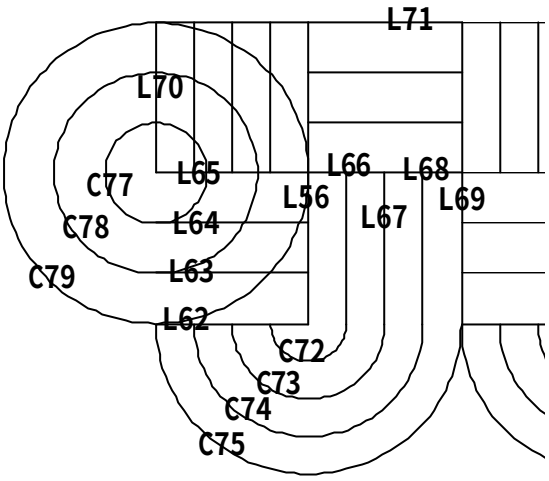

04、 ROtate, L50, nod, P51, C, 90, →L56
05、 DIVide, L50, 4, →P57~P59
06、 DIVide, L56, 3, →P60~P61
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
07、 COpy, L50, nod, P52, nod, P51, →L62, nod, P61, →L63, nod, P60, →L64, end, L56 **上**, →L65
08、 COpy, L56, nod, P51, nod, P57, →L66, nod, P58, →L67, nod, P59, →L68, nod, P52, →L69



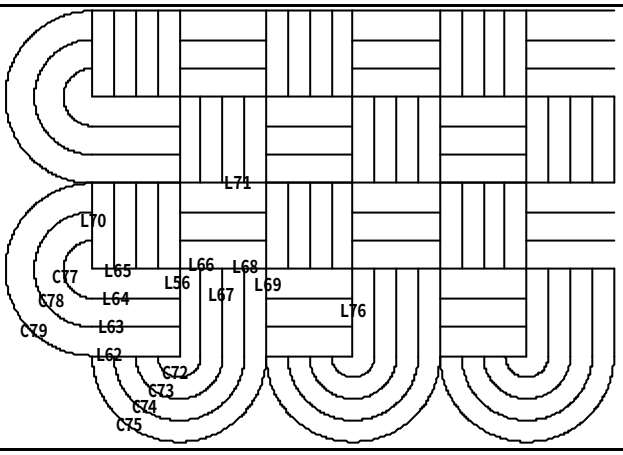

09、 Erase, **所有點**
10、 ROtate, **所有線段物件**, end, L56 **上**, C, 180, →L70;L71 **等線段物件**
11、 Circle, end, L56 **下**, end, L66 **下**, →C72
12、 Circle, @(**鎖定上次點**), end, L67 **下**, →C73
13、 Circle, @(**鎖定上次點**), end, L68 **下**, →C74
14、 Circle, @(**鎖定上次點**), end, L69 **下**, →C75



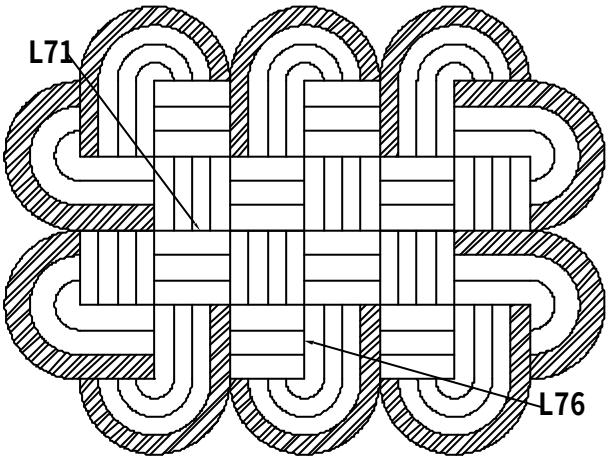

15、 TRim,
16、 Erase,
17、 ARray, **矩形陣列**, **列**, 1, **行**, 3, **列偏移**, 0, **行偏移**, end, L62 **左**, end, L69 **下**, **物件**, **所有物件**, →3 **組相同物件**(L76)



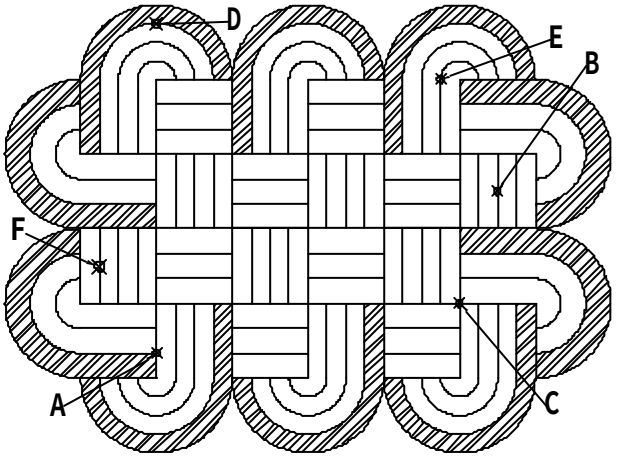

18、 Circle, end, L65 **左**, end, L64 **左**, →C77
19、 Circle, @(**鎖定上次點**), end, L63 **左**, →C78
20、 Circle, @(**鎖定上次點**), end, L62 **左**, →C79




21、 TRim,
22、 COpy, **所有物件(除下方** 3 **組** U **形半圓外**), end, L62 **左**, end, L70 **上**, →**上下** 2 **組物件**

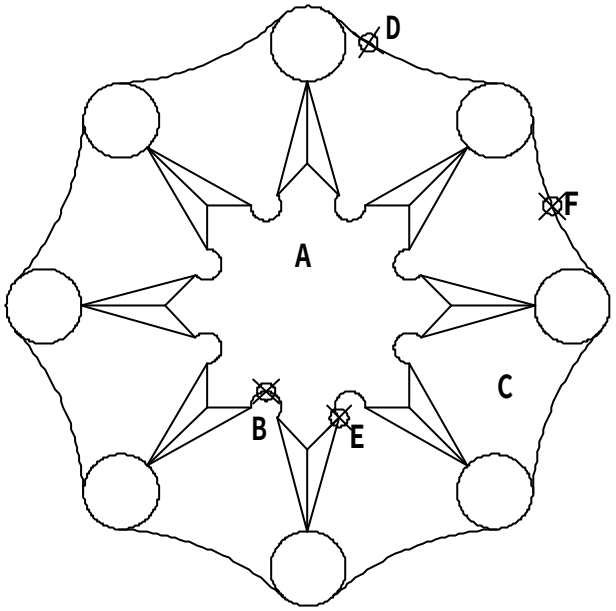



23、 MIrror, **下方** 3 **組** U **形半圓**, end, L71 **左**, end, L71 **右**, →**上方** 3 **組** U **形半圓**
24、 MIrror, **左側** 3 **組** U **形半圓**, end, L76 **上**, end, L76 **下**, →**右側** 3 **組** U **形半圓**
25、 Hatch, →**斜線區域**



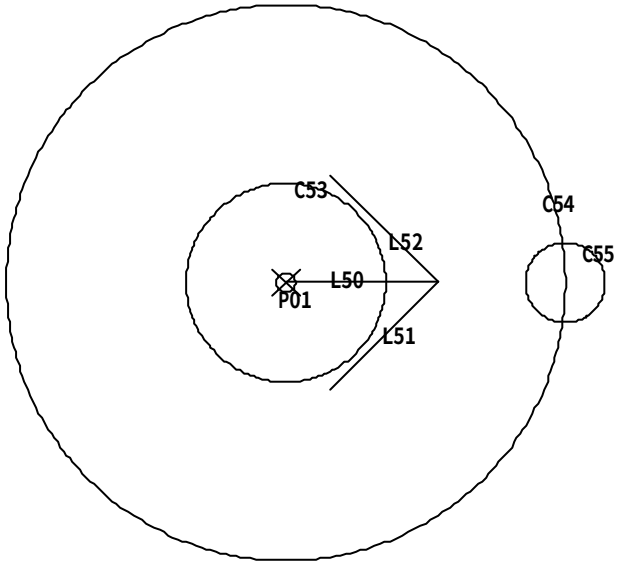

26、 DIst, int, A, mid, B, →**距離**=102.8023
27、 DIst, int, C, qua, D, →XY **平面內角度**= 136.8476
28、 DIst, int, E, mid, F, →**距離**= 105.9592
29、 **點選斜線區域**, CRTL+1, →**查幾何面積值**= 4426.2590
30、 **產生只有外圍邊緣圖形**, Hatch,
31、 CTRL+1, →**查幾何面積值**= 16823.2162

## 2.1.6 AutoCAD2D2008-106

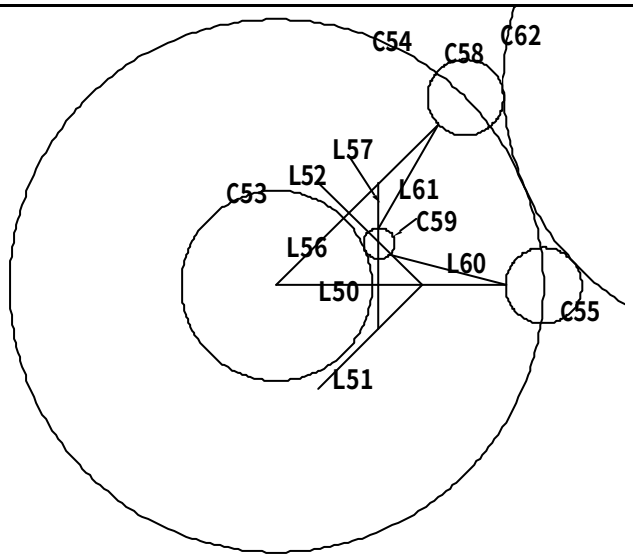



1、 A **區域周長為何？** 314.7996
2、 **弧** B **半徑為何？** 4.0839
3、 C **區域面積為何？** 1264.6938
4、 **弧** D **包含角為何？** 39.5696
5、 **交點** E **至中點** F **距離為何？** 79.8855

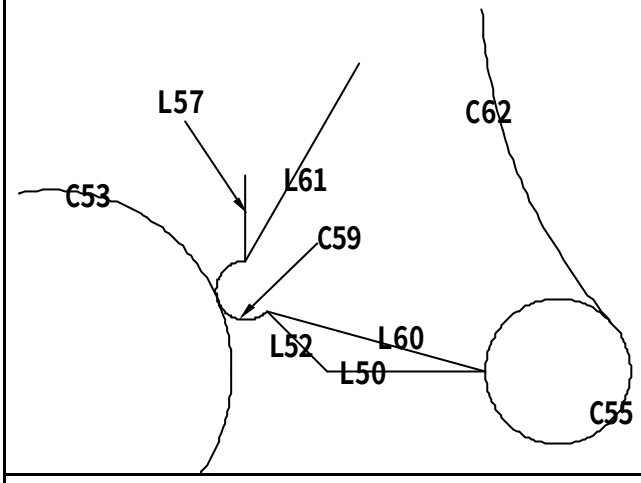
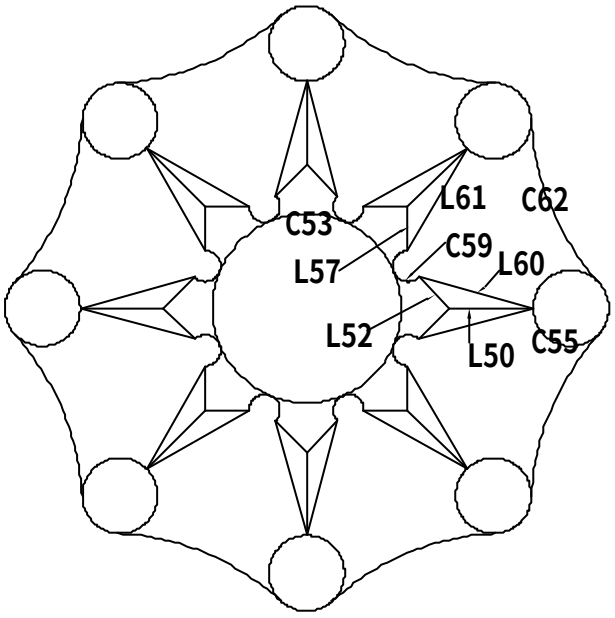
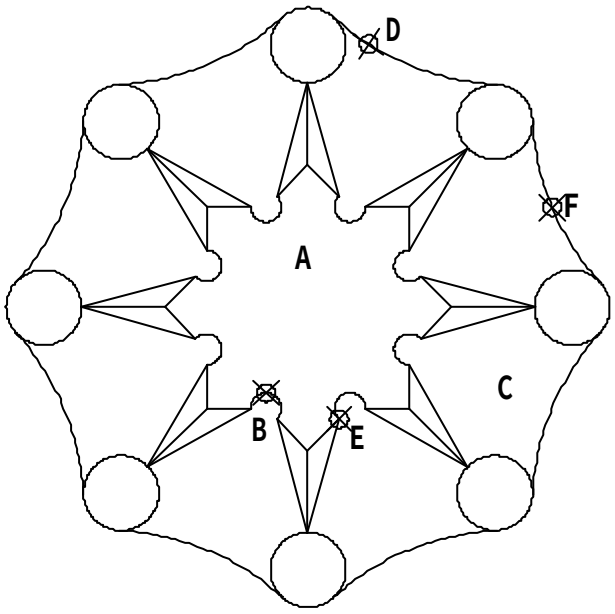
**解題步驟：**



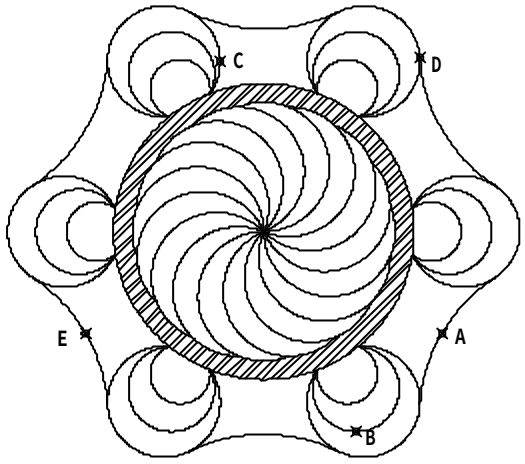

01、 Line, **任一點** P01, (**正交模式**), **向右**, 38(=76/2), →L50
02、 ROtate, L50, end, L50 **右**, C, 45, →L51
03、 ROtate, L50, end, L50 **右**, C, -45, →L52
04、 Circle, end, L50 **左**, D, 50, →C53
05、 Circle, @(**鎖定上次點**), D, 140, →C54
06、 Circle, qua, C54 **右**, D, 20, →C55




07、 EXtend, C55, L50 **右**, →L50 **向右延伸**
08、 ROtate, L50;L51;C55, cen, C53, C, 45, →L56;L57;C58
09、 Circle, int, L52 & L57, tan, C53, →C59
10、 Line, int, L50 & C55, int, L52 & C59, →L60
11、 Line, int, L56 & C58, int, L57 & C59, →L61
12、 Circle, 3P, tan, C55 **右上**, tan, C54 **右上**, tan, C58 **右下**, →C62

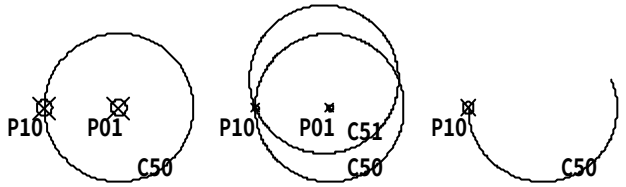
| | |
|---|---|
|  | 13、 TRim,<br>14、 Erase, |
|  | 15、 ARray, **環形陣列**, **中心點**, cen, C53, **項目總數**, 8, **佈滿角度**, 360, **物件**, **除** C53 **外的所有物件**, →**全圓佈滿** 8 **組物件** |
|  | 16、 BOundary, A **區**, **點選** A **區邊緣**, CTRL+1, → **查幾何長度值**= 314.7996<br>17、 **點選弧** B, CTRL+1, →**查幾何半徑值**= 4.0839<br>18、 BOundary, C **區**,<br>19、 **點選** C **區邊緣**, CTRL+1, →**查幾何面積值**= 1264.6938<br>20、 **點選弧** D, CTRL+1, →**查幾何總角度值**= 39.5696<br>21、 DIst, int, E, mid, F, →**距離**=79.8855 |

## 2.1.7 AutoCAD2D2008-107




1、 **弧** A **長度為何？** 41.2919
2、 **弧** B **中心點至四分點** C **距離為何？** 95.6925
3、 **斜線區域面積為何？** 1227.1739
4、 **交點** D **至弧** E **中心點水平距離為何？** 127.9872
5、 **圖形最外圍的面積為何？** 11666.4562

**解題步驟：**



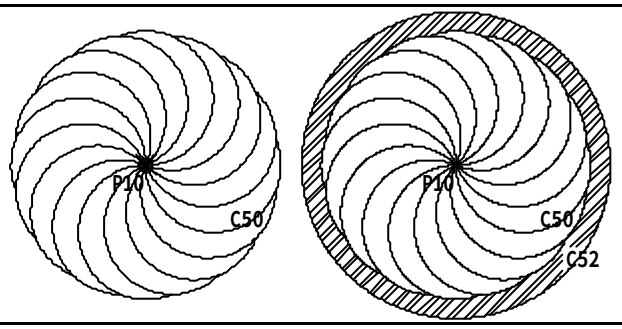

01、 Circle, **任一點** P01, D,35(=70/2), →C50
02、 POint, qua, C50 **左**, →P10

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

03、 ROtate, C50, nod, P10, c, ‘ cal, 360/16, →C51

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

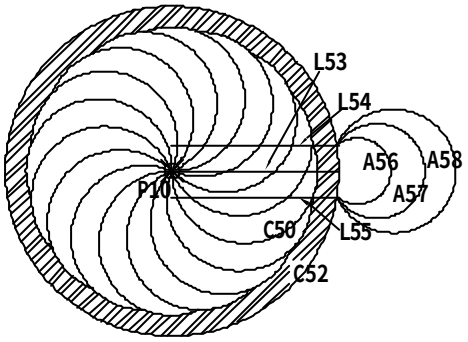
04、 TRim, Erase,



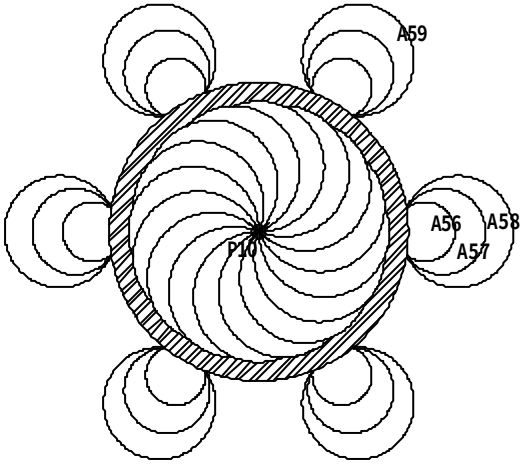

05、 ARray, **環形陣列**, **中心點**, nod, P10, **項目總數**, 16, **佈滿角度**, 360, **物件**, C50, →16 **個** C50

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

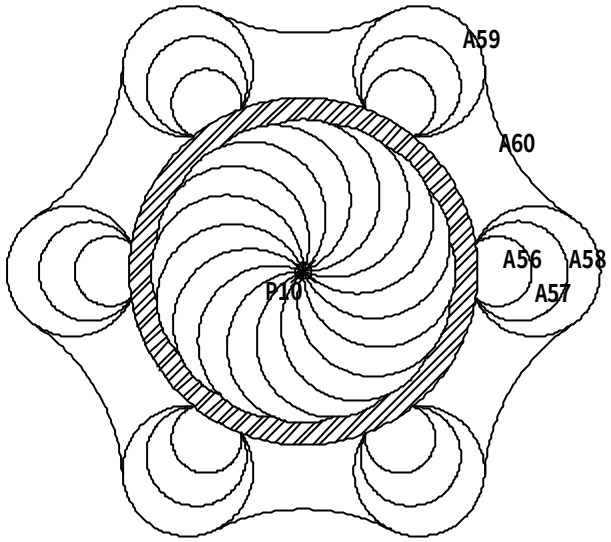
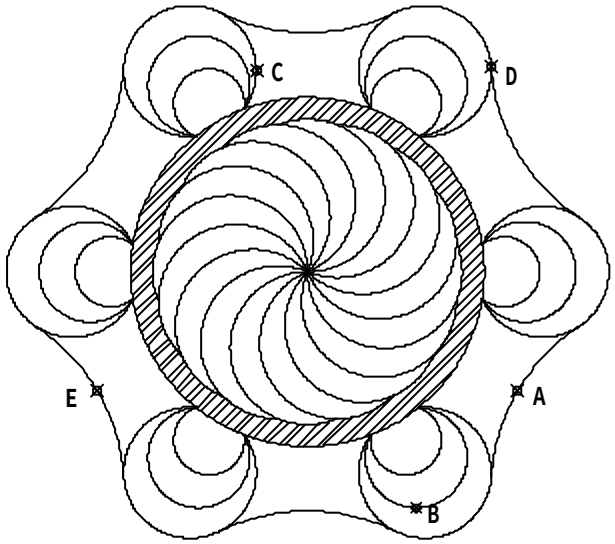
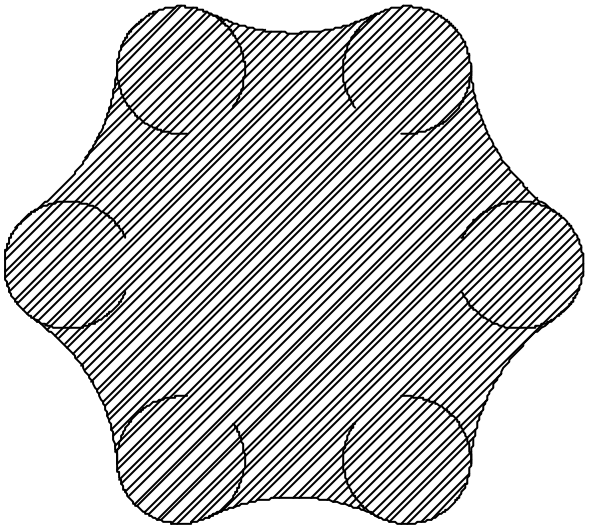
06、 Circle, nod, P10, D, 80, →C52
07、 Hatch, →**產生斜線區域**
08、 **點選斜線區域**, CTRL+1, →**查詢幾何面積值** =1227.1739, **即第** 3 **題**



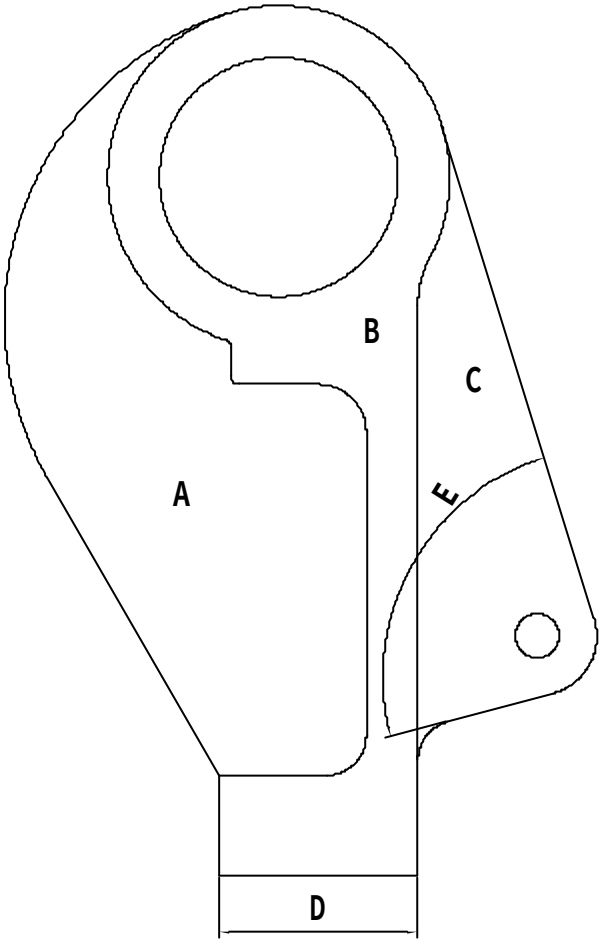

09、 Line, nod, P10, qua, C52 **右**, →L53
10、 Offset, 6.25(=12.5/2), L53, **向上點**, →L54, L53, **向下點**, →L55
11、 Arc, int, L55 & C52, E, int, L54 & C52, R, -8, →A56
12、 Arc, int, L55 & C52, E, int, L54 & C52, R, -11.5, →A57
13、 Arc, int, L55 & C52, E, int, L54 & C52, R, -15, →A58




14、 Erase,
15、 ARray, **環形陣列**, **中心點**, nod, P10, **項目總數**, 6, **佈滿角度** 360, **物件**, A56;A57;A58, →A59 **等物件**

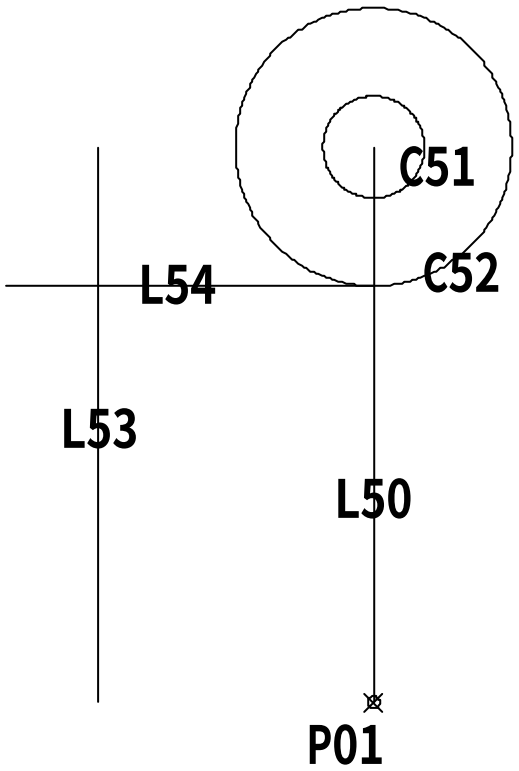
| | |
|---|---|
|  | 16、 Fillet, T, N, R, 45, A58 **左上**, A59 **左**, →A60<br>17、 ARray, **環形陣列**, **中心點**, nod, P10, **項目總數**, 6, **佈滿角度** 360, **物件**, A60, →6 **個** A60 **相同物件** |
|  | 18、 **選弧** A, CRTL+1, →**查詢幾何弧長值** =41.2919<br>19、 DIst, cen, **弧** B, qua, **弧** C, →**距離**=95.6925<br>20、 **點選斜線區域**, CTRL+1, →**查詢幾何面積值** =1227.1739<br>21、 DIst, int, D, cen, **弧** E, →X **差值**=127.9872 |
|  | 22、 COpy, **所有物件**,<br>23、 Erase, **成右圖形**<br>24、 Hatch, →**產生斜線區域**<br>25、 **點選斜線區域**, CTRL+1, →**查詢幾何面積值** =11666.4562, **即第** 5 **題** |

## 2.1.8 AutoCAD2D2008-108

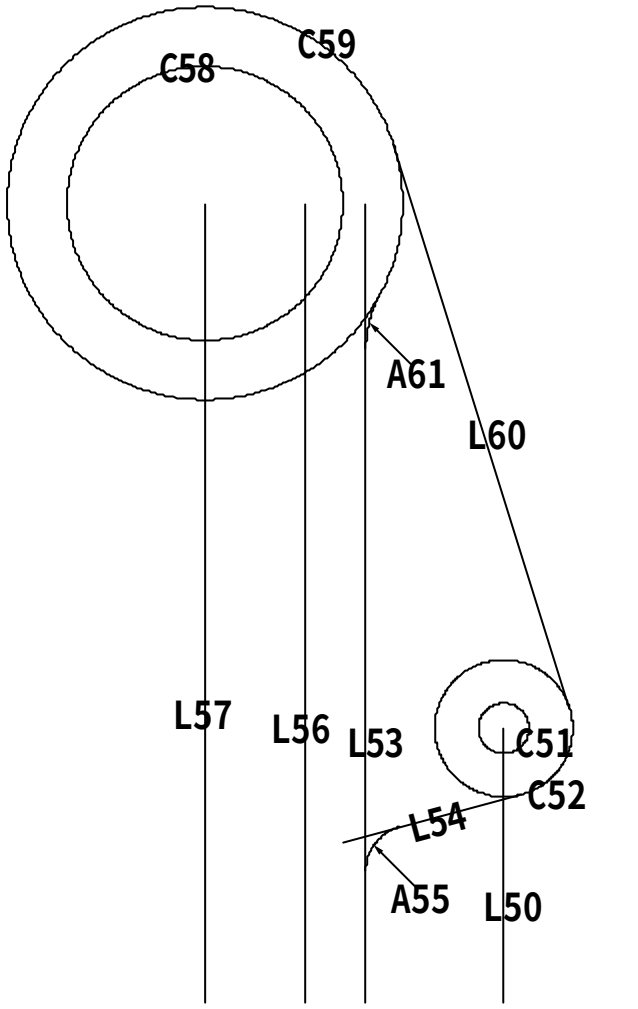



1、 A **區域周長為何？** 508.7433
2、 B **區域扣除內孔面積為何？** 6183.94855
3、 C **區域扣除內孔面積為何？** 3752.2578
4、 **尺寸** D **長度為何？** 50.1696
5、 E **角度為何？** 87.7615

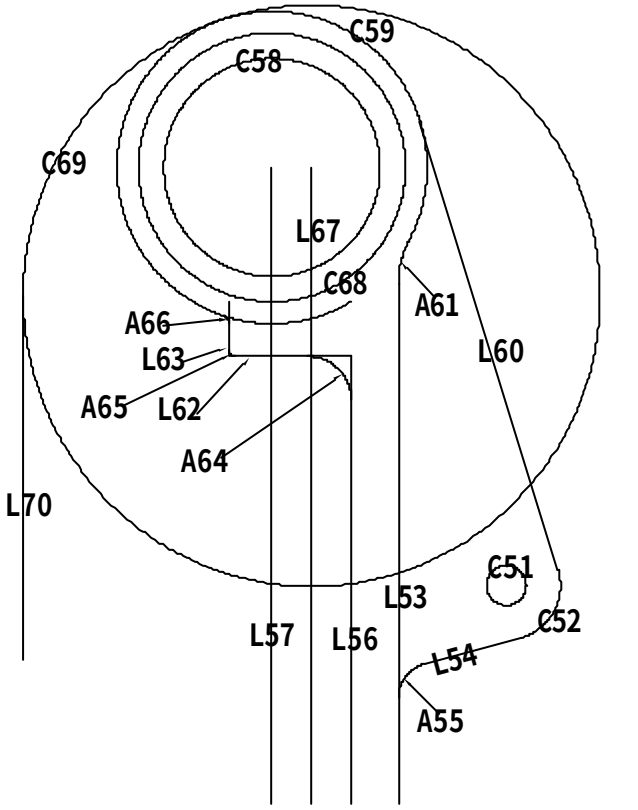
**解題步驟：**



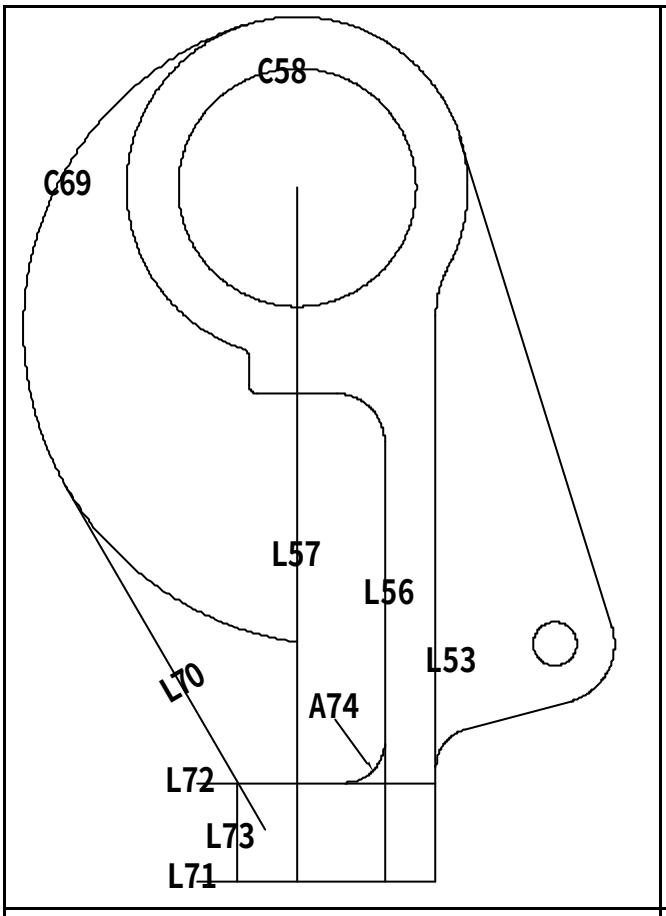

01、 Line, **任一點** P01, (**正交模式**), **向上** 60, →L50
02、 Circle, end, L50 **上**, D, 11, →C51
03、 Circle, @(**鎖定上次點**), 15, →C52
04、 Offset, 30, L50, **向左點**, →L53
05、 Line, qua, C52 **下**, (**正交模式**), **向左** 40, →L54



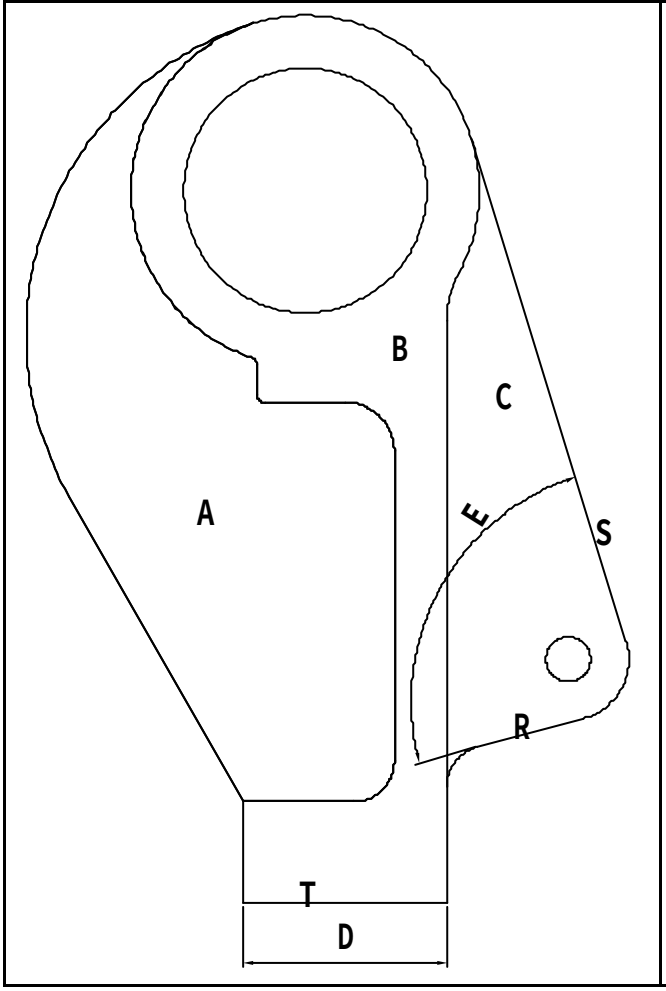

06、 ROtate, L54, cen, C52, 15, →L54 轉 15 度
07、 Fillet, T, N, R, 10, L53 下, L54 左下, →A55
08、 Offset, 13, L53, 向左點, →L56
09、 Offset, 22, L56, 向左點, →L57
10、 LENgthen, T, 175, L57 上, → L57 向上延伸, L56 上, → L56 向上延伸, L53 上, → L53 向上延伸
11、 Circle, end, L57 上, D, 60, →C58
12、 Circle, @(鎖定上次點), D, 86, →C59
13、 Line, tan, C59 右, tan, C52 右, →L60
14、 Fillet, T, N, R, 25, C59 右下, L53 上, →A61




15、 LENgthen, DE, -52, L56 上, →L56 向下縮短
16、 Line, end, L56 上, (正交模式), 向左 34(=22+12), →L62, 向上 15, →L63
17、 Fillet, T, N, R, 12, L56 上, L62 右, →A64
18、 Fillet, T, N, R, 2, L62 左, L63 下, →A65
19、 Fillet, T, N, R, 2, L63 上, C59 左下, →A66
20、 Offset, 11, L57, 向右點, →L67
21、 Circle, cen, C59, 37(=80-86/2), →C68
22、 Circle, int, L67 & C68, 80, →C69
23、 Line, qua, C69 左, (正交模式), 向下 100, →L70

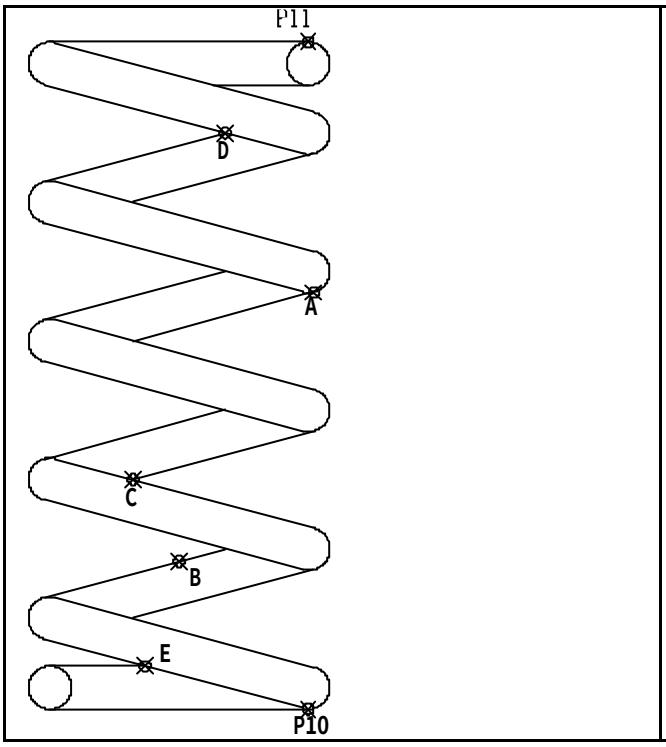



24、 TRim,
25、 Erase,
26、 ROtate, L70, cen, C69, 30(=120-90), →L70**轉** 30 **度**
27、 Line, end, L53 **下**, (**正交模式**), **向左** 60, →L71
28、 Offset, 25, L71, **向上點**, →L72
29、 Line, int, L70 & L72, (**正交模式**), **向下** 25, →L73
30、 Fillet, T, N, R, 10, L56 **下**, L72 **右**, →A74



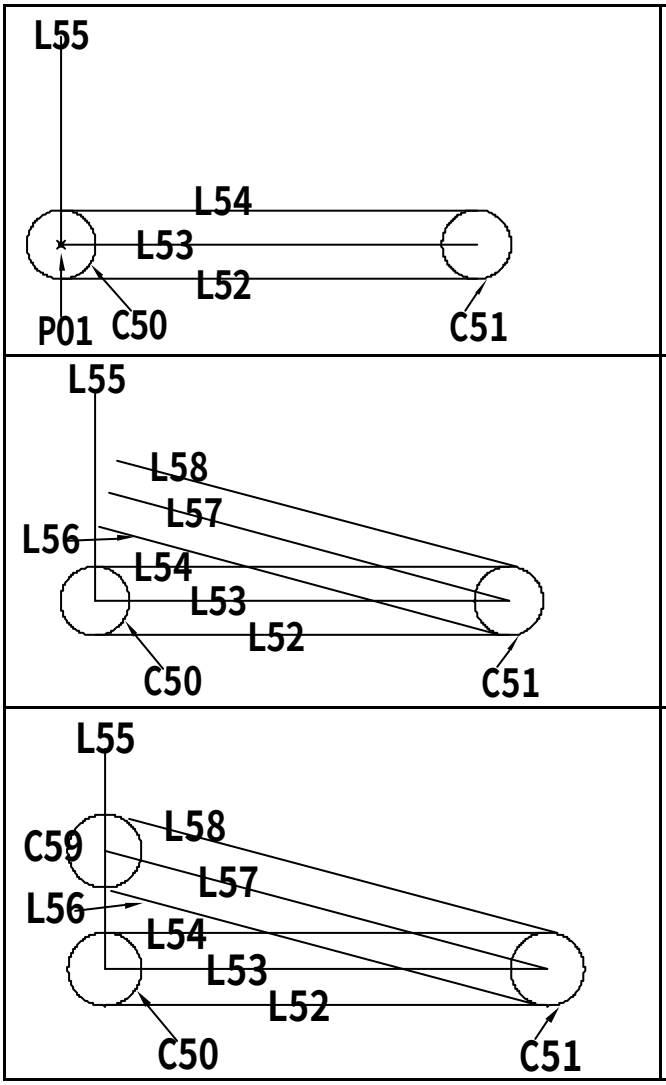

31、 TRim,
32、 Erase,
33、 BOundary, **點** A **區域**,
34、 **點** A **區域邊緣**, CTRL+1, **→查幾何長度值** =508.7433
35、 Hatch, **點** B **區域**, **→斜線區域**
36、 **點** B **區域斜線**, CTRL+1, **→查幾何面積值** =6183.9485
37、 Hatch, **點** C **區域**, **→斜線區域**
38、 **點** C **區域斜線**, CTRL+1, **→查幾何面積值** =3752.2578
39、 **點** T, CTRL+1, **→查幾何長度值**=50.1696
40、 DimANgular, S, R, **→查標註值**=87.7615

## 2.1.9 AutoCAD2D2008-109




1、 **彈簧總高為何？**154.6926
2、 **交點** A **至交點** B **距離為何？**69.7541
3、 **圖形最外圍圍成面積為何？** 6076.6275
4、 **當交點** C **座標** 30,40**, 交點** E **的** X**座標為何？** 32.7027
5、 **當交點** C **座標** 30,40**, 交點** D **的** Y **座標為何？** 120.3848

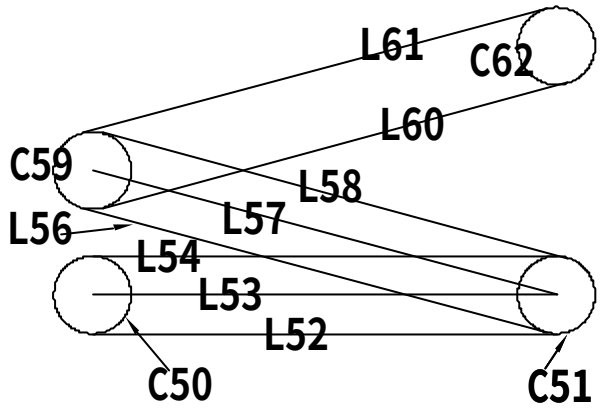
**解題步驟：**



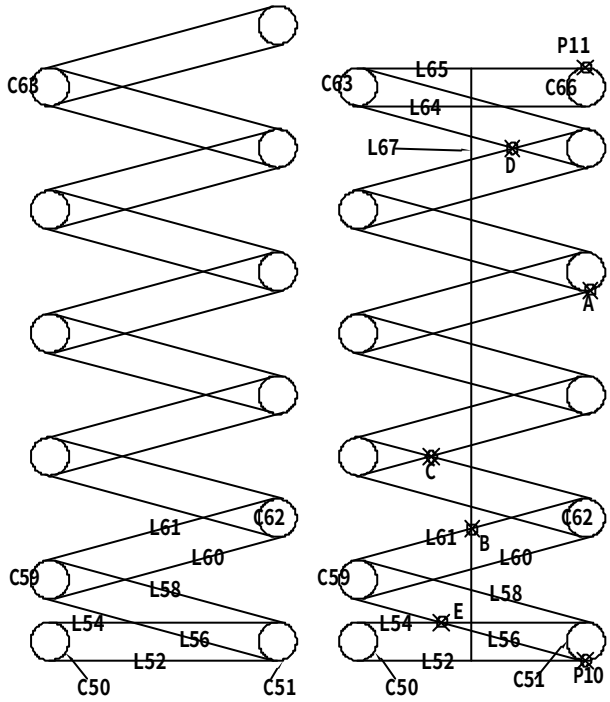

01、 Circle, **任一點** P01, D, 10, →C50
02、 COpy, C50, **任一點**, (**正交模式**), **向右** 60, →C51
03、 Line, qua, C50 **下**, qua, C51 **下**, →L52
04、 Offset, 5, L52, **向上點**, →L53, L53, **向上點**, →L54,
05、 Line, cen, C50, (**正交模式**), **向上** 30, →L55

06、 ROtate, L52;L53;L54, cen, C51, C, -15, →L56;L57;L58

07、 EXtend, L55 **左上**, L57, →L57 **向左上延伸**
08、 Circle, int, L57 & L55, D, 10, →C59
09、 EXtend, C59, L56 **左上**;L58 **左上**, →L56;L58 **向左上延伸**



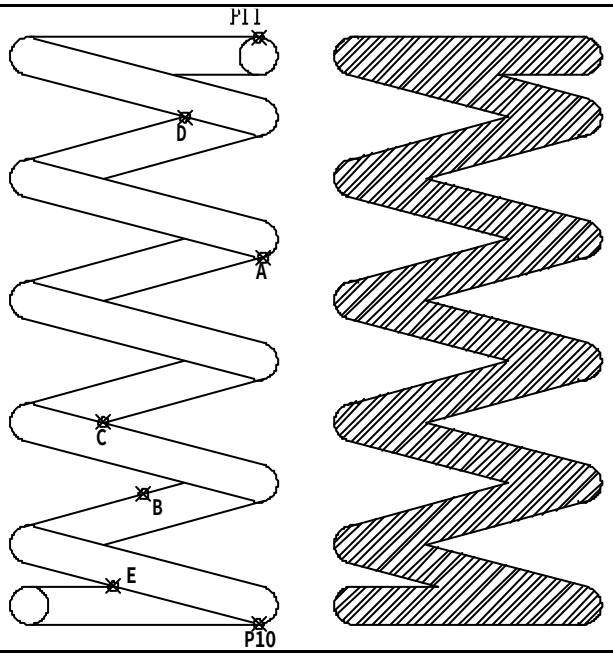

10、 Erase,
11、 ROtate, L56;L58;C51, cen, C59, C, 30, →L60;L61;C62




12、 Erase,
13、 ARray, **矩形陣列**, **列**, 5, **行**, 1, **列偏移**, cen, C51, cen ,C62, **行偏移**, 0, **物件**, L56;L58;C59;L60;L61;C62, →C63 **等物件**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

14、 COpy, C51;L52;L54, cen, C50, cen, C63, →L64;L65;C66
15、 Line, mid, L52, mid, L65, →L67
16、 POint, int, →A;B;C;D;E;P10;P11

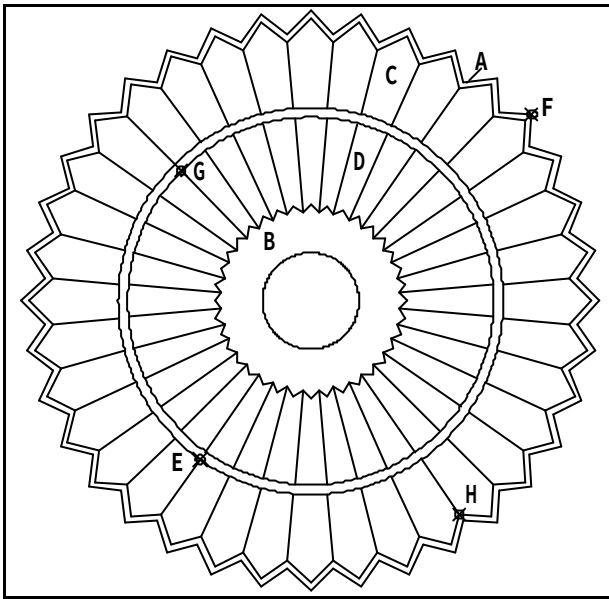



17、 TRim,
18、 Erase,
19、 DIst, nod, P10, nod, P11, →Y **差值**=154.6926
20、 DIst, nod, A, nod, B, →**距離**= 69.7541

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

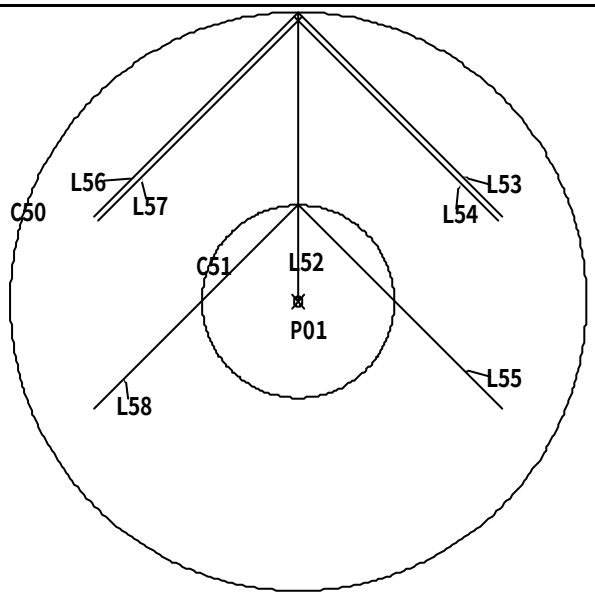
21、 COpy, **所有物件**
22、 TRim,
23、 Erase,
24、 Hatch, →**斜線區域**
25、 **點斜線區域**, CTRL+1, →**查幾何面積值** =6076.6275
26、 Move, **選所有物件**, nod, C, 30,40, →**所有物件點皆移動**
27、 **雙點滑鼠滾輪**, →**所有物件縮回**
28、 ID, nod, E, →X=32.7027
29、 ID, nod, D, →Y=120.3848

## 2.1.10 AutoCAD2D2008-110

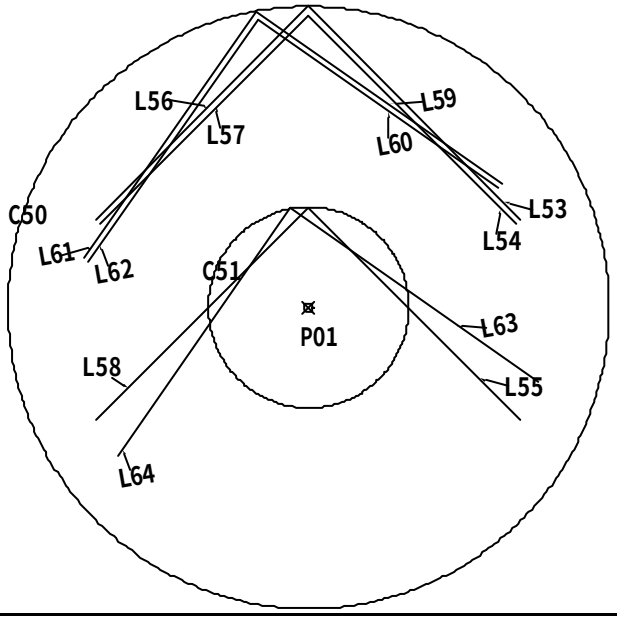



1、 A **區域面積為何？**3634.1375
2、 B **區域扣除內孔面積為何？**5277.0212
3、 C **與** D **區域面積和為何？**1422.9706
4、 **交點** E **至交點** F **角度為何？**39.9612
5、 **交點** G **至交點** H **距離為何？** 228.6956

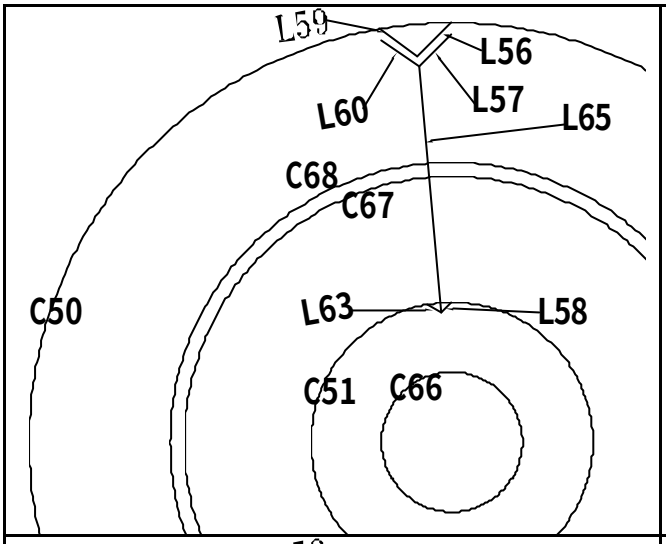
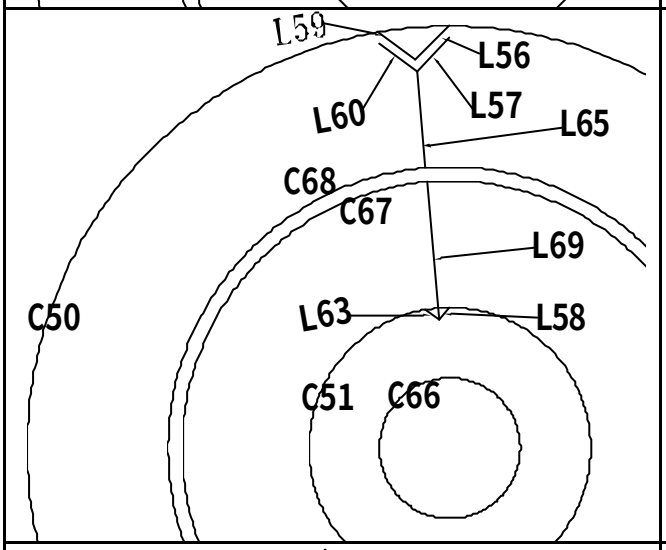
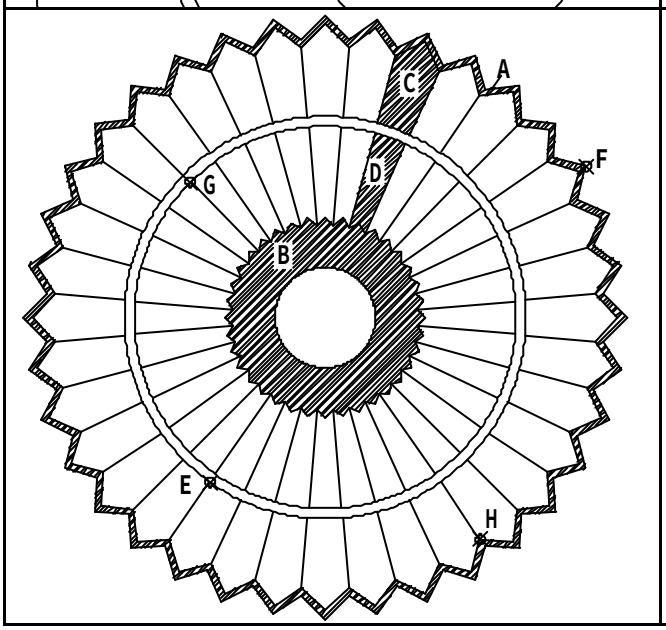
**解題步驟：**



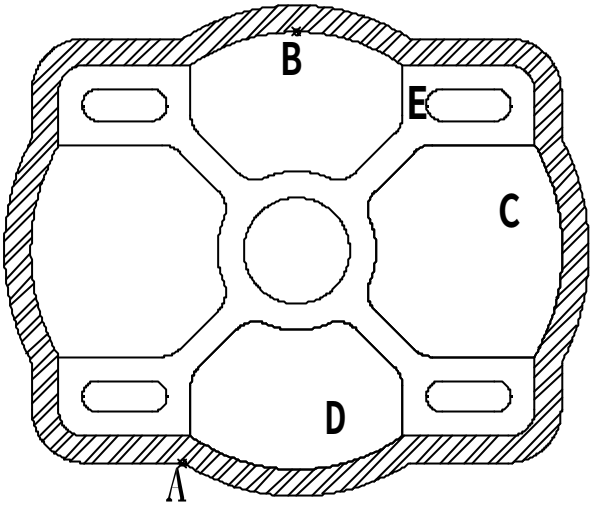

01、 Circle, **任一點** P01, D, 300, →C50
02、 Circle, @(**鎖定上次點**), D, 100, →C51
03、 Line, cen, C51, qua, C50 **上**, →L52
04、 ROtate, L52, int, C50 & L52, C, 45, →L53
05、 Offset, 3, L53, **向下點**, →L54
06、 COpy, L53, int, C50 & L52, int, C51 & L52, →L55
07、 MIrror, L53;L54;L55, qua, C51 **上**, qua, C51 **下**, →L56;L57;L58




08、 Erase,
09、 TRim,
10、 ROtate, L53~L58, cen, C51, C, 10, →L59~64

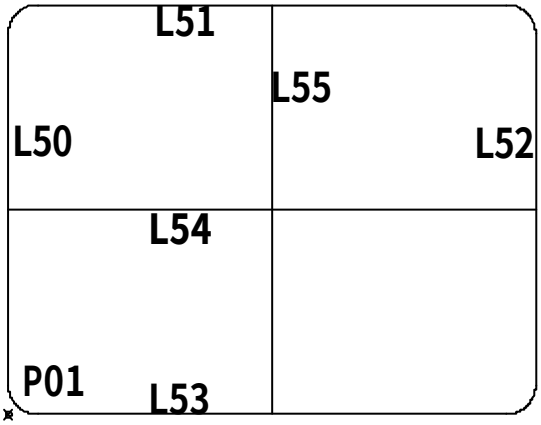
| | |
|---|---|
|  | 11、 TRim,<br>12、 Erase,<br>13、 Line, int, L57 & L60, int, L58 & L63, →L65<br>14、 Circle, cen, C51, D, 50, →C66<br>15、 Circle, @(**鎖定上次點**), D, 190, →C67<br>16、 Circle, @(**鎖定上次點**), D, 200, →C68 |
|  | 17、 TRim, →L65 **分成** L65 **與** L69 |
|  | 18、 ARray, **環形陣列**, **中心點**, cen, C51, **項目總數**, 36, **佈滿角度**, 360, **物件**, L56~L60;L63;L65;L69, →**全圓佈滿** 36 **個**<br>19、 Hatch, →**斜線區域**<br>20、 **點斜線** A **區域**, CTRL+1, →**查幾何面積值** =3634.1375<br>21、 **點斜線** B **區域**, CTRL+1, →**查幾何面積值** =5277.0212<br>22、 **點斜線** C **與** D **區域**, CTRL+1, →**查幾何面積值**=1422.9706<br>23、 DIst, nod, E, nod, F, →XY **平面內角度** =39.9612<br>24、 DIst, nod, G, nod, H, →**距離**=228.6956 |

## 2.1.11 AutoCAD2D2008-111

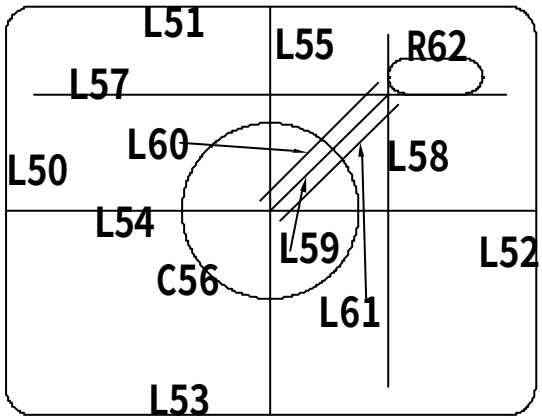



1、 **交點** A **至中點** B **距離為何？** 84.0759
2、 C **區域周長為何？** 132.6652
3、 D **區域周長為何？** 113.4574
4、 E **區域扣除內孔面積為何？** 1833.9378
5、 **斜線區域面積為何？** 1677.8025

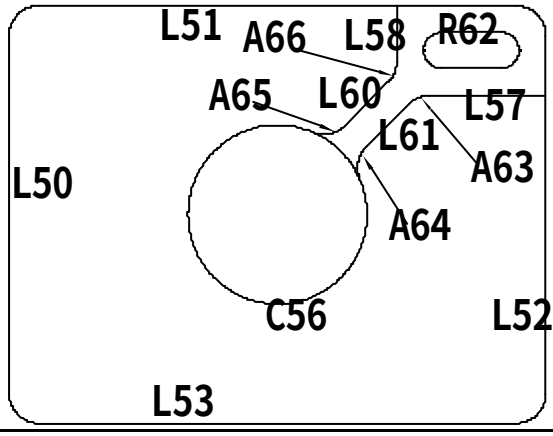
**解題步驟：**



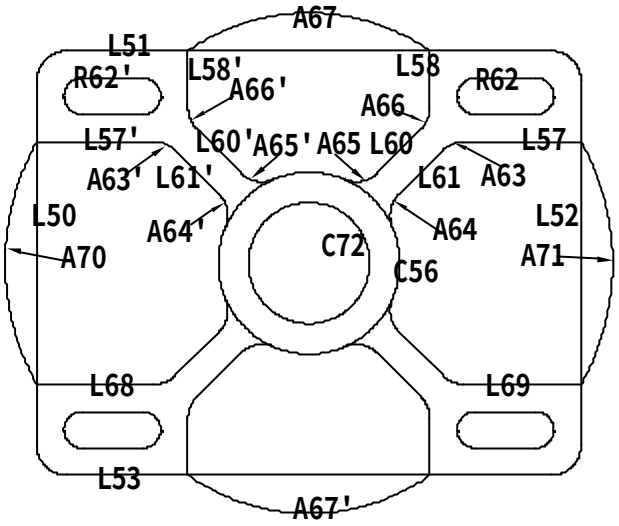
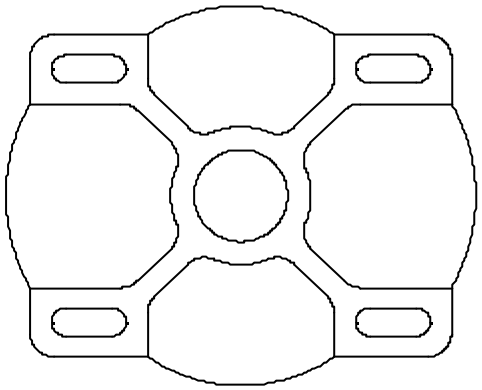
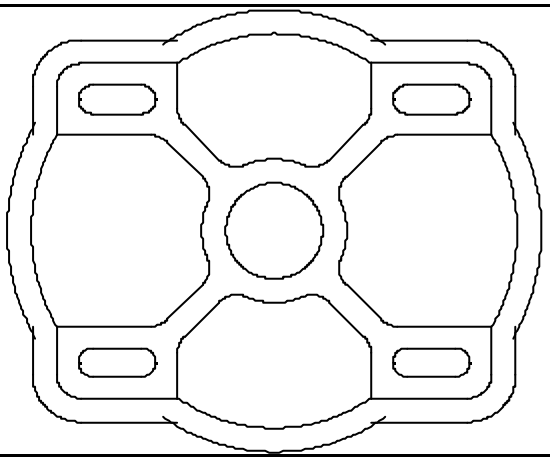
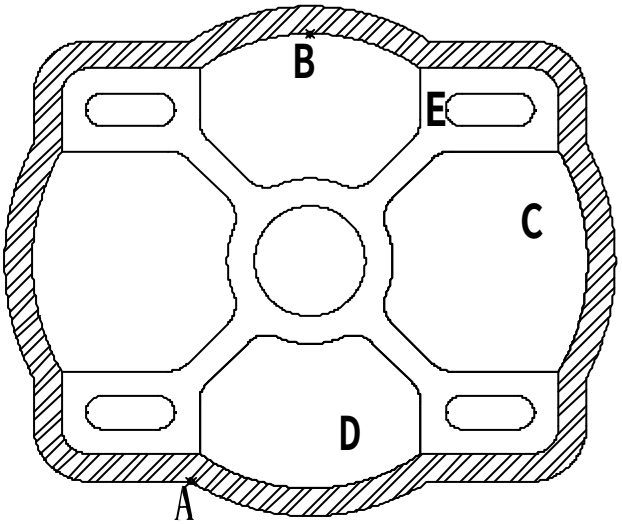

01、 RECtangle, F, 5, **任一點** P01, D, 90, 70, **向右上點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Line, mid, L50, mid, L52, →L54
04、 Line, mid, L51, mid, L53, →L55



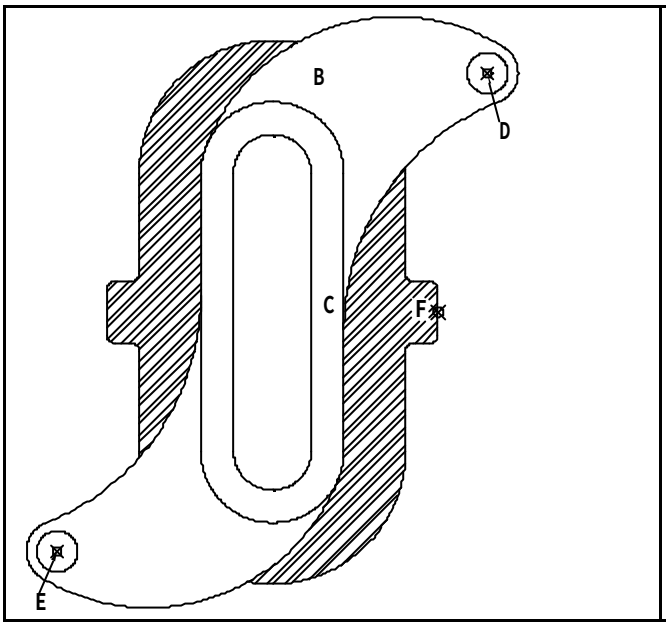

05、 Circle, int, L54 & L55, D, 30, →C56
06、 Offset, 20(=40/2), L54, **向上點**, →L57, L55, **向右點**, →L58
07、 Line, cen, C56, int, L57 & L58, →L59
08、 Offset, 2.5(=5/2), L59, **向上點**, →L60, L59, **向下點**, →L61
09、 RECtangle, F, 3, end, L59 **右上**, D, 16, 6, →R62




10、 Move, R62, end, L59 **右上**, @4.5,4.5, →R62 **向右上移動** (4.5=)
11、 Fillet, T, T, R, 5, M, L57 **左**, L61 **右上**, →A63, C56 **右上**, L61 **左下**, →A64, C56 **上**, L60 **左下**, →A65, L58 **下**, L60 **右上**, →A66

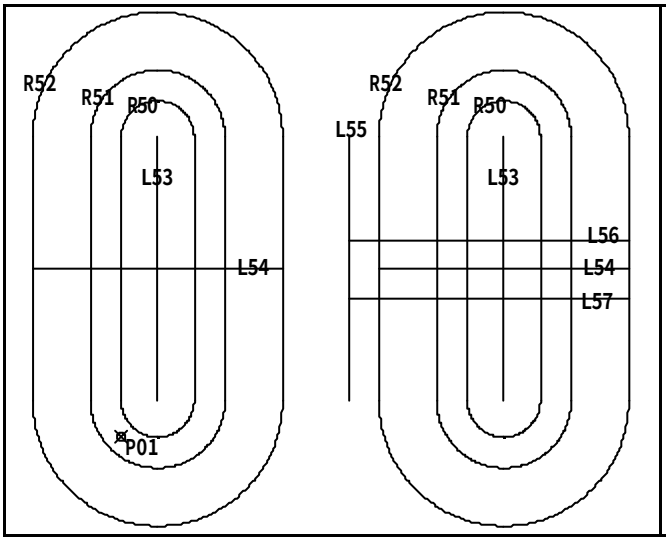
| | |
|---|---|
|  | 12、 MIrror, L60~A66, qua, C56 **上**, qua, C56 **下**, →L60' ~A66'<br>13、 Arc, int, L58 & L51, E, int, L58' & L51, R, 35, →A67<br>14、 MIrror, L60~A66;L60' ~A66' ;A67, qua, C56 **左**, qua, C56 **右**, →A67' ;L68;L69 **等物件**<br>15、 Arc, int, L57' & L50, E, int, L68 & L50, R, 41, →A70<br>16、 Arc, int, L69 & L52, E, int, L57 & L52, R, 41, →A71<br>17、 Circle, cen, C56, D, 20, →C72 |
|  | 18、 TRim,<br>19、 Erase, |
|  | 20、 Offset, 5, **依序點圖形外部邊緣各物件**, **向外點**, **→圖形向外擴大** |
|  | 21、 TRim,<br>22、 Erase,<br>23、 DIst, int, A, mid, B, **→距離**=84.0759<br>24、 BOundary, **點** C **區域**<br>25、 **點** C **區域邊緣**, CRTL+1, **→查幾何長度值** =132.6652<br>26、 BOundary, **點** D **區域**<br>27、 **點** D **區域邊緣**, CRTL+1, **→查幾何長度值** =113.4574<br>28、 Hatch, **點** E **區域**, **→斜線區域**<br>29、 **點** E **區域斜線**, CRTL+1, **→查幾何面積值** =1833.9378<br>30、 Hatch, **點斜線區域**, **→斜線區域**<br>31、 **點斜線區域**, CRTL+1, **→查幾何面積值** =1677.8025 |

## 2.1.12 AutoCAD2D2008-112




1、 **斜線區域面積為何？** 2995.4698
2、 B **區域扣除內孔面積為何？** 1691.4287
3、 C **區域扣除內孔面積為何？** 1766.5840
4、 **中心點** D **至** E **角度為何？** 228.0641
5、 **中心點** D **至中點** F **垂直距離為何？** 58.5119

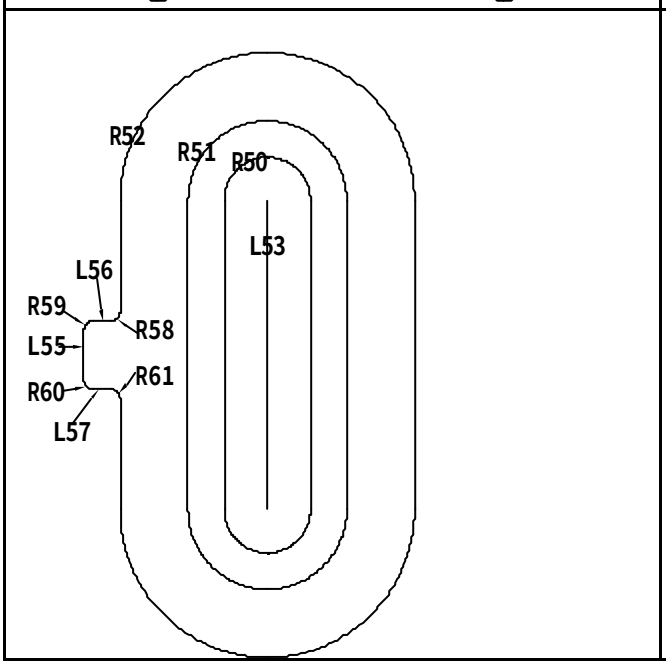
**解題步驟：**



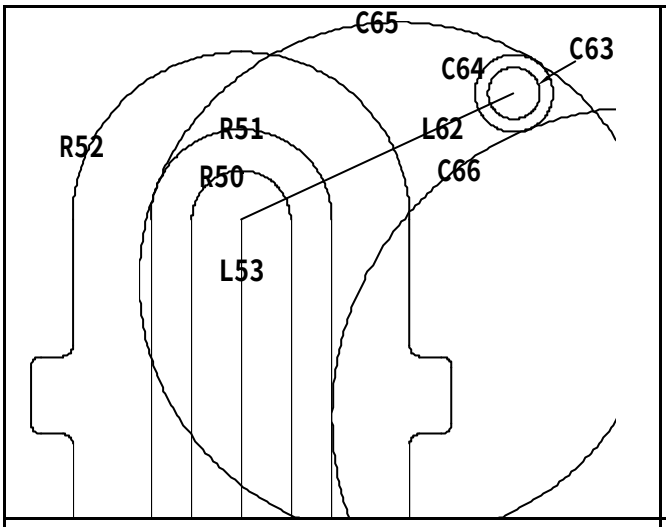

01、 RECtangle, F, 9.5(=19/2), **任一點** P01, D, 19, 87(=68+19), **向右上點**, →R50
02、 Offset, 8, R50, **向外點**, →R51
03、 Offset, 15, R51, **向外點**, →R52
04、 Line, cen, R50 **上半圓**, cen, R50 **下半圓**, →L53
05、 Line, mid, R52 **左直線**, mid, R52 **右直線**, →L54

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

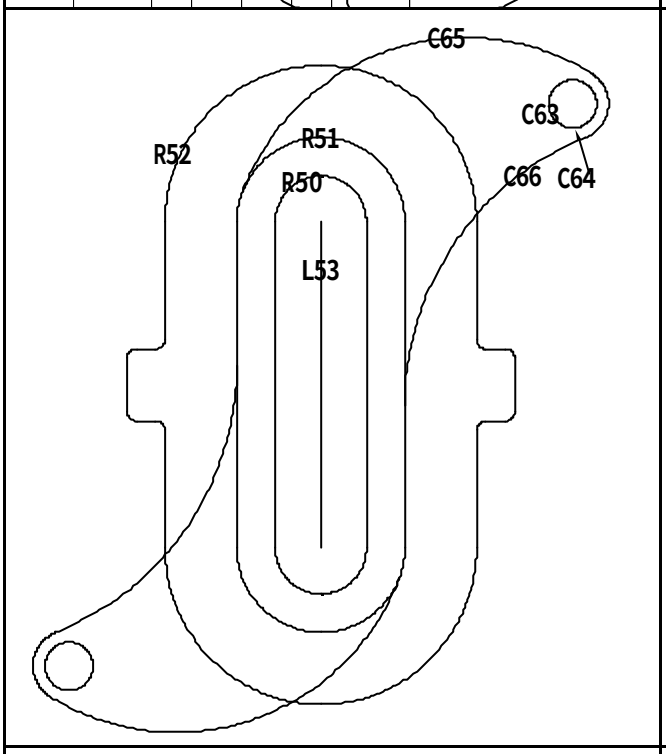
06、 Offset, 40.5(=81/2), L53, **向左點**, →L55
07、 Offset, 7.5(=15/2), L54, **向上點**, →L56
08、 Offset, 7.5(=15/2), L54, **向下點**, →L57
09、 EXtend, L55, L56 **左**;L57 **左**, →L56;L57 **向左延伸**



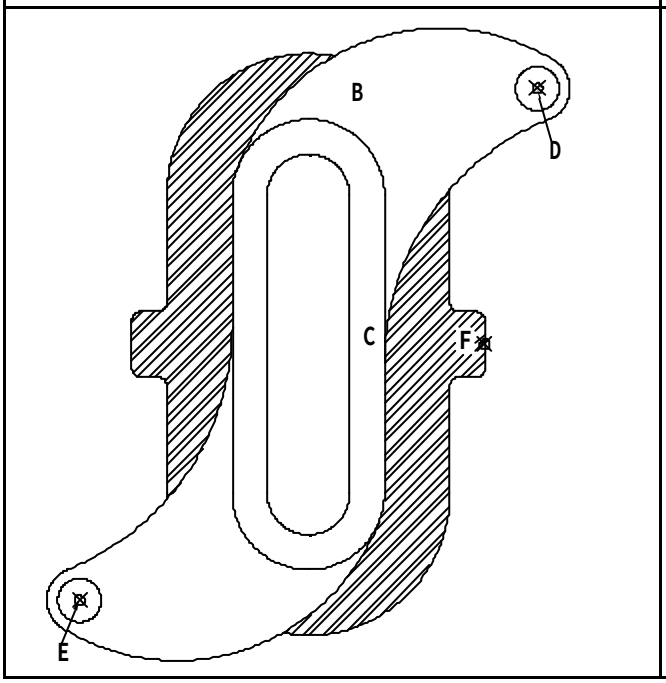

10、 Fillet, T, N, R, 2, M, R52 **左直線上**, L56 **左**, →R58, L55 **上**, L56 **左**, →R59, L55 **下**, L57 **左**, →R60, R52 **左直線下**, L57 **左**, →R61




11、 MIrror, L55~L57; R58~R61, end, L53 **上**, end, L53 **下**, →**右側相同物件**
12、 TRim,
13、 Line, end, L53 **上**, @58<25, →L62
14、 Circle, end, L62 **右上**, D, 10, →C63
15、 Circle, @(**鎖定上次點**), 7.5, →C64
16、 Circle, T, C64 **右上**, R51 **上半圓左**, 50, →C65
17、 Circle, T, C64 **右下**, R51 **右直線中**, 60, →C66

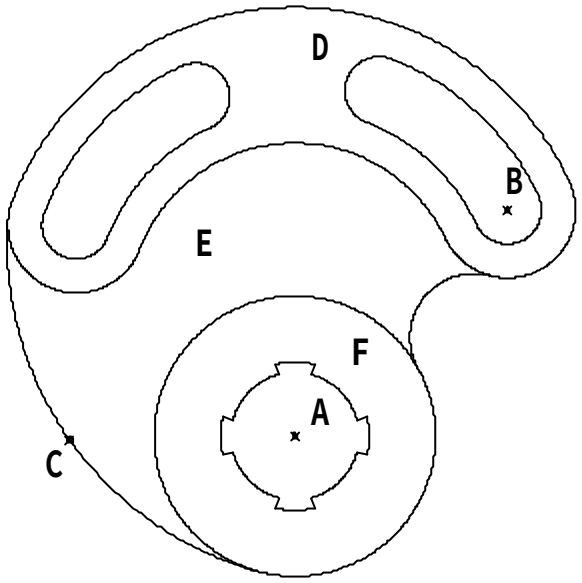



18、 TRim,
19、 Erase,
20、 ROtate, C63~C66, mid, L53, C, 180, →**左下相同物件**



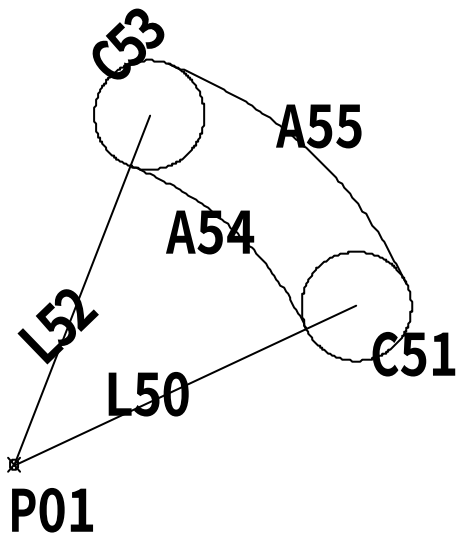

21、 TRim,
22、 Erase,
23、 Hatch, →**斜線區域**
24、 **點斜線區域**, CRTL+1, →**查幾何面積值** =2995.4698
25、 Hatch, **點** B **區域** →B **區域斜線**
26、 **點** B **區域斜線**, CRTL+1, →**查幾何面積值** =1691.4287
27、 Hatch, **點** C **區域** →C **區域斜線**
28、 **點** C **區域斜線**, CRTL+1, →**查幾何面積值** =1766.5840
29、 DIst, cen, D, cen, E, →XY **平面內角度** =228.0641
30、 DIst, cen, D, mid, F, →Y **差值**=58.5119

## 2.1.13 AutoCAD2D2008-113

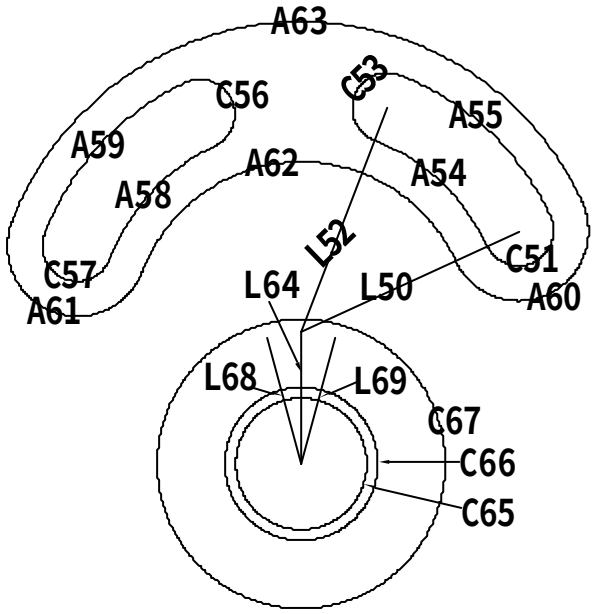



1、 **中心點** A **至中心點** B **角度為何？** 46.8873
2、 **弧** C **長度為何？** 109.6340
3、 D **區域扣除內孔面積為何？**3166.7254
4、 E **區域周長為何？** 425.3248
5、 F **區域扣除內孔面積為何？** 2629.2513

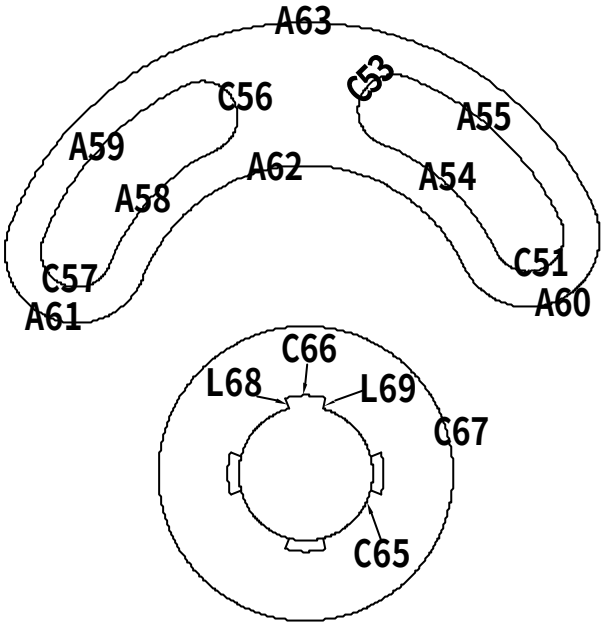
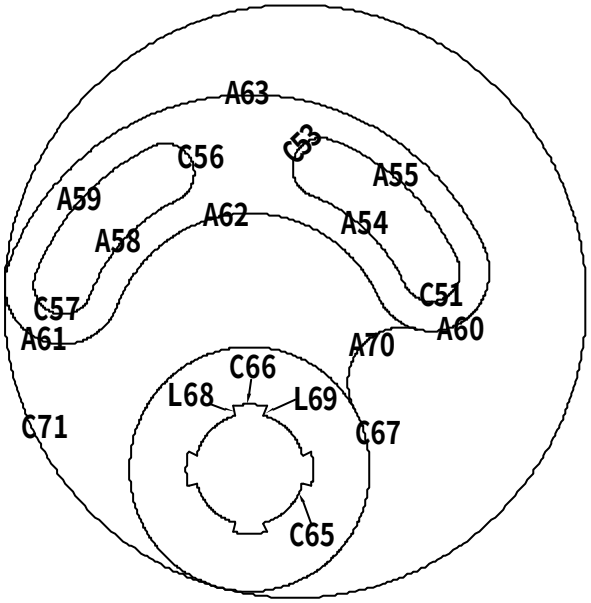
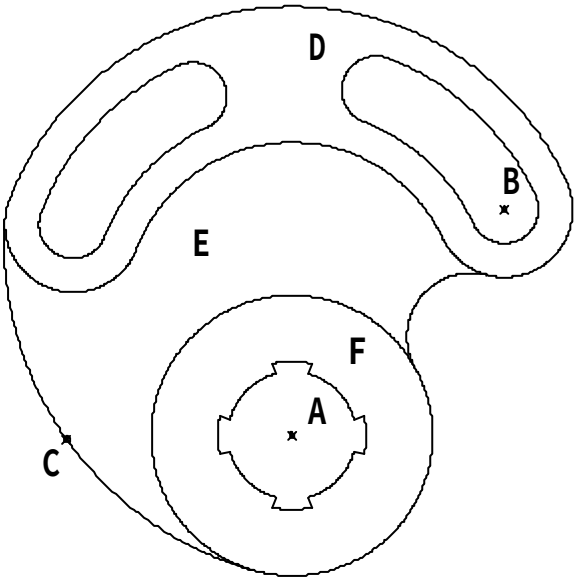
**解題步驟：**



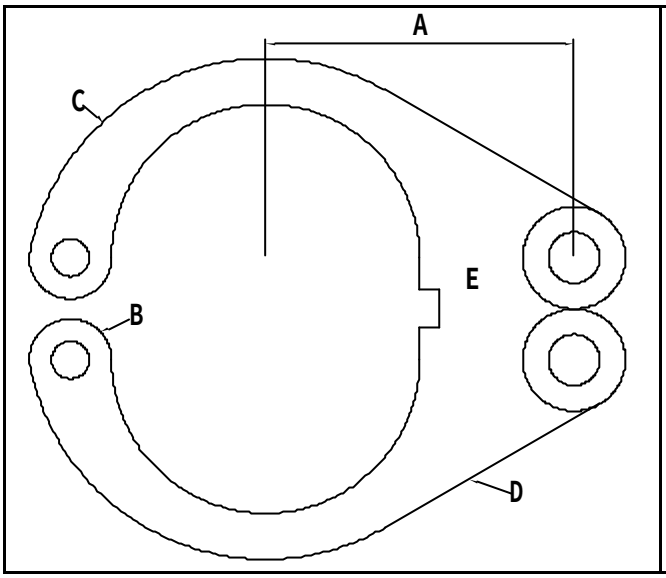

01、 Line, **任一點** P01, @55<25, →L50
02、 Circle, end, L50 **右上**, 8, →C51
03、 ROtate, L50;C51, end, L50 **左下**, C, 44, →L52;C53
04、 Arc, C, int, L50 & L52, int, L50 & C51, int, L52 & C53, →A54
05、 Offset, 16(=8*2), A54, **向外點**, →A55




06、 TRim,
07、 ROtate, C51;C53;A54;A55, int, L50 & L52, C, 90(=44+46), →C56;C57;A58;A59
08、 Offset, 8, C51, **向外點**, →A60, C57, **向外點**, →A61
09、 Arc, C, int, L50 & L52, int, L50 & A60, end, A61 **右**, →A62
10、 Offset, 32(=16*2), A62, **向外點**, →A63
11、 Line, int, L50 & L52, (**正交模式**), **向下** 30,→L64
12、 Circle, end, L64 **下**, D, 30, →C65
13、 Circle, @(**鎖定上次點**), D, 35, →C66
14、 Circle, @(**鎖定上次點**), D, 66, →C67
15、 ROtate, L64, cen, C67, C, 15(=30/2), →L68
16、 ROtate, L64, cen, C67, C, -15(=30/2), →L69

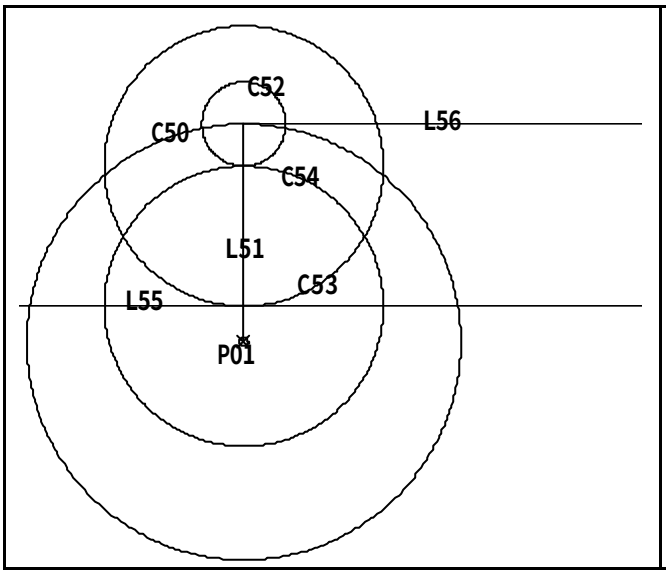
| | |
|---|---|
|  | 17、 TRim,<br>18、 Erase,<br>19、 ARray, **環形陣列**, **中心點**, cen, C67, **項目總數**, 4, **佈滿角度**, 360, **物件**, C66;L68;L69, **→全圓佈滿** 4 **組相同物件** |
|  | 20、 TRim,<br>21、 Fillet, T, N, R, 15, A60 **左下**, C67 **右上**, →A70<br>22、 Circle, T, A61 **左上**, C67 **左下**, 80, →C71 |
|  | 23、 TRim,<br>24、 DIst, cen, A, cen, B, →XY **平面內角度** =46.8873<br>25、 **點選弧** C, CTRL+1, **→查詢幾何弧長值** =109.6340<br>26、 Hatch, D**區域**, **點選斜線**, CTRL+1, **→查詢幾何面積值**=3166.7254<br>27、 BOundary, E **區域**, **點選邊緣**, CTRL+1, **→查詢幾何長度值**=425.3248<br>28、 Hatch, F **區域**, **點選斜線**, CTRL+1, **→查詢幾何面積值**=2629.2513 |

## 2.1.14 AutoCAD2D2008-114

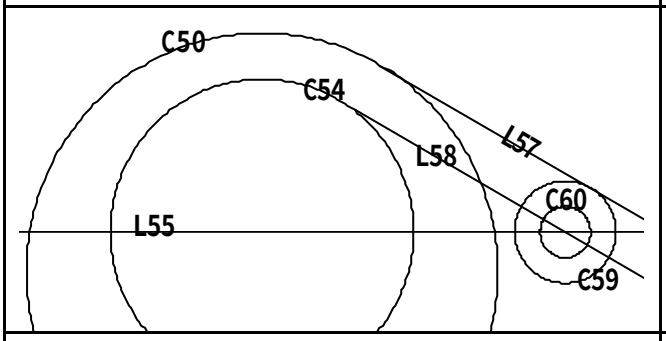



1、 **尺寸** A **長度為何？** 90.0814
2、 **弧** B **半徑為何？** 12.0244
3、 **弧** C **長度為何？** 135.0553
4、 **斜線** D **長度為何？** 72.2628
5、 E **區域扣除內孔面積為何？** 8135.9266

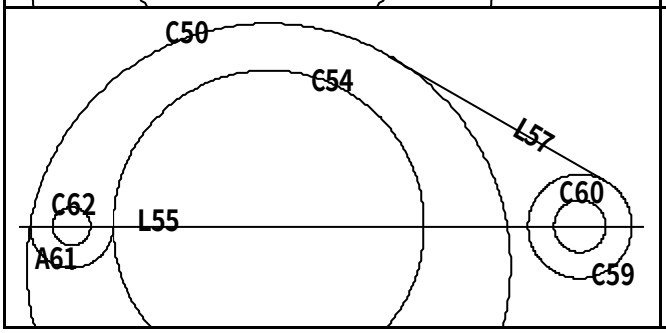
**解題步驟：**



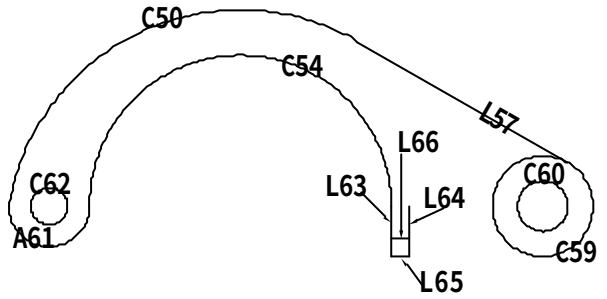
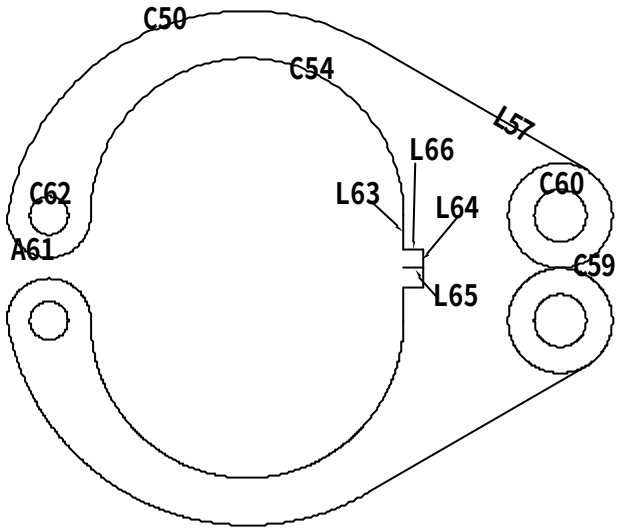
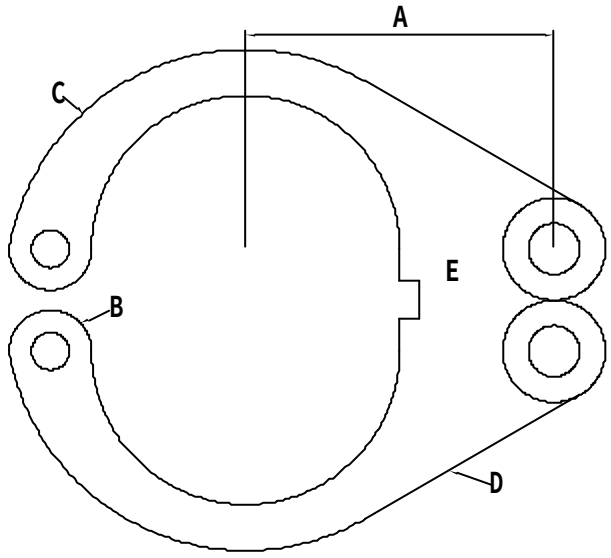

01、 Circle, **任一點** P01, 70, →C50
02、 Line, cen, C50, qua, C50 **上**, →L51
03、 Circle, qua, C50 **上**, 13.5, →C52
04、 Circel, qua, C52 **下**, 45, →C53
05、 Circel, qua, C53 **下**, 45, →C54
06、 XLine, qua, C54 **左**, qua, C54 **右**, →L55
07、 Line, qua, C50 **上**, (**正交模式**), **向右** 150, →L56



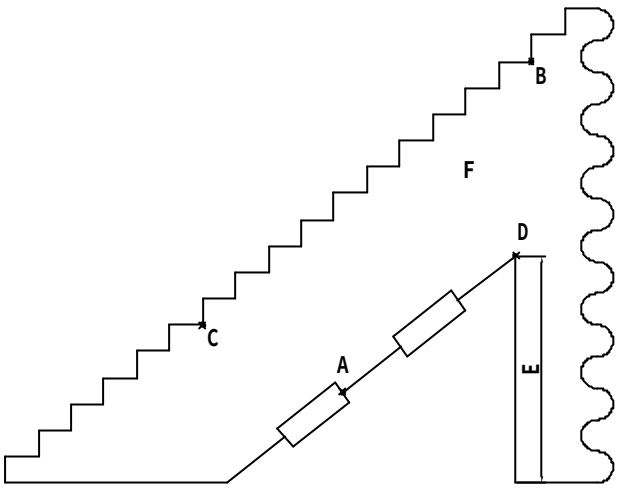

08、 Erase,
09、 ROtate, L56, cen, C50, -30, →L57
10、 Offset, 15(=30-15), L57, **向左下點**, →L58
11、 Circle, int, L58 & L55, D, 30, →C59
12、 Circle, @(**鎖定上次點**), D, 15, →C60




13、 TRim,
14、 Erase,
15、 Arc, C, m2p(mtp), int, L55 & C50, int, L55 & C54, int, L55 & C50, int, L55 & C54,→A61
16、 Circle, cen, A61, D, 11, →C62

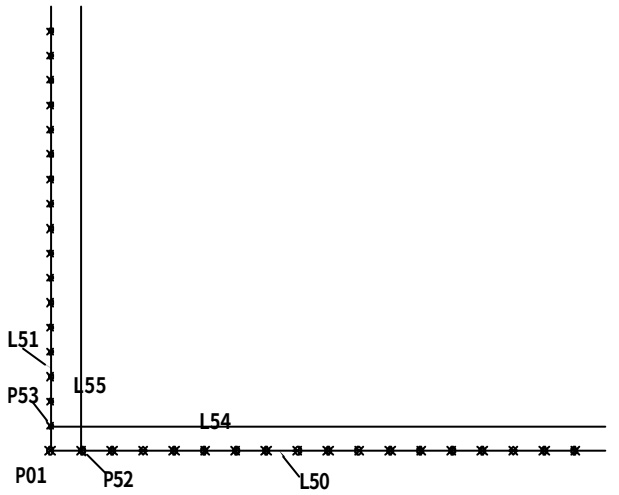
| | |
|---|---|
|  | 17、 TRim,<br>18、 Erase,<br>19、 Line, end, C54 **右**, (**正交模式**), **向下** 15, →L63<br>20、 Offset, 5.5, L63, **向右點**, →L64<br>21、 Line, end, L63 **下**, end, L64 **下**, →L65<br>22、 Offset, 5.5, L65, **向上點**, →L66 |
|  | 23、 TRim,<br>24、 Erase,<br>25、 MIrror, **所有物件**, end, L65 **左**, end, L65 **右**, **→下方對稱物件** |
|  | 26、 Erase,<br>27、 DIst, cen, C54, cen, C60, →X **差值**=90.0814<br>28、 **點選弧** B, CTRL+1, **→查幾何半徑值** =12.0244<br>29、 **點選弧** C, CTRL+1, **→查幾何弧長值** =135.0553<br>30、 **點選直線** D, CTRL+1, **→查幾何長度值** =72.2628<br>31、 Hatch, **點選** E **區域**, **→產生斜線區域**<br>32、 **點選** E **斜線區域**, CTRL+1, **→查幾何面積值** =8135.9266 |

## 2.1.15 AutoCAD2D2008-115



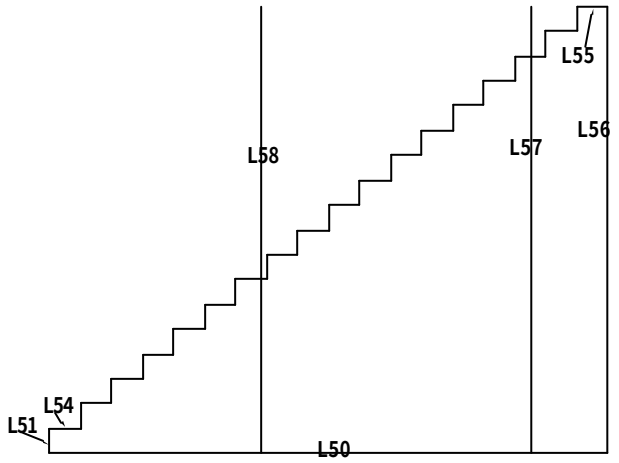

1、 交點 A 至交點 B 距離為何？ 140.5017
2、 15 個半圓弧總長為何？ 274.8894
3、 交點 C 至交點 D 水平距離為何？ 115.6667
4、 尺寸 E 值為何？ 83.9799
5、 F 區域的面積為何？ 15508.7748

**解題步驟：**

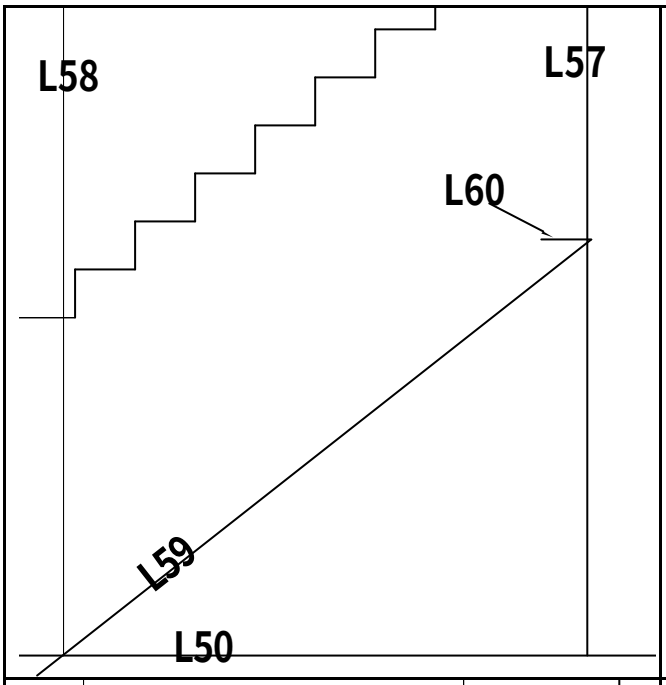



01、 Line, 任一點 P01, (正交模式), 向右 218.5, →L50
02、 Line, end, L50 左, (正交模式), 向上 175, →L51
03、 DIVide, L50, 18, →P52 等 17 個點(水平方向)
04、 DIVide, L51, 18, →P53 等 17 個點(垂直方向)
05、 Offset, T, L50, nod, P53, →L54
06、 Offset, T, L51, nod, P52, →L55

07、 TRim,
08、 Erase,

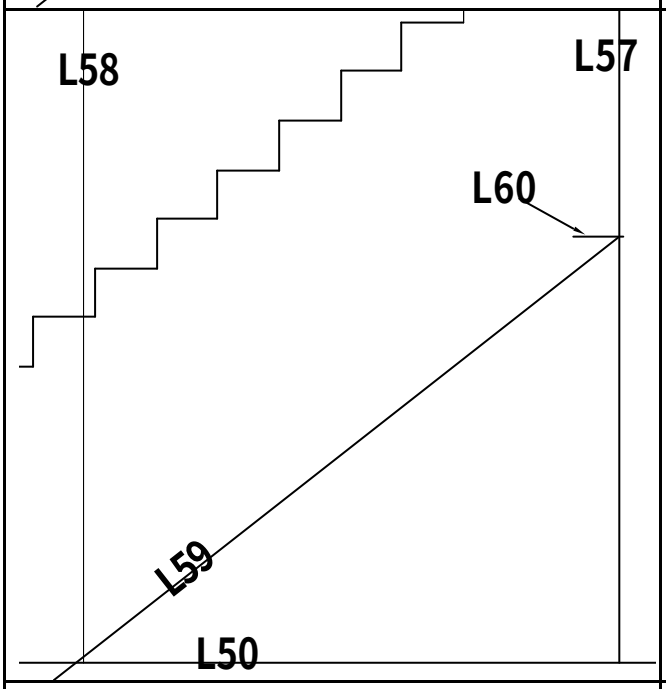



09、 ARray, 矩形陣列, 列, 1, 行, 18, 列偏移, 0, 行偏移, end, L51 下, end, L54 右, 陣列角度, end, L51 下, end, L54 右, 物件, L51;L54, →L55 等 18 組物件
10、 Line, end, L55 右, end, L50 右, →L56 (點選 L56, CTRL+1, →查幾何長度值=175)
11、 Offset, 30, L56, 向左點,→L57
12、 Offset, 135.8, L56, 向左點,→L58

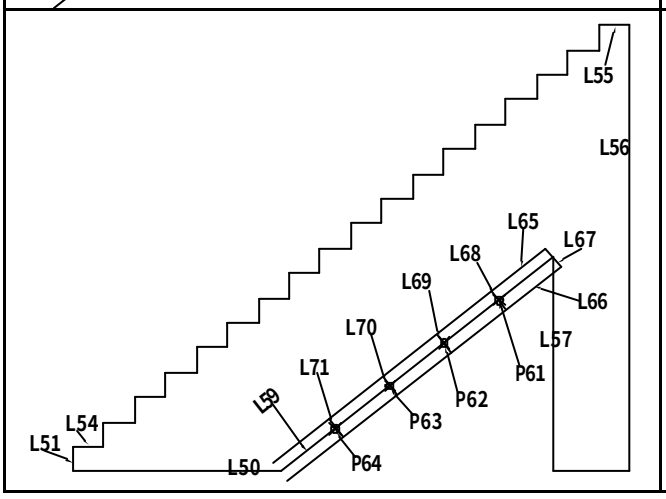



13、 ROtate, L50, int, L58 & L50, C, 38.2(=90-51.8), →L59

14、 Line, end, L59 **上**, (**正交模式**), **向左** 10, →L60




15、 Move, L59, end, L60 **右**, int, L57 & L60, →L59 **向左移**




16、 TRim,

17、 Erase,

18、 DIVide, L59, 5, →P61~P64

19、 Offset, 4.5(=9/2), L59, **向上點**, →L65

20、 Offset, 4.5(=9/2), L59, **向下點**, →L66

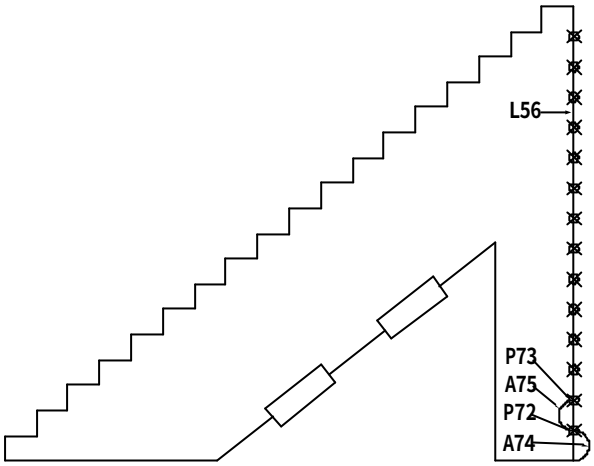
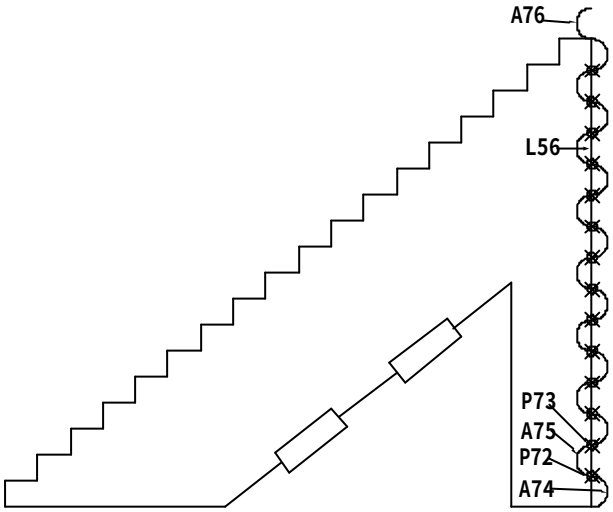
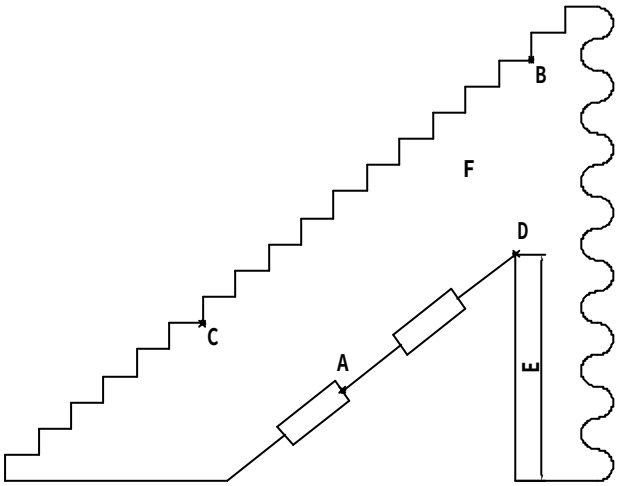
21、 Line, end, L65 **上**, end, L66 **上**, →L67

22、 Offset, T, L67, nod, P61, →L68

23、 Offset, T, L67, nod, P62, →L69

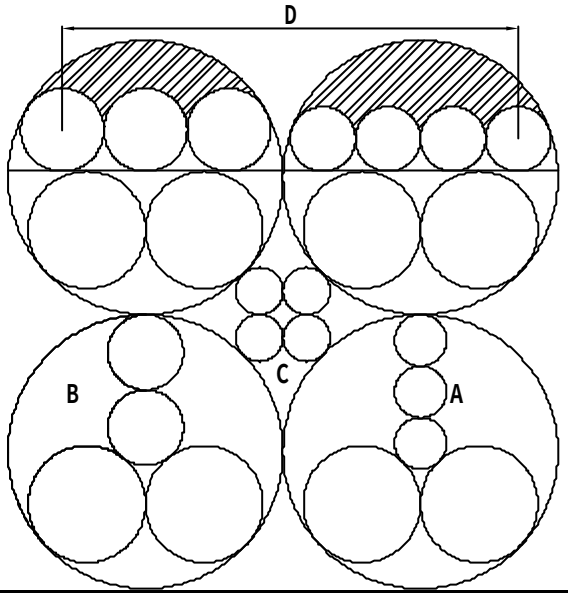
24、 Offset, T, L67, nod, P63, →L70

25、 Offset, T, L67, nod, P64, →L71

| | |
|---|---|
|  | 26、 TRim,<br>27、 Erase,<br>28、 DIVide, L56, 15, →P72;P73 **等各點**<br>29、 Arc, C, m2p, end, L56 **下**, nod, P72, end, L56 **下**, nod, P72, →A74<br>30、 Arc, C, m2p, nod, P73, nod, P72, nod, P73, nod, P72, →A75 |
|  | 31、 ARray, **矩形陣列**, **列**, 8(=15/2), **行**, 1, **列偏移**, end, L56 **下**, nod, P73, **行偏移**, 0, **陣列角度**, 0, **物件**, A74;A75, →A76 **等** 8 **組物件** |
|  | 32、 Erase,<br>33、 DIst, int, A, int, B, →**距離**=140.5017<br>34、 PEdit, A74, Y, J, A75, **及其他半圓物件**, →**成為聚合線**<br>35、 **點選聚合線**, CTRL+1, →**查幾何長度值**=274.8894<br>36、 DIst, int, C, int, D, →X **差值**=115.6667<br>37、 **點** E **直線**, CTRL+1, →**查幾何長度值**=83.9799<br>38、 Hatch, **點** F **區域**, →**產生斜線區域**<br>39、 **點選** F **斜線區域**, CTRL+1, →**查幾何面積值**=15508.7748 |

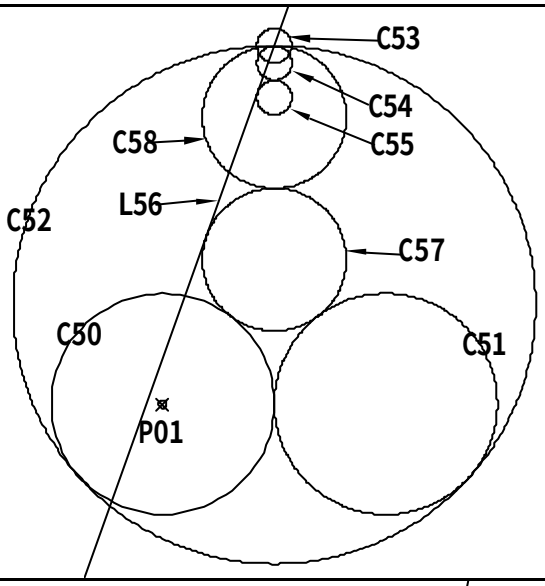
## 2.2 第二類：綜合應用二

### 2.2.1 AutoCAD2D2008-201

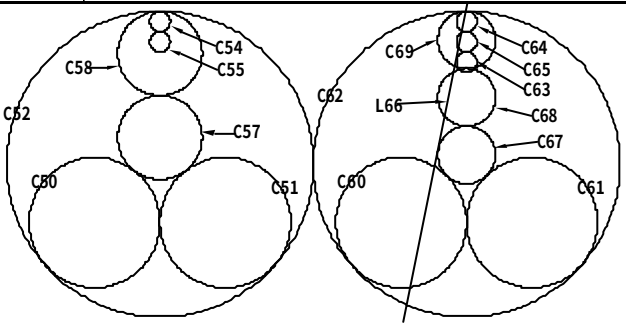



1、 **圓** A **直徑值為何？** 11.2848
2、 **區域** B **周長為何？** 153.3837
3、 **區域** C **面積為何？** 104.2242(**課本**:104.2243)
4、 **水平距離** D **值為何？** 99.4580
5、 **斜線區域總面積為何？** 1010.0215

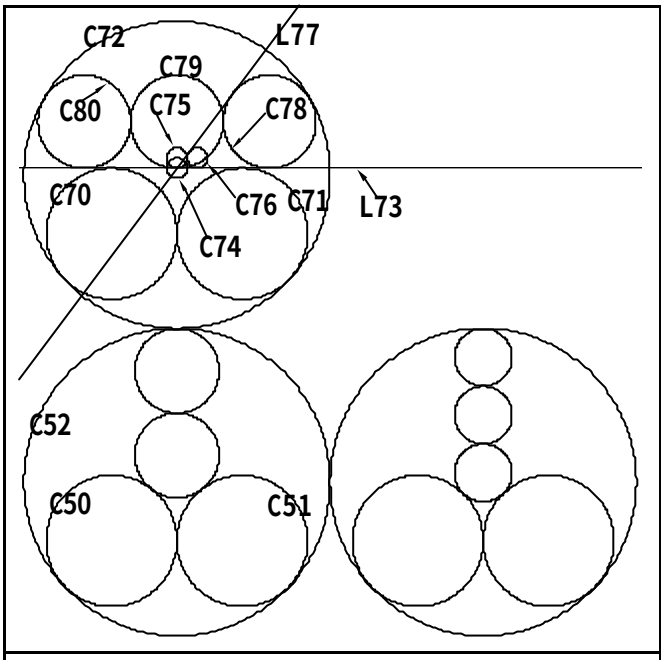
**解題步驟：**



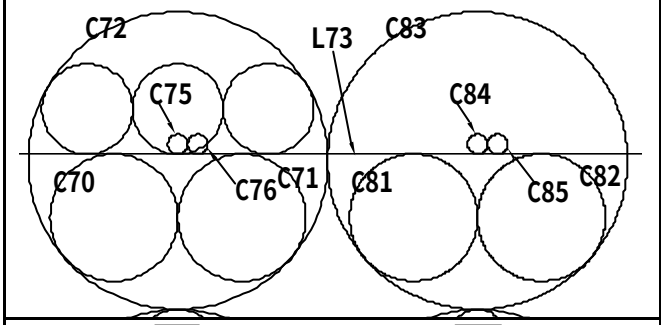

01、 Circle, **任一點** P01, D, 25.6, →C50
02、 COpy, C50, qua, C50 **左**, qua, C50 **右**, →C51
03、 Circle, T, C50 **左下**, C51 **右下**, 30, →C52
04、 Circle, qua, C52 **上**, 2(**大約**), →C53
05、 COpy, C53, qua, C53 **上**, qua, C52 **上**, →C54, qua, C54 **下**, →C55
06、 XLine, qua, C52 **上**, tan, C55 **左**, →L56
07、 Circle, 3P, tan, L56, tan, C50**右上**, tan, C51**左上**, →C57
08、 COpy, C57, qua, C57 **下**, qua, C57 **上**, →C58



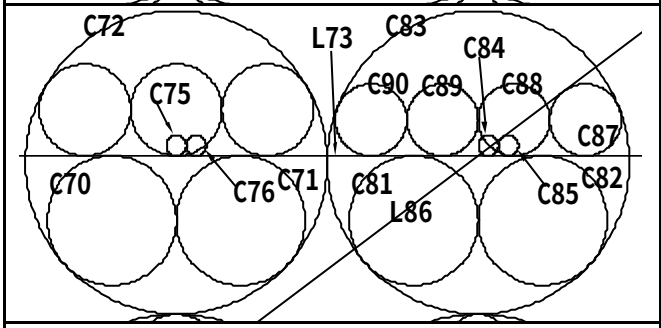

09、 Erase,
10、 COpy, C50;C51;C52;C54;C55, qua, C52 **左**, qua, C52 **右**, →C60;C61;C62;C64;C65
11、 COpy, C65, qua, C65 **上**, qua, C65 **下**, →C63
12、 XLine, qua, C62 **上**, tan, C63 **左**, →L66
13、 Circle, 3P, tan, L66, tan, C60**右上**, tan, C61**左上**, →C67
14、 COpy, C67, qua, C67 **下**, qua, C67 **上**, →C68, qua, C68 **上**, →C69



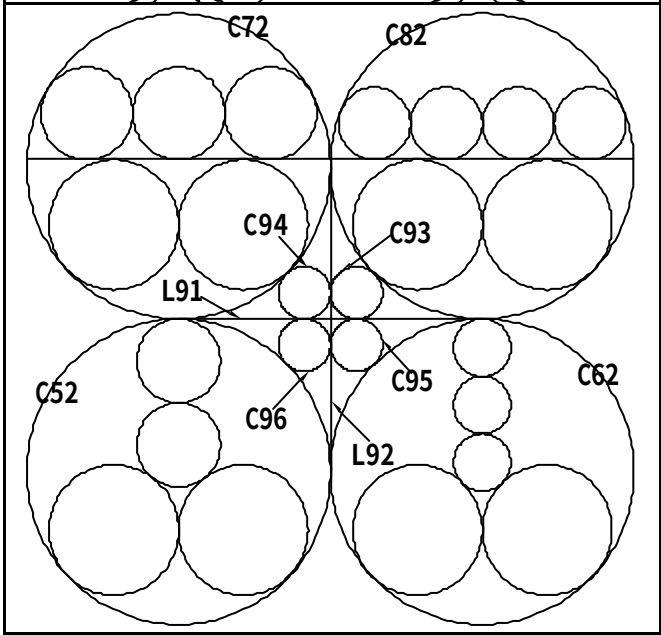

15、 Erase,
16、 COpy, C50;C51;C52, qıa, C52 下, qua, C52 上, →C70;C71;C72
17、 XLine, qua, C70 上, qua, C71 上, →L73
18、 Circle, m2p, qua, C70 上, qua, C71 上, 2, →C74
19、 COpy, C74, cen, C74, qua, C74 上, →C75
20、 COpy, C75, qua, C75 左, qua, C75 右, →C76
21、 XLine, cen, C74, tan, C76 左上, →L77
22、 Circle, 3P, tan, L77, tan, L73, tan, C72 右上, →C78
23、 COpy, C78, qua, C78 右, qua, C78 左, →C79, qua, C79 左, →C80,



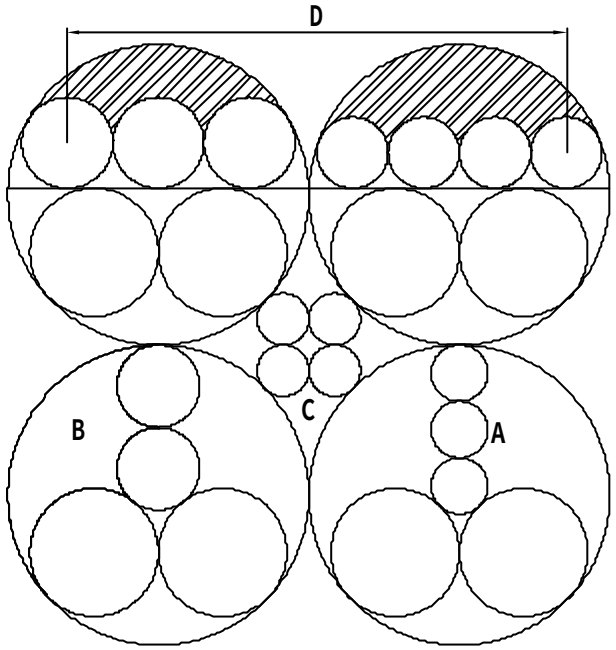

24、 Erase,
25、 COpy, C70;C71;C72;C75;C76, qua, C72 左, qua, C72 右, →C81;C82;C83;C84;C85



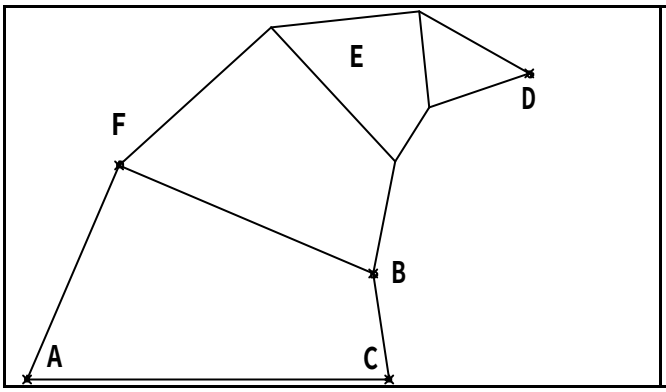

26、 Move, C84;C85, qua, C84 左, cen, C84, →C84;C85 向右移
27、 XLine, m2p, qua, C81 上, qua, C82 上, tan, C85 左上, →L86
28、 Circle, 3P, tan, L86, tan, L73, tan, C83 右上, →C87
29、 COpy, C87, qua, C87 右, qua, C87 左, →C88, qua, C88 左, →C89, qua, C89 左, →C90




30、 Earse,
31、 Line, int, C62 & C82, int, C52 & C72, →L91
32、 Line, int, C72 & C82, int, C52 & C62, →L92
33、 Circle, 3P, tan, L91, tan, L92, tan, C82 左下, →C93
34、 MIrror, C93, end, L92 上, end, L92 下, →C94
35、 MIrror, C93;C94, end, L91 左, end, L91 右, →C95;C96

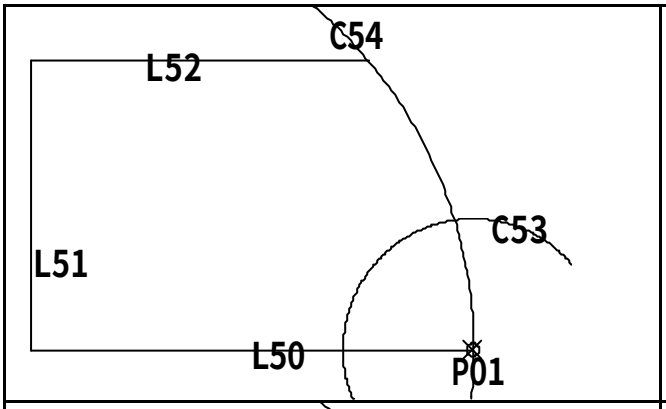
| | |
|---|---|
|  | 36、 **選** A, CRTL+1, **→查幾何直徑值**=11.2848<br>37、 BOundary, **點** B **區域**<br>38、 **點** B **區域邊緣**, CRTL+1, **→查幾何長度值**=153.3837<br>39、 Hatch, **點** C **區域**, **→產生斜線區域**<br>40、 **點** C **區域斜線**, CRTL+1, **→查幾何面積值**=104.2243<br>41、 DIst, cen, **左側圓**, cen, **右側圓**, →X **差值**=99.4580<br>42、 Hatch, **點左上斜線區域**, **點右上斜線區域**<br>43、 **點斜線區**, CRTL+1, **→查幾何面積值**=1010.0215 |
| | 44、 **如果無法正常產生斜線區域**, COpy **物件**,<br>45、 TRim, **成左圖**<br>46、 Hatch, **以選取物件方式**, **產生斜線區域** |

## 2.2.2 AutoCAD2D2008-202

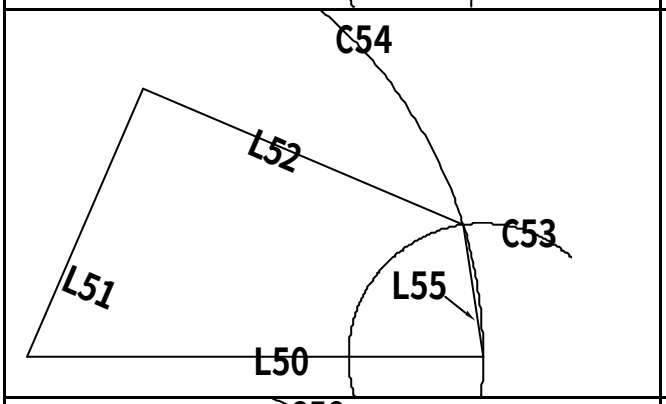



1、 **交點** A **至交點** B **距離為何？** 85.1469
2、 **交點** C **至交點** D **高度為何？** 72.0247
3、 E **區域面積為何？** 714.9200
4、 **交點** F **至交點** D **距離為何？** 98.6386
5、 **圖形所圍成的面積為何？** 5721.0237

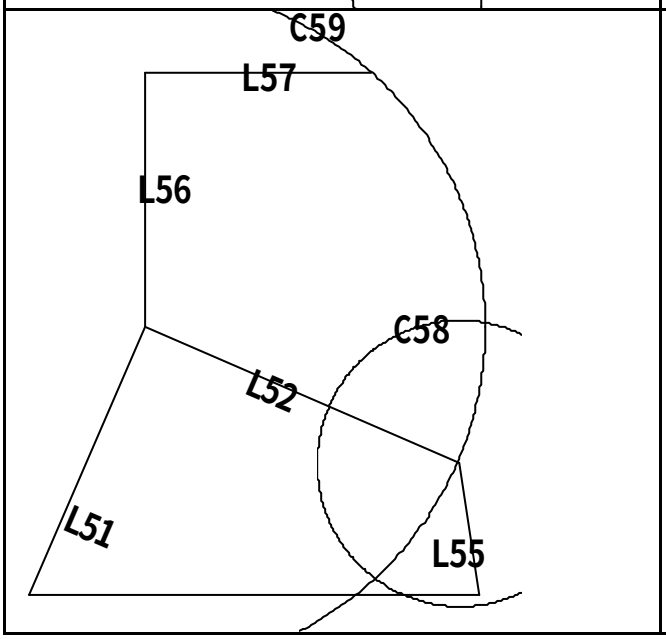
**解題步驟：**



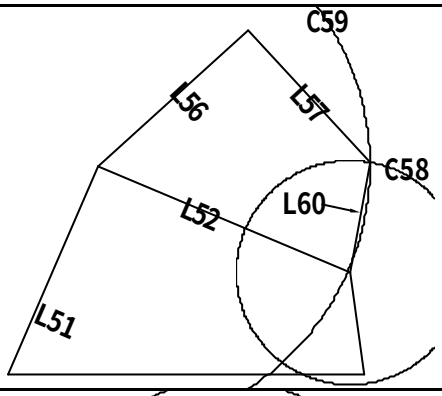
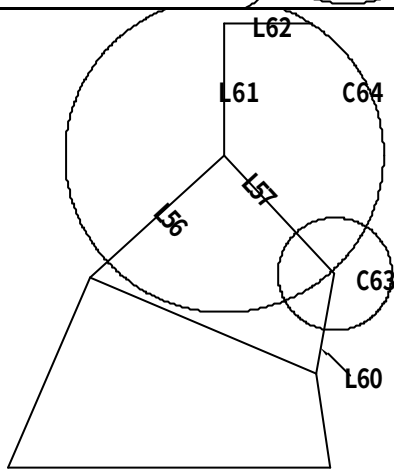
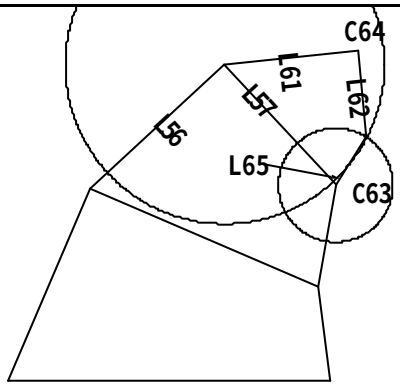
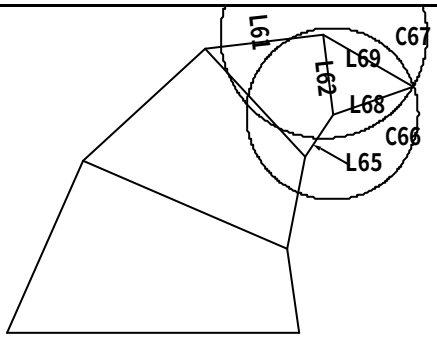
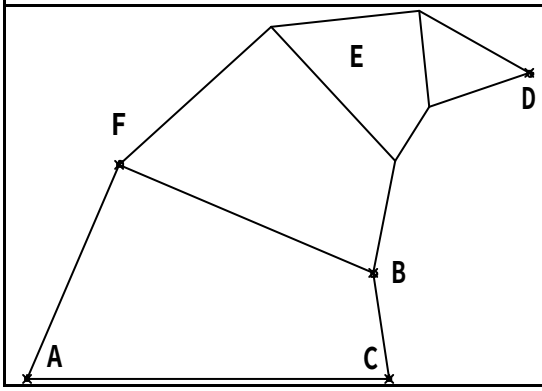

01、 Line, **任一點** P01, (**正交模式**), **向左** 85, →L50, **向上** 55, →L51, **向右** 65, →L52
02、 Circle, end, L50 **右**, 25, →C53
03、 Circle, int, L50 & L51, end, L52 **右**, →C54



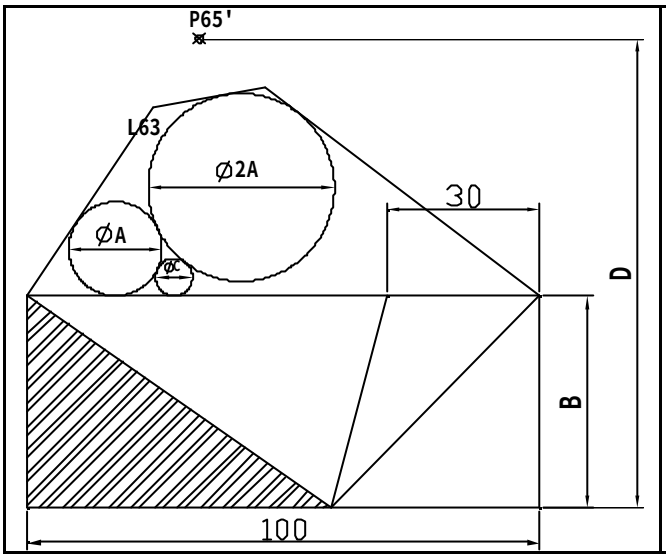

04、 ROtate, L51;L52, int, L50 & L51, C, R, int, L50 & L51, end, L52 **右**, int, C53 & C54, →L51;L52 **轉動**
05、 Line, end, L50 **右**, int, C53 & C54, →L55




06、 Erase,
07、 Line, int, L51 & L52, (**正交模式**), **向上** 48, →L56, **向右** 43, →L57
08、 Circle, int, L52 & L55, 27, →C58
09、 Circle, int, L51 & L52, end, L57 **右**, →C59

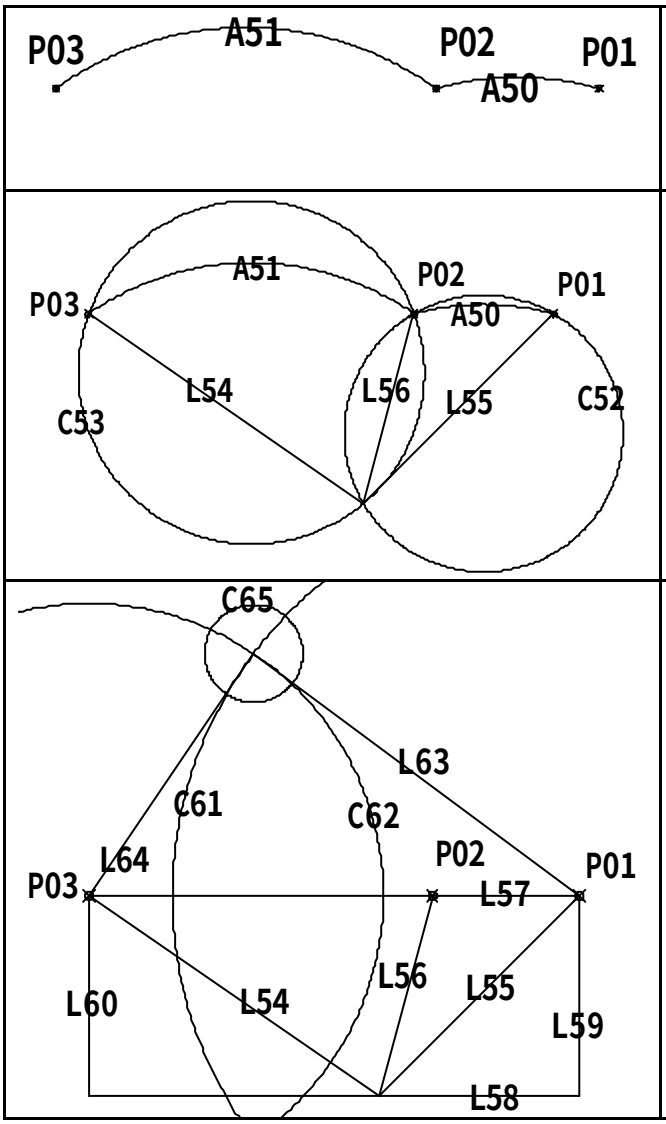
| | |
|---|---|
|  | 10、 ROtate, L56;L57, int, L51 & L52, C, R, int, L51 & L52, end, L57 **右**, int, C58 & C59, →L56;L57 **轉動**<br>11、 Line, end, L52 **右**, int, C58 & C59, →L60 |
|  | 12、 Erase,<br>13、 Line, int, L56 & L57, (**正交模式**), **向上** 35, →L61, **向右** 23, →L62<br>14、 Circle, int, L57 & L60, 15, →C63<br>15、 Circle, int, L56 & L57, end, L62 **右**, →C64 |
|  | 16、 ROtate, L61;L62, int, L56 & L57, C, R, int, L56 & L57, end, L62 **右**, int, C63 & C64, →L61;L62 **轉動**<br>17、 Line, end, L57 **下**, int, C63 & C64, →L65 |
|  | 18、 Erase,<br>19、 Circle, int, L62 & L65, 25, →C66<br>20、 Circle, int, L61 & L62, 30, →C67<br>21、 Line, int, L62 & L65, int, C66 & C67, →L68, int, L61 & L62, →L69 |
|  | 22、 Erase,<br>23、 DIst, int, A, int, B, →**距離**=85.1469<br>24、 DIst, int, C, int, D, →Y **差值**=72.0247<br>25、 Hatch, **點** E **區域**, →**斜線區域**<br>26、 **點** E **區域**, CRTL+1, →**查幾何面積值** =714.9200<br>27、 DIst, int, F, int, D, →**距離**= 98.6386<br>28、 Hatch, **分別點** 4 **個區域**<br>29、 **點** 4 **個區域斜線**, CRTL+1, →**查幾何面積值** =5721.0237 |

## 2.2.3 AutoCAD2D2008-203

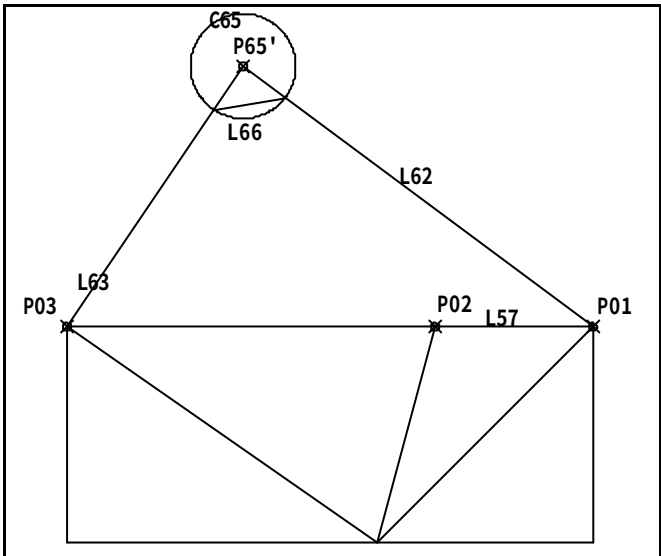
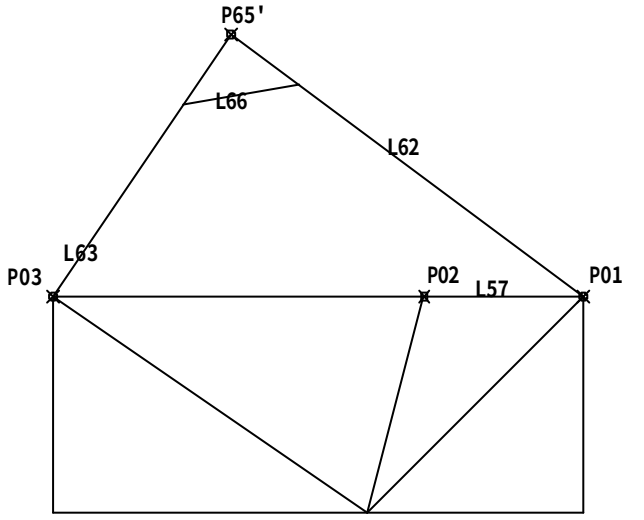
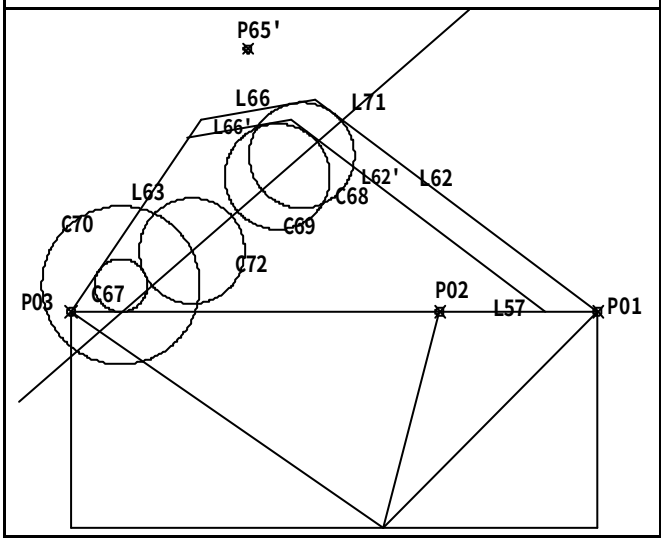


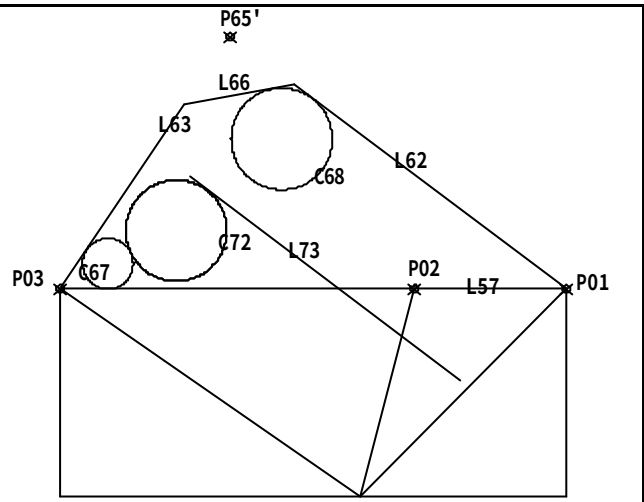
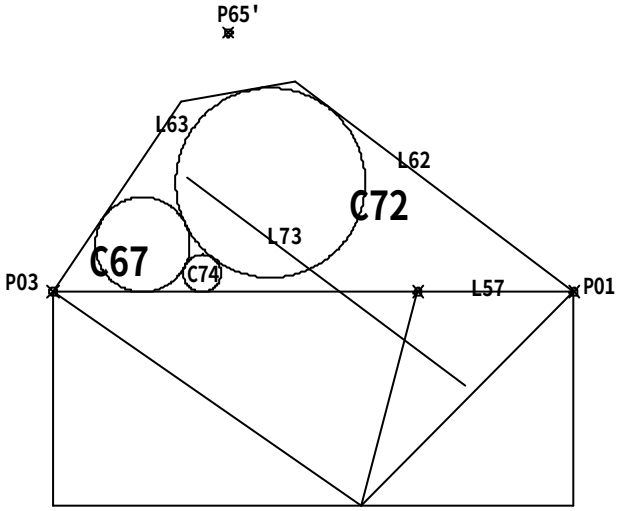
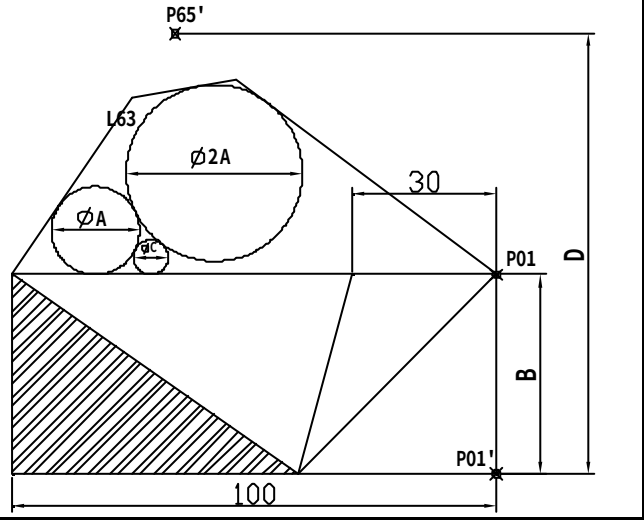

1、 **直徑** A **值為何？** 18.1854
2、 B **尺寸值為何？** 41.1316
3、 **直徑** C **值為何？** 7.1774
4、 **尺寸** D **垂直高度為何？** 90.8715
5、 **斜線區域面積為何？** 1215.5789

**解題步驟：**

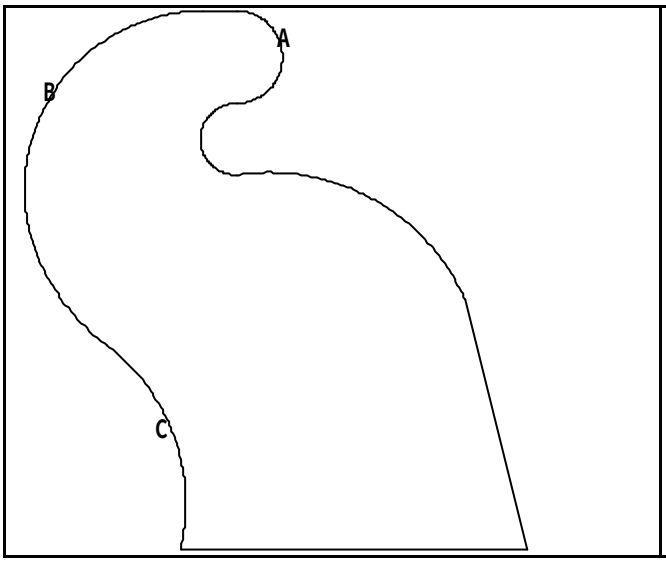



01、 Point, **任一點**, →P01
02、 COpy, P01, **任一點**,**正交模式**, **向左** 30, →P02, **向左** 100, →P03
03、 Arc, P01, E, P02, A, 30, →A50
04、 Arc, P02, E, P03, A, 70, →A51
05、 Circle, 3P, nod, P01, nod, P02, cen, A50, →C52(**圓周角相等**)
06、 Circle, 3P, nod, P02, nod, P03, cen, A51, →C53(**圓周角相等**)
07、 Line, nod, P03, int, C52 & C53, →L54, nod, P01, →L55
08、 Line, nod, P02, int, C52 & C53, →L56
09、 Line, nod, P03, nod, P01, →L57
10、 Offset, T, L57, int, L54 & L55, →L58
11、 Line, nod, P01, end, L58 **右**, →L59
12、 Line, nod, P03, end, L58 **左**, →L60
13、 Circle, nod, P01, 83, →C61
14、 Circle, nod, P03, 60, →C62
15、 Line, nod, P01, int, C61 & C62, →L63
16、 Line, nod, P03, int, C61 & C62, →L64
17、 Circle, int, C61 & C62, 10(**大約**), →C65

| | |
|---|---|
|  | 18、 Erase,<br>19、 POint, int, L62 & L63, →P65’<br>20、 Line, int, L62 & C64, int, L63 & C64, →L66 |
|  | 21、 Erase,<br>22、 SCale, L66, nod, P65’ , R, int, L63 & L66, int, L62 & L66, 22, →L66 **改變** |
|  | 23、 TRim,<br>24、 Erase,<br>25、 Circle, T, tan, L57, tan, L63, 5(**大約**), →C67<br>26、 Circle, T, tan, L62, tan, L66, 10(C67 **半徑的** 2 **倍**), →C68<br>27、 SCale, L62; L66, nod, P03, C, 0.9, →L62’ ; L66’<br>28、 Circle, T, tan, L62’ , tan, L66’ , 10(C67 **半徑的** 2 **倍**), →C69<br>29、 Offset, 10(C67 **直徑**), C67, **向外**, →C70<br>30、 XLine, cen, C68, cen, C69, →L71<br>31、 Circle, int, C70 & L71, 10(C67 **半徑的** 2 **倍**), →C72 |

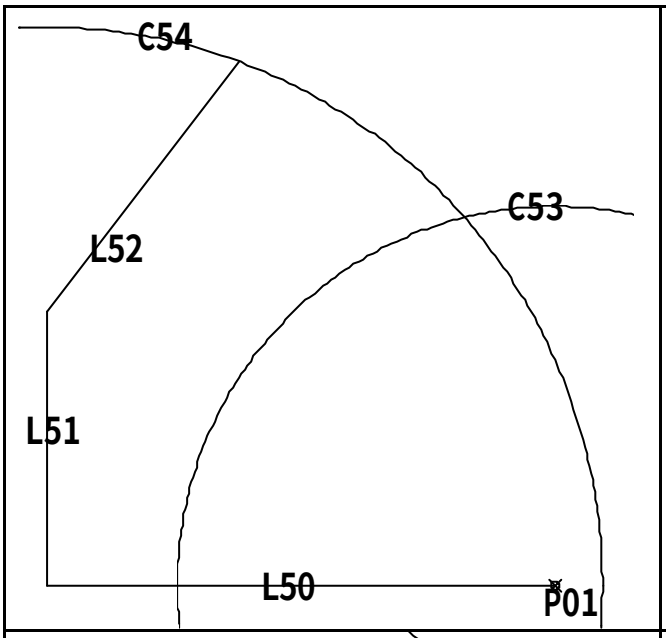
| | |
|---|---|
|  | 32、 Erase,<br>33、 COpy, L62, cen, C68, cen, C72, →L73 |
|  | 34、 Erase,<br>35、 SCale, C67;C72, nod, P03, R, nod, P03, int, L53 & L73, nod, P01, →C67 **與** C72 **皆放大**<br>36、 Circle, 3P, tan, L57, tan, C67, tan, C72, →C74 |
|  | 37、 Erase,<br>38、 **選** A, CRTL+1 →**查幾何直徑值**=18.1854<br>39、 DIst, nod, P01' , nod, P01, →Y **差值**=41.1316<br>40、 **選** C, CRTL+1, →**查幾何直徑值**=7.1774<br>41、 DIst, nod, P01' , nod, P65, →Y **差值**=90.8715<br>42、 Hatch, **點左邊斜線區域**, →**斜線區域**<br>43、 **點左邊斜線**, CRTL+1, →**查幾何面積值** =1215.5789 |

## 2.2.4 AutoCAD2D2008-204

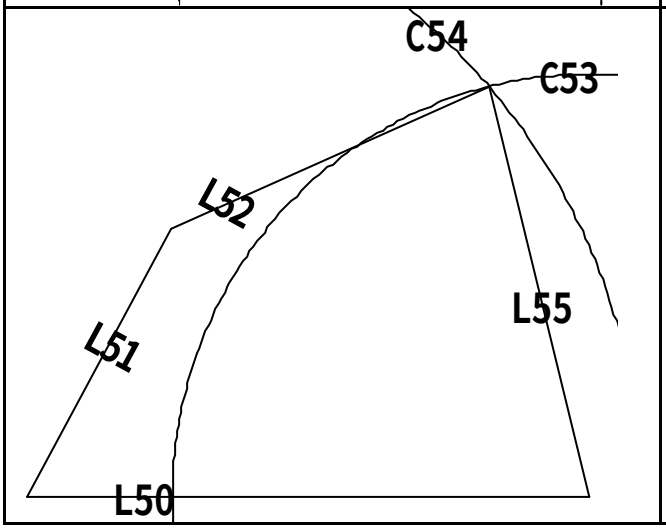



1、 **尺寸** A **半徑為何？**12.9802
2、 **弧** B **長度為何？** 99.3324
3、 **弧** C **至弧** B **中心點距離為何？** 106.3866
4、 **弧** C **長度為何？** 64.0974
5、 **圖形所圍成的面積為何？** 12737.8749

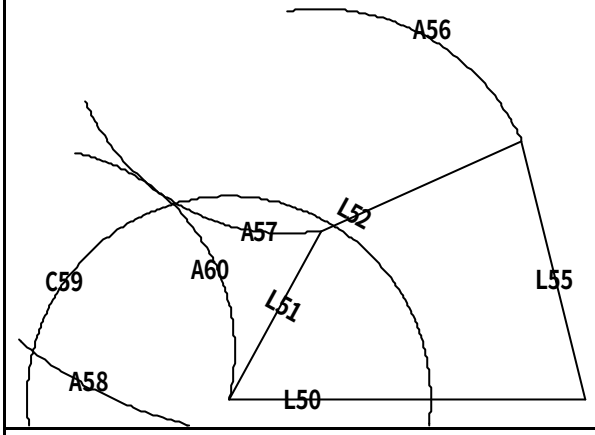
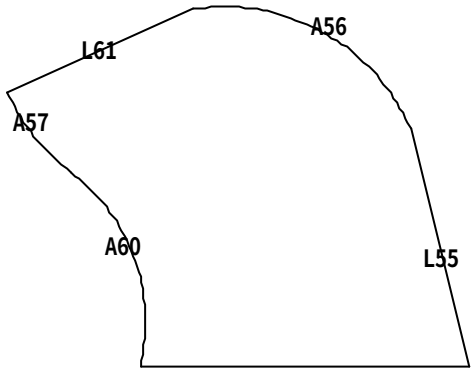
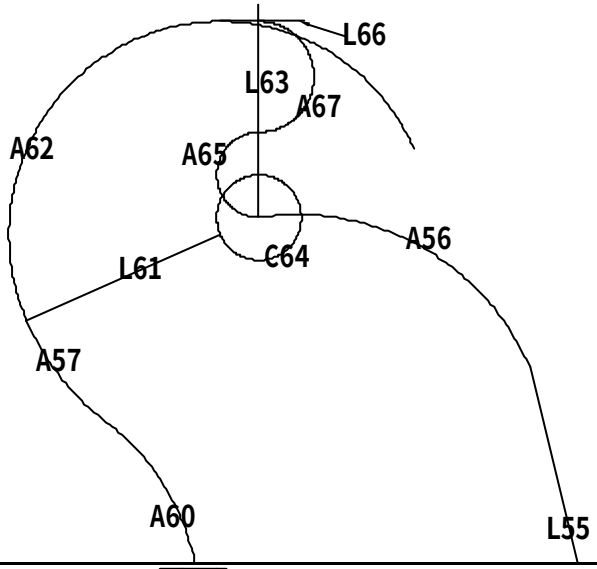
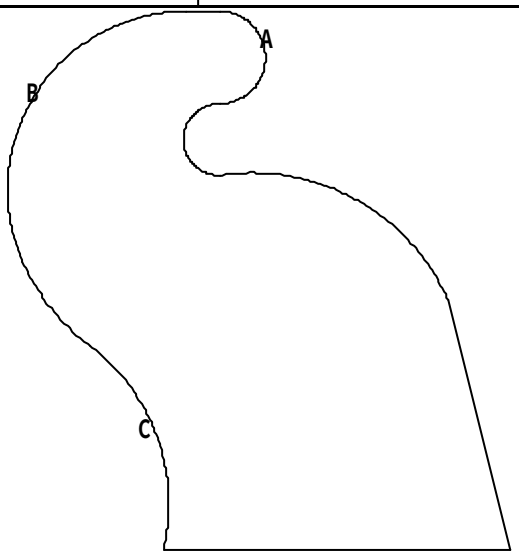
**解題步驟：**



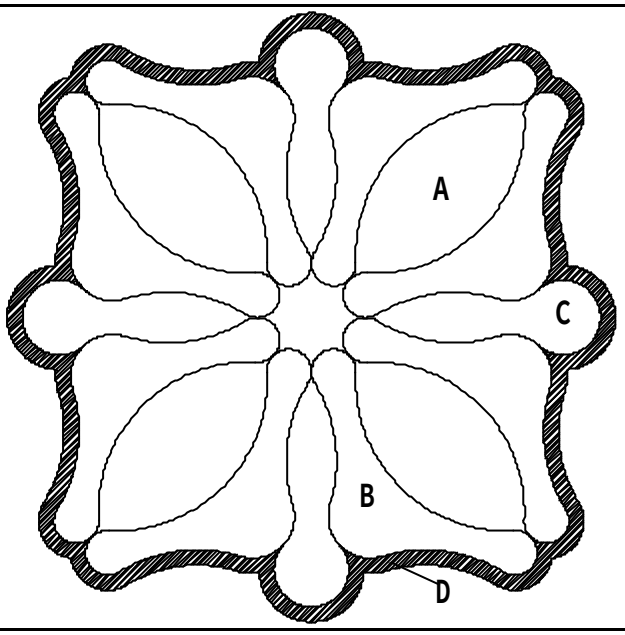

01、 Line, **任一點** P01, (**正交模式**), **向左** 97, →L50, **向上** 52, →L51, @60<52.4, →L52
02、 Circle, end, L50 **右**, 72, →C53
03、 Circle, int, L50 & L51, end, L52 **上**, →C54




04、 ROtate, L51;L52, int, L50 & L51, C, R, int, L50 & L51, end, L52 **上**, int, C53 & C54, →L51;L52 **轉動**
05、 Line, end, L50 **右**, int, C53 & C54, →L55

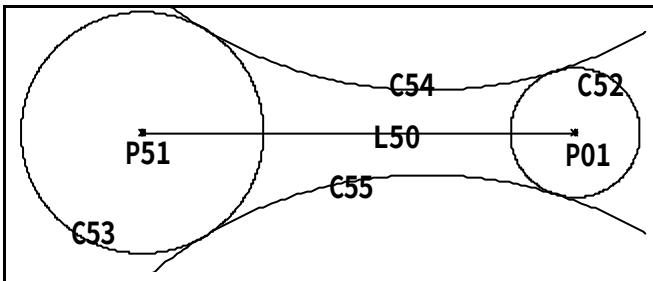
| | |
|---|---|
|  | 06、 Erase,<br>07、 Arc, C, int, L51 & L52, int, L52 & L55, A, 75, →A56<br>08、 Arc, C, end, A56 上, int, L51 & L52, A, -75, →A57<br>09、 Offset, 55, A57, **向外點**, →A58<br>10、 Circle, int, L50 & L51, 55, →C59<br>11、 Arc, C, int, A58 & C59, int, L50 & L51, end, A57 上, →A60 |
|  | 12、 TRim,<br>13、 Erase,<br>14、 Line, end, A56 上, end, A57 上, →L61 |
|  | 15、 LENgthen, T, 50, L61 上, →L61 **縮短**<br>16、 Arc, C, end, L61 上, int, A57 & L61, A, -180, →A62<br>17、 Line, end, A56 上, (**正交模式**), **向上** 50, →L63<br>18、 Circle, int, A56 & L63, 10, →C64<br>19、 Arc, C, int, C64 & L63, cen, C64, A, -180, →A65<br>20、 Line, qua, A62 上, (**正交模式**), **向右** 20, →L66<br>21、 Arc, C, m2p, int, L63 & A65, int, L63 & L66, int, L63 & A65, int, L63 & L66, →A67 |
|  | 22、 TRim,<br>23、 Erase,<br>24、 **點選弧** A, CTRL+1, →**查幾何半徑值** =12.9802<br>25、 **點選弧** B, CTRL+1, →**查幾何弧長值** =99.3324<br>26、 DIst, cen, C, cen, B, →**距離**=106.3866<br>27、 **點選弧** C, CTRL+1, →**查幾何弧長值** =64.0974<br>28、 Hatch, **點選內部**, →**斜線區域**<br>29、 **點選斜線區域**, CTRL+1, →**查幾何面積值** =12737.8749 |

## 2.2.5 AutoCAD2D2008-205

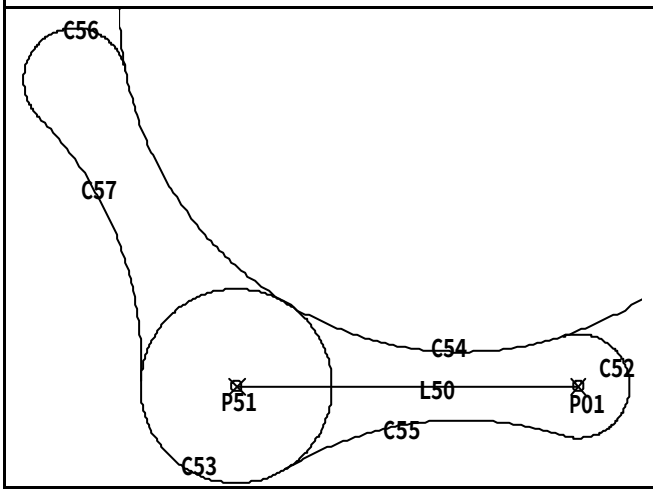



1、 A **區域周長為何？** 387.2618
2、 B **區域周長為何？** 509.5828
3、 C **區域周長為何？** 357.7635
4、 **斜線區域** D **面積為何？** 15686.2961
5、 **圖形所圍成外圍周長為何？** 1584.2512

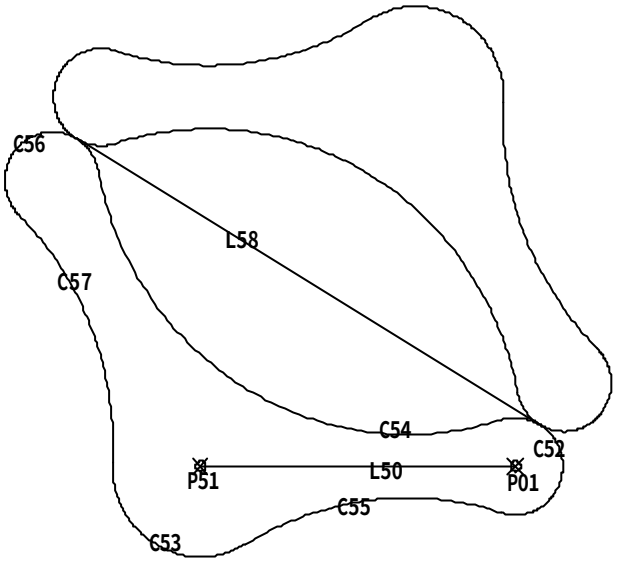
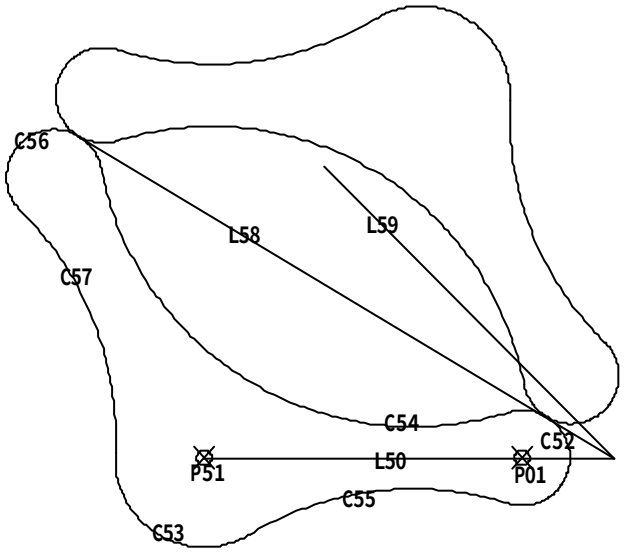
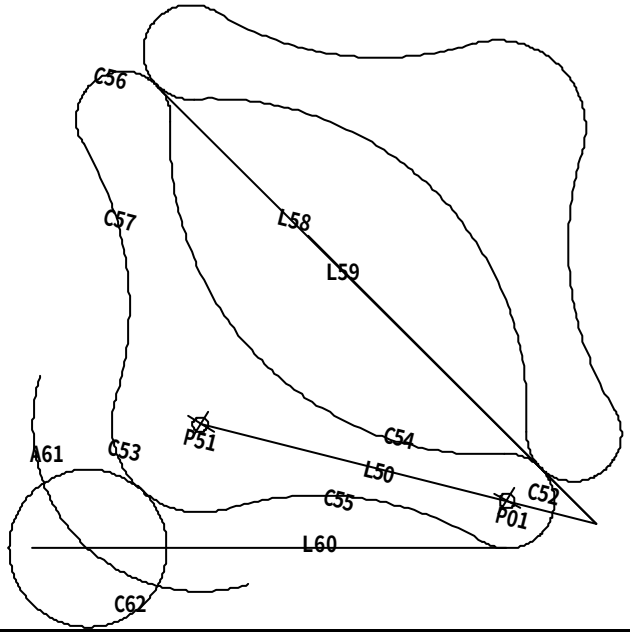
**解題步驟：**

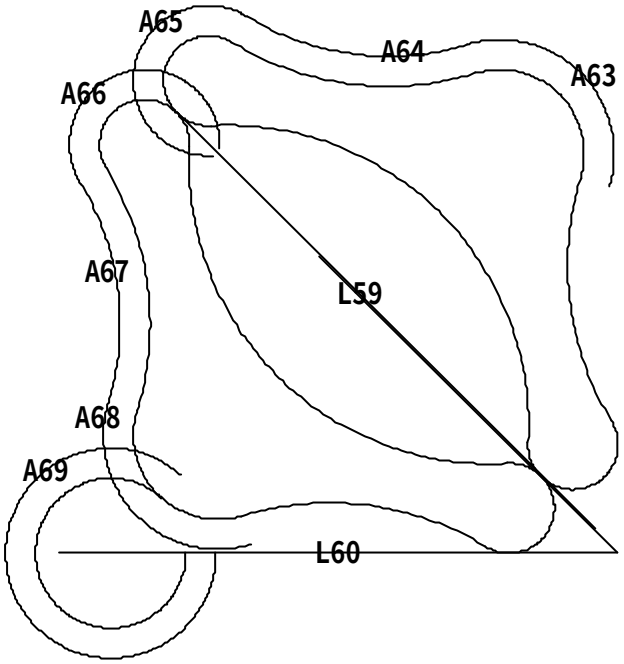
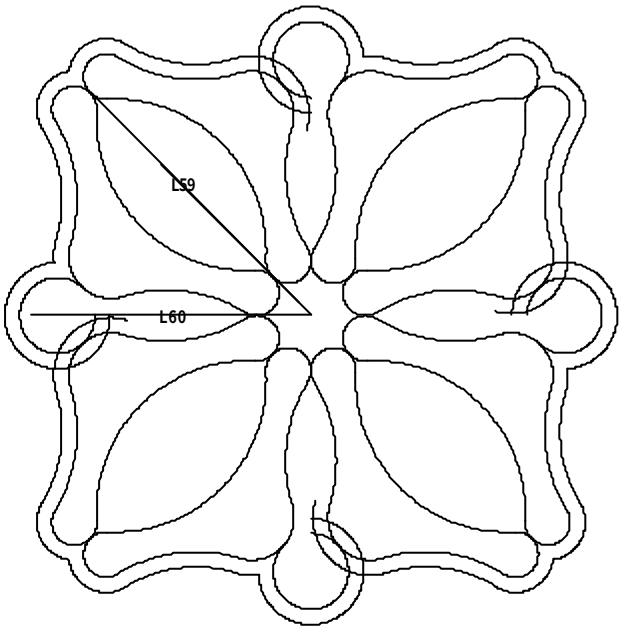
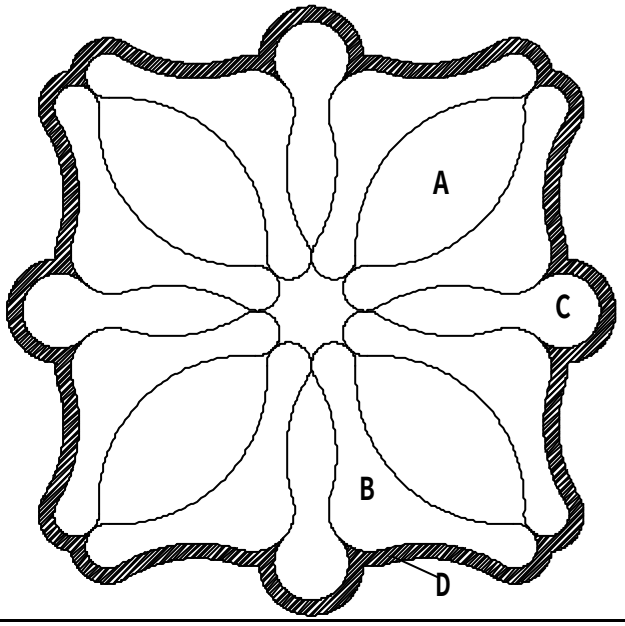



01、 Line, **任一點** P01, (**正交模式**), **向左**, 100, →L50
02、 POint, end, L50 **左**, →P51
03、 Circle, end, L50 **右**, 15, →C52
04、 Circle, end, L50 **左**, 28, →C53
05、 Circle, T, tan, C52 **左上**, tan, C53 **右上**, 100, →C54
06、 Circle, T, tan, C52 **左下**, tan, C53 **右下**, 100, →C55

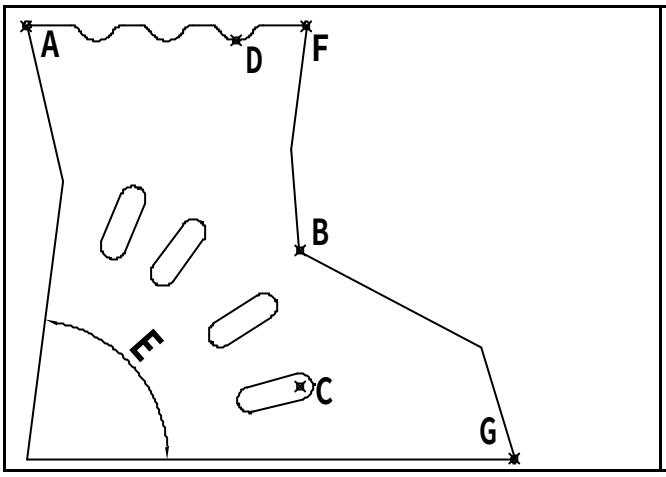



07、 TRim,
08、 MIrror, C52;C55, cen, C53, int, C53 & C54, →C56;C57

| | |
|---|---|
|  | 09、 TRim,<br>10、 Line, tan, C52 **上**, tan, C56 **右上**, →L58<br>11、 MIrror, C52~C57, end, L58 **上**, end, L58 **下**, **→對稱物件** |
|  | 12、 Fillet, T, T, R, 0, L50**右**, L58 **下**, →L50**與** L58 **相交於右下**<br>13、 ROtate, L50, int, L50 & L58, C, -45, →L59 |
|  | 14、 ROtate, **所有物件**(L59**除外**), int, L50 & L58, R, int, L50 & L58, mid, L58, end, L59 **上**, → **所有物件轉動**(L58 **與** L59 **重疊**)<br>15、 Line, qua, C52 **下**, (**正交模式**), **向左**, 150, →L60<br>16、 Offset, 25, C53, **向左下點**, →A61<br>17、 Circle, int, L60 & A61, 25, →C62 |

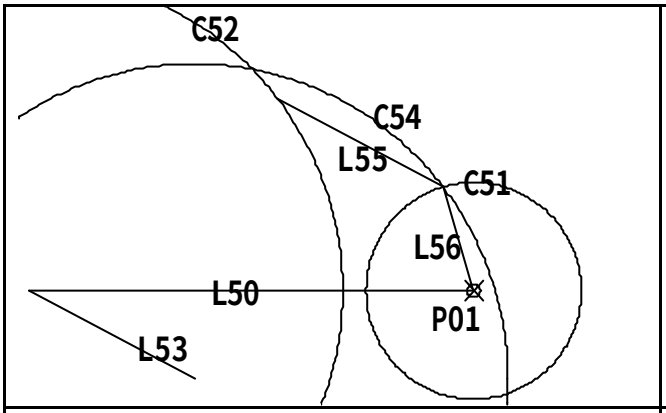
| | |
|---|---|
|  | 18、 TRim,<br>19、 Erase,<br>20、 Fillet, T, T, R, 0, L60 **右**, L59 **下**, →L60 **與** L59 **相交於右下**<br>21、 Offset, 10, **外圍物件**, →A63~A69 |
|  | 22、 TRim,<br>23、 ARray, **環形陣列**, **中心點**, int, L59 & L60, **項目總數**, 4, **佈滿角度**, 360, **物件**, **所有物件**(L59 **與** L60 **除外**), →**全圓佈滿** 4 **組相同物件** |
|  | 24、 TRim,<br>25、 Erase,<br>26、 BOundary, **點** A **區域**,<br>27、 **點** A **區域邊緣**, CRTL+1, →**查幾何長度值** =387.2618<br>28、 BOundary, **點** B **區域**,<br>29、 **點** B **區域邊緣**, CRTL+1, →**查幾何長度值** =509.5828<br>30、 BOundary, **點** C **區域**,<br>31、 **點** C **區域邊緣**, CRTL+1, →**查幾何長度值** =357.7635<br>32、 Hatch, **點** D **區域**, →**斜線區域**<br>33、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值** =15686.2961<br>34、 BOundary, **點** D **區域**,<br>35、 **點** D **區域邊緣**, CRTL+1, →**查幾何長度值** =1548.2512 |

## 2.2.6 AutoCAD2D2008-206

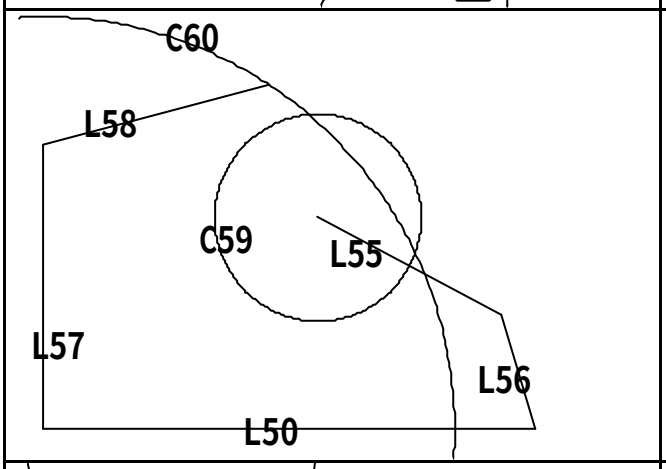



1、 **交點** A **至交點** B **距離為何？** 69.2049
2、 **中心點** C **至弧** D **中心點距離為何？** 72.5736
3、 **尺寸** E **角度為何？** 82.8841
4、 **交點** F **至交點** G **水平距離為何？** 40.5846
5、 **圖形扣除內孔的面積為何？** 5058.2505

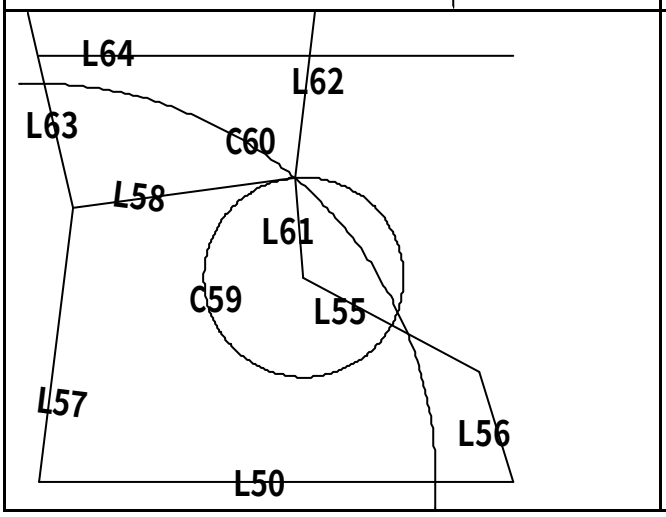
**解題步驟：**



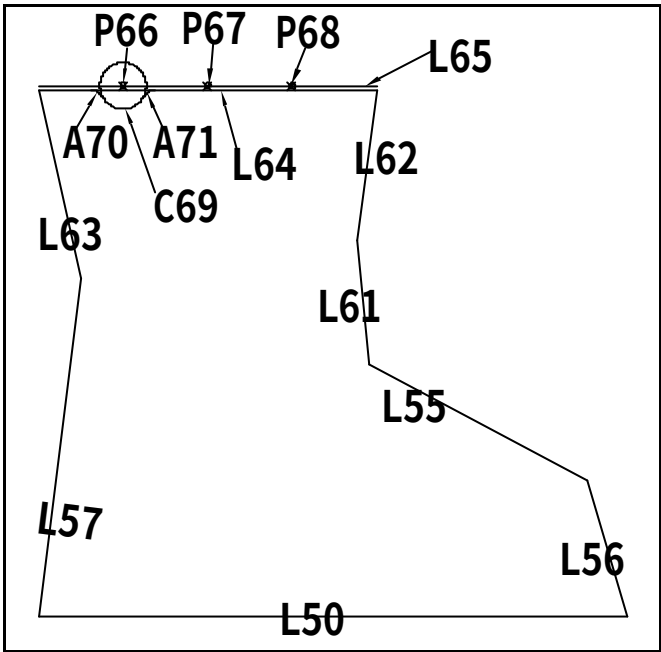
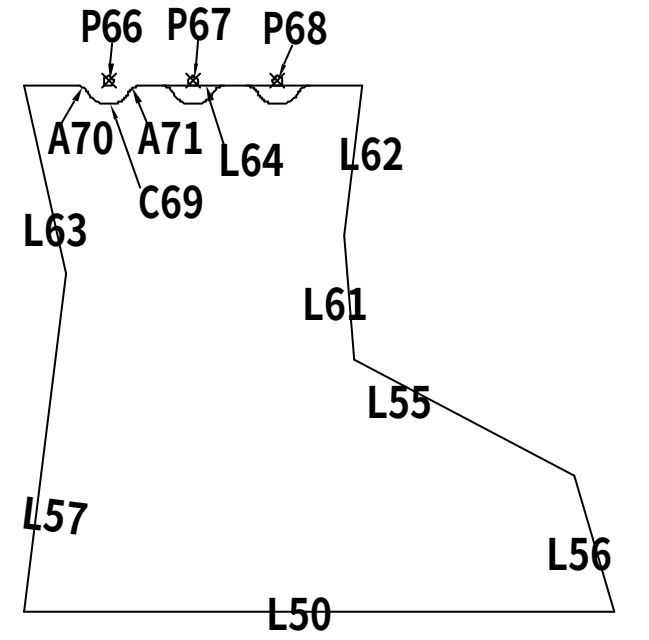
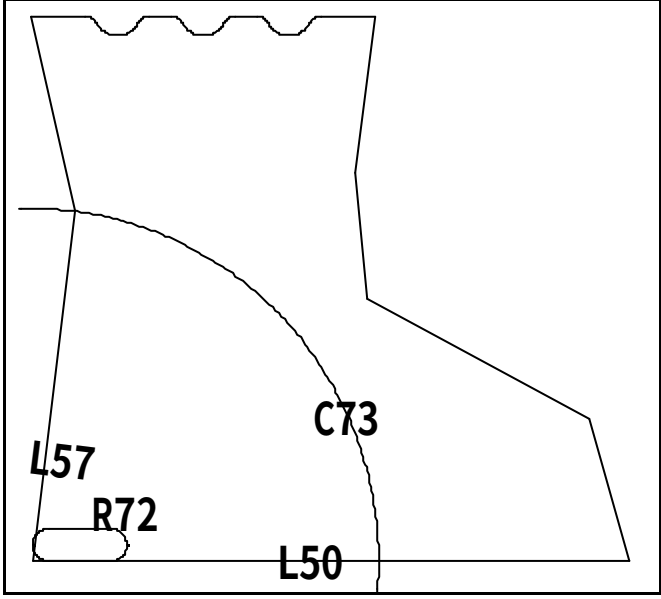

01、 Line, **任一點** P01, (**正交模式**), **向左** 95, →L50
02、 Circle, end, L50 **右**, 23, →C51
03、 Circle, end, L50 **左**, 67, →C52
04、 Line, end, L50 **左**, @40<-28, →L53
05、 Circle, end, L53 **下**, 67, →C54
06、 COpy, L53, end, L53 **下**, int, C51 & C54, →L55
07、 Line, end, L50 **右**, int, C51 & C54, →L56

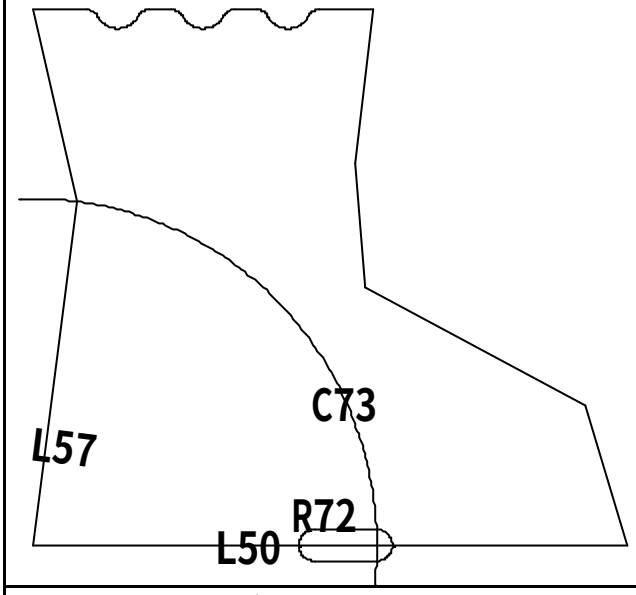
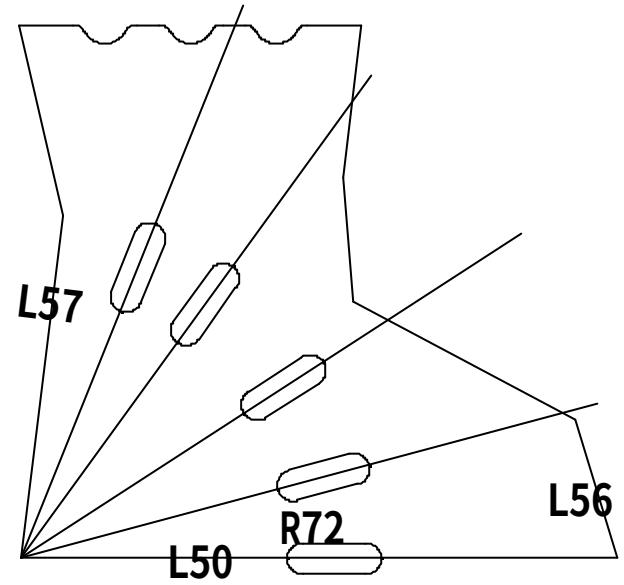
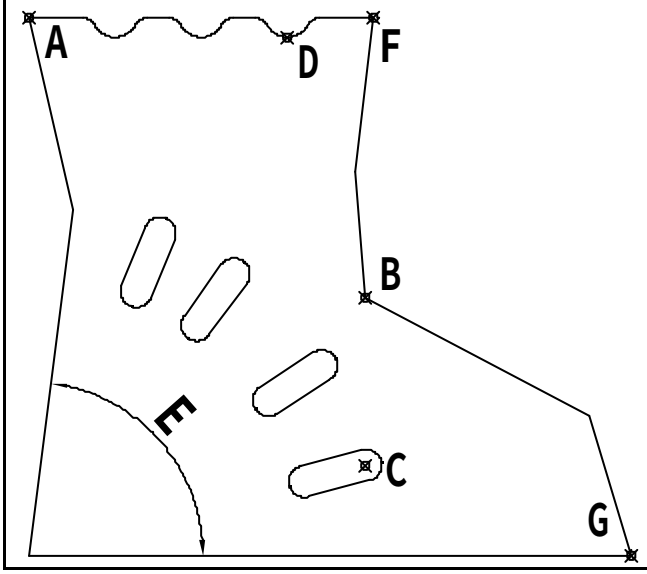



08、 Erase,
09、 Line, end, L50 **左**, (**正交模式**), **向上** 55, →L57, @45<15, →L58
10、 Circle, end, L55 **上**, 20, →C59
11、 Circle, int, L50 & L57, L58 **右**, →C60

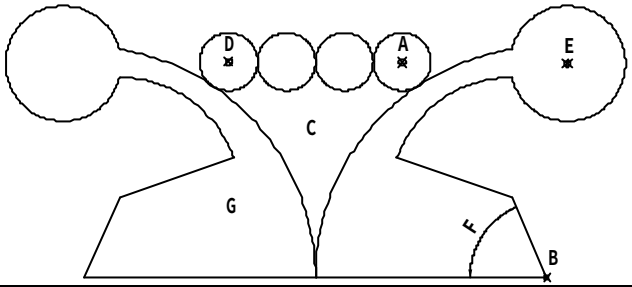



12、 ROtate, L57;L58, int, L50 & L57, R, int, L50 & L57, end, L58 **上**, int, C59 & C60, →L57;L58 **轉動**
13、 Line, end, L55 **左**, int, C59 & C60, →L61
14、 ROtate, L58, int, C59 & C60, C, -105, →L62
15、 ROtate, L58, int, L57 & L58, C, 95, →L63
16、 Offset, 85, L50, **向上點**, →L64

| | |
|---|---|
|  | 17、 Erase,<br>18、 Fillet, T, T, R, 0, L62, L64, →L62 **與** L64**相交**<br>19、 Fillet, T, T, R, 0, L63, L64, →L63 **與** L64**相交**<br>20、 Offset, 0.75(=3.75-3), L64, **向上點**, →L65<br>21、 DIVide, L65, 4, →P66~P68<br>22、 Circle, nod, P66, 3.75, →C69<br>23、 Fillet, T, N, R, 2, L64, C69 **左下**, →A70<br>24、 Fillet, T, N, R, 2, L64, C69 **右下**, →A71 |
|  | 25、 TRim,<br>26、 Erase,<br>27、 COpy, C69;A70;A71, nod, P66, nod, P67, → **相同物件**, nod, P68, →**相同物件** |
|  | 28、 TRim,<br>29、 RECtangle, F, 2.5, int, L50 & L57, D, 15, 5, →R72<br>30、 Circle, int, L50 & L57, 55, →C73 |

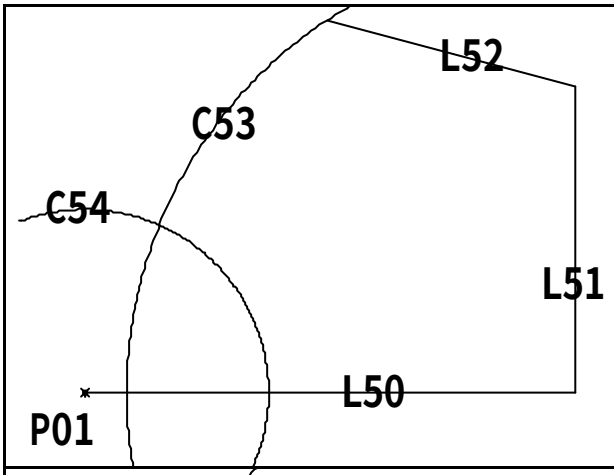
| | |
|---|---|
|  | 31、 Move, R72, cen, R72 **右**, int, L50 & C73, →R72 **向下與向右移動** |
|  | 32、 **點選** L50; R72, **再點** L50 **左側**(**成為紅色點**), **按滑鼠左鍵**, **選旋轉**, C, 15, →**第** 1 **個**, 33(=18+15),→**第** 2**個**, 54(=18+15+21), →**第** 3 **個**, 68(=18+15+21+14),→**第** 4 **個**, |
|  | 33、 Erase,<br>34、 DIst, int, A, int, B, →**距離**=69.2049<br>35、 DIst, cen, C, cen, D, →**距離**=72.5736<br>36、 DinANgular, E, →82.8841<br>37、 DIst, int, F, int, G, →X **差值**=40.5846<br>38、 Hatch, **點選內部**, →**斜線區域**<br>39、 **點選斜線區域**, CTRL+1, →**查幾何面積值** =5058.2505 |

## 2.2.7 AutoCAD2D2008-207

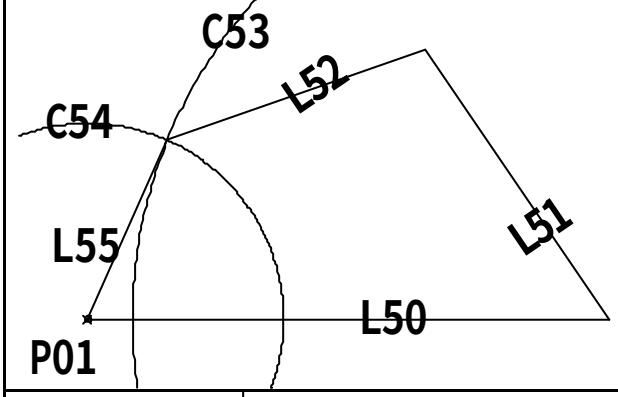



1、 **中心點** A **至中心點** B **高度為何？** 74.8331
2、 C **區域面積為何？** 118.2060
3、 **中心點** D **至中心點** E **距離為何？** 117.2296
4、 **尺寸** F **角度為何？** 66.0441
5、 G **區域周長為何？** 433.1723

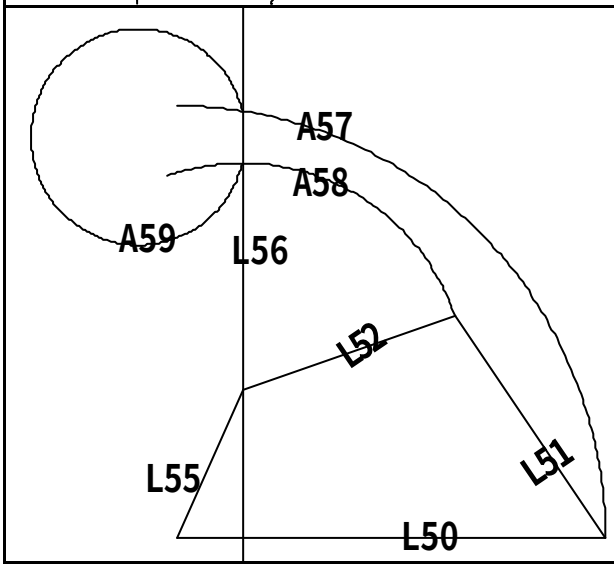
**解題步驟：**



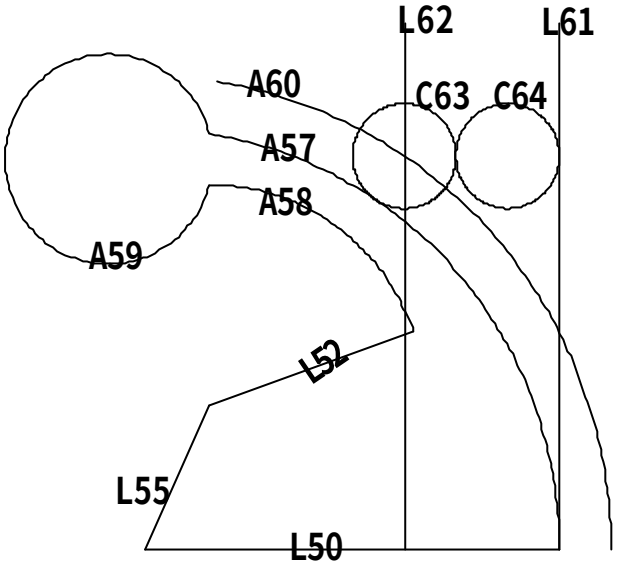
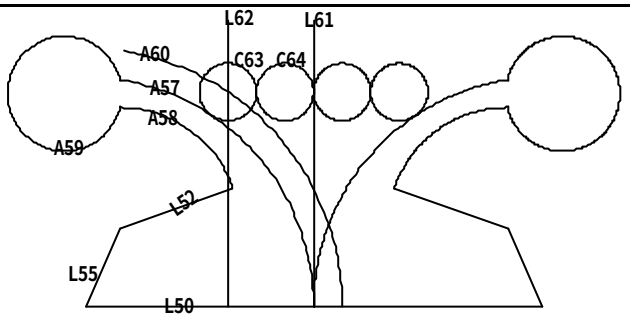
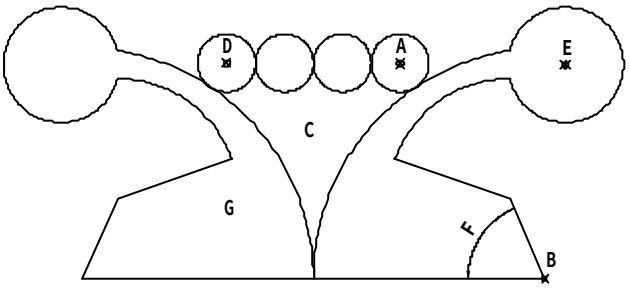

01、 Line, **任一點** P01, (**正交模式**), **向右** 80, →L50, **向上** 50, →L51, @42<165, →L52
02、 Circle, int, L50 & L51, end, L52 **上**, →C53
03、 Circle, end, L50 **左**, 30, →C54



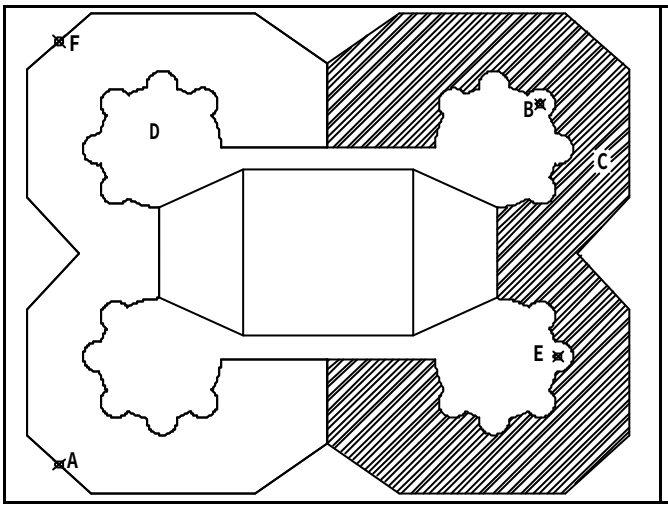

04、 ROtate, L51;L52, int, L50 & L51, R, int, L50 & L51, end, L52 **上**, int, C53 & C54, →L51;L52 **轉動**
05、 Line, end, L50 **左**, int, C53 & C54, →L55




06、 XLine, V, int, L52 & L55, →L56
07、 Arc, C, end, L50 **左**, end, L50 **右**, A, 90, →A57
08、 Arc, C, end, L52 **左**, end, L52 **右**, A, 90, →A58
09、 Arc, int, L56 & A57, E, int, L56 & A58, R, -20, →A59

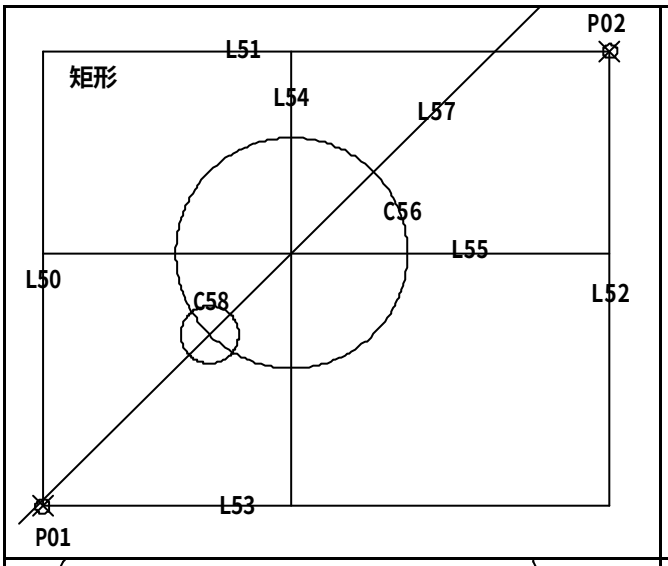
| | |
|---|---|
|  | 10、 TRim,<br>11、 Erase,<br>12、 Offset, 10, A57, **向外點**, →A60<br>13、 Line, int, L50 & A57, (**正交模式**), **向上** 100, →L61<br>14、 Offset, 30(=20+10), L61, **向左點**, →L62,<br>15、 Circle, int, A60 & L62, D, 20, →C63<br>16、 COpy, C63, qua, C63 **左**, qua, C63 **右**, →C64 |
|  | 17、 MIrror, **所有物件**(A60;L61;L62 **除外**), end, L61 **上**, end, L61 **下**, →**對稱物件** |
|  | 18、 Erase,<br>19、 DIst, cen, A, int, B, →Y **差值**=74.8331<br>20、 Hatch, **點選** C **區域**, →**斜線區域**<br>21、 **點選** C **斜線區域**, CTRL+1, →**查幾何面積值** =1418.2060<br>22、 DIst, cen, D, cen, E, →**距離**=117.2296<br>23、 DimANgular, F, →66.0441<br>24、 BOundary, **點選** G **區域**<br>25、 **點選** G **區域邊緣**, CTRL+1, →**查幾何長度值** =433.1723 |

## 2.2.8 AutoCAD2D2008-208

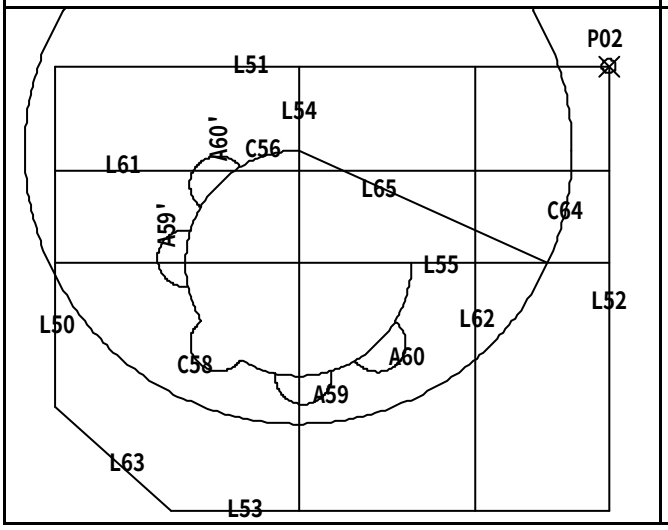



1、 **中點** A **至中心點** B **距離為何？** 196.8514
2、 C **區域周長為何？** 767.3276
3、 **中央正方形區域面積為何？** 3017.1099
4、 **中心點** E **至中點** F **距離為何？** 193.0594
5、 D **區域面積為何？** 3470.6788

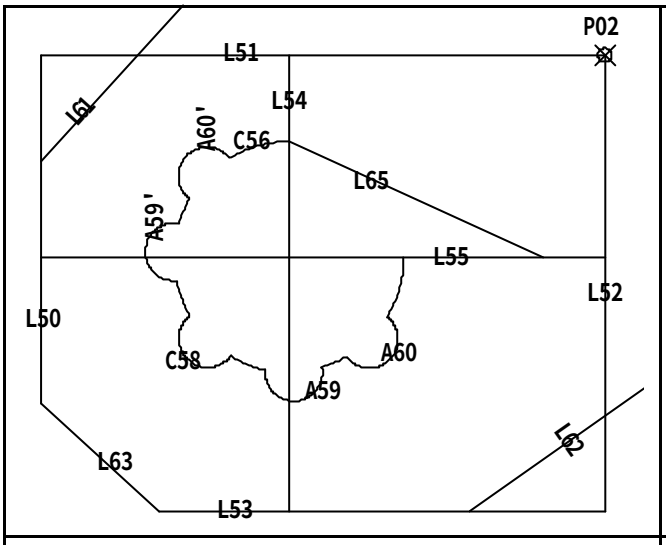
**解題步驟：**



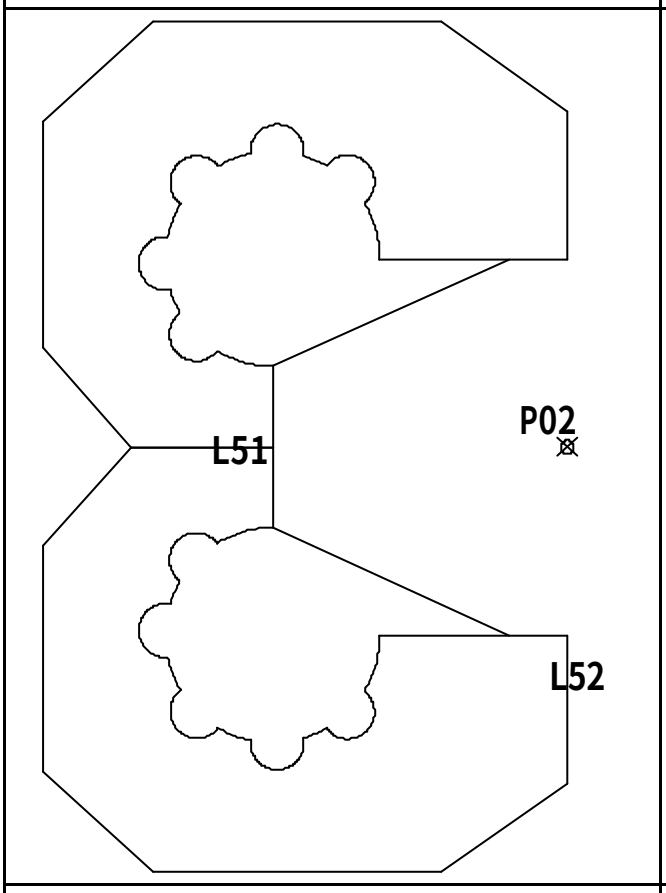

01、 RECtangle, **任一點** P01, D, 98(=196/2), 79(=158/2), **向右上點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 POint, int, L51 & L52, →P02
04、 Offset, 43, L50, **向右點**, →L54
05、 Offset, 44, L53, **向上點**, →L55
06、 Circle, int, L54 & L55, 20, →C56
07、 XLine, int, L54 & L55, @20<45, →L57
08、 Circle, int, C56 & L57, 5, →C58



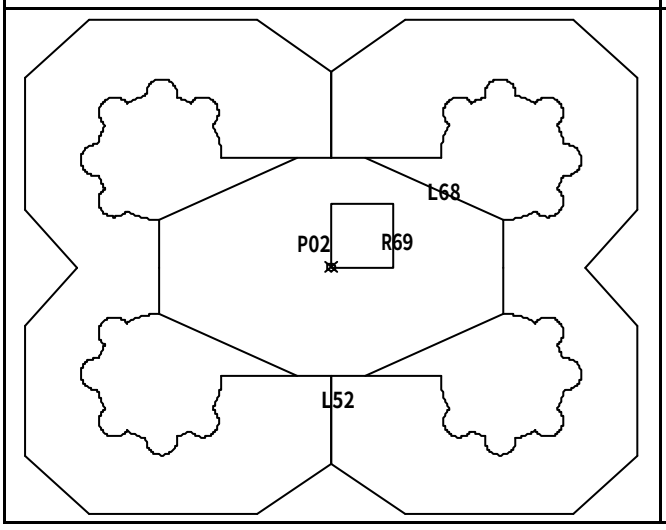

09、 Erase,
10、 TRim,
11、 ROtate, C58, cen, C56, C, 47.5, →C59
12、 ROtate, C59, cen, C56, C, 42, →C60
13、 MIrror, C59; C60, cen, C56, cen, C58, →C59' ;C60'
14、 Offset, 18.5, L51, **向下點**, →L61
15、 Offset, 23.5, L52, **向左點**, →L62
16、 CHAmfer, D, 20.5, 18.5, L53 **左**, L50 **下**, →L63
17、 Circle, int, C56 & L54, 48.5, →C64
18、 Line, cen, C64, int, C64 & L55, →L65




19、 Erase,
20、 TRim,
21、 ROtate, L61, int, L50 & L61, 48(=138-90), →L61 **轉動**
22、 ROtate, L62, int, L53 & L62, -55(=-(145-90)), →L62 **轉動**




23、 Erase,
24、 TRim,
25、 MIrror, **所有物件**, end, L51 **左**, end, L51 **右**, **→上部圖形**




26、 MIrror, **所有物件**, end, L52 **上**, end, L52 **下**, **→上右部圖形**
27、 RECtangle, nod, P02, D, 20(**大約**), 20, **向右上點**, →R69

| | |
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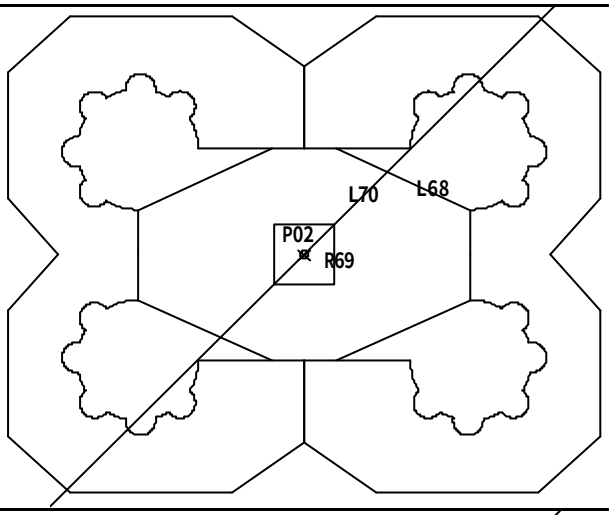
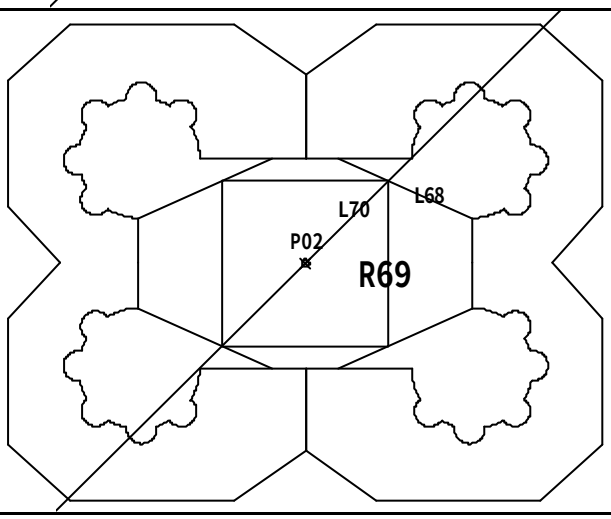
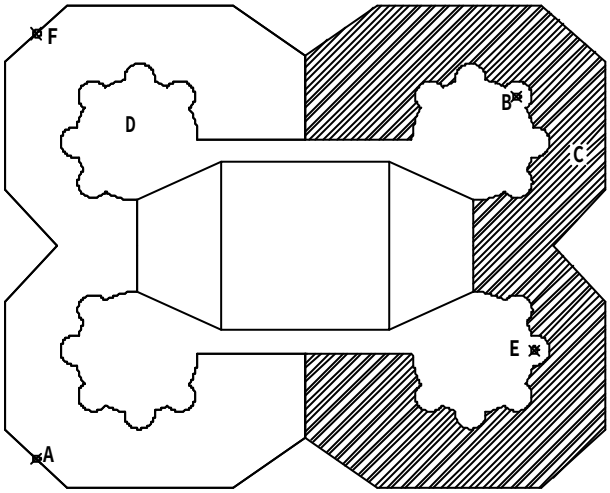
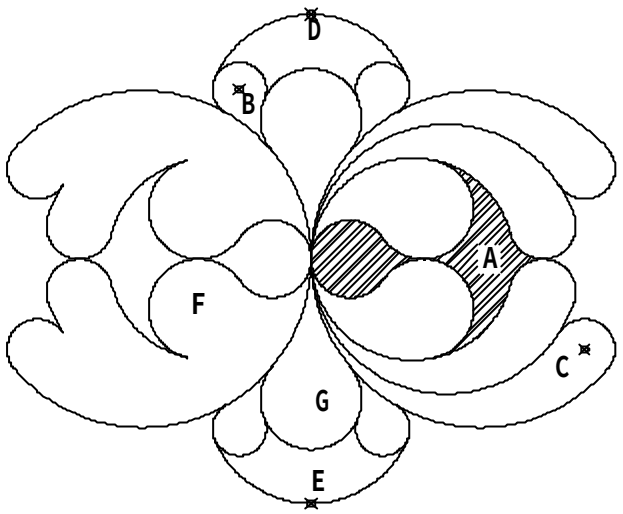
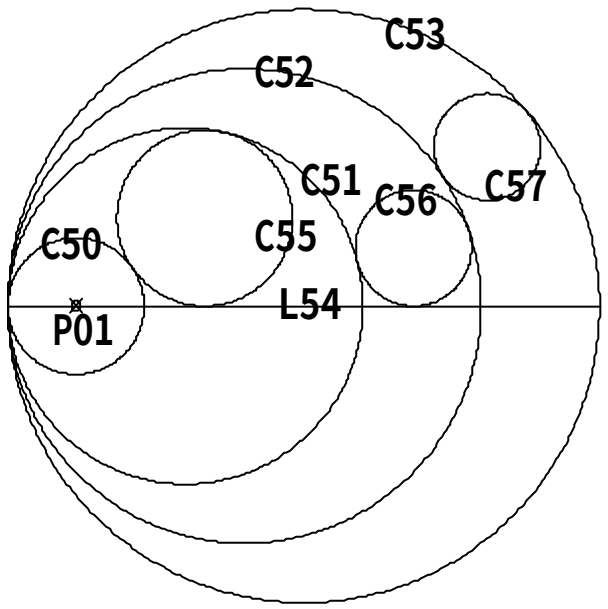
|  | 28、 Move, R69, **任一點**, @-10,-10, →R69 **向左下移動**<br>29、 XLine, nod, P02, @20<45, →L70 |
|  | 30、 SCale, R69, nod, P02, R, nod, P02, int, R69 & L70, int, L68 & L70, →R69 **放大**<br>31、 Erase,<br>32、 TRim, |
|  | 33、 DIst, mid, A, cen, B, →**距離**=196.8514<br>34、 BOundary, **點** C **區域**<br>35、 **點** C **區域邊緣** CRTL+1, →**查幾何長度值**=767.3276<br>36、 Hatch, **點中央正方形區域**, →**斜線區域**<br>37、 **點中央正方形區域斜線**, CRTL+1, →**查幾何面積值**=3017.1099<br>38、 DIst, cen, E, mid, F, →**距離**=193.0594<br>39、 Hatch, **點** D **區域**, →**斜線區域**<br>40、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值**=3470.6788 |

## 2.2.9 AutoCAD2D2008-209

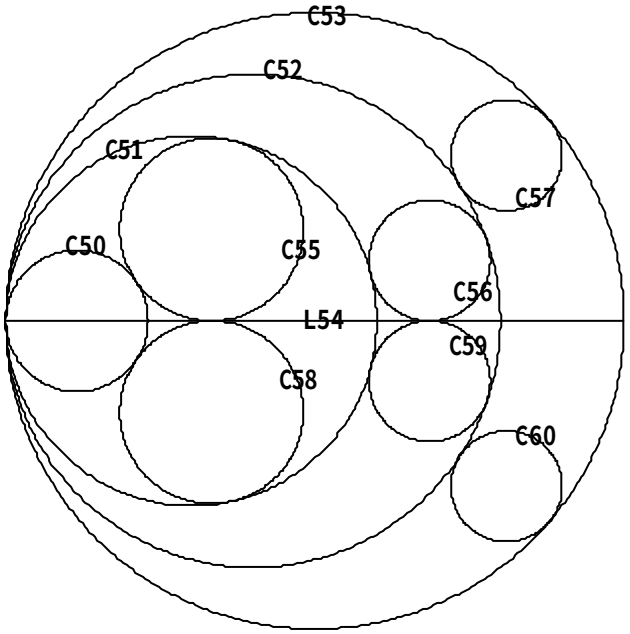



1、 **斜線** A **區域面積為何？** 1000.4467
2、 **中心點** B **至中心點** C **距離為何？** 128.1088
3、 **四分點** D **至四分點** E **高度為何？** 145.0662
4、 F **區域面積為何？** 2963.9509
5、 G **區域周長為何？** 137.2347

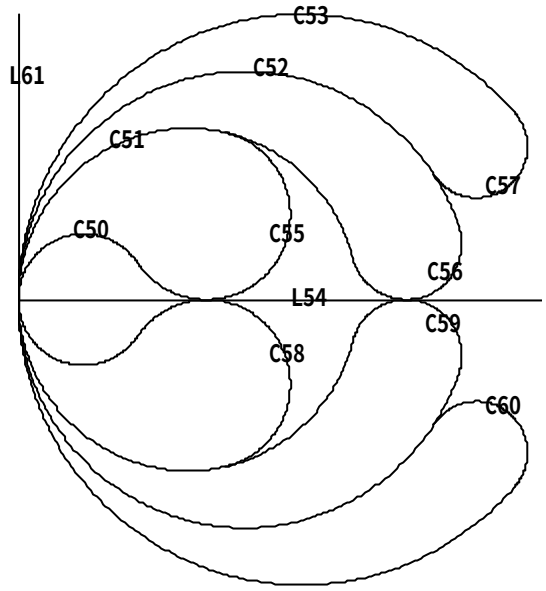
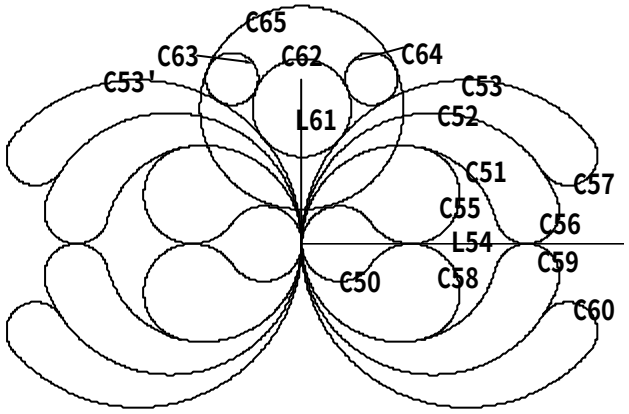
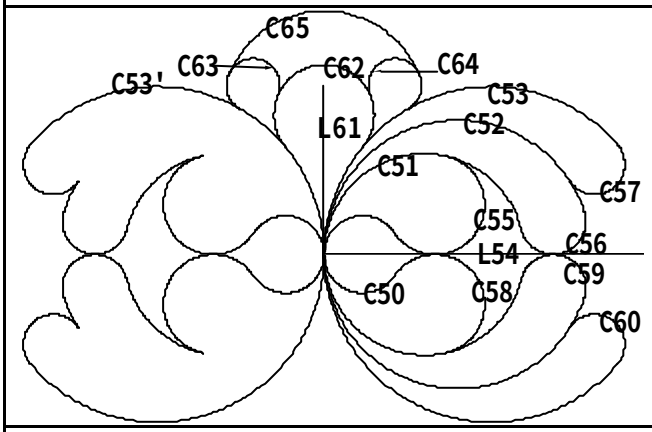
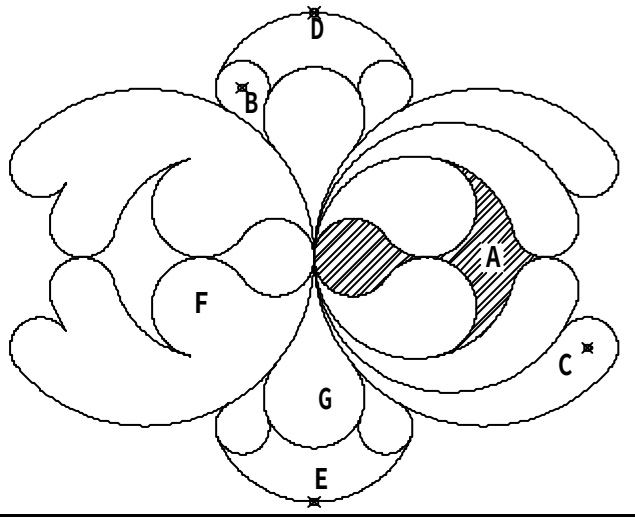
**解題步驟：**



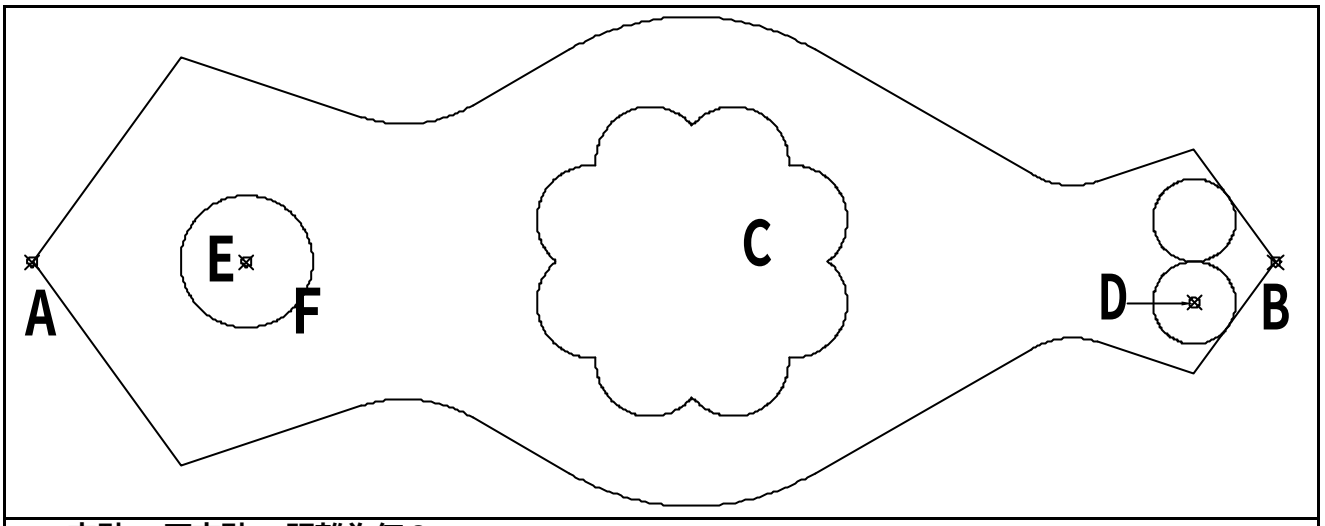

01、 Circle, **任一點** P01, D, 23, →C50
02、 **點選** C50, qua, C50 **左**, (**成為紅色掣點**), **按滑鼠右鍵**, (**出現視窗**), **選比例**, C, R, qua, C50 **左**, qua, C50 **右**, (**正交模式**), **向右** 60, →C51, **向右** 80, →C52, **向右** 100, →C53
03、 Line, qua, C50 **左**, qua, C53 **右**, →L54
04、 Circle, 3P, tan, C50 **右上**, tan, L54, tan, C51 **右上**, →C55
05、 Circle, 3P, tan, C51 **右上**, tan, L54, tan, C52 **右上**, →C56
06、 Circle, T, C52 **右上**, C53 **右上**, 9, →C57




07、 MIrror, C55~C57, end, L54 **左**, end, L54 **右**, →C58~C60

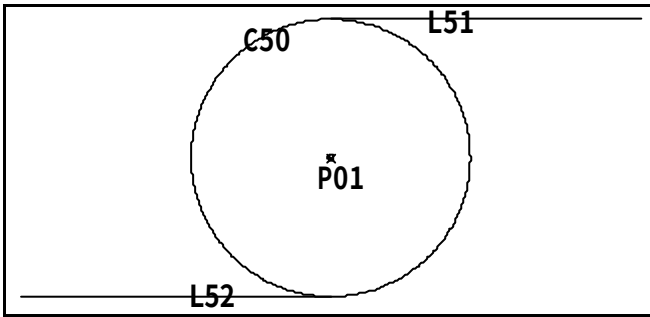
| | |
|---|---|
|  | 08、 TRim,<br>09、 Line, qua, C50 **左**, (**正交模式**), **向上** 50, →L61 |
|  | 10、 MIrror, **所有物件**(L54**與** L61 **除外**), end, L61 **上**, qua, C50 **左**, →**左側對稱圖形**<br>11、 Circle, T, C53 **左上**, C53'**右上**, 15(=30/2), →C62<br>12、 Circle, T, C62 **左上**, C53'**右上**, 8, →C63<br>13、 Circle, T, C62 **右上**, C53 **左上**, 8, →C64<br>14、 Circle, T, C64 **右上**, C63 **左上**, 31, →C65 |
|  | 15、 TRim,<br>16、 Erase, |
|  | 17、 MIrror, **所有物件**(L54**與** L61 **除外**), end, L54 **左**, end, L54 **右**, →**下方對稱圖形**<br>18、 Hatch, A **區域**, →**斜線區域**<br>19、 **點選** A **斜線區域**, CTRL+1, →**查幾何面積值** =1000.4467<br>20、 DIst, cen, B, cen, C, →**距離**=128.1088<br>21、 DIst, qua, D, qua, E, →Y **差值**=145.0662<br>22、 Hatch, **點選** F **區域**, →**斜線區域**<br>23、 **點選** F **斜線區域**, CTRL+1, →**查幾何面積值** =2963.9509<br>24、 BOundary, **點選** G **區域**,<br>25、 **點選** G **區域邊緣**, CTRL+1, →**查幾何長度值** =137.2347 |

## 2.2.10 AutoCAD2D2008-210

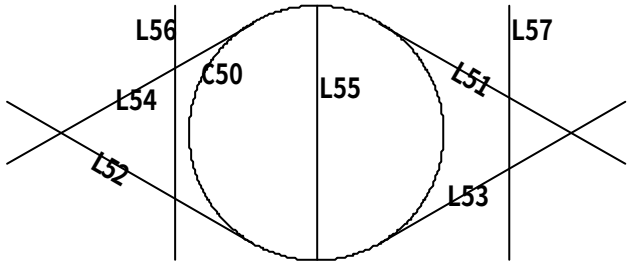



1、 **交點** A **至交點** B **距離為何？** 288.1679
2、 C **區域周長為何？** 201.6344
3、 **中心點** D **至中心點** E **距離為何？** 174.1126
4、 **圓** F **直徑為何？** 24.2825
5、 **圖形扣除內孔面積為何？** 9389.9414

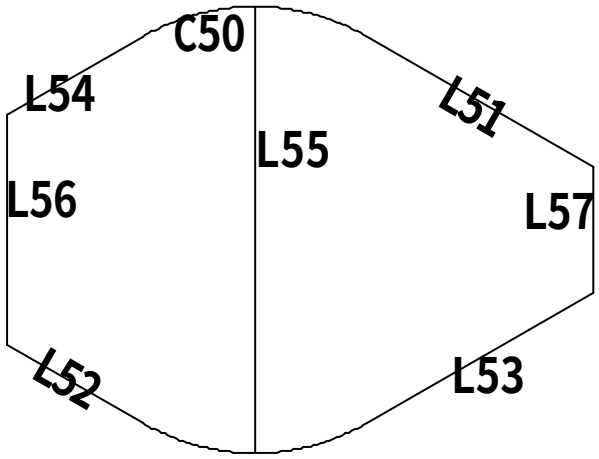
**解題步驟：**



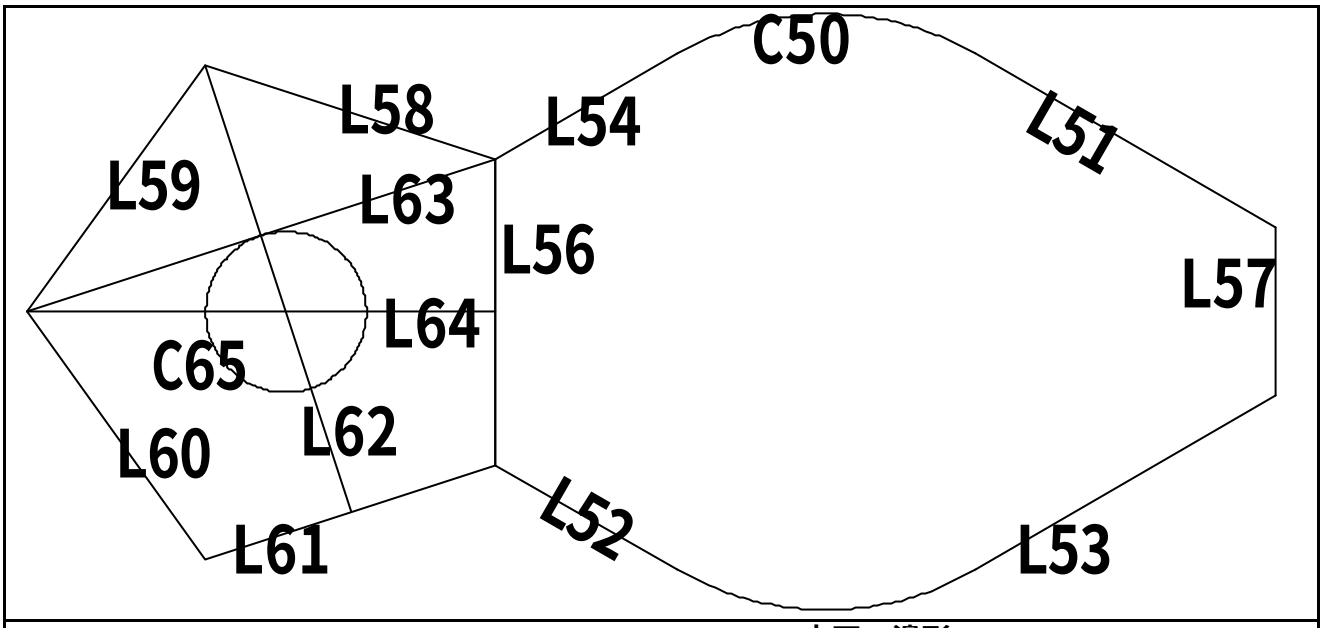

01、 Circle, **任一點** P01, D, 90, →C50
02、 Line, qua, C50 **上**, (**正交模式**), **向右** 100, →L51
03、 Line, qua, C50 **下**, (**正交模式**), **向左** 100, →L52



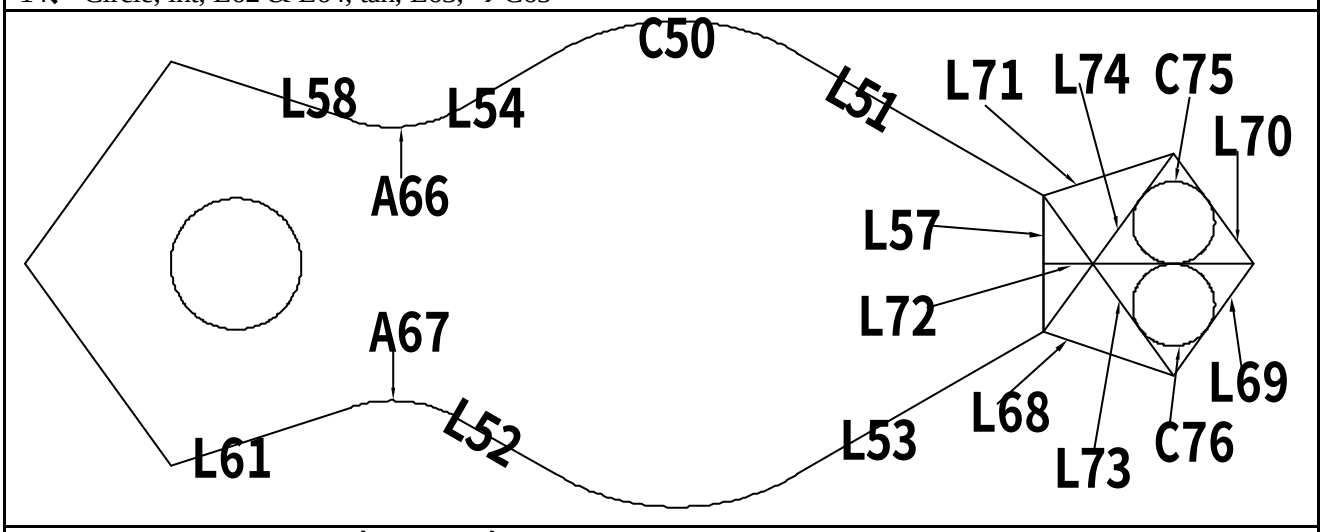

04、 ROtate, L51;L52, cen, C50, -30, →L51 **與** L52 **轉動**
05、 MIrror, L51;L52, qua, C50 **左**, qua, C50 **右**, →L53;L54
06、 Line, qua, C50 **上**, qua, C50 **下**, →L55
07、 Offset, 50, L55, **向左點**, →L56
08、 Offset, 68, L55, **向右點**, →L57



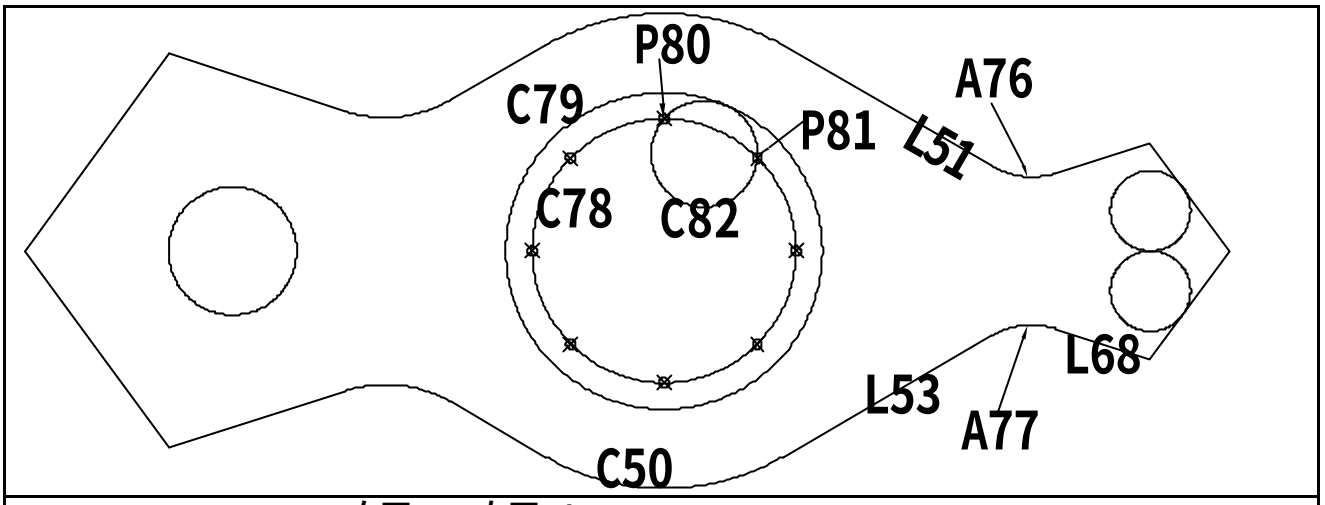

09、 TRim,
10、 Erase,



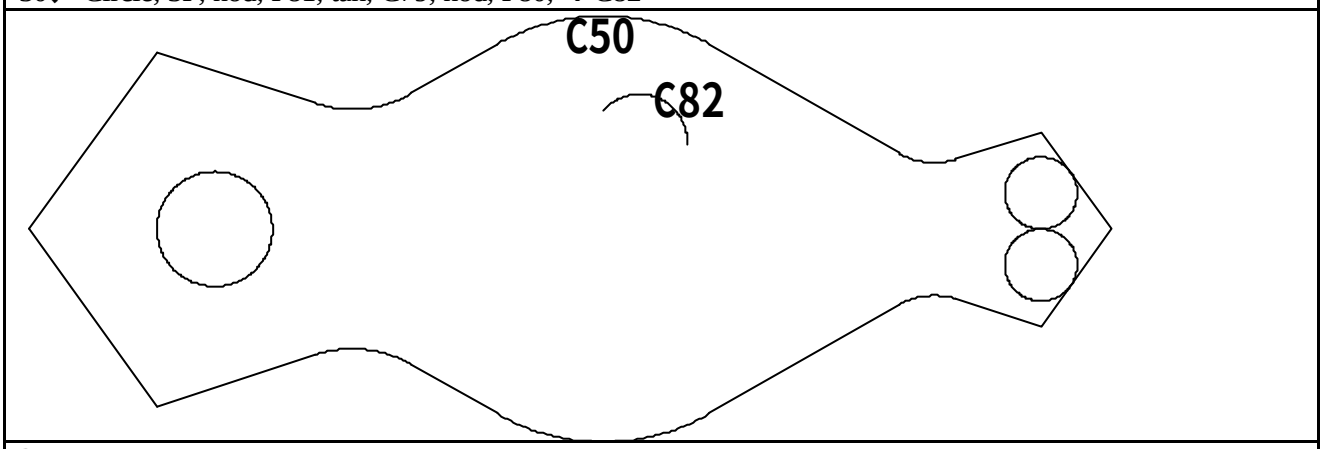

11、 POLygon, 5, E, int, L52 & L56, int, L54 & L56, →L58~L61 **之正** 5**邊形**
12、 Line, mid, L61, int, L58 & L59, →L62
13、 Line, int, L54 & L56, int, L59 & L60 ,→L63, mid, L56, →L64
14、 Circle, int, L62 & L64, tan, L63, →C65



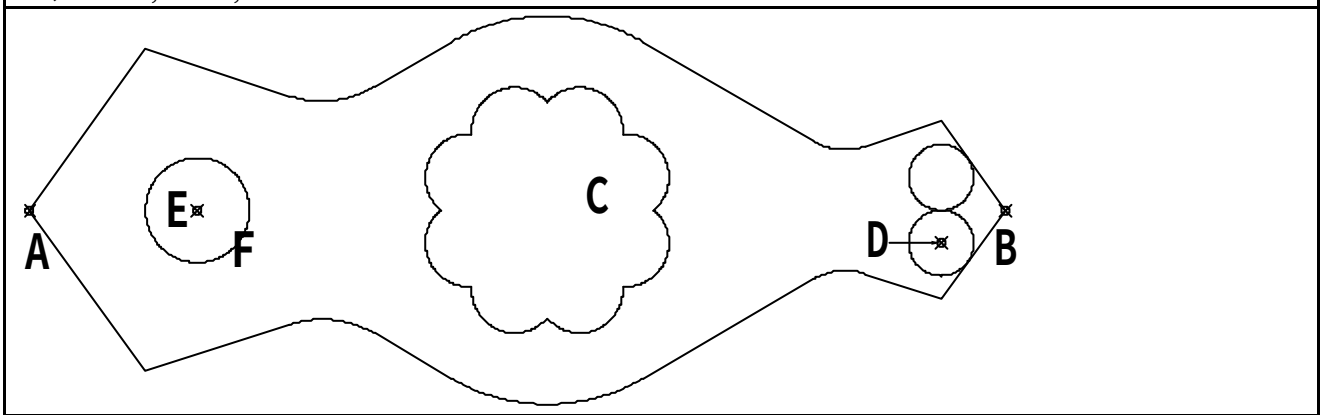

15、 Fillet, T, T, R, 25, L54 **左下**, L58 **右下**, →A66
16、 Fillet, T, T, R, 25, L52 **左上**, L61 **右上**, →A67
17、 Erase,
18、 POLygon, 5, E, int, L51 & L57, int, L53 & L57, →L68~L71 **之正** 5**邊形**
19、 Line, mid, L57, int, L69 & L70, →L72
20、 Line, int, L51 & L57, int, L68 & L69, →L73
21、 Line, int, L53 & L57, int, L70 & L71, →L74
22、 Circle, 3P, tan, L70, tan, L74, tan, L72, →C75
23、 Circle, 3P, tan, L70, tan, L73, tan, L69, →C76




24、 Fillet, T, T, R, 15, L51 **右下**, L71 **左下**, →A76
25、 Fillet, T, T, R, 15, L53 **右上**, L68 **左上**, →A77
26、 Erase,
27、 Circle, cen, C50, D, 50, →C78
28、 Circle, @(**鎖定上次點**), D, 60, →C79
29、 DIVide, C78, 8, →P80;P81 **等點**
30、 Circle, 3P, nod, P81, tan, C79, nod, P80, →C82

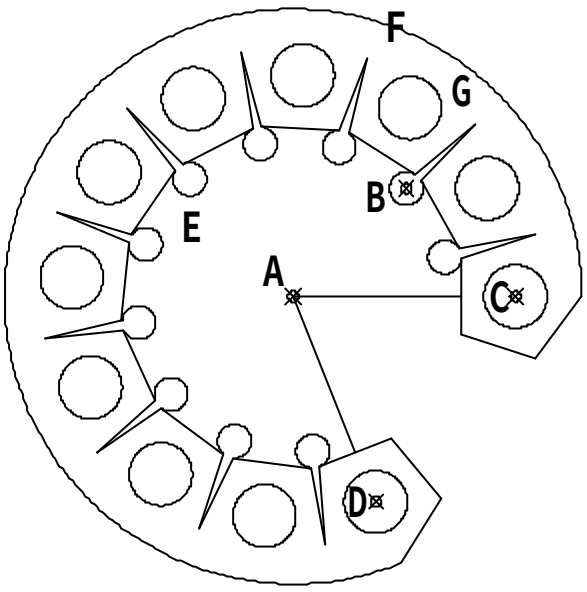



31、 TRim, Erase,



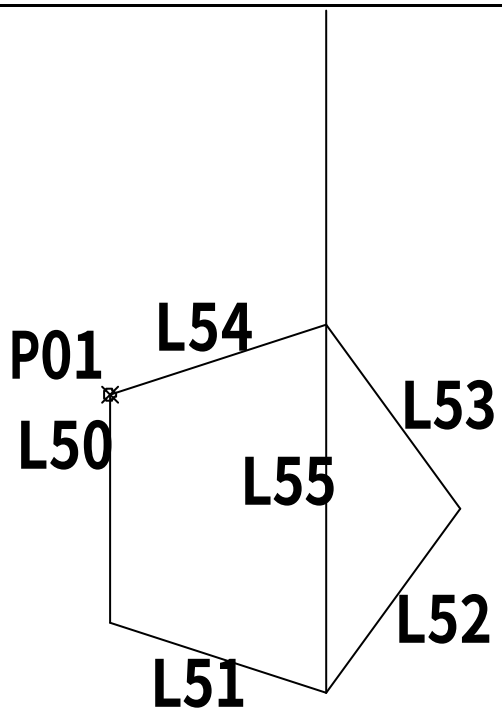

32、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 8, **佈滿角度**, 360, **物件**, C82, →**全圓佈滿** 8 **個相同物件**
33、 DIst, int, A, int, B, →**距離**=228.1679
34、 BOundary, **點選** C **區域**
35、 **點選** C **區域邊緣**, CTRL+1,→**查幾何長度值**=201.6344
36、 DIst, cen, D, cen, E, →**距離**=174.1126
37、 **點選** F **圓**, CTRL+1, →**查幾何直徑值**=24.2825
38、 Hatch, **點選內部**, →**斜線區域**
39、 **點選斜線區域**, CTRL+1, →**查幾何面積值** =9389.9414

## 2.2.11 AutoCAD2D2008-211

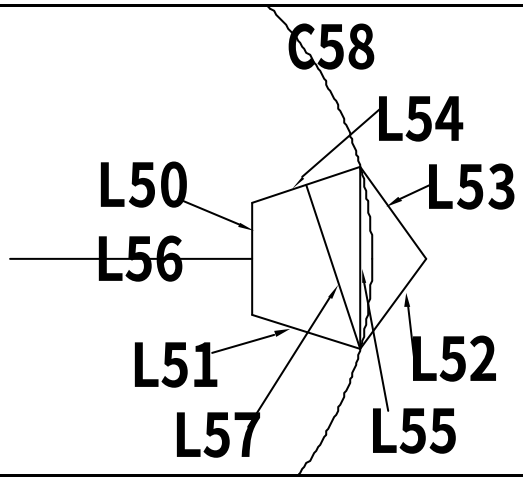



1、 交點 A 至中心點 B 距離為何？ 37.2006
2、 中心點 C 至中心點 D 距離為何？ 59.1822
3、 E 區域面積為何？ 3660.7093
4、 弧 F 半徑為何？ 68.5317
5、 G 區域扣除內孔面積為何？ 6240.5997

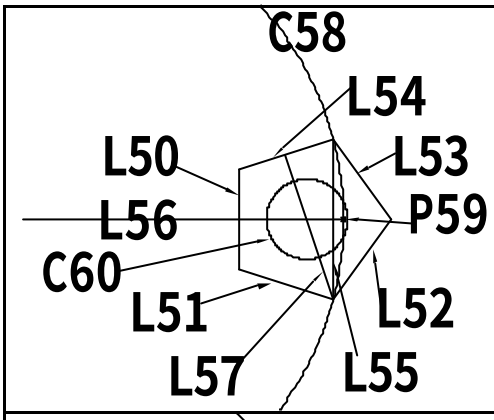
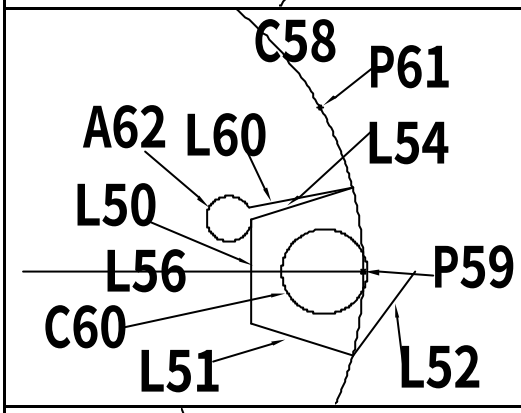
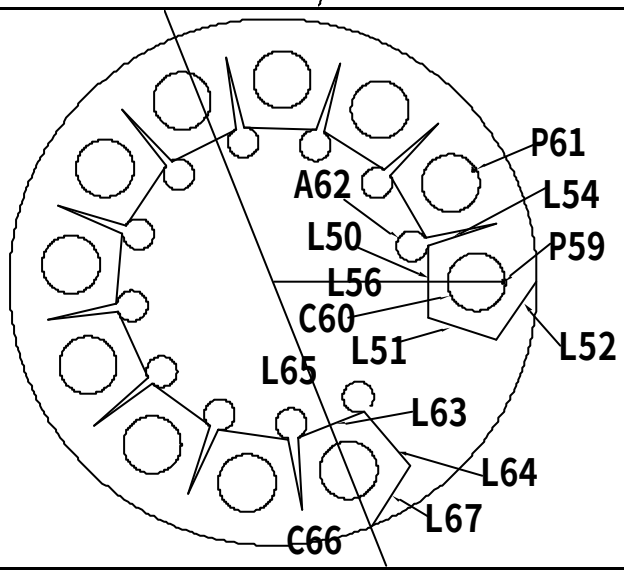
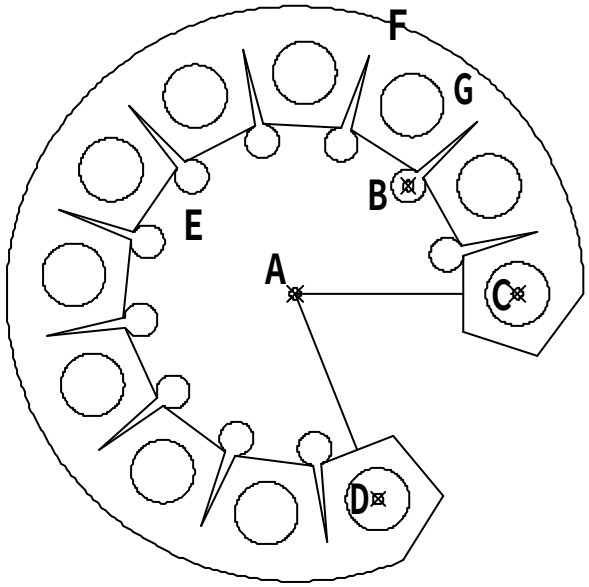
**解題步驟：**



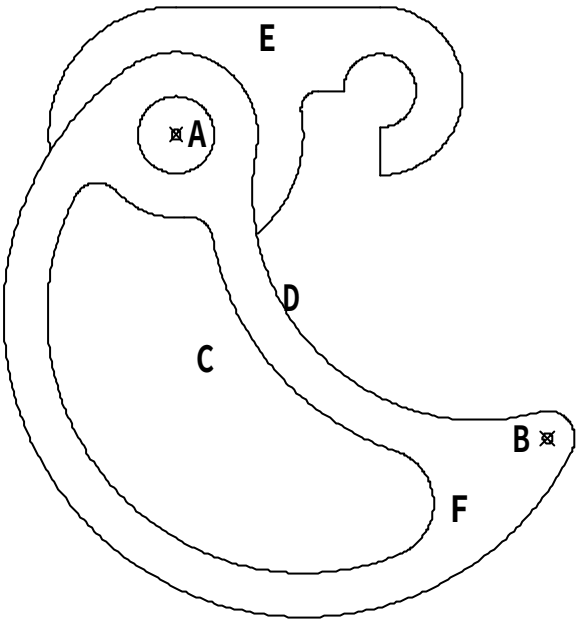

01、 POLygon, 5, E, 任一點 P01, (正交模式), 向下 10, →L50~L54 之正 5 邊形
02、 Line, int, L51 & L52, (正交模式), 向上 30, →L55




03、 SCale, L50~L54, int, L51 & L52, R, int, L51 & L52, int, L53 & L54, end, L55 上, →L50~L54 放大
04、 Line, mid, L50, (正交模式), 向左 40, →L56
05、 Line, mid, L54, int, L51 & L52, →L57
06、 Circle, end, L56 左, int, L51 & L52, →C58

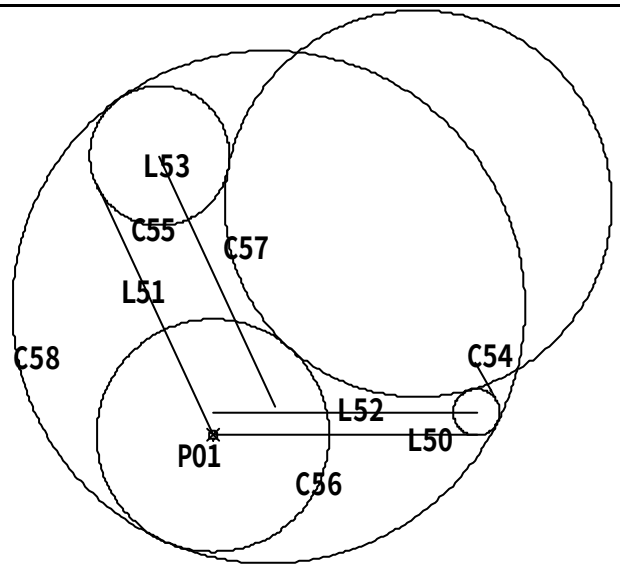
| | |
|---|---|
|  | 07、 EXtend, C58, L56 **右**, →L56 **向右延伸**<br>08、 POint, int, L56 & C58, →P59<br>09、 Circle, int, L56 & L57, D, 15, →C60<br>10、 eXplode, L50~L54 **之正** 5 **邊形**, →L50~L54 |
|  | 11、 Erase,<br>12、 ROtate, L51; P59, cen, C58, C, R, cen, C58, int, L51 & C58, int, L54 & C58, →L60;P61<br>13、 Arc, end, L60 **左**, E, end, L54 **左**, R, -4(=8/2), →A62 |
|  | 14、 Erase,<br>15、 ARray, **環形陣列**, **中心點**, end, L56 **左**, **項目總數**, 11, **項目間角度**, nod, P61, **物件**, L50;L51; L54;C60;A62, →L63;L64 **等** 11 **組物件**<br>16、 XLine, end, L56 **左**, mid, L63, →L65<br>17、 Circle, end, L56 **左**, end, L52 **上**, →C66<br>18、 Line, end, L64 **下**, int, L65 & C66, →L67 |
|  | 19、 Erase,<br>20、 TRim,<br>21、 DIst, int, A, cen, B, →**距離**=37.2006<br>22、 DIst, cen, C, cen, D, →**距離**=59.1822<br>23、 Hatch, **點選** E **區域**, →**斜線區域**<br>24、 **點選** E **斜線區域**, CTRL+1, →**查幾何面積值** =3660.7093<br>25、 **點選** F **弧**, CTRL+1, →**查幾何半徑值** =68.5317<br>26、 Hatch, **點選** G **區域**, →**斜線區域**<br>27、 **點選** G **斜線區域**, CTRL+1, →**查幾何面積值** =6240.5997 |

## 2.2.12 AutoCAD2D2008-212

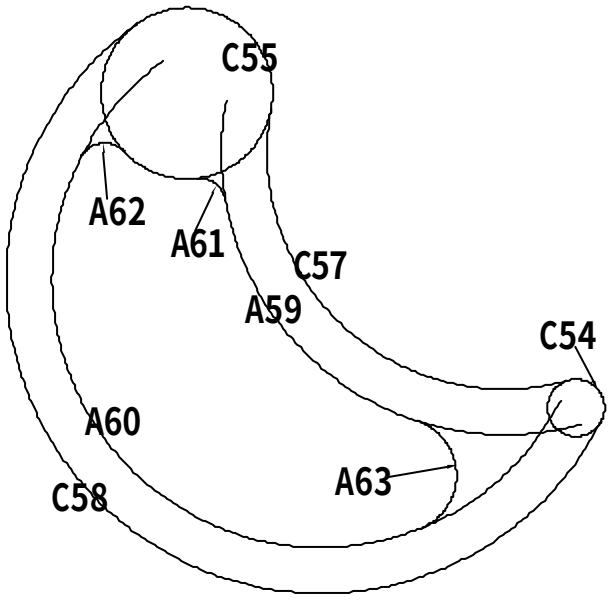



1、 **中心點** A **至中心點** B **高度距離為何？** 82.6703
2、 C **區域周長為何？** 333.3582
3、 **弧** D **半徑為何？** 62.3105
4、 E **區域面積為何？** 2670.9164
5、 F **區域扣除內孔面積為何？** 6243.3390

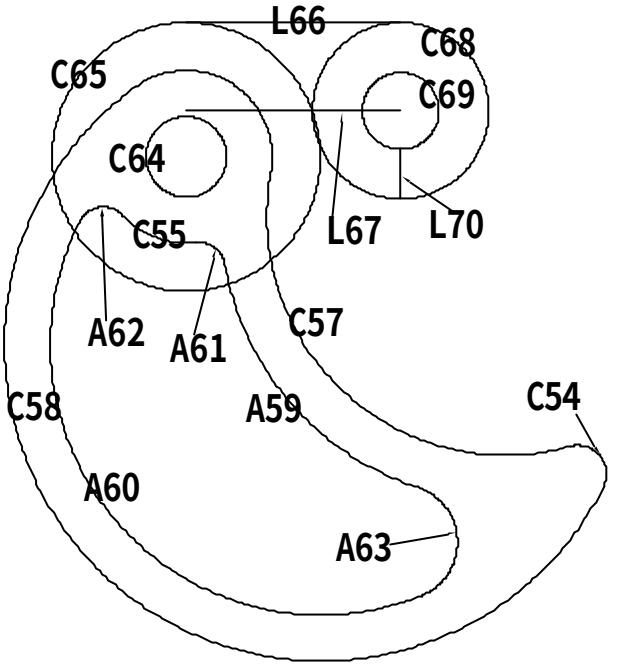
**解題步驟：**



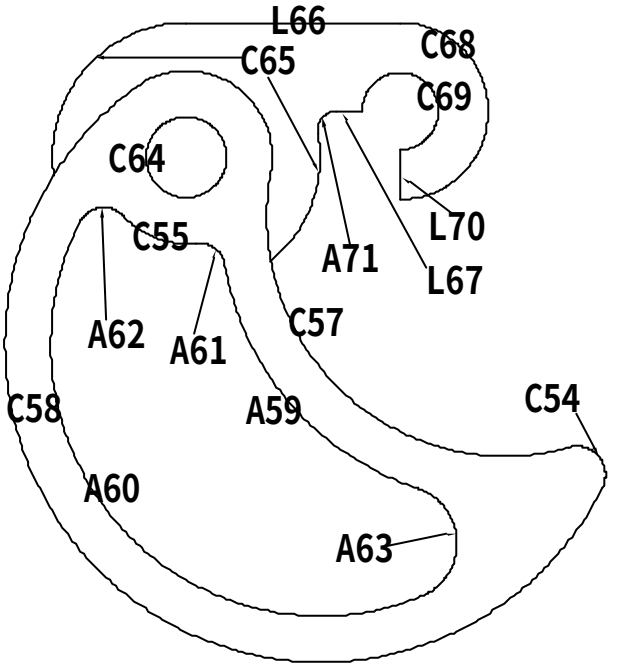

01、 Line, **任一點** P01, (**正交模式**), **向右** 85, →L50
02、 Line, end, L51 **左**, @89<115, →L51
03、 Offset, 7.5, L50, **向上點**, →L52
04、 Offset, 22.5, L51, **向右點**, →L53
05、 Circle, end, L52 **右**, 7.5, →C54
06、 Circle, end, L53 **上**, 22.5, →C55
07、 Circle, int, L50 & L51, 37.5, →C56
08、 Circle, 3P, tan, C55 **右**, tan, C56 **右上**, tan, C54 **上**, →C57
09、 Circle, 3P, tan, C55 **左上**, tan, C56 **左下**, tan, C54 **右下**, →C58



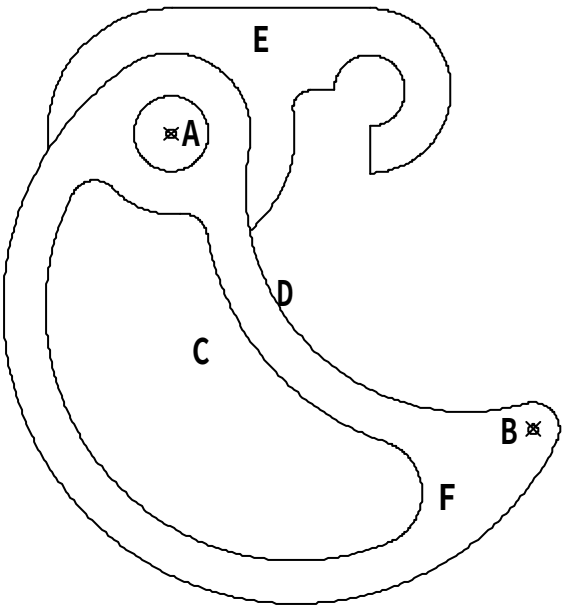

10、 TRim,
11、 Erase,
12、 Offset, 12, C57, **向下點**, →A59
13、 Offset, 12, C58, **向上點**, →A60
14、 Fillet, T, N, R, 7, C55 **下**, A59 **左**, →A61
15、 Fillet, T, N, R, 7, C55 **左下**, A60 **左上**, →A62
16、 Fillet, T, N, R, 15, A59 **左下**, A60 **下**, →A63




17、 TRim,
18、 Erase,
19、 Circle, cen, C55, D, 21, →C64
20、 Circle, @(**鎖定上次點**), 35, →C65
21、 Line, qua, C65 **上**, (**正交模式**), **向右** 56, →L66
22、 Offset, 23, L66, **向下點**, →L67
23、 Circle, end, L67 **右**, 23, →C68
24、 Circle, @(**鎖定上次點**), 10, →C69
25、 Line, qua, C69 **下**, qua, C68 **下**, →L70

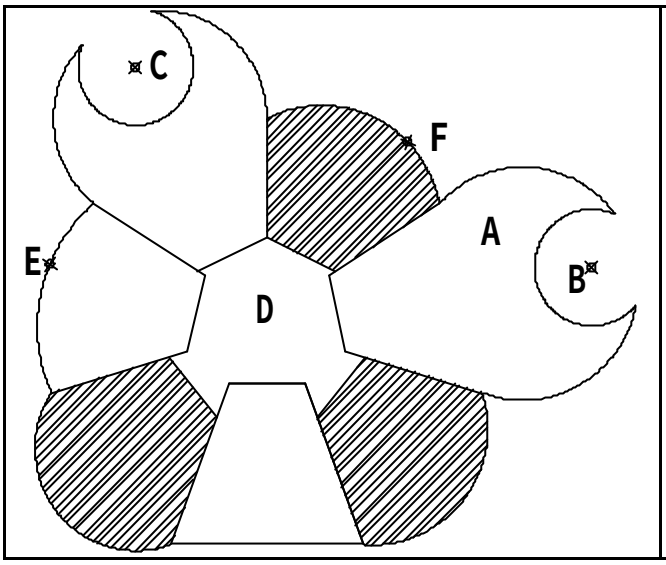



26、 TRim,
27、 Erase,
28、 Fillet, T, T, R, 5, C65 **右上**, L67 **右**, →A71



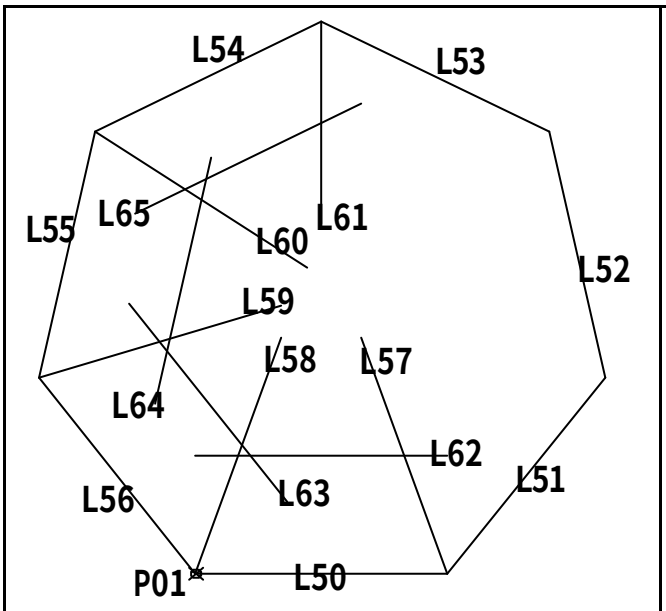

29、 DIst, cen, A, cen, B, →Y **差值**=82.6703
30、 BOundary, **點選** C **區域**,
31、 **點選** C **區域邊緣**, CTRL+1, →**查幾何長度值** =333.3582
32、 **點選** D **弧**, CTRL+1, →**查幾何半徑值** =62.3105
33、 Hatch, **點選** E **區域**, →**斜線區域**
34、 **點選** E **斜線區域**, CTRL+1, →**查幾何面積值** =2670.9164
35、 Hatch, **點選** F **區域**, →**斜線區域**
36、 **點選** F **斜線區域**, CTRL+1, →**查幾何面積值** =6243.3390

## 2.2.13 AutoCAD2D2008-213

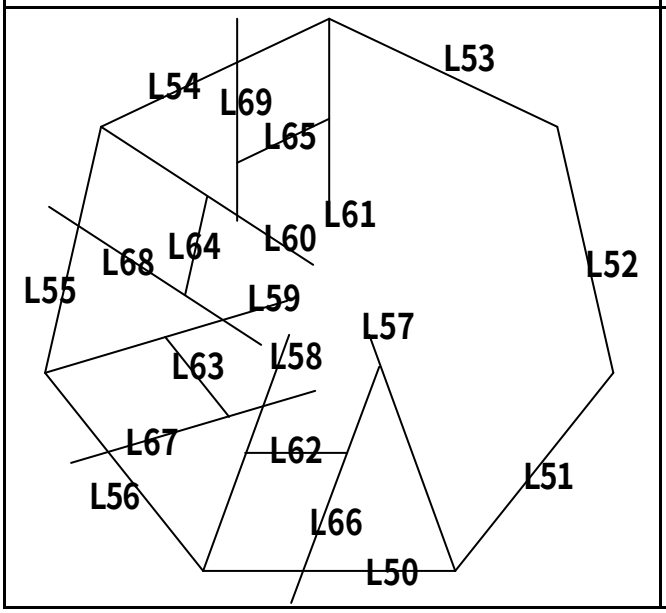



1、 A 區域周長為何？ 262.6383
2、 中心點 B 至中心點 C 距離為何？ 129.7450
3、 D 區域周長為何？ 173.3665
4、 弧 E 至弧 F 中心點水平距離為何？ 32.6560
5、 斜線區域面積總和為何？ 4596.7732

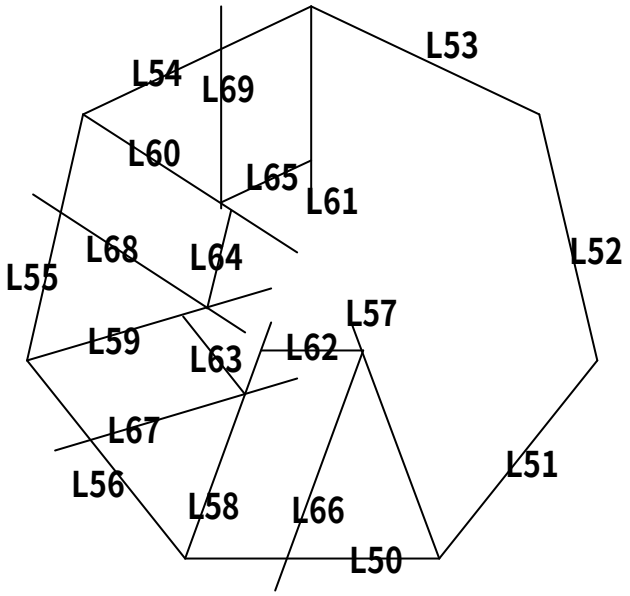
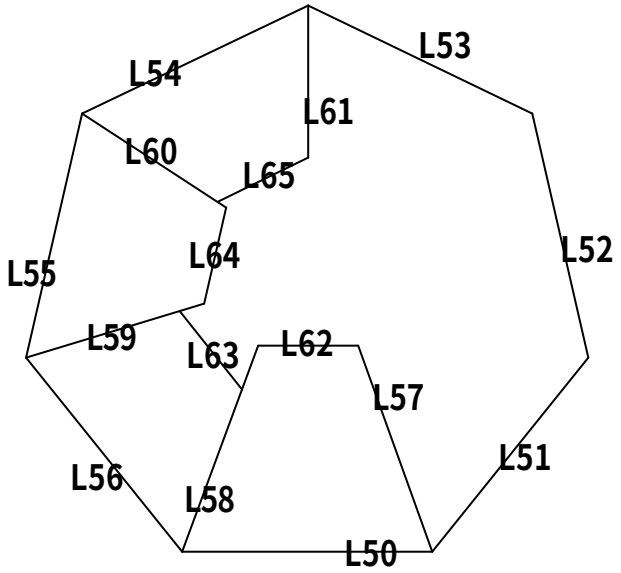
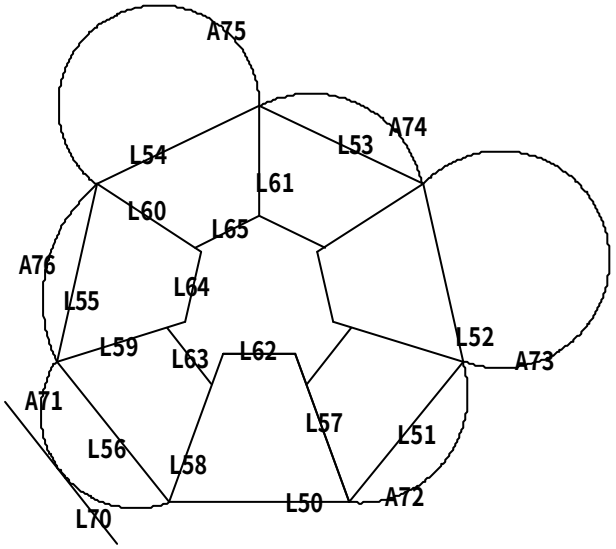
**解題步驟：**

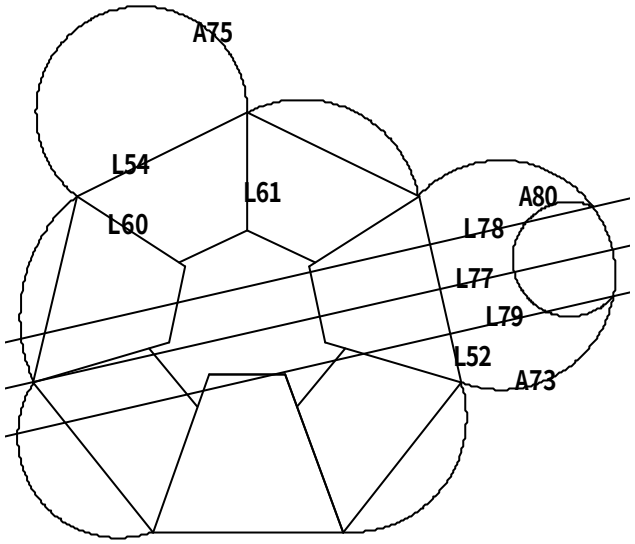
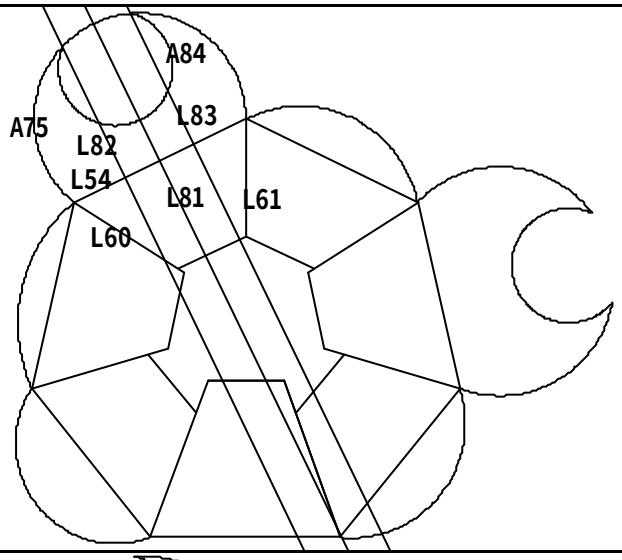
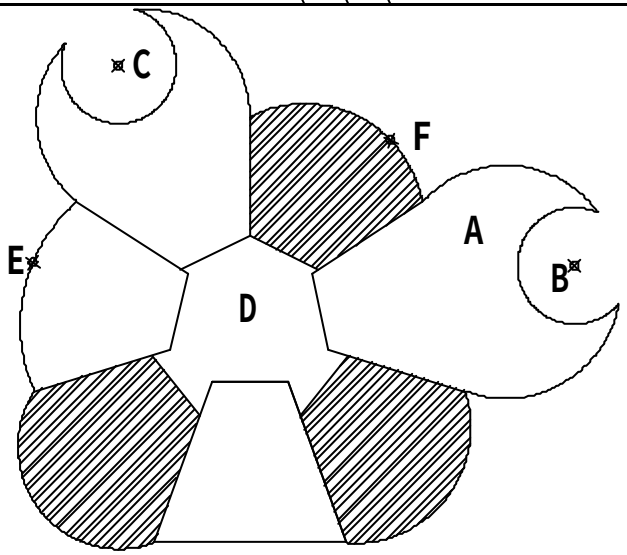



01、 POLygon, 7, E, 任一點 P01, (正交模式), 向右 50, →L50~L56 之正 7 邊形
02、 eXplode, L50~L56 之正 7 邊形, →L50~L56
03、 ROtate, L50, int,L50 & L51, C, -70, →L57
04、 ROtate, L50, int,L50 & L56, C, 70, →L58
05、 ROtate, L56, int,L55 & L56, C, 68, →L59
06、 ROtate, L55, int,L54 & L55, C, 70, →L60
07、 Line, int, L53 & L54, (正交模式), 向下 40, →L61
08、 Offset, T, L50, mid, L58, →L62, L56, mid, L59, →L63, L55, mid, L60, →L64, L54, mid, L61, →L65

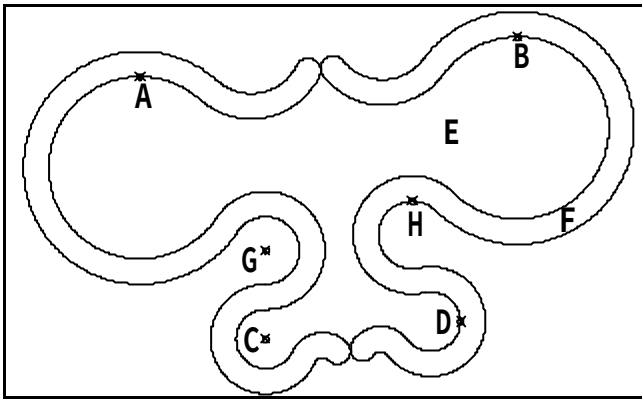



09、 TRim,
10、 LENgthen, T, 20, L62 右, →L62 縮短, L63 下, →L63 縮短, L64 下, →L64 縮短, L65 下, →L65 縮短
11、 Offset, T, L58, end, L62 右, →L66, L59, end, L63 右, →L67, L60, end, L64 右, →L68, L61, end, L65 右, →L69

| | |
|---|---|
|  | 12、 EXtend, L57, L66 **上**, →L66 **向上延伸**<br>13、 Move, L62, int, L62 & L66, int L66 & L57, →L62 **向上移動**<br>14、 Move, L63, int, L63 & L67, int L67 & L58, →L63 **向右上移動**<br>15、 Move, L64, int, L64 & L68, int L68 & L59, →L64 **向右移動**<br>16、 Move, L65, int, L65 & L69, int L69 & L60, →L54 **向下移動** |
|  | 17、 Erase,<br>18、 TRim, |
|  | 19、 MIrror, L58;L59;L60;L63:L64;L65, mid, L50, int, L54 & L53, →**右側對稱圖形**<br>20、 Offset, 18, L56, **向左下點**, →L70<br>21、 Arc, int, L55 & L56, mid, L70, int, L56 & L50, →A71<br>22、 Arc, int, L50 & L51, E, int, L51 & L52, R, 30, →A72<br>23、 Arc, int, L51 & L52, E, int, L52 & L53, R, -30, →A73<br>24、 Arc, int, L52 & L53, E, int, L53 & L54, R, 31.25, →A74<br>25、 Arc, int, L53 & L54, E, int, L54 & L55, R, -27.8, →A75<br>26、 Arc, int, L54 & L55, E, int, L55 & L56, A, 74, →A76 |

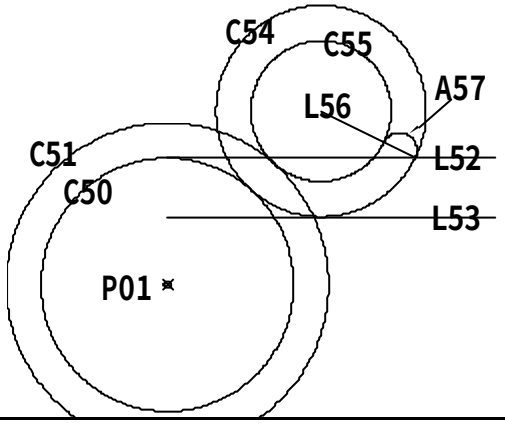
| | |
|---|---|
|  | 27、 Erase,<br>28、 XLine, mid, L52, cen, A73, →L77,<br>29、 Offset, 12(=24/2), L77, **向上點**, →L78, L77, **向下點**, →L79<br>30、 Arc, int, A73 & L78, E, int, A73 & L79, R, -15, →A80 |
|  | 31、 TRim,<br>32、 Erase,<br>33、 XLine, mid, L54, cen, A75, →L81,<br>34、 Offset, 10(=20/2), L81, **向左點**, →L82, L81, **向右點**, →L83<br>35、 Arc, int, A75 & L82, E, int, A75 & L83 R, -15, →A84 |
|  | 36、 TRim,<br>37、 Erase,<br>38、 BOundary, **點選** A **區域**,<br>39、 **點選** A **區域邊緣**, CTRL+1, →**查幾何長度值** =262.6383<br>40、 DIst, cen, B, cen, C, →**距離**=129.7450<br>41、 BOundary, **點選** D **區域**,<br>42、 **點選** D **區域邊緣**, CTRL+1, →**查幾何長度值** =173.3665<br>43、 DIst, cen, E, cen, F, →X **差值**=32.6560<br>44、 Hatch, **點選** 3 **區域**, →**斜線區域**<br>45、 **點選斜線區域**, CTRL+1, →**查幾何面積值** =4596.7732 |

## 2.2.14 AutoCAD2D2008-214

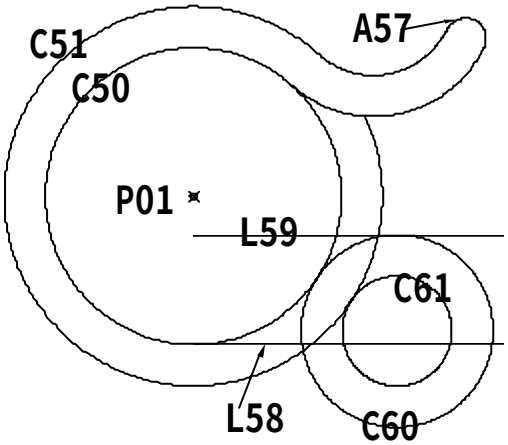



1、 **弧** A **至弧** B **中心點距為何？** 112.3486
2、 **中心點** C **至弧** D **中心點距離為何？** 49.1678
3、 E **區域面積為何？** 7447.6085
4、 F **區域周長為何？** 635.1898
5、 **中心點** G **至弧** H **中心點距離為何？** 43.6324

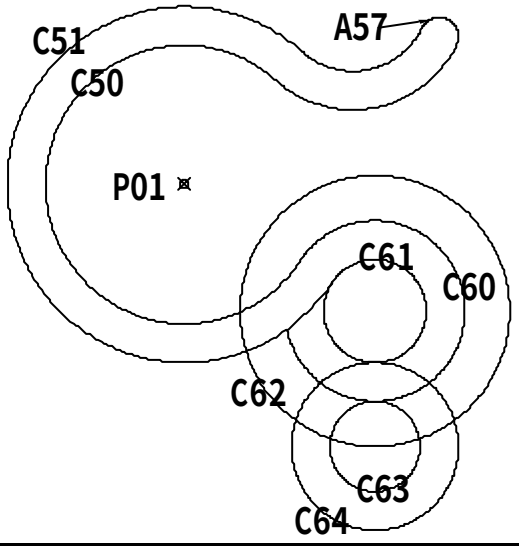
**解題步驟：**



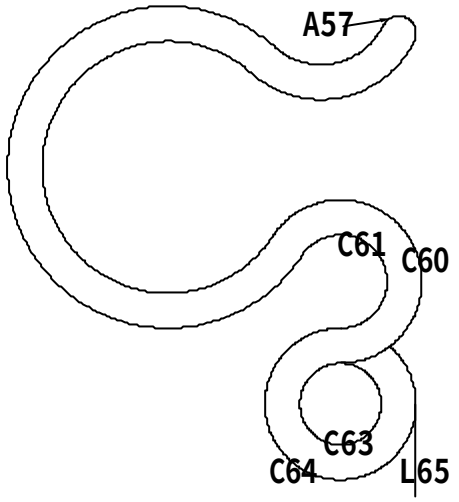
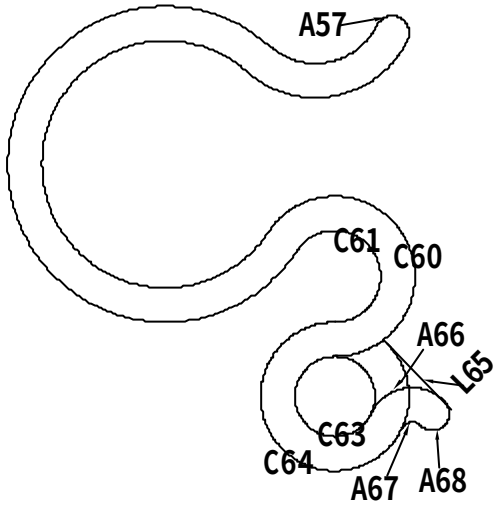
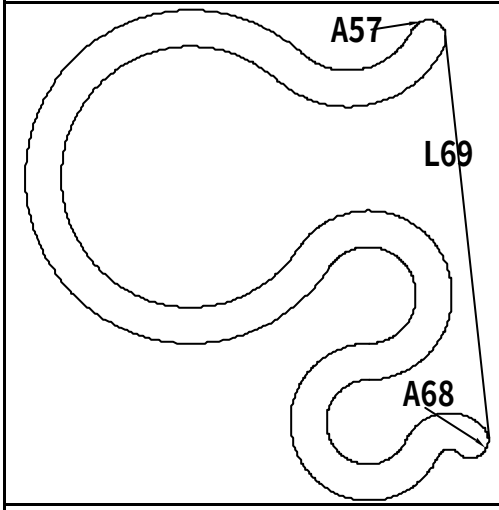
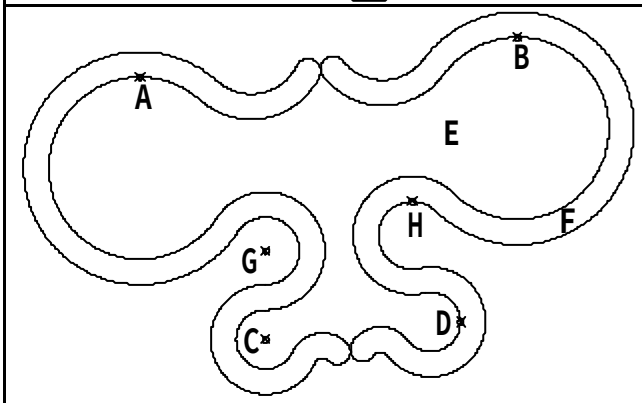

01、 Circle, **任一點** P01, 27, →C50
02、 Offset, 7.5, C50, **向外點**, →C51
03、 Line, qua, C50 **上**, (**正交模式**), **向右** 70, →L52
04、 Offset, 12.5, L52, **向下點**, →L53
05、 Circle, T, C50, L53, 22.5, →C54
06、 Offset, 7.5, C54, **向內點**, →C55
07、 Line, int, L52 & C54, cen, C54, →L56
08、 Arc, int, L52 & C54, E, int, C55 & L56, 3.75(=7.5/2), →A57



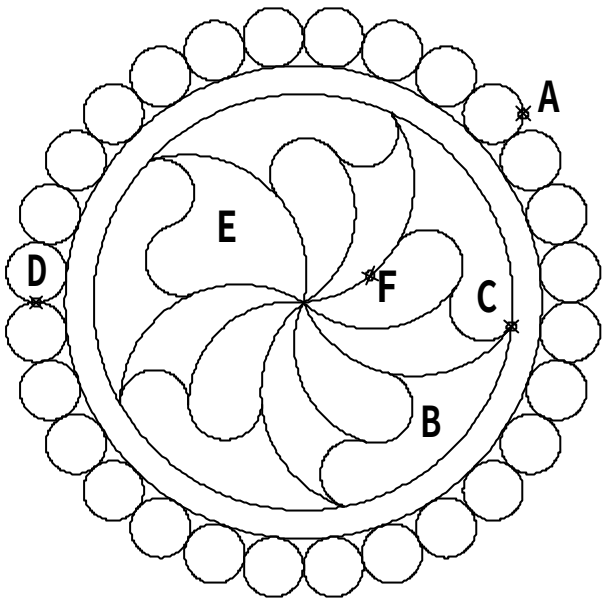

09、 Erase,
10、 TRim,
11、 Line, qua, C50 **下**, (**正交模式**), **向右** 70, →L58
12、 Offset, 20, L58, **向上點**, →L59
13、 Circle, T, C50, L59, 17.5, →C60
14、 Offset, 7.5, C60, **向內點**, →C61




15、 Erase,
16、 TRim,
17、 Offset, 8.75, C60, **向外點**, →C62
18、 Circle, qua, C62 **下**,8.75, →C63
19、 Offset, 7.5, C63, **向外點**, →C64

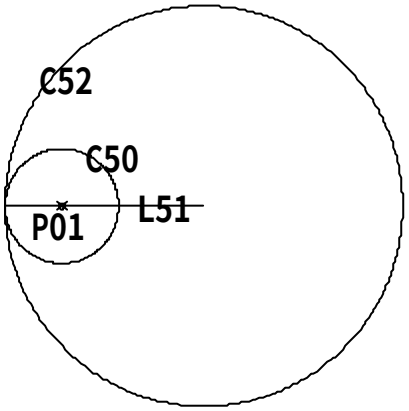
| | |
|---|---|
|  | 20、 Erase,<br>21、 TRim,<br>22、 Line, qua, C64 **右**, (**正交模式**), **向下** 20, →L65 |
|  | 23、 ROtate, L65, cen, C64, 45, →L65 **轉動**<br>24、 Fillet, T, N, R, 10, C63 **右下**, L65, →A66<br>25、 Offset, 7.5, R66, **向下點**, →A67<br>26、 Arc, C, m2p, end, A67 **右**, end, A66 **右**, end, A67 **右**, end, A66 **右**, →A68 |
|  | 27、 Erase,<br>28、 TRim,<br>29、 Line, tan, A57 **右下**, tan, A68 **右上**, →L69 |
|  | 30、 MIrror, **所有物件**(L69 **除外**), end, L69 **上**, end, L69 **下**, →**右部圖形**<br>31、 DIst, cen, A, cen, B, →**距離**=112.3486<br>32、 DIst, cen, C, cen, D, →**距離**=49.1678<br>33、 Hatch, **點** E **區域**, →**斜線區域**<br>34、 **點** E **斜線區域**, CRTL+1, →**查幾何面積值**=7447.6085<br>35、 BOundary, **點** F **區域**<br>36、 **點** F **區域邊緣** CRTL+1, →**查幾何長度值**=635.1898<br>37、 DIst, cen, G, cen, H, →**距離**=43.6324 |

## 2.2.15 AutoCAD2D2008-215

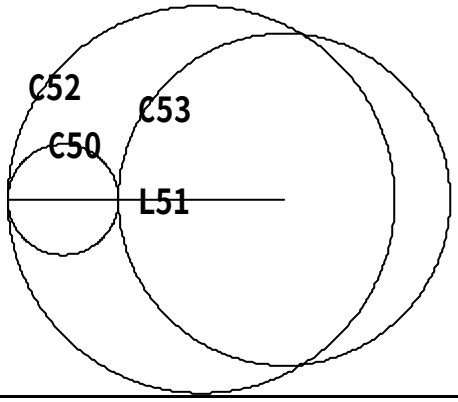



1、 **圓** A **半徑為何？** 15.7578
2、 B **區域周長為何？** 328.0658
3、 **交點** C **至交點** D **垂直距離為何？** 13.0110
4、 E **區域面積為何？** 4294.1761
5、 **弧** F **夾角為何？** 121.5881

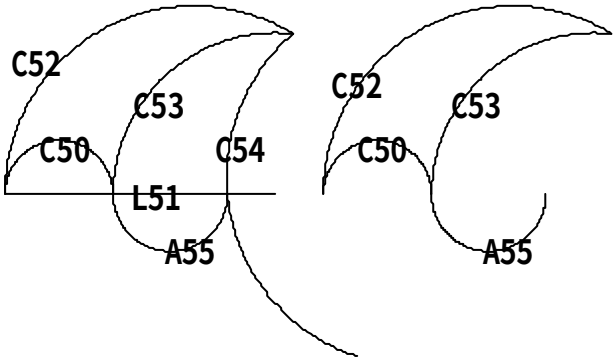
**解題步驟：**




01、 Circle, **任一點** P01, 18, →C50
02、 Line, qua, C50 **左**, (**正交模式**), **向右** 63, →L51
03、 Circle, end, L51 **右**, 63, →C52



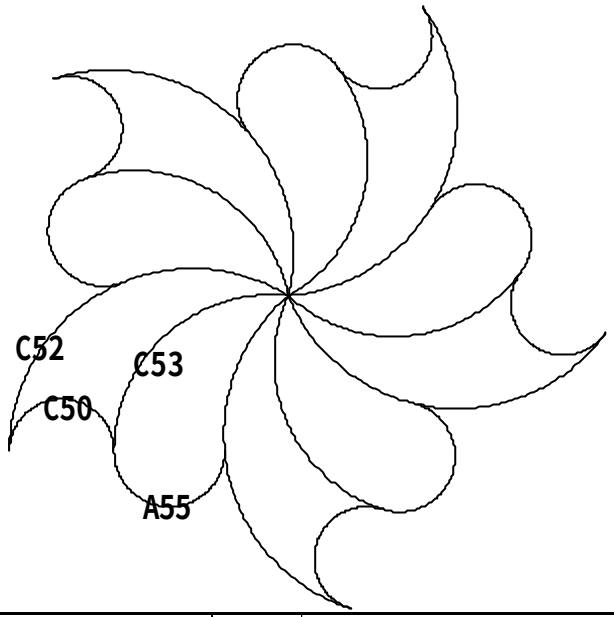

04、 LENgthen, T, 90(=18+18+54), L51 **右**, →L51 **向右延伸**
05、 Circle, end, L51 **右**, 54, →C53



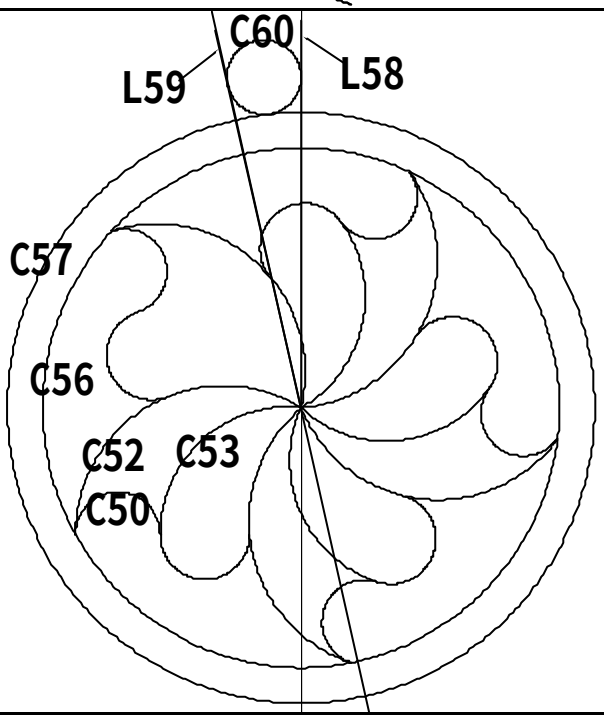

06、 TRim,
07、 Erase,
08、 ROtate, C52, int, C52 & C53, C, 72(=360/5), →C54
09、 Arc, C, m2p, int, L51 & C53, int, L51 & C54, int, L51 & C53, int, L51 & C54, →A55

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

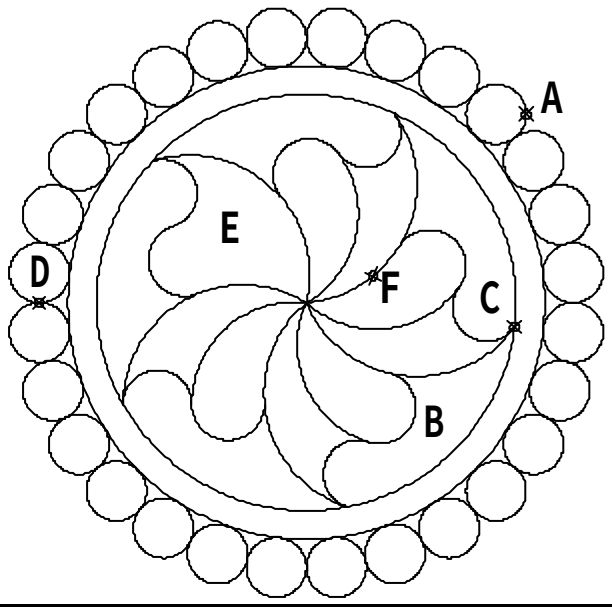
10、 Erase,
11、 TRim,




12、 ARray, **環形陣列**, **中心點**, int, C52 & C53, **項目總數**, 5, **佈滿角度**, 360, **物件**, C50;C52;C53;A55, →**全圓** 5 **組物件**



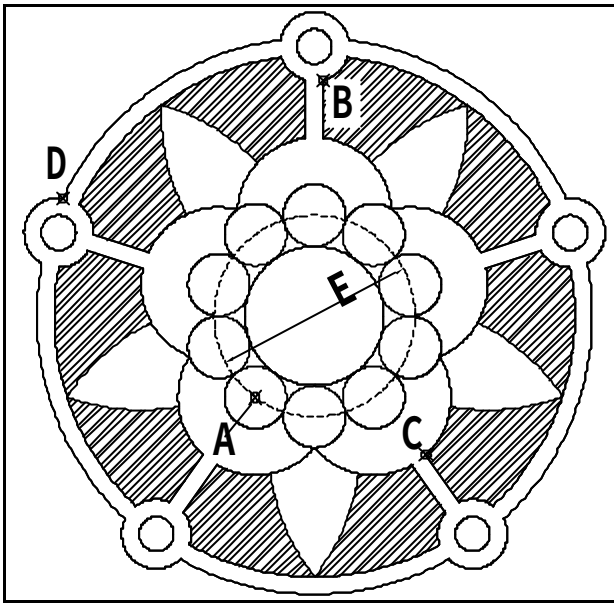

13、 Circle, int, C52 & C53, int, C50 & C52, →C56
14、 Offset, 15, C56, **向外點**, →C57
15、 XLine, V, cen, C57, →L58
16、 ROtate, L58, cen, C57, C, ‘ cal, 360/28, →L59
17、 Circle, 3P, tan, C57 **上**, tan, L58, tan, L59, →C60




18、 Erase,
19、 ARray, **環形陣列**, **中心點**, cen, C57, **項目總數**, 28, **佈滿角度**, 360, **物件**, C60, →**全圓** 28 **組物件**
20、 **點選圓** A, CTRL+1, →**查幾何半徑值** =15.7578
21、 BOundary, **點** B **區域**
22、 **點** B **區域邊緣** CRTL+1, →**查幾何長度值** =328.0658
23、 DIst, int, C, int, D, →Y **差值**=13.0110
24、 Hatch, **點** E **區域**, →**斜線區域**
25、 **點** E **斜線區域**, CRTL+1, →**查幾何面積值** =4294.1761
26、 **點選弧** F, CTRL+1, →**查幾何總角度值** =121.5881

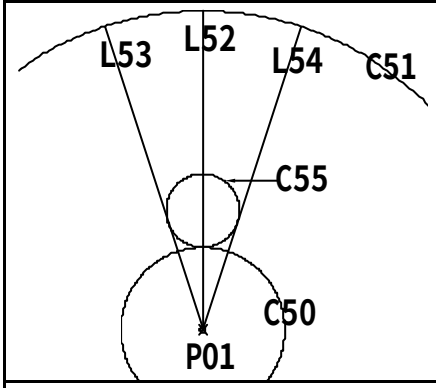
## 2.3 第三類：綜合應用三

### 2.3.1 AutoCAD2D2008-301

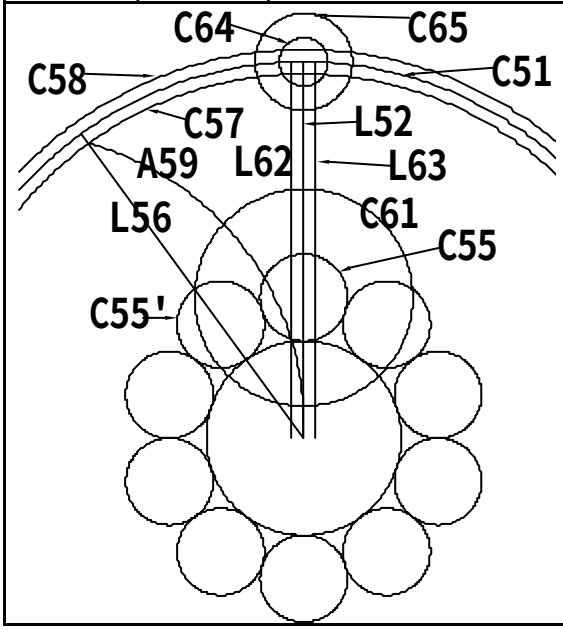



1、 **中心點** A **至交點** B **垂直距離為何？** 91.2339
2、 **交點** C **至交點** D **水平距離為何？** 104.5568
3、 **斜線區域面積總和為何？** 7091.1787
4、 **尺寸** E **值為何？** 57.8885
5、 **圖形最外圍周長為何？** 542.2316

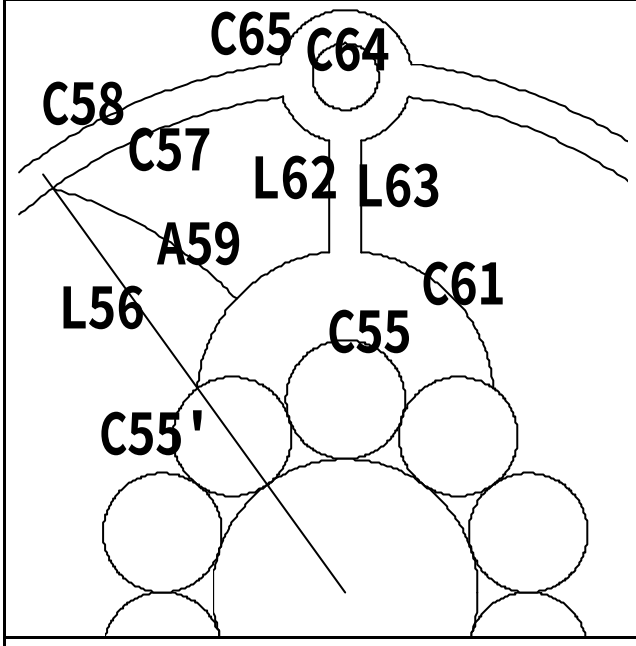
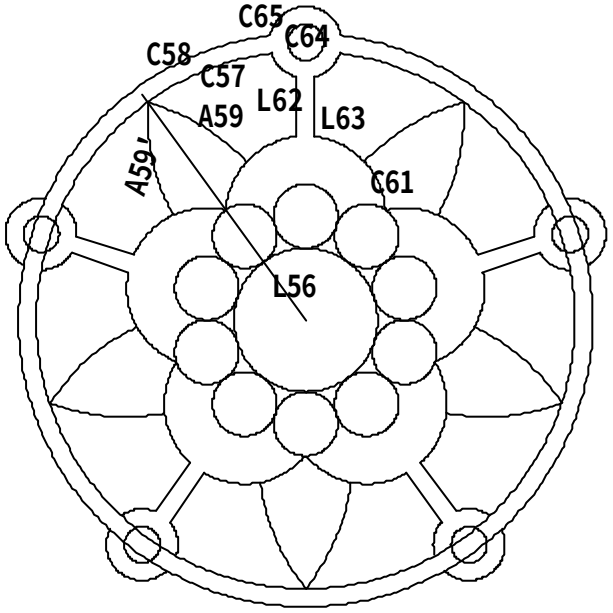
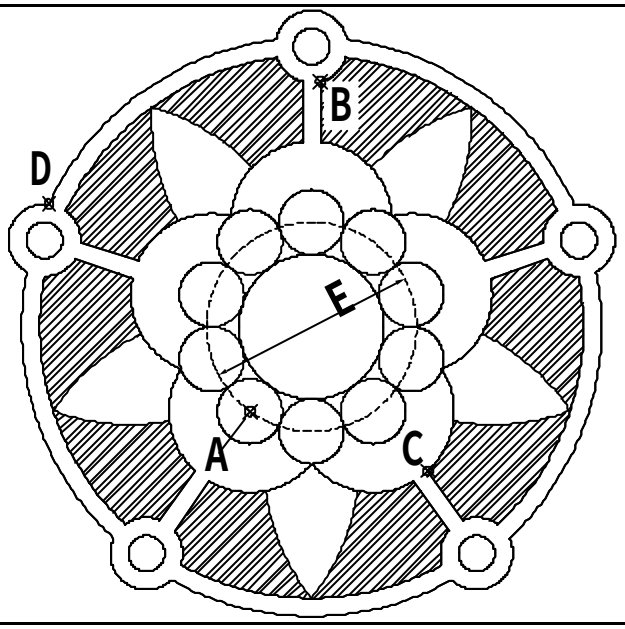
**解題步驟：**



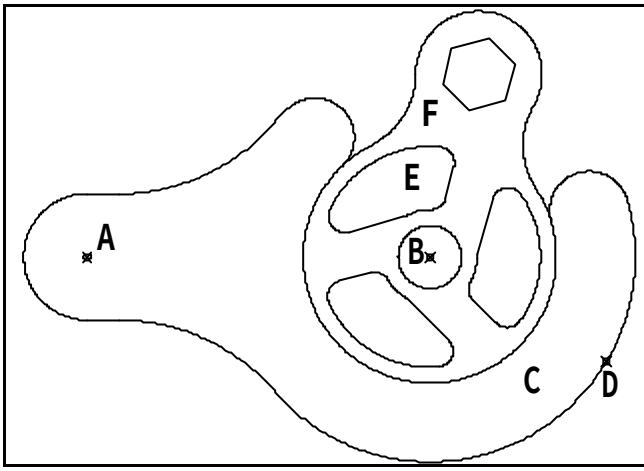

01、 Circle, **任一點** P01, D, 40, →C50
02、 Circle, @(**鎖住上次點**), D, 155, →C51
03、 Line, cen, C51, qua, C51 **上**, →L52
04、 ROtate, L52, cen, C50, C, 18 **度**(=(360/10)/2), →L53
05、 ROtate, L52, cen, C50, C, -18 **度**, →L54
06、 Circle, 3P, tan, C50, tan, L53, tan, L54, →C55




07、 Erase
08、 ARray, **環形陣列**, **中心點**, cen, C51, **項目總數**, 10, **佈滿角度**, 360, **物件**, C55, →C55' **等** 9 **個**
09、 ROtate, L52, cen, C51, C, 36 **度**(=(360/5)/2), →L56
10、 Offset, 2.5, C51, **向內點**, →C57
11、 Offset, 2.5, C51, **向外點**, →C58
12、 Arc, cen, C51, E, int, L56 & C57, R, 60, →A59
13、 Circle, cen, C55, int, C55' & L56, →C61
14、 Offset, 2.5, L52, **向左點**, →L62
15、 Offset, 2.5, L52, **向右點**, →L63
16、 Circle, int, C51 & L52, D, 10, →C64
17、 Circle, @(**鎖住上次點**), D, 20, →C65

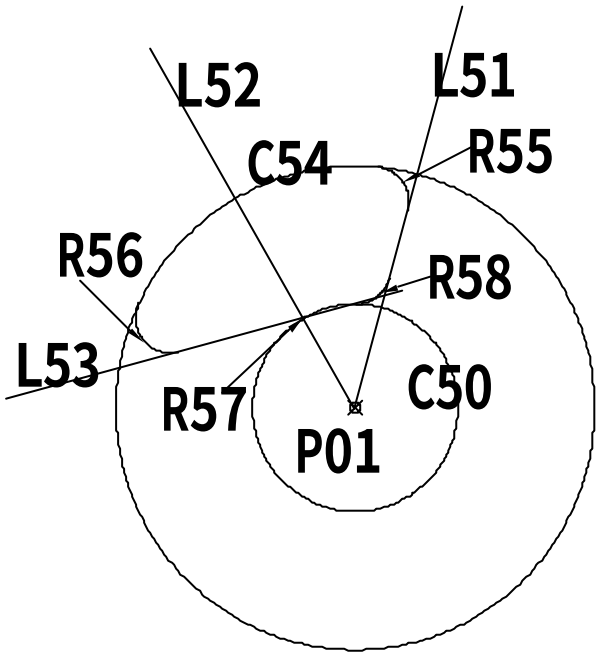
| | |
|---|---|
|  | 18、 TRim,<br>19、 Erase, |
|  | 20、 MIrror, A59, cen, C57, int, L59 & C57, →A59'<br>21、 ARray, **環形陣列**, **中心點**, cen, C57, **項目總數**, 5, **佈滿角度**, 360, **物件**, A59;A59' ;**等物件**, →**整圓** 5 **組物件**<br>22、 Erase, |
|  | 23、 DIst, cen, A, int, B, →Y **差值**=91.2339<br>24、 DIst, int, C, int, D, →X **差值**=104.5568<br>25、 Hatch, **點所有斜線區域**, →**斜線區域**<br>26、 **點斜線區域**, CRTL+1, →**查幾何面積值** =7091.1787<br>27、 Circle, cen, C58, cen, C55, →**虛圓**<br>28、 **點虛圓**, CRTL+1, →**查幾何直徑值**=57.8885<br>29、 BOundary, **斜線區域外至最外圓之間**<br>30、 **點最外圓邊緣**, CRTL+1, →**查幾何長度值** =542.2316 |

## 2.3.2 AutoCAD2D2008-302

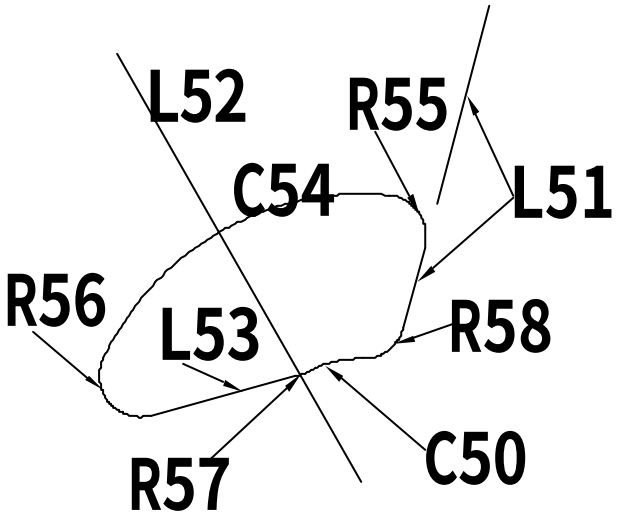



1、 **中心點** A **至中心** B **距離為何？** 108.3616
2、 C **區域周長為何？** 624.6004
3、 **弧** D **包含角為何？** 155.3936
4、 E **區域周長為何？** 106.4915
5、 F **區域扣除內孔面積為何？** 3721.4081

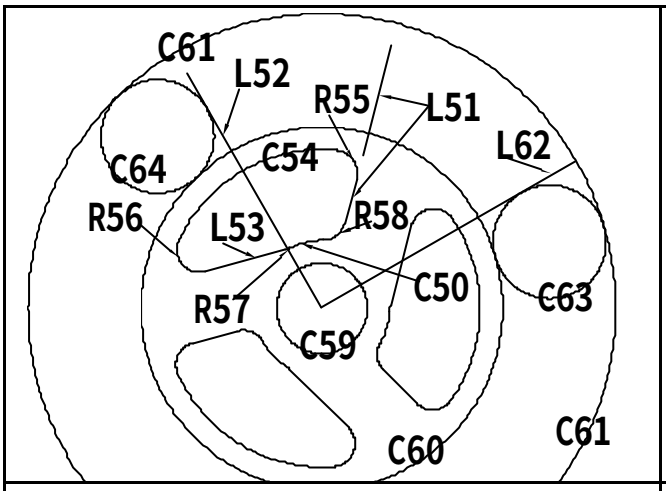
**解題步驟：**



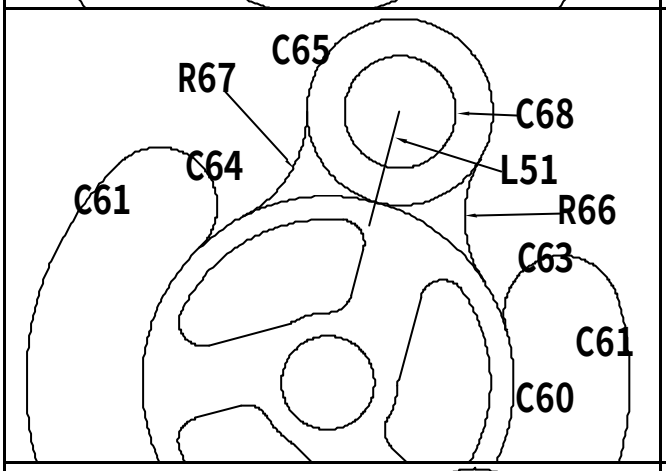

01、 Circle, **任一點** P01, D, 30, →C50
02、 Line, cen, C50, @60<75, →L51
03、 ROtate, L51, cen, C50, C, 45(=90/2), →L52
04、 ROtate, L52, int, C50 & L52, C, 75, →L53
05、 Circle, cen, C50, D, 70, →C54
06、 Fillet, T, N, R, 5, M, L51, C54, →R55, C54, L53, →R56, L53, C50, →R57, L51, C50, →R58



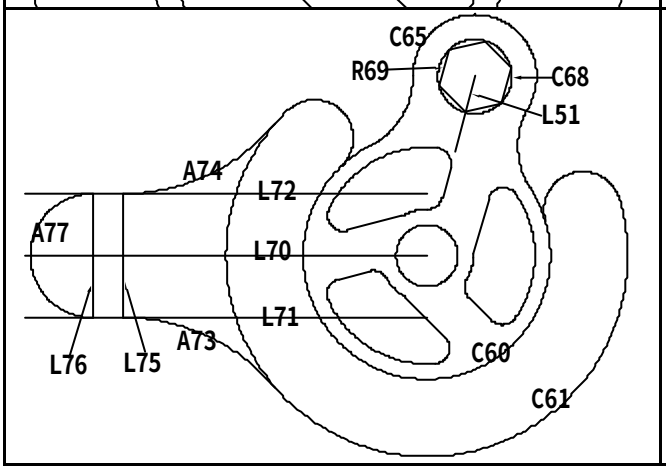

07、 TRim,
08、 Erase,



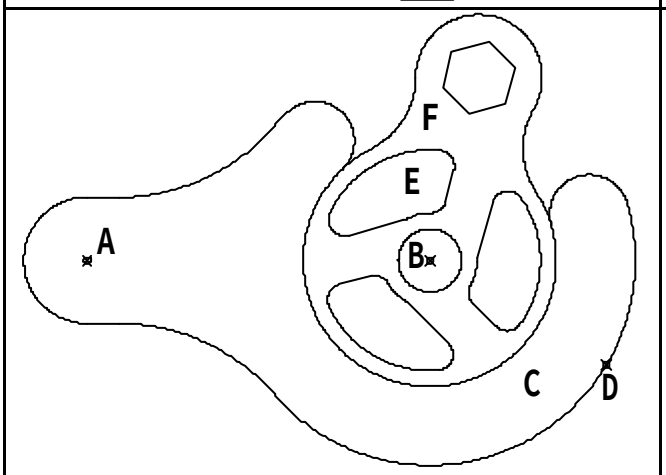

09、 Circle, cen, C50, D, 20, →C59
10、 Circle, @(**鎖定上次點**), D, 80, →C60
11、 Circle, @(**鎖定上次點**), 65, →C61
12、 ARray, **環形陣列**, **中心點**, cen, C59, **項目總數**, 3, **佈滿角度**, 360, **物件**, C50;R59;L51;R55;C54;R56;L53;R57, →**整圓** 3 **組物件**
13、 Line, cen, C59, @65<30, →L62
14、 Circle, 3P, tan, C61, tan, L62, tan, C60, →C63
15、 Circle, 3P, tan, C61, tan, L52, tan, C60, →C64




16、 TRim,
17、 Erase,
18、 Circle, end, L51 **上**, 20, →C65
19、 Fillet, T, N, R, 25, M, C60 **右上**, C65 **右下**, →R66, C60 **左上**, C65 **左下**, →R67
20、 Circle, cen, C65, D, 24, →C68

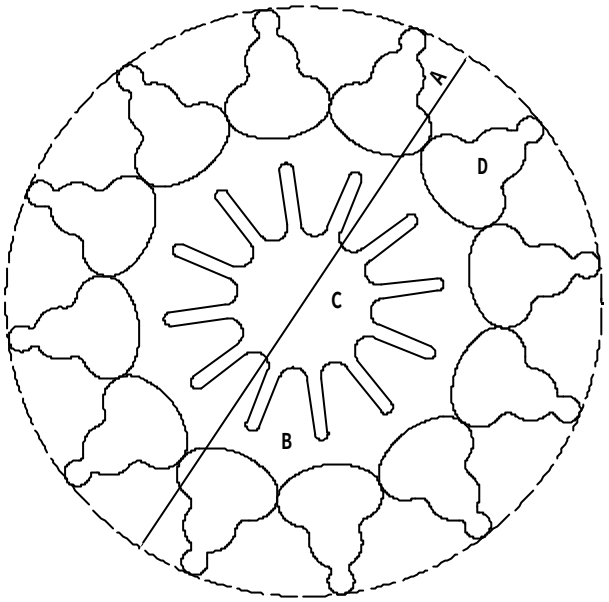



21、 POLygon, 6, cen, C68, I, int, L51 & C68, →R69
22、 Line, cen, C61, (**正交模式**), **向左** 130, →L70
23、 Offset, 20, L70, **向下點**, →L71
24、 Offset, 20, L70, **向上點**, →L72
25、 Fillet, T, N, R, 65, C61 **左下**, L71, →A73
26、 Fillet, T, N, R, 65, C61 **左上**, L72, →A74
27、 Line, int, L71 & A73, int, L72 & A74, →L75
28、 Offset, 10, L75, **向左點**, →L76
29、 Arc, C, int L70 & L76, int, L72 & L76, int, L71 & L76, →A77



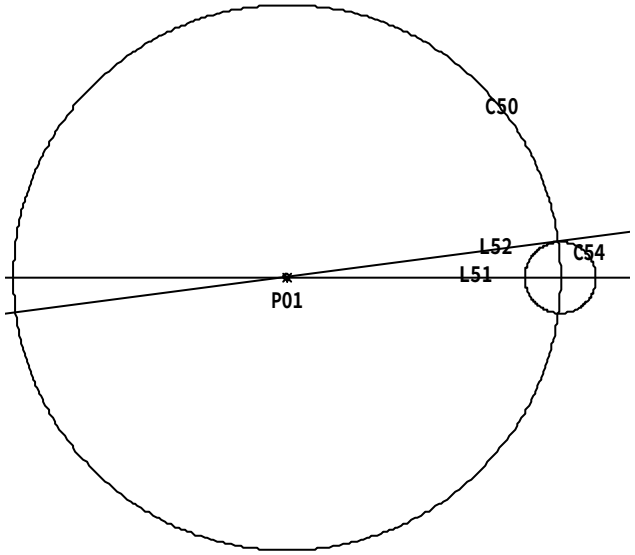

30、 TRim,
31、 Erase,
32、 DIst, cen, A, cen, B, →**距離**=108.3616
33、 Boundary, **點** C **區域**
34、 **點選** C **區域邊緣**, CRTL+1, →**查幾何長度值** =624.6004
35、 **點選弧** D, CRTL+1, →**查幾何總角度值** =155.3936
36、 Boundary, **點** E **區域**
37、 **點選** E **區域邊緣**, CRTL+1, →**查幾何長度值** =106.4915
38、 Hatch, **點** F **區域**, →**斜線區域**
39、 **點** F **區域斜線**, CRTL+1, →**查幾何面積值** =3721.4081

## 2.3.3 AutoCAD2D2008-303

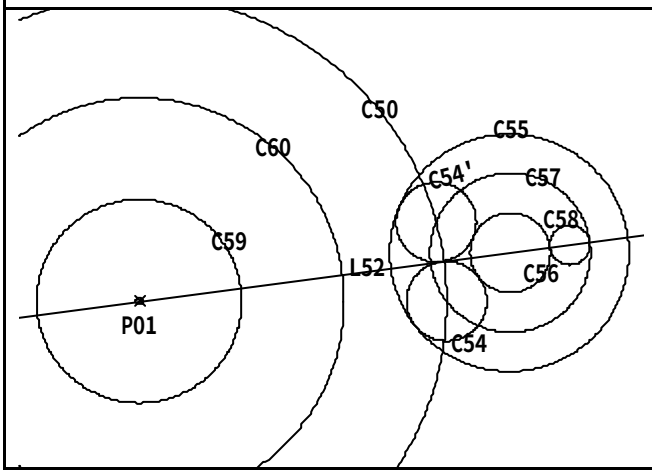



1、 **圖形最大直徑** A **值為何？** 266.1435
2、 B **區域扣除內孔面積為何？** 13621.8213
3、 C **區域周長為何？** 910.5599
4、 D **區域周長為何？** 174.4251
5、 **圖形所圍成最外圍面積為何？** 37908.0931

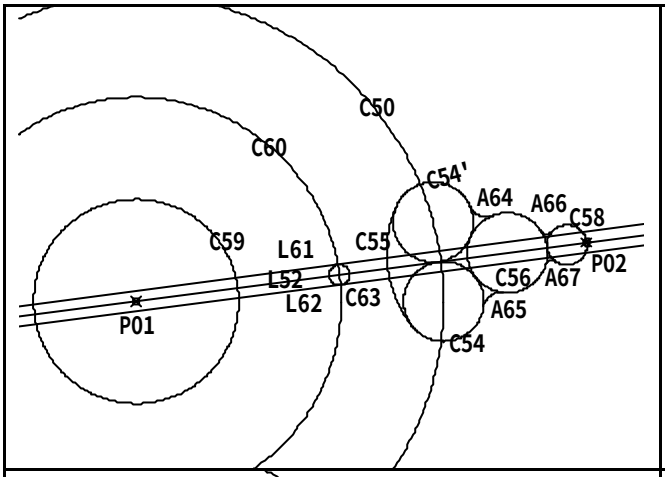
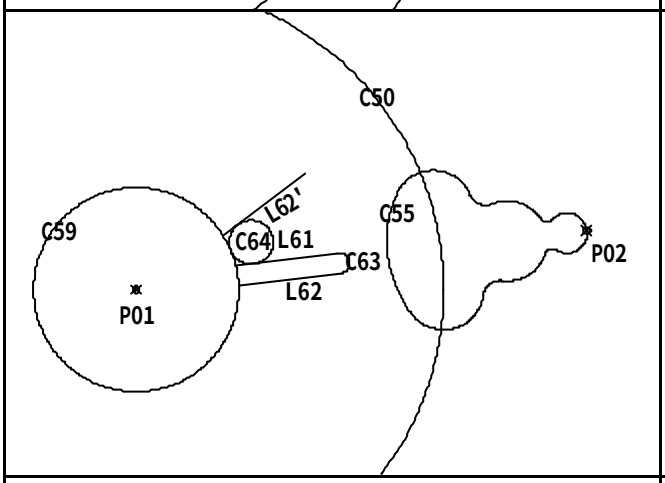
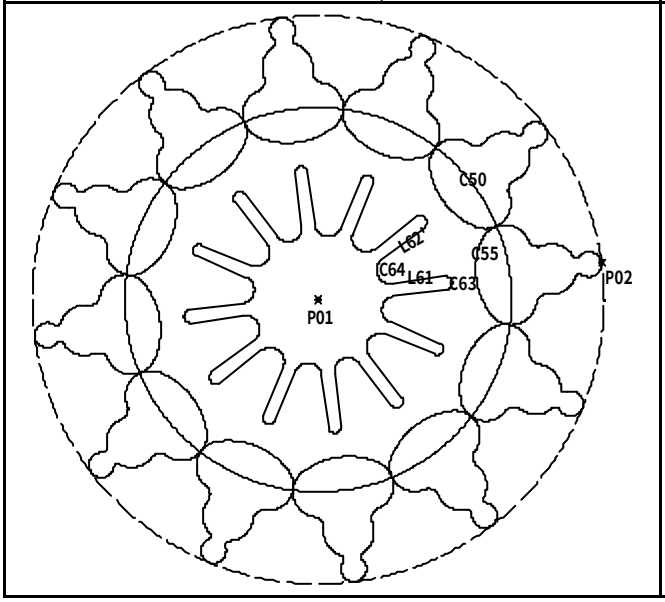
**解題步驟：**

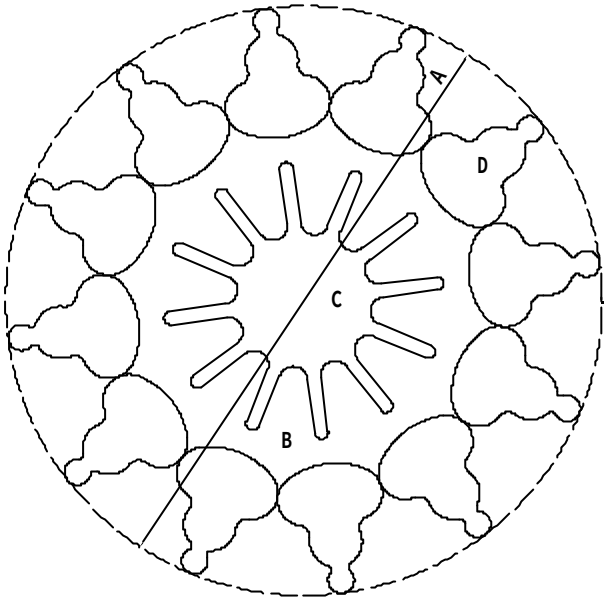
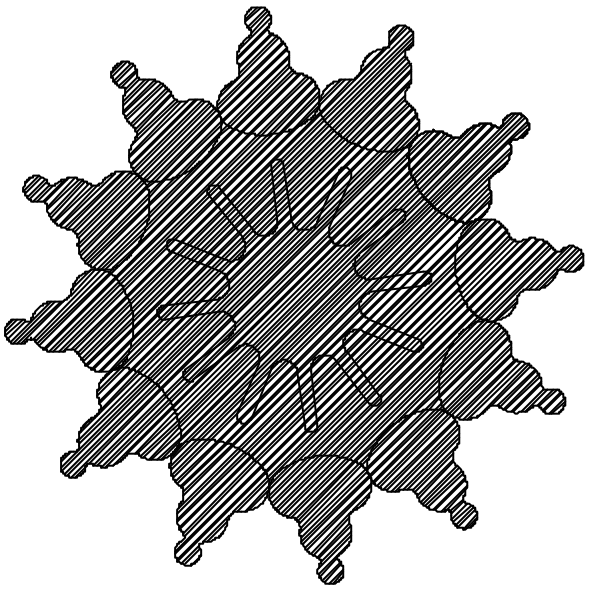



01、 Circle, **任一點** P01, D, 180, →C50
02、 XLine, @(**鎖定上次點**), qua, C50 **右**, →L51
03、 ROtate, L51, cen, C50, C, 7.5(=360/12/2/2), →L52
04、 Circle, int, C50 & L51, tan, L52, →C54

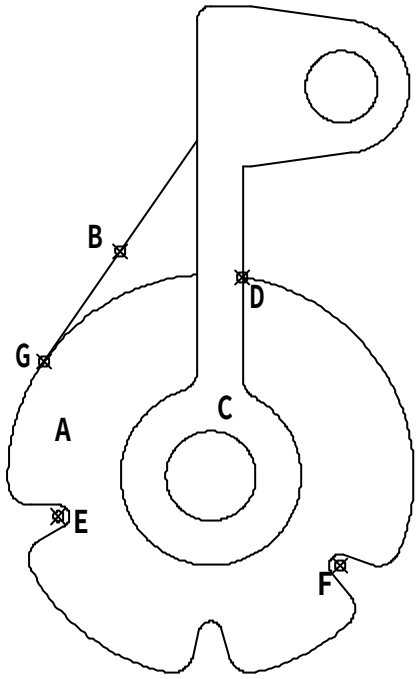



05、 Erase
06、 ROtate, C54, cen, C50, C, 15 **度**, →C54'
07、 Circle, T, C54 **左下**, C54'**左上**, 35, →C55
08、 Circle, T, C54 **右上**, C54'**右下**, cen, C54, qua, C54 **右**, →C56
09、 Offset, cen, C56, qua, C56 **右**, C56, **向外點**, →C57
10、 Circle, 2P, int, L52 & C56, int, L52 & C57, →C58
11、 Circle, cen, C50, D, 60, →C59
12、 Circle, @(**鎖定上次點**), D, 120, →C60

| | |
|---|---|
|  | 13、 TRim,<br>14、 Erase,<br>15、 Offset, 3, L52, **向上點**, →L61<br>16、 Offset, 3, L52, **向下點**, →L62<br>17、 Circle, int, L52 & C60, D, 6, →C63<br>18、 Fillet, R, 5, C54' , C56, →A64<br>19、 Fillet, R, 5, C54, C56, →A65<br>20、 Fillet, R, 2, C56, C67, →A66<br>21、 Fillet, R, 2, C56, C67, →A67<br>22、 POint, int, L52 & C58, →P02 |
|  | 23、 Erase,<br>24、 TRim,<br>25、 ROtate, L62, cen, C59, C, 30 **度**(=360/12), →L62'<br>26、 Circle, 3P, tan, L61, tan, C59, tan, L62', →C64 |
|  | 27、 TRim,<br>28、 Erase,<br>29、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 12, **佈滿角度**, 360, **物件**, C55;C64;L62 **等**, →**全圓佈滿**<br>30、 Circle, cen, C50, nod, P02, →**外圍虛圓** |

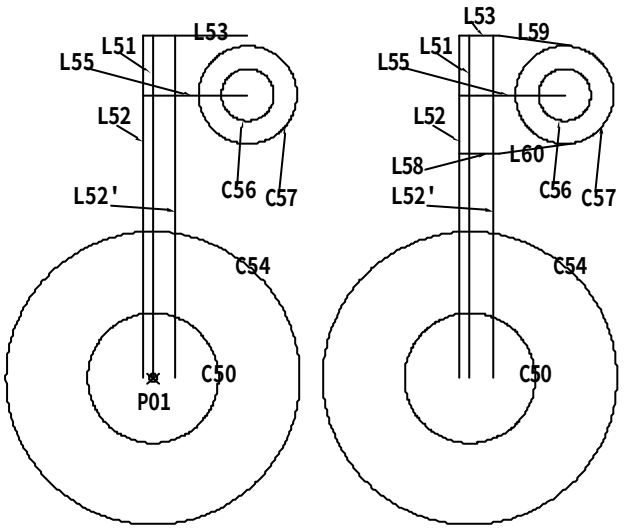
| | |
|---|---|
|  | 31、 **選外圍虛圓**, CRTL+1, **→查幾何直徑值** =266.1435<br>32、 Hatch, **點** B **區域**, **→斜線區域**<br>33、 **選** B **區域斜線**, CRTL+1, **→查幾何面積值** =13621.8213<br>34、 Boundary, **點** C **區域**<br>35、 **點** C **區域邊緣**, CRTL+1, **→查幾何長度值** =910.5599<br>36、 Boundary, **點** D **區域**<br>37、 **點** D **區域邊緣**, CRTL+1, **→查幾何長度值** =174.4251 |
|  | 38、 Hatch, **點所有封閉區域**, **→斜線區域**<br>39、 **點斜線**, CRTL+1, **→查幾何面積值** =37908.0931 |

## 2.3.4 AutoCAD2D2008-304




1、 A **區域周長為何？** 479.5603
2、 **線段** B **長度為何？** 60.0750
3、 C **區域扣除內孔面積為何？** 2685.6922
4、 **交點** D **至中心點** E **距離為何？** 67.2953
5、 **中心點** F **至交點** G **水平距離為何？** 65.7360

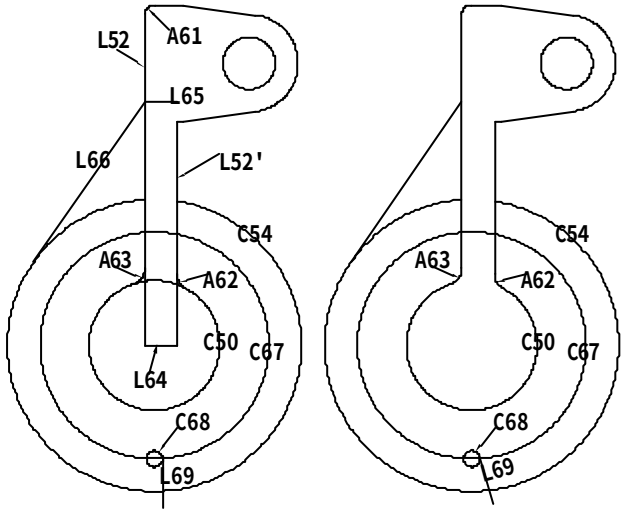
**解題步驟：**




01、 Circle, **任一點** P01, D, 40, →C50
02、 Line, CEN, C50, (**正交模式**), **向上** 105, →L51
03、 Offset, 3, L51, **向左點**, →L52
04、 Offset, 10, L52, **向右點**, →L52'
05、 Line, end, L52 **上**, (**正交模式**), **向右** 32(=12+20), →L53
06、 Circle, cen, C50, D, 90, →C54
07、 Offset, 18, L53, **向下點**, →L55
08、 Circle, end, L55 **右**, D, 16, →C56
09、 Circle, @(**鎖定上次點**), 15, →C57

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

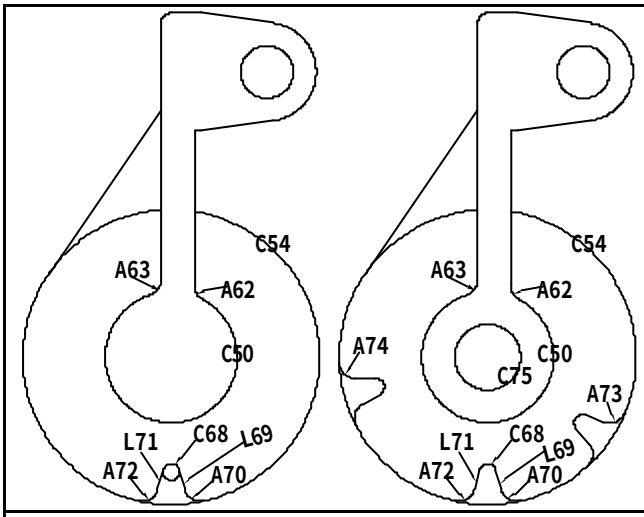
10、 LENgthen, T, 12, L53 **右**, →L53 **向左縮短**
11、 Offset, 36(=18*2), L53, **向下點**, →L58
12、 Line, end, L53 **右**, tan, C57 **上**,→L59
13、 Line, end, L58 **右**, tan, C57 **下**,→L60




14、 TRim, Erase,
15、 Fillet, T, T, R, 3, L52, L53, →A61
16、 Fillet, T, N, R, 5, L50, L52', →A62
17、 Fillet, T, N, R, 5, L50, L52, →A63
18、 Line, end, L52 **下**, end, L52'**下**, →L64
19、 Offset, 75, L64, **向上點**, →L65
20、 Line, int, L52 & L65, tan, C54 **左上**, →L66
21、 Circle, cen, C50, 35, →C67
22、 Circle, qua, C67 **下**, 2.5, →C68
23、 Line, qua, C68 **右**, (**正交模式**), **向下** 15, →L69

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

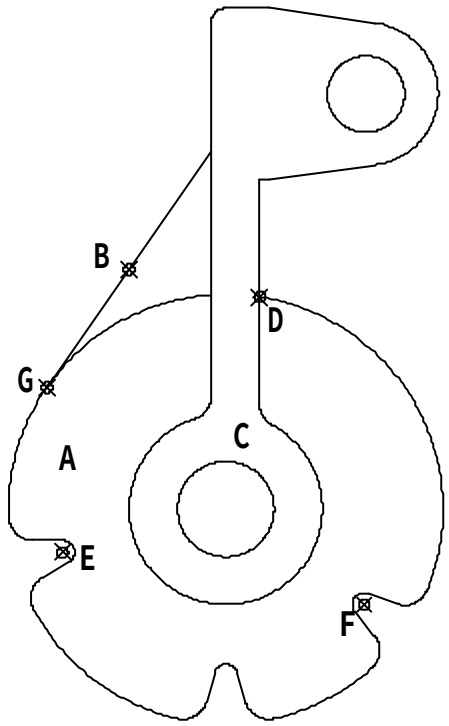
24、 TRim,
25、 Erase,
26、 ROtate, L69, cen, C68, 15,→L69 **旋轉**




27、 Fillet, T, N, R, 5, L69, C54, →A70
28、 MIrror, L69;A70, qua, C68 **上**, qua, C68 **下**, →L71;A72

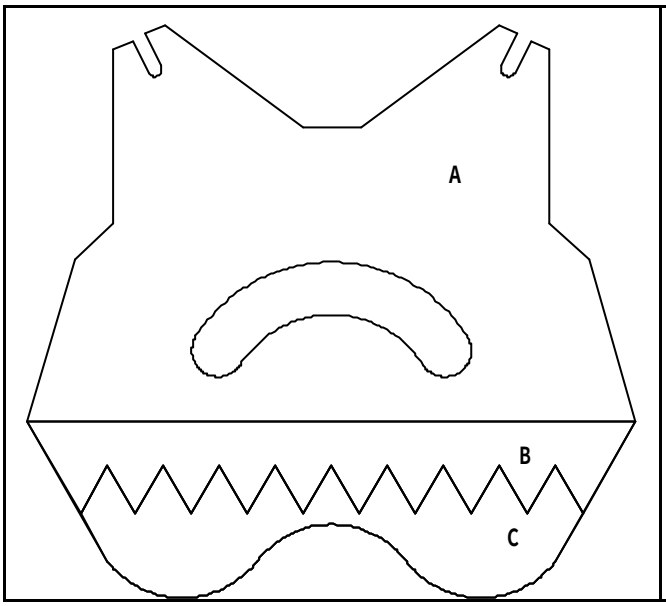
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

29、 TRim,
30、 ROtate, C68;L69;A70;L71;A72, cen, C50, C, 55, →A73 **等物件**
31、 ROtate, C68;L69;A70;L71;A72, cen, C50, C, -75, →A74 **等物件**
32、 TRim,
33、 Circle, cen, C50, D, 20, →C75



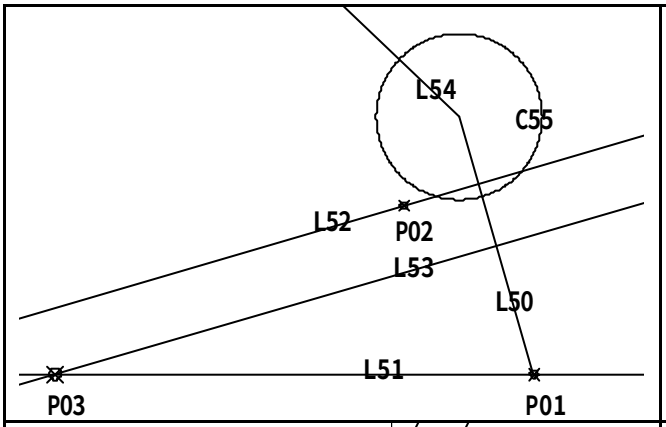

34、 BOundary, **點** A **區域**
35、 **點** A **區域邊緣**, CRTL+1, →**查幾何長度值** =479.5603
36、 **點選線段** B, CRTL+1, →**查幾何長度值** =60.0750
37、 Hatch, **點** C **區域**, →**斜線區域**
38、 **點** C **區域斜線**, CRTL+1, →**查幾何面積值** =2685.6922
39、 DIst, int, D, cen, E, →**距離**=67.2953
40、 DIst, cen, F, int, G, →X **差值**=65.7360

## 2.3.5 AutoCAD2D2008-305

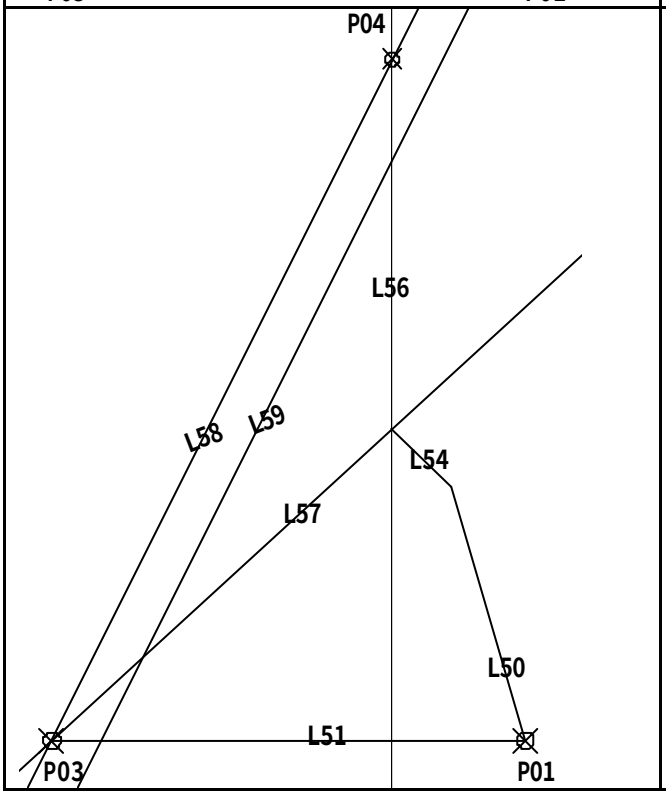



1、 **圖形最大高度值為何？** 239.6524
2、 **圖形最大寬度值為何？** 253.7264
3、 A **區域扣除內孔面積為何？** 27235.0738
4、 B **區域周長為何？** 761.1791
5、 C **區域面積為何？** 6711.3034

**解題步驟：**




01、 Line, **任一點** P01, @71<106.25 (=180-((180-32.5)/2)), →L50
02、 XLine, end, L50 **下**, (**正交模式**), **向左**, **任一點**, →L51
03、 XLine, **任一點** P02, per, L50, →L52
04、 Offset, T, L52, mid, L50, →L53
05、 POint, int, L51 & L53, →P03
06、 ROtate, L50, end, L50 **上**, C, -150,→L54
07、 Circle, int, L50 & L54, 22 →C55




08、 TRm,
09、 Erase,
10、 XLine, end, L54 **上**, (**正交模式**), **向上任一點**, →L56
11、 XLine, end, L54 **上**, nod, P03, →L57
12、 ROtate, L57, nod, P03, C, 21, →L58
13、 Offset, 12, L58, **向右點**, →L59
14、 POint, int, L56 & L58, →P04

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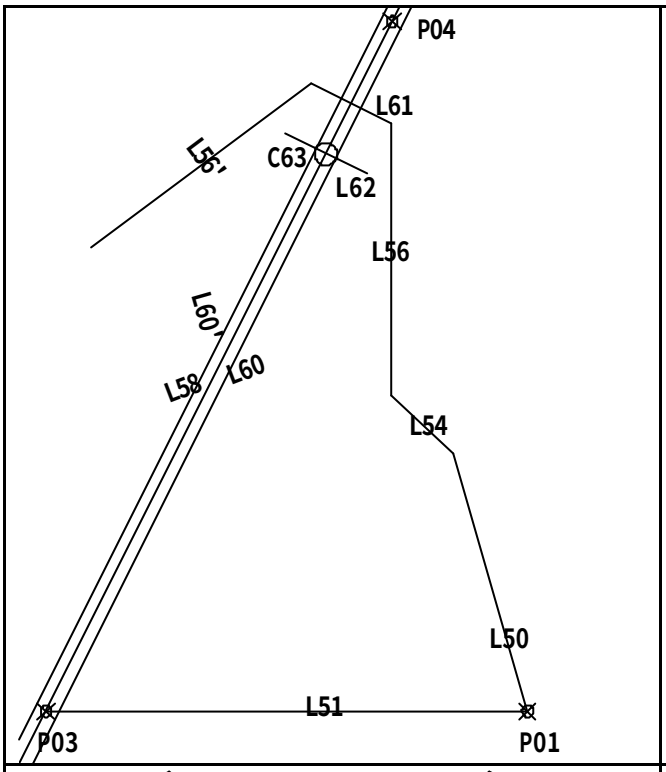
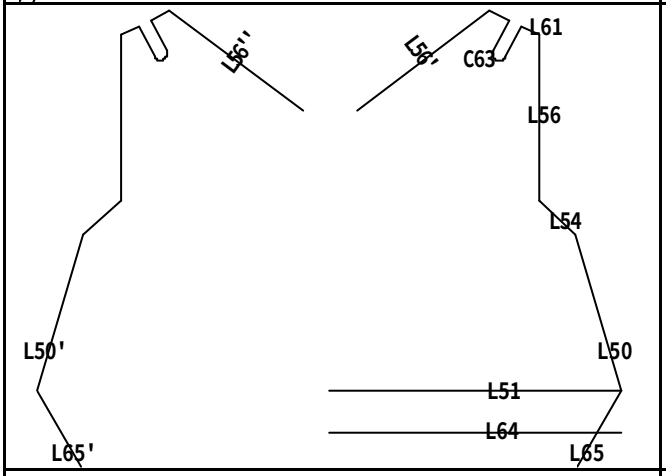
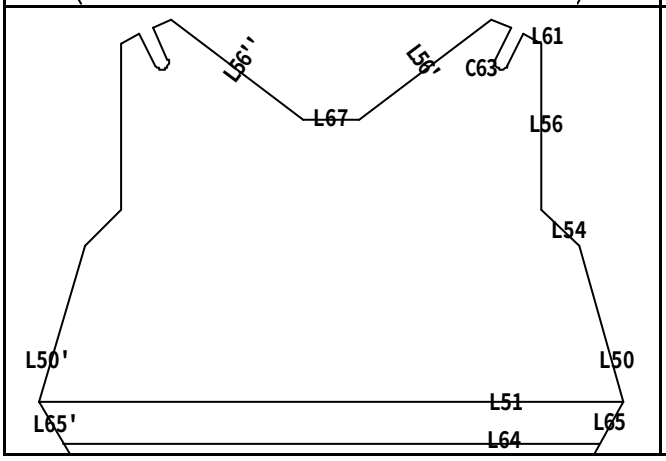
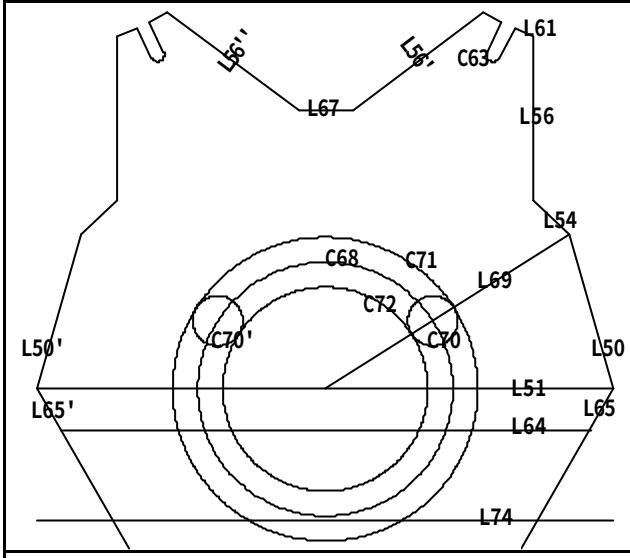
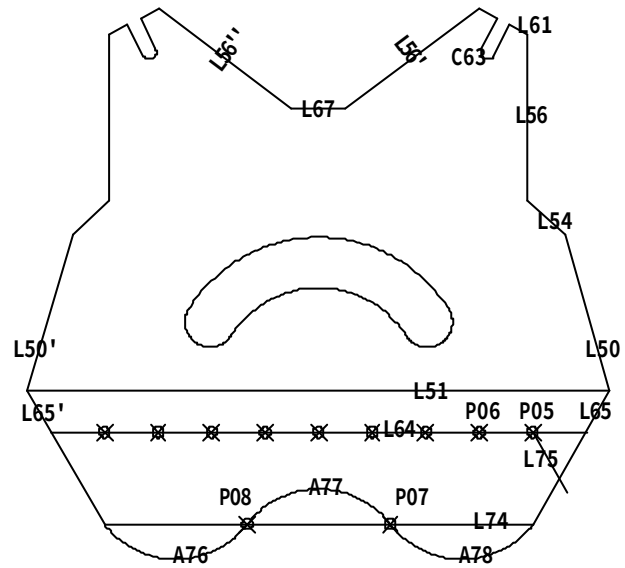
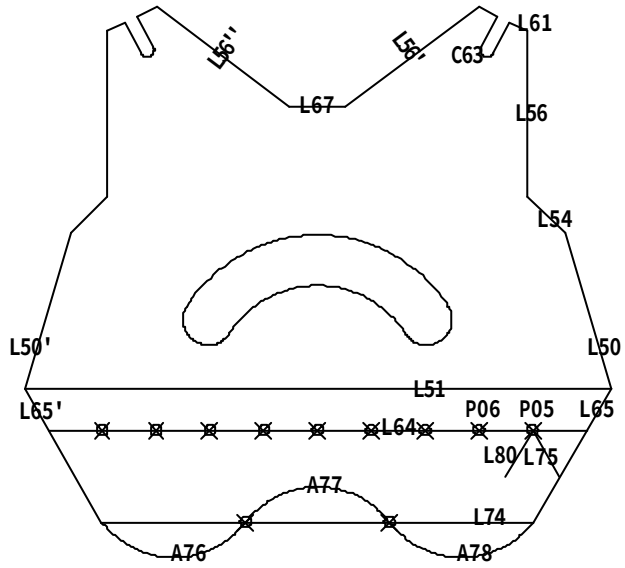
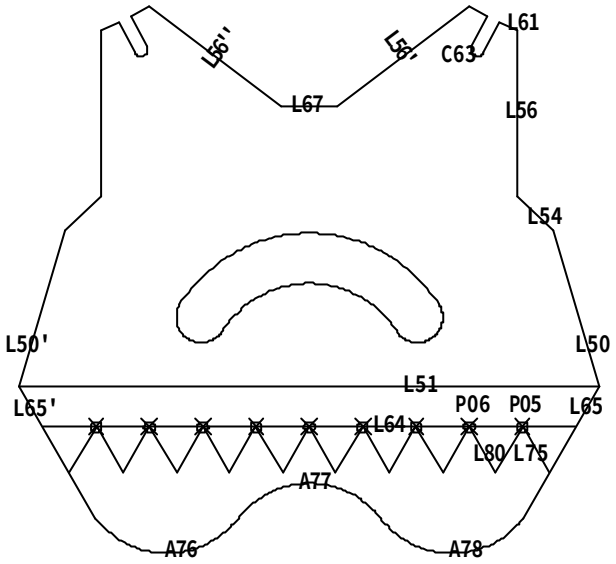
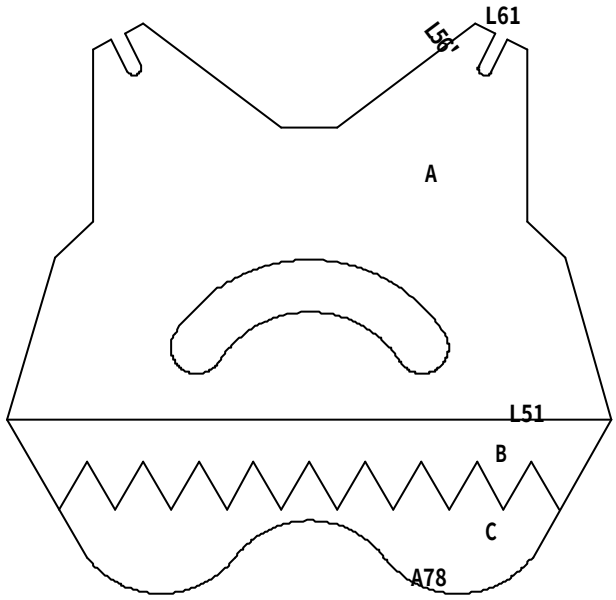
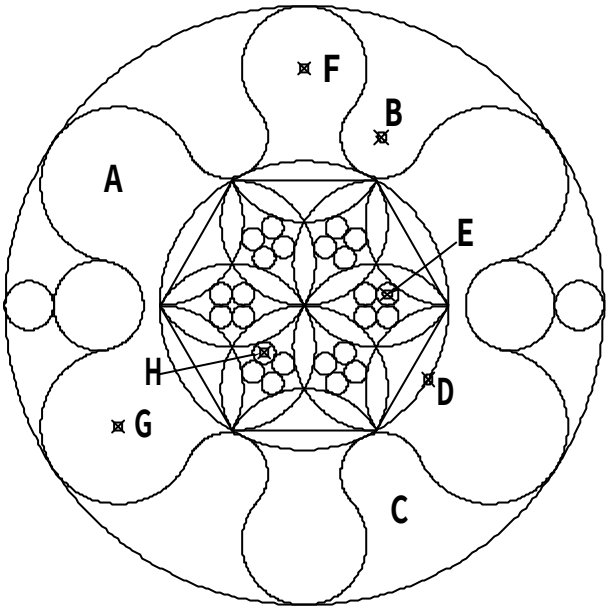
|---|---|
|  | 15、 TRim,<br>16、 Erase,<br>17、 Offset, 3, L58, **向下點**, →L60<br>18、 MIrror, L56;L60, nod, P03, nod, P04, →L56' ;L60'<br>19、 Line, end, L56 **上**, end, L56'**上**, →L61<br>20、 Offset, 15, L61, **向下點**, →L62<br>21、 Circle, int, L58 & L62, 3,→C63 |
|  | 22、 TRim,<br>23、 Erase,<br>24、 Offset, 18, L51, **向下點**, →L64<br>25、 ROtate, L51, int, L50 & L51, C, 60 **度**, →L65<br>26、 MIrror, **所有物件**(**除** L51;L64), end, L51 **左**, end, L64 **左**, →L56' ' ;L51' ;L65'**等物件** |
|  | 27、 EXtend, L50' ;L65' , L51 **左** ;L64 **左** , →L51;L64 **分別與** L50' ;L65'**相交**<br>28、 TRim,<br>29、 Line, end, L56' , end, L56' ' , →L67 |

| | |
|---|---|
|  | 30、 Circle, mid, L51, 56, →C68<br>31、 Line, cen, C68, int, L50 & L54, →L69<br>32、 Circle, int, C68 & L69, 11, →C70<br>33、 Circle, cen, C68, int, L69 & C70 **右上**, →C71<br>34、 Circle, @(**鎖定上次點**), int, L69 & C70 **左下**, →C72<br>35、 MIrror, C70, qua, C71 **上**, qua, C71 **下**, →C70'<br>36、 Offset, 58, L51, **向下點**, →L74 |
|  | 37、 TRim,<br>38、 Erase,<br>39、 DIVide, L64, 10, →P05;P06 **等** 9 **個點**<br>40、 DIVide, L74, 3, →P07 **及** P08 **兩個點**<br>41、 Line, nod, P05, @30<-60, →L75<br>42、 Arc, int, L65' & L74, E, nod, P08, A, 105, →A76<br>43、 Arc, nod, P07, E, nod, P08, A, 105, →A77<br>44、 Arc, nod, P07, E, int, L65 & L74, A, 105, →A78 |
|  | 45、 TRim,<br>46、 Erase,<br>47、 ROtate, L75, nod, P05, C, -60, →L80 |

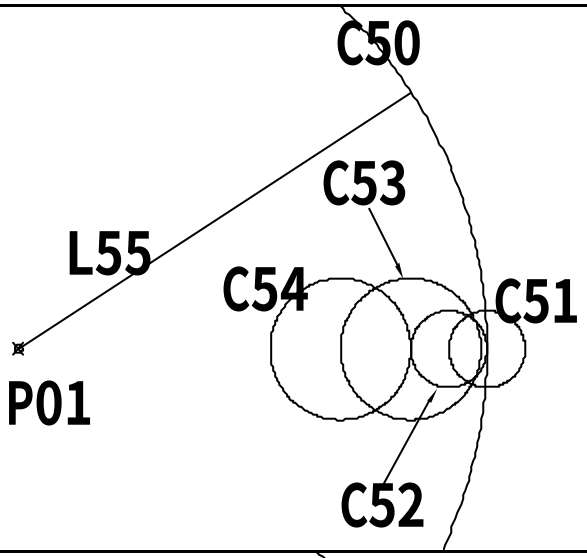
| | |
|---|---|
|  | 48、 Erase,<br>49、 ARray, **矩形陣列**, **列**, 1, **行**, 9, **列偏移**, 0, **行偏移**, nod, P05, nod, P06, **物件**, L75;L80, →9 **組等同於** L75 **與** L80 **之物件**<br>50、 Erase, |
|  | 51、 DIst, int, L56’ & L61, qua, A78, →Y **差值** =239.6524<br>52、 **選** L51, CRTL+1, →**查幾何長度值**=253.7264<br>53、 Hatch, **點** A **區域**, →**斜線區域**<br>54、 **選** A **區域斜線**, CRTL+1, →**查幾何面積值** =27235.0738<br>55、 Boundary, **點** B **區域**<br>56、 **點** B **區域邊緣**, CRTL+1, →**查幾何長度值** =761.1791<br>57、 Hatch, **點** C **區域**, →**斜線區域**<br>58、 **選** C **區域斜線**, CRTL+1, →**查幾何面積值** =6711.3034 |

## 2.3.6 AutoCAD2D2008-306

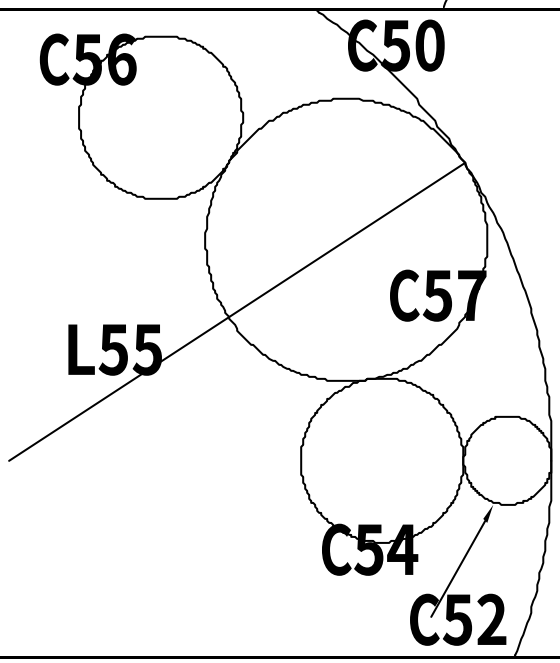



1、 A **區域面積為何？** 3031.4522
2、 **中心點** B **至中心點** H **距離為何？** 65.6283
3、 C **區域周長為何？** 200.3394
4、 **圓** D **半徑值為何？** 38.5662
5、 ∠FEG **角度為何？** 95.8438

**解題步驟：**




01、 Circle, **任一點** P01, D, 160, →C50
02、 Circle, qua, C50 **右**, 6.5, →C51
03、 Circle, qua, C51 **左**, 6.5, →C52
04、 Circle, qua, C52 **左**, 12, →C53
05、 Circle, qua, C53 **左**, 12, →C54
06、 Line, cen, C50, @80<33, →L55




07、 Erase,
08、 MIrror, C54, end, L55 **左下**, end, L55 **右上**, →C56
09、 Circle, 3P, tan, C50, tan, C54, tan, C56, →C57

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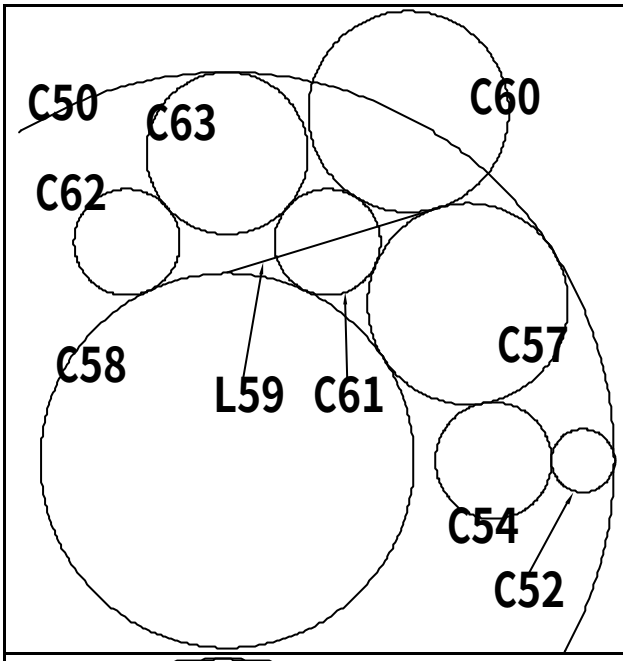
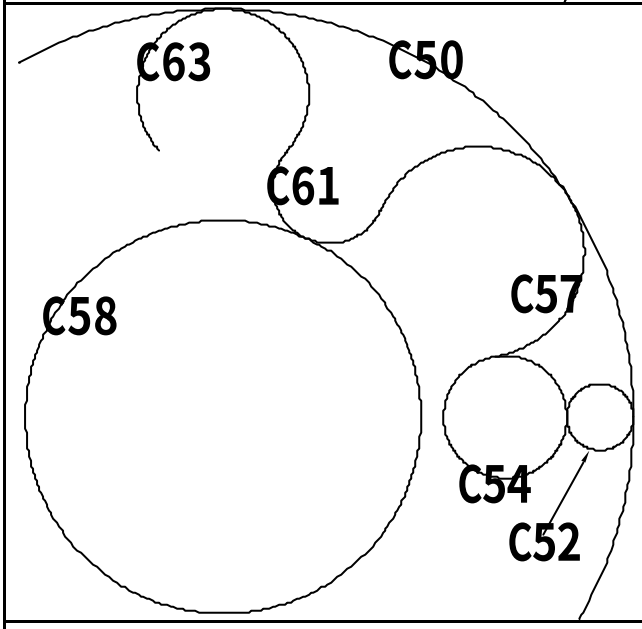
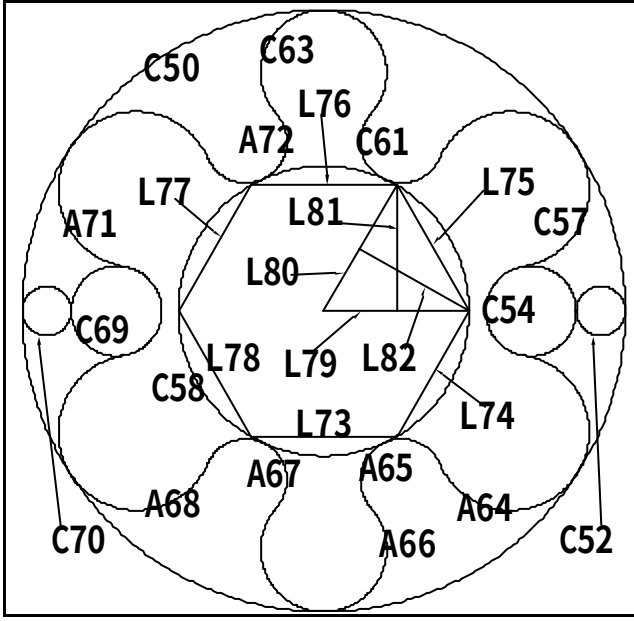
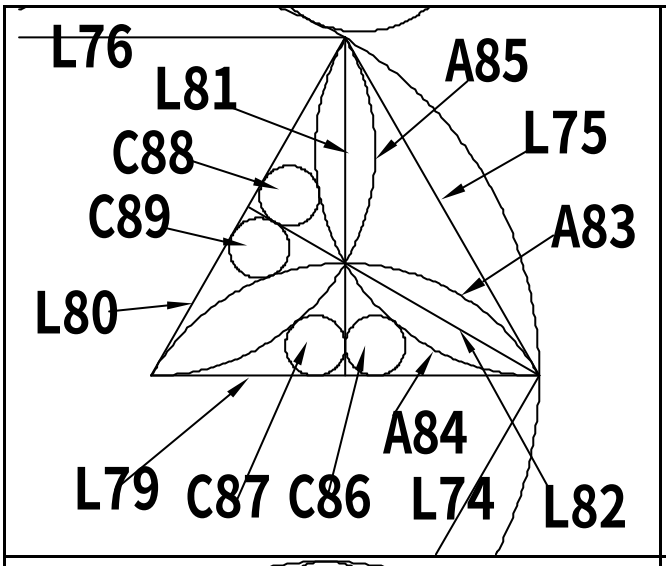
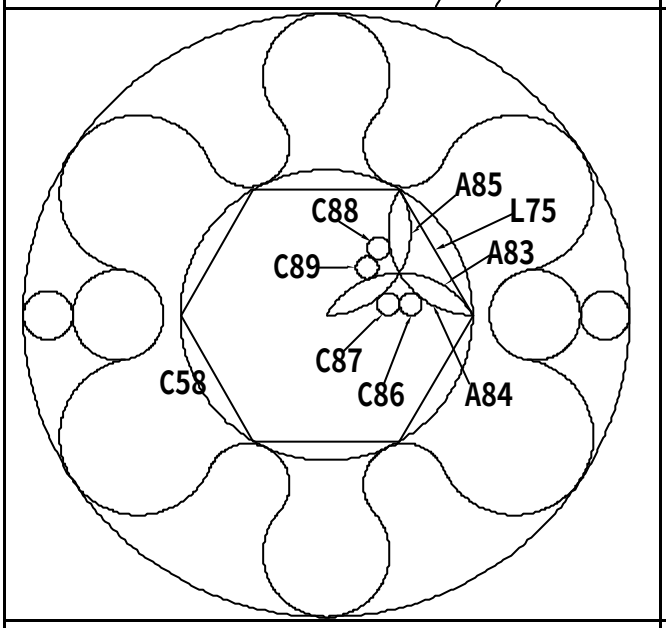
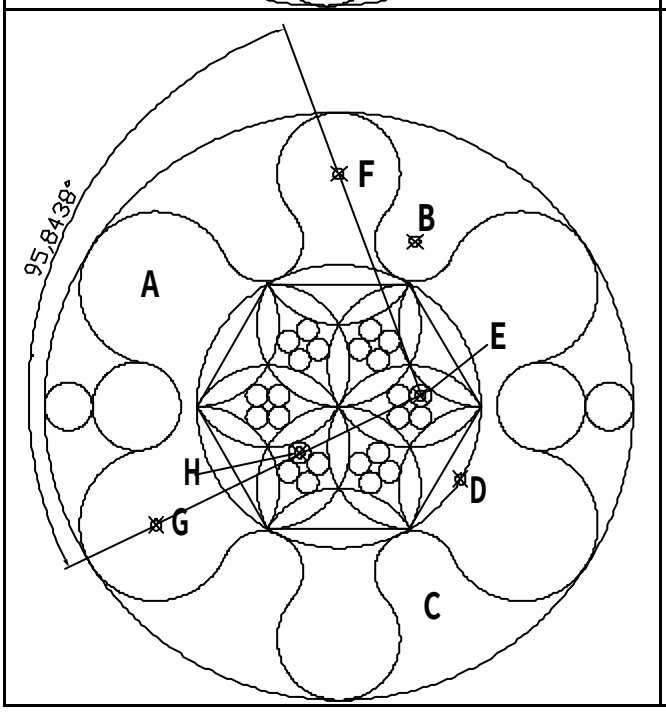
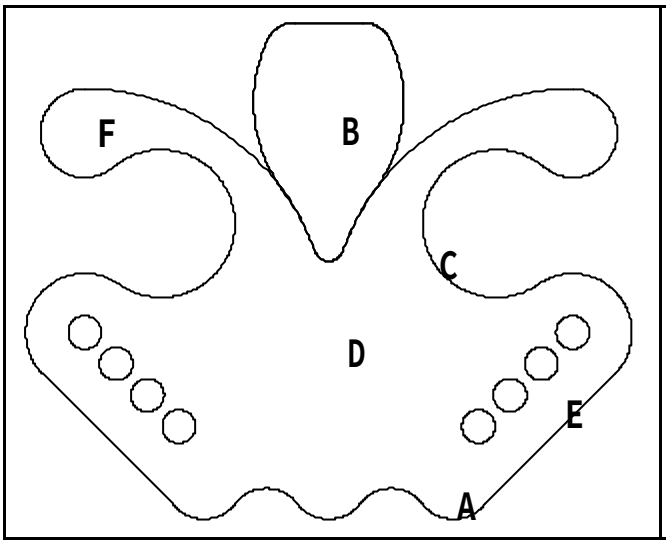
|---|---|
|  | 10、 Erase,<br>11、 Circle, cen, C50, tan, C57, →C58<br>12、 Line, Qua, C58 上, tan, C57 左上, →L59<br>13、 MIrror, C57, end, L59 左下, end, L59 右上, →C60<br>14、 Circle, 3P, tan, C57, tan, C60, tan, C58, →C61<br>15、 MIrror, C61, qua, C50 上, qua, C58 上, →C62<br>16、 Circle, 3P, tan, C50, tan, C61, tan, C62, →C63 |
|  | 17、 TRim,<br>18、 Erase, |
|  | 19、 MIrror, C57;C61;C63, qua, C58 左, qua, C58 右, →A64;A65;A66<br>20、 MIrror, C57;C61;C52;C54;A64;A65, qua, C58 上, qua, C58 下, →A67~A72<br>21、 POLgyon, 6, cen, C58, I, qua, C58 右, →L73~L78 之正 6 邊形<br>22、 Line, int, L74 & L75, cen, C58, →L79, int, L75 & L76, →L80, mid, L79, →L81<br>23、 Line, int, L74 & L75, mid, L80, →L82 |

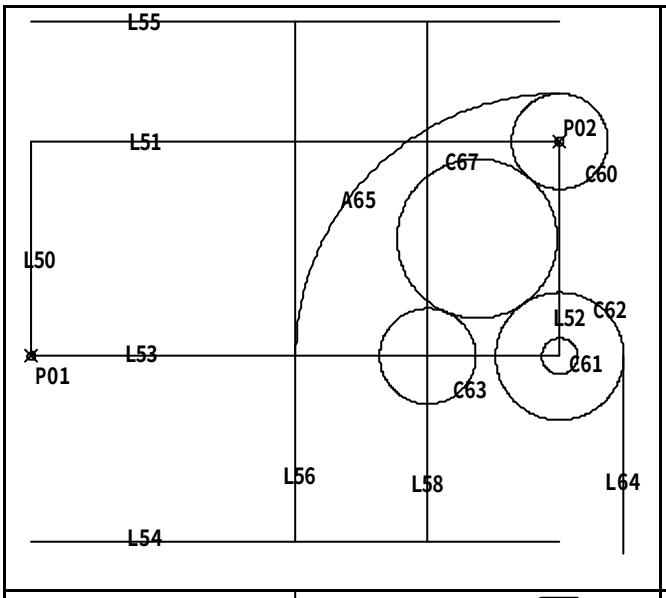
| | |
|---|---|
|  | 24、 Arc, int, L74 & L75, int, L81 & L82, int, L79 & L80, →A83<br>25、 Arc, int, L76 & L75, int, L81 & L82, int, L75 & L74, →A84<br>26、 Arc, int, L79 & L80, int, L81 & L82, int, L75 & L76, →85<br>27、 Circle, 3P, tan, A84, tan, L81, tan, L79, →C86<br>28、 Circle, 3P, tan, A85, tan, L81, tan, L79, →C87<br>29、 Circle, 3P, tan, A84, tan, L82, tan, L80, →C88<br>30、 Circle, 3P, tan, A83, tan, L82, tan, L80, →C89 |
|  | 31、 Erase, |
|  | 32、 ARray, **環形陣列**, **中心點**, cen, C58, **項目總數**, 6, **佈滿角度**, 360, **物件**, A83~A85;C86~C89, →6 **邊形內佈滿物件**<br>33、 Hatch, **點** A **區域**, →**斜線區域**<br>34、 **選** A **斜線區域**, CRTL+1, →**查幾何面積值** =3031.4522<br>35、 DIst, cen, B, cen, H, →**距離**=65.6283<br>36、 BOundary, **點** C **區域**,<br>37、 **點** C **區域邊緣**, CRTL+1, →**查幾何長度值** =200.3394<br>38、 **點選圓** D, CRTL+1, →**查幾何半徑值** =38.5662<br>39、 Line, cen, F, cen, E, cen, G<br>40、 DimANgular, **線段** FE, **線段** EG, **點圓外** →95.8438 |

## 2.3.7 AutoCAD2D2008-307

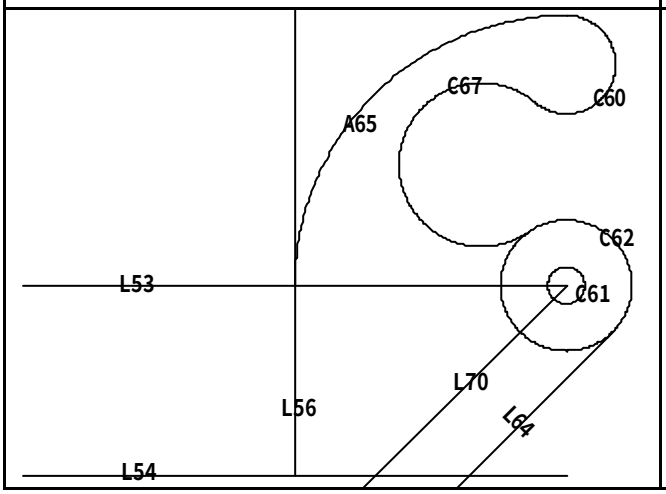



1、 **弧** A **半徑為何？** 13.3066
2、 B **區域周長為何？** 212.8395
3、 **弧** C **長度為何** 111.3373
4、 D **區域扣除內孔之面積為何？** 16648.2194
5、 **中點** E **至中心點** F **距離為何？** 188.2630

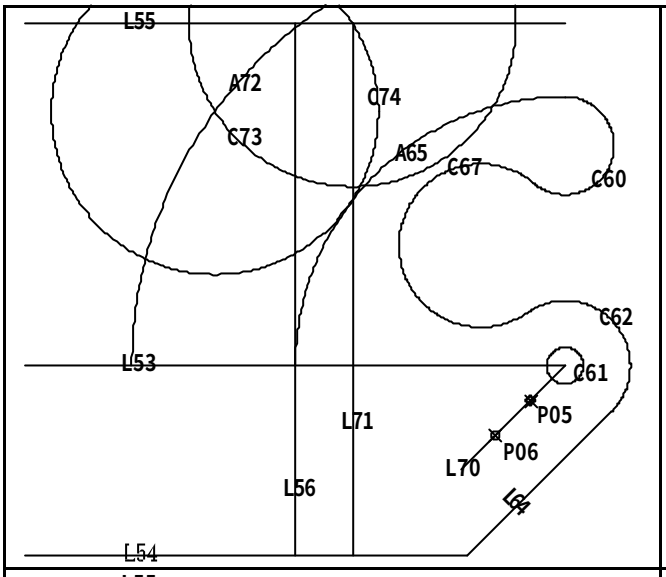
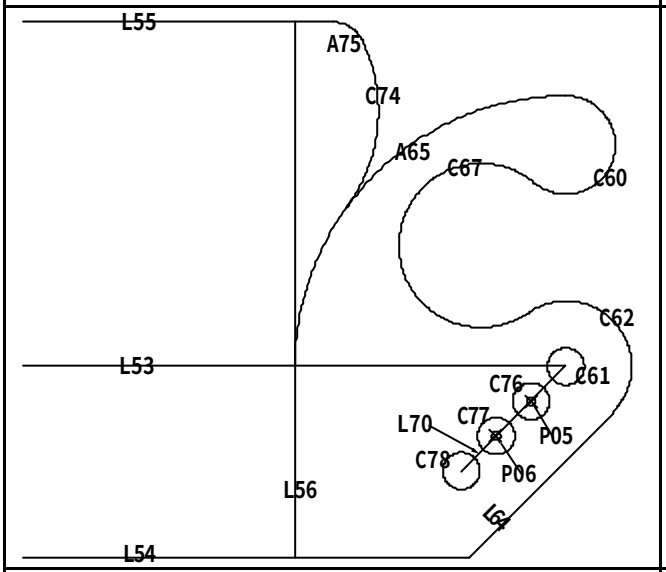
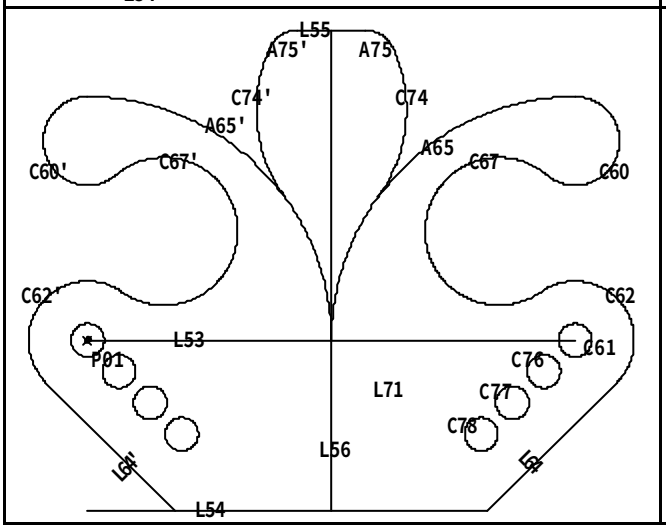
**解題步驟：**

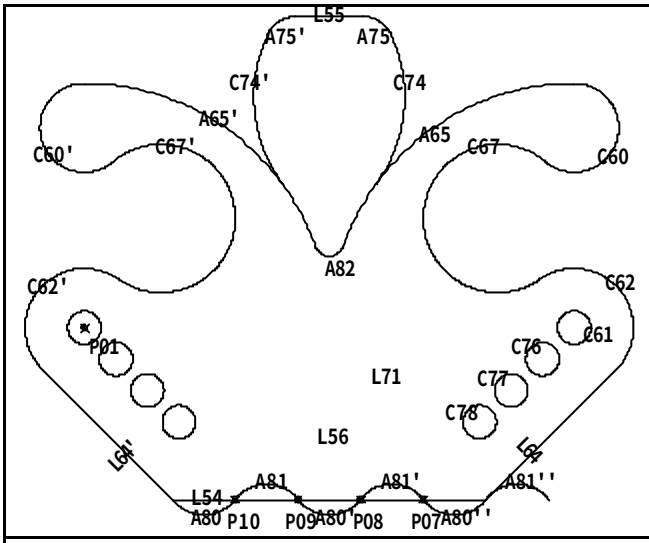
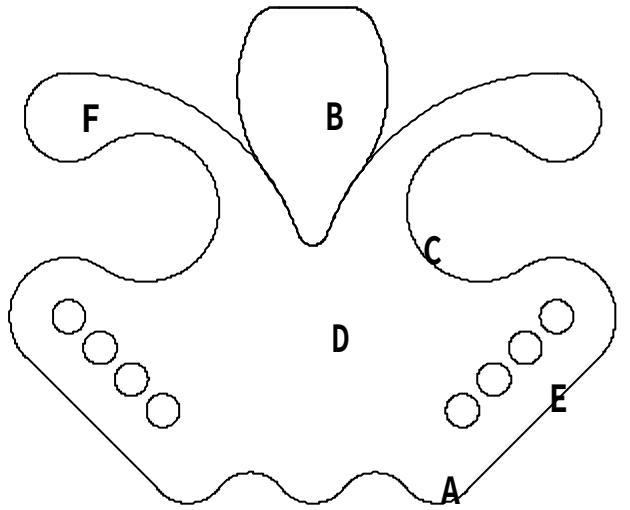



01、 RECtangle，**任一點** P01, D, 165, 67.5, **向右上點**→L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 58, L53, **向下點**, →L54
04、 Offset, 163, L54, **向上點**, →L55
05、 Line, mid, L54, mid, L55 →L56
06、 Offset, 41.25, L56, **向右點**, →L58
07、 Circle, int, L52 & L51, 15, →C60
08、 Circle, int, L52 & L53, D, 11.25, →C61
09、 Circle,@(**鎖定上次點**), 20, →C62
10、 Circle, int, L58 & L53, D, 30, →C63
11、 Line, qua, C62 **右**, (**正交模式**), **向下**,100, →L64
12、 Arc, C, cen, C62, qua, C60 **上**, int, L56 &L53, →A65
13、 Circle, 3P, tan, C60, tan, C63, tan, C62, →C67

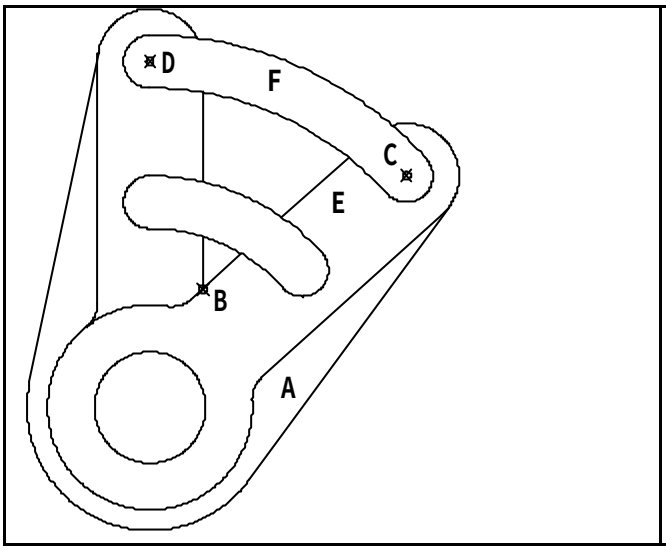



14、 TRim,
15、 Erase,
16、 ROtate, L64, cen, C62, -45, →L64 **轉動**
17、 Offset, T, L64, cen, C62, →L70

| | |
|---|---|
|  | 18、 TRim,<br>19、 Erase,<br>20、 Offset, 17.5(=35/2), L56, **向右點**, →L71<br>21、 Offset, 50, A65, **向外點**, →A72<br>22、 Circle, int, L71 & L55, 50, →C73<br>23、 Circle, int, A72 & C73, 50, →C74<br>24、 LENgthen, T, 45, L70, →L70 **總長為** 45<br>25、 DIVide, L70, 3, →P05;P06 |
|  | 26、 TRim,<br>27、 Erase,<br>28、 Fillet, T, T, R, 10, C74, L55, →A75<br>29、 COpy, C61, cen, C61, end, L70, →C76, nod, P06, →C77, nod, P05, →C78 |
|  | 30、 TRim,<br>31、 Erase,<br>32、 MIrror, L64;C62;C61 **等物件**, int, L56 & L54, int, L56 & L53, →L64' ;C62' **等物件** |

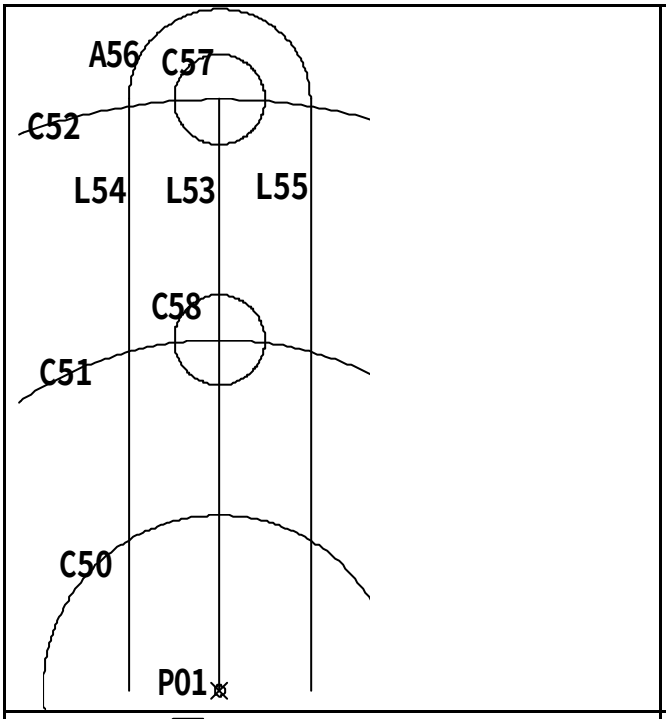
| | |
|---|---|
|  | 33、 TRim,<br>34、 Erase,<br>35、 DIVide, L54, 5, →P07~P10<br>36、 Arc, int, L64' & L54, E, nod, P10, A, 105, →A80<br>37、 Arc, nod, P09, E, nod, P10, A, 105, →A81<br>38、 COpy, A80;A81, int, L64' & L54, nod, P09, →A80' ;A81' , nod, P07, →A80' ' ;A81' '<br>39、 Fillet, T, T, R, 5, A65, A65' , →A82 |
|  | 40、 TRim,<br>41、 Erase,<br>42、 **選弧** A, CRTL+1, →**查幾何半徑值**=13.3066<br>43、 BOundary, **點** B **區域**,<br>44、 **點** B **區域邊緣**, CRTL+1, →**查幾何長度值** =212.8395<br>45、 **選弧** C, CRTL+1, →**查幾何弧長值**=111.3373<br>46、 Hatch, **點** D **區域**, →**斜線區域**<br>47、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值** =16648.2194<br>48、 DIst, mid, E, cen, F, →**距離**=188.2630 |

## 2.3.8 AutoCAD2D2008-308

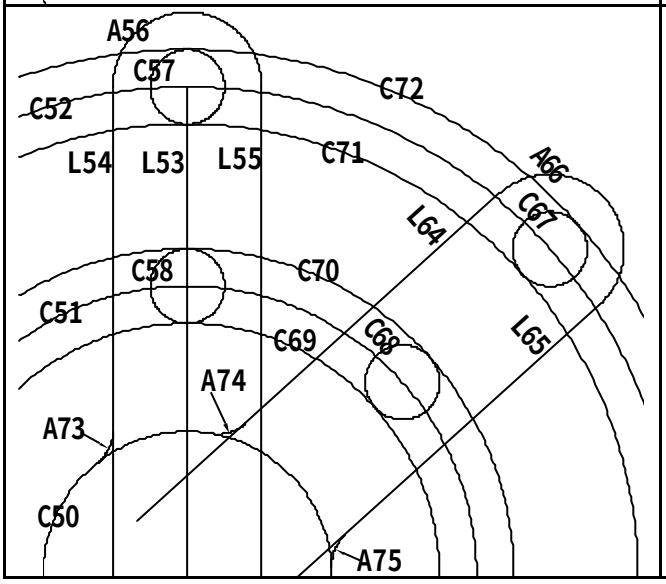



1、 A **區域面積為何？** 2870.0375
2、 **交點** B **至中心點** C **水平距離為何？** 69.6911
3、 **中心點** C **至中心點** D **距離為何？** 95.9898
4、 **區域** E **周長為何？** 502.8327
5、 **區域** F **面積為何？** 2033.8671

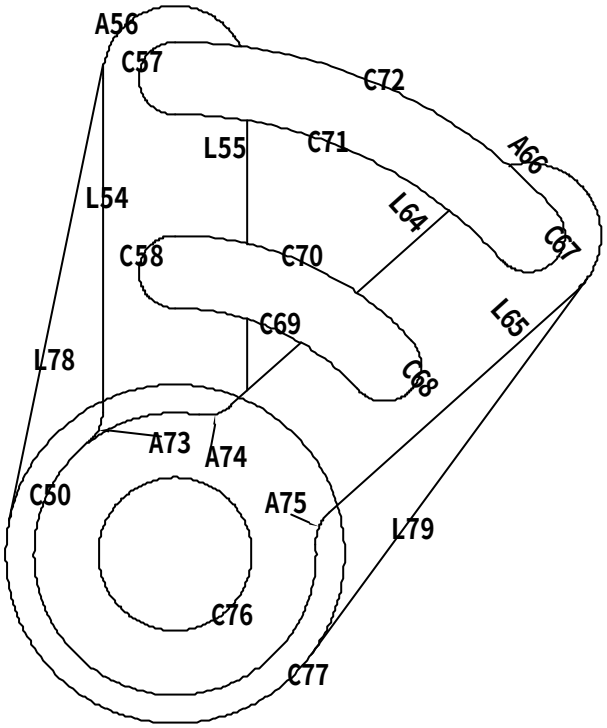
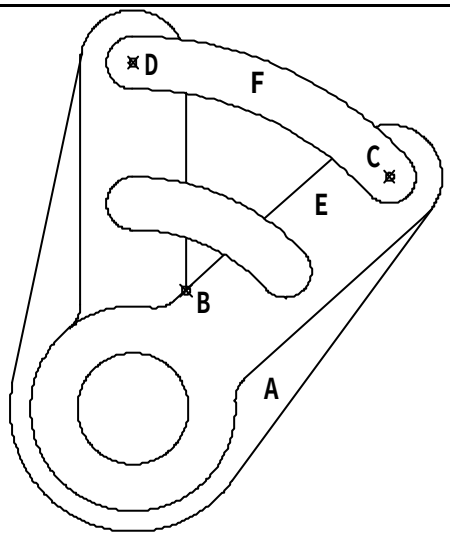
**解題步驟：**



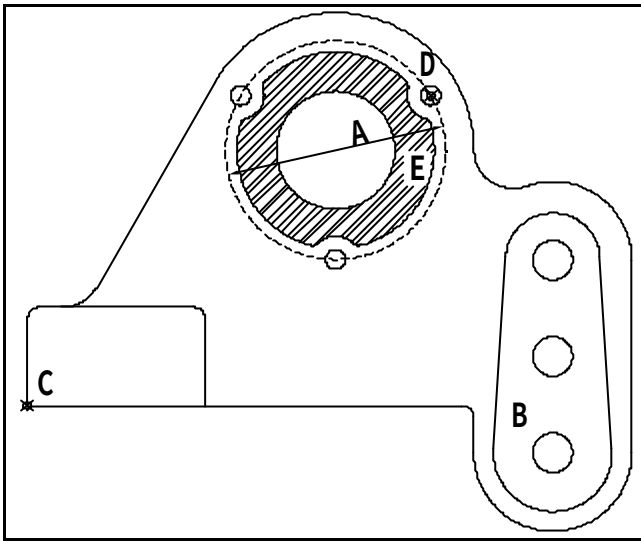

01、 Circle, **任一點** P01, D, 70, →C50
02、 Circle, @(**鎖定上次點**),, 70, →C51
03、 Offset, 48, C51, **向外點**, →C52
04、 Line, cen, C50, qua, C52 **上**, →L53
05、 Offset, 18, L53, **向左點**, →L54
06、 Offset, 18, L53, **向右點**, →L55
07、 Arc, C, int, C52 & L53, end, L55 **上**, end, L54 **上**, →A56
08、 Circle, int, C52 & L53, 9, →C57
09、 Circle, int, C51 & L53, 9, →C58




10、 ROtate, L54;L55;A56;C57;C58, cen, C50, C, -48, →L64;L65;A66;C67;C68
11、 Offset, 9, C51, **向內點**, →C69, C51, **向外點**, →C70, C52, **向內點**, →C71, C52, **向外點**, →C72
12、 Fillet, T, N, R, 10, L54, C50, →A73
13、 Fillet, T, N, R, 10, L64, C50, →A74
14、 Fillet, T, N, R, 10, L65, C50, →A75

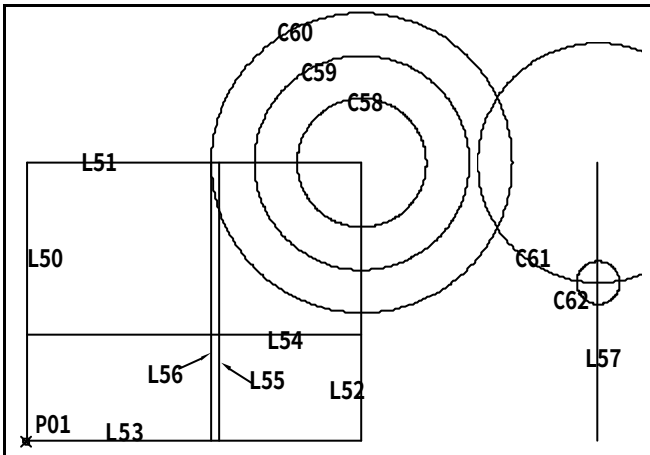
| | |
|---|---|
|  | 15、 TRim,<br>16、 Erase,<br>17、 Circle, cen, C50, D, 38, →C76<br>18、 Circle, @(**鎖定上次點**), 42, →C77<br>19、 Line, tan, A56 **左**, tan, C77 **左上**, →L78<br>20、 Line, tan, A66 **右**, tan, C77 **右下**, →L79 |
|  | 21、 TRim<br>22、 Erase<br>23、 Hatch, **點** A **區域**, **→斜線區域**<br>24、 **點** A **區域斜線**, CRTL+1, **→查幾何面積值** =2870.0375<br>25、 DIst, int, B, cen, C, **→**X **差值**=69.6911<br>26、 DIst, cen, C, cen, D, **→距離**=95.9898<br>27、 Boundary, **點** E **區域**<br>28、 **點** E **區域邊緣**, CRTL+1, **→查幾何長度值** =502.8327<br>29、 Hatch, **點** F **區域**, **→斜線區域**<br>30、 **點** F **區域斜線**, CRTL+1, **→查幾何面積值** =2033.8671 |

## 2.3.9 AutoCAD2D2008-309

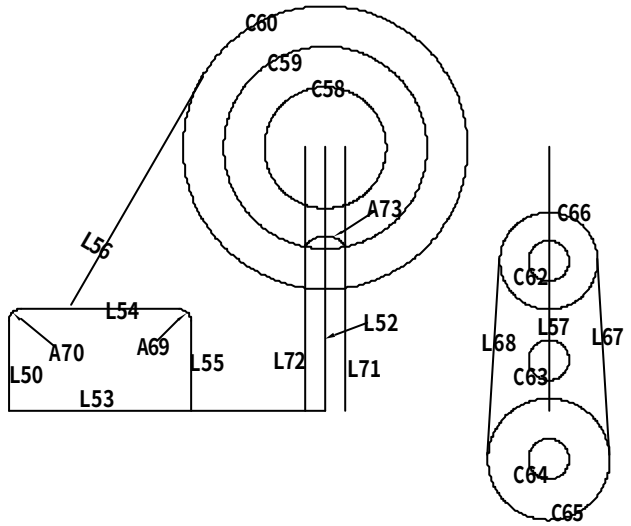



1、 A **的直徑值為何？** 55.6230
2、 B **區域扣除內孔面積為何？** 1669.4848
3、 **交點** C **至中心點** D **角度值為何？** 37.7018
4、 **斜線區域** E **區域面積為何？** 1188.3760
5、 **圖形所圍成最外圍周長為何？**487.7341

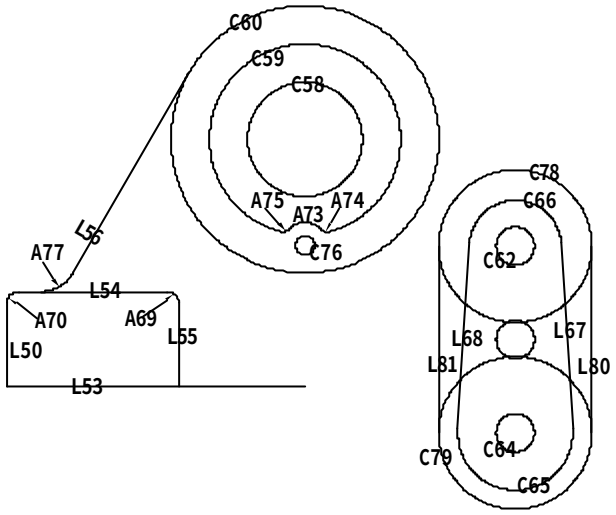
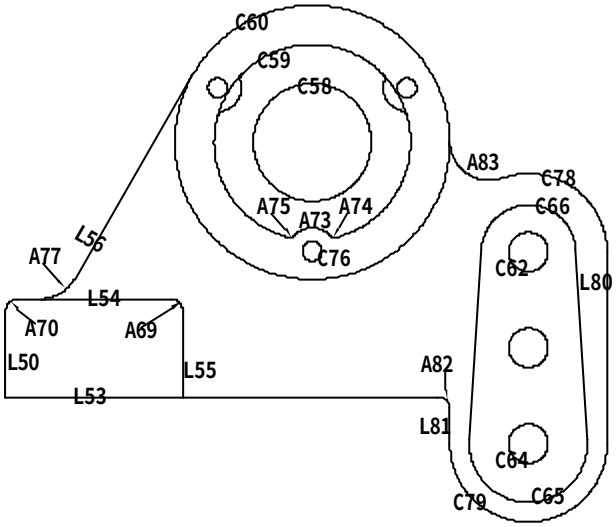
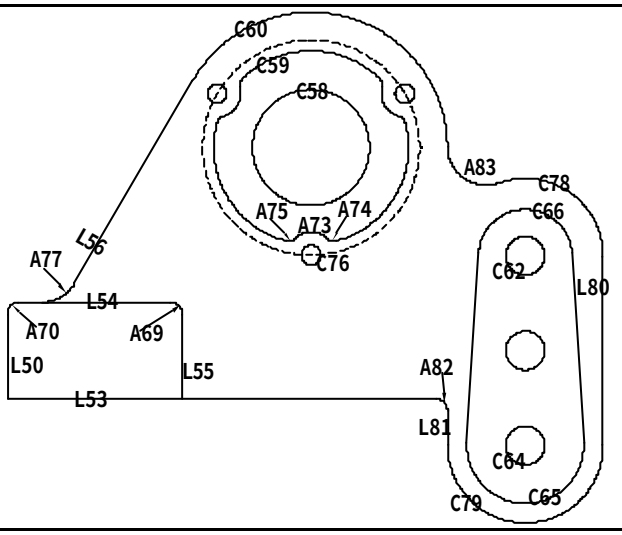
**解題步驟：**

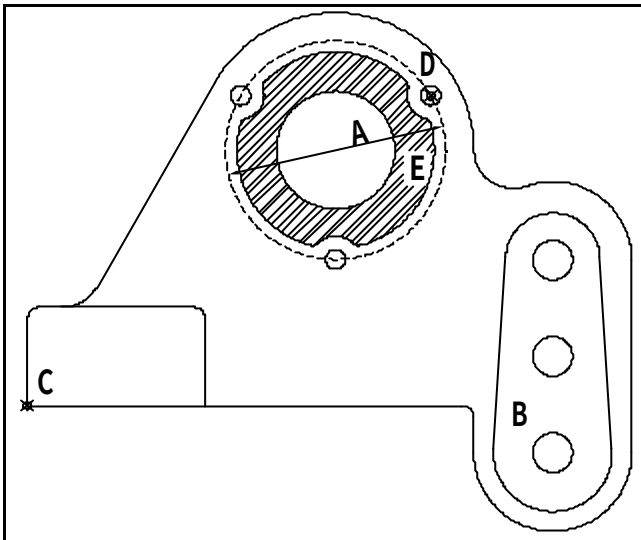
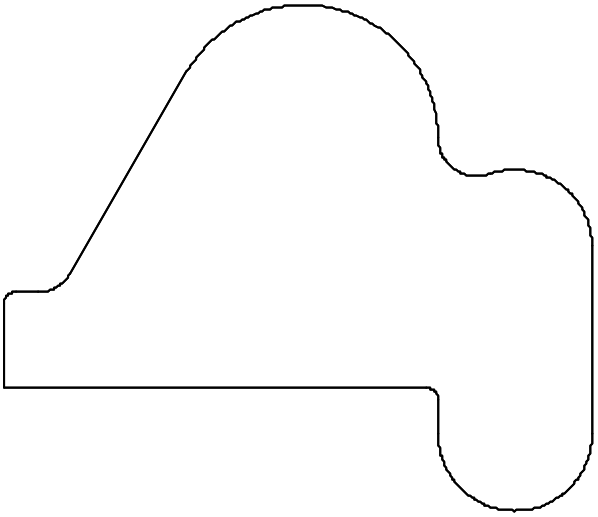



01、 RECtangle, **任一點** P01, D, 78, 65, **向右上點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 25, L53, **向上點**, →L54
04、 Offset, 33, L52, **向左點**, →L55
05、 Offset, 35, L52, **向左點**, →L56
06、 Offset, 55, L52, **向右點**, →L57
07、 Circle, int, L51 & L52, D, 30, →C58
08、 Circle,@(**鎖定上次點**), D, 50, →C59
09、 Circle,@(**鎖定上次點**), 35, →C60
10、 Circle, end, L57 **上**, 28, →C61
11、 Circle, int, L57 & C61, D, 10, →C62

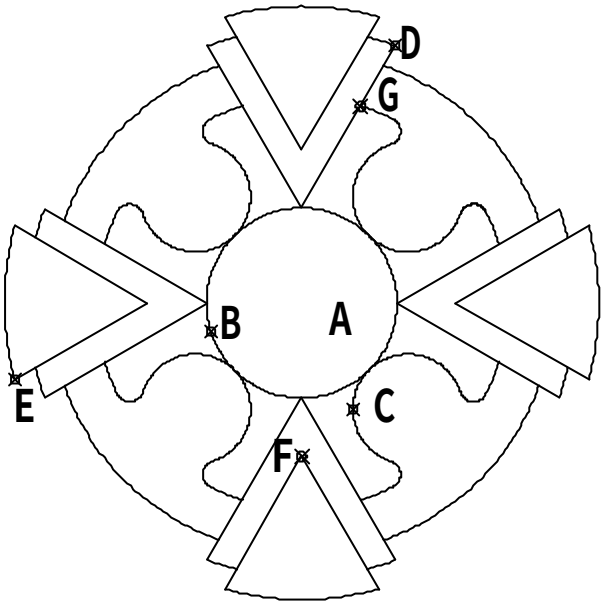



12、 Erase,
13、 COpy, C62, cen, C62, (**正交模式**), **向下**, 24.5, →C63, **向下**, 49, →C64
14、 Circle, cen, C64, 15, →C65
15、 Circle, cen, C62, 12, →C66
16、 Line, tan, C66 **右**, tan, C65 **右**, →L67
17、 Line, tan, C66 **左**, tan, C65 **左**, →L68
18、 ROtate, L56, cen, C60, -30, →L56 **轉動**
19、 Fillet, T, T, R, 3, L55, L54, →A69
20、 Fillet, T, T, R, 3, L50, L54, →A70
21、 Offset, 5, L52, **向右點**, →L71
22、 Offset, 5, L52, **向左點**, →L72
23、 Arc, int, C59 & L71, E, int, C59 & L72, R, 6, →A73

| | |
|---|---|
|  | 24、 Erase,<br>25、 TRim,<br>26、 Fillet, T, T, R, 2, C59, A73, →A74<br>27、 Fillet, T, T, R, 2, C59, A73, →A75<br>28、 Circle, cen, A73, D, 5, →C76<br>29、 Fillet, T, N, R, 10, L54, L56, →A77<br>30、 Circle, cen, C62, 20, →C78<br>31、 Circle, cen, C64, 20, →C79<br>32、 Line, tan, C78 **右**, tan, C79 **右**, →L80<br>33、 Line, tan, C78 **左**, tan, C79 **左**, →L81 |
|  | 34、 Fillet, T, T, R, 3, L53, L81, →A82<br>35、 Fillet, T, T, R, 10, C60, C78, →A83<br>36、 TRim,<br>37、 ARray, **環形陣列**, **中心點**, cen, C58, **項目總數**, 3, **佈滿角度**, 360, **物件**, A73~A75;C76, →**全圓佈滿** 3 **個** |
|  | 38、 TRim,<br>39、 Erase,<br>40、 Circle, cen, C58, cen, C76, →**虛圓** |

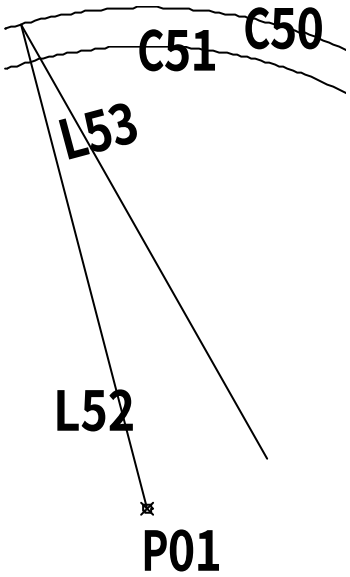
| | |
|---|---|
|  | 41、 Hatch, **點** E **區域**, **→斜線區域**<br>42、 **選虛圓**, CRTL+1, **→查幾何直徑值**=55.6230<br>43、 Hatch, **點** B **區域**, **→斜線區域**<br>44、 **點** B **區域斜線**, CRTL+1, **→查幾何面積值** =1669.4848<br>45、 DIst, int, C, cen, D, →XY **平面內角度** =37.7018<br>46、 **點** E **區域斜線**, CRTL+1, **→查幾何面積值** =1188.3760 |
|  | 47、 COpy, **所有物件**<br>48、 Erase,<br>49、 TRim,<br>50、 BOundary, **點內面**<br>51、 **點邊緣**, CRTL+1, **→查幾何長度值** =487.7341 |

## 2.3.10 AutoCAD2D2008-310

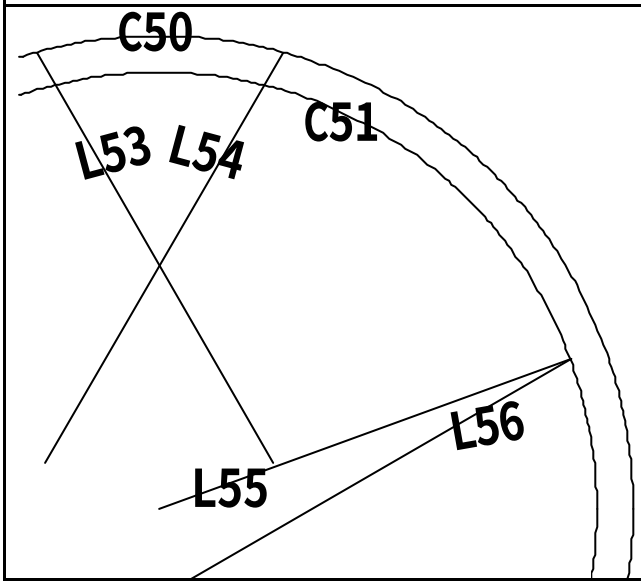



1、 A **區域周長為何？** 521.5896
2、 **圓** B **半徑為何？** 20.8378
3、 **弧** C **夾角為何？** 234.2769
4、 **交點** D **至交點** E **距離為何？** 110.9004
5、 **交點** F **至交點** G **高度為何？** 76.7681

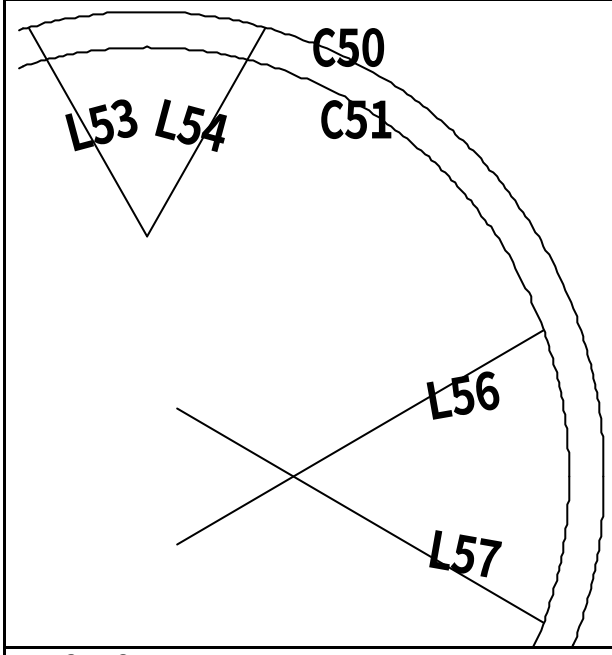
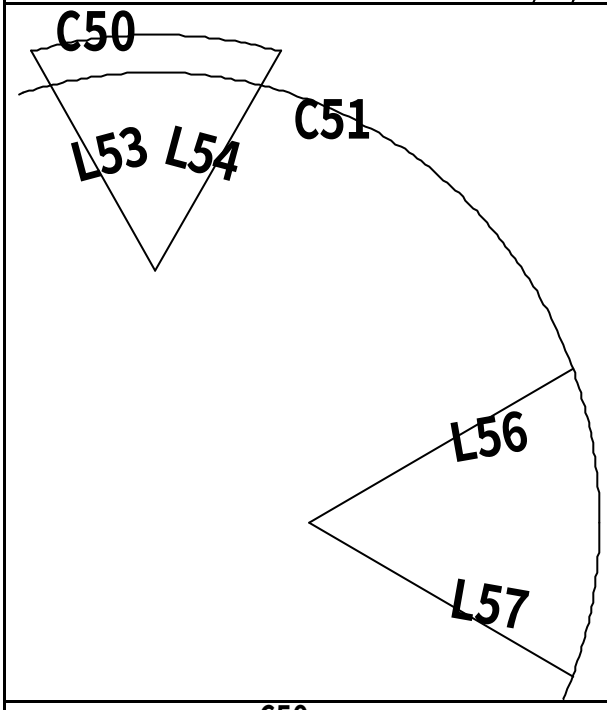
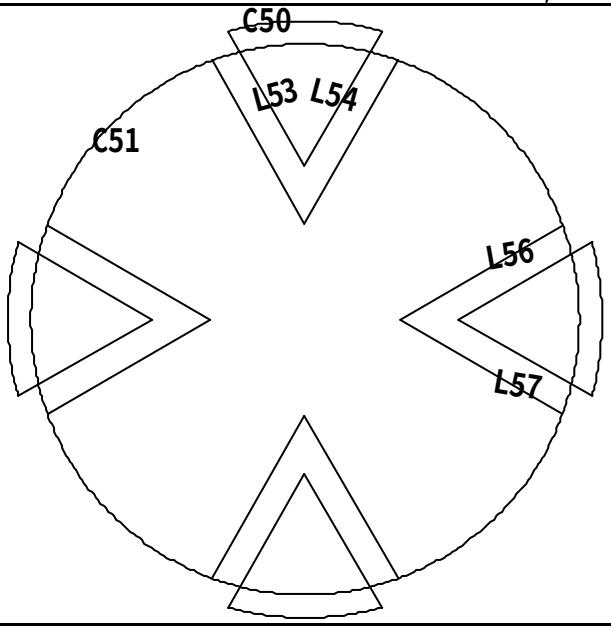
**解題步驟：**

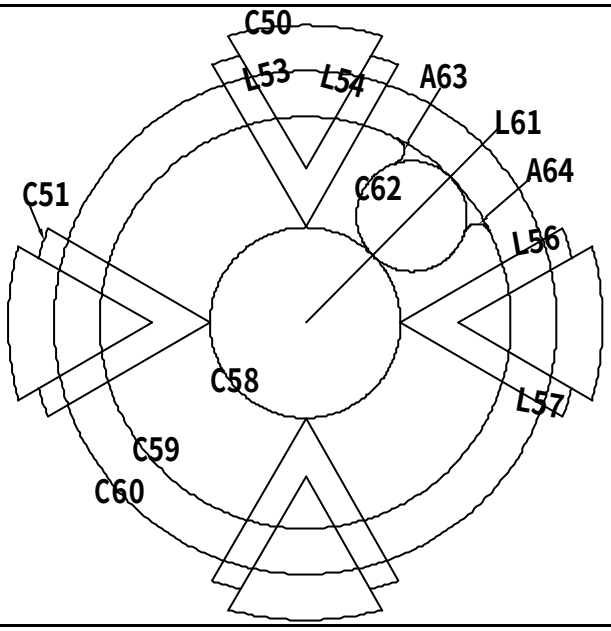
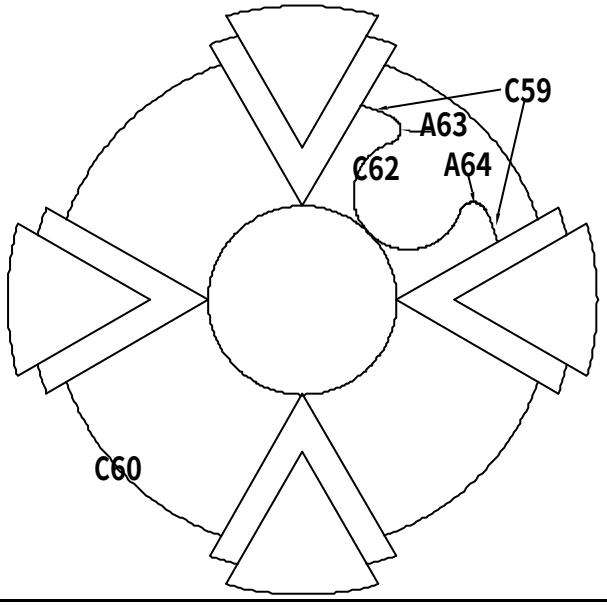
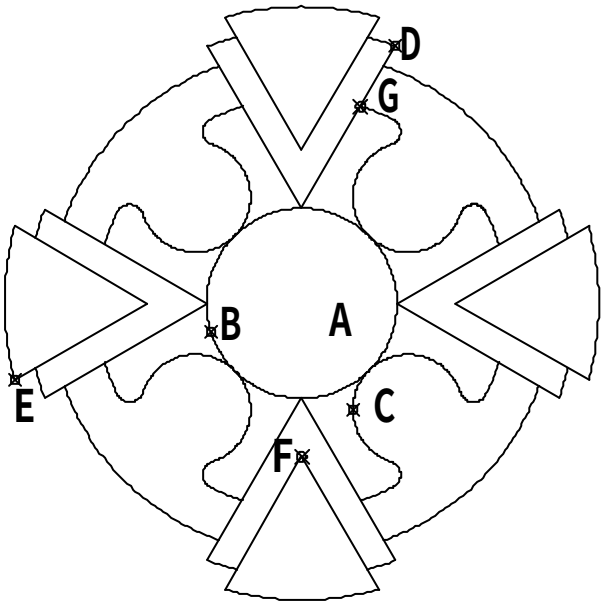



01、 Circle, **任一點** P01, D, 130, →C50
02、 Circle, @(**鎖定上次點**), D, 120, →C51
03、 Line, cen, C50, @65<105, →L52
04、 ROtate, L52, int, L52 & C50, C, 15, →L53

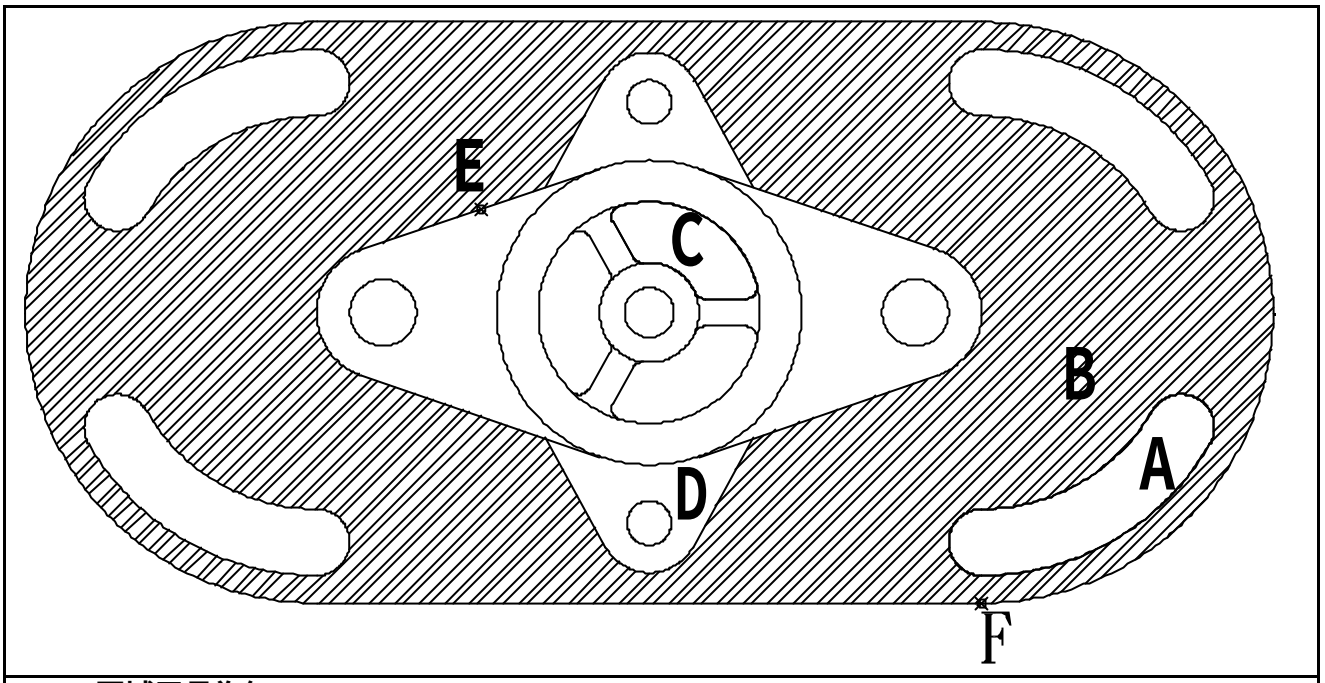



05、 Erase,
06、 MIrror, L53, qua, C50 **上**, qua, C51 **上**, →L54
07、 Line, cen, C50, @60<20, →L55
08、 ROtate, L55, int, L55 & C51, C, 10, →L56

| | |
|---|---|
|  | 09、 Erase,<br>10、 TRim,<br>11、 MIrror, L56, qua, C50 **右**, qua, C51 **右**, →L57 |
|  | 12、 TRim, |
|  | 13、 ARray, **環形陣列**, **中心點**, cen, C51, **項目總數**, 4, **佈滿角度**, 360, **物件**, C50;L53;L54;L56;57, →**全圓佈滿** 4 **組物件** |

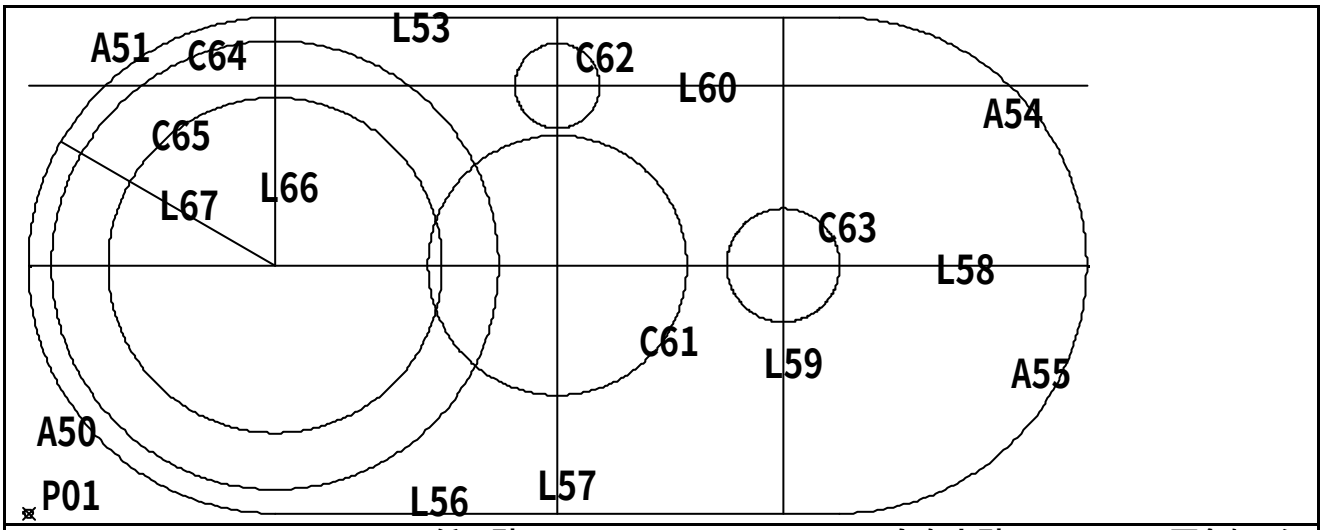
| | |
|---|---|
|  | 14、 Circle, cen, C50, int, L56 & L57, →C58<br>15、 Circle, @(**鎖定上次點**), D, 90, →C59<br>16、 Circle, @(**鎖定上次點**), D, 110, →C60<br>17、 Line, cen, C58, @60<45, →L61<br>18、 Circle, m2p, int, C59 & L61, int, C60 & L61, int, C59 & L61, →C62<br>19、 Fillet, T, N, R, 3, C62 **上**, C59, →A63<br>20、 Fillet, T, N, R, 3, C62 **右**, C59, →A64 |
|  | 21、 TRim,<br>22、 Erase, |
|  | 23、 ARray, **環形陣列**, **中心點**, cen, C60, **項目總數**, 4, **佈滿角度**, 360, **物件**, C59;C62;A63;A64, →**全圓佈滿** 4 **組物件**<br>24、 BOundary, **點** A **區域**<br>25、 **點** A **區域邊緣**, CRTL+1, →**查幾何長度值**=521.5896<br>26、 **點圓** B, CRTL+1, →**查幾何半徑值**=20.8378<br>27、 **點弧** C, CRTL+1, →**查幾何總角度值**=234.2769<br>28、 DIst, int, D, int, E, →**距離**=110.9004<br>29、 DIst, int, F, int, G, →Y **差值**=76.7681 |

## 2.3.11 AutoCAD2D2008-311

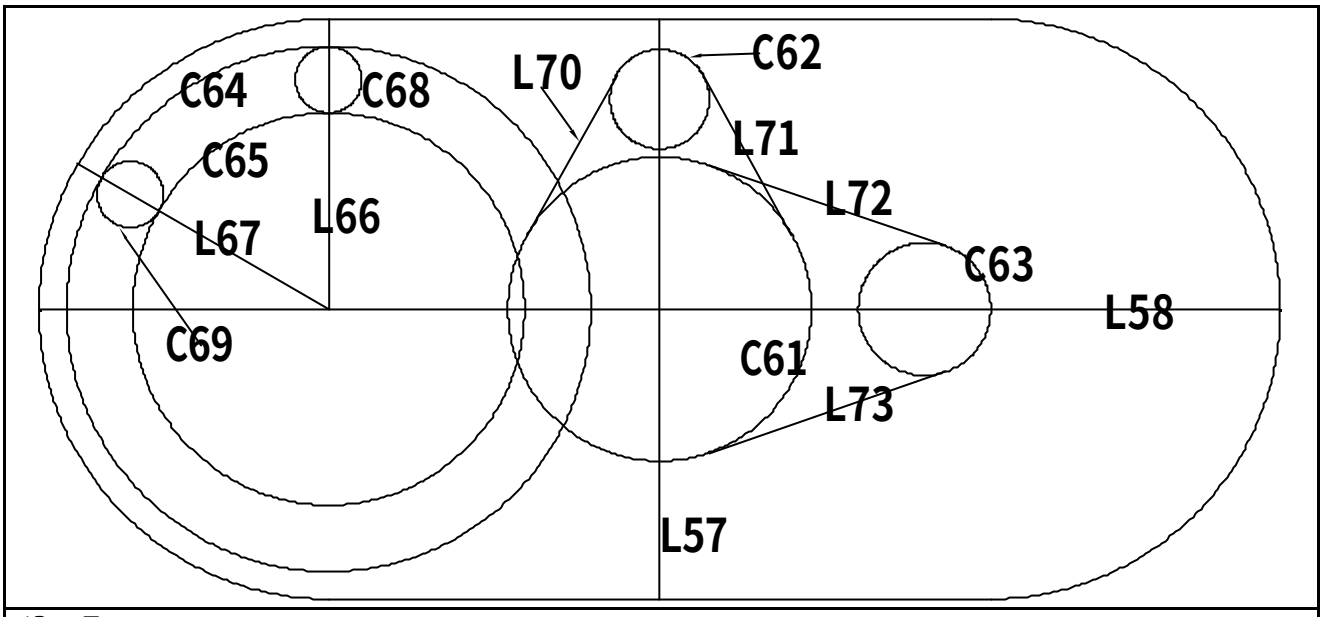



1、 A **區域周長為何？** 124.6165
2、 **斜線** B **區域面積為何？** 13340.9226
3、 C **區域周長為何？** 69.7173
4、 D **區域扣除內孔面積為何？** 417.6424
5、 **中點** E **至交點** F **之垂直距離為何？** 71.1919

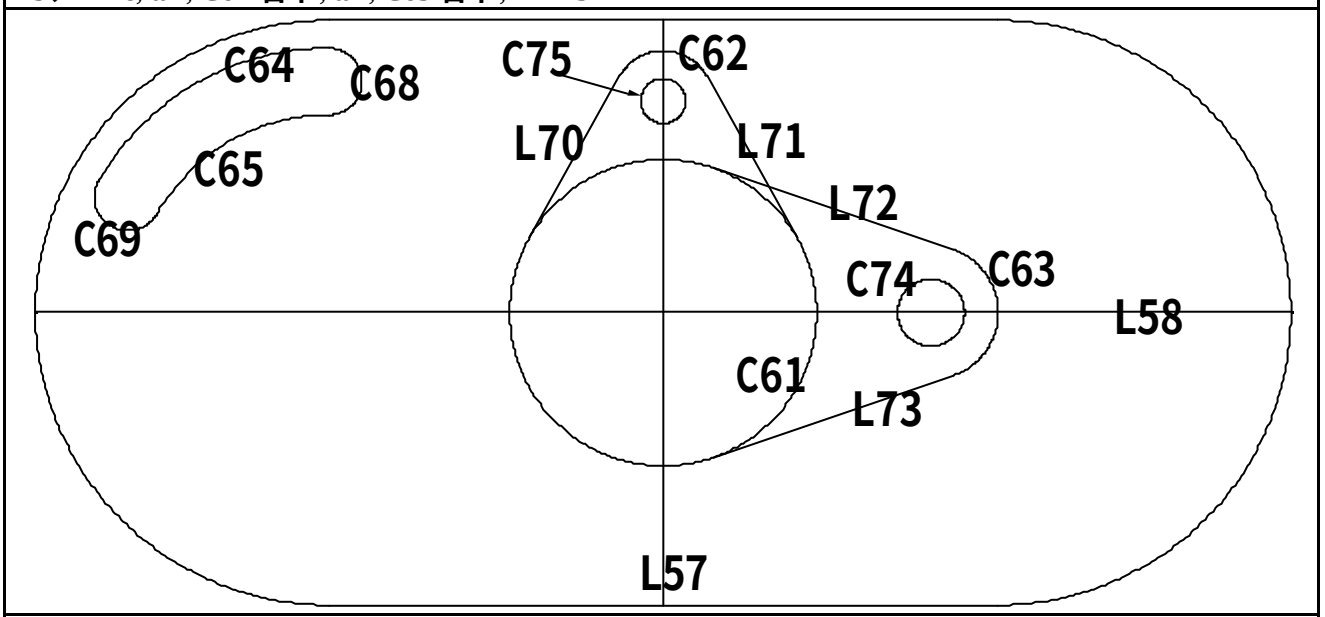
**解題步驟：**



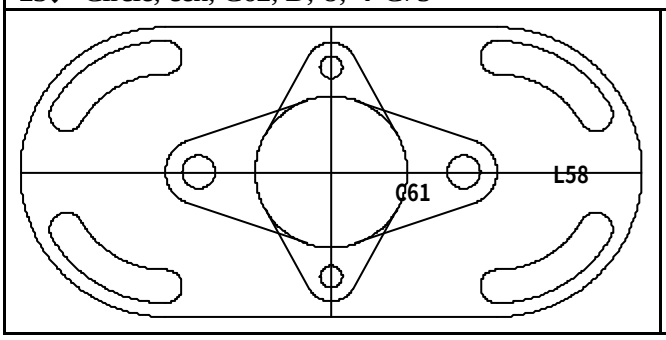

01、 RECtangle, F, 52.5(=105/2), **任一點** P01, D, 225(=120+105), 105, **向右上點**, →A50~L56 **圓角矩形**
02、 eXplode, A50~L56 **圓角矩形**, →A50~L56
03、 Line, mid, L56, mid, L53, →L57
04、 Line, end, A50 **上**, end, A55 **上**, →L58
05、 Offset, 48, L57, **向右點**, →L59
06、 Offset, 38, L58, **向上點**, →L60
07、 Circle, int, L57 & L58, D, 55, →C61
08、 Circle, int, L57 & L60, 9, →C62
09、 Circle, int, L59 & L58, 12, →C63
10、 Circle, cen, A51, 47.5, →C64
11、 Circle, cen, A51, 35.5, →C65
12、 Line, end, L53 **左**, cen, C64, →L66, @52.5<150, →L67



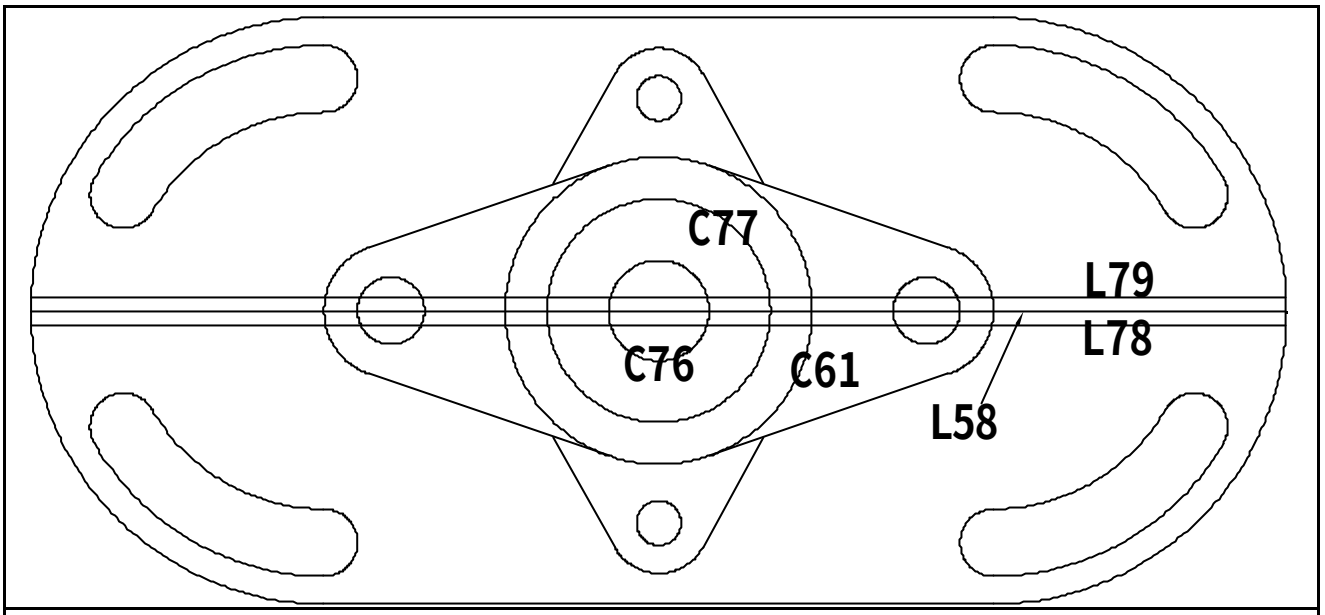

13、 Erase,
14、 Circle, 2P, int, C64 & L66, int, C65 & L66, →C68
15、 Circle, 2P, int, C64 & L67, int, C65 & L67, →C69
16、 Line, tan, C61 **左上**, tan, C62 **左上**, →L70
17、 Line, tan, C61 **右上**, tan, C62 **右上**, →L71
18、 Line, tan, C61 **右上**, tan, C63 **右上**, →L72
19、 Line, tan, C61 **右下**, tan, C63 **右下**, →L73



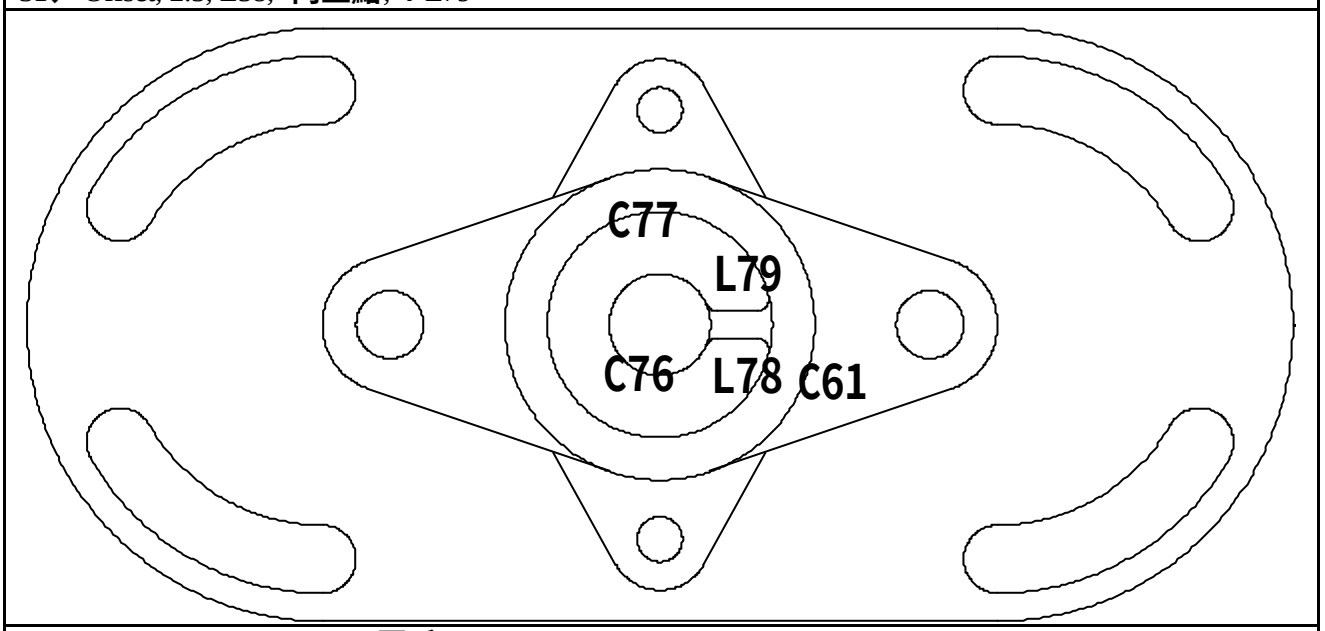

20、 TRim,
21、 Erase,
22、 Circle, cen, C63, D, 12, →C74
23、 Circle, cen, C62, D, 8, →C75



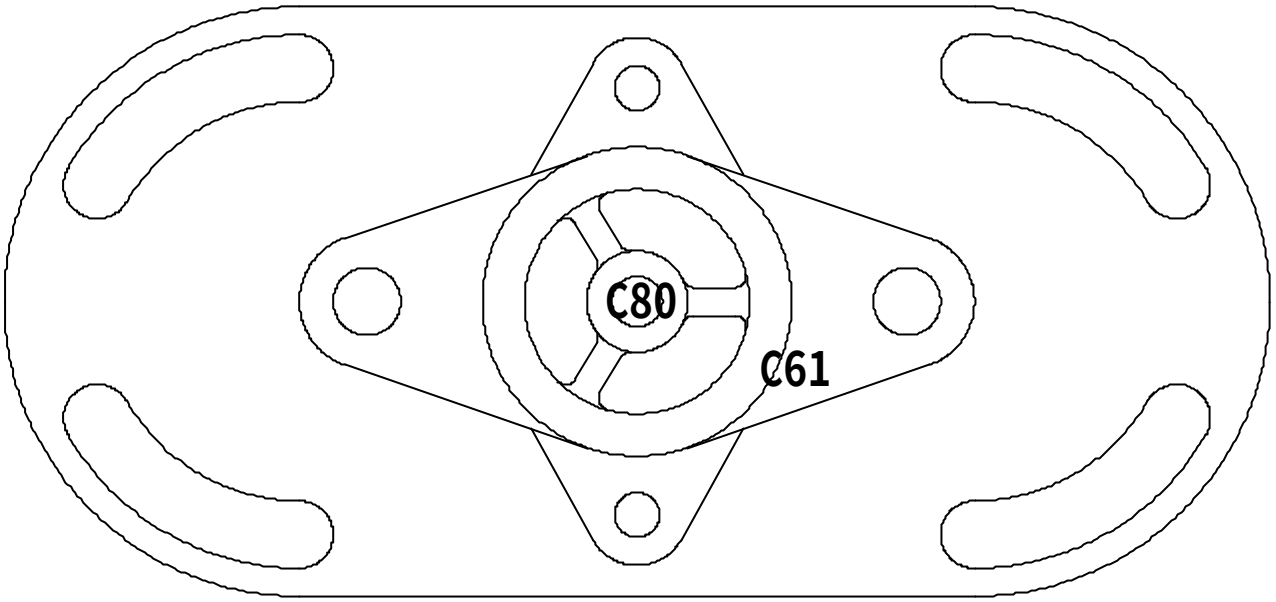

24、 MIrror, **上下**, →**上下對稱物件**
25、 MIrror, **左右**, →**左右對稱物件**




26、 TRim,
27、 Erase,
28、 Circle, cen, C63, D, 18, →C76
29、 Circle, cen, C63, D, 40, →C77
30、 Offset, 2.5, L58, **向下點**, →L78
31、 Offset, 2.5, L58, **向上點**, →L79

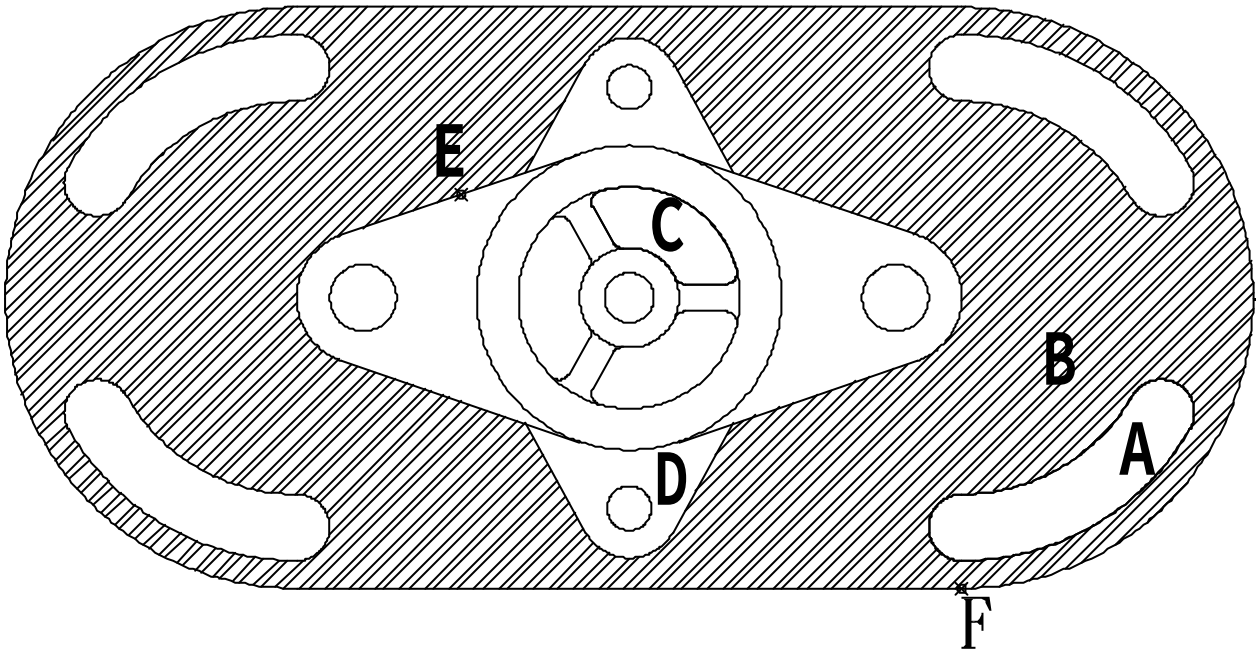



32、 Fillet, R, 2, C77, L78, →**圓弧**
33、 Fillet, R, 2, C77, L79, →**圓弧**
34、 Fillet, R, 2, C76, L78, →**圓弧**
35、 Fillet, R, 2, C76, L79, →**圓弧**




36、 ARray, **環形陣列**, **中心點**, cen, C61, **項目總數**, 3, **佈滿角度**, 360, **物件**, 4 **個圓弧及** 2 **個線**, →**全圓佈滿** 3 **組物件**

37、 Circle, cen , C61, D, 9, →C80




38、 Hatch, **點** B **區域**, B(**移除邊界**), **點內部圓**, →**斜線區域**

39、 BOundary, **點** A **區域**,

40、 **點** A **區域邊緣**, CRTL+1, →**查幾何長度值**=124.6165

41、 **點** B **區域斜線**, CRTL+1, →**查幾何面積值**=13340.9226

42、 BOundary, **點** C **區域**,

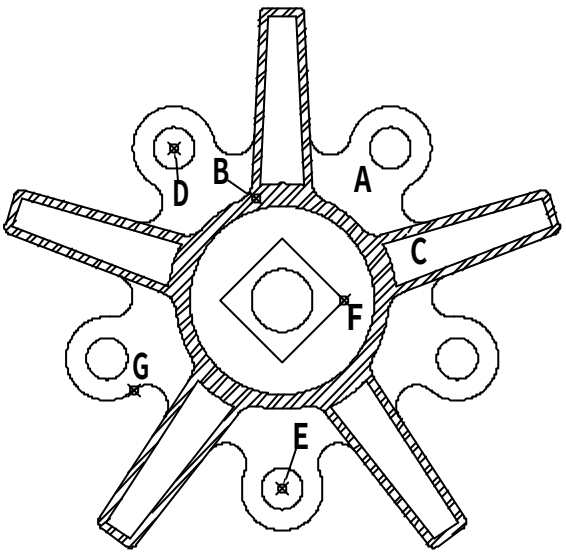
43、 **點** C **區域邊緣**, CRTL+1, →**查幾何長度值**=69.7173

44、 Hatch, **點** D **區域**, →**斜線區域**

45、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值**=417.6424

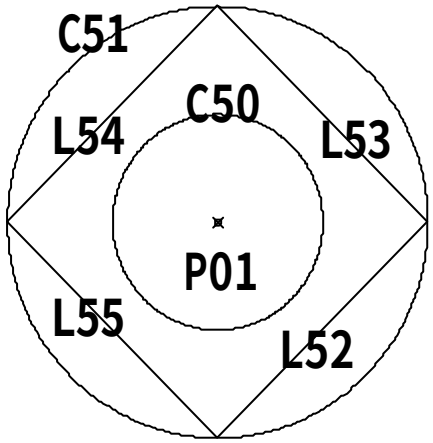
46、 DIst, mid, E, int, F, →Y **差值**=71.1919

## 2.3.12 AutoCAD2D2008-312

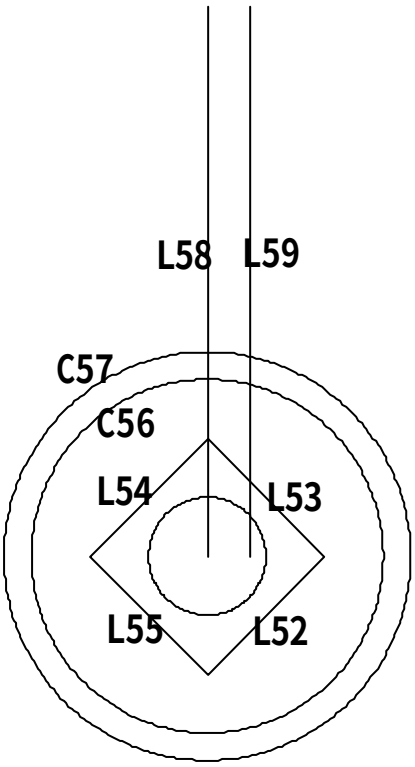



1、 A **區域扣除內孔面積為何？** 507.1112
2、 **斜線** B **區面積為何？** 1620.9285
3、 C **區域周長為何？** 99.0567
4、 **中心點** D **至** E **之距離為何？** 85.5951
5、 **交點** F **至** G **之水平距離為何？** 51.2170

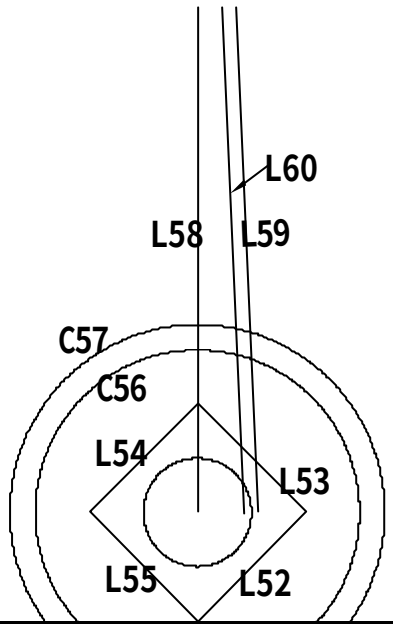
**解題步驟：**




01、 Circle, **任一點** P01, D, 15, →C50
02、 Circle, @(**鎖定上次點**), D, 30, →C51
03、 POLygon, 4, cen, C51, I, qua, C51 **右**, →L52~L55 **之正** 4 **邊形**

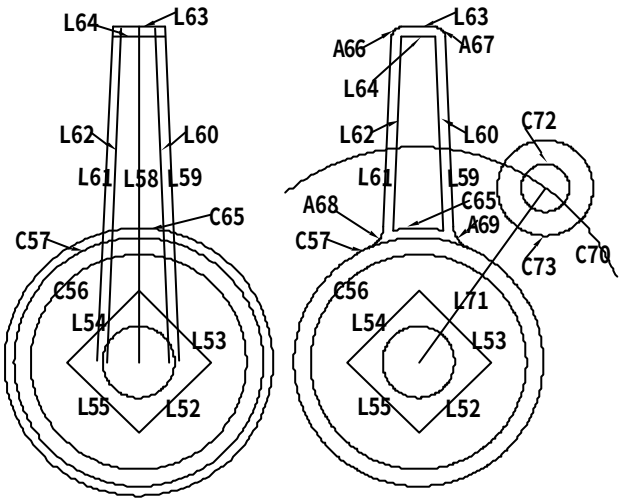



04、 Erase,
05、 Circle, cen, C50, D, 45, →C56
06、 Circle, @(**鎖定上次點**), 26, →C57
07、 Line, cen, C57, (**正交模式**), **向上** 70, →L58
08、 Offset, 5.5(=11/2), L58, **向右點**, →L59




09、 ROtate, L59, end, L59 **上**, 2.5(=5/2), →L59 **轉動**

10、 Offset, 2, L59, **向左點**, →L60




11、 MIrror, L59;L60, end, L58 **上**, cen, C56, →L61;L62

12、 Line, end, L59 **上**, end, L61 **上**, →L63

13、 Offset, 2, L63, **向下點**, →L64

14、 Circle, cen, C56, 28, →C65

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

15、 TRim,

16、 Erase,

17、 Fillet, T, T, R, 2, L61, L63, →A66

18、 Fillet, T, T, R, 2, L69, L63, →A67

19、 Fillet, T, N, R, 5, L61, C57, →A68

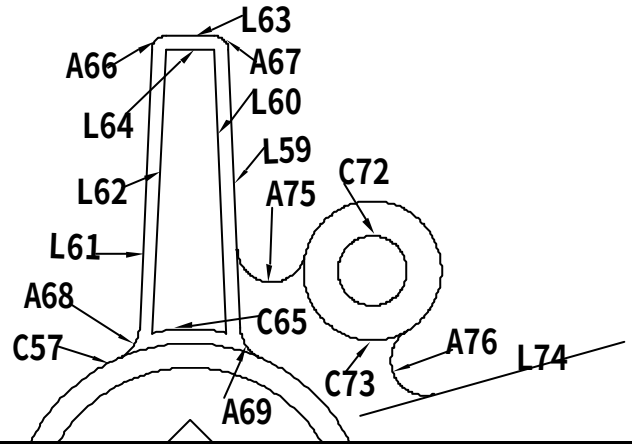
20、 Fillet, T, N, R, 5, L59, C57, →A69

21、 Circle, cen, C56, D, 90, →C70

22、 Line, cen, C70, @45<54(=90-36), →L71

23、 Circle, int, C70 & L71, D, 10, →C72

24、 Circle, @(**鎖定上次點**), D, 20, →C73

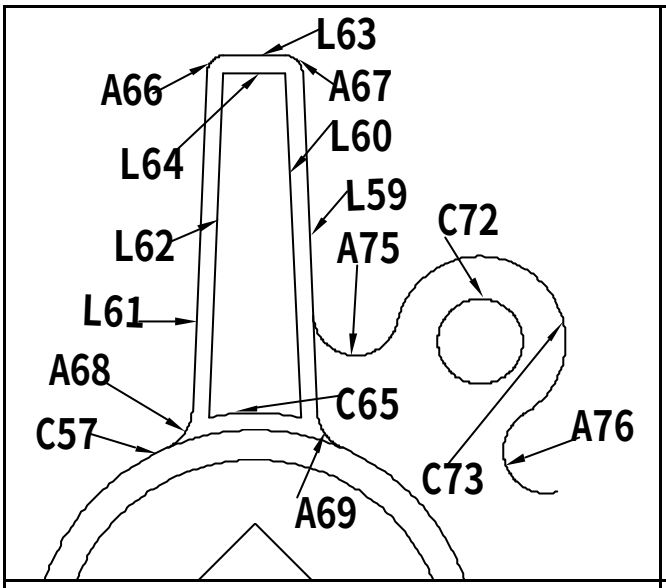
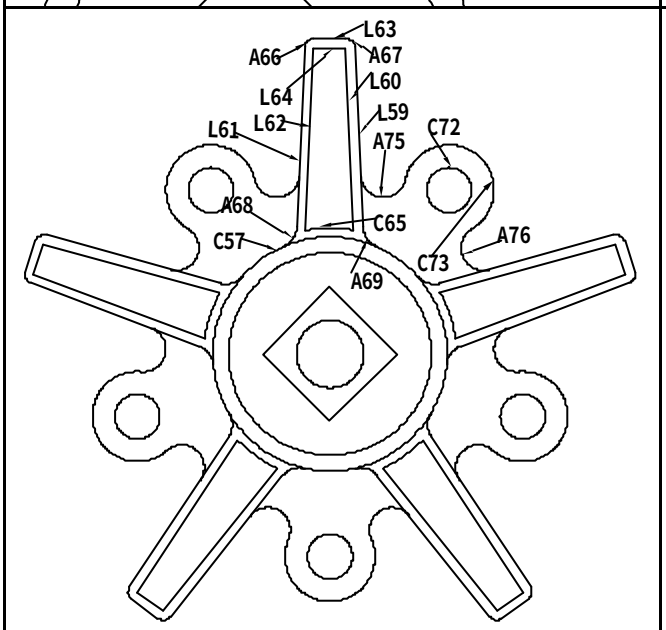
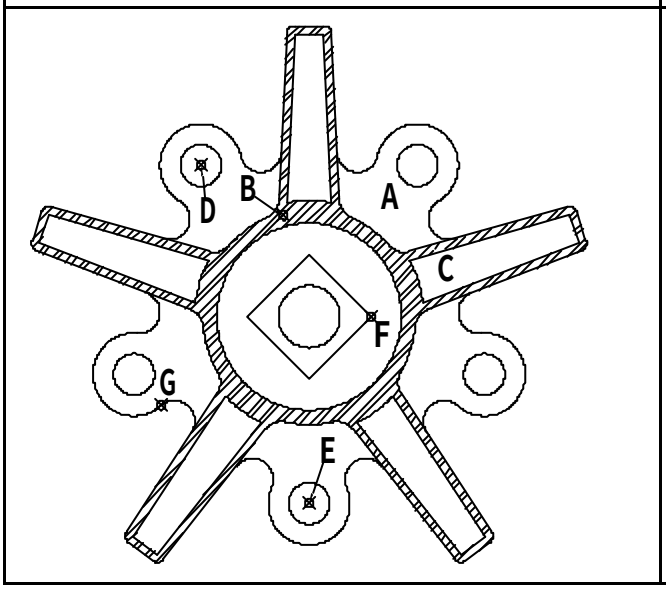



25、 Erase,

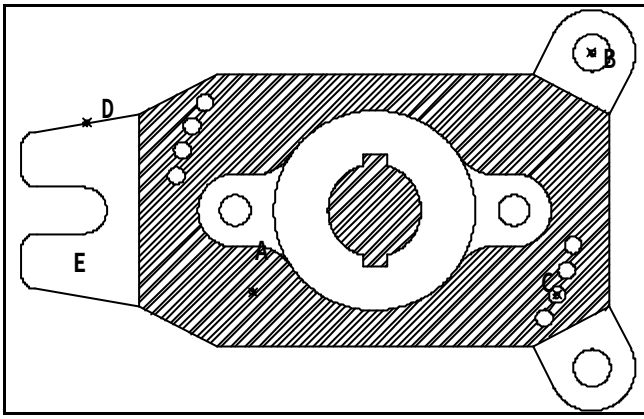
26、 ROtate, L61, cen, C57, C, -72, →L74

27、 Fillet, T, N, R, 5, L59, C73, →A75

28、 Fillet, T, N, R, 5, L74, C73, →A76

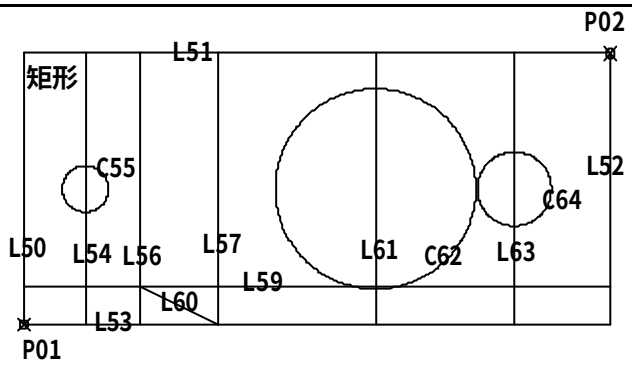
| | |
|---|---|
|  | 29、 TRim,<br>30、 Erase, |
|  | 31、 ARray, **環形陣列**, **中心點**, cen, C57, **項目總數**, 5, **佈滿角度**, 360, **物件** ,L59;L60;A67;L63;L64;A66;L61;L62;C65;A68;A69;A75;C72;C73;A76, →**全圓佈滿** 5**組物件** |
|  | 32、 Hatch, **點** A **區域**, →**斜線區域**<br>33、 **點** A **區域斜線**, CRTL+1, →**查幾何面積值** =507.1112<br>34、 Hatch, **點** B **區域**, B, **點內面圓及四方形**, →**斜線區域**<br>35、 **點** B **區域斜線**, CRTL+1, →**查幾何面積值** =1620.9285<br>36、 BOundary, **點** C **區域**,<br>37、 **點** C **區域邊緣**, CRTL+1, →**查幾何長度值** =99.0567<br>38、 DIst, cen, D, cen, E, →**距離**=85.5951<br>39、 DIst, int, F, int, G, →X **差值**=51.2170 |

## 2.3.13 AutoCAD2D2008-313

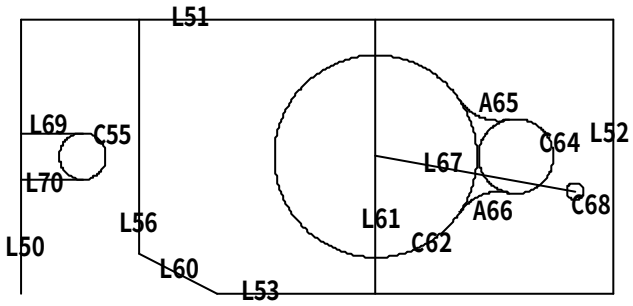



1、 **弧** A **中心點至中心點** B **距離為何？** 134.3503
2、 **中心點** C **至中點** D **角度為何？** 159.7613
3、 E **區域面積為何？** 1733.2481
4、 **斜線區域扣除** 8 **個內孔面積為何？** 8887.1545
5、 **圖形最外圍所圍成的周長為何？** 638.9340

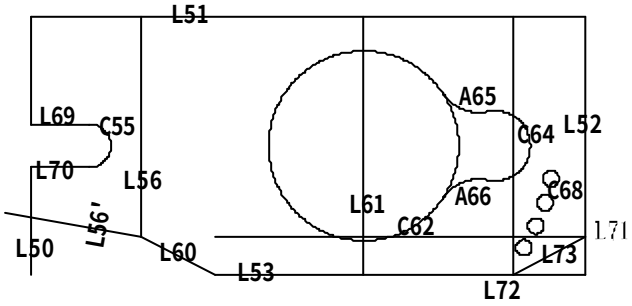
**解題步驟：**



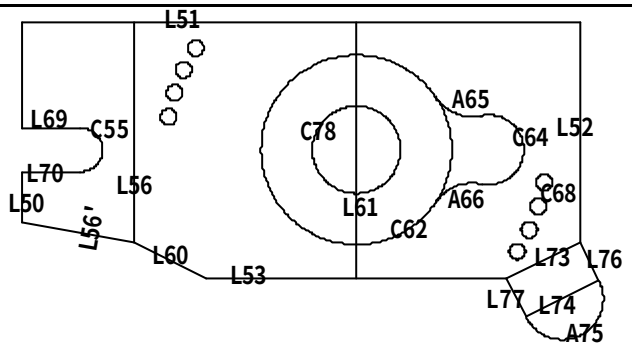
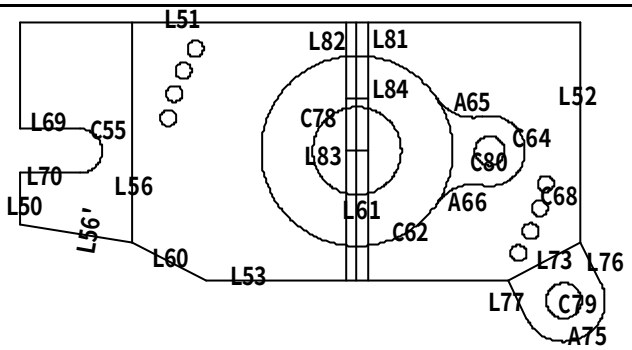
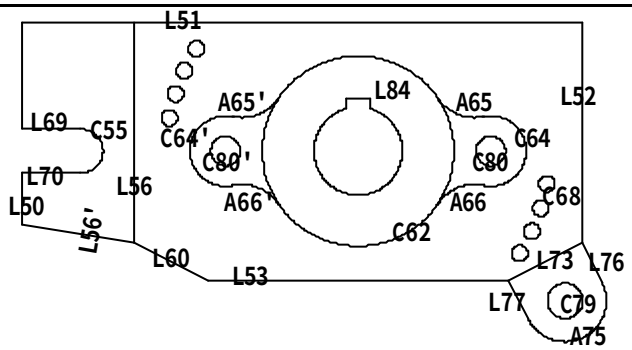
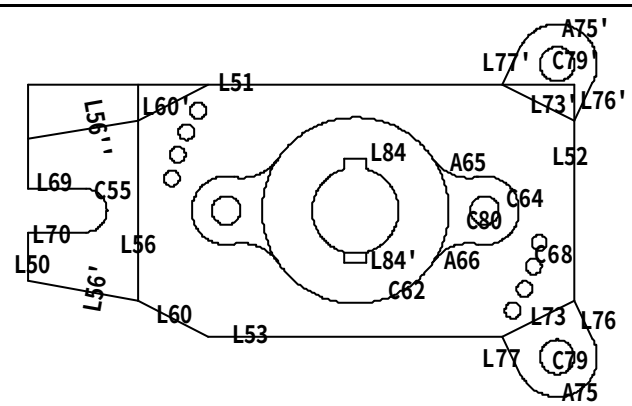
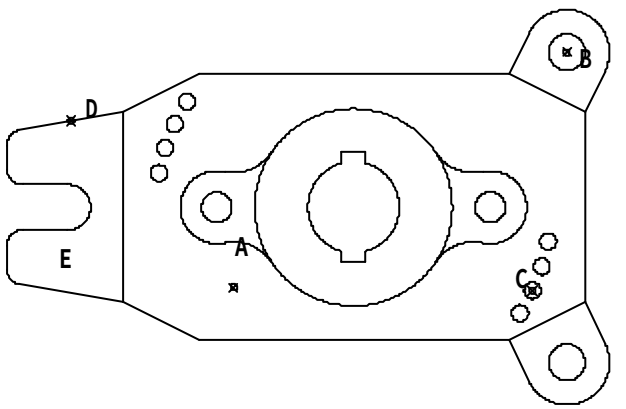

01、 RECtangle, **任一點** P01, D, 190, 88, **向右上點**, **任一點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 20, L50, **向右點**, →L54
04、 Circle, mid, L54, D, 15, →C55
05、 Offset, 38, L50, **向右點**, →L56
06、 Offset, 25, L56, **向右點**, →L57
07、 Offset, 12.5, L53, **向上點**, →L59
08、 Line, int, L56 & L59, int, L53 & L57, →L60
09、 Offset, 76, L56, **向右點**, →L61
10、 Circle, mid, L61, D, 65, →C62
11、 Offset, 45, L61, **向右點**, →L63
12、 Circle, mid, L63, 12, →C64

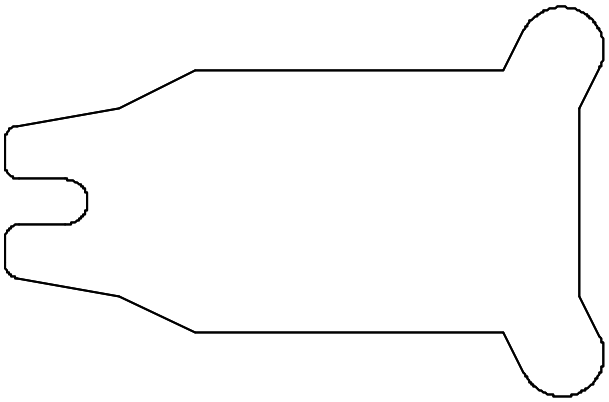



13、 Erase,
14、 TRim,
15、 Fillet, T, T, R, 15, C64 **上**, C62 **上**, →A65
16、 Fillet, T, T, R, 15, C64 **下**, C62 **下**, →A66
17、 Line, cen, C62, @65<-10, →L67
18、 Circle, end, L67, D, 5.5, →C68
19、 Line, qua, C55 **上**, (**正交模式**), **向左**, 20, →L69
20、 Line, qua, C55 **下**, (**正交模式**), **向左**, 20, →L70

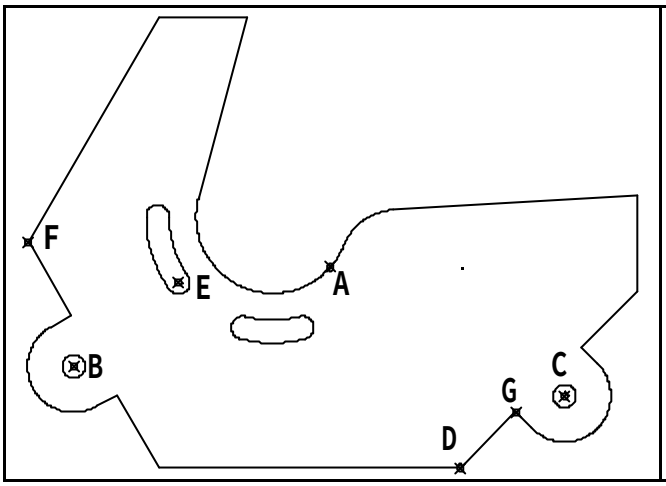



21、 TRim,
22、 Erase,
23、 ROtate, L56, int, L56 & L60, C, 80, →L56'
24、 Offset, 12.5, L53, **向上點**, →L71
25、 Offset, 25, L52, **向右點**, →L72
26、 Line, int, L52 & L71, int, L53 & L72, →L73
27、 ARray, **環形陣列**, **中心點**, cen, C62, **項目總數**, 4, **佈滿角度**, -22.5, **物件**, C68, →4 **個小圓**

| | |
|---|---|
|  | 28、 TRim,<br>29、 Erase,<br>30、 ROtate, C68**等** 4**小圓**, cen, C62, C, 180, →**對向** 4**個小圓**<br>31、 Offset, 15, L73, **向下點**, →L74<br>32、 Arc, C, mid, L74, end, L74 **下**, end, L74 **上**, →A75<br>33、 Line, int, L52 & L73, end, L74 **上**, →L76<br>34、 Line, int, L53 & L73, end, L74 **下**, →L77 |
|  | 35、 TRim,<br>36、 Erase,<br>37、 Circle, cen, c62, D, 30, →C78<br>38、 Circle, cen, A75, D, 12, →C79<br>39、 Circle, cen, C64, D, 10, →C80<br>40、 Offset, 3.75, L61, **向右點**, →L81<br>41、 Offset, 3.75, L61, **向左點**, →L82<br>42、 Line, mid, L81, mid, L82, →L83<br>43、 Offset, 18, L83, **向上點**, →L84 |
|  | 44、 TRim,<br>45、 Erase,<br>46、 MIrror, C64;C80;A65;A66, qua, C62 **上**, qua, C62 **下**, →C64' ;C80' ;A65' ;A66' |
|  | 47、 MIrror, L73;A75;L76;L77;L60;L56' ;L84 **等**, qua, C62 **左**, qua, C62 **右**, →L73' ;A75' ;L76' ;L77' ;L60' ;L56' ' ;L84'**等** |
|  | 48、 TRim,<br>49、 Erase,<br>50、 Fillet, R, 5, L56' ' , L50, →**圓弧**<br>51、 Fillet, R, 5, L69, L50, →**圓弧**<br>52、 Fillet, R, 5, L70, L50, →**圓弧**<br>53、 Fillet, R, 5, L56' , L50, →**圓弧**<br>54、 Hatch, **點** A/C **區域**, B(**移除邊界**), **點內兩側小圓**, →**斜線區域**<br>55、 DIst, cen, A, cen , B, →**距離**=134.3503<br>56、 DIst, cen, C, mid , D, →XY **平面內角度** =159.7613 |

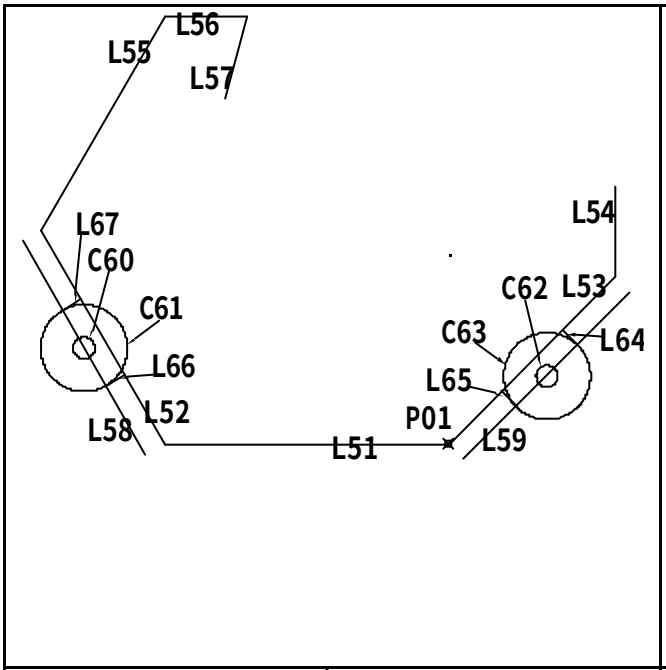
| | |
|---|---|
|  | 57、 Hatch, **點** E **區域**, **→斜線區域**<br>58、 **點** E **區域斜線**, CRTL+1, **→查幾何面積值** =1733.2481<br>59、 **點** A/C **區域斜線**, CRTL+1, **→查幾何面積值** =8887.1545<br>60、 COpy,<br>61、 BOundary, **點內區域**,<br>62、 **點邊緣**, CRTL+1, **→查幾何長度值** =638.9340 |

## 2.3.14 AutoCAD2D2008-314

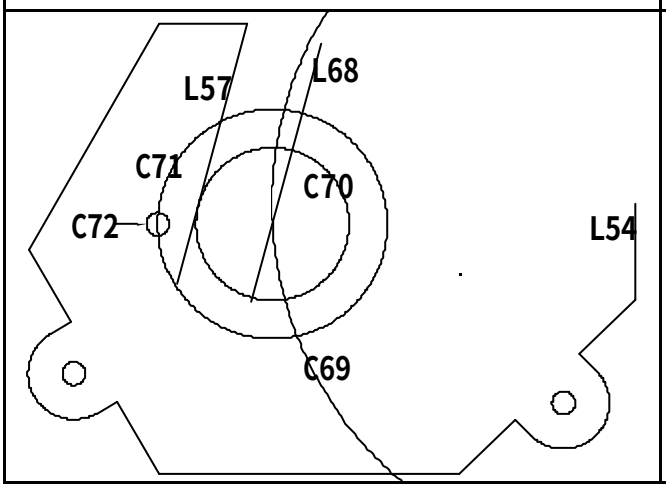



1、 **弧** A **夾角為何？** 172.8786
2、 **中心點** B **至中心點** C **距離為何？** 128.2515
3、 **交點** D **至中心點** E **角度為何？** 146.6423
4、 **交點** F **至交點** G **距離為何？** 134.6259
5、 **圖形扣除內孔面積為何？** 9967.5897

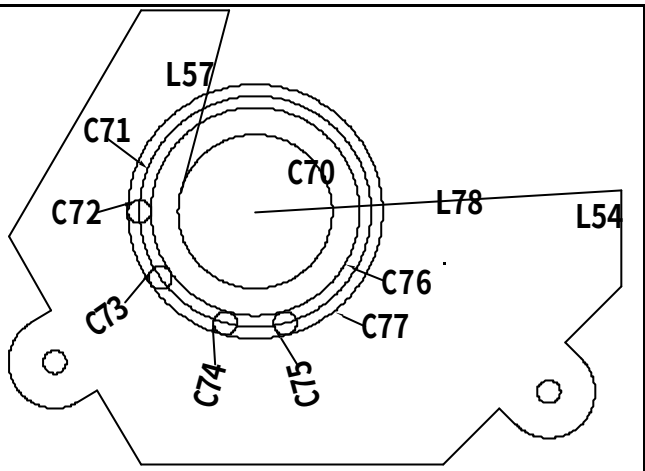
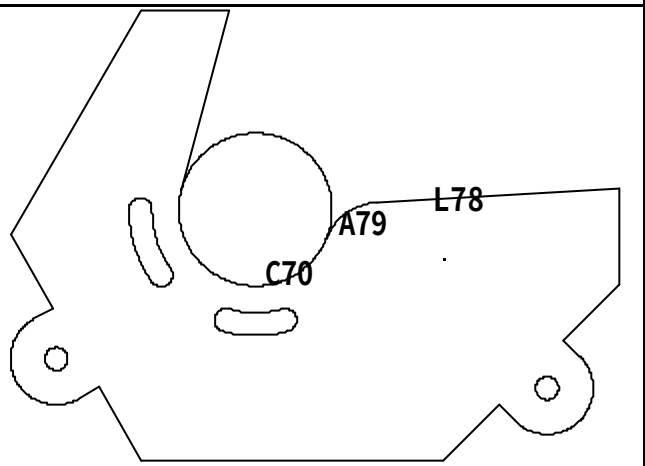
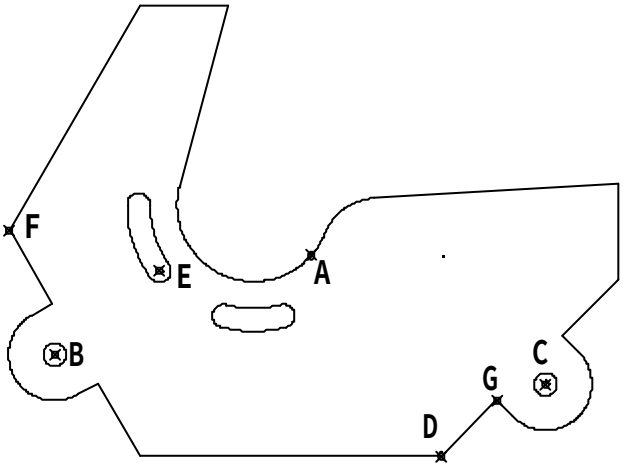
**解題步驟：**



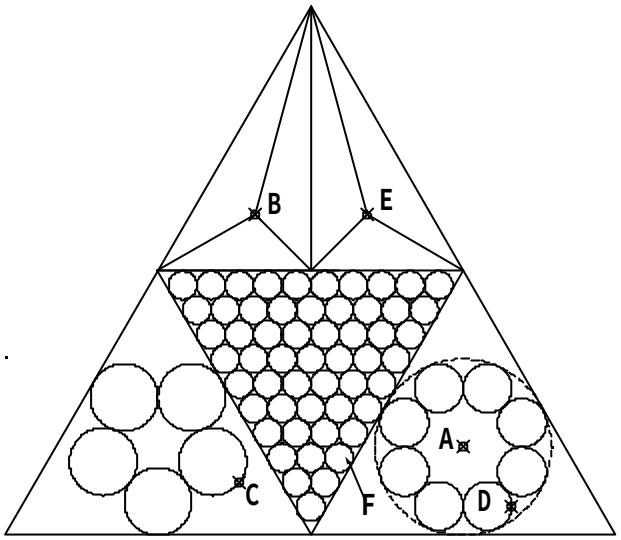

01、 Line, **任一點** P01, (**正交模式**), **向左** 78.6, →L51, @68<120, →L52
02、 Line, end, L51 **右**, @65<45, →L53, (**正交模式**), **向上** 25, →L54
03、 ROtate, L52, end, L52 **上**, C, 120, →L55
04、 Line, end, L55 **上**, (**正交模式**), **向右** 23, →L56
05、 ROtate, L56, end, L56 **右**, C, 75, →L57
06、 Offset, 6, L52, **向下點**, →L58, L53, **向下點**, →L59
07、 Circle, mid, L58, D, 6, →C60
08、 Circle, @(**鎖定上次點**), 12, →C61
09、 COpy, C60;C61, cen, C61, mid, L59, →C62;C63
10、 Line, int, L59 & C63 **右**, per, L53, →L64
11、 Line, int, L59 & C63 **左**, per, L53, →L65
12、 Line, int, L58 & C61 **下**, per, L52, →L66
13、 Line, int, L58 & C61 **上**, per, L52, →L67




14、 Erase,
15、 TRim,
16、 LENgthen, T, 70, L57 **下**, →L57 **向下延伸**
17、 Offset, 20, L57, **向右點**, →L68
18、 Circle, end, L54 **上**, 95(=75+20), →C69
19、 Circle, int, L68 & C69, 20, →C70
20、 Circle, @(**鎖定上次點**), 30, →C71
21、 Circle, qua, C71 **左**, D, 6, →C72

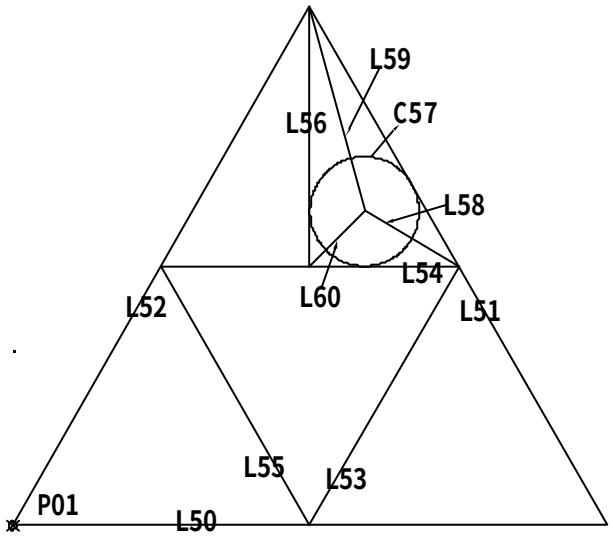
| | |
|---|---|
|  | 22、 TRim,<br>23、 Erase,<br>24、 ROtate, C72, cen, C70, C, 35, →C73<br>25、 ROtate, C73, cen, C70, C, 40, →C74<br>26、 ROtate, C74, cen, C70, C, 30, →C75<br>27、 Offset, 3(=6/2), C71, **向內點**, →C76, C71, **向外點**, →C77<br>28、 Line, end, L54 **上**, cen, C70, →L78 |
|  | 29、 TRim,<br>30、 Erase,<br>31、 Fillet, T, N, R, 15, L78, C70 **右**, →A79 |
|  | 32、 TRim,<br>33、 **點選弧** A, CTRL+1, **→查幾何總角度值** =172.8786<br>34、 DIst, cen, B, cen , C, **→距離**=128.2515<br>35、 DIst, int, D, cen , E, →XY **平面內角度** =146.6423<br>36、 DIst, int, F, int , G, **→距離**=134.6259<br>37、 Hatch, **點內部區域**, **→斜線區域**<br>38、 **點內部區域斜線**, CRTL+1, **→查幾何面積值** =9967.5897 |

## 2.3.15 AutoCAD2D2008-315

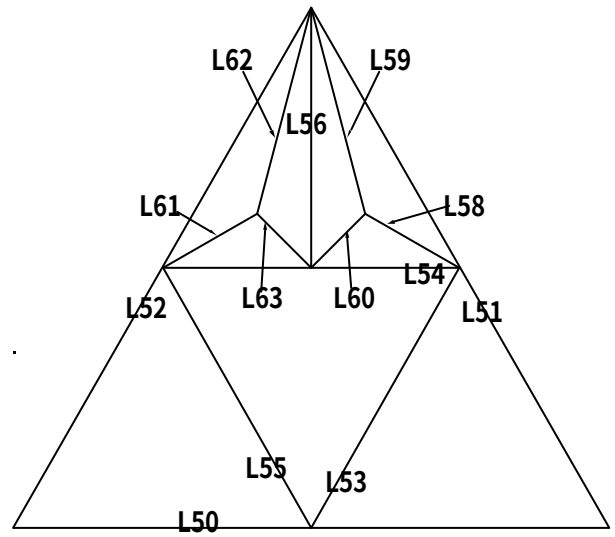



1、 **中心點** A **至交點** B **距離為何？** 91.9877
2、 **圓** C **半徑為何？** 9.9801
3、 **圓** D **半徑為何？** 7.1907
4、 **交點** B **至交點** E **距離為何？** 32.9423
5、 **圓** F **直徑為何？** 8.3861

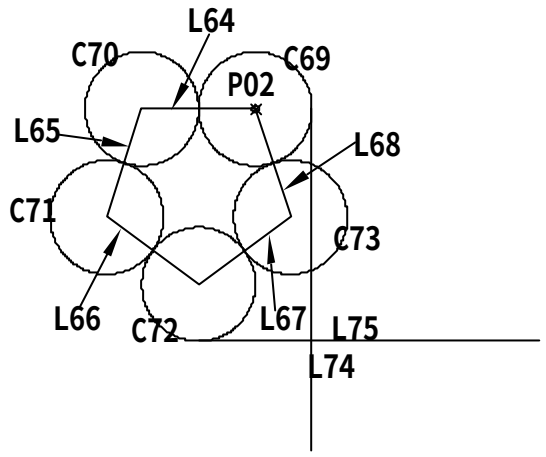
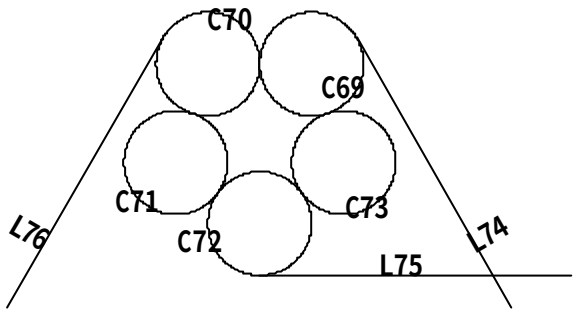
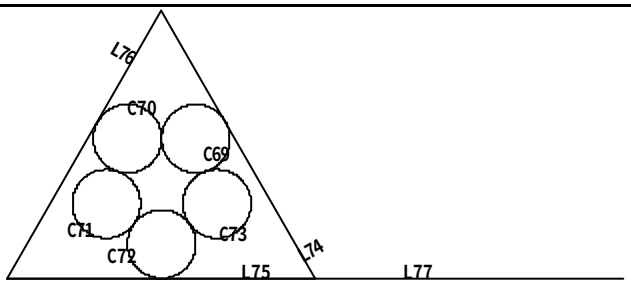
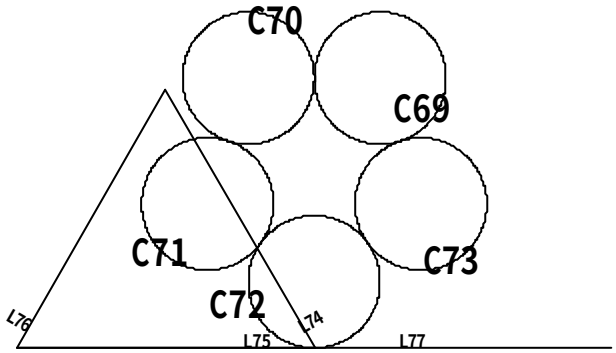
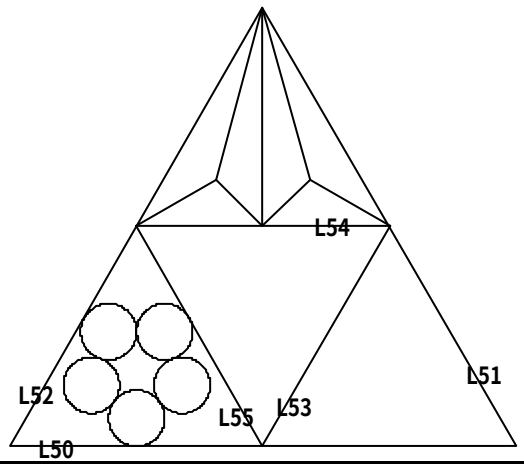
**解題步驟：**

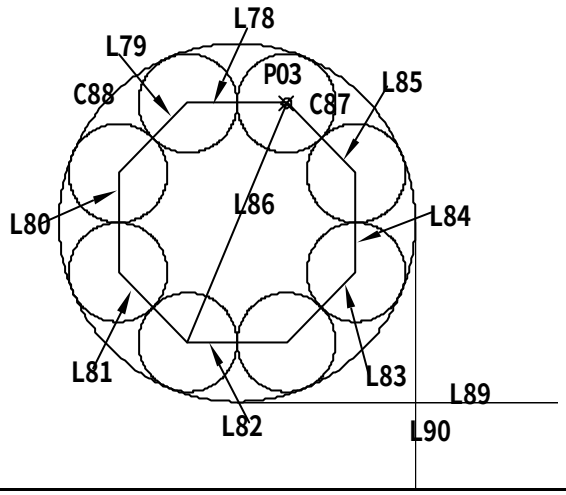
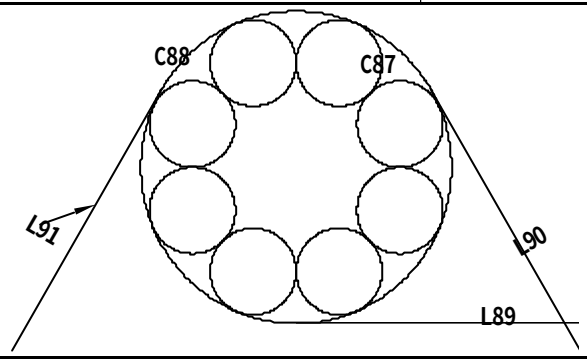
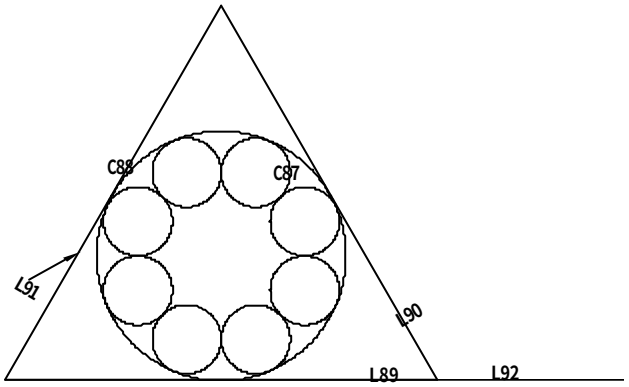
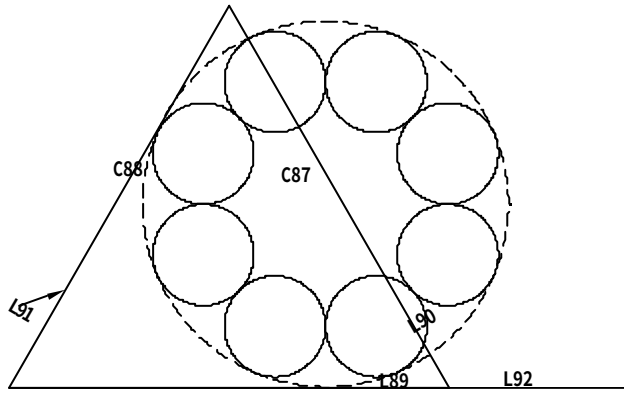


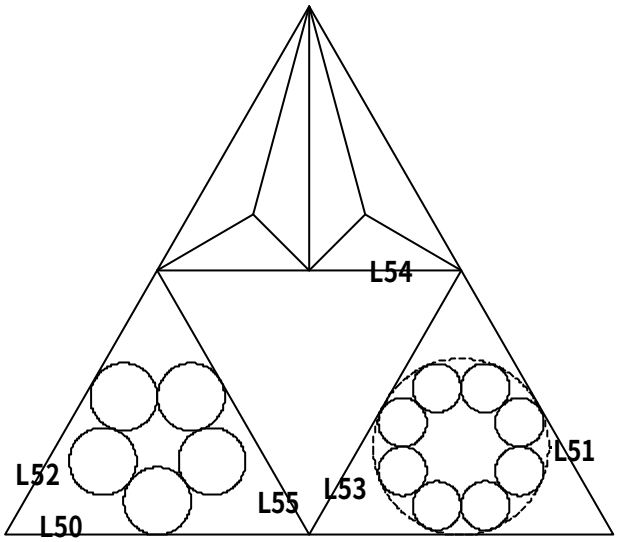
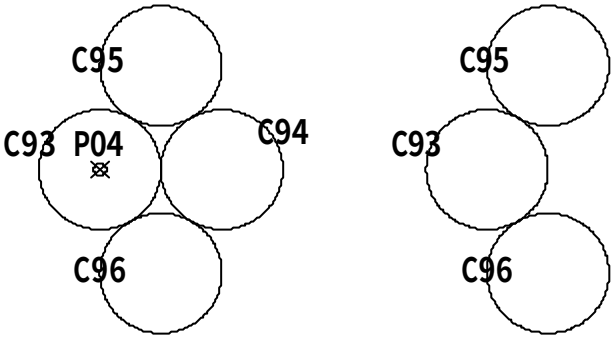
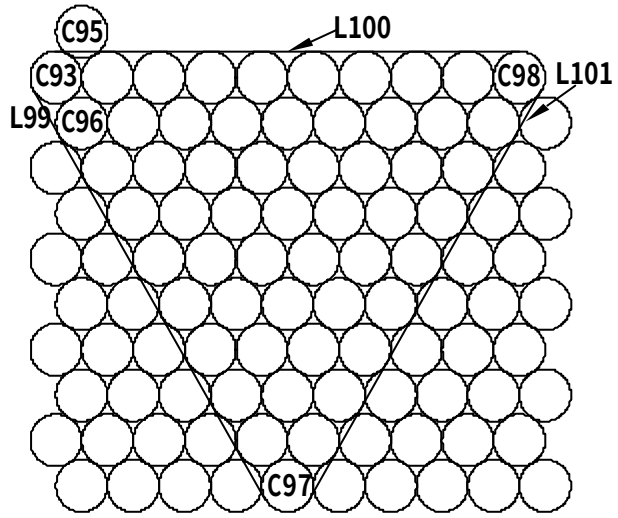
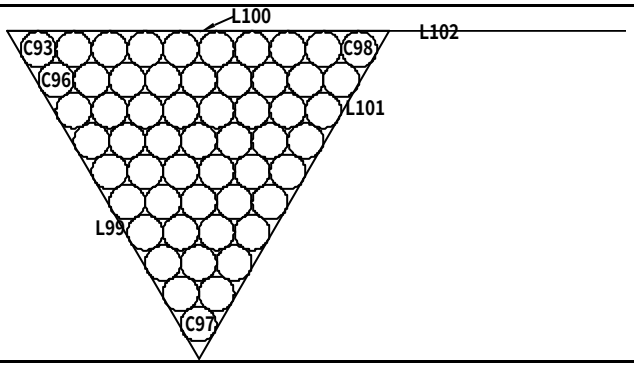

01、 POLygon, 3, E, **任一點** P01, (**正交模式**), **向右** 180, →L51~L53 **之正** 3 **邊形**
02、 Line, mid, L50, mid, L51, →L53, mid, L52, →L54, mid, L50, →L55
03、 Line, mid, L54, int, L51 & L52, →L56
04、 Circle, 3P, tan, L51, tan, L56, tan, L54, →C57
05、 Line, cen, C57, int, L54 & L51, →L58
06、 Line, cen, C57, int, L56 & L51, →L59
07、 Line, cen, C57, int, L54 & L56, →L60

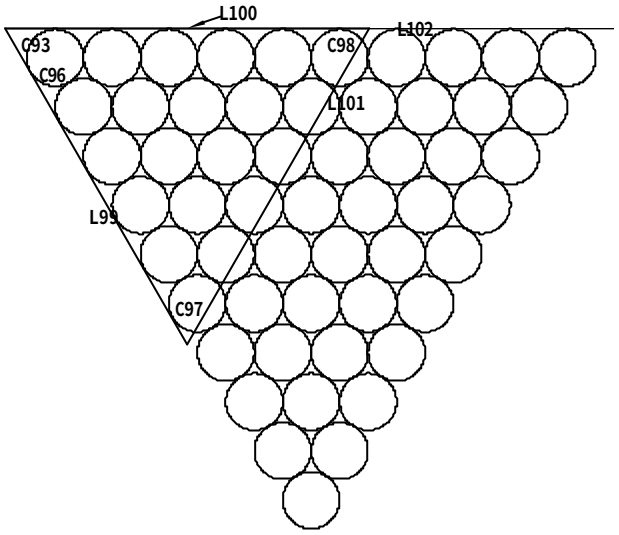
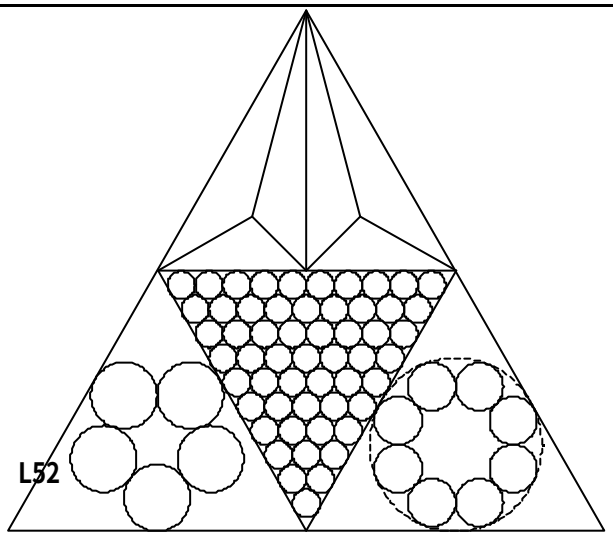
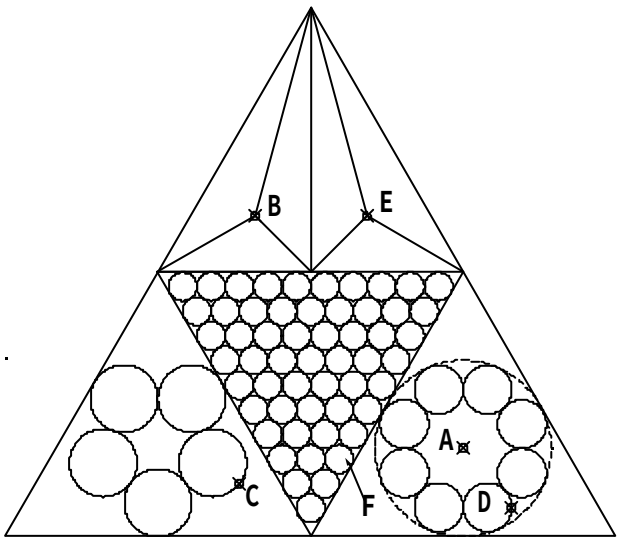



08、 Erase,
09、 MIrror, L58~L60, int, L51 & L52, mid, L54, →L61~L63

| | |
|---|---|
|  | 10、 POLygon, 5, E, **任一點** P02, (**正交模式**), **向左** 10, →L64~L68 **之正** 5 **邊形**<br>11、 Circle, int, L64 & L68, mid, L64, →C69<br>12、 COpy, C69, cen, C69, int, L64 & L65, →C70, int, L65& L66, →C71, int, L66 & L67, →C72, int, L67 & L68, →C73<br>13、 Line, qua, C69, (**正交模式**), **向下** 30, →L74<br>14、 Line, qua, C72, (**正交模式**), **向右** 30, →L75 |
|  | 15、 Erase,<br>16、 ROtate, L74, cen, C69, 30, →L74 **轉動**<br>17、 MIrror, L74, int, C69 & C70, int, C72 & L75, →L76 |
|  | 18、 Fillet, T, T, R, 0, L74, L75, →L74 **與** L75**相交**<br>19、 Fillet, T, T, R, 0, L76, L75, →L76 **與** L75**相交**<br>20、 Fillet, T, T, R, 0, L74, L76, →L74 **與** L76**相交**<br>21、 Line, int, L76 & L75, (**正交模式**), **向右** 90(=180/2), →L77 |
|  | 22、 SCale, C69~C73, int, L75 & L76, R, int, L75 & L76, int, L75 & L74, end, L77 **右**, →C69~C73 **放大** |
|  | 23、 Move, C69~C73, int, L75 & L76, int, L52 & L50, →C69~C73 **放置於正** 3 **角形內** |

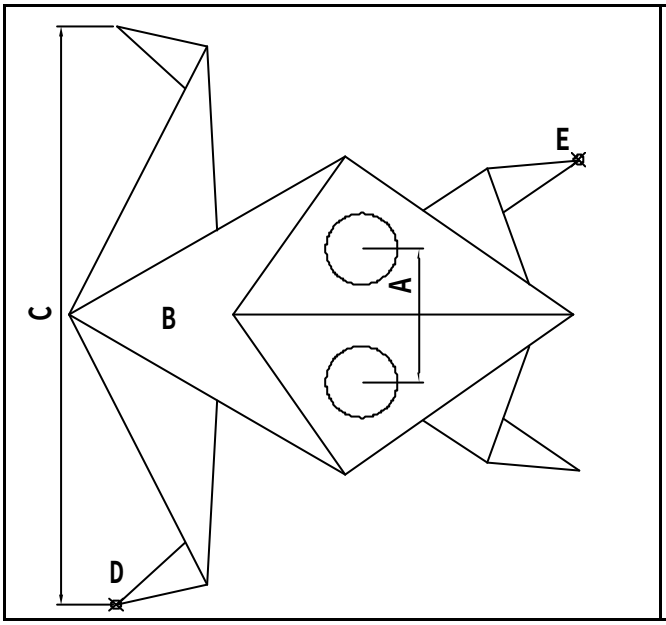
| | |
|---|---|
|  | 24、 POLygon, 8, E, **任一點** P03, (**正交模式**), **向左** 10, →L6478~L85 **之正** 8 **邊形**<br>25、 Line, int, L78 & L85, int, L81 & L82, →L86<br>26、 Circle, int, L78 & L85, mid, L78, →C87<br>27、 ARray, **環形陣列**, **中心點**, mid, L86, **項目總數**, 8, **佈滿角度**, 360, **物件**, C87, →**正** 8**邊形角點佈滿圓**<br>28、 Circle, mid, L86, tan, C87 **右上**, →C88<br>29、 Line, qua, C88 **下**, (**正交模式**), **向右** 30, →L89<br>30、 Line, qua, C88 **右**, (**正交模式**), **向下** 30, →L90 |
|  | 31、 Erase,<br>32、 ROtate, L90, cen, C88, 30, →L90 **轉動**<br>33、 MIrror, L90, qua, C88 **上**, qua, C88 **下**, →L91 |
|  | 34、 Fillet, T, T, R, 0, L89, L90, →L89 **與** L90**相交**<br>35、 Fillet, T, T, R, 0, L90, L91, →L90 **與** L91**相交**<br>36、 Fillet, T, T, R, 0, L89, L91, →L89 **與** L91**相交**<br>37、 Line, int, L89 & L91, (**正交模式**), **向右** 90(=180/2), →L92 |
|  | 38、 SCale, C87 **等** 8**個圓**, int, L89 & L91, R, int, L89 & L91, int, L89 & L90 end, L92 **右**, →C87 **等** 8 **個圓放大** |

| | |
|---|---|
|  | 39、 Move, C87 等 8 個圓, int, L89 & L91, int, L53 & L55, →C87 等 8 個圓放置於正 3 角形內 |
|  | 40、 Circle, 任一點 P04, D, 5, →C93<br>41、 COpy, C93, qua, C93 左, qua, C93 右, →C94<br>42、 Circle, T, C93 右上, C94 左上, 2.5(=5/2), →C95<br>43、 Circle, T, C93 右下, C94 左下, 2.5(=5/2), →C96<br>- - - - - - - - - - - - - - - - - - - - - - - - - - - - - -<br>44、 Erase, |
|  | 45、 ARray, 矩形陣列, 列, 5, 行, 10, 列偏移, qua, C95 下, qua, C96 下, 行偏移, qua, C93 左, qua, C93 右, 物件, C93;C96, →C97;C98 等 100 個物件<br>46、 Line, tan, C93 左下, tan, C97 左下, →L99<br>47、 Line, tan, C93 上, tan, C98 上, →L100<br>48、 Line, tan, C97 右下, tan, C98 右下, →L101 |
|  | 49、 Fillet, T, T, R, 0, L99, L100, →L99 與 L100 相交<br>50、 Fillet, T, T, R, 0, L99, L101, →L99 與 L101 相交<br>51、 Fillet, T, T, R, 0, L100, L101, →L100 與 L101 相交<br>52、 Line, int, L99 & L100, (正交模式), 向右 90(=180/2), →L102 |

| | |
|---|---|
|  | 53、 SCale, C93;C97;C98 **等圓**, int, L99 & L100, R, int, L99 & L100, int, L100 & L101 end, L102 **右**, →C93;C97;C98 **等圓放大** |
|  | 54、 Move, C93;C97;C98 **等圓**, int, L99 & L100, mid, L52, →C93;C97;C98 **等圓放置於正** 3 **角形內** |
|  | 55、 DIst, cen, A, int, B, →**距離**=91.9877<br>56、 **點選圓** C, CTRL+1, →**查幾何半徑值**=9.9801<br>57、 **點選圓** D, CTRL+1, →**查幾何半徑值**=7.1907<br>58、 DIst, int, B, int, E, →**距離**=32.9423<br>59、 **點選圓** F, CTRL+1, →**查幾何直徑值**=8.3861 |

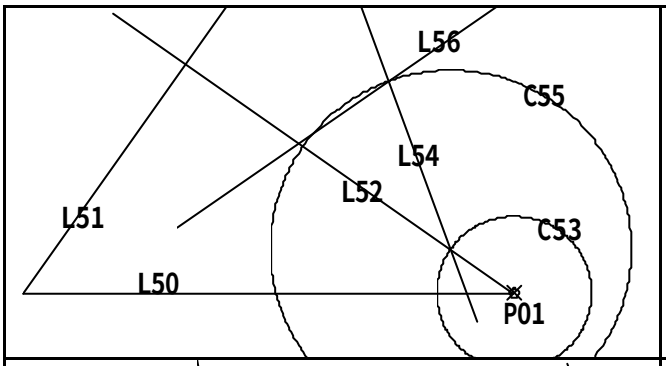
## 2.4 第四類：玩具&生活用品應用

### 2.4.1 AutoCAD2D2008-401

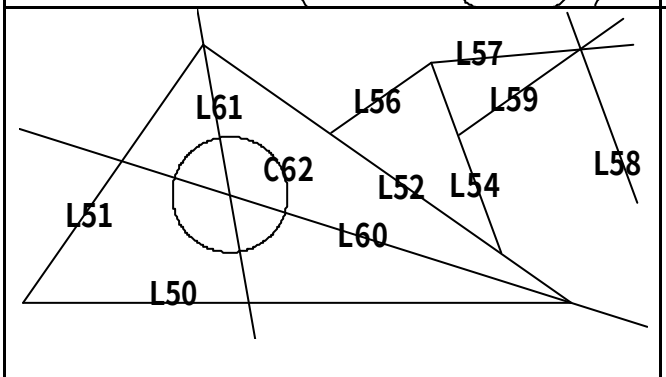



1、 **尺寸** A **長度為何？** 37.3092
2、 B **區域面積為何？** 2055.7608
3、 **尺寸** C **長度為何？** 162.2334
4、 **端點** D **至端點** E **之距離為何？** 179.7561
5、 **圖形所圍成最外圍面積為何？** 9896.4295

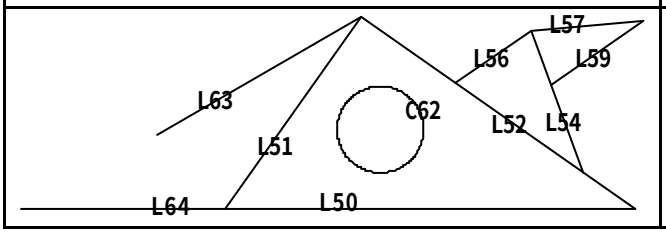
**解題步驟：**



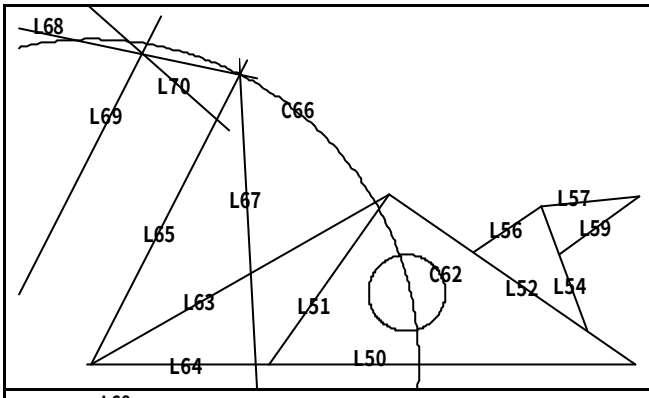
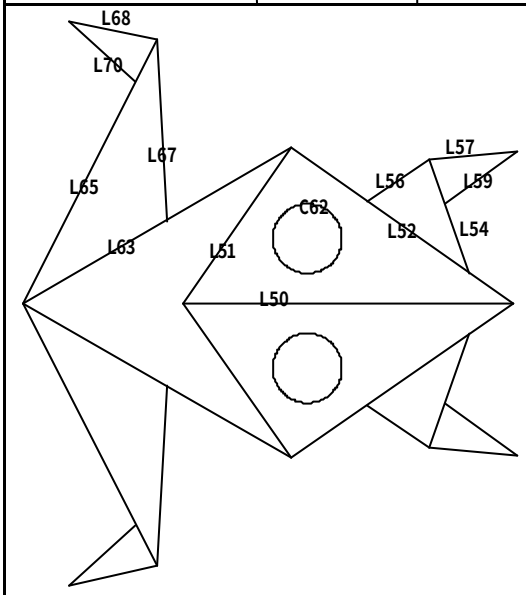
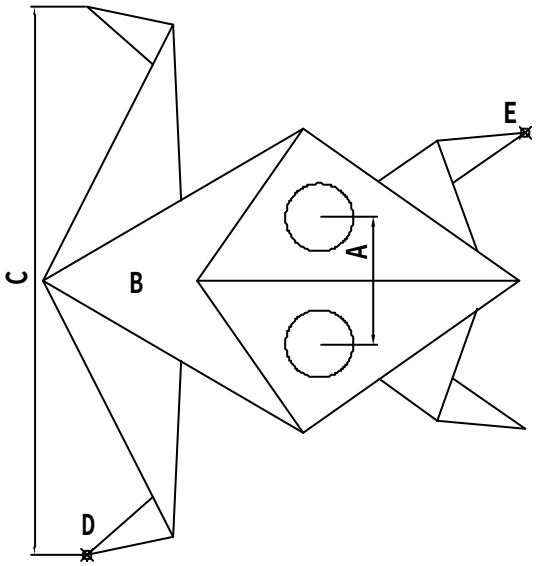
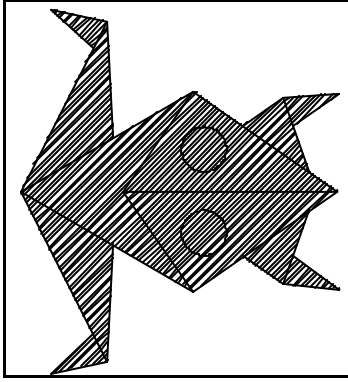

01、 Line, **任一點** P01, (**正交模式**), **向左**, 95, →L50, @95<55, →L51
02、 ROtate, L50, end, L50 **右**, C, -35, →L52
03、 Circle, int, L50 & L52, 15, →C53
04、 ROtate, L52, int, L52 & C53, C, -35, →L54
05、 Circle, int, L52 & L54, 35, →C55
06、 ROtate, L54, int, L54 & C55, C, -75,→L56



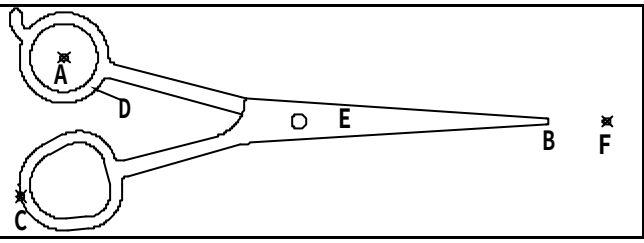

07、 TRim,
08、 Erase,
09、 ROtate, L54, int, L54 & L56, C, 75(=180-105), →L57
10、 Offset, 25, L54, **向右點**, →L58
11、 ROtate, L57, int, L57 & L58, C, 30, →L59
12、 XLine, B, int, L52 & L50, int, L52 & L51, int, L50 & L51, →L60
13、 XLine, B, int, L52 & L51, int, L50 & L51, int, L50 & L52, →L61
14、 Circle, int, L60 & L61, D, 20, →C62



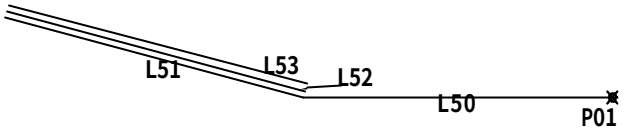
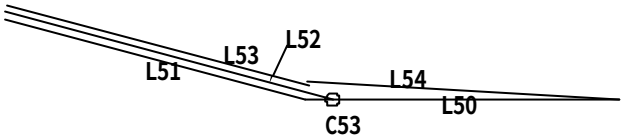
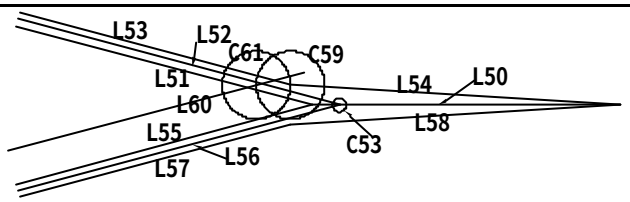
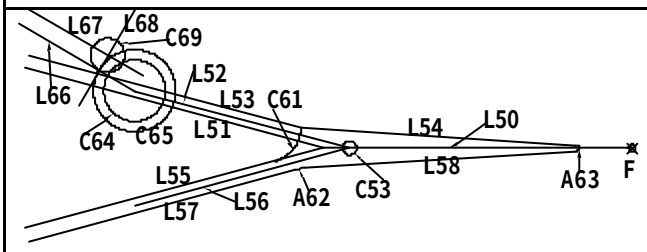
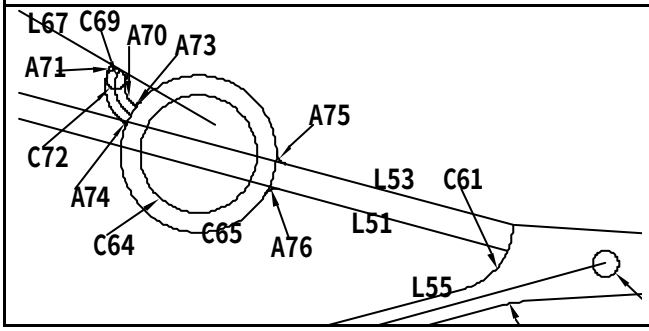

15、 TRim,
16、 Erase,
17、 ROtate, L51, int, L51 & L52, C, -25, →L63
18、 Line, int, L50 & L51, (**正交模式**), **向左**, 50(**大約**), →L64

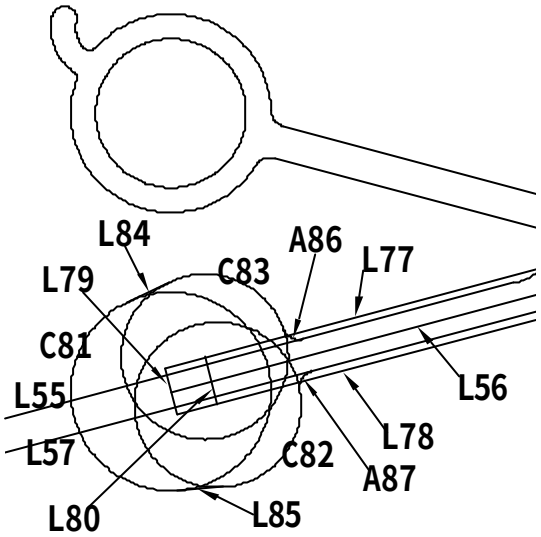
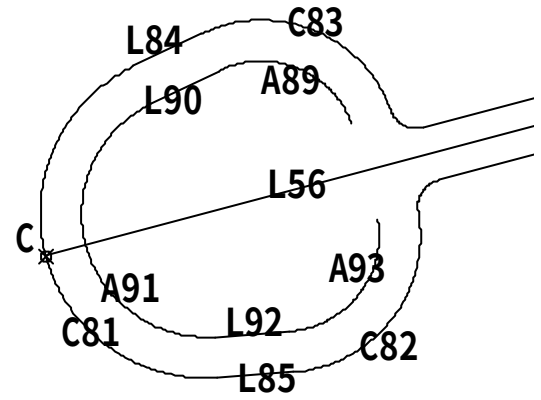
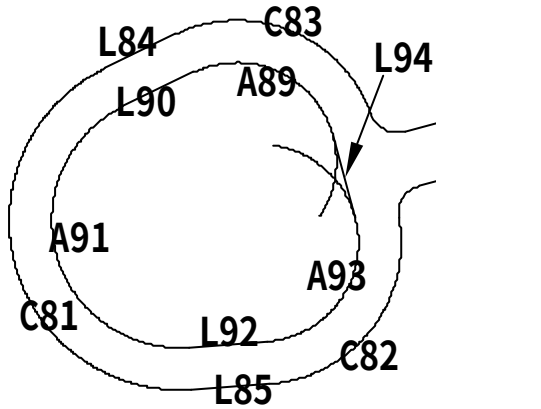
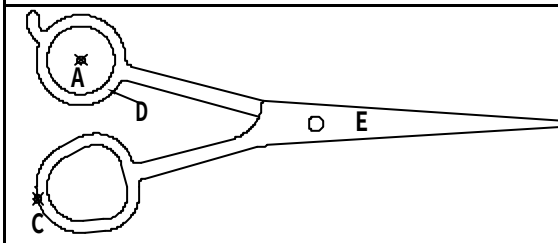
| | |
|---|---|
|  | 19、 EXtend, L64, L63, →L63 **交於** L64<br>20、 ROtate, L63, int, L63 & L64, C, 33, →L65<br>21、 Circle, int, L63 & L64, 85, →C66<br>22、 ROtate, L65, int, L65 & C66, C, 30, →L67<br>23、 ROtate, L65, int, L65 & C66, C, -75(=180-73-30), →L68<br>24、 Offset, 25, L65, **向左點**, →L69<br>25、 ROtate, L68, int, L68 & L69, C, -30, →L70 |
|  | 26、 TRim,<br>27、 Erase,<br>28、 MIrror, **所有物件**, end, L50 **左**, end, L50 **右**, **→下部圖形** |
|  | 29、 DIst, cen, **上圓**, cen, **下圓**, **→距離**=37.3092<br>30、 Hatch, **點** B **區域**, **→斜線區域**<br>31、 **點** B **區域斜線**, CRTL+1 **→查幾何面積值** =2055.7608<br>32、 DIst, int, **左上端**, int, D **左下端**, **→距離** =162.2334<br>33、 DIst, int, D **左下端**, int, E **右上端**, **→距離** =179.7561 |
|  | 34、 Hatch, **點所有區域**, **→斜線區域**<br>35、 **點斜線**, CRTL+1, **→查幾何面積值** =9896.4295 |

## 2.4.2 AutoCAD2D2008-402

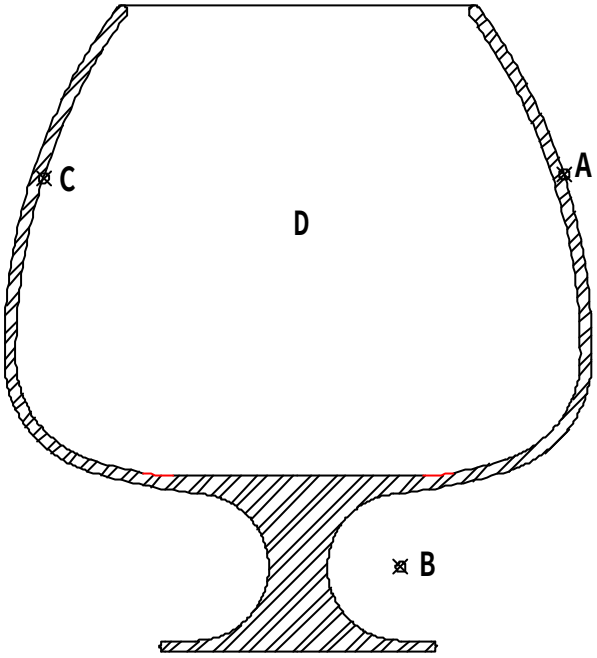
| | |
|---|---|
|  | 1、**中心點** A **至四分點** B **水平距離為何？** 129.6774<br>2、**中心點** A **至交點** C **垂直距離為何？** 36.7523<br>3、B **區域扣除內孔面積為何？** 373.6039<br>4、E **區域扣除內孔面積為何？** 924.5497<br>5、**交點** C **至端點** F **距離為何？** 157.9115 |

**解題步驟：**

| | |
|---|---|
|  | 01、 Line, **任一點** P01, (**正交模式**), **向左**, 90, →L50, @90<165, →L51<br>02、 Offset, 2(=4/2), L51, **向上點**, →L52, L52, **向上點**, →L53 |
|  | 03、 EXtend, L50, L52 **下**, →L52 **向下延伸**<br>04、 Circle, int, L50 & L52, D, 4, →C53<br>05、 ROtate, L50, end, L50 **右**, C, -3.5(=7/2), →L54 |
|  | 06、 Fillet, T, T, R, 0, L53, L54, →L53 **與** L54 **相交**<br>07、 MIrror, L51~L54, end, L50 **左**, end, L50 **右**, →L55~L58<br>08、 Circle, int, L53 & L54, 10, →C59<br>09、 Offset, 10, L55, **向上點**, →L60<br>10、 Circle, int, C59 & L60, 10, →C61 |
|  | 11、 Fillet, T, T, R, 15, L57, L58, →A62<br>12、 Fillet, T, T, R, 1, L54, L58, →A63<br>13、 POint, end, L50 **右**, →F<br>14、 LENgthen, T, 65, L52 **左**, →L52 **縮短**<br>15、 LENgthen, T, 65, L56 **左**, →L56 **縮短**<br>16、 Circle, end, L52 **左**, D, 18, →C64<br>17、 Circle, @(**鎖定上次點**), D, 24, →C65<br>18、 ROtate, L52, end, L52 **左**, C, 165(=180-15), →L66<br>19、 Offset, 5, L66, **向上點**, →L67<br>20、 ROtate, L66, int, L66 & C69, C, -90, →L68<br>21、 Circle, int, L67 & L68, 5, →C69 |
|  | 22、 TRim,<br>23、 Erase,<br>24、 Offset, 1.5(=3/2), C69, **向右點**, →A70<br>25、 Offset, 1.5(=3/2), C69, **向左點**, →A71<br>26、 Circle, 3P, tan, A70, tan, C69, tan, A71, →C72<br>27、 Fillet, T, N, R, 1, C65, A70, →A73<br>28、 Fillet, T, N, R, 1, C65, A71, →A74<br>29、 Fillet, T, N, R, 2, C65, L53, →A75<br>30、 Fillet, T, N, R, 2, C65, L51, →A76 |

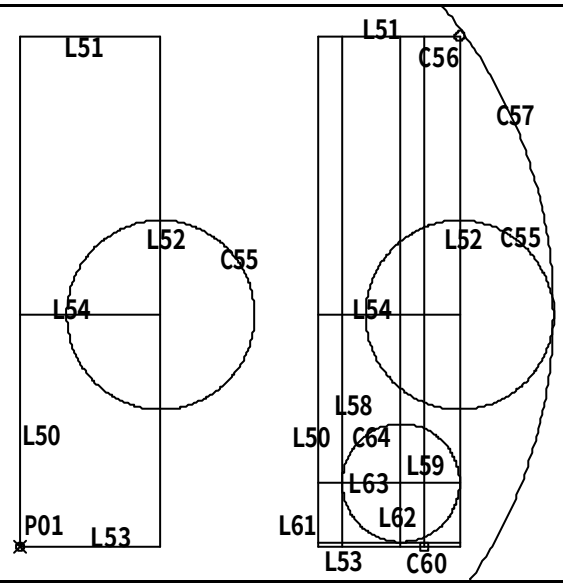
| | |
|---|---|
|  | 31、 TRim,<br>32、 Erase,<br>33、 Offset, 3(6/2), L56, **向上點**, →L77<br>34、 Offset, 3(6/2), L56, **向下點**, →L78<br>35、 Line, end, L77 **左**, end, L78 **左**, →L79<br>36、 Offset, 5, L79, **向右點**, →L80<br>37、 Circle, int, L56 & L79, 12, →C81<br>38、 Circle, int, L78 & L80, 10, →C82<br>39、 Circle, int, L77 & L80, 10, →C83<br>40、 Line, tan, C81, tan, C83, →L84<br>41、 Line, tan, C81, tan, C82, →L85<br>42、 Fillet, T, N, R, 2, L55, C83, →A86<br>43、 Fillet, T, N, R, 2, L57, C82, →A87 |
|  | 44、 TRim,<br>45、 Erase,<br>46、 Offset, 3, C83, **向內點**, →A89, L84, **向內點**, →L90, C81, **向內點**, →A91, L85, **向內點**, →L92, C82, **向內點**, →A98,<br>47、 EXtend, C81, L56 **左**, →L56 **向左延伸**<br>48、 POint, int, C81 & L56, →C |
|  | 49、 LENgthen, DY, A89, **向下拉伸**, →A89 **延伸**<br>50、 LENgthen, DY, A93, **向上拉伸**, →A93 **延伸**<br>51、 Line, tan, A89, tan, A93, →L94 |
|  | 52、 TRim,<br>53、 Erase,<br>54、 DIst, cen, A, qua, B, →X **差值**=129.6774<br>55、 DIst, cen, A, int, C, →Y **差值**=36.7523<br>56、 Hatch, **點** D **區域**, →**斜線區域**<br>57、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值** =373.6039<br>58、 Hatch, **點** E **區域**, →**斜線區域**<br>59、 **點** E **區域斜線**, CRTL+1, →**查幾何面積值** =924.5497<br>60、 DIst, nod, C, nod, F, →**距離**=157.9115 |

## 2.4.3 AutoCAD2D2008-403




1、 **弧** A **包含角度為何？** 36.5795
2、 **中心點** B **至弧** C **中心點距離為何？** 48.6303
3、 D **區域面積為何？** 6709.8799
4、 **斜線區域面積為何？** 1014.8311
5、 **圖形所圍成最外圍周長為何？** 417.5565

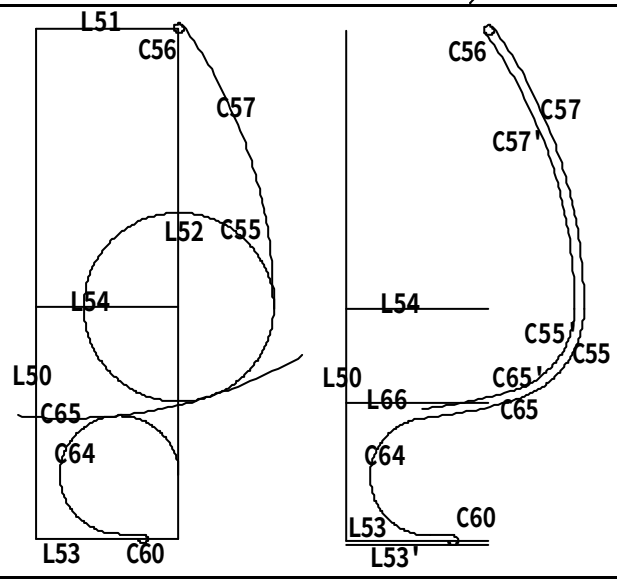
**解題步驟：**




01、 RECtangle, **任一點** P01, D, 30, 108(=49+59), **向右上點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 59, L51, **向下點**, →L54
04、 Circle, int, L54 & L52, 20, →C55

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

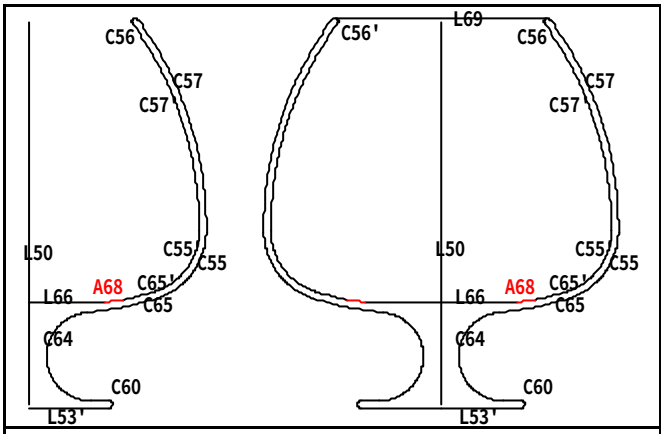
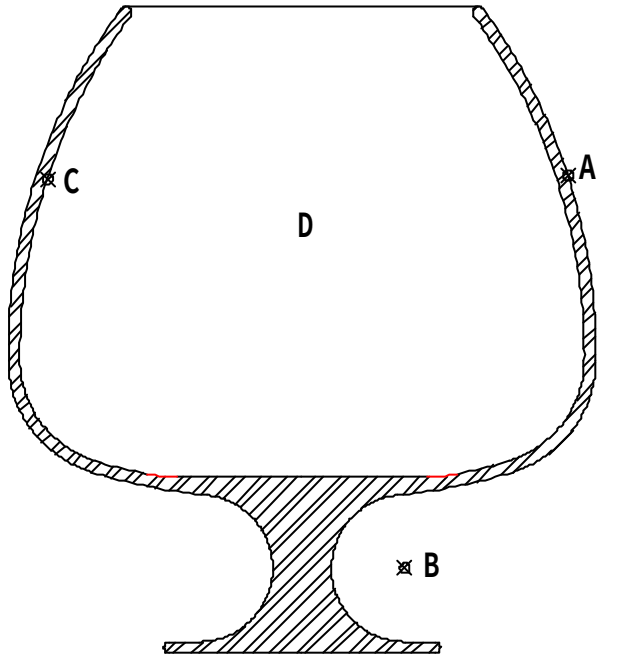
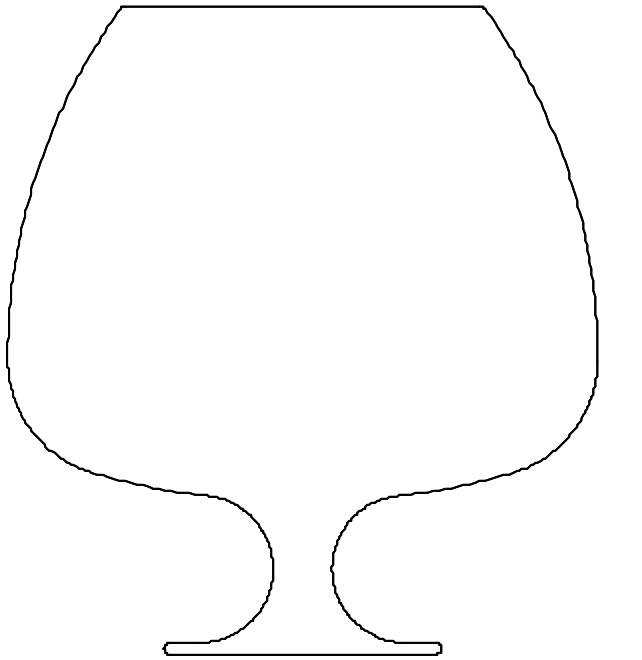
05、 Circle, int, L51 & L52, 1, →C56
06、 Circle, T, C56 **右上**, C66 **右**, 100, →C57
07、 Offset, 5, L50, **向右點**, →L58
08、 Offset, 22.5(=45/2), L50, **向右點**, →L59
09、 Circle, int, L59 & L53, 1, →C60
10、 Offset, 1, L53, **向上點**, →L61
11、 Offset, 12.5, L58, **向右點**, →L62
12、 Offset, 12.5, L61, **向上點**, →L63
13、 Circle, int, L62 & L63, 12.5, →C64



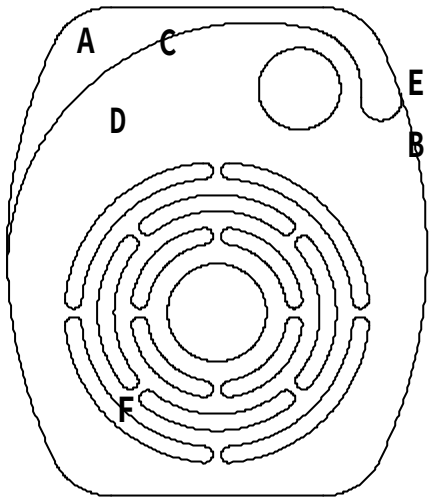

14、 TRim,
15、 Erase,
16、 Circle, T, C55 **正下**, C64 **正上**, 100, →C65

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

17、 TRim,
18、 Erase,
19、 Offset, 20, L54, **向下點**, →L66
20、 Offset, 2, C57, **向左點**, →C57’, C55, **向左點**, →C55’, C65, **向左點**, →C65’
21、 Offset, 1, L53, **向下點**, →L53’

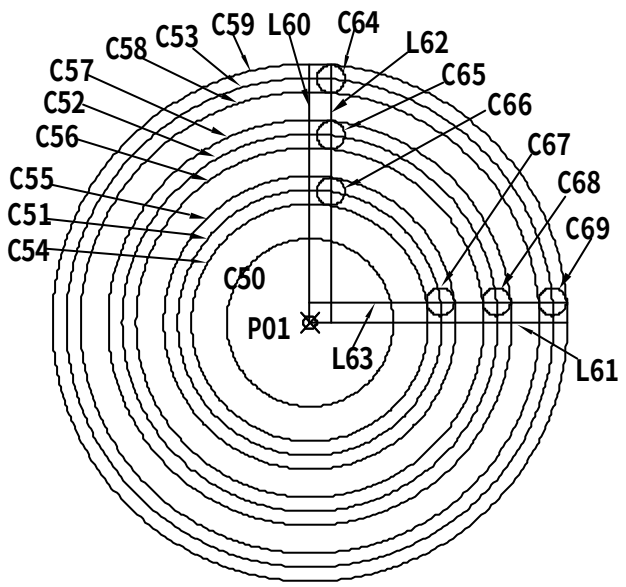
| | |
|---|---|
|  | 22、 TRim,<br>23、 Erase,<br>24、 Fillet, T, T, R, 25, L66 **右**, C65'**下**, →A68<br>- - - - - - - - - - - - - - - - - - - - - - - - - - - - - -<br>25、 MIirror, **所有物件**, end, L50 **上**, end, L50 **下**, **→左邊所有物件**<br>26、 Line, tan, C56 **上**, tan, C56'**上**, →L69 |
|  | 27、 TRim,<br>28、 Erase,<br>29、 Hatch, **點斜線區域**, **→區域斜線**<br>30、 **選** A, CRTL+1, **→查幾何總角度值**=36.5795<br>31、 DIst, cen, C, cen , B, **→距離**=48.6303<br>32、 Hatch, **點** D **區域**, **→區域斜線**<br>33、 **點** D **區域斜線**, CRTL+1, **→查幾何面積值** =6709.8799<br>34、 **點斜線區域**, CRTL+1, **→查幾何面積值** =1014.8311 |
|  | 35、 TRim,<br>36、 Erase,<br>37、 BOundary, **點內部**, **選邊緣**, CRTL+1, **→查幾何長度值** =417.5565 |

## 2.4.4 AutoCAD2D2008-404

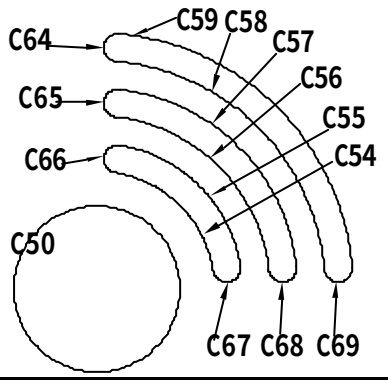



1、 A **區域面積為何？** 992.9484
2、 **弧** B **半徑為何？** 120.9339
3、 **弧** C **夾角為何？** 107.3779
4、 D **區域扣除內孔面積為何？** 7270.1962
5、 **交點** E **至中點** F **高度為何？** 77.1400

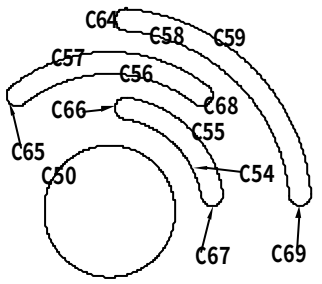
**解題步驟：**



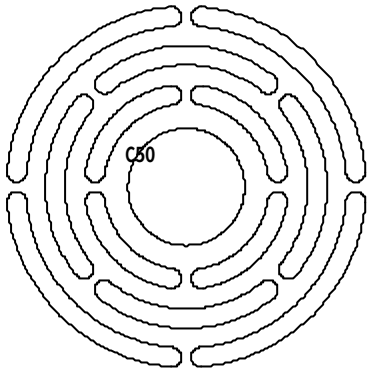

01、 Circle, **任一點** P01, D, 24, →C50
02、 Circle, @(**鎖定上次點**), D, 38, →C51
03、 Circle, @(**鎖定上次點**), D, 54, →C52
04、 Circle, @(**鎖定上次點**), D, 70, →C53
05、 Offset, 2(=4/2), C51, **向下點**, →C54, C51, **向上點**, →C55, C52, **向下點**, →C56, C52, **向上點**, →C57, C53, **向下點**, →C58, C53, **向上點**, →C59
06、 Line, qua, C59 **上**, cen, C50, →L60, qua, C59 **右**, →L61
07、 Offset, 3(=6/2), L60, **向右點**, →L62, L61, **向上點**, →L63
08、 Circle, int, C53 & L62, D, 4, →C64
09、 COpy, C64, cen, C64, int, L62 & C52, →C65, int, L62 & C51, →C66, int, L63 & C51, →C67, int, L63 & C52, →C68, int, L63 & C53, →C69



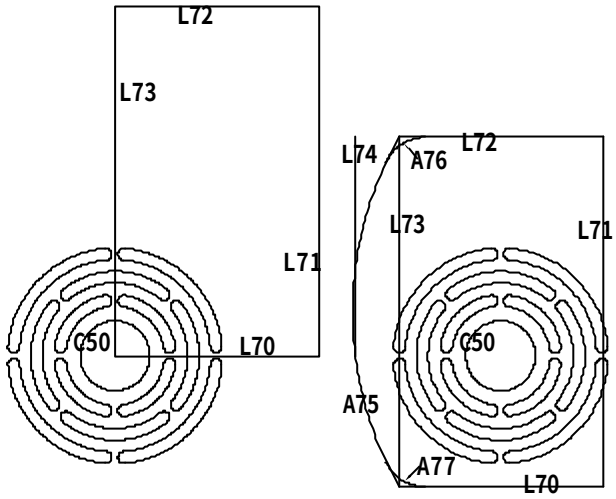

10、 Erase,
11、 TRim,




12、 ROtate, C56;C57;C65;C68, cen, C50, 45, →C56;C57;C65;C68 **轉動**




13、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 4, **佈滿角度**, 360, **物件**, C54~C59;C64~C69, →**全圓佈滿** 4 **組**




14、 RECtangle, cen, C50, D, 70, 120, →L70~L73 **之矩形**

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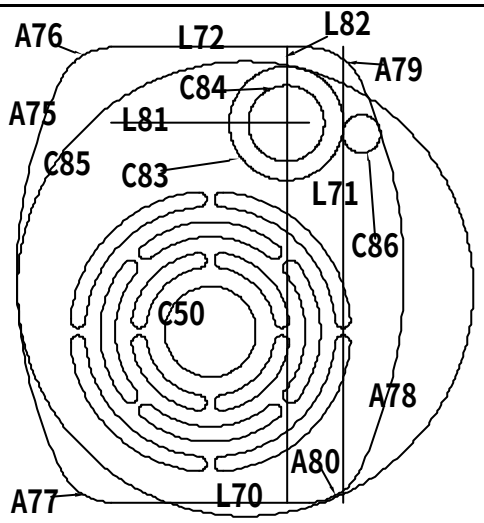
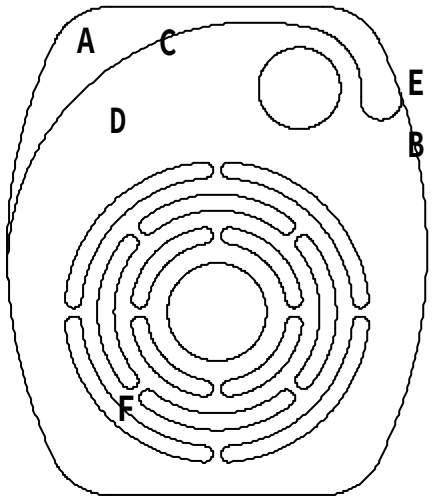
15、 Move, L70~L73 **之矩形**, cen, C50, @-35,-45 , →L70~L73 **之矩形向左下移動**
16、 eXplode, L70~L73 **之矩形**, →L70~L73
17、 Offset, 15(=50-35), L73, **向左點**, →L74
18、 LENgthen, DE, -45, L74 **下**, →L74 **向上縮短**
19、 Arc, int, L72 & L73, end, L74 **下**, int, L70 & L73, →A75
20、 Fillet, T, N, R, 15, L72, A75, →A76
21、 Fillet, T, N, R, 15, L70, A75, →A77

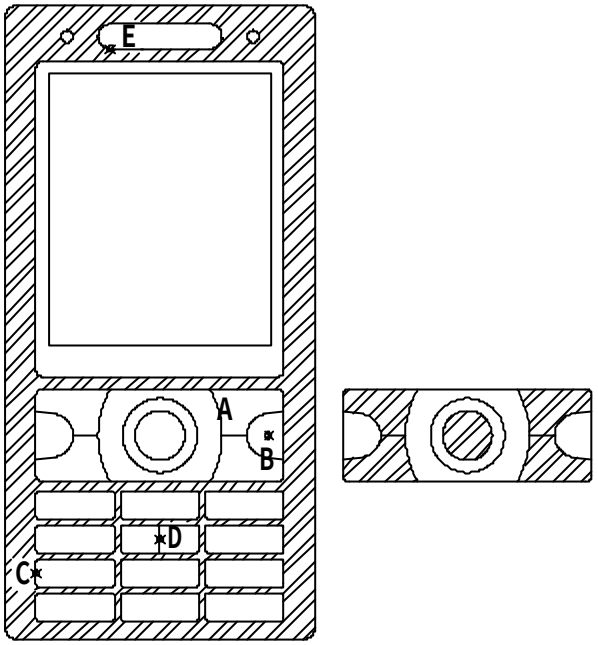



22、 Erase,
23、 TRim,
24、 MIrror, A75~A77, qua, C50 **上**, qua, C50 **下**, →A78~A80
25、 Offset, 20, L72, **向下點**, →L81
26、 Offset, 15(=(70/2)-20), L71, **向左點**, →L82
27、 Circle, int, L81 & L82, 15, →C83
28、 Circle, @(**鎖定上次點**), D, 20, →C84
29、 Circle, T, C83 **上**, A75 **左**, 60, →C85
30、 Circle, T, C83 **右**, A78 **右**, 5, →C86



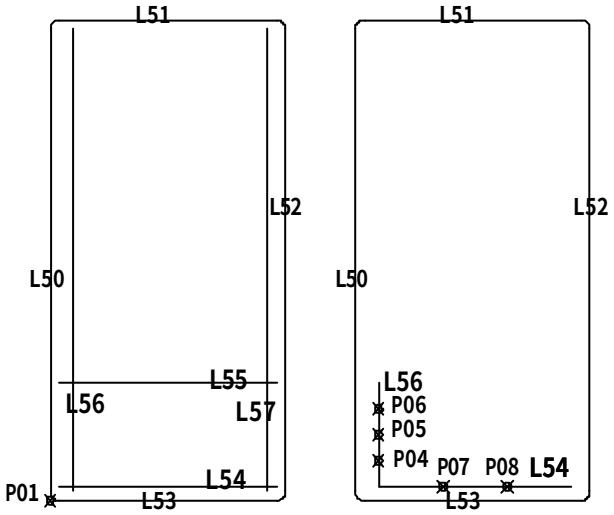

31、 TRim,
32、 Erase,
33、 Hatch, **點** A **區域**, →**斜線區域**
34、 **點** A **區域斜線**, CRTL+1, →**查幾何面積值** =992.9484
35、 **點選弧** B, CRTL+1, →**查幾何半徑值** =120.9339
36、 **點選弧** C, CRTL+1, →**查幾何總角度值** =107.3779
37、 Hatch, **點** D **區域**, →**斜線區域**
38、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值** =7270.1962
39、 DIst, int, E, mid, F, →Y **差值**=77.1400

## 2.4.5 AutoCAD2D2008-405

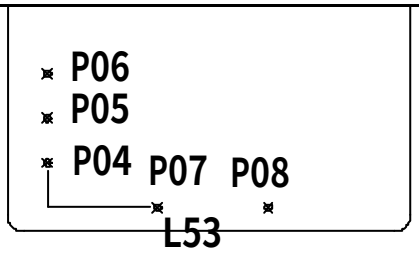



1、 **弧** A **包含角為何？** 59.7257
2、 **中心點** B **至中點** C **距離為何？** 43.6094
3、 **中點** D **至交點** E **距離為何？** 79.6528
4、 **左側圖斜線區域面積為何？** 1719.0089
5、 **右側局部視圖斜線區域面積為何？** 292.4712

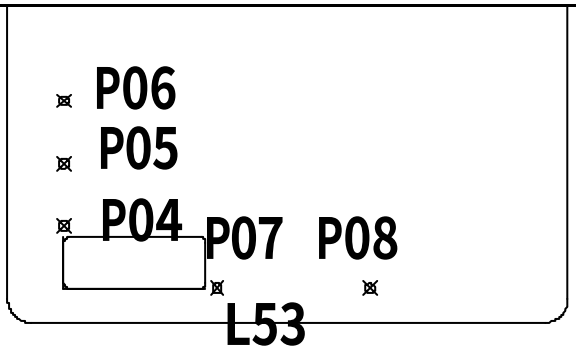
**解題步驟：**



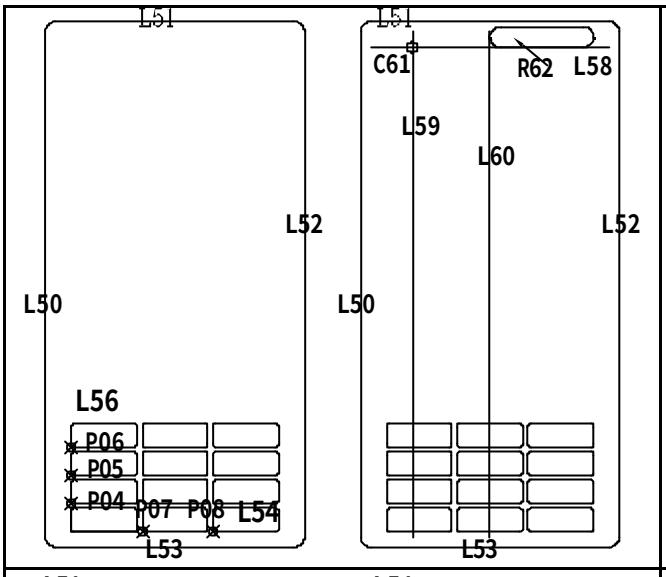

01、 RECtangle, F, 2, **任一點** P01, D, 50, 102.5, **向右上點**, →L50~L53 **之圓角矩形**
02、 eXplode, L50~L53 **之圓角矩形**, →L50~L53
03、 Offset, 3, L53, **向上點**, →L54
04、 Offset, 22(=21+1), L54, **向上點**, →L55
05、 Offset, 5, L50, **向右點**, →L56
06、 Offset, 41(=40+1), L56, **向右點**, →L57
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
07、 TRim,
08、 Erase,
09、 DIVide, L56, 4, →P04~P06
10、 DIVide, L54, 3, →P07;P08




11、 LENgthen, DY, L56 **上**, nod, P04, →L56 **縮短**, L54 **右**, nod, P07, →L54 **縮短**
12、 LENgthen, DE, -1, L56 **上**, →L56 **再縮短** 1, L54 **右**, →L54 **再縮短** 1



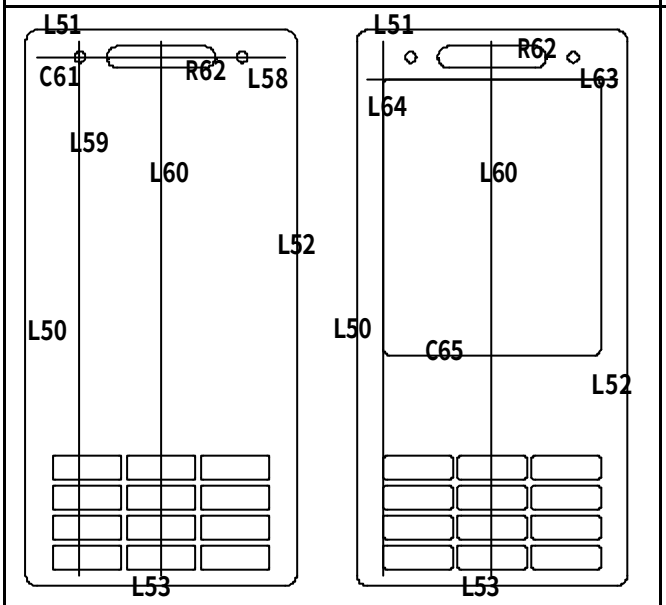

13、 RECtangle, F, 0.5, int, L54 & L56, D, int, L54 & L56, end, L54 **右**, int, L54 & L56, end, L56 **上**, **向右上點**, →**小圓角矩形**




14、 Erase,
15、 ARray, **矩形陣列**, **列**, 4,**行**, 3,**列偏移**, nod, P04, nod, P05, **行偏移**, nod, P07, nod, P08, **物件**, **小圓角矩形**, →12 **個小圓角矩形**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

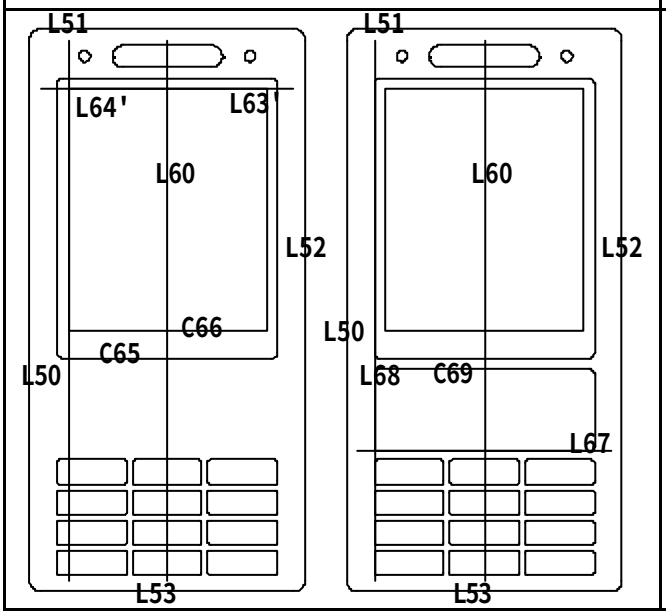
16、 Offset, 5, L51, **向下點**, →L58
17、 Offset, 10, L50, **向右點**, →L59
18、 Offset, 15, L59, **向右點**, →L60
19、 Circle, int, L58 & L59, D, 2, →C61
20、 RECtangle, F, 2(=4/2), int, L58 & L60, D, 20(=2+8+8+2), 4, **向右上點**, →R62




21、 Move, R62, **任一點**, @-10,-2 →R62 **向左下移動**
22、 COpy, C61, **任一點**, @30,0 →**右側出現小圓**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

23、 Offset, 9, L51, **向下點**, →L63
24、 Offset, 5, L50, **向右點**, →L64
25、 RECtangle, F, 1, int, L63 & L64, D, 40(=20+20), 51, **向右下點**, →C65 **圓角矩形**




26、 Move, L63, **任一點**, @0,-2, →L63' (L63 **向下移動**)
27、 Move, L64, **任一點**, @2,0, →L64' (L64 **向右移動**)
28、 RECtangle, F, 0, int, L63' & L64', D, 36(=18+18), 44, **向右下點**, →C66 **矩形**
29、 Erase,

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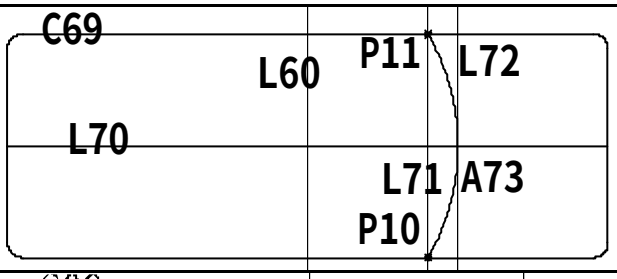
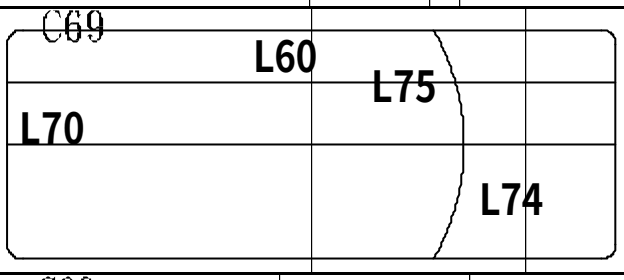
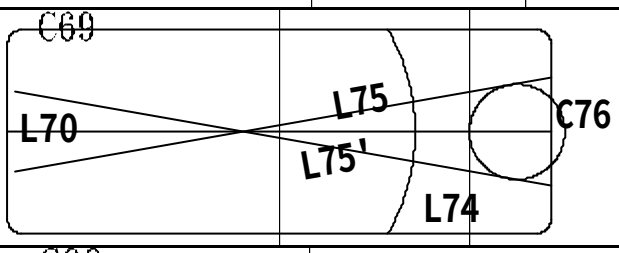
30、 Offset, 77(=60+17), L51, **向下點**, →L67
31、 Offset, 5, L50, **向右點**, →L68
32、 RECtangle, F, 1, int, L67 & L68, D, 40, 15, **向右上點**, →C69 **圓角矩形**



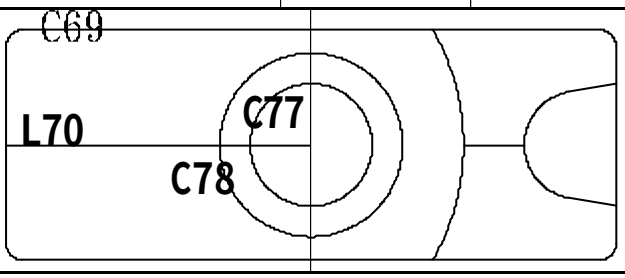

33、 Erase,
34、 Line, mid, C69 **左直線**, mid, C69 **右直線**, →L70
35、 Offset, 8(=16/2), L60, **向右點**, →L71
36、 Offset, 10(=20/2), L60, **向右點**, →L72
37、 Arc, int, C69 **下** & L71(P10), int, L70 & L72, int, C69 **上** & L71(P11), →A73



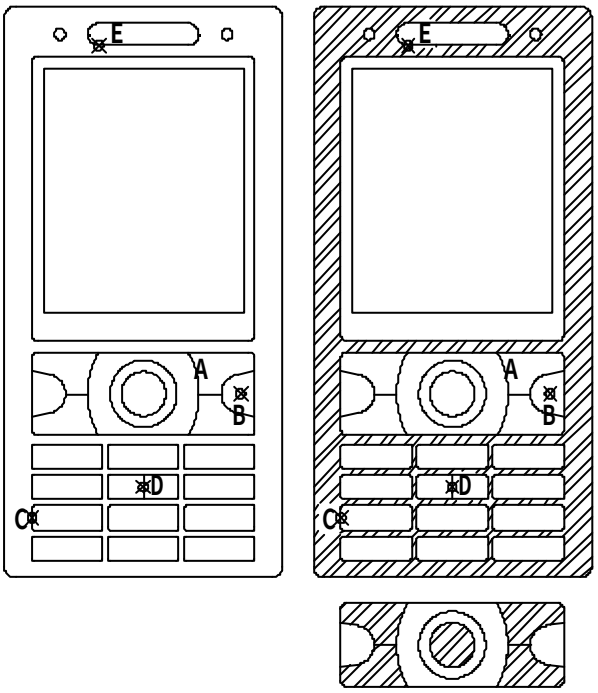

38、 Erase,
39、 Offset, 14(=28/2), L60, **向右點**, →L74
40、 Offset, 4(=8/2), L70, **向上點**, →L75




41、 ROtate, L75, int, L75 & C69 **右直線**, 10, →L75 **轉動**
42、 MIrror, L75, end, L70 **左**, end, L70 **右**, →L75'
43、 Circle, 3P, tan, L75, tan, L74, tan, L75', →C76




44、 TRim,
45、 Erase,
46、 Circle, int, L60 & L70, D, 8, →C77
47、 Circle, @(**鎖定上次點**), D, 12, →C78

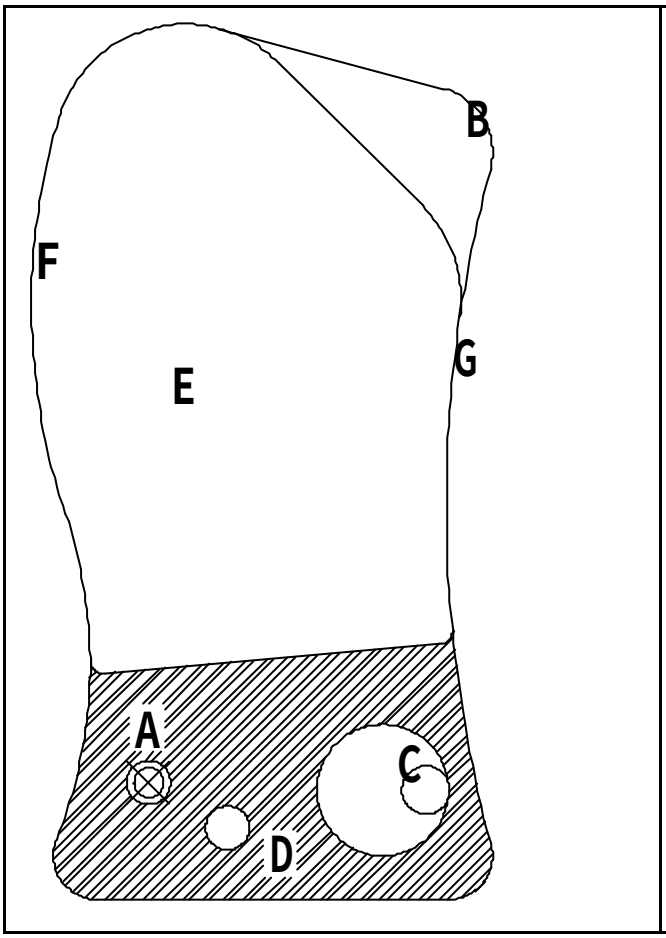



48、 MIrror, **左側物件**, qua, C78 **上**, qua, C78 **下**, →**左側物件** TRim
49、 Erase,

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

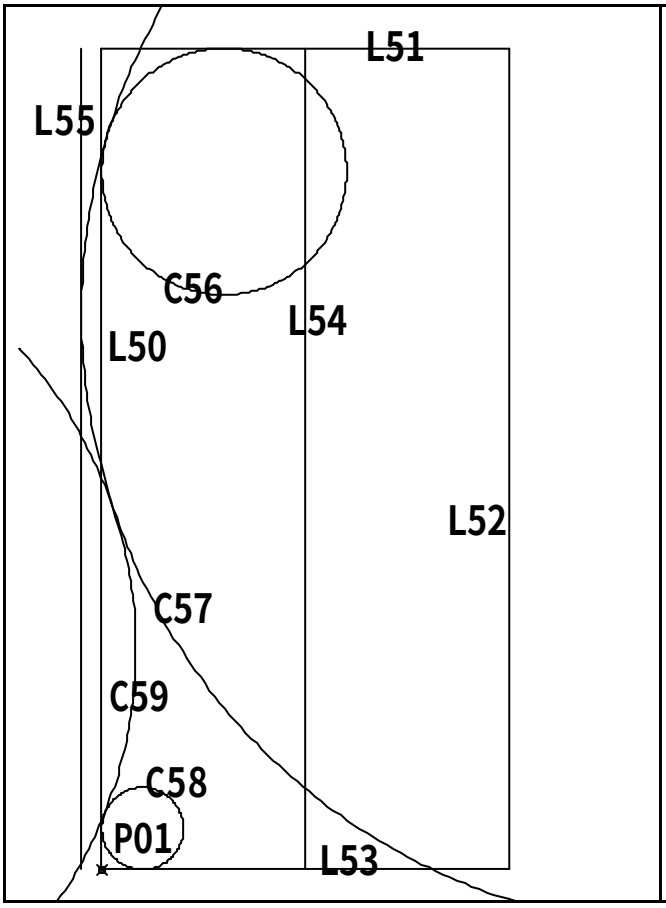
50、 Hatch, **點斜線區域**, →**斜線區域**
51、 **選** A **弧**, CTRL+1, →**查幾何總角度值** =59.7257
52、 Line, mid, **上**, mid, **下**, →D
53、 DIst, cen, B, mid, C, →**距離**=43.6094
54、 DIst, mid, D, int, E, →**距離**=79.6528
55、 **點右上圖斜線區域**, CTRL+1, →**查幾何總角度值**=1719.0089
56、 **點右下圖斜線區域**, CTRL+1, →**查幾何總角度值**=292.4712

## 2.4.6 AutoCAD2D2008-406

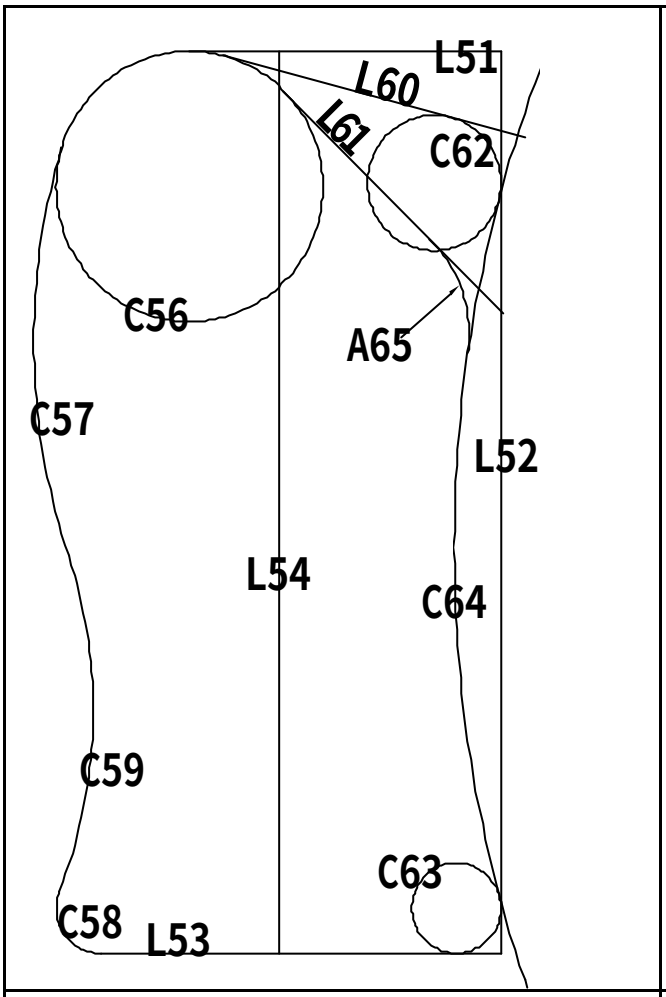



1、 **中心點** A **至弧** B **中心點距離為何？** 157.3622
2、 **弧** B **長度為何？** 23.5223
3、 **斜線** D **區域面積為何？** 4140.6799
4、 E **區域周長為何？** 420.2203
5、 **弧** F **至弧** G **中心點距離為何？** 248.7071

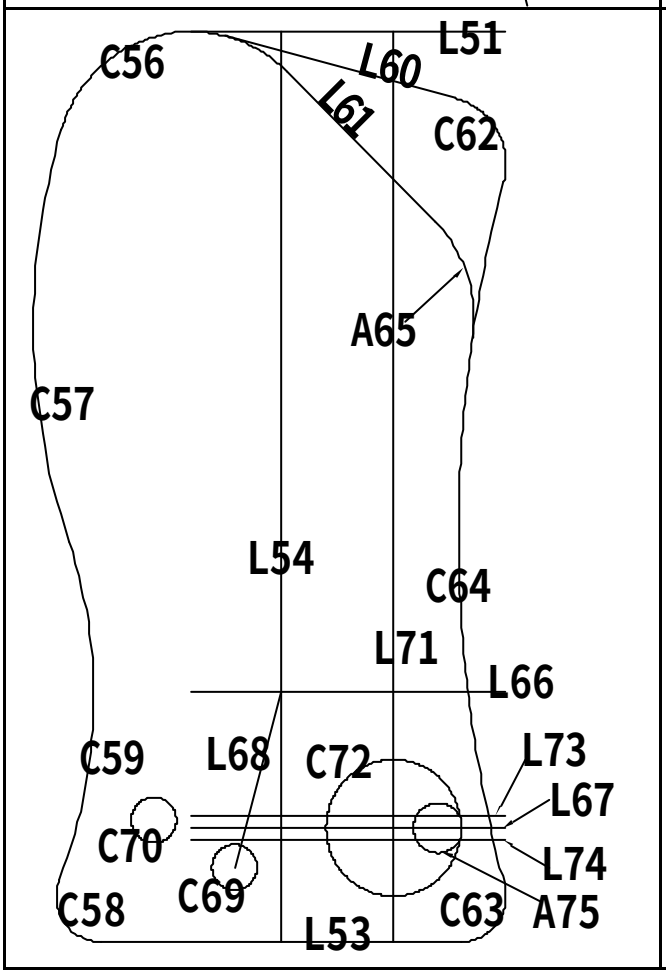
**解題步驟：**



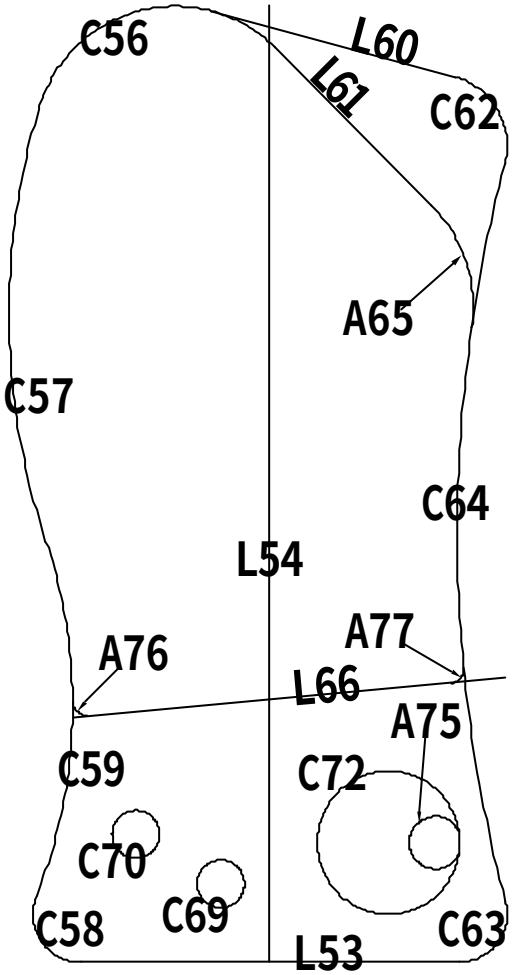

01、 RECtangle，**任一點** P01, D, 100, 200, **向右上點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Line, mid, L51, mid, L53, →L54
04、 Offset, 55, L54, **向左點**, →L55
05、 Circle, T, L50, L51, 30, →C56
06、 Circle, T, L55, C56, 150, →C57
07、 Circle, T, L50, L53, 10, →C58
08、 Circle, T, C57, C58, 108, →C59



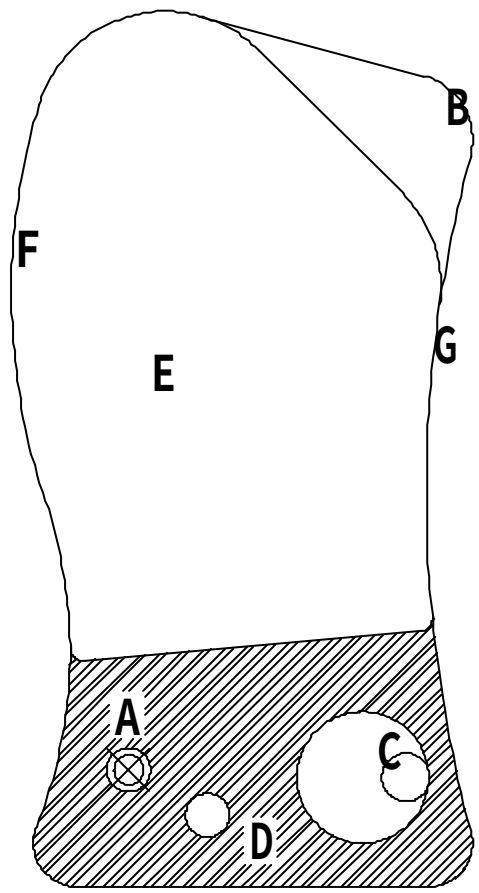

09、 TRim,
10、 Erase,
11、 ROtate, L51, cen, C56, C, -15, →L60
12、 ROtate, L51, cen, C56, C, -45, →L61
13、 Circle, T, L52, L60, 15, →C62
14、 Circle, T, L52, L53, 10, →C63
15、 Circle, T, C62, C63, 300, →C64
16、 Fillet, T, N, R, 30, L61, C64, →A65




17、 TRim,
18、 Erase,
19、 Offset, 145, L51, **向下點**, →L66
20、 Offset, 30, L66, **向下點**, →L67
21、 Line, int, L54 & L66, @40<-105, →L68
22、 Circle, end, L68 **下**, D, 10, →C69
23、 ROtate, C69, int, L54 & L66, C, -30, →C70
24、 Offset, 25, L54, **向右點**, →L71
25、 Circle, int, L67 & L71, D, 30, →C72
26、 Offset, 2.5(=5/2), L67, **向上點**, →L73
27、 Offset, 2.5(=5/2), L67, **向下點**, →L74
28、 Arc, int, C72 & L73, E, int, C72 & L74, R, -5.5(=11/2), →A75

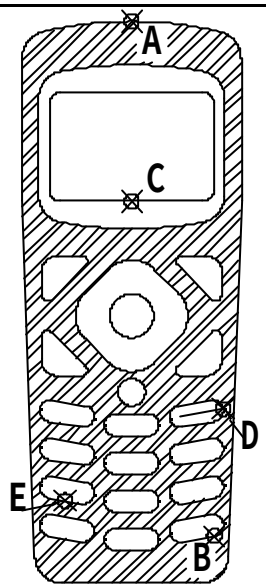



29、 ROtate, L66, int, L66 & L54, 5, →L66 **轉動**
30、 EXtend, C59, L66 **左**, →L66 **向左延伸**
31、 Fillet, T, N, R, 3, L66 & C59, →A76, L66 & C64, →A77



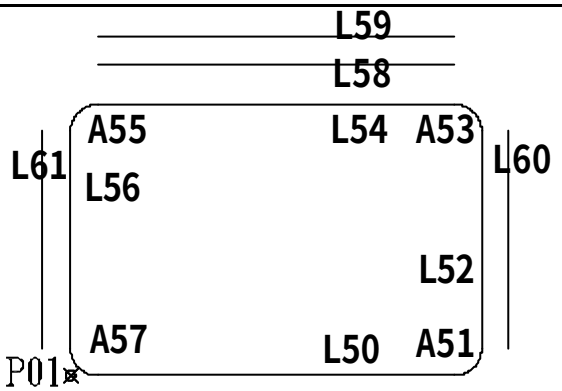

32、 TRim,
33、 Erase,
34、 Hatch, **點** D **區域**, →**斜線區域**
35、 DIst, cen, A, cen, B, →**距離**=157.3622
36、 **點選弧** B, CRTL+1, →**查幾何弧長值** =23.5223
37、 **點** D **區域斜線**, CRTL+1, →**查幾何面積值** =4140.6799
38、 BOundary, **點** E **區域**,
39、 **點** E **區域邊緣**, CRTL+1, →**查幾何長度值** =420.2203
40、 DIst, cen, F, cen, G, →**距離**= 248.7071

## 2.4.7 AutoCAD2D2008-407

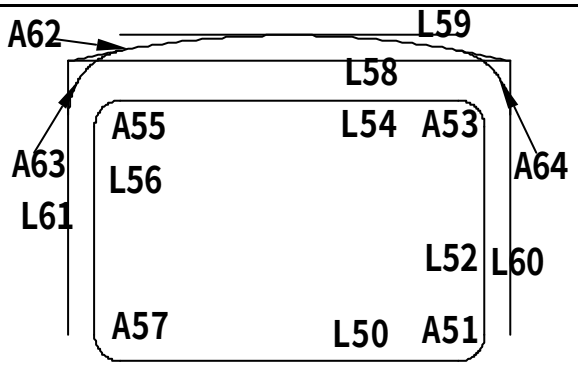



1、 **弧** A **半徑為何？** 68.1667
2、 **交點** B **至交點** C **角度為何？** 103.8412
3、 **交點** D **至中點** E **距離為何？** 33.4531
4、 **圖形最外圍周長為何？** 269.5254
5、 **斜線區域面積為何？** 2001.4971

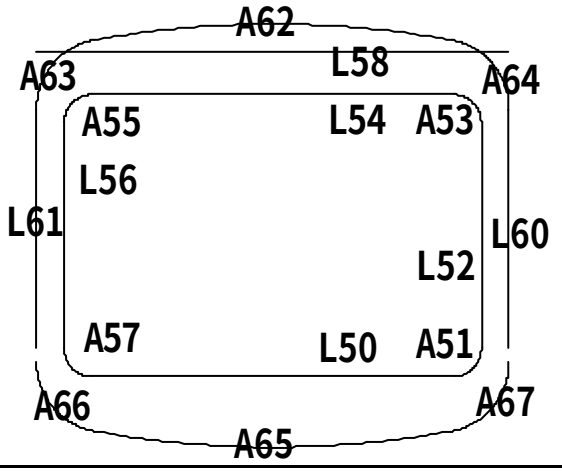
**解題步驟：**



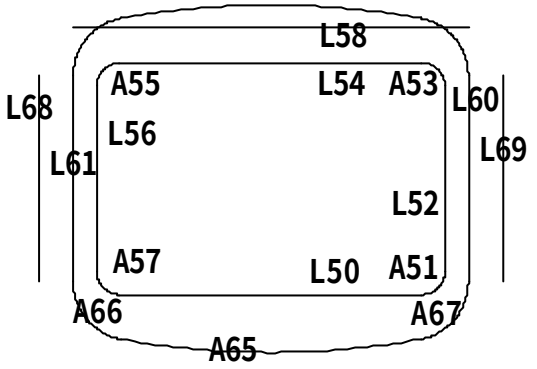

01、 RECtangle, F, 2, **任一點** P01, D, 30, 20, **向右上點**, →L50~A57 **之圓角矩形**
02、 eXplode, L50~A57 **之圓角矩形**, →L50~A57
03、 Offset, 3(=5-2), L54, **向上點**, →L58
04、 Offset, 2, L58, **向上點**, →L59, L52, **向右點**, →L60, L56, **向左點**, →L61




05、 Fillet, T, T, R,0, L58, L60, →L58 **與** L60 **相交**,
06、 Fillet, T, T, R,0, L58, L61, →L58 **與** L61 **相交**
07、 Arc, int, L58 & L60, mid, L59, int, L58 & L61, →A62
08、 Fillet, T, N, R, 5, A62, L61, →A63
09、 Fillet, T, N, R, 5, A62, L60, →A64

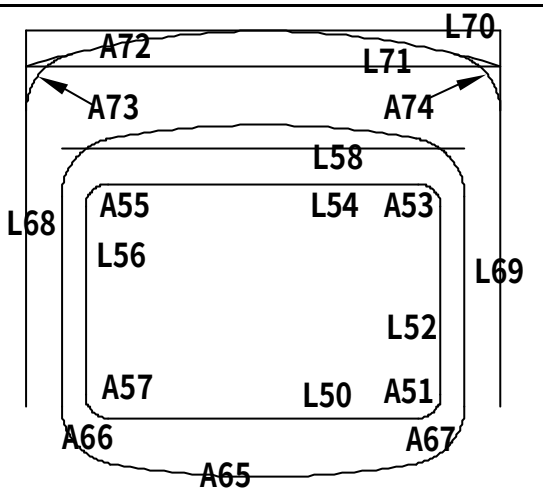



10、 TRim,
11、 Erase,
12、 MIrror, A62~A64, mid, L52, mid, L56, →A65~A67




13、 EXtend, A66;A67, L60 **下**; L61 **下**, →L60 **與** L61 **分別向下延伸**

14、 Offset, 3(=(40-34)/2), L61, **向左點**, →L68, L60, **向右點**, →L69




15、 Offset, 10, L58, **向上點**, →L70

16、 Offset, 3, L70, **向下點**, →L71

17、 Fillet, T, T, R, 0, L71, L68, →L71 **與** L68**相交**

18、 Fillet, T, T, R, 0, L71, L69, →L71 **與** L69**相交**

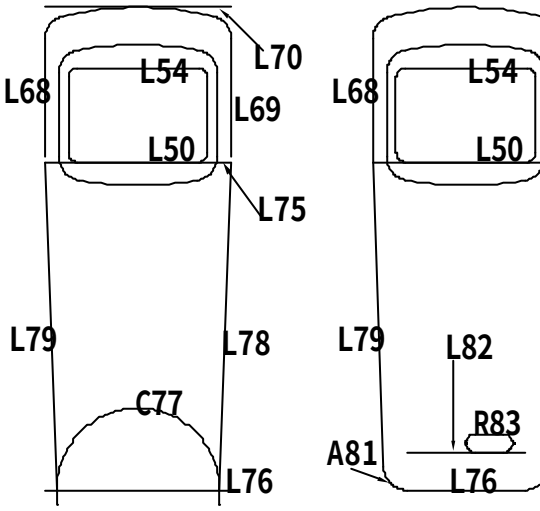
19、 Fillet, T, T, R, 0, L70, L68, →L70 **與** L68**相交**

20、 Fillet, T, T, R, 0, L70, L69, →L70 **與** L69**相交**

21、 Arc, int, L69 & L71, mid, L70, int, L68 & L71, →A72

22、 Fillet, T, T, R, 5, A72, L68, →A73

23、 Fillet, T, T, R, 5, A72, L69, →A74




24、 TRim, Erase,

25、 Offset, 33, L70, **向下點**, →L75

26、 Offset, 70, L75, **向下點**, →L76

27、 Circle, mid, L76, D, 35, →C77

28、 Line, end, L75 **右**, int, L76 & C77 **右**, →L78

29、 Line, end, L75 **左**, int, L76 & C77 **左**, →L79

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

30、 EXtend, L75, L68 **下**; L69 **下**, →L68 **與** L69 **分別向下延伸**

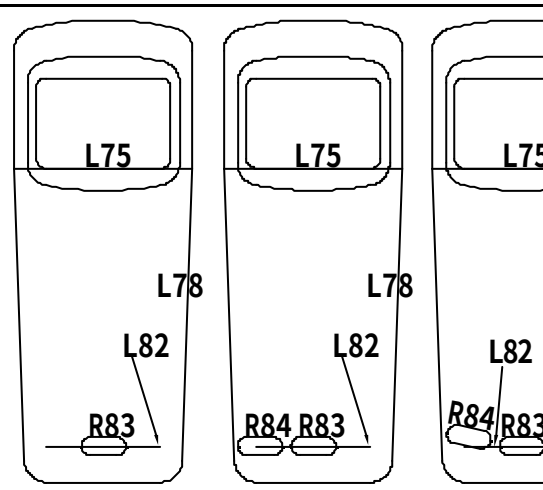
31、 Fillet, T, T, R, 5, L76, L78, →A80

32、 Fillet, T, T, R, 5, L76, L79, →A81

33、 Erase,

34、 Offset, 8, L76, **向上點**, →L82

35、 RECtangle, F, 2, mid, L82, D, 10, 4, **向右上點**, →R83 **之圓角矩形**




36、 Move, R83, mid, L82, @-5,-2, →R83 **向左下移動**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

37、 COpy, R83, mid, L82, @-12,0, →R84

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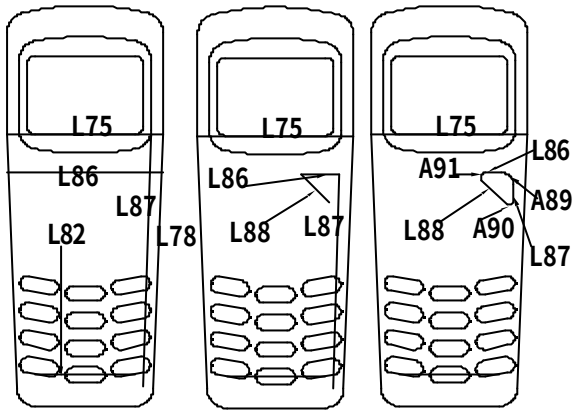
38、 ROtate, R84, mid, L82, -10, →R84 **轉動**

39、 MIrror, R84, mid, R83 **上水平線**, mid, R83 **下水平線**, →R85



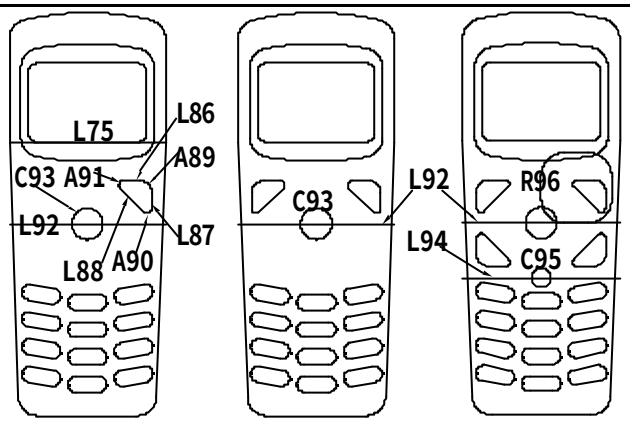

40、 ARray, **矩形陣列**, **列**, 4, **行**, 1, **列偏移**, 7(=21/3), **行偏移**, 0, **物件**, R83~R85, →**產生** 4**組** R83~R85 **相同物件**
41、 Offset, 10, L75, **向下點**, →L86
42、 Offset, 3, L78, **向左點**, →L87

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

43、 TRim,
44、 LENgthen, T, 10, L86 **左**, →L86 **向右縮短**
45、 ROtate, L86, end, L86 **左**, C, -45, →L88

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

46、 Fillet, T, T, R, 3, L86, L87, →A89
47、 Fillet, T, T, R, 1, L87, L88, →A90
48、 Fillet, T, T, R, 1, L86, L88, →A91



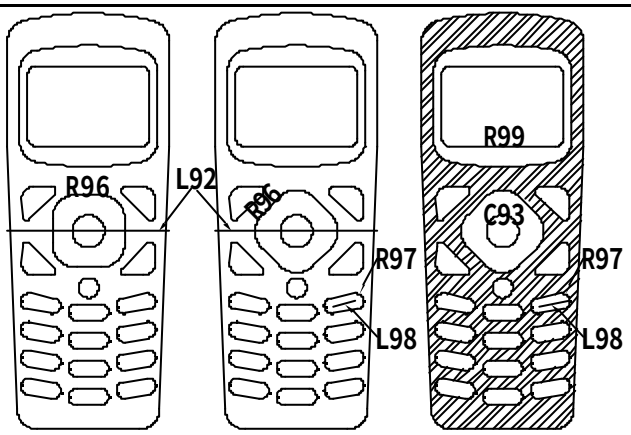

49、 Offset, 21(=10+11), L75, **向下點**, →L92
50、 Circle, mid, L92, D, 8, →C93

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

51、 MIrror, L86~L88;A89~A91, qua, C93 **上**, qua, C93 **下**, →**左側相同物件**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

52、 MIrror, L86~L88;A89~A91(**左右側相同物件**), qua, C93 **左**, qua, C93 **右**, →**下側相同物件**
53、 Offset, 14(=11+3), L92, **向下點**, →L94
54、 Circle, mid, L94, D, 5, →C95
55、 RECtangle, F, 6, mid, L92, D, 18, 18, **向右上點**, →R96 **之圓角矩形**



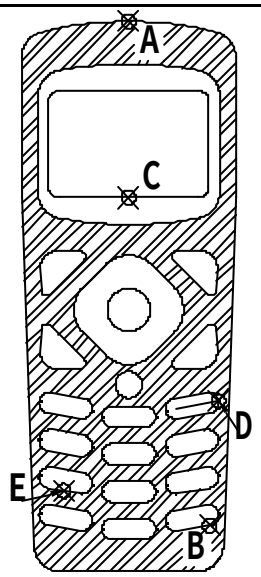

56、 Erase,
57、 Move, R96, mid, L92, @-9,-9, →R96 **向左下移動**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

58、 ROtate, R96, mid, L92, 45, →R96 **轉動**
59、 Line, cen, R97 **左**, cen, R97 **右**, →L98

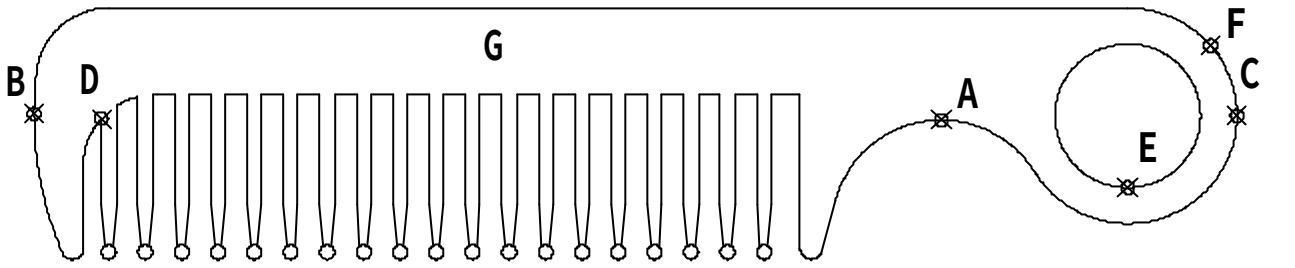
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

60、 Erase,
61、 EXtend, R97 **右**, L98 **右**, →L98 **向右延伸**(**以利找到交點** D)
62、 Hatch, **點外區域**, B, **點選** C93;R99, →**斜線區域**



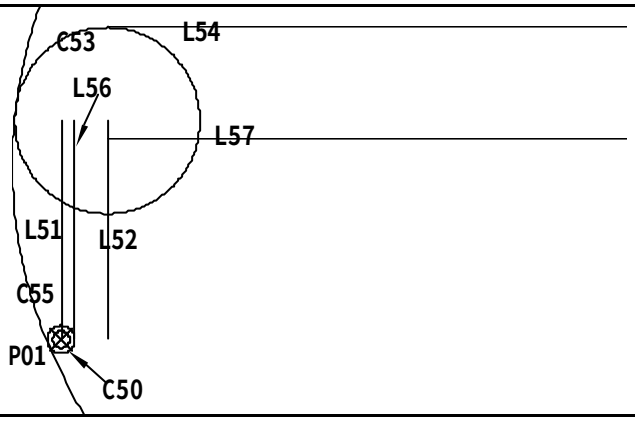

63、 **點選弧** A, CRTL+1, →**查幾何半徑值** =68.1667
64、 DIst, int, B, mid, C, →XY **平面內角度** =103.8412
65、 DIst, int, D, mid, E, →**距離**=33.4531
66、 BOundary, **點斜線區域**, (**先解除斜線區域才能執行** BOundary)
67、 **點外圍邊緣**, CRTL+1, →**查幾何長度值** =269.5254
68、 Hatch, **點斜線區域內**, →**斜線區域**
69、 **點斜線區域**, CRTL+1, →**查幾何面積值** =2001.4971

## 2.4.8 AutoCAD2D2008-408

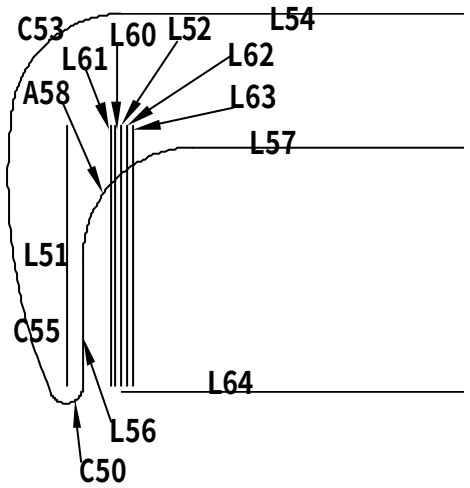



1、 **弧** A **夾角為何？** 133.7083
2、 **四分點** B **至四分點** C **水平距離為何？** 165.2804
3、 **交點** D **至交點** E **距離為何？** 141.3274
4、 **弧** F **長度為何？** 62.4937
5、 G **區域扣除內孔面積為何？** 3180.8710

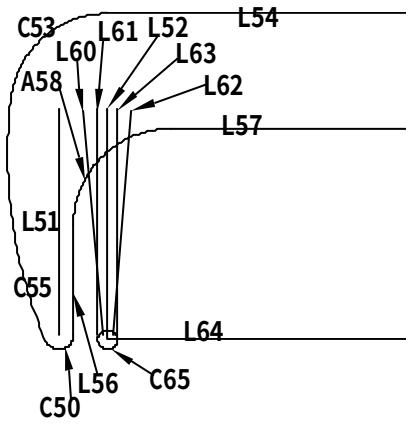
**解題步驟：**



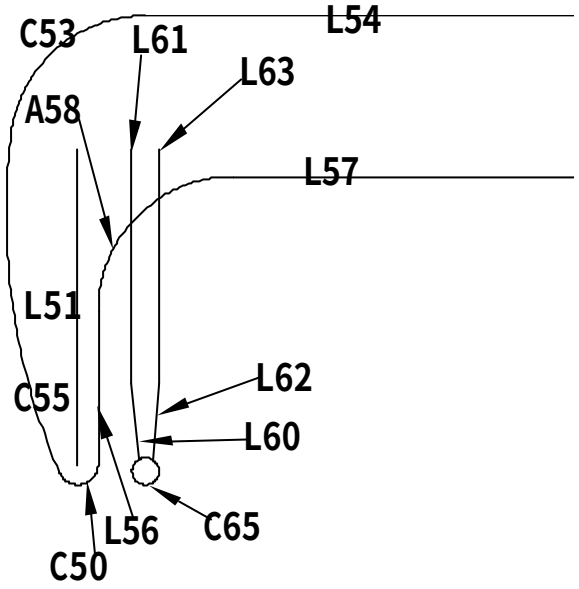

01、 Circle, **任一點** P01, 1.5, →C50
02、 Line, cen, C50, (**正交模式**), **向上**, 23.5, →L51
03、 Offset, 5, L51, **向右點**, →L52
04、 Circle, end, L52 **上**, 10, →C53
05、 Line, qua, C53 **上**, (**正交模式**), **向右**, 140, →L54
06、 Circle, T, C50, C53, 50, →C55
07、 Offset, 1.5, L51, **向右點**, →L56
08、 Offset, 12, L54, **向下點**, →L57



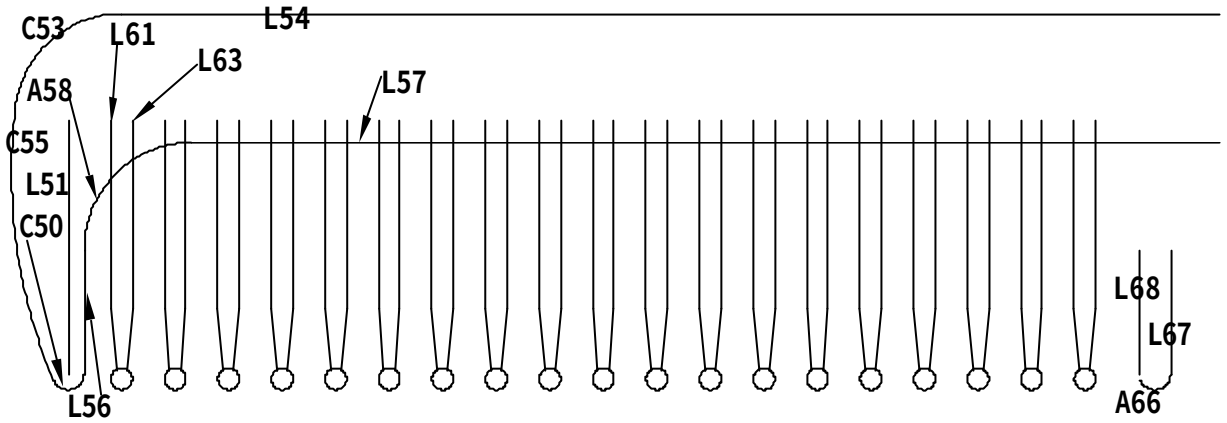

09、 TRim,
10、 Fillet, T, T, R, 10, L56, L57, →A58
11、 Offset, 0.5, L52, **向左點**, →L60, L60, **向左點**, →L61, L52, **向右點**, →L62, L62, **向右點**, →L63
12、 Offset, 34(=35-(2/2)), L54, **向下點**, →L64



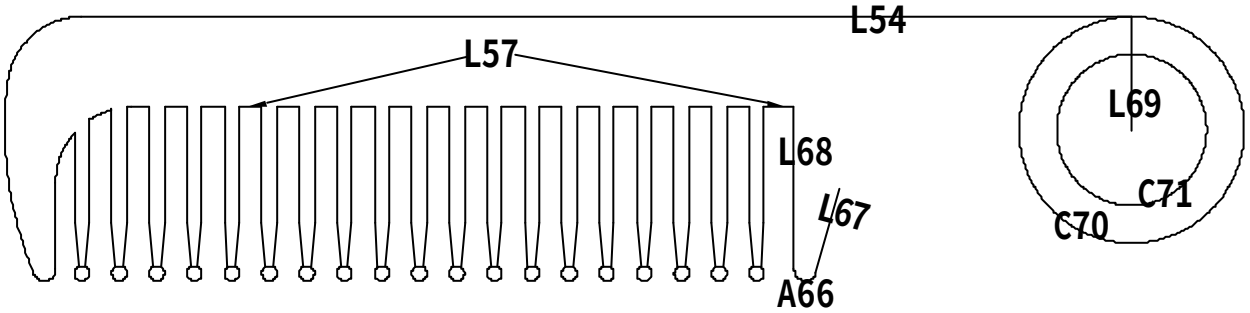

13、 Fillet, T, T, R, 0, L52, L64, →L52 **與** L64**相交**
14、 Circle, int, L52 & L64, D, 2, →C65
15、 ROtate, L60, int, L60 & C65, 5(=10/2), →L60 **轉動**
16、 ROtate, L60, int, L62 & C65, -5(=10/2),→L62 **轉動**



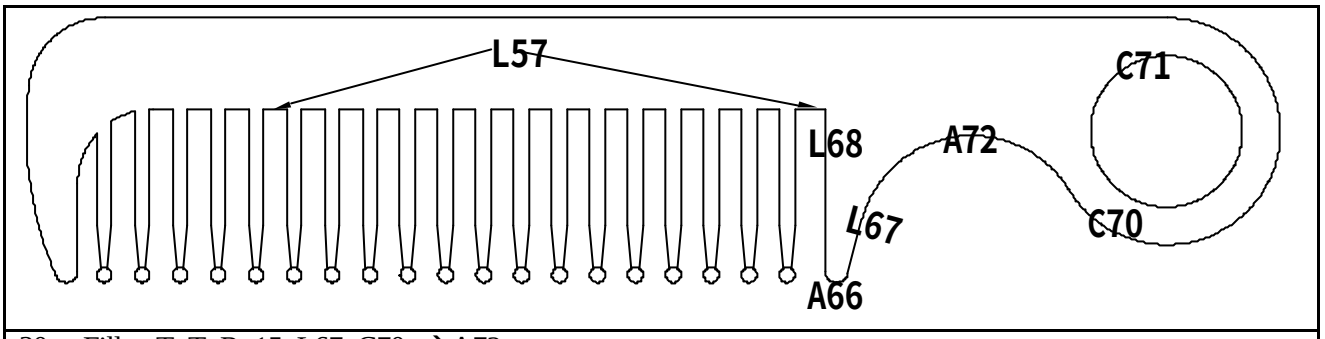

17、 TRim,
18、 Erase,



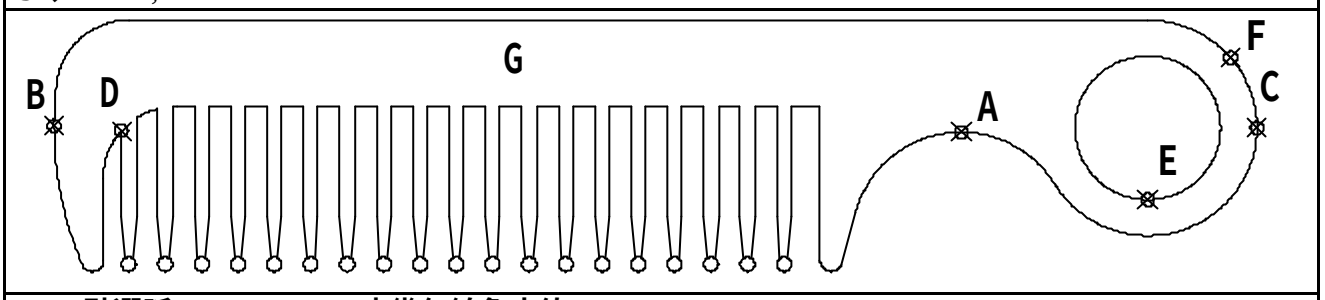

19、 ARray, **矩形陣列**, **列**, 1, **行**, 19, **列偏移**, 0, **行偏移**, 5(=(95-5)/(19-1)), **物件**, L60~L63;C65, →19 **組相同物件**
20、 COpy, C50;L56, cen, C50, (**正交模式**), **向右**, 101.5(=95+6.5), →A66;L67
21、 Offset, 3(=1.5x2), L67, **向左點**, →L68




22、 EXtend, L68, A66 **左上**, →A66 **向上延伸**
23、 EXtend, L57, L68 **上**, →L68 **向上延伸**
24、 ROtate, L67, cen, A66, -15, →L67 **轉動**
25、 TRim,
26、 Erase,
27、 Line, end, L54 **右**, (**正交模式**), **向下**, 15, →L69
28、 Circle, end, L69 **下**, D, 30, →C70
29、 Circle, @(**鎖定上次點**), D, 20, →C71

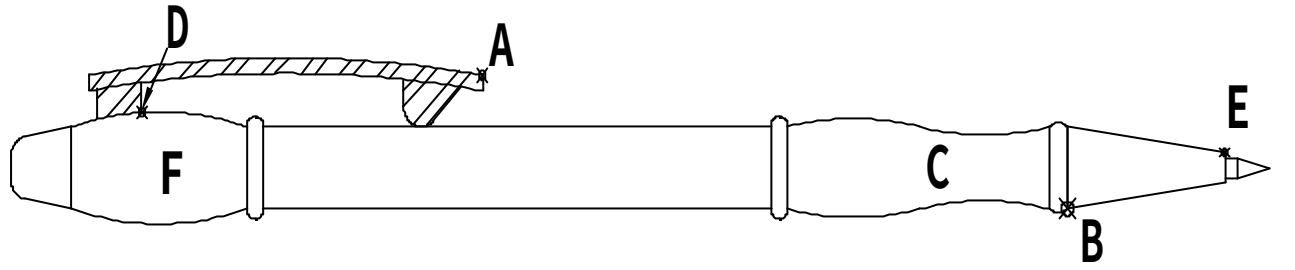



30、 Fillet, T, T, R, 15, L67, C70, →A72
31、 TRim,



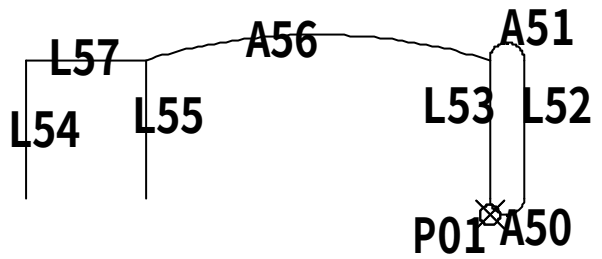

32、 **點選弧** A, CRTL+1, →**查幾何總角度值**=133.7083
33、 DIst, qua, B, qua, C,→X **差值**=165.2804
34、 DIst, int, D, qua, E, →**距離**=141.3274
35、 **點選弧** F, CRTL+1, →**查幾何弧長度值**=62.4937
36、 Hatch, **點** G **區域**, →**斜線區域**
37、 **點** G **斜線區域**, CRTL+1, →**查幾何面積值**=3180.8710

## 2.4.9 AutoCAD2D2008-409

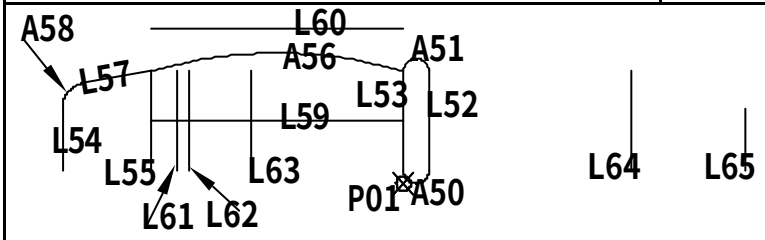



1、 **交點** A **至交點** B **距離為何？** 68.2528
2、 C **周長為何？** 76.4689
3、 **交點** D **至交點** E **距離為何？** 122.0624
4、 F **區域面積為何？** 245.3035
5、 **斜線區域面積為何？** 101.9084

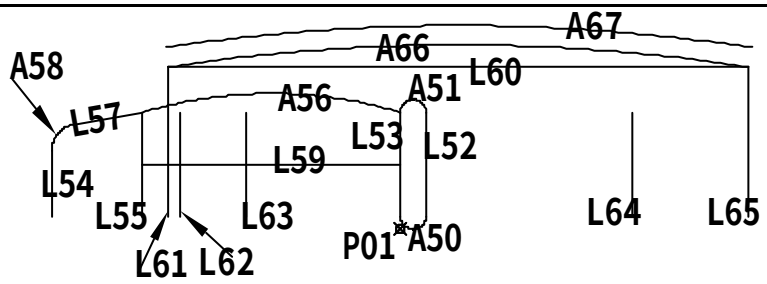
**解題步驟：**



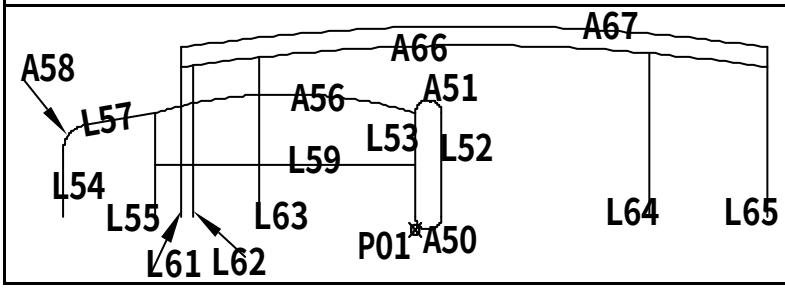

01、 RECtangle, F, 1, **任一點** P01, D, 2, 10(=1+8+1), **向右上點**, →A50~L53 **之圓角矩形**
02、 eXplode, A50~L53 **之圓角矩形**, →A50~L53
03、 Offset, 27, L53, **向左點**, →L54
04、 Offset, 7, L54, **向右點**, →L55
05、 Arc, end, L53 **上**, E, end, L55 **上**, R, 35, →A56
06、 Line, end, L54 **上**, end, L55 **上**, →L57



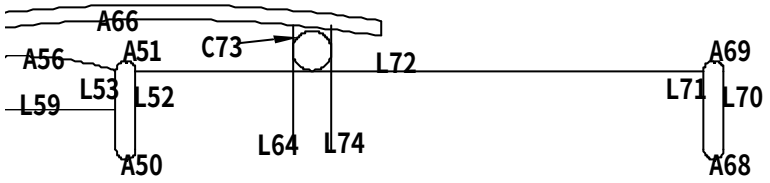

07、 ROtate, L57, int, L55 & A56, 10, →L57 **轉動**
08、 Fillet, T, T, R, 2, L54, L57, →A58
09、 Line, mid, L55, mid, L53, →L59
10、 Offset, 7.5, L59, **向上點**, →L60
11、 Offset, 2, L55, **向右點**, →L61
12、 Offset, 3, L55, **向右點**, →L62
13、 Offset, 5, L62, **向右點**, →L63
14、 Offset, 30, L63, **向右點**, →L64
15、 Offset, 54, L54, **向右點**, →L65



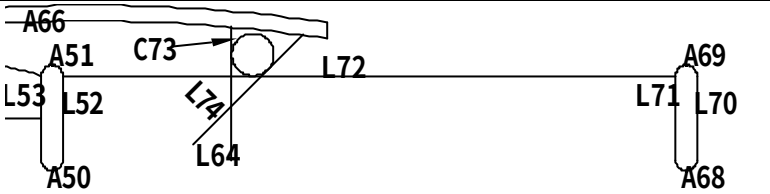

16、 Fillet, T, T, R, 0, L60, L61, →L60 **與** L61 **相交**
17、 Fillet, T, T, R, 0, L60, L65, →L60 **與** L65 **相交**
18、 Arc, int, L60 & L65, E, int, L60 & L61, R, 150, →A66
19、 Offset, 1.5, A66, **向上點**, →A67



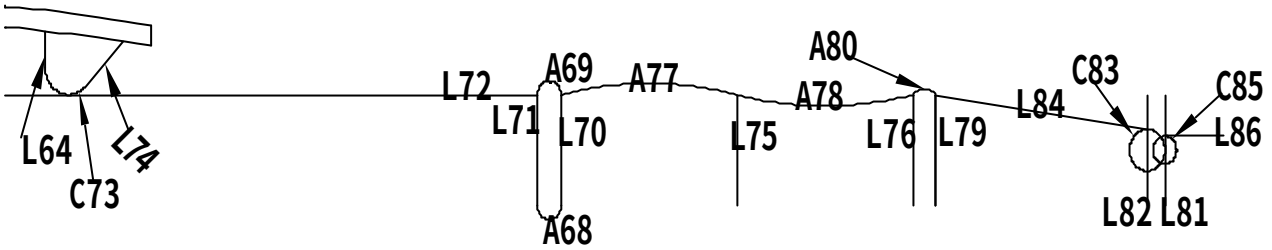

20、 Fillet, T, T, R, 0, L61, A67, →L61 **與** A67 **相交**
21、 Fillet, T, T, R, 0, L65, A67, →L65 **與** A67 **相交**
22、 EXtend, A66, L62;L63;L64, →L62;L63 **及** L64 **皆向上延伸**



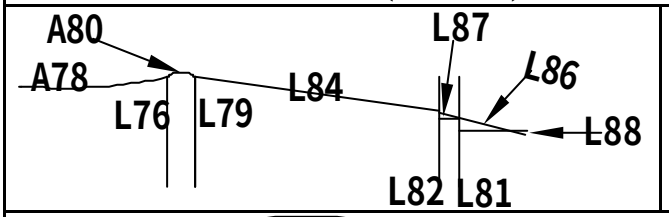

23、 COpy, A50~L53, int, L53 &L59, (**正交模式**), **向右**, 60(=62-2), →A68~L71
24、 Line, int, A51 & L52, int, A69 & L70, →L72
25、 Circle, T, L64, L72, 2, →C73
26、 Offset, 4, L64, **向右點**, →L74



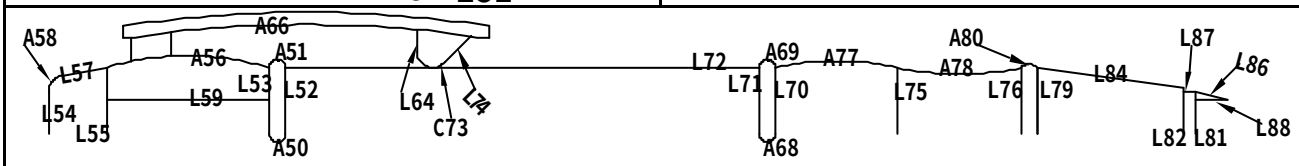

27、 ROtate, L74, cen, C73, -45, →L74 **轉動**
28、 EXtend, A66, L74 **上**, →L74 **向上延伸**



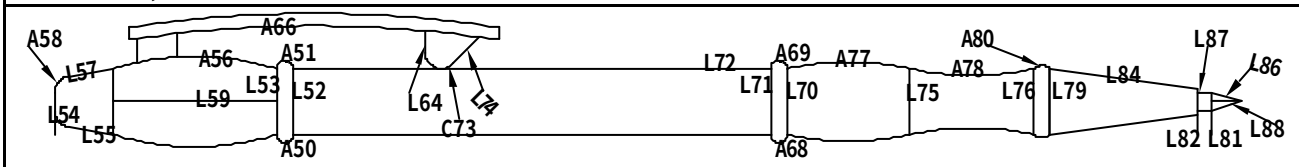

29、 Erase,
30、 TRim,
31、 Offset, 15, L70, **向右點**, →L75
32、 Offset, 30, L70, **向右點**, →L76
33、 Arc, end, L75 **上**, E, int, A69 & L70, R, 35, →A77
34、 Arc, end, L75 **上**, E, end, L76, R, 35, →A78
35、 Offset, 2, L76, **向右點**, →L79
36、 Arc, end, L79 **上**, E, int, L76 & A78, R, 1.5, →A80
37、 Offset, 19.5, L76, **向右點**, →L81
38、 Offset, 1.5, L81, **向左點**, →L82
39、 Circle, mid, L82, D, 3, →C83
40、 Line, int,L79 & A80, int, L82 & C83, →L84
41、 Circle, mid, L81, D, 2, →C85
42、 Line, int, L81 & C85 **上**, (**正交模式**), **向右** 5, →L86



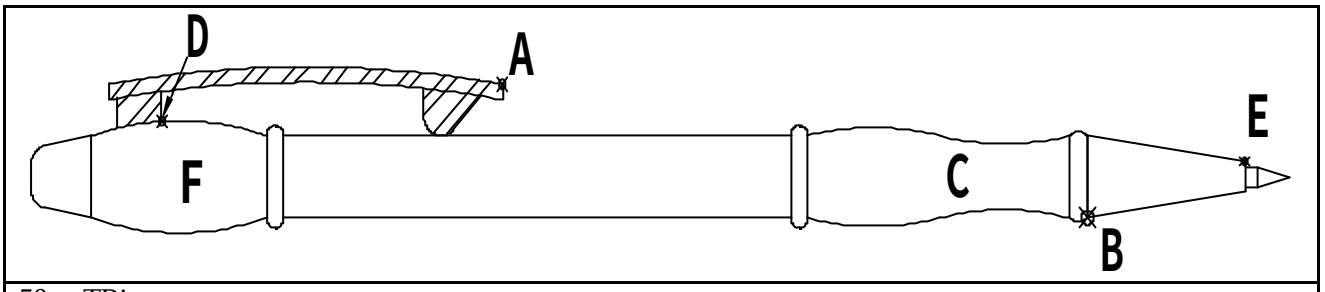

43、 Erase,
44、 ROtate, L86, int, L86 & L81, -15, →L86 **轉動**
45、 Line, int, L86 & L81, (**正交模式**), **向左**, 1.5, →L87
46、 Line, mid, L81, (**正交模式**), **向右**, 5, →L88




47、 TRim,
48、 Erase,

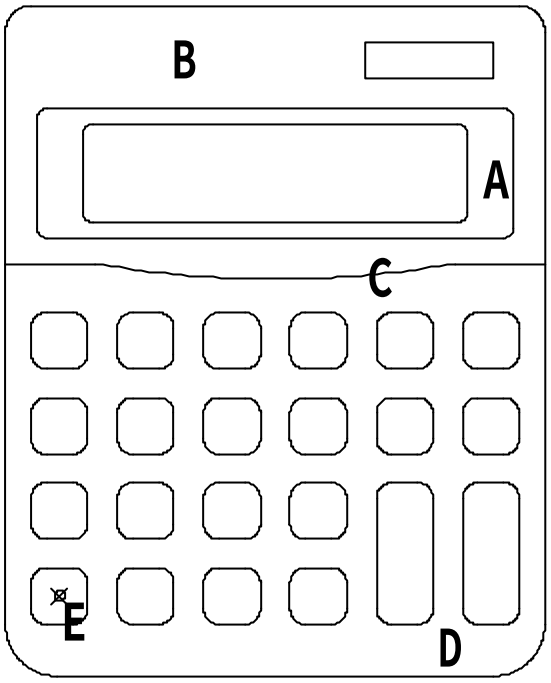



49、 MIrror, A58;L57;A56;L72;A77;A78;A80;L84;L87;L86, mid, L52, mid, L53, →**下側相同物件**



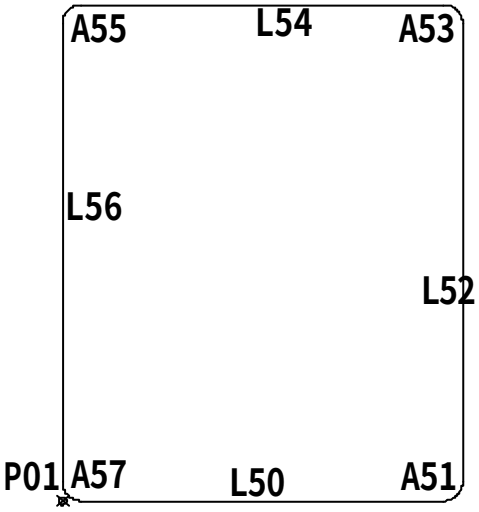

50、 TRim,
51、 Erase,
52、 DIst, int, A, int, B, →**距離**=68.2528
53、 BOundary, **點** C **區域**,
54、 **點選** C **區域邊緣**, CRTL+1, →**查幾何弧長度值**=76.4689
55、 DIst, int, D, int, E, →**距離**=122.0624
56、 Hatch, **點** F **區域**, →**斜線區域**
57、 **點** F **斜線區域**, CRTL+1, →**查幾何面積值**=245.3035
58、 Hatch, **點斜線區域內**, →**斜線區域**
59、 **點斜線區域**, CRTL+1, →**查幾何面積值**=101.9084

## 2.4.10 AutoCAD2D2008-410

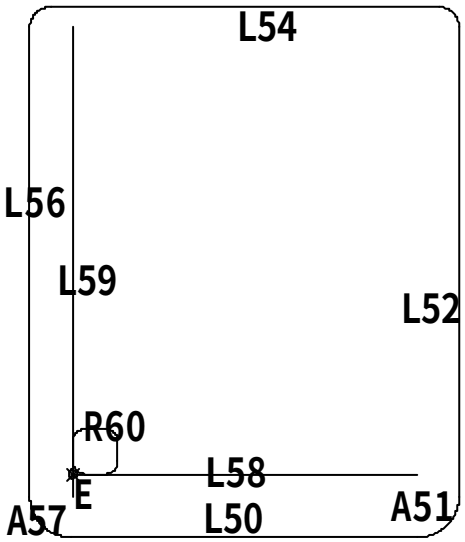



1、 A **區域扣除內孔面積為何？** 900.0000
2、 B **區域扣除內孔面積為何？** 2882.9090
3、 **弧** C **半徑為何？** 205.6667
4、 D **區域扣除按鍵面積為何？** 5361.8388
5、 **交點** E **至弧** C **中心點距離為何？** 270.4478

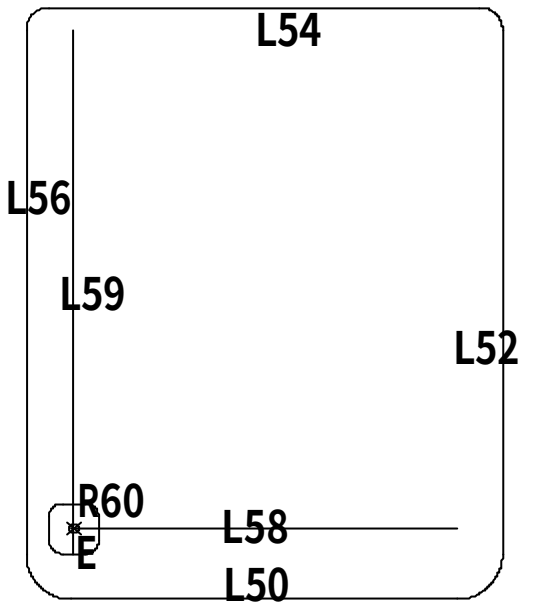
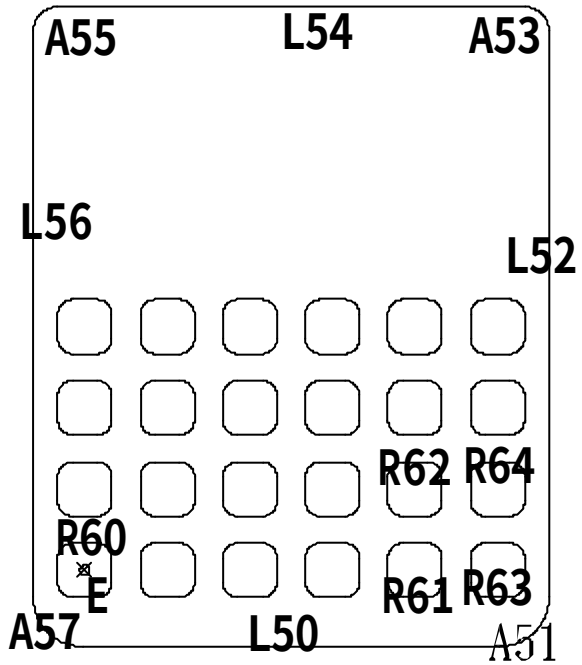
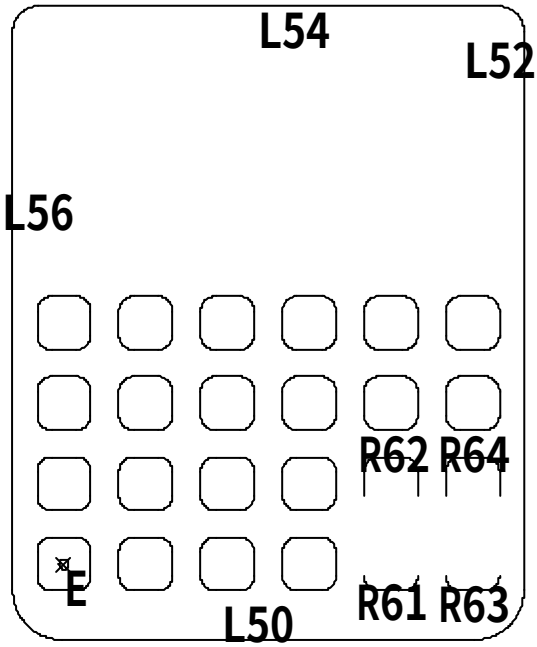
**解題步驟：**

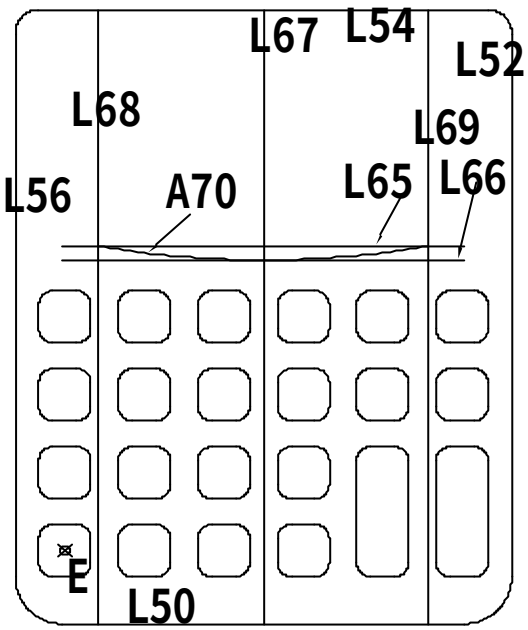



01、 RECtangle, F, 5, **任一點** P01, D, 105, 130, **向右上點**, →L50~A57 **之圓角矩形**
02、 eXplode, L50~A57 **之圓角矩形**, →L50~A57

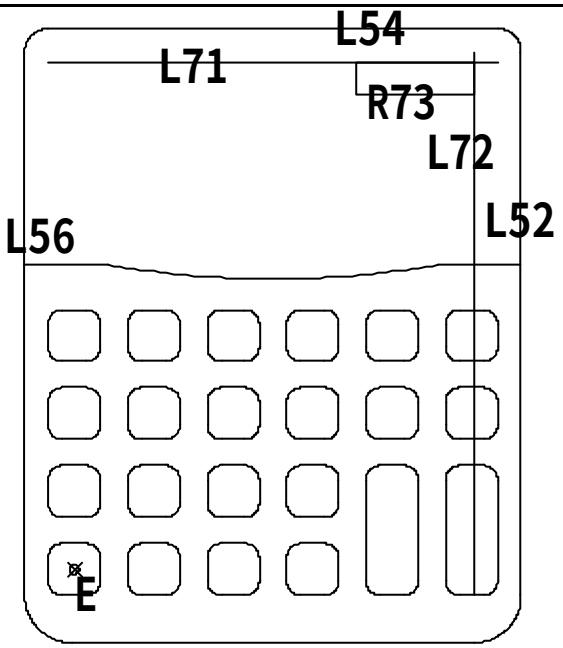



03、 Erase, A57;A51
04、 Fillet, T, T, R,10, L50, L52, →A51(R5 **改** R10)
05、 Fillet, T, T, R,10, L50, L56, →A57(R5 **改** R10)
06、 Offset, 15.5, L50, **向上點**, →L58
07、 Offset, 10.5, L56, **向右點**, →L59
08、 RECtangle, F, 3, int, L58 & L59, D, 11, 11, **向右上點**, →R60 **之圓角矩形**

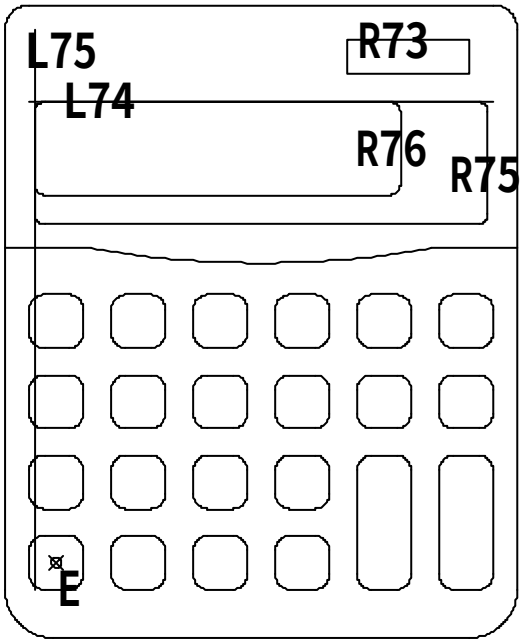
| | |
|---|---|
|  | 09、 Move, R60, int, L58 & L59, @-5.5,-5.5, →R60 **向左下移動**<br>10、 POint, int, L58 & L59, →E **點** |
|  | 11、 Erase,<br>12、 ARray, **矩形陣列**, **列**, 4, **行**, 6, **列偏移**, 16.5, **行偏移**, 16.8, **物件**, R60, →R61~R64 **等** 24 **個相同物件**<br>13、 eXplode, R61~R64 **之圓角矩形**, →R61~R64 **等物件** |
|  | 14、 Erase, R61~R64 **之直線與弧** |



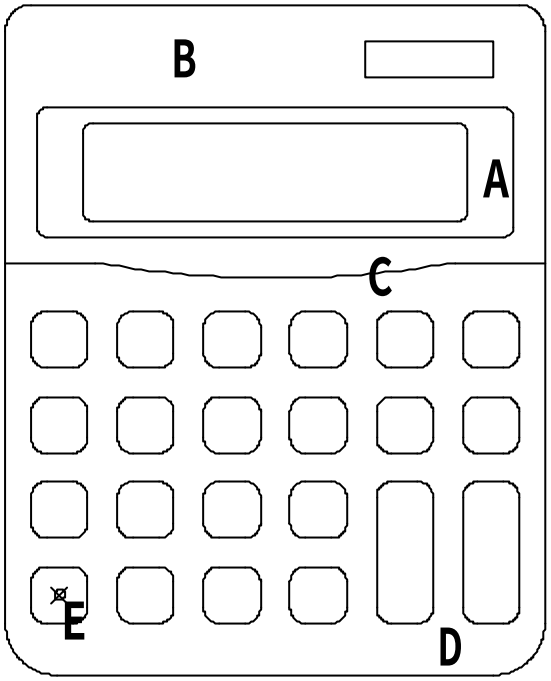

15、 EXtend, R61 弧;R63 弧, R62 直線;R64 直線, →R61~R64 互相相交
16、 Offset, 80, L50, 向上點, →L65
17、 Offset, 3, L65, 向下點, →L66
18、 Line, mid, L50, mid, L54, →L67
19、 Offset, 35(=70/2), L67, 向左點, →L68
20、 Offset, 35(=70/2), L67, 向右點, →L69
21、 Arc, int, L65 & L68, int, L66 & L67, int, L65 & L69, →A70




22、 Erase,
23、 EXtend, L52;L56, L65 左;L65 右, →L65 向左右延伸
24、 TRim,
25、 Offset, 7, L54, 向下點, →L71
26、 Offset, 10, L52, 向左點, →L72
27、 RECtangle, F, 0, int, L71 & L72, D, 25, 7, 向左下點, →R73 之矩形




28、 Erase,
29、 Offset, 20, L54, 向下點, →L74
30、 Offset, 6, L56, 向右點, →L75
31、 RECtangle, F, 2, int, L74 & L75, D, 93(=105-6-6), 25, 向右下點, →R75 之圓角矩形
32、 RECtangle, F, 2, int, L74 & L75, D, 75, 19, 向右下點, →R76 之圓角矩形




33、 Move, R76, int, L74 & L75, @9,-3, →R76 **向右下移動**

34、 Erase,

35、 Hatch, **點** A **區域**, **→斜線區域**

36、 **點** A **斜線區域**, CRTL+1, **→查幾何面積值** =900.0000

37、 Hatch, **點** B **區域**, B, **點** R76, **→斜線區域**

38、 **點** B **斜線區域**, CRTL+1, **→查幾何面積值** =2882.9090

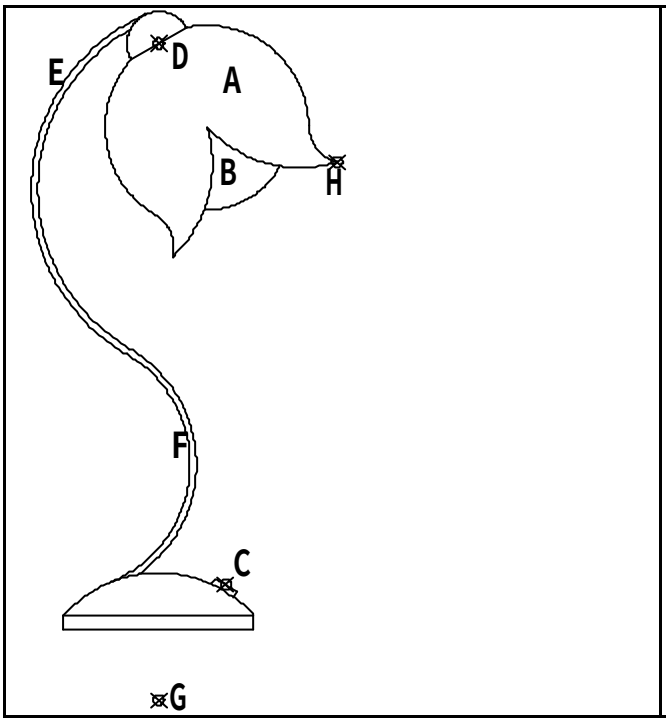
39、 **點選弧** C, CRTL+1, **→查幾何半徑值** =205.6667

40、 Hatch, **點** D **區域**, **→斜線區域**

41、 **點** D **斜線區域**, CRTL+1, **→查幾何面積值** =5361.8388

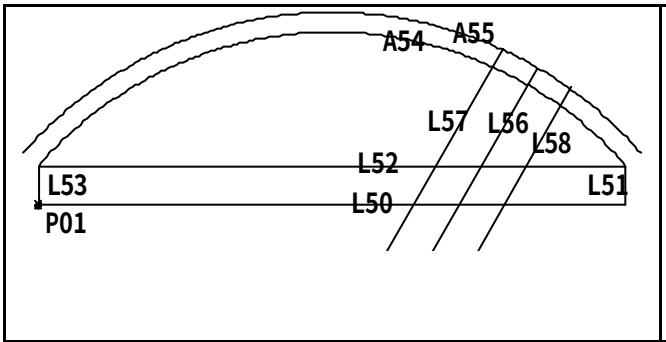
42、 DIst, nod, E, cen, C, **→距離**=270.4478

## 2.4.11 AutoCAD2D2008-411

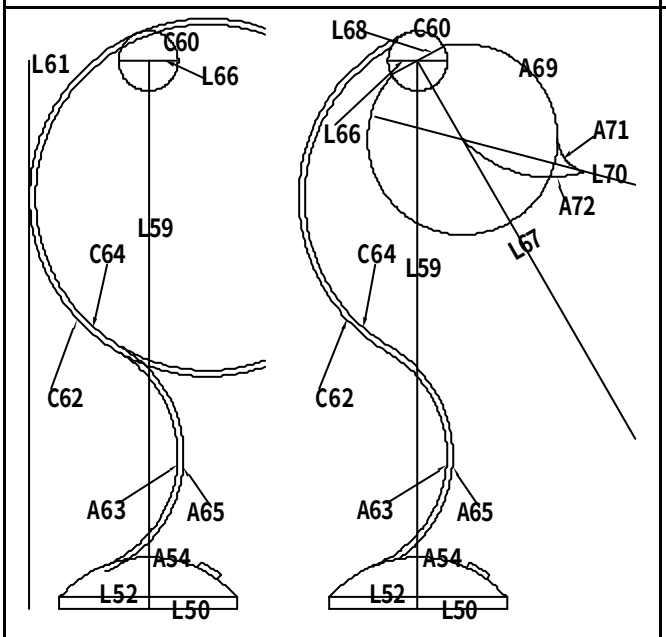



1、 A **區域面積為何？** 17359.3458
2、 B **區域周長為何？** 205.0501
3、 **中點** C **至中心點** D **距離為何？** 428.4399
4、 **中心點** E **至中心點** F **高度為何？** 216.3112
5、 **中心點** G **至交點** H **距離為何？** 445.5791

**解題步驟：**



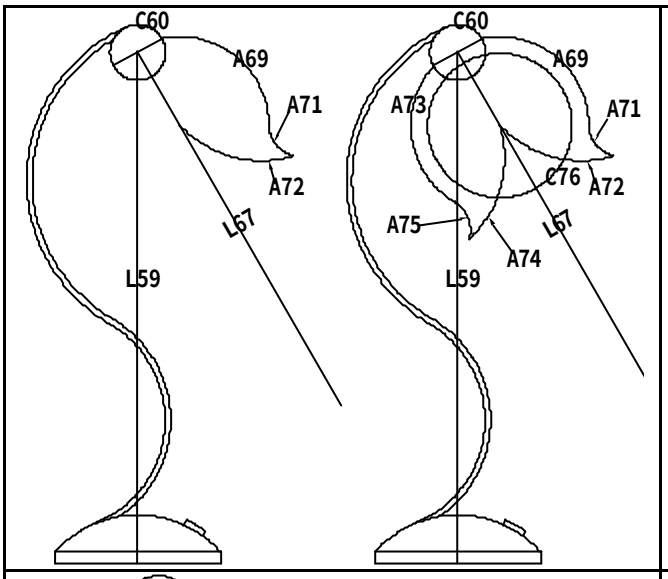
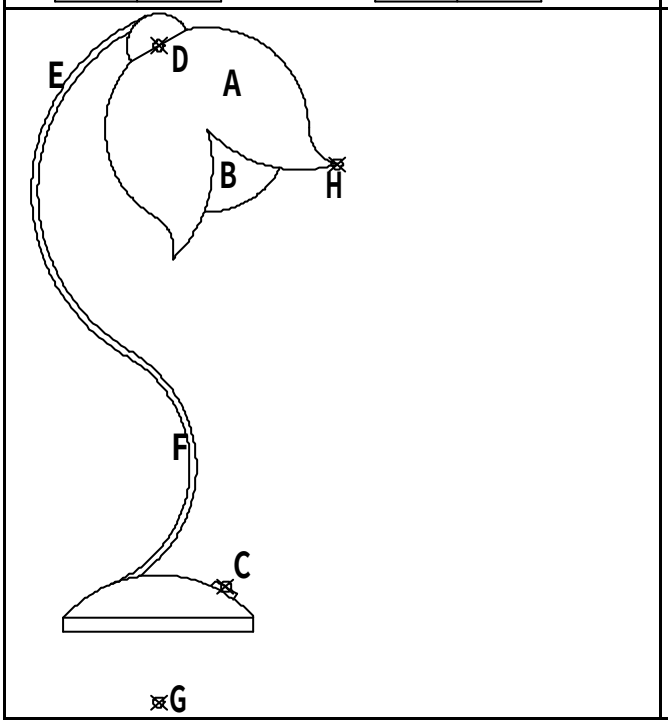

01、 RECtangle, F, 0, **任一點** P01, D, 150(=75X2), 10, **向右上點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Arc, end, L51 **上**, E, end, L53 **上**, R, 100, →A54
04、 Offset, 5, A54, **向上點**, →A55
05、 Line, cen, A54, @105<60, →L56
06、 Offset, 10(=20/2), L56, **向上點**, →L57, L56, **向下點**, →L58



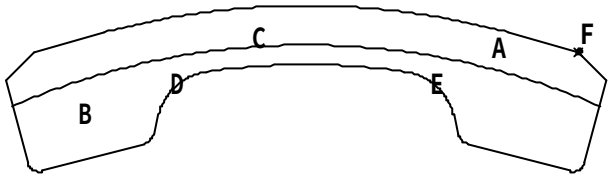

07、 TRim,
08、 Erase,
09、 Line, mid, L50, (**正交模式**), **向上** 460, →L59
10、 Circle, end, L59 **上**, 25, →C60
11、 Offset, 100, L59, **向左點**, →L61
12、 Circle, T, C60 **左上**, L61, 150, →C62
13、 Fillet, T, N, R, 100(=105-5), C62 **左下**, A54 **左上**, →A63
14、 Offset, 5, C62, **向右點**, →C64
15、 Offset, 5, A63, **向右點**, →A65
16、 Line, qua, C60 **左**, qua, C60 **右**, →L66

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

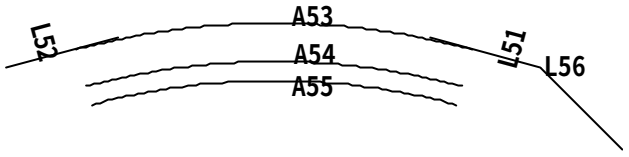
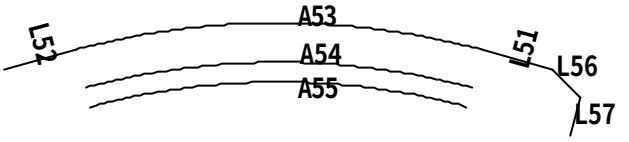
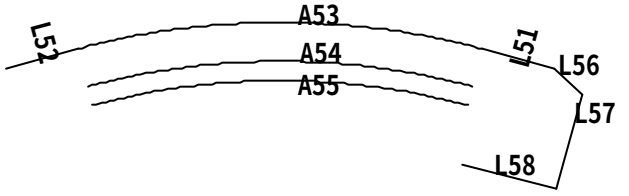
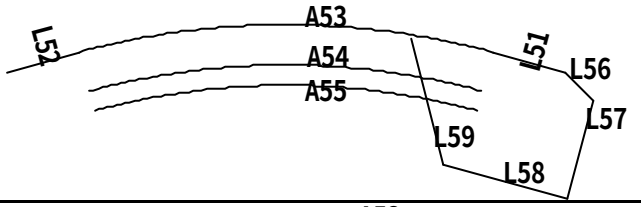
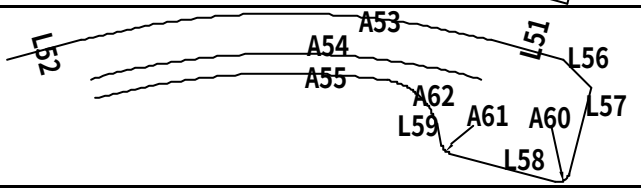
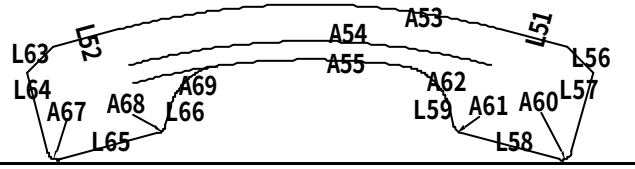
17、 TRim,
18、 Erase,
19、 ROtate, L59;L66, cen, C60, C, 30, →L67;L68
20、 Arc, end, L68 **下**, E, end, L68 **上**, R, -80(=160/2) →A69
21、 ROtate, L67, cen, A69, C, 45, →L70
22、 Fillet, T, N, R, 30, A69 **右下**, L70, →A71
23、 Arc, cen, A69, E, int, L70 & A71, R, 100 →A72

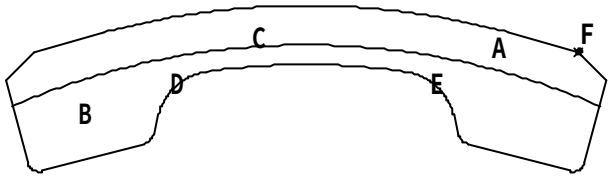
| | |
|---|---|
|  | 24、 TRim,<br>25、 Erase,<br>- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -<br>26、 MIrror, A69;A71;A72, cen, C60, int, L67 & A72, →A73~A75<br>27、 Circle, cen, A69, 65, →C76 |
|  | 28、 TRim,<br>29、 Erase,<br>30、 Hatch, **點** A **區域**, **→斜線區域**<br>31、 **點** A **斜線區域**, CRTL+1, **→查幾何面積值** =17359.3458<br>32、 BOundary, **點** B **區域**,<br>33、 **點** B **區域邊緣**, CRTL+1, **→查幾何長度值** =205.0501<br>34、 DIst, mid, C, cen, D, **→距離**=428.4399<br>35、 DIst, cen, E, cen, F, **→**Y **差值**=216.3112<br>36、 DIst, cen, G, int, H, **→距離**=445.5791 |

## 2.4.12 AutoCAD2D2008-412

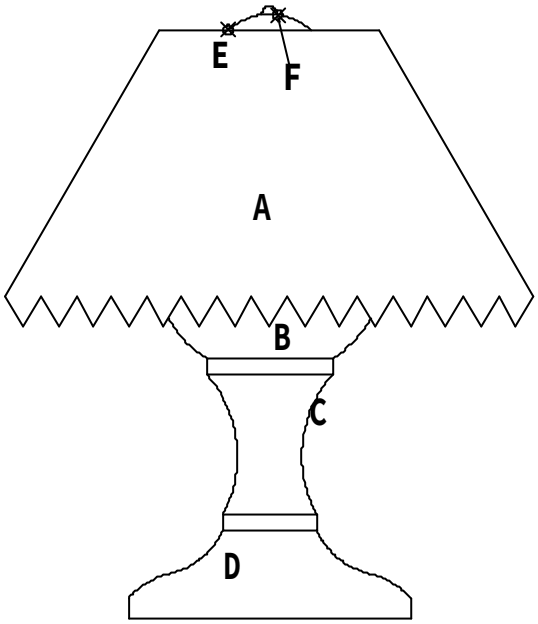
| | |
|---|---|
|  | 1、 A **區域面積為何？** 1565.2718<br>2、 B **區域周長為何？** 351.9501<br>3、 **弧** C **夾角為何？** 46.6683<br>4、 **弧** D **至弧** E **中心點距離為何？** 46.7015<br>5、 **交點** F **至弧** D **中心點距離為何？** 95.4504 |

**解題步驟：**

| | |
|---|---|
| | 01、 Line, **任一點** P01, (**正交模式**), **向上** 30, →L51<br>02、 Offset, 140, L51, **向左點**, →L52 |
|  | 03、 ROtate, L51, end, L51 **下**, 75, → L51 **轉動**<br>04、 ROtate, L52, end, L52 **下**, -75, → L52 **轉動**<br>05、 Fillet, T, T, R, 200, L51, L52, →A53<br>06、 Offset, 10, A53, **向下點**, →A54<br>07、 Offset, 5, A54, **向下點**, →A55<br>08、 ROtate, L51, end, L51 **右**, C, 150(=180-30), →L56 |
|  | 09、 LENgthen, T, 10, L56 **右**, →L56 **縮短**<br>10、 ROtate, L56, end, L56 **右**, C, 120, →L57 |
|  | 11、 LENgthen, T, 25, L57 **下**, →L57 **向下延伸**<br>12、 ROtate, L57, end, L57 **下**, C, 90, →L58 |
|  | 13、 LENgthen, T, 32, L58 **左**, →L58 **向左延伸**<br>14、 ROtate, L58, end, L58 **左**, C, 120, →L59 |
|  | 15、 Fillet, T, T, R, 3, L57, L58, →A60, L58, L59, →A61,<br>16、 Fillet, T, T, R, 15, A55, L59, →A62 |
|  | 17、 MIrror, L56~L59;A60~A61, qua, A53, qua, A55, →L63~L66;A67~A69 |

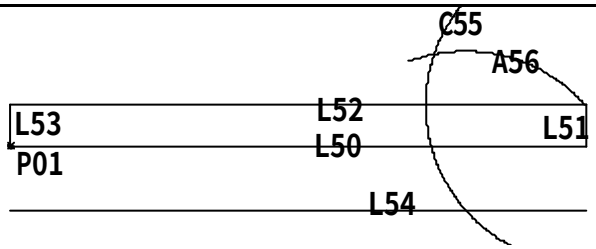
| | |
|---|---|
|  | 18、 TRim,<br>19、 Erase,<br>20、 EXtend, L57;L64, A54 **左**;A54 **右**, →A54 **左側及右側分別與** L64 **及** L57 **相交**<br>21、 Hatch, **點** A **區域**, →**斜線區域**<br>22、 **點** A **斜線區域**, CRTL+1, →**查幾何面積值** =1565.2718<br>23、 BOundary, **點** B **區域**,<br>24、 **點** B **區域邊緣**, CRTL+1, →**查幾何長度值** =351.9501<br>25、 **點選弧** C, CRTL+1, →**查幾何總角度值** =46.6683<br>26、 DIst, cen, D, cen, E, →**距離**=46.7015<br>27、 DIst, int, F, cen, D, →**距離**=95.4504 |

## 2.4.13 AutoCAD2D2008-413

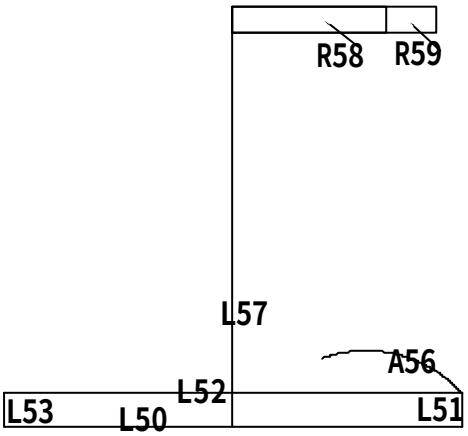



1、 A **區域面積為何？** 43750.2542
2、 B **區域周長為何？** 409.3573
3、 **弧** C **長度為何？** 96.2559
4、 D **區域面積為何？** 7121.8668
5、 **交點** E **至交點** F **距離為何？** 32.9060

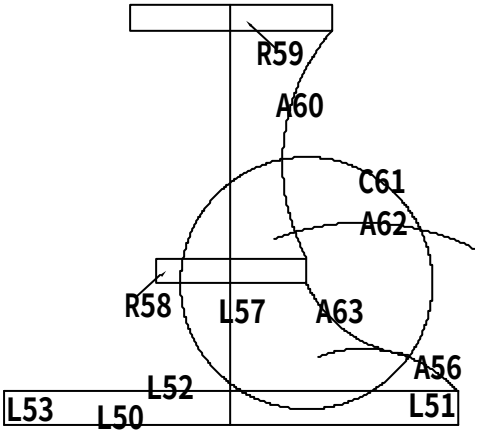
**解題步驟：**



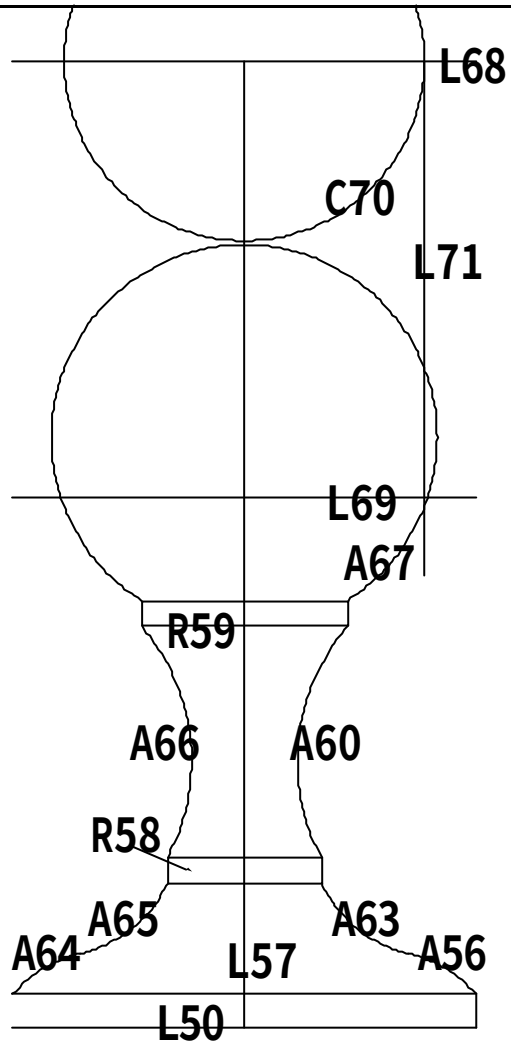

01、 RECtangle, F, 0, **任一點** P01, D, 180, 13, **向右上點**, →L50~ L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 20(=50-30), L50, **向下點**, →L54
04、 Circle, int, L51 & L52, 50, →C55
05、 Arc, C, int, L54 & C55, int, L51 & L52, int, C55 & L52, →A56



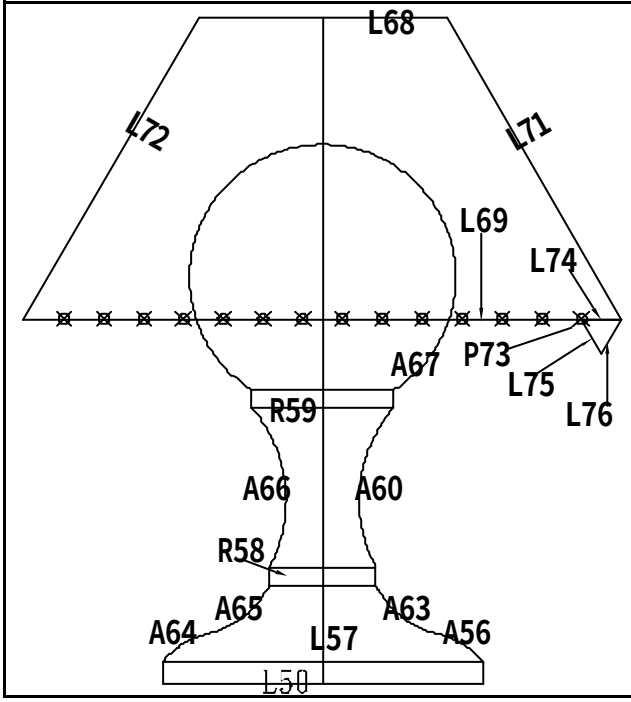

06、 Erase,
07、 Line, mid, L50, (**正交模式**), **向上** 166, →L57
08、 RECtangle, F, 0, end, L57 **上**, D, 60, 10, **向右下點**, →R58 **之矩形**
09、 RECtangle, F, 0, end, L57 **上**, D, 80, 10, **向右下點**, →R59 **之矩形**



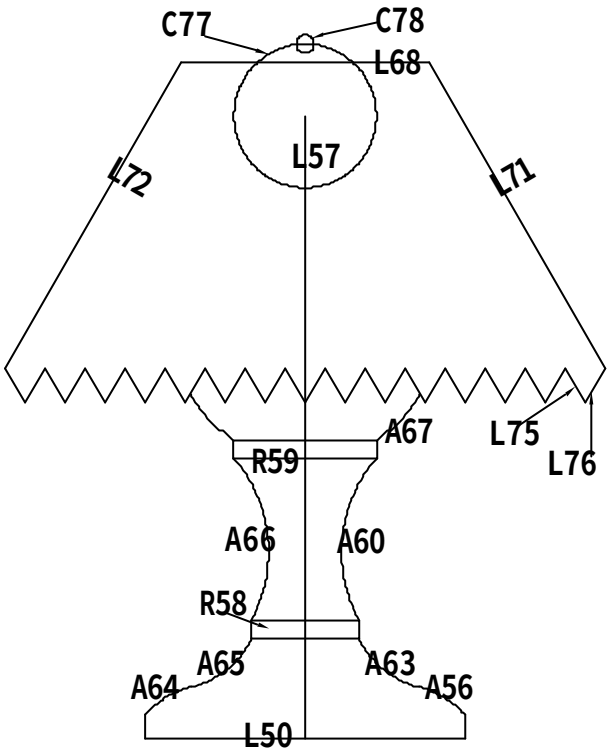
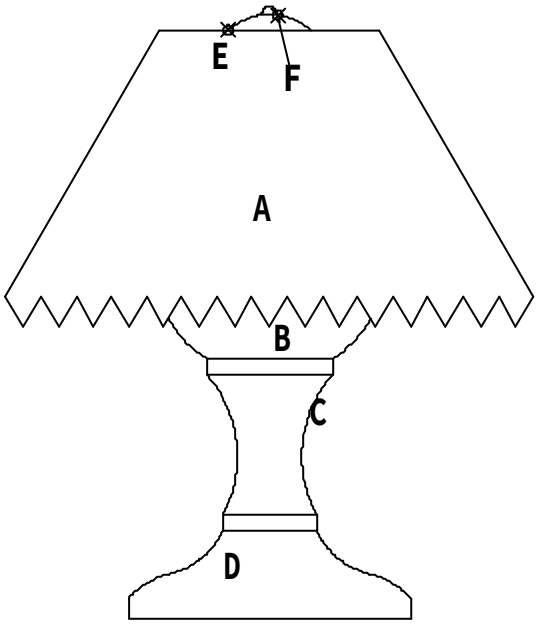

10、 Move, R58, end, L57 **上**, @-30,-100, →R58 **向左下移動**
11、 Move, R59, end, L57 **上**, @-40,0, →R58 **向左移動**
12、 Arc, end, R59 **右下**, E, end, R58 **右上**, R, 80, →A60
13、 Circle, end, R58 **右下**, 50, →C61
14、 Offset, 50, A56, **向上點**, →A62
15、 Arc, C, int, C61 & A62, end, R59 **右下**, int, L51 & L52, →A63



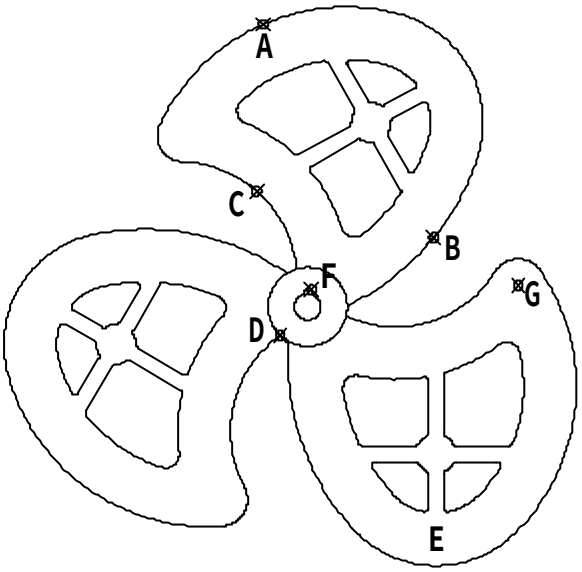

16、 Erase,
17、 TRim,
18、 MIrror, A56;A63;A60, mid, R58 上, mid, R59 上, →A64~A66
19、 Arc, end, R59 右上, E, end, R59 左上, R, -75, →A67
20、 Offset, 376, L50, 向上點, →L68
21、 Offset, 170, L68, 向下點, →L69
22、 EXtend, L68, L57, →L57 與 L68 相交
23、 Circle, int, L57 & L68, D, 140, →C70
24、 Line, qua, C70 右, (正交模式), 向下 200, →L71




25、 Erase,
26、 TRim,
27、 ROtate, L71, int, L71 & L68, 30,→L71 轉動
28、 MIrror, L71, mid, L69, mid, L68, →L72
29、 Fillet, T, T, R, 0, L69, L71 →L69 與 L71 閉合
30、 Fillet, T, T, R, 0, L68, L71 →L68 與 L71 閉合
31、 Fillet, T, T, R, 0, L68, L72 →L68 與 L72 閉合
32、 Fillet, T, T, R, 0, L68, L72 →L68 與 L72 閉合
33、 DIVide, L69, 15, →P73 等 14 個點
34、 POLygon, 3, E, int, L71 & L69, nod, P73, →L74~L76 之正 3 角形
35、 eXplode, L74~L76 之正 3 角形, →L74~L76

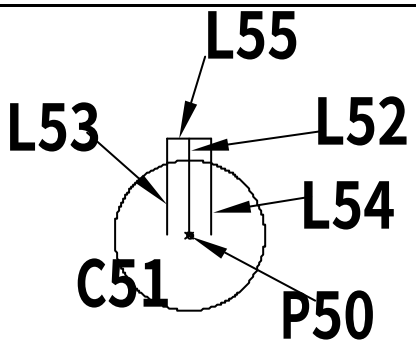
| | |
|---|---|
|  | 36、 ARray, **矩形陣列**, **列**, 1, **行**, 15, **列偏移**, 0, **行偏移**, int, L71 & L69, nod, P73, **物件**, L75;L76, →L75;L76 **相同物件** 15 **組**<br>37、 Erase,<br>38、 LENgthen, DE, -30(=40-10), **點** L57 **上**, →L57 **向下縮短**<br>39、 Circle, end, L57 **上**, 40, →C77<br>40、 Circle, Qua, C77 **上**, 5, →C78 |
|  | 41、 TRim,<br>42、 Erase,<br>43、 Hatch, **點** A **區域**, →**斜線區域**<br>44、 **點** A **斜線區域**, CRTL+1, →**查幾何面積值** =43750.2542<br>45、 BOundary, **點** B **區域**,<br>46、 **點** B **區域邊緣**, CRTL+1, →**查幾何長度值** =409.3573<br>47、 **點選弧** C, CRTL+1, →**查幾何弧長值** =96.2559<br>48、 Hatch, **點** D **區域**, →**斜線區域**<br>49、 **點** D **斜線區域**, CRTL+1, →**查幾何面積值** =7121.8668<br>50、 DIst, int, E, int, F, →**距離**=32.9060 |

## 2.4.14 AutoCAD2D2008-414

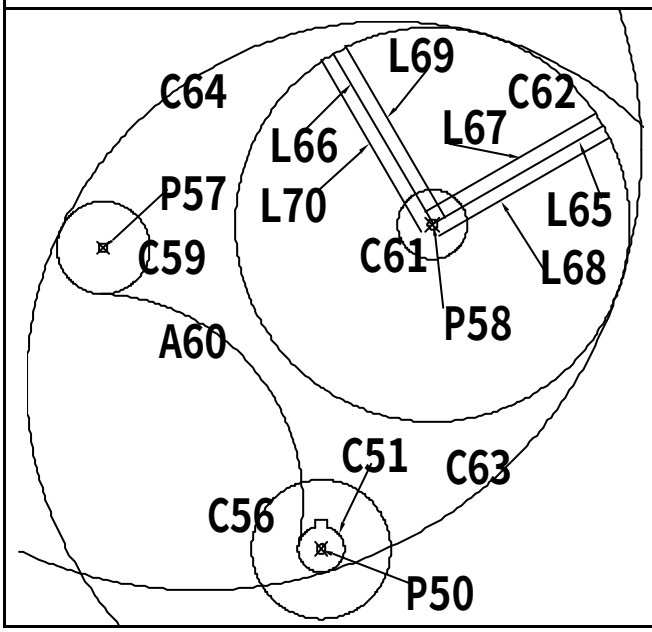



1、 **弧** A **至弧** B **中心點距離為何？** 64.5416
2、 **弧** B **包含角為何？** 46.5243
3、 **弧** C **中心點至交點** D **距離為何？** 43.8872
4、 E **區域扣除內孔面積為何？** 6494.0320
5、 **交點** F **至中心點** G **距離為何？** 78.3102

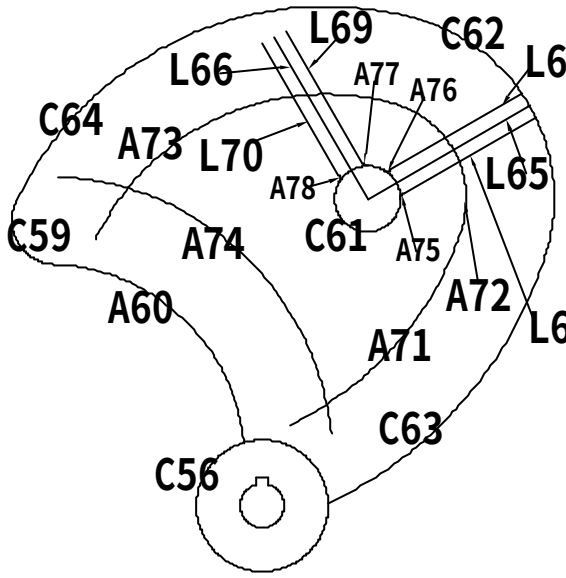
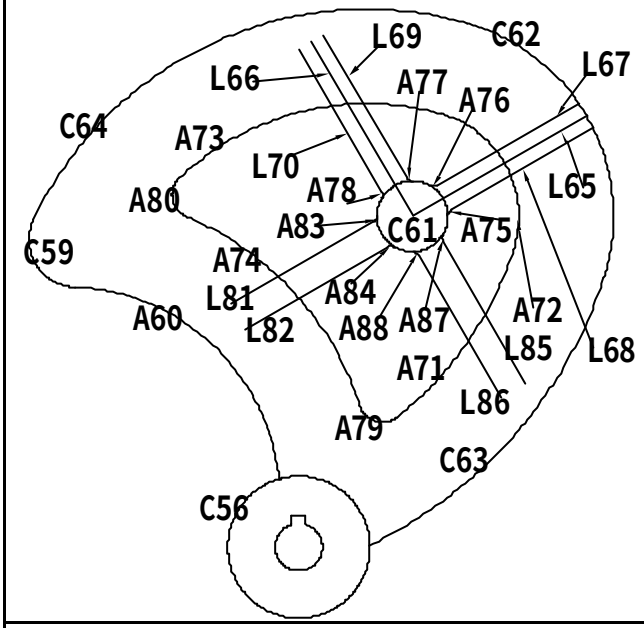
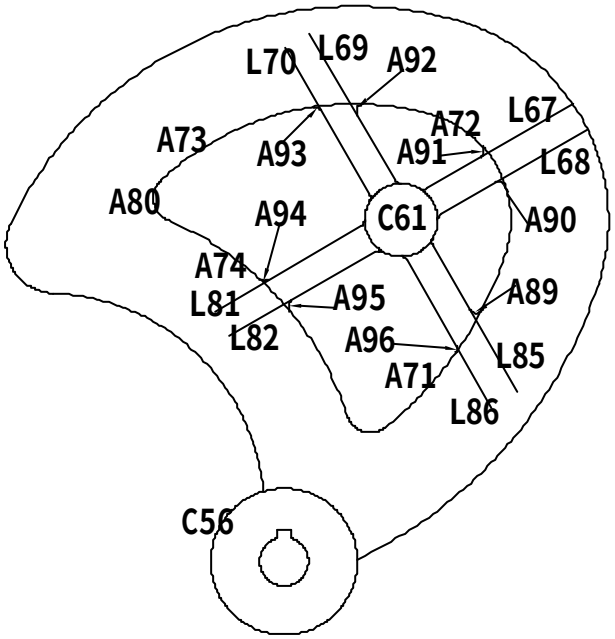
**解題步驟：**

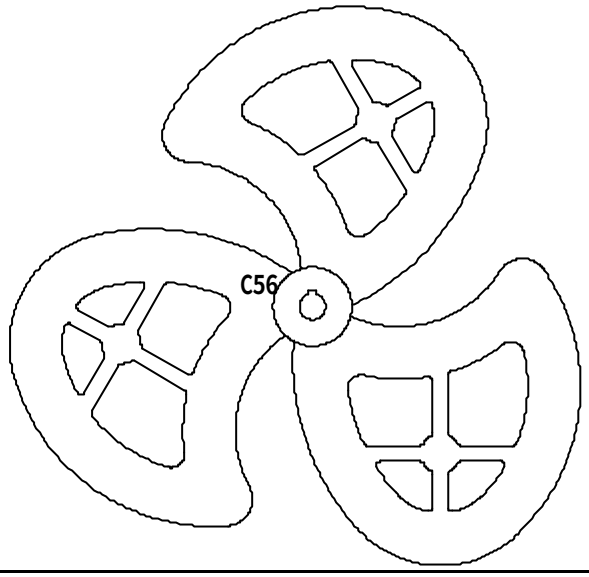
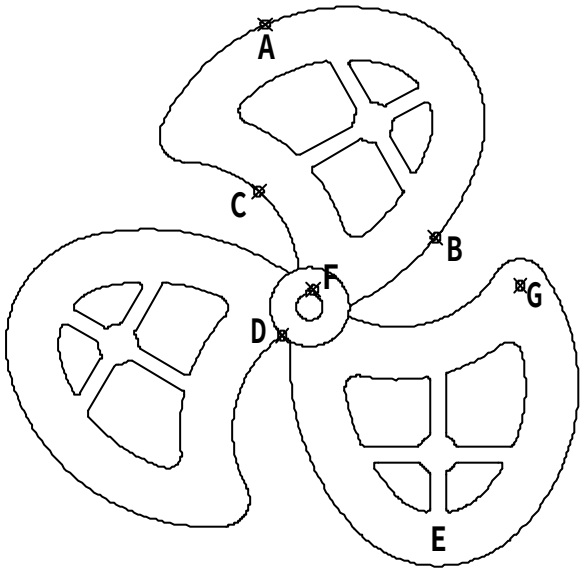



01、 POint, **任一點**, →P50
02、 Circle , nod, P50, D, 10, →C51
03、 Line, nod, P50, (**正交模式**), **向上** 6.5, →L52
04、 Offset, 1.5(=3/2), L52, **向左點**, →L53, L52, **向右點**, →L54
05、 Line, end, L52 **上**, end, L54 **上**, →L55

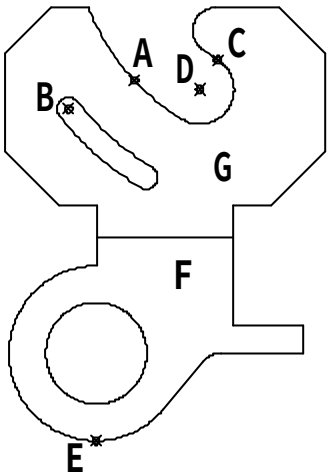



06、 TRim,
07、 Erase,
08、 Circle, cen, C51, D, 30, →C56
09、 COpy, P50, nod, P50, @-47,65, →P57, @24,70, →P58
10、 Circle, nod, P57, 10, →C59
11、 Fillet, T, N, R, 45, C51 **左**, C59 **下**, →A60
12、 Circle, nod, P58, D, 15, →C61
13、 Circle, @(**鎖定上次點**), 42.5, →C62
14、 Circle, T, C51 **右下**, C62 **右下**, 95, →C63
15、 Circle, T, C59 **左上**, C62 **上**, 78, →C64
16、 Line, cen, C61, @42.5<30, →L65
17、 Line, cen, C61, @42.5<120, →L66
18、 Offset, 3(=6/2), L65, **向上點**, →L67, L65, **向下點**, →L68, L66, **向上點**, →L69, L66, **向下點**, →L70

| | |
|---|---|
|  | 19、 TRim,<br>20、 Erase,<br>21、 Offset, 20, C63, **向上點**, →A71, C62, **向下點**, →A72, C64, **向下點**, →A73, A60, **向上點**, →A74,<br>22、 Fillet, T, T, R, 1.5, C61, L68, →A75<br>23、 Fillet, T, T, R, 1.5, C61, L67, →A76<br>24、 Fillet, T, T, R, 1.5, C61, L69, →A77<br>25、 Fillet, T, T, R, 1.5, C61, L70, →A78 |
|  | 26、 Fillet, T, T, R, 5, A71, A74, →A79,<br>27、 Fillet, T, T, R, 5, A74, A73, →A80<br>28、 MIrror, L67;L68;A75;A76, end, L66 **下**, int, C61 & L66, →L81;L82;A83;A84<br>29、 MIrror, L69;L70;A77;A78, end, L65 **下**, int, C61 & L65, →L85;L86;A87;A88 |
|  | 30、 Fillet, T, N, R, 1.5, L85, A71, →A89<br>31、 Fillet, T, N, R, 1.5, L68, A72, →A90<br>32、 Fillet, T, N, R, 1.5, L67, A72, →A91<br>33、 Fillet, T, N, R, 1.5, L69, A73, →A92<br>34、 Fillet, T, N, R, 1.5, L70, A73, →A93<br>35、 Fillet, T, N, R, 1.5, L81, A74, →A94<br>36、 Fillet, T, N, R, 1.5, L82, A74, →A95<br>37、 Fillet, T, N, R, 1.5, L86, A71, →A96 |

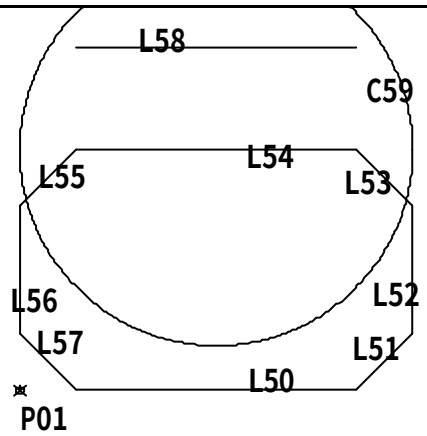
| | |
|---|---|
|  | 38、 TRim,<br>39、 Erase,<br>40、 ARray, **環形陣列**, **中心點**, cen, C56, **項目總數**, 3, **佈滿角度**, 360, **物件**, **所有物件**(**除** C56 **圓內物件外**), →**全圓相同物件** 3 **組** |
|  | 41、 DIst, cen, A, cen, B, →**距離**=64.5416<br>42、 **點選弧** B, CRTL+1, →**查幾何總角度值** =46.5243<br>43、 DIst, cen, C, int, D, →**距離**=43.8872<br>44、 Hatch, **點** E **區域**, →**斜線區域**<br>45、 **點** E **斜線區域**, CRTL+1, →**查幾何面積值** =6494.0320<br>46、 DIst, int, F, cen, G, →**距離**=78.3102 |

## 2.4.15 AutoCAD2D2008-415

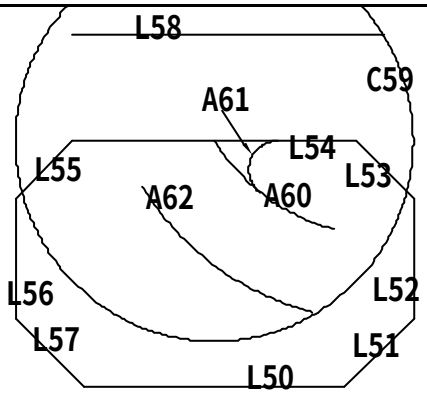



1、 **弧** A **至長度為何？** 32.1984
2、 **中心點** B **至交點** C **距離為何？** 34.1748
3、 **中心點** D **至四分點** E **高度為何？** 76.2193
4、 F **區域扣除內孔面積為何？** 1336.6476
5、 G **區域扣除內孔面積為何？** 2359.7822

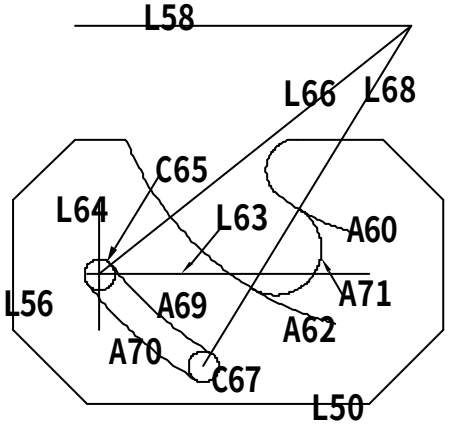
**解題步驟：**



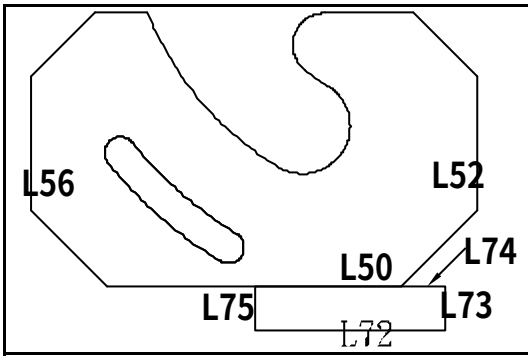
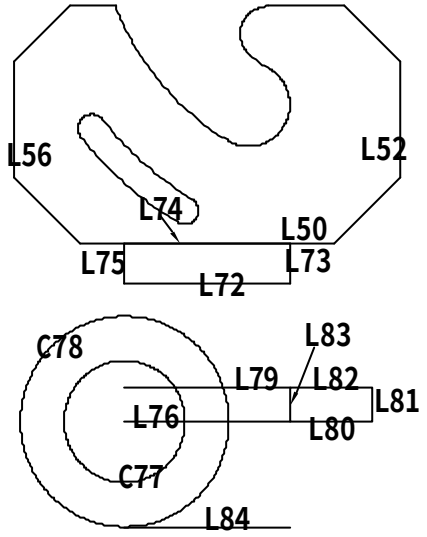
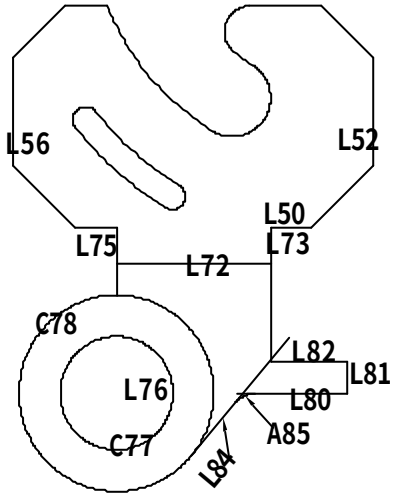
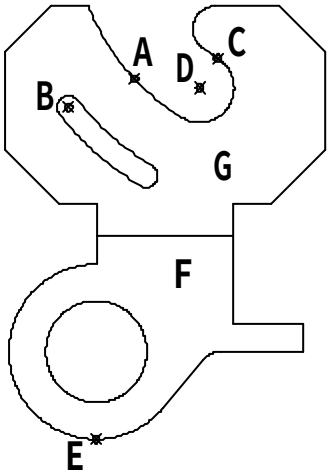

01、 RECtangle, C, 10, 10, **任一點** P01, D, 70, 43, **向右上方點**, →L50~L57 **之倒角矩形**
02、 eXplode, L50~L57 **之倒角矩形**, →L50~L57
03、 Offset, 18.5, L54, **向上點**, →L58
04、 Circle, mid, L54, 35, →C59




05、 Erase, L51;L57
06、 CHAmfer, T, T, D, 12, 12, L50 **右**, L52 **下**, →L51, L50 **左**, L56 **下**, →L57
07、 EXtend, C59, L58 **右**, →L58 **向右延伸**
08、 Arc, C, int, L58 & C59, mid, L54, int, L53 & L54, →A60
09、 Fillet, T, N, R, 5, L54, A60, →A61
10、 Offset, 15(=50-35), A60, **向下點**, →A62

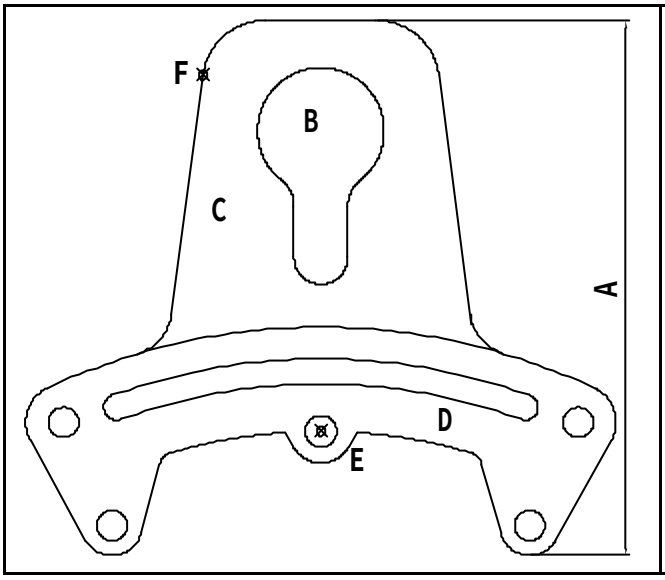



11、 Erase,
12、 EXtend, L54, A62, →L54 **與** A62 **相交**
13、 Offset, 21, L50, **向上點**, →L63
14、 Offset, 14, L56, **向右點**, →L64
15、 Circle, int, L63 & L64, 2.5, →C65
16、 Line, cen, C65, end, L58 **右**, →L66
17、 ROtate, C65;L66, end, L58 **右**, C, 20, →C67;L68
18、 Arc, C, end, L58 **右**, int, C65 & L66, int, C67 & L68, →A69
19、 Offset, 5, A69, **向下點**, →A70
20、 Arc, C, m2p, int, L68 & A62, int, L68 & A60, int, L68 & A62, int, L68 & A60, →A71

| | |
|---|---|
|  | 21、 Erase,<br>22、 TRim,<br>23、 RECtangle, C, 0, 0, F, 0, mid, L50, D, 30, 7, **向右下方點**, →L72~L75 **之矩形** |
|  | 24、 Move, L72~L75 **之矩形**, mid, L50, @-15,0, →L72~L75 **之矩形向左移動**<br>25、 eXplode, L72~L75 **之矩形**, →L72~L75<br>26、 Offset, 25(=32-7), L72, **向下點**, →L76<br>27、 Circle, end, L76 **左**, D, 22, →C77<br>28、 Circle, @(**鎖定上次點**), D, 38, →C78<br>29、 Offset, 19(=26-7), L72, **向下點**, →L79<br>30、 RECtangle, C, 0, 0, F, 0, end, L79 **右**, D, 15, 6, **向右下方點**, →L81~L83 **之矩形**<br>31、 eXplode, L81~L83 **之矩形**, →L81~L83<br>32、 Line, qua, C78 **下**, (**正交模式**), **向右** 30, →L84 |
|  | 33、 TRim,<br>34、 Erase,<br>35、 Fillet, T, T, R, 0, C78, L75, →C78 **與** L75 **相交**<br>36、 Fillet, T, T, R, 0, L82, L73, →L82 **與** L73 **相交**<br>37、 ROtate, L84, cen, C78, 50, →L84<br>38、 Fillet, T, T, R, 5, L80, L84, →A85 |
|  | 39、 TRim,<br>40、 Erase,<br>41、 **點選弧** A, CRTL+1, → **查幾何弧長值** =32.1984<br>42、 DIst, cen, B, int, C, →**距離**=34.1748<br>43、 DIst, cen, D, qua, E, →Y **差值**=76.2193<br>44、 Hatch, **點** F **區域**, →**斜線區域**<br>45、 **點** F **斜線區域**, CRTL+1, →**查幾何面積值** =1336.6476<br>46、 Hatch, **點** G **區域**, →**斜線區域**<br>47、 **點** G **斜線區域**, CRTL+1, →**查幾何面積值** =2359.7822 |

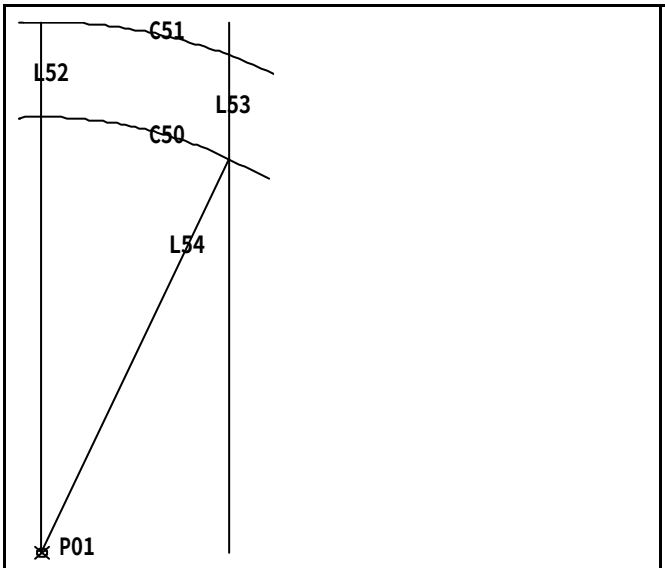
## 2.5 第五類：機械設計應用

### 2.5.1 AutoCAD2D2008-501

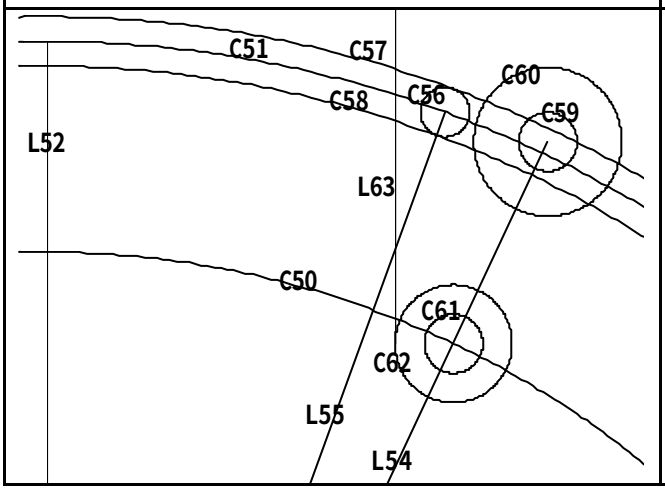



1、 **尺寸** A **長度為何？** 169.2538
2、 B **區域周長為何？** 176.6796
3、 C **區域扣除內孔面積為何？** 6393.3638
4、 D **區域扣除內孔面積為何？** 5941.5174
5、 **中心點** E **至交點** F **距離為何？** 118.6239

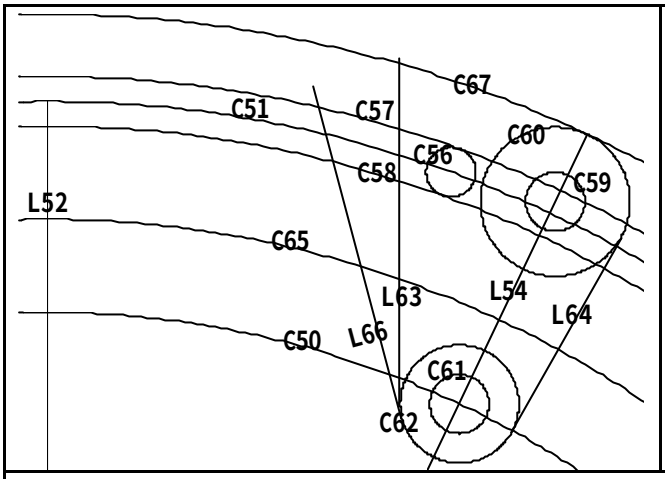
**解題步驟：**



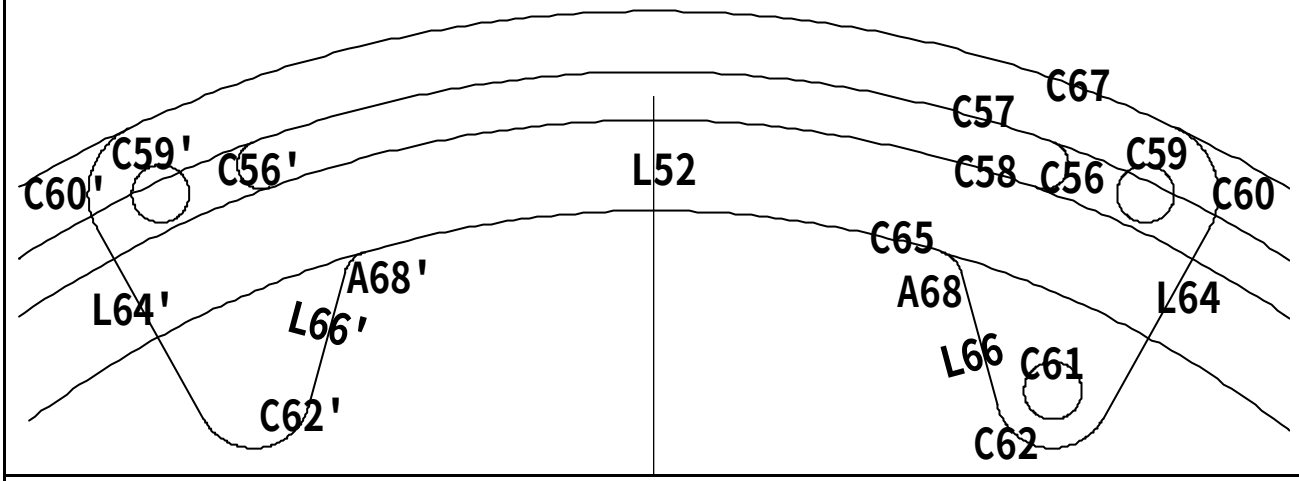

01、 Circle, **任一點** P01, 155, →C50,
02、 Circle, @(**鎖上次點**), 189, →C51,
03、 Line, cen, C50, qua, C51 **上**, →L52
04、 Offset, 66, L52, **向右點**, →L53
05、 Line, cen, C50, int, C50 & L53, →L54



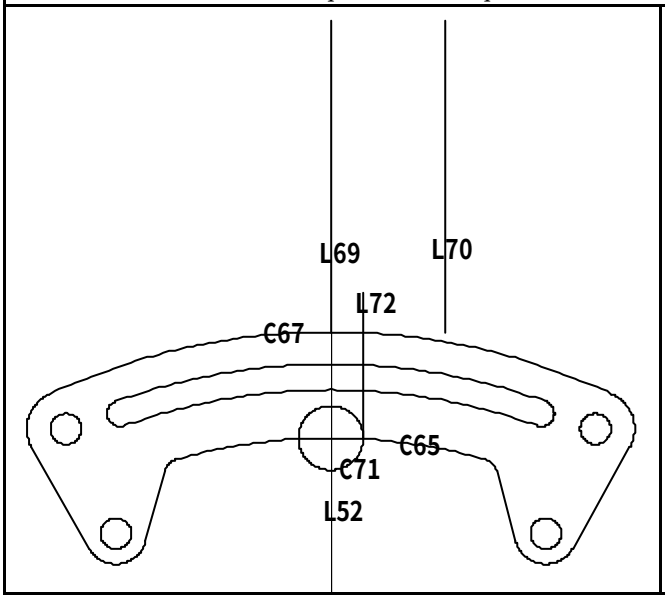

06、 Erase, L53
07、 ROtate, L52, cen, C50, C, -20, →L55
08、 Circle, int, L55 & C51, D, 8, →C56
09、 Offset, 4(=8/2), C51, **向上點**, →C57, **向下點**, →C58
10、 LENgthen, DE, 36, **點** L54 **上**, →L54 **向上延伸**
11、 Circle, end, L54 **上**, D, 9.5, →C59
12、 Circle, @(**鎖上次點**), 12, →C60
13、 Circle, int, C50 & L54, D, 9.5, →C61
14、 Circle, @(**鎖上次點**), 9.5, →C62
15、 Line, qua, C62 **左**, (**正交模式**), **向上** 50, →L63



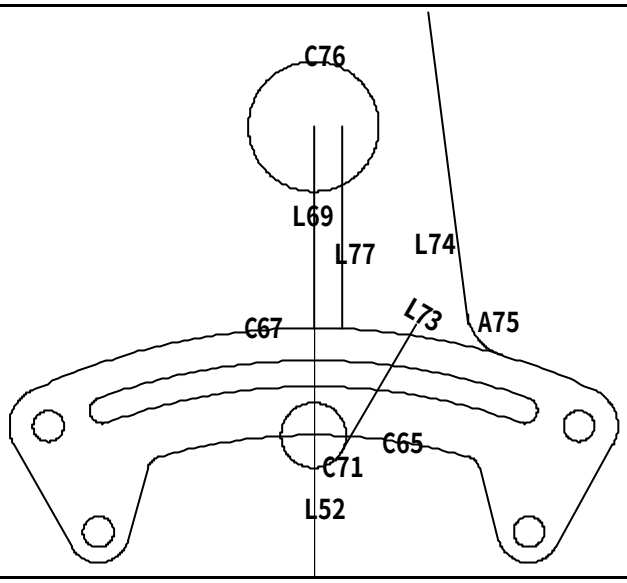
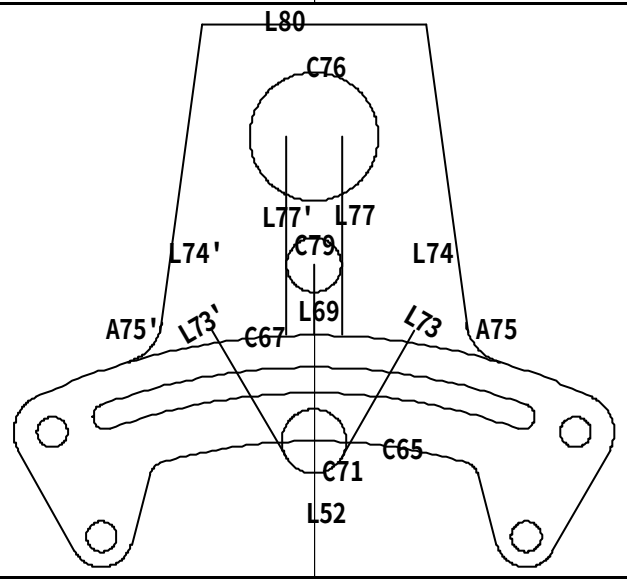
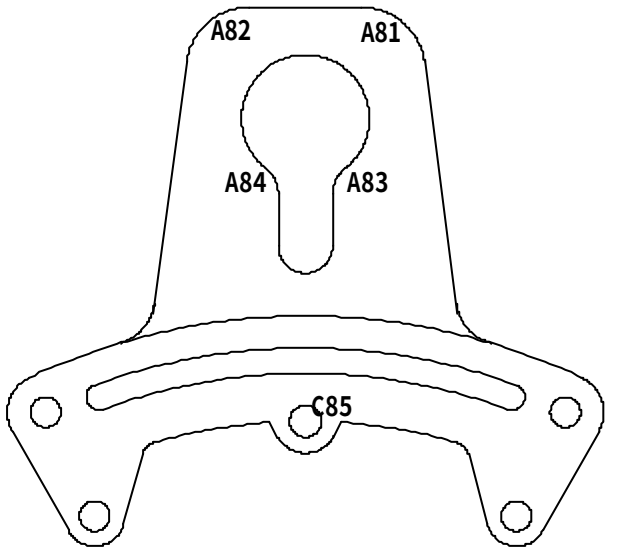

16、 Erase,
17、 Line, tan, C60 **右**, tan, C62 **右**, →L64
18、 Circle, cen, C50, 170, →C65
19、 ROtate, L63, cen, C62, C, 15, →L66
20、 EXtend, C60, L54 **上**, →L54 **與** C60 **相交**
21、 Offset, T, C57 int, L54 & C60, →C67

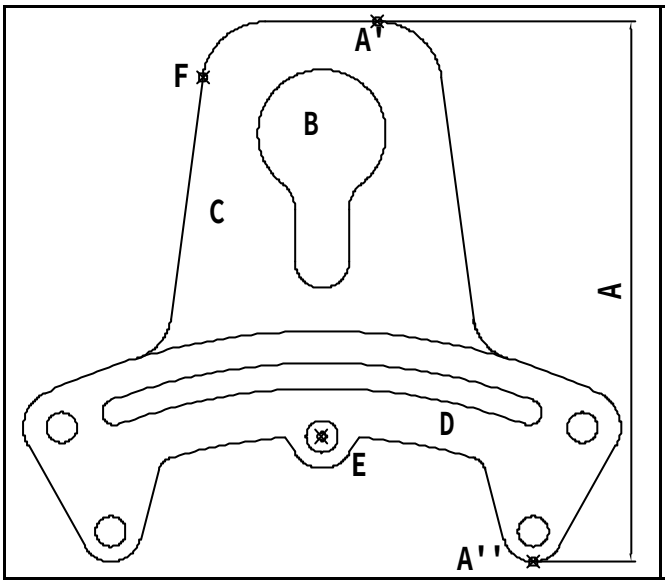



22、 Fillet, T, T, R, 5, C65, L66, →A68
23、 TRim,
24、 Erase,
25、 MIrror, **右側物件**, qua, C67 **上**, qua, C57 **上**, →**左側物件**




26、 TRim,
27、 Erase,
28、 Line, qua, C67 **上**, (**正交模式**), **向上** 97, →L69
29、 Offset, 35, L69, **向右點**, →L70
30、 Circle, int, L52 & C65, 10, →C71
31、 Line, qua, C71 **右**, (**正交模式**), **向上** 45, →L72

| | |
|---|---|
|  | 32、 ROtate, L72, cen, C71, -30, →L73<br>33、 ROtate, L70, end, L70 上, 7.5, →L74<br>34、 Fillet, T, N, R, 15, L74, C67, →A75<br>35、 LENgthen, DE, -35, 點 L69 上, →L69 縮短<br>36、 Circle, end, L69 上, 20, →C76<br>37、 Offset, 8.5, L69, 向右點, →L77 |
|  | 38、 MIrror, L73;L74;A75;L77, qua, C76 上, qua, C76 下, →L73';L74';A75';L77'<br>39、 LENgthen, DE, -40, 點 L69上, →L69再縮短<br>40、 Circle, end, L69 上, 8.5, →C79<br>41、 Line, end, L74'上, end, L74 上, →L80 |
|  | 42、 Fillet, T, T, R, 20, L74, L80, →A81<br>43、 Fillet, T, T, R, 20, L74' , L80, →A82<br>44、 Fillet, T, T, R, 10, L77, C76, →A83<br>45、 Fillet, T, T, R, 10, L77' , C76, →A84<br>46、 TRim,<br>47、 Erase,<br>48、 Circle, cen, C71, D, 10, →C85 |




49、 DIst, end, A' , qua, A' ' , →Y **差值**=169.2538

50、 BOundary, **點** B **區域**,

51、 **選** B **區域邊緣**, CTRL+1, →**查幾何長度值** =176.6796

52、 Hatch, **點** C **區域**, →**斜線區域**

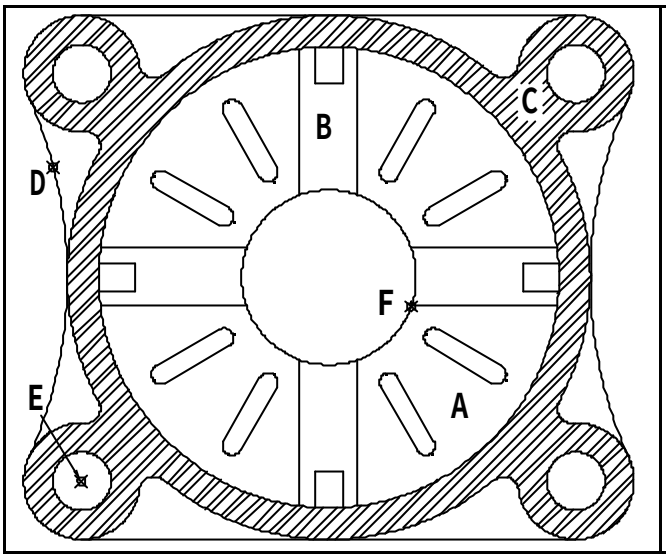
53、 **選** C **區域斜線**, CTRL+1, →**查幾何面積值** =6393.3638

54、 Hatch, **點** D **區域**, →**斜線區域**

55、 **選** D **區域斜線**, CTRL+1, →**查幾何面積值** =5941.5174

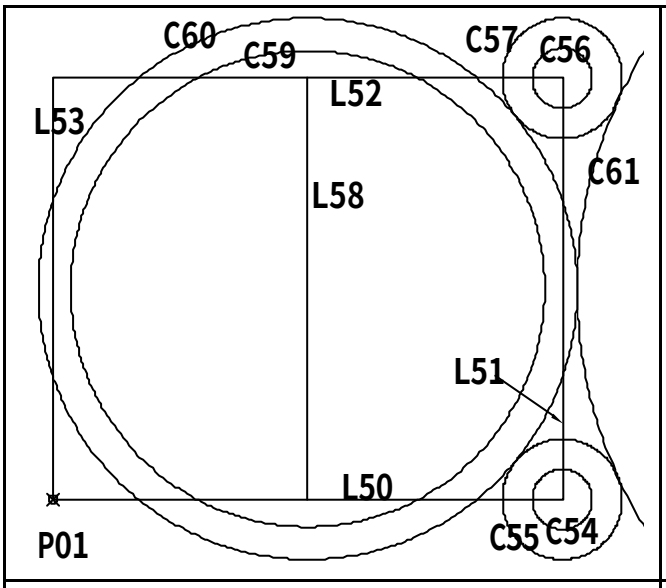
56、 Dist, cen, E, int, F, →**距離**=118.6239

## 2.5.2 AutoCAD2D2008-502

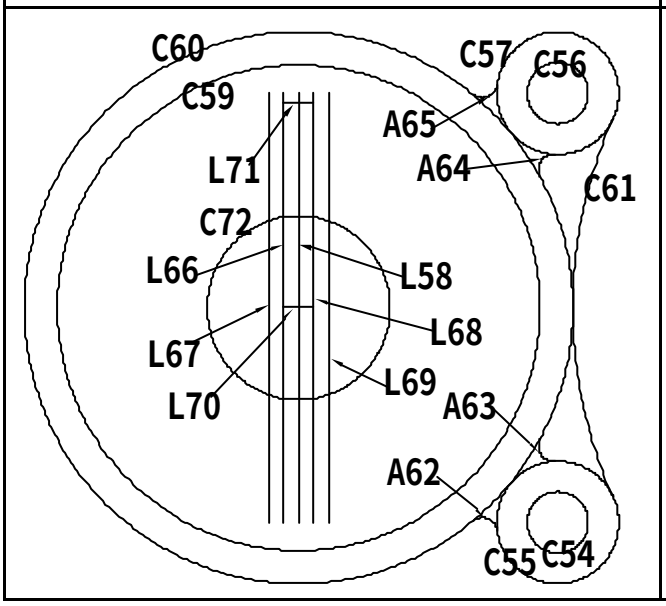



1、 A **區域扣除內孔面積為何？** 1015.9028
2、 B **區域周長為何？** 99.3705
3、 **斜線** C **區域面積為何？** 3521.1822
4、 **弧** D **夾角為何？** 48.3790
5、 **中心點** E **至交點** F **距離為何？** 76.9155

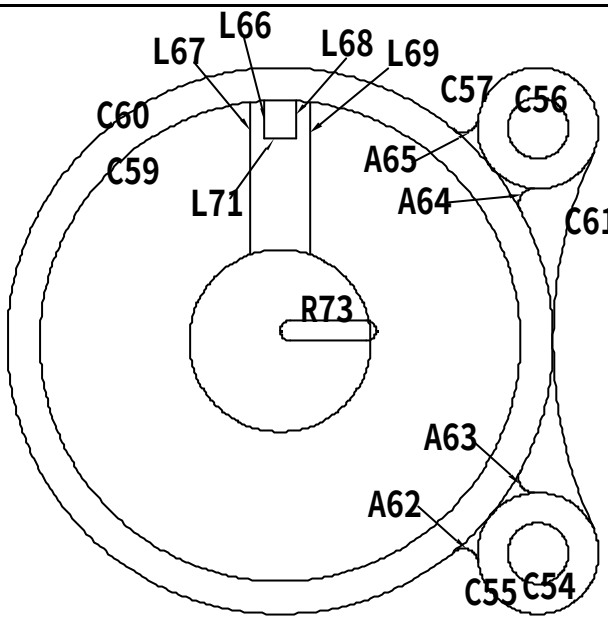
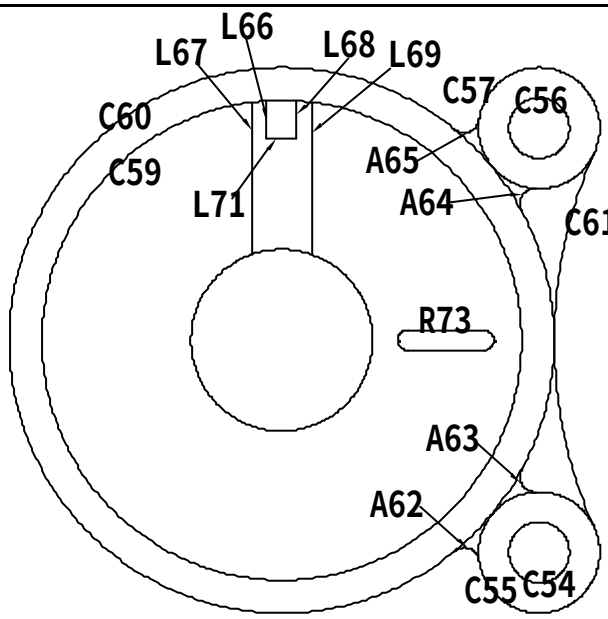
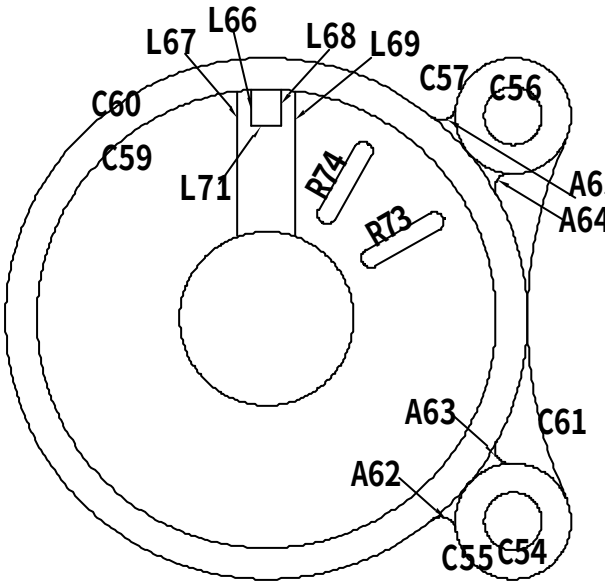
**解題步驟：**

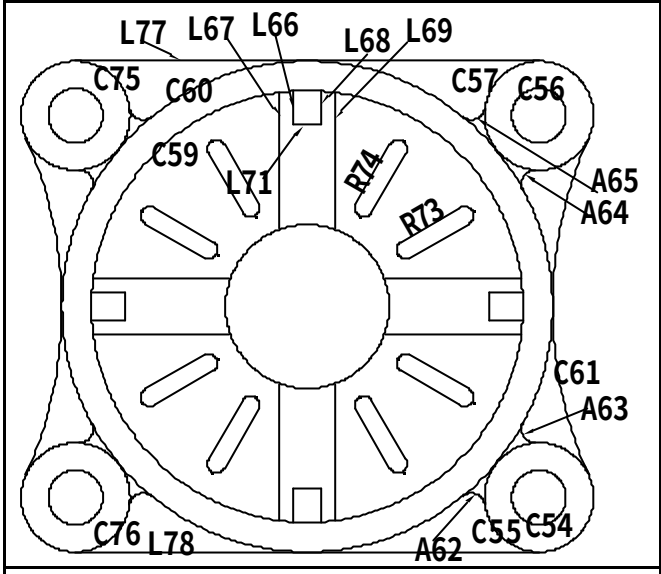



01、 RECtangle，**任一點** P01, D, 102, 84, **向右上方點**, →L50~L53 **之矩形**,
02、 Circle, int, L50 & L51, D, 12, →C54
03、 Circle, @(**鎖上次點**), D, 24, →C55
04、 COpy, C54;C55, cen, C55, int, L51 & L52, →C56;C57
05、 Line, mid, L50, mid, L52, →L58
06、 Circle, Mid, L58, D, 95, →C59
07、 Circle, @(**鎖上次點**), D, 108, →C60




08、 Erase, L50~L53 **之矩形**
09、 Circle, 3P, tan, C55, tan, C60, tan, C57, →C61
10、 Fillet, T, T, R, 3, C55 **左**, C60 **右下**, →A62, C55 **上**, C60 **右下**, →A63, C57 **下**, C60 **右上**, →A64, C57 **左**, C60 **右上**, →A65
11、 Offset, 3, L58, **向左點**, →L66, L66, **向左點**, →L67, L58, **向右點**, →L68, L68, **向右點**, →L69
12、 Line, mid, L66, mid, L68, →L70
13、 Offset, 40, L70, **向上點**, →L71
14、 Circle, cen, C59, D, 36, →C72

| | |
|---|---|
|  | 15、 EXtend, C59, L66~L69, →L66~L69 **分別與** C59 **相交**<br>16、 TRim,<br>17、 Erase,<br>18、 RECtangle, F, 2, cen, C59, D, 19, 4, **向右上方點**, →R73 **之圓角矩形** |
|  | 19、 Move, R73 **之圓角矩形**, cen, C59, @23,-2, →R73 **之圓角矩形向右下移動** |
|  | 20、 ROtate, R73, cen, C59, 30, →R73 **轉動**<br>21、 ROtate, R73, cen, C59, C, 30, →R74 |

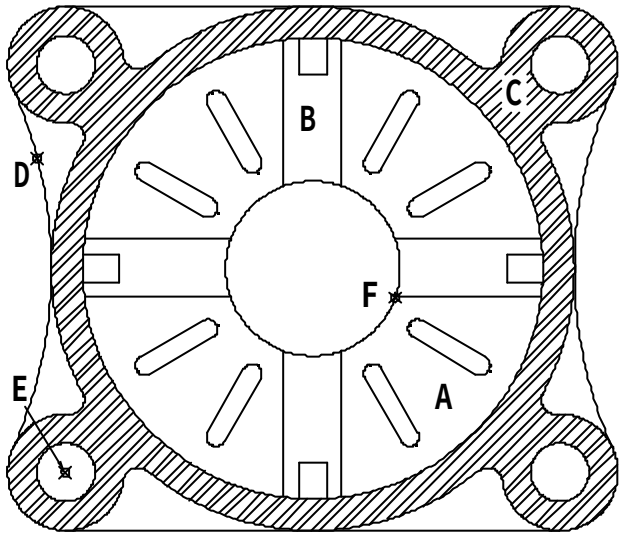



22、 ARray, **環形陣列**, **中心點**, cen, C59, **項目總數**, 4, **佈滿角度**, 360, **物件**, L66~L69;L71;R73;R74, →**全圓佈滿** 4 **組相同物件**

23、 MIrror, C54~C57;C61~A65, qua, C60 **上**, qua, C60 **下**, →C75~C76 **等左側物件**

24、 Line, qua, C57 **上**, qua, C75 **上**, →L77

25、 Line, qua, C55 **下**, qua, C76 **下**, →L78




26、 TRim,

27、 Hatch, **點** A **區域**, →**斜線區域**

28、 **選** A **區域斜線**, CTRL+1, →**查幾何面積值** =1015.9028

29、 BOundary, **點** B **區域**,

30、 **選** B **區域邊緣**, CTRL+1, →**查幾何長度值** =99.3705

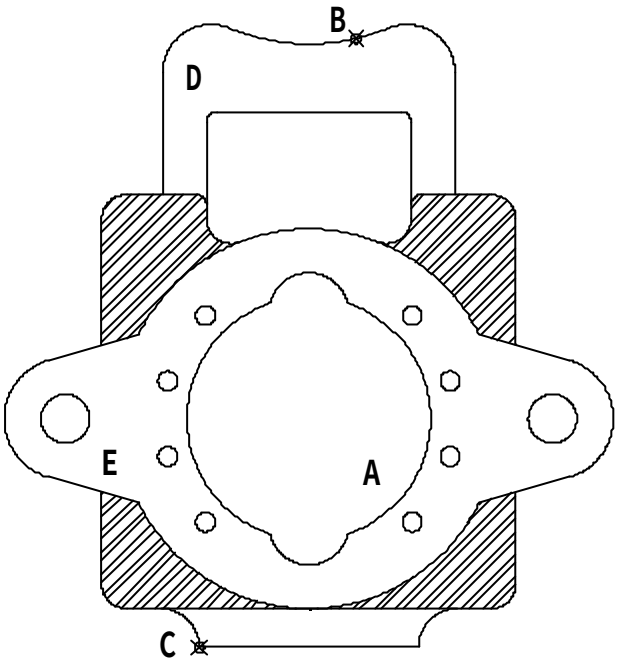
31、 Hatch, **點** C **區域**, B, **點** R73 **等** 8 **個圓角矩形**; **點** C72, →**斜線區域**

32、 **選** C **區域斜線**, CTRL+1, →**查幾何面積值** =3521.1822

33、 **選弧** D, CTRL+1, →**查幾何總角度值** =48.3790

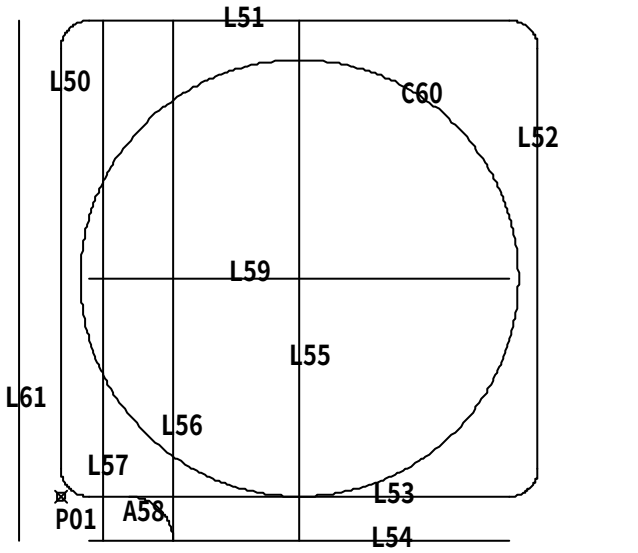
34、 Dist, cen, E, int, F, →**距離**=76.9155

## 2.5.3 AutoCAD2D2008-503

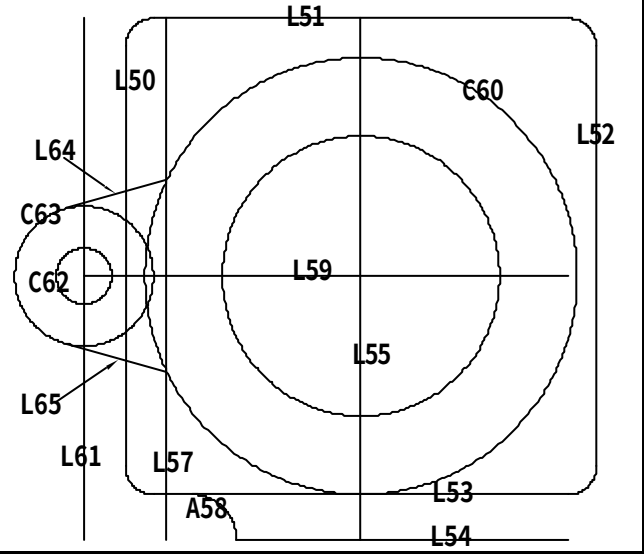



1、 A **區域周長為何？** 168.4735
2、 **弧** B **中心點至交點** C **距離值為何？** 167.5179
3、 D **區域面積為何？** 1249.5808
4、 E **區域扣除內孔面積為何？** 3648.5020
5、 **斜線區域總面積為何？** 1775.6296

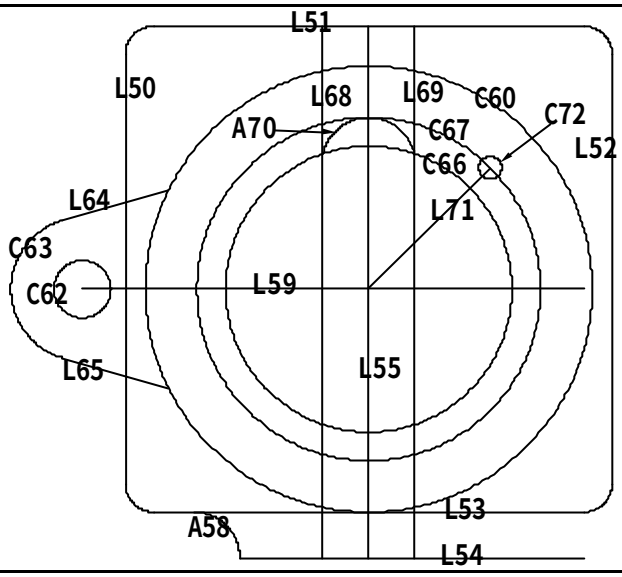
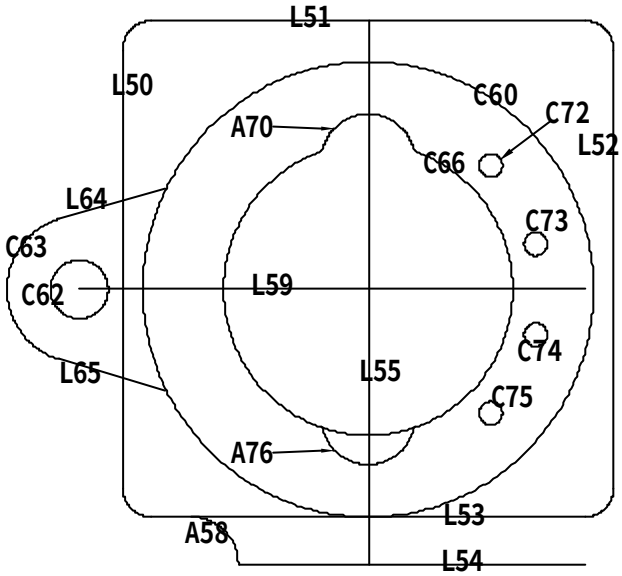
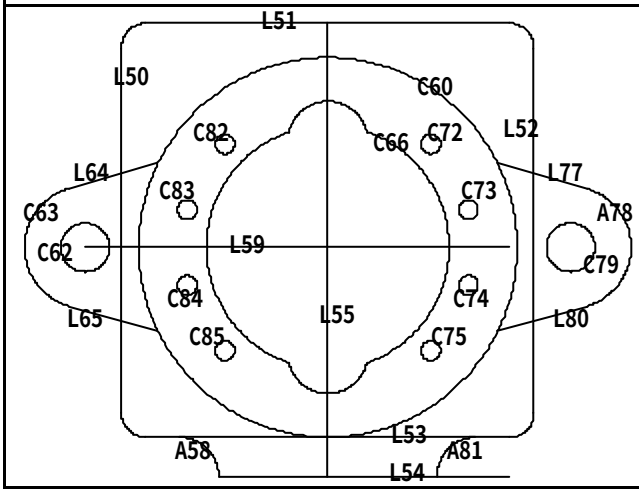
**解題步驟：**

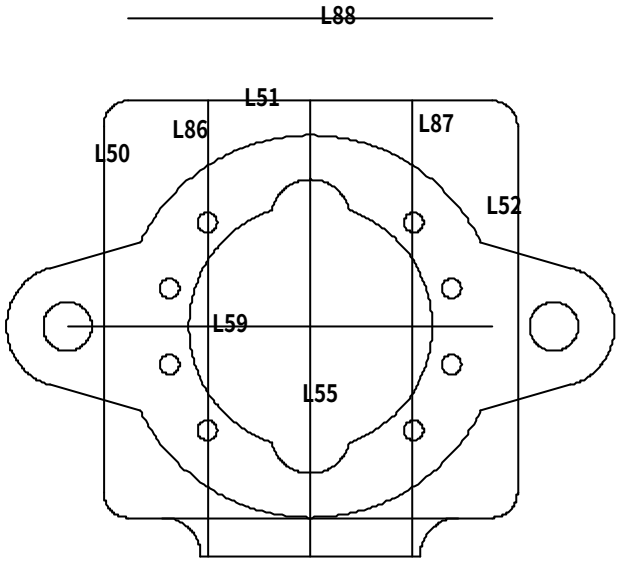
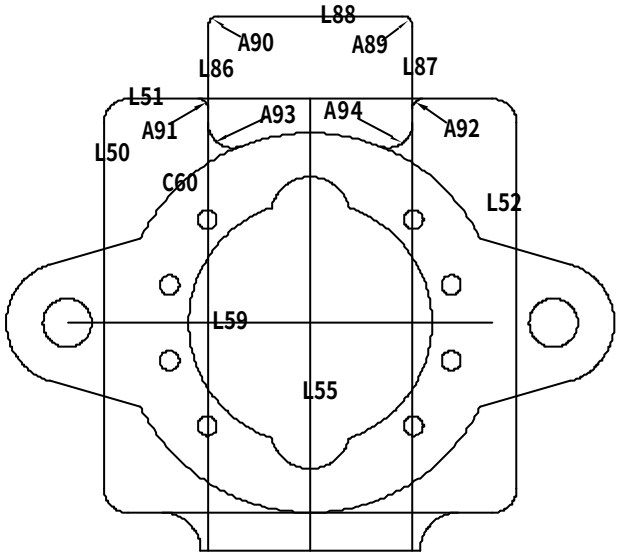
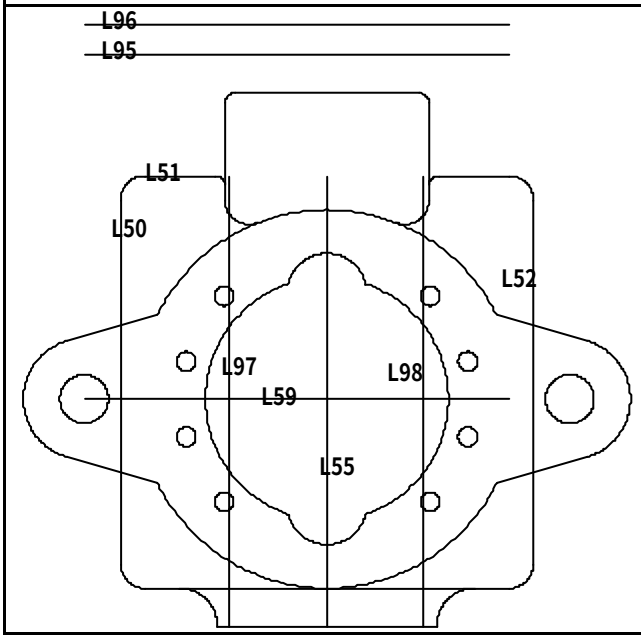


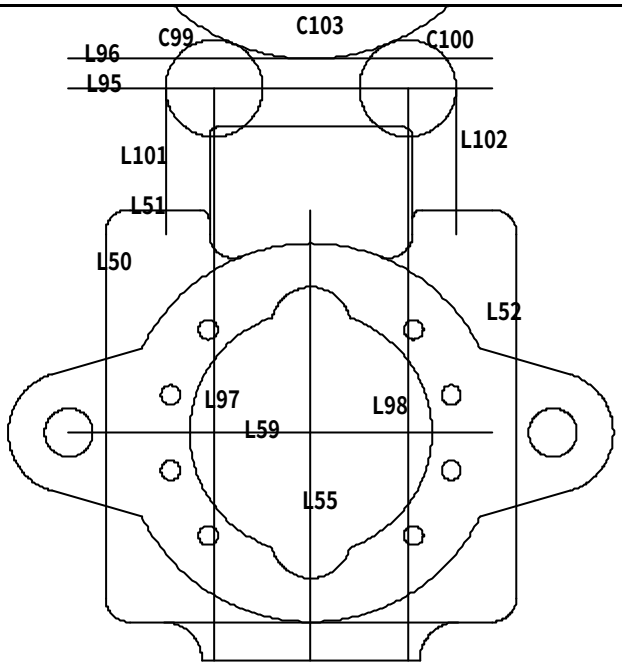

01、 RECtangle, F, 5, **任一點** P01, D, 85, 85, **向右上點**, →L50~L53 **之圓角矩形**
02、 eXplode, L50~L53 **之圓角矩形**, →L50~L53
03、 Offset, 8, L53, **向下點**, →L54
04、 Line, mid, L54, mid, L51, →L55
05、 Offset, 22.5(=45/2), L55, **向左點**, →L56
06、 Offset, 35, L55, **向左點**, →L57
07、 Fillet, T, N, R, 8, L53, L56, →A58
08、 Offset, 39(=78/2), L53, **向上點**, →L59
09、 Circle, int, L55 & L59, D, 78, →C60



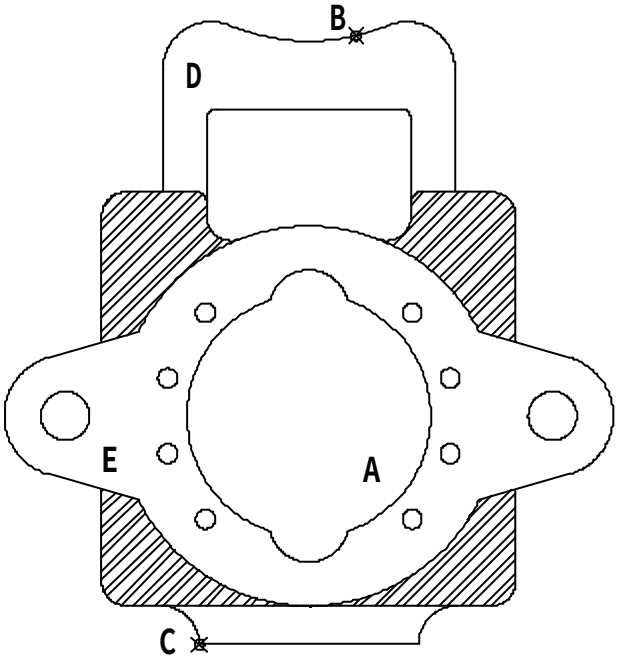

10、 Erase,
11、 TRim,
12、 Offset, 50, L55, **向左點**, →L61
13、 EXtend, L61, L59, →L59 **向左延伸**
14、 Circle, int, L59 & L61, D, 10, →C62
15、 Circle, @(**鎖定上次點**), 12.5, →C63
16、 Line, int, L57 & C60, tan, C63 **上**, →L64
17、 Line, int, L57 & C60, tan, C63 **下**, →L65

| | |
|---|---|
|  | 18、 Erase,<br>19、 TRim,<br>20、 Circle, int, L55 & L59, D, 50, →C66<br>21、 Circle, @(**鎖定上次點**), D, 60, →C67<br>22、 Offset, 8(=16/2), L55, **向左點**, →L68<br>23、 Offset, 8(=16/2), L55, **向右點**, →L69<br>24、 Arc, int, C66 & L69, int, C67 & L55, int, C66 & L68, →A70<br>25、 Line, cen, C66, @30<45, →L71<br>26、 Circle, int, C67 & L71, D, 4, →C72 |
|  | 27、 Erase,<br>28、 TRim,<br>29、 ARray, **環形陣列**, **中心點**, cen, C66, **項目總數**, 4, **佈滿角度**, -90, **物件**, C72, →C73~C75<br>30、 MIrror, A70, qua, C60 **左**, qua, C60 **右**, →A76 |
|  | 31、 TRim,<br>32、 MIrror, A58;C62~L65;C72~C75, qua, C60 **上**, qua, C60 **下**, →L77~C85 |

| | |
|---|---|
|  | 33、 TRim,<br>34、 Offset, 21(=42/2), L55, **向左點**, →L86, L55, **向右點**, →L87<br>35、 Offset, 17, L51, **向上點**, →L88 |
|  | 36、 Fillet, T, T, R, 2, L88, L87, →A89<br>37、 Fillet, T, T, R, 2, L88, L86, →A90<br>38、 Fillet, T, N, R, 2, L51, L86, →A91<br>39、 Fillet, T, N, R, 2, L51, L87, →A92<br>40、 Fillet, T, N, R, 5, C60, L86, →A93<br>41、 Fillet, T, N, R, 5, C60, L87, →A94 |
|  | 42、 TRim,<br>43、 Erase,<br>44、 Offset, 71, L59, **向上點**, →L95<br>45、 Offset, 6, L95, **向上點**, →L96<br>46、 Offset, 20(=40/2), L55, **向左點**, →L97, L55, **向右點**, →L98 |

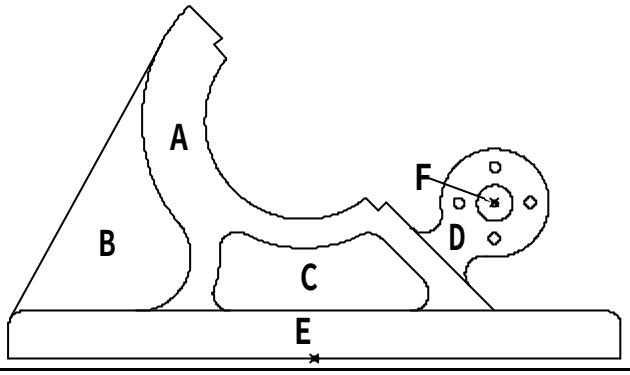



47、 EXtend, L95, L97 上;L98 上, →L97;L98 向上延伸
48、 Circle, int, L95 & L97, 10, →C99
49、 Circle, int, L95 & L98, 10, →C100
50、 Line, qua, C99 左,(正交模式), 向下 30, →L101
51、 Line, qua, C100 右,(正交模式), 向下 30, →L102
52、 Circle, 3P, tan, C99 右上, tan, L96 中, tan, C100 左上, →C103



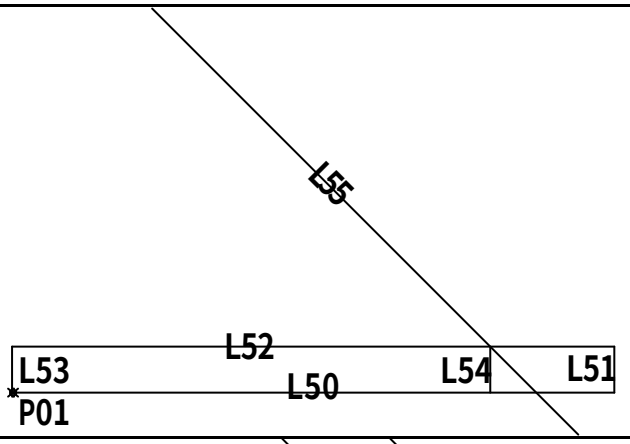

53、 TRim,
54、 Erase
55、 BOundary, 點 A 區域,
56、 點 A 區域邊緣, CTRL+1, →查幾何長度值 =168.4735
57、 DIst, cen, B, int, C, →距離=167.5179
58、 Hatch, 點 D 區域, →斜線區域
59、 點 D 區域斜線, CTRL+1, →查幾何面積值 =1249.5808
60、 Hatch, 點 E 區域, →斜線區域
61、 點 E 區域斜線, CTRL+1, →查幾何面積值 =3648.5020
62、 Hatch, 點斜線區域, →斜線區域
63、 點斜線區域, CTRL+1, →查幾何面積值 =1775.6296

## 2.5.4 AutoCAD2D2008-504

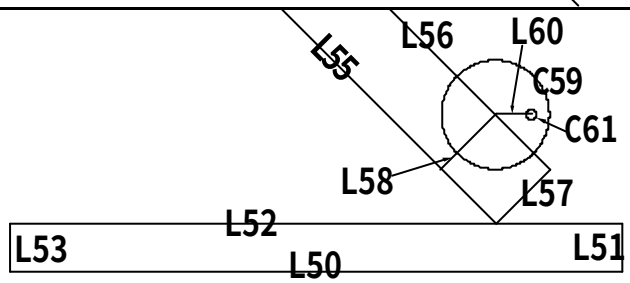



1、 A **區域周長為何？** 399.3165
2、 B **區域面積為何？** 1526.1634
3、 C **區域面積為何？** 1060.2966
4、 D **區域扣除內孔面積為何？** 703.0853
5、 **中點** E **至中心點** F **距離為何？** 65.9809

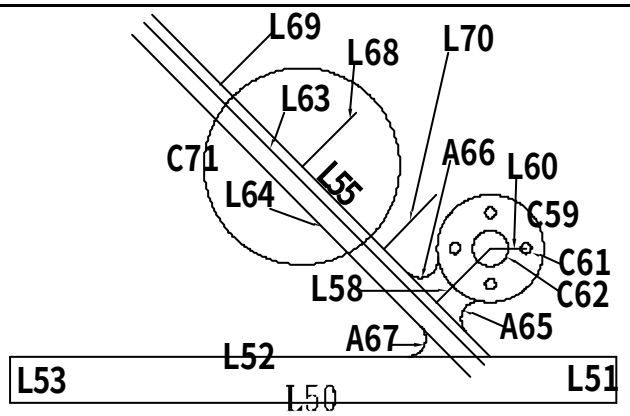
**解題步驟：**



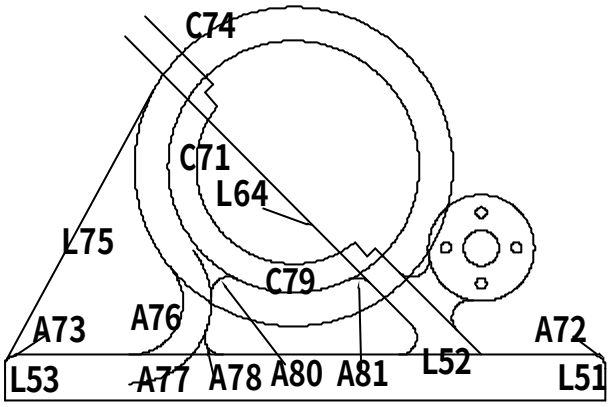

01、 RECtangle, **任一點** P01, D, 170, 13, **向右上方點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 35, L51, **向左點**, →L54
04、 ROtate, L52, int, L52 & L54, C, -45, →L55




05、 Erase, TRim,
06、 Offset, 21.25, L55, **向右上點**, →L56
07、 Line, int, L52 & L55, end, L56 **下**, →L57
08、 Offset, 21.25, L57, **向左上點**, →L58
09、 Circle, int, L56 & L58, D, 30, →C59
10、 Line, cen, C59, (**正交模式**), **向右** 10, →L60
11、 Circle, end, L60 **右**, D, 3, →C61




12、 Erase,
13、 ARray, **環形陣列**, **中心點**, cen, C59, **項目總數**, 4, **佈滿角度**, 360, **物件**, C61, **→全圓佈滿** 4 **個** C61 **相同物件**
14、 Circle, cen, C59, D, 10, →C62
15、 Offset, 3, L55, **向左下點**, →L63
16、 Offset, 8, L55, **向左下點**, →L64
17、 Fillet, T, N, R, 5, C59 **下**, L55, →A65
18、 Fillet, T, N, R, 5, C59 **左**, L55, →A66
19、 Fillet, T, N, R, 5, L52, L64, →A67
20、 Offset, 53.75, L58, **向左上點**, →L68
21、 Offset, 32.5, L68, **向左上點**, →L69, L68, **向右下點**, →L70
22、 Circle, int, L55 & L68, 27.5, →C71

<table>
<tr>
<td> 
</td>
<td>23、 EXtend, L63, L69;L70, →L69;L70 分別與 L63 相交<br>24、 TRim, Erase,<br>25、 Fillet, T, T, R, 3.5, L51, L52, →A72<br>26、 Fillet, T, T, R, 3.5, L52, L53, →A73<br>27、 Circle, cen, C71, 45, →C74<br>28、 Line, tan, A73 左上, tan, C74 左上, →L75<br>29、 Fillet, T, N, R, 15, L52, C74, →A76<br>30、 Offset, 8, A76, 向右點, →A77<br>31、 Fillet, T, N, R, 5, A77, L52, →A78<br>32、 Circle, cen, C74, 35.5, →C79<br>33、 Fillet, T, N, R, 5, A77, C79, →A80<br>34、 Fillet, T, N, R, 5, L64, C79, →A81</td>
</tr>
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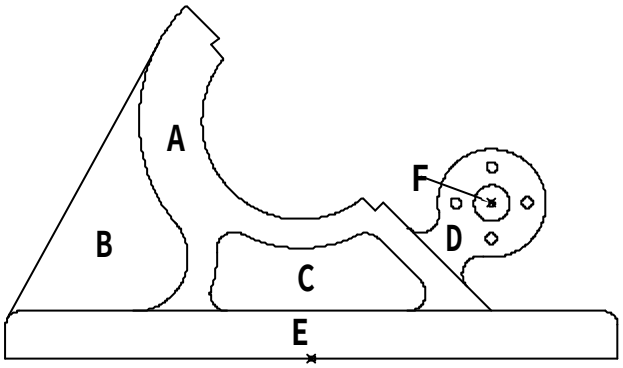
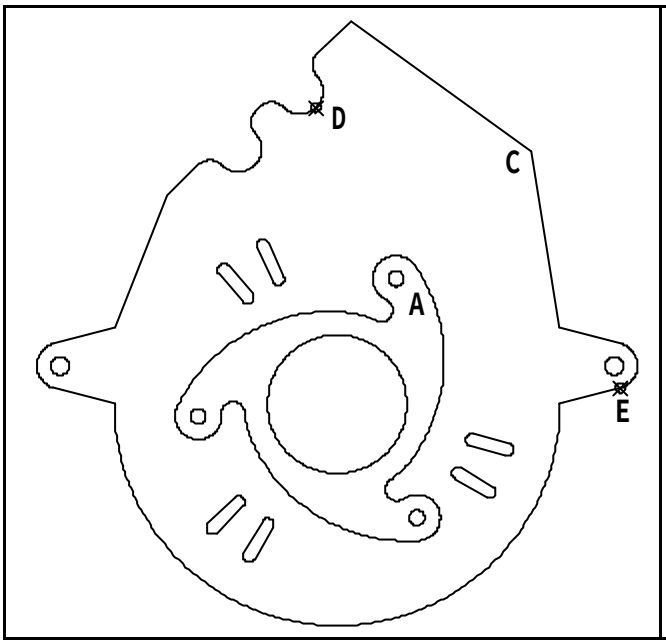
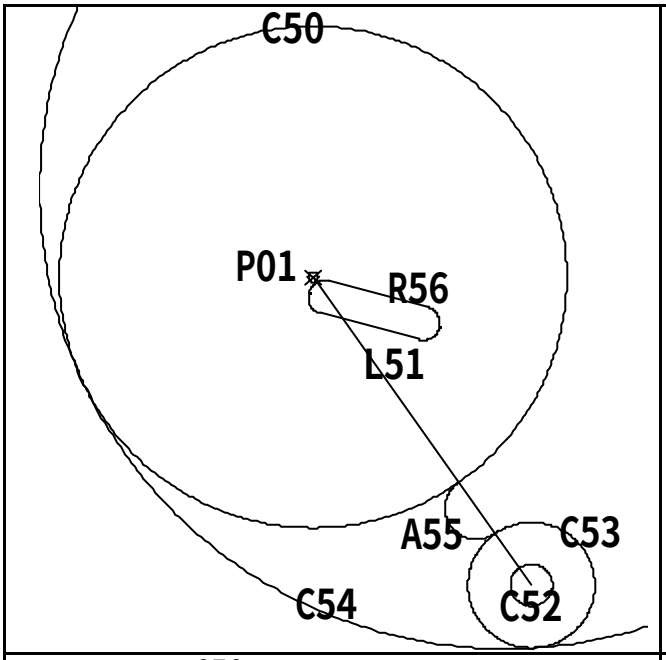
<td> 
</td>
<td>35、 TRim,<br>36、 Erase,<br>37、 BOundary, 點 A 區域,<br>38、 選 A 區域邊緣, CTRL+1, →查幾何長度值 =399.3165<br>39、 Hatch, 點 B 區域, →斜線區域<br>40、 選 B 區域斜線, CTRL+1, →查幾何面積值 =1526.1634<br>41、 Hatch, 點 C 區域, →斜線區域<br>42、 選 C 區域斜線, CTRL+1, →查幾何面積值 =1060.2966<br>43、 Hatch, 點 D 區域, →斜線區域<br>44、 選 D 區域斜線, CTRL+1, →查幾何面積值 =703.0853<br>45、 Dist, mid, E, cen, F, →距離=65.9809</td>
</tr>
</table>

## 2.5.5 AutoCAD2D2008-505

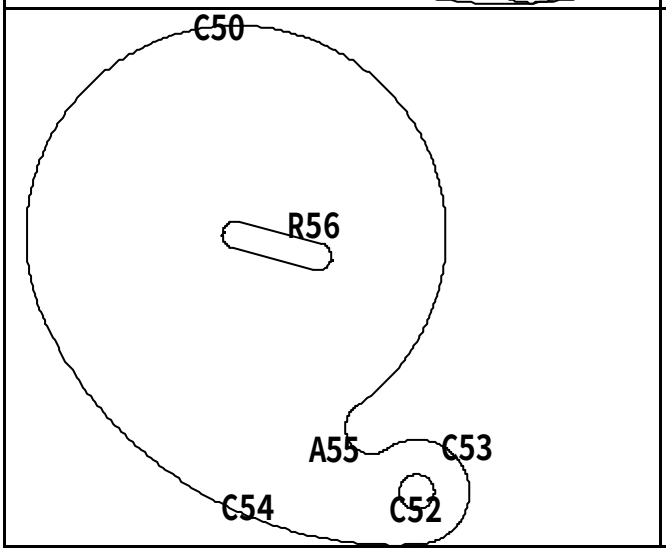



1、 **圖形最大高度值為何？** 196.3148
2、 A **區域扣除內孔面積為何？** 2857.2691
3、 **中心點** B **至交點** C **距離為何？** 160.1087
4、 **中點** D **至交點** E **距離為何？** 134.1428
5、 **圖形最外圍面積為何？** 21705.8414

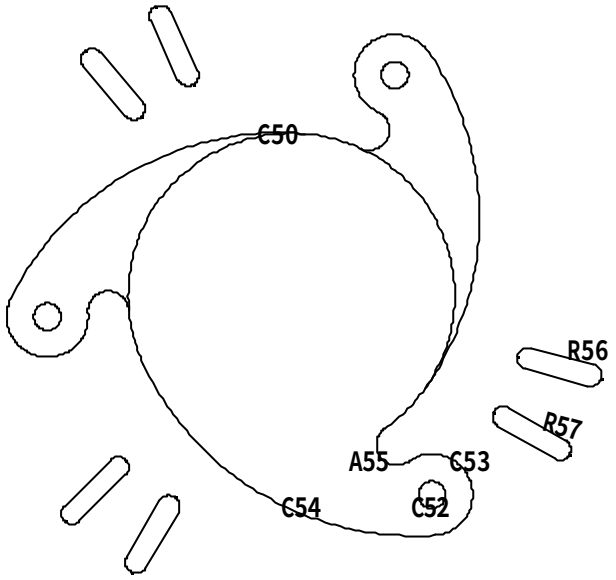
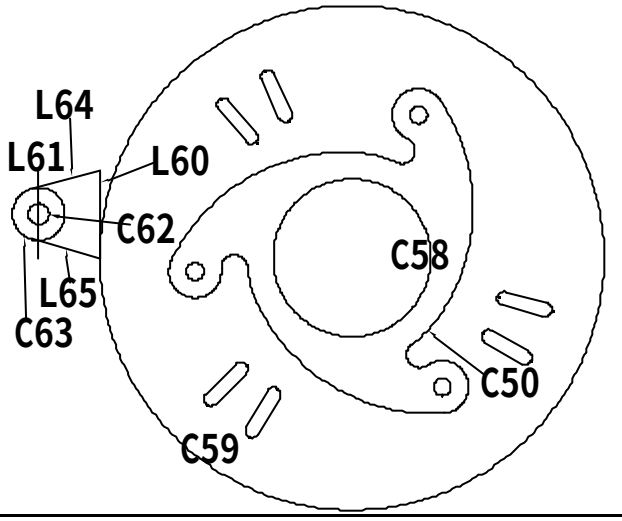
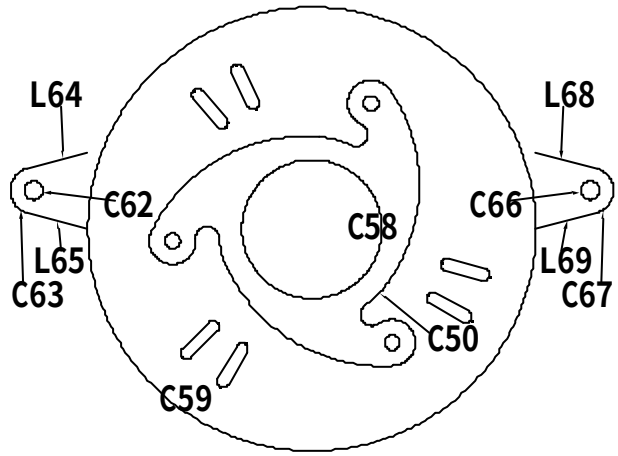
**解題步驟：**

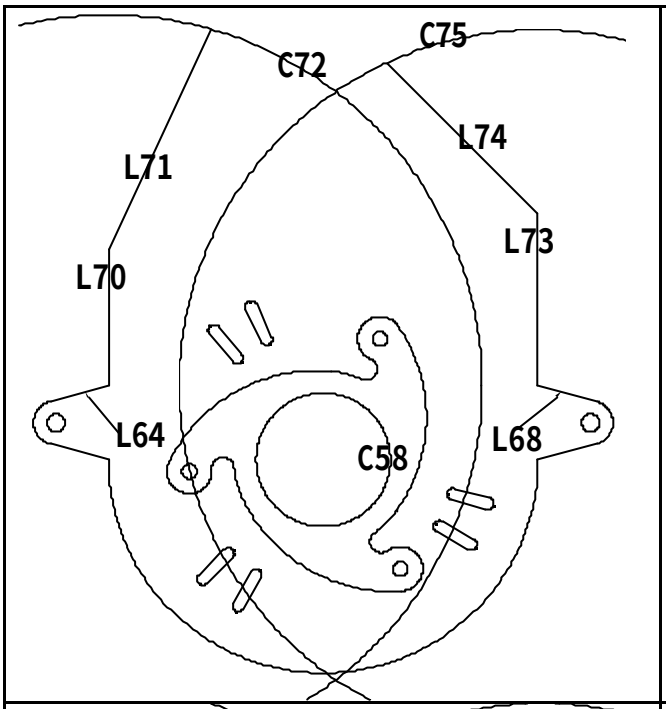



01、 Circle, **任一點** P01, D, 60, →C50
02、 Line, cen, C50, @45<-55, →L51
03、 Circle, end, L51 **右下**, D, 5, →C52
04、 Circle, @(**鎖定上次點**), 7.5, →C53
05、 Circle, T, C50 **左下**, C53 **下**,55, →C54
06、 Arc, C, mtp, int, C50 & L51, int, C53 & L51, int, C50 & L51, int, C53 & L51, →A55
07、 RECtangle, F, 2, cen, C50, R, -15, D, 16, 4, **向右下點**, →R56

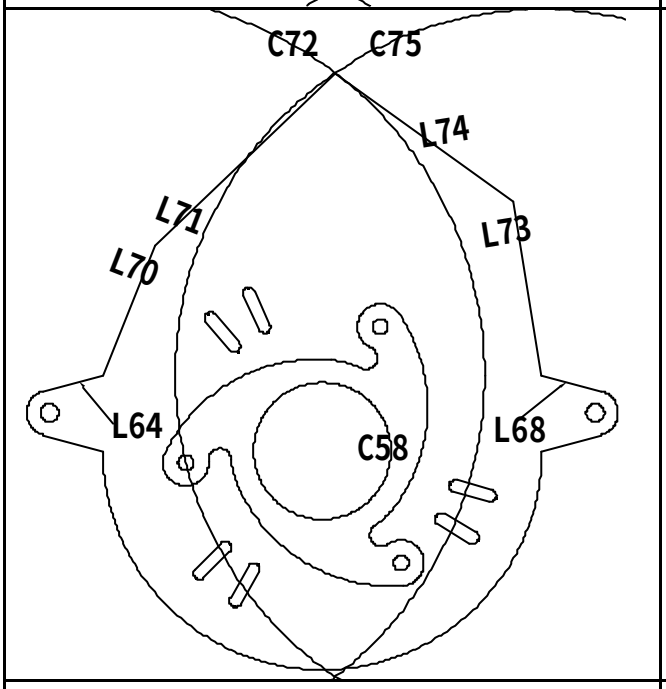



08、 Erase,
09、 TRim,
10、 Move, R56, cen, R56 **左**, cen, C50, →R56 **左側中心移至** C50 **中心**

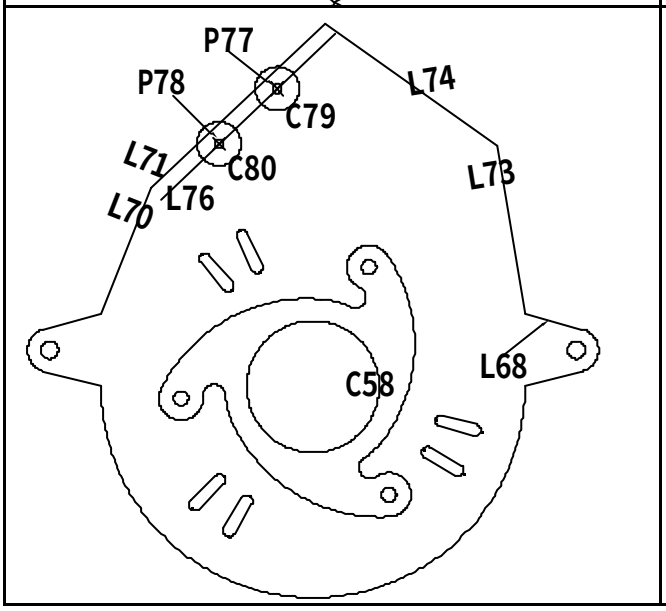
| | |
|---|---|
|  | 11、 Move, R56, cen, C50, @45<-15, →R56 **向右下移動**<br>12、 ROtate, R56, cen, C50, C, -15, →R57<br>13、 ARray, **環形陣列**, **中心點**, cen ,C50, **項目總數**, 3, **佈滿角度**, 360, **物件**, C52~A55, → **全圓佈滿** 3 **組相同物件**<br>14、 ROtate, R56;R57, cen, C50, C, -105(=90+(30-15)), →**左下方相同物件**<br>15、 ROtate, R56;R57, cen, C50, C, 145(=30+90+25), →**左上方相同物件** |
|  | 16、 Circle, cen, C50, D, 45, →C58<br>17、 Circle, @(**鎖定上次點**), 72, →C59<br>18、 TRim,<br>19、 Line, qua, C59 **左**, (**正交模式**), **向上** 25, →L60<br>20、 Offset, 17.5, L60, **向左點**, →L61<br>21、 Circle, mid, L61, D, 6, →C62<br>22、 Circle, @(**鎖定上次點**), 7.5, →C63<br>23、 Line, end, L60 **上**, tan, C63 **上**, →L64<br>24、 Line, int, L60 & C59, tan, C63 **下**, →L65 |
|  | 25、 Erase,<br>26、 TRim,<br>27、 MIrror, C62~L65, qua, C59 **上**, qua, C59 **下**, →C66~L68 |



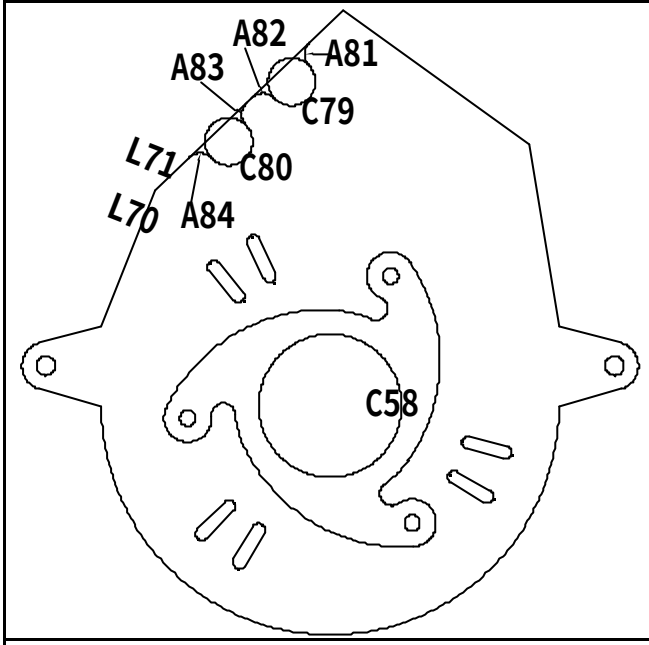
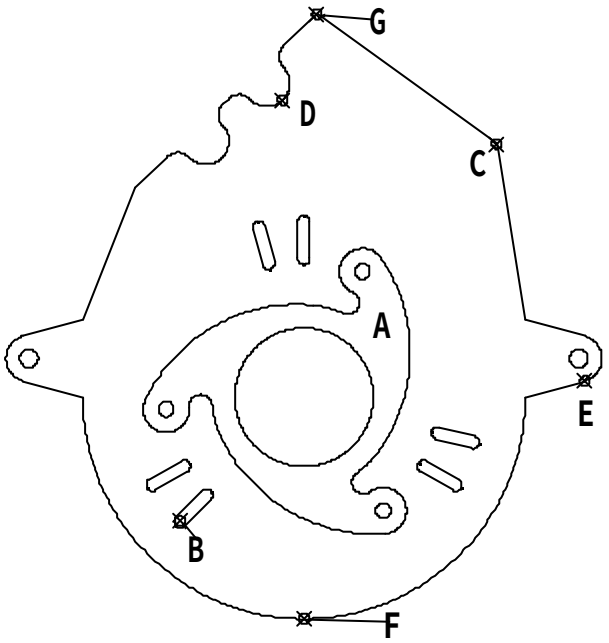

28、 TRim,
29、 Line, end, L64 **上**, (**正交模式**), **向上** 46, →L70, @82<65, →L71
30、 Circle, int, L64 & L70, end, L71 **上**, →C72
31、 Line, end, L68 **上**, (**正交模式**), **向上** 58, →L73, @72<135, →L74
32、 Circle, int, L68 & L73, end, L74 **上**, →C75



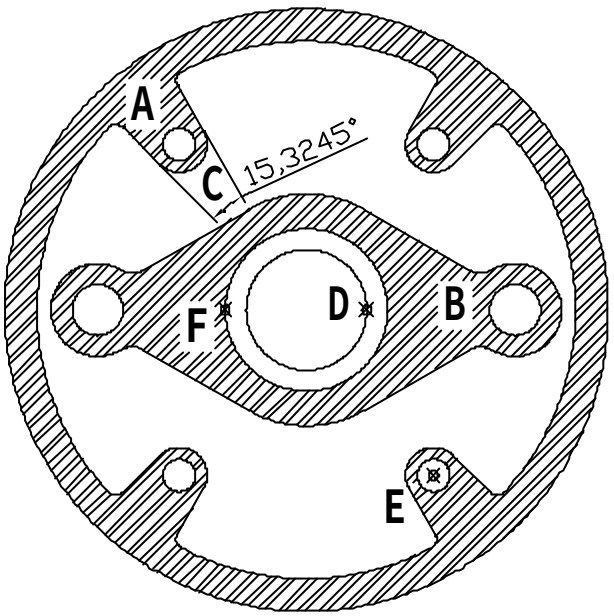

33、 ROtate, L70;L71, cen, C72, R, cen, C72, int, L71 & C72, int, C72 & C75, →L70;L71 **轉動**
34、 ROtate, L73;L74, cen, C75, R, cen, C75, int, L74& C75, int, C72 & C75, →L73;L74 **轉動**




35、 Erase,
36、 Offset, 5, L71, **向下點**, →L76
37、 DIVide, L76, 3, →P77;P78
38、 Circle, nod, P77, 7.5, →C79
39、 Circle, nod, P78, 7.5, →C80

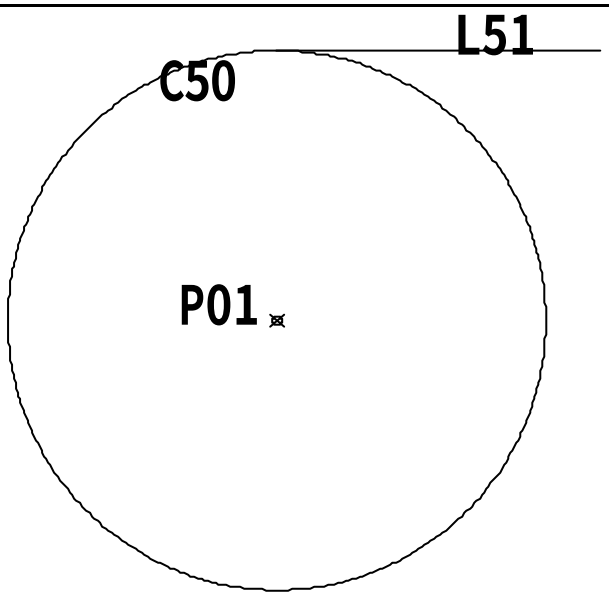
| | |
|---|---|
|  | 40、 Erase,<br>41、 Fillet, T, N, R, 5, L71, C79 **右上**, →A81<br>42、 Fillet, T, N, R, 5, L71, C79 **左下**, →A82<br>43、 Fillet, T, N, R, 5, L71, C80 **右上**, →A83<br>44、 Fillet, T, N, R, 5, L71, C80 **左下**, →A84 |
|  | 45、 TRim,<br>46、 Erase<br>47、 DIst, qua, F, int, G, →Y **差值**=196.3148<br>48、 Hatch, **點** A **區域**, →**斜線區域**<br>49、 **點選** A **斜線區域**, CTRL+1, →**查幾何面積值** =2857.2691<br>50、 DIst, cen, B, int, C, →**距離**=160.1087<br>51、 DIst, mid, D, int, E, →**距離**=134.1428<br>52、 BOundary, **點** D **或** C **附近空白位置**,<br>53、 **點選外圍邊界**, CTRL+1, →**查幾何面積值** =21705.8414 |

## 2.5.6 AutoCAD2D2008-506

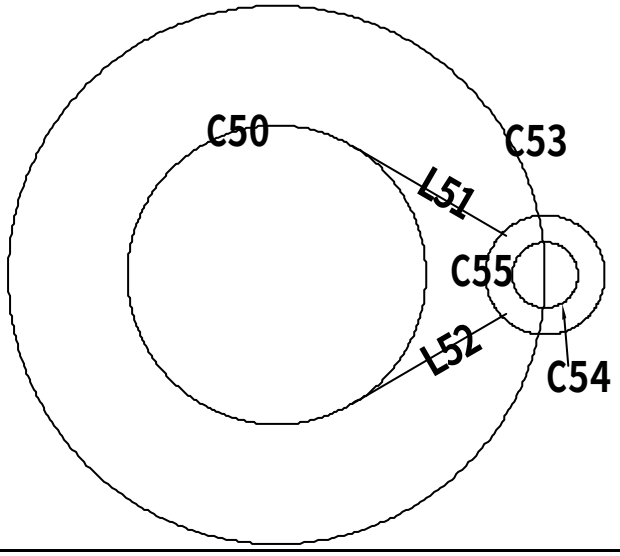



1、 **斜線** A **區域扣除內孔面積為何？** 3494.3010
2、 **斜線** B **區域面積為何？** 2077.2033
3、 C **夾角為何？** 15.3245
4、 **四分點** D **至中心點** E **高度為何？** 35.7009
5、 **中心點** E **至四分點** F **距離為何？** 57.3590

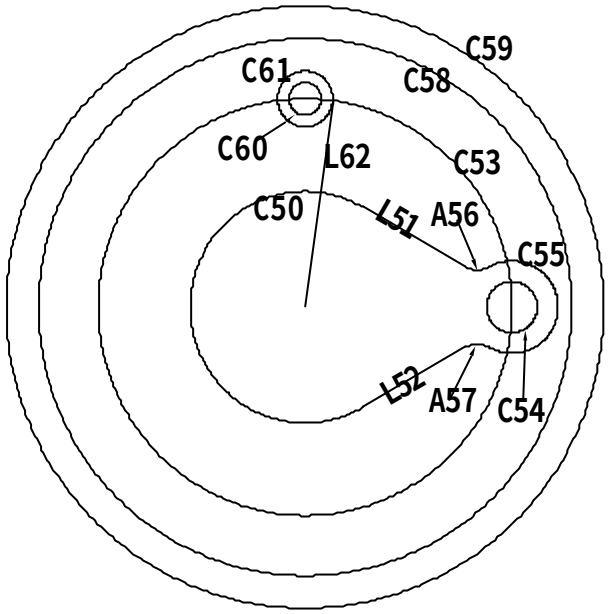
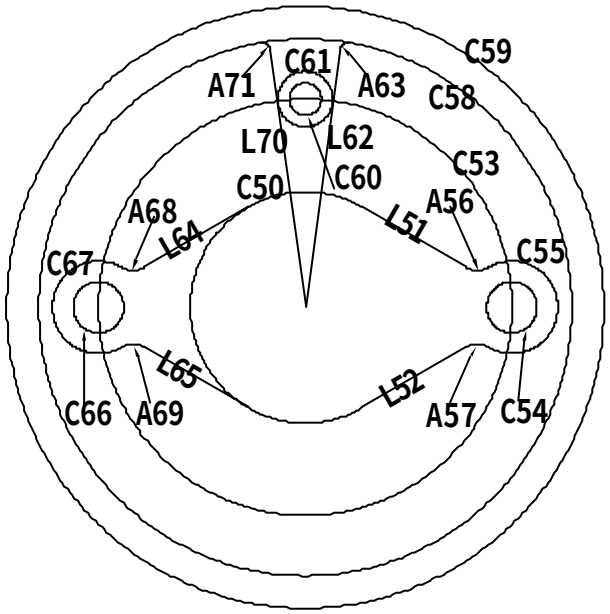
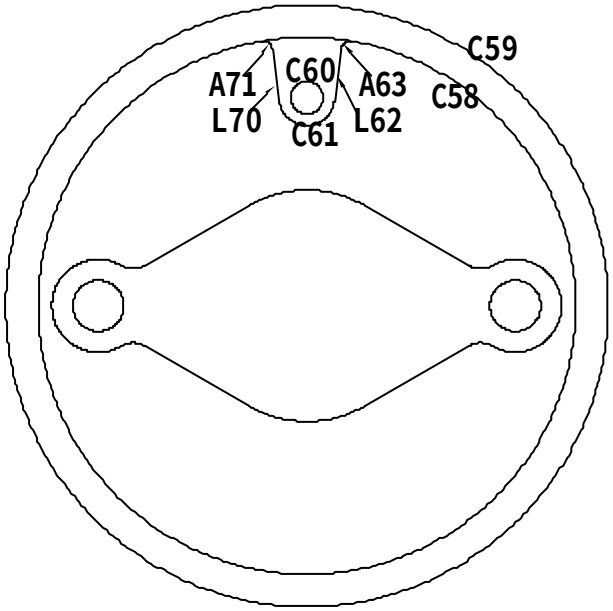
**解題步驟：**

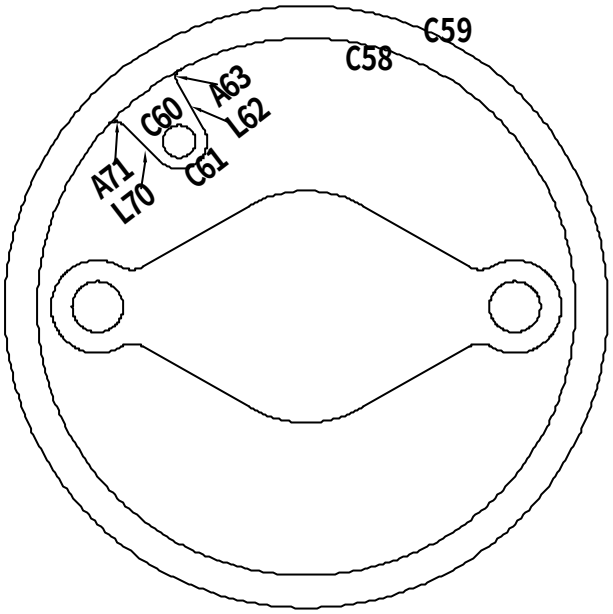
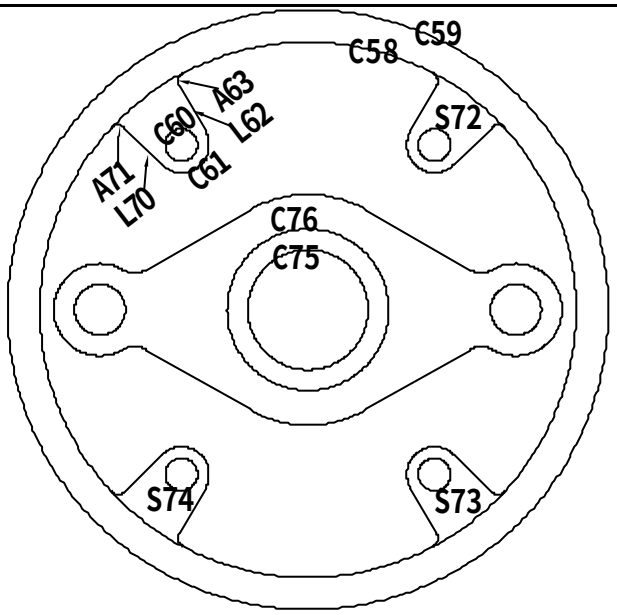
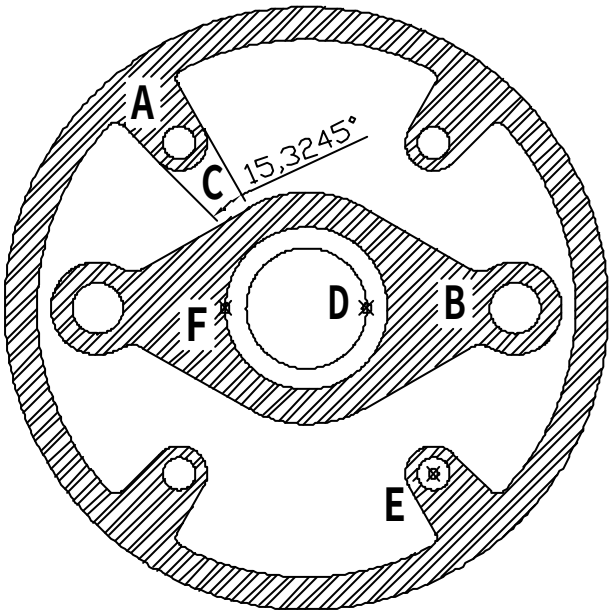



01、 Circle, **任一點** P01, D, 50, →C50
02、 Line, qua,, C50 **上**, (**正交模式**), **向右** 30, →L51

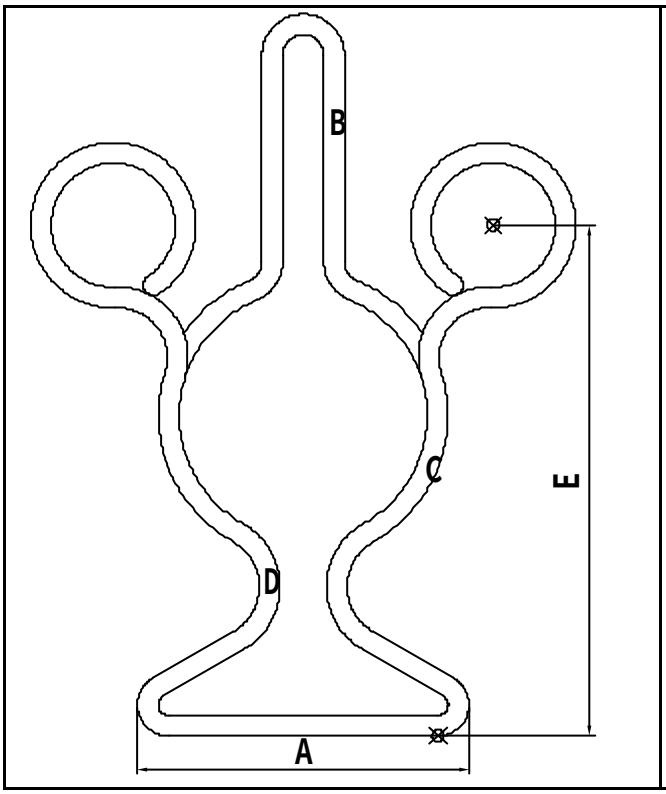



03、 ROtate, L51, cen, C50, -30, →L51 **轉動**
04、 MIrror, L51, qua, C50 **左**, qua, C50 **右**, →L52
05、 Circle, cen, C50, D, 90, →C53
06、 Circle, qua, C53 **右**, D, 11, →C54
07、 Circle, @(**鎖定上次點位**), D, 20, →C55

| | |
|---|---|
|  | 08、 Fillet, T, T, R, 5, L51, C55 **左上**, →A56, L52, C55 **左下**, →A57<br>09、 Circle, cen, C53, D, 116, →C58<br>10、 Circle, @(**鎖定上次點位**), D, 130, →C59<br>11、 Circle, qua, C53 **上**, D, 7, →C60<br>12、 Circle, @(**鎖定上次點位**), 6, →C61<br>13、 Line, cen, C53, tan, C61, →L62 |
|  | 14、 Fillet, T, T, R, 2, C58, L62, →A63<br>15、 MIrror, L51;L52;C54~A57;L62;A63, qua, C59 **上**, qua, C59 **下**, →L64~A71 |
|  | 16、 TRim,<br>17、 Erase, |

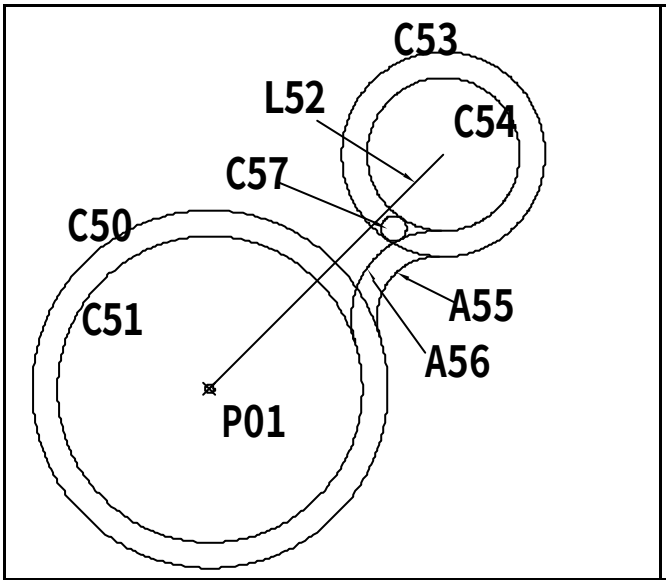
| | |
|---|---|
|  | 18、 ROtate, C60~A63;L70;A71, cen ,C59, 37.5, →C60~A63;L70;A71 **轉動** |
|  | 19、 MIrror, C60~A63;L70;A71, qua, C59 **上**, qua, C59 **下**, →**右上方** S72 **相同物件組**<br>20、 MIrror, C60~A63;L70;A71;S72, qua, C59 **左**, qua, C59 **右**, →**下方** S73;S74 **相同物件組**<br>21、 Circle, cen, C58, D, 26, →C75<br>22、 Circle, @(**鎖定上次點位**), D, 35, →C76 |
|  | 23、 TRim,<br>24、 Hatch, **點** A **區域**, →**斜線區域**<br>25、 **點選** A **斜線區域**, CTRL+1, →**查幾何面積值** =3494.3010<br>26、 Hatch, **點** B **區域**, →**斜線區域**<br>27、 **點選** B **斜線區域**, CTRL+1, →**查幾何面積值** =2077.2033<br>28、 DimANgular, L62, L70, →15.3245<br>29、 DIst, qua, D, cen, E, →Y **差值**=35.7009<br>30、 DIst, cen, E, qua, F, →**距離**=57.3590 |

## 2.5.7 AutoCAD2D2008-507

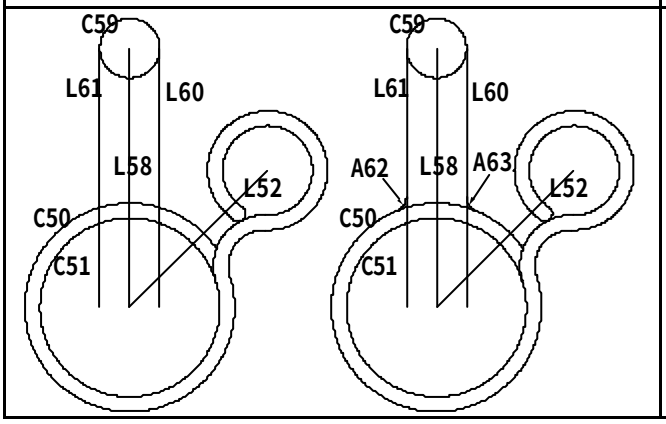



1、 **尺寸** A **長度為何？** 80.1995
2、 B **區域面積為何？**923.7420
3、 C **區域周長為何？**1103.8020
4、 **弧** D **長度為何？** 36.9212
5、 **尺寸** E **長度為何？** 123.3241

**解題步驟：**



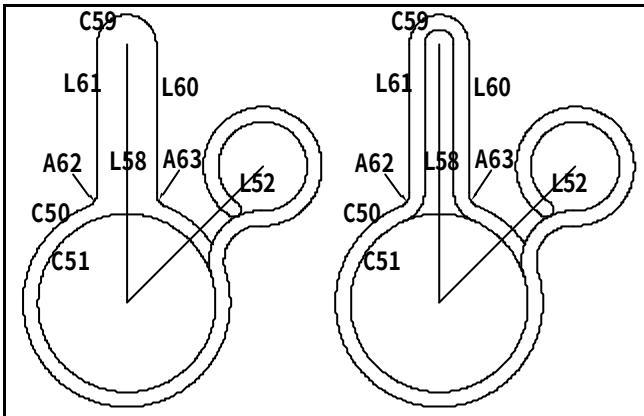
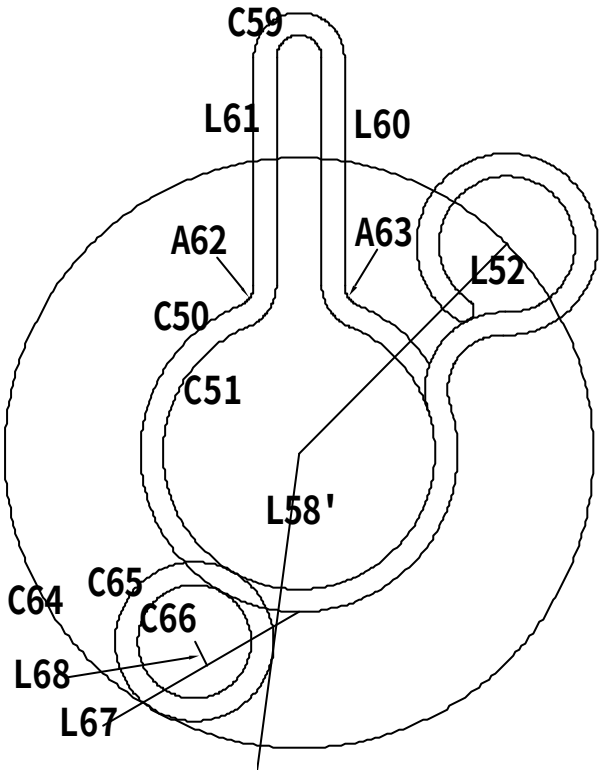
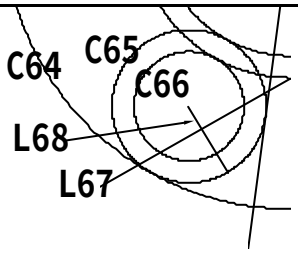
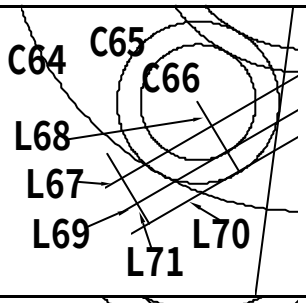
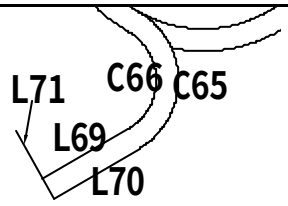
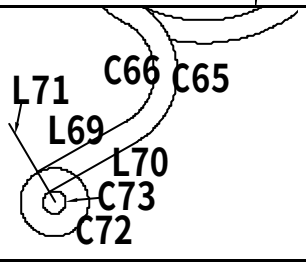

01、 Circle, **任一點** P01, D, 70(=65+5), →C50
02、 Offset, 5, C50, **向內點**, →C51
03、 Line, cen, C50, @65<45, →L52
04、 Circle, end, L52 **右上**, 20, →C53
05、 Offset, 5, C53, **向內點**, →C54
06、 Fillet, T, T, R, 12, C50 **右上**,C53 **左下**, →A55
07、 Offset, 5, A55, **向上點**, →A56
08、 Circle, 3P, tan, C54 **左下**, tan, C53 **左下**, tan, A56 **左上**, →C57

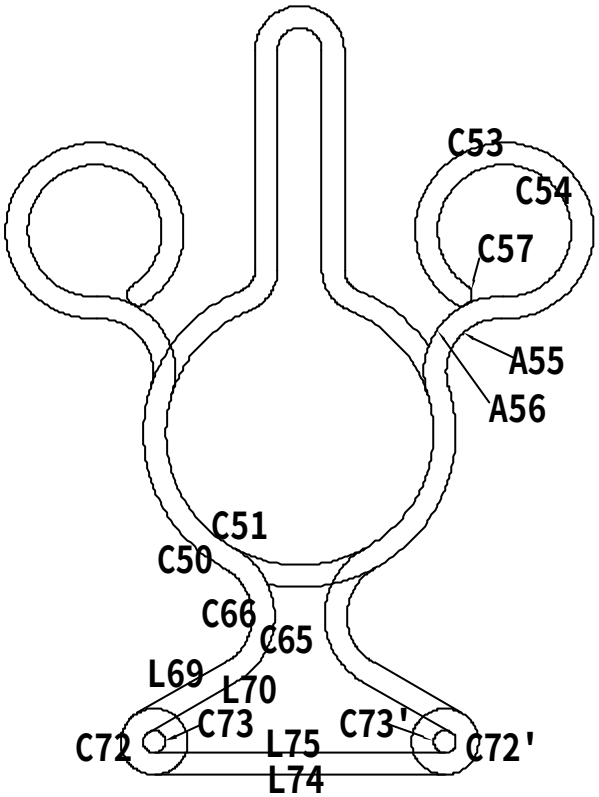



09、 TRim,
10、 Line, cen, C50, (**正交模式**), **向上** 87, →L58
11、 Circle, end, L58 **上**, 10, →C59
12、 Offset, 10, L58, **向右點**, →L60, **向左點**, →L61

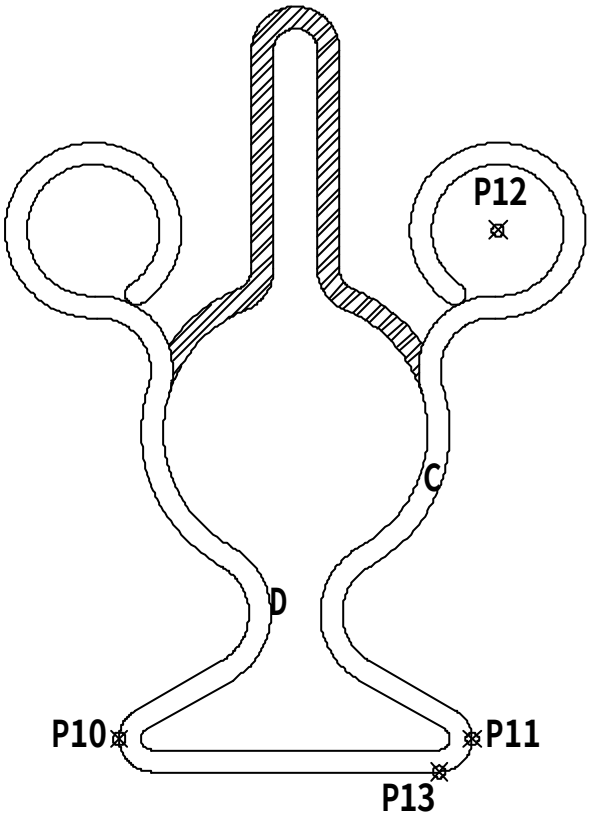
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

13、 Fillet, T, N, R, 5, L61, C50, →A62
14、 Fillet, T, N, R, 5, L60, C50, →A63

| | |
|---|---|
|  | 15、 TRim,<br>- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -<br>16、 Offset, 5, C59, **向內點**, L60, **向內點**, L61, **向內點**, A62, **向內點**, A63, **向內點** |
|  | 17、 TRim,<br>18、 ROtate, L58, cen, C50, 172.5(=180-15/2) →L58'<br>19、 Circle, cen, C50, end, L52 **上**, →C64<br>20、 Circle, 3P, tan , C51, tan , L58', tan, C64, →C65<br>21、 Offset, 5, C65, **向內點**, →C66<br>22、 Line, qua, C50 **下**, @50<-150, →L67<br>23、 Line, cen, C66, per, L67, →L68 |
|   | 24、 EXtend, C65, L68, →L68 **延長至** C65<br>25、 Offset, T, L67, int, C66 & L68, →L69, L67, int, C66 & L65, →L70<br>- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -<br>26、 Offset, 22.5, L68, **向左下點**, →L71 |
|   | 27、 TRim,<br>28、 Erase,<br>- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -<br>29、 LENgthen, DE, 2.5(=7.5-5), **點** L71 **下端** →L71 **向下延伸**<br>30、 Circle, end, L71 **下**, 7.5, →C72<br>31、 Offset, 5, C72, **向內點**, →C73 |

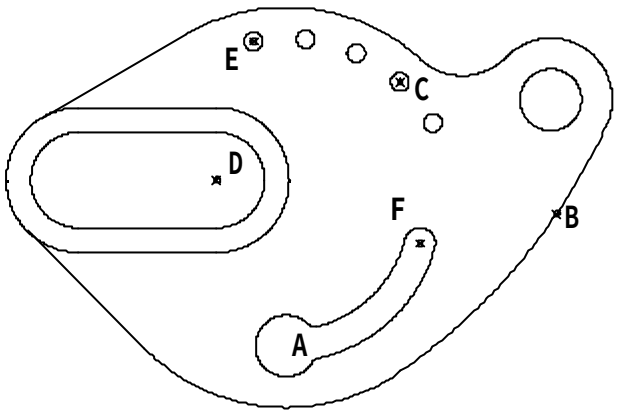



32、 Erase,
33、 MIrror, C53;C54;A55;A56;C57;C65;C66;L69;L70;L72;L73, qua, C50 下, qua, C51 下, →L72' ;L73' 等物件
34、 Line, qua, C72 下, qua, C72'下, →L74
35、 Line, qua, C73 下, qua, C73'下, →L75



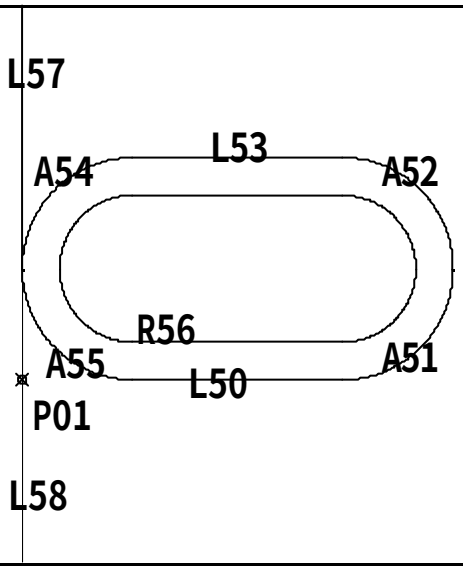

36、 TRim,
37、 Erase,
38、 DIst, qua, P10, qua, P11, →尺寸 A 長度=80.1995
39、 Hatch, 點斜線區域, →斜線區域
40、 點斜線區域, CTRL+1, →查幾何面積值=923.7420
41、 Boundary, 點 C 區域,
42、 點 C 區域邊緣, CTRL+1,→查幾何長度值=1103.8020
43、 點弧 D, CTRL+1,→查幾何弧長值=36.9212
44、 Dist, cen, P12, end, P13, →尺寸 E 長度=123.5293

## 2.5.8 AutoCAD2D2008-508

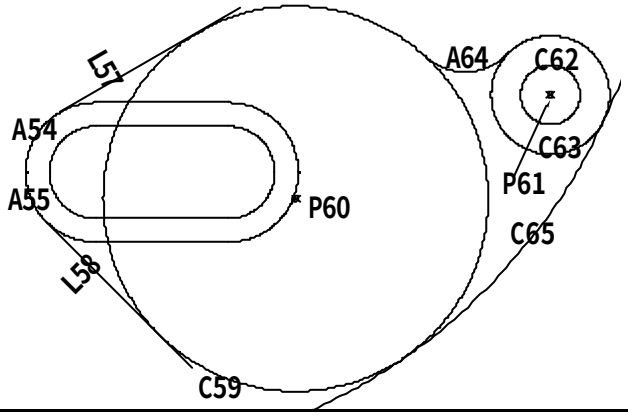



1、 A **區域周長為何？** 167.9482
2、 **弧段** B **夾角為何？** 30.1994
3、 **中心點** C **至中心點** D **距離為何？** 67.5152
4、 **中心點** E **至中心點** F **距離為何？** 85.0615
5、 **圖形最外圍周長為何？** 516.3625

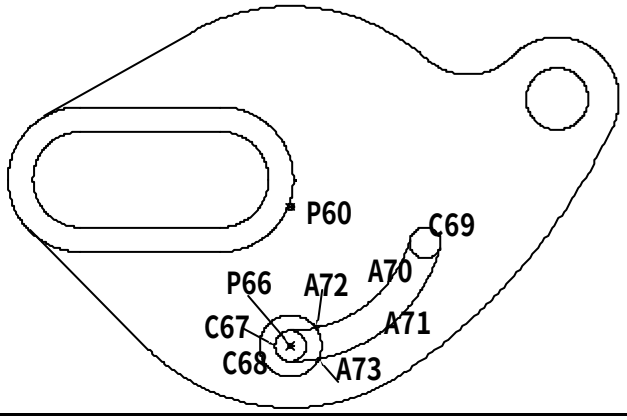
**解題步驟：**



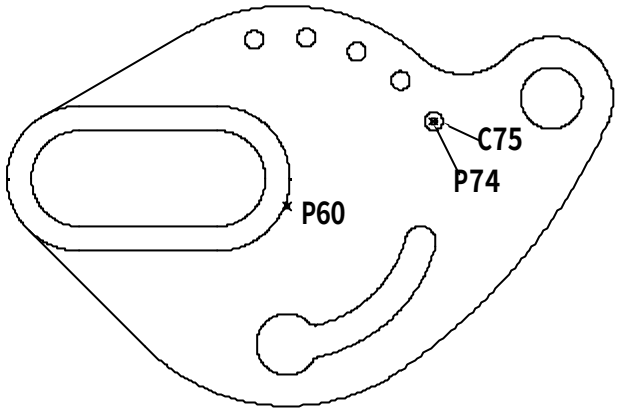
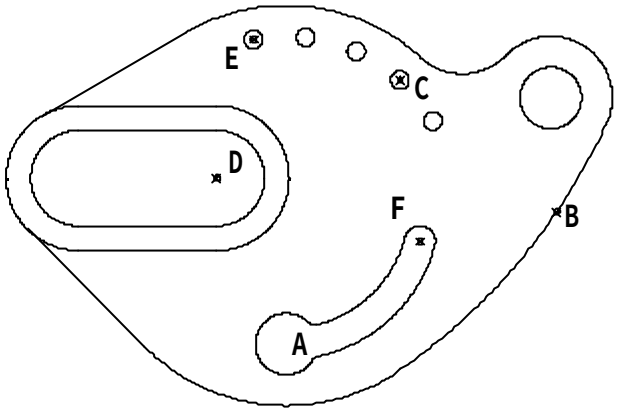
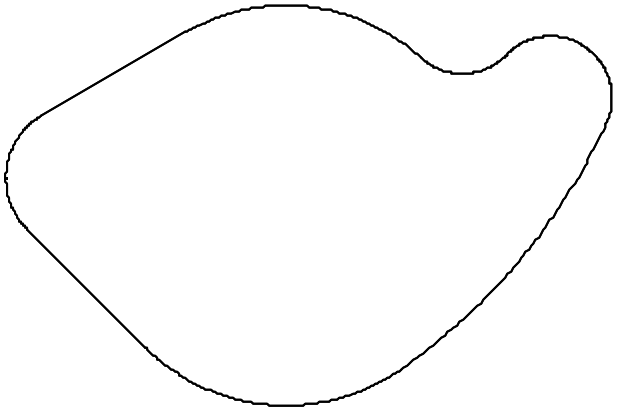

01、 RECtangle, F, 23.5, **任一點** P01, D, 92, 47, **向右上方點**, →L50~A55 **之圓角矩形**
02、 Offset, 8, L50~A55 **之圓角矩形**, **向內點**, →R56 **之圓角矩形**
03、 Line, qua, A54 **左**, (**正交模式**), **向上** 70, →L57
04、 Line, qua, A54 **左**, (**正交模式**), **向下** 70, →L58



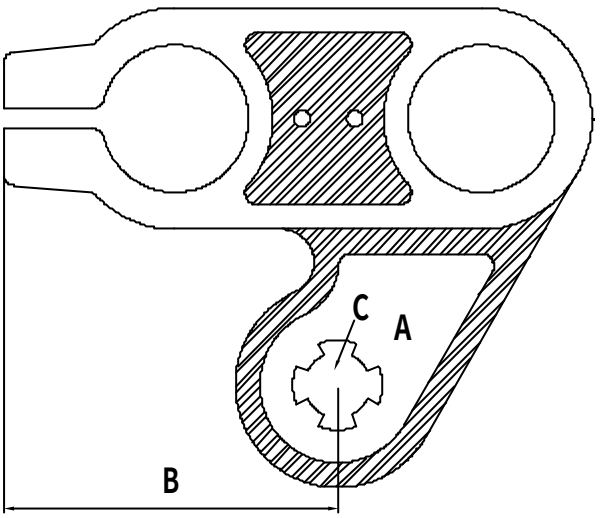

05、 ROtate, L57, cen, A54, -60, →L57 **轉動**
06、 ROtate, L58, cen, A54, 45, →L58 **轉動**
07、 Circle, T, L57, L58, 65(=130/2), →C59
08、 POint, cen, C59, →P60
09、 COpy, P60, nod, P60, @86,35, →P61
10、 Circle, nod, P61, D, 20, →C62
11、 Circle, @(**鎖定上次點位**), D, 40, →C63
12、 Fillet, T, T, R, 20, C59 **右上**, C63 **左上**→A64
13、 Circle, T, C59 **右下**, C63 **右下**, 200, →C65




14、 TRim,
15、 Erase,
16、 COpy, P60, nod, P60, (**正交模式**), **向下** 45, →P66
17、 Circle, nod, P66, 5, →C67
18、 Circle, @(**鎖定上次點位**), D, 20, →C68
19、 ROtate, C67, nod, P60, C, 75, →C69
20、 Arc, C, nod, P60, qua, C67 **上**, A, 75, →A70
21、 Offset, 10, A70, **向下點**, →A71
22、 Fillet, T, T, R, 3, C68 **右上**, A70, →A72, C68 **右下**, A71, →A73

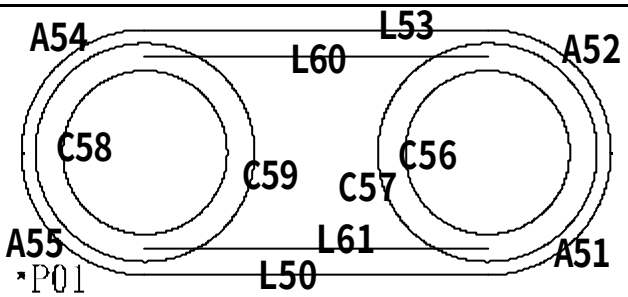
| | |
|---|---|
|  | 23、 TRim,<br>24、 Erase,<br>25、 COpy, P60, nod, P60, @55<30, →P74<br>26、 Circle, nod, P74, D, 6, →C75<br>27、 ARray, **環形陣列**, **中心點**, nod, P60, **項目總數**, 5, **項目間角度**, 17.8, **物件**, C75, →5 **個相同物件** |
|  | 28、 Erase,<br>29、 BOundary, **點** A **區域**<br>30、 **點** A **區邊緣**, CTRL+1, →**查幾何長度值** =167.9482<br>31、 **點弧** B, CTRL+1, →**查幾何總角度值** =30.1994<br>32、 DIst, cen, C, cen, D, →**距離**=67.5152<br>33、 DIst, cen, E, cen, F, →**距離**=85.0615 |
|  | 34、 COpy, TRim, Erase, →**左側圖形**<br>35、 BOundary, **點中間區域**<br>36、 **點邊緣**, CTRL+1, →**查幾何長度值** =516.3625 |

## 2.5.9 AutoCAD2D2008-509

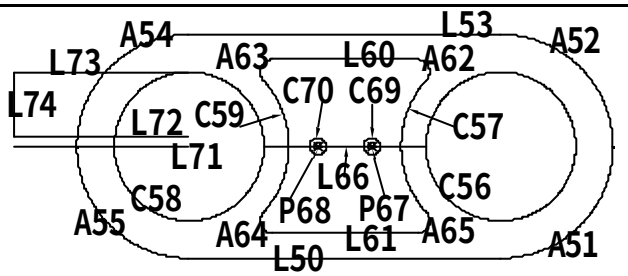



1、 A **區域扣除內孔面積為何？** 1565.7087
2、 **尺寸** B **長度為何？** 81.7816
3、 **斜線區域總面積為何？** 2630.9468
4、 C **區周長為何？** 84.7375
5、 **圖形所圍成最外圍周長為何？** 616.4273

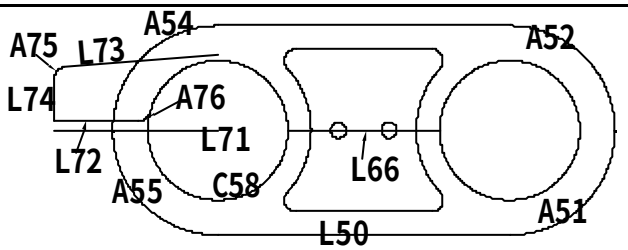
**解題步驟：**



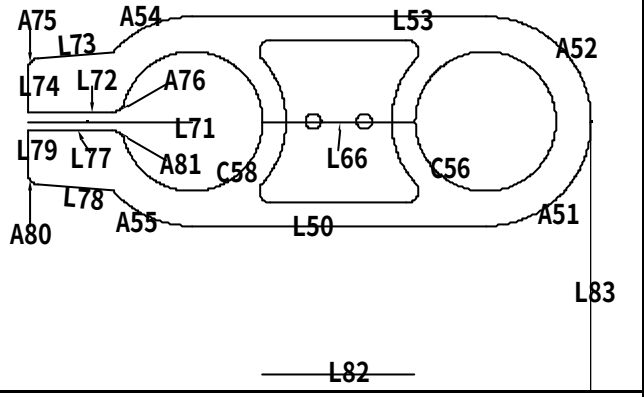
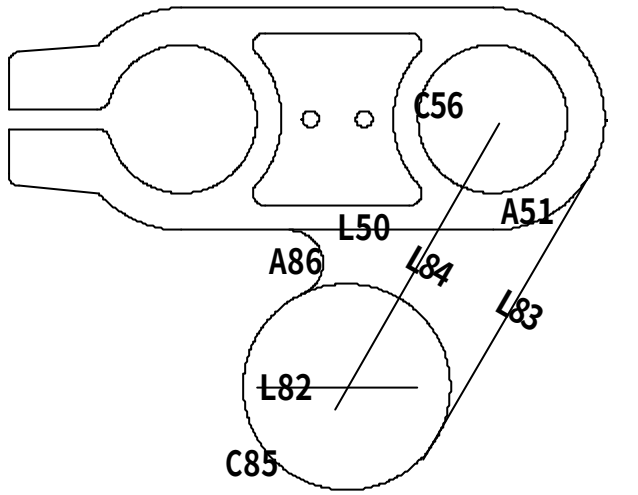
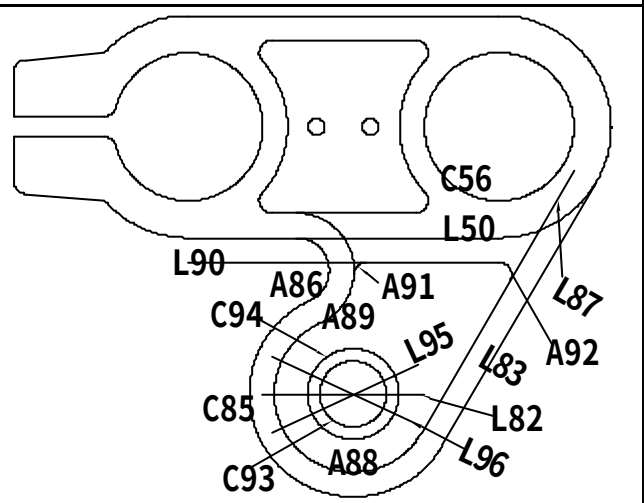
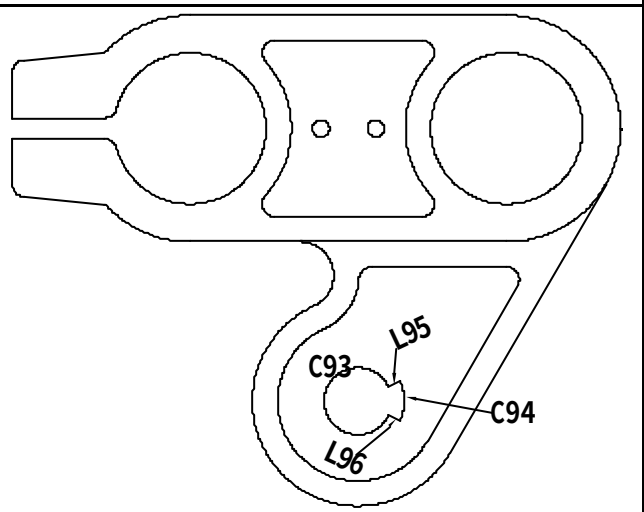

01、 RECtangle, F, 27, **任一點** P01, D, 129, 54, **向右上方點**, →L50~A55 **之圓角矩形**
02、 eXplode, L50~A55 **之圓角矩形**, →L50~A55
03、 Circle, cen, A51, D, 36, →C56
04、 Circle, @(**鎖定上次點**), 24, →C57
05、 COpy, C56;C57, cen, A51, cen, A55, →C58;C59
06、 Offset, 6, L53, **向下點**, →L60, L50, **向上點**, →L61

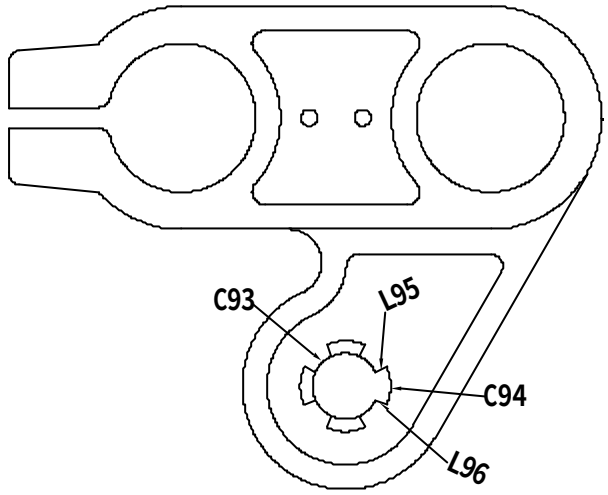
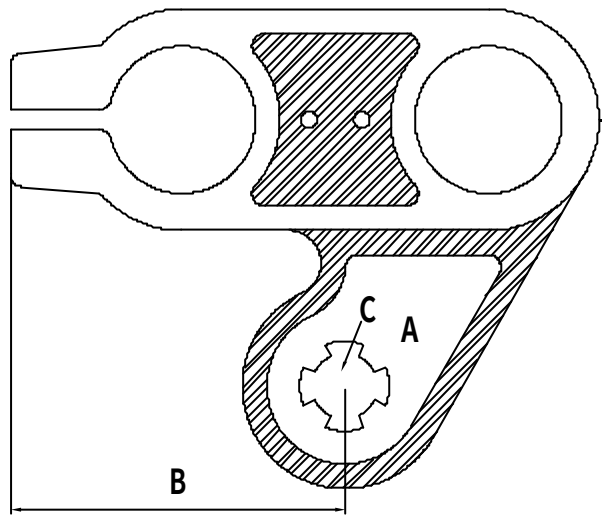
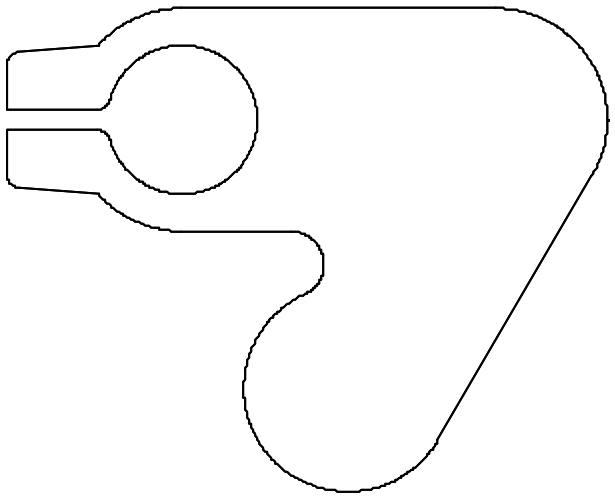



07、 Fillet, T, T, R, 3, C57, L60, →A62, C59, L60, →A63, C59, L61, →A64, C57, L61, →A65
08、 TRim,
09、 Line, qua, C58 **右**, qua, C56 **左**, →L66
10、 DIVide, L66, 3, →P67;P68
11、 Circle, nod, P67, D, 4, →C69
12、 Circle, nod, P68, D, 4, →C70
13、 Line, cen, C58, (**正交模式**), **向左** 42, →L71
14、 Offset, 2.5(=5/2), L71, **向上點**, →L72
15、 Offset, 18, L71, **向上點**, →L73
16、 Line, end, L72 **左**, end, L73 **左**, →L74

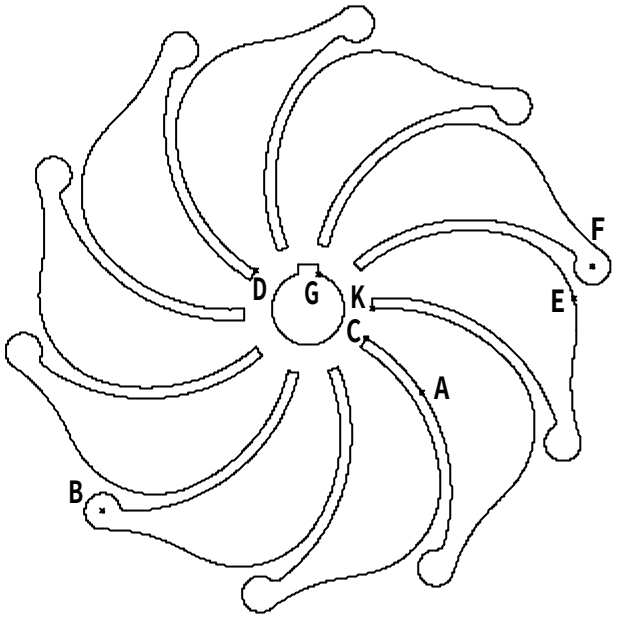



17、 ROtate, L73, int, L73 & A54, 5, →L73 **轉動**
18、 Fillet, T, T, R, 3, L73, L74, →A75, L72, C58, →A76

| | |
|---|---|
|  | 19、 TRim,<br>20、 Erase,<br>21、 MIrror, L72~A76, end, L71 **左**, end, L71 **右**, →L77~A81<br>22、 Offset, 65, L66, **向下點**, →L82<br>23、 Line, qua, A51 **右**, (**正交模式**), **向下** 80, →L83 |
|  | 24、 TRim,<br>25、 Erase,<br>26、 ROtate, L83, cen, C56, -30, →L83 **轉動**<br>27、 Offset, 25, L83, **向左點**, →L84<br>28、 Circle, int, L82 & L84, 25, →C85<br>29、 Fillet, T, N, R, 8, L50, C85 **上**, →A86 |
|  | 30、 Erase,<br>31、 Offset, 6, L83, **向左點**, →L87, C85, **向上點**, →A88, A86, **向右點**, →A89, L50, **向下點**, →L90<br>32、 Fillet, T, T, R, 3, A89, L90, →A91, L90, L87, →A92<br>33、 Circle, cen, C85, D, 16, →C93<br>34、 Circle, cen, C85, D, 22, →C94<br>35、 ROtate, L82, cen, C85, C, 25, →L95<br>36、 ROtate, L82, cen, C85, C, -25, →L96 |
|  | 37、 TRim,<br>38、 Erase, |

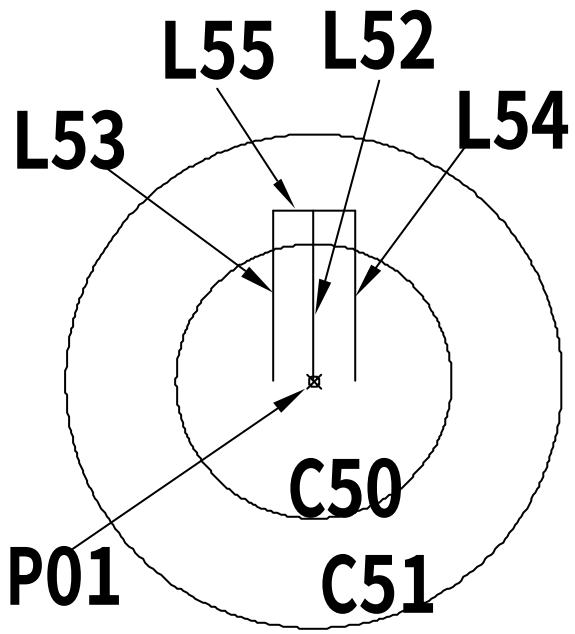
| | |
|---|---|
|  | 39、 ARray, **環形陣列**, **中心點**, cen, C93, **項目總數**, 4, **佈滿角度**, 360, **物件**, C94~L96, **→全圓佈滿** 4 **組相同物件** |
|  | 40、 TRim,<br>41、 Hatch, **點** A **區域**, **→斜線區域**<br>42、 **點** A **區域斜線**, CTRL+1, **→查幾何面積值** =1565.7087<br>43、 DimLInear, int, L79 & A80, cen, C85, **→**81.7816<br>44、 Hatch, **分別點斜線區域**, **→斜線區域**<br>45、 **點斜線區域**, CTRL+1, **→查幾何面積值** =2630.9468<br>46、 BOundary, **點** C **區域**<br>47、 **點** C **區邊緣**, CTRL+1, **→查幾何長度值** =84.7375 |
|  | 48、 COpy, TRim, Erase, **→左側圖形**<br>49、 BOundary, **點中間區域**<br>50、 **點邊緣**, CTRL+1, **→查幾何長度值** =616.4273 |

## 2.5.10 AutoCAD2D2008-510

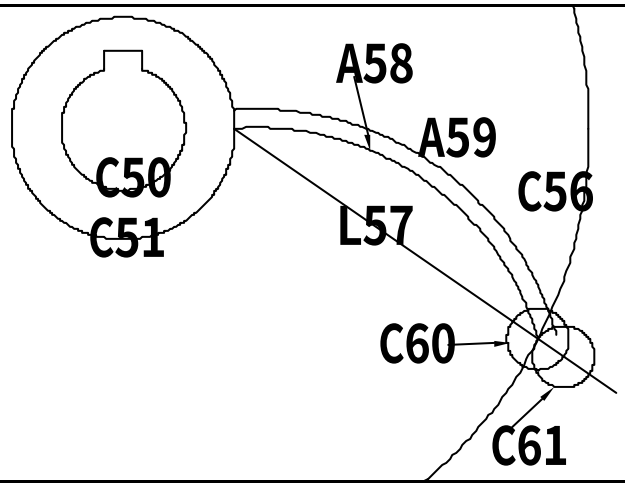



1、 **弧** A **夾角為何？** 75.3370
2、 **中心點** B **至交點** C **距離為何？** 87.5254
3、 **交點** D **至弧** E **中心點距離為何？** 66.0053
4、 **中心點** F **至交點** G **距離為何？** 76.1246
5、 **圖形扣除內孔面積為何？** 16173.2946

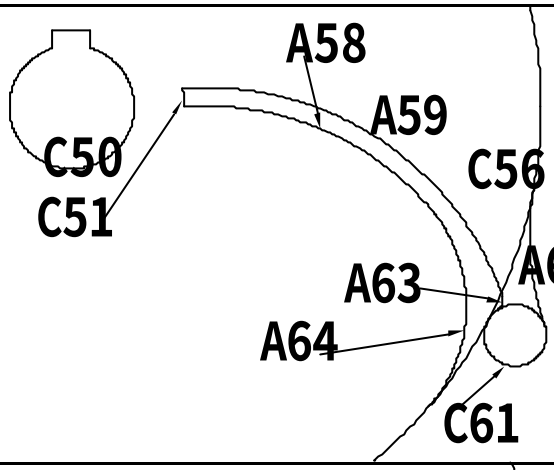
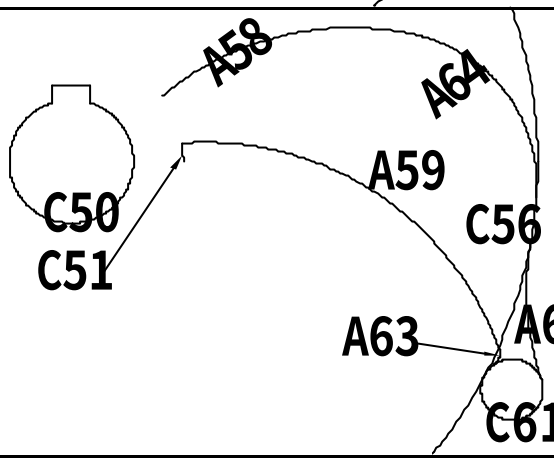
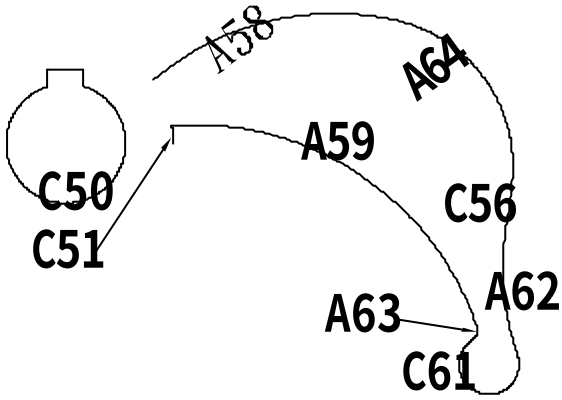
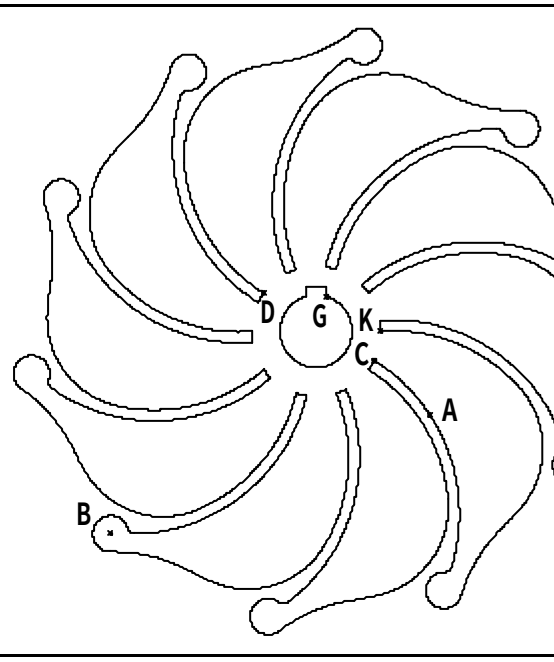
**解題步驟：**



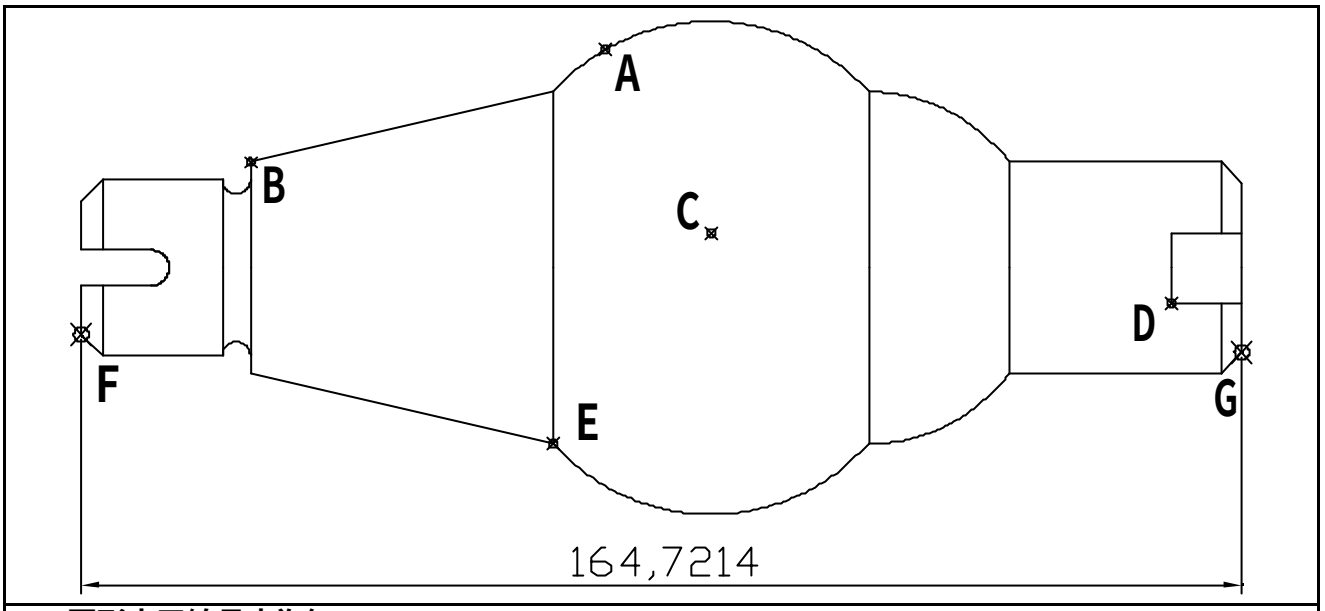

01、 Circle, **任一點** P01, D, 20, →C50
02、 Circle, @(**鎖定上次點位**), D, 36, →C51
03、 Line, cen, C50, (**正交模式**), **向上** 12.5, →L52
04、 Offset, 3(=6/2), L52, **向左點**, →L53, L52, **向右點**, →L54
05、 Line, end, L53 **上**, end, L54 **上**, →L55




06、 TRim,
07、 Erase,
08、 Circle cen, C51, D, 150, →C56
09、 Line, qua, C51 **右**, @75<-35, →L57
10、 Arc, int, C56 & L57, E, int, C51 & L57, R, 47, →A58
11、 Offset, 3(=50-47), A58, **向上點**, →A59
12、 Circle, int, C56 & L57, 5, →C60
13、 Circle, int, C60 & L57, 5, →C61

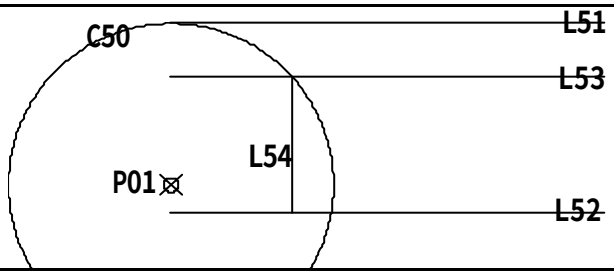
| | |
|---|---|
|  | 14、 Fillet, T, T, R, 0, C51, A59, →使 A59 與 C51 接合<br>15、 TRim,<br>16、 Erase,<br>17、 Fillet, T, T, R, 45, C61 右上, C56 右, →A62<br>18、 Fillet, T, T, R, 2, A59 下, C61 左上, →A63<br>19、 Fillet, T, T, R, 25, A58 下, C56 右下, →A64 |
|  | 20、 ROtate, A58;A64, cen, C50, 36, →A58;A64 轉動 |
|  | 21、 TRim,<br>22、 Erase, |
|  | 23、 ARray, 環形陣列, 中心點, cen, C50, 項目總數, 10, 佈滿角度, 360, 物件, C51;C56;A58;A59;C61~C64, →全圓佈滿 10 組相同物件<br>24、 點弧 A, CTRL+1, →查幾何總角度值 =75.3370<br>25、 DIst, cen, B, int, C, →距離=87.5254<br>26、 DIst, int, D, cen, E, →距離=66.0053<br>27、 DIst, cen, F, int, G, →距離=76.1246<br>28、 Hatch, , 點圖內區域, →斜線區域<br>29、 點斜線區域, CTRL+1, →查幾何面積值 =16173.2946 |

## 2.5.11 AutoCAD2D2008-511

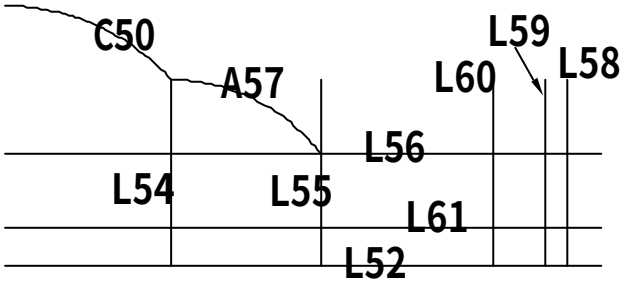



1、 **圖形水平總長度為何？** 164.7214
2、 **弧** A **夾角為何？** 96.3794
3、 **交點** B **至中心點** C **距離為何？** 66.1212
4、 **交點** D **至交點** E **距離為何？** 89.9724
5、 **圖形最外圍所圍成的面積為何？** 6954.7401

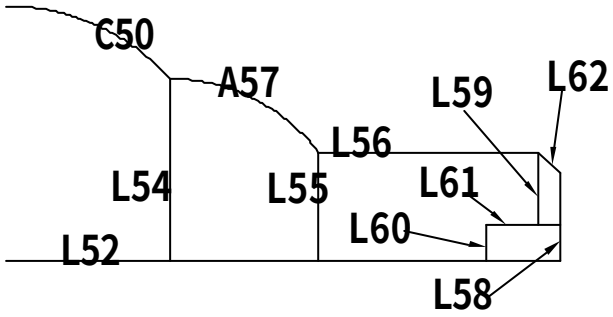
**解題步驟：**



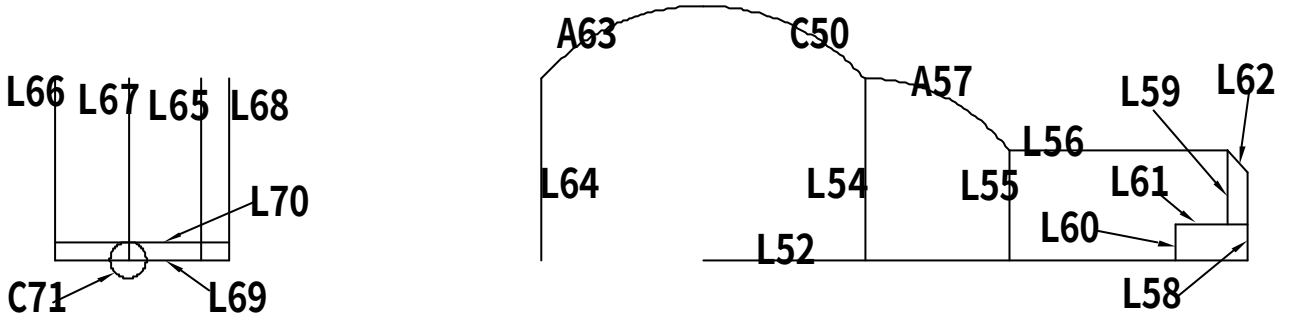

01、 Circle, **任一點** P01, 30, →C50
02、 Line, qua, C50 **上**, (**正交模式**), **向右** 80, →L51
03、 Offset, 35(=70/2), L51, **向下點**, →L52
04、 Offset, 10(=(70-50)/2), L51, **向下點**, →L53
05、 Line, int, C50 & L53, (**正交模式**), **向下** 25, →L54



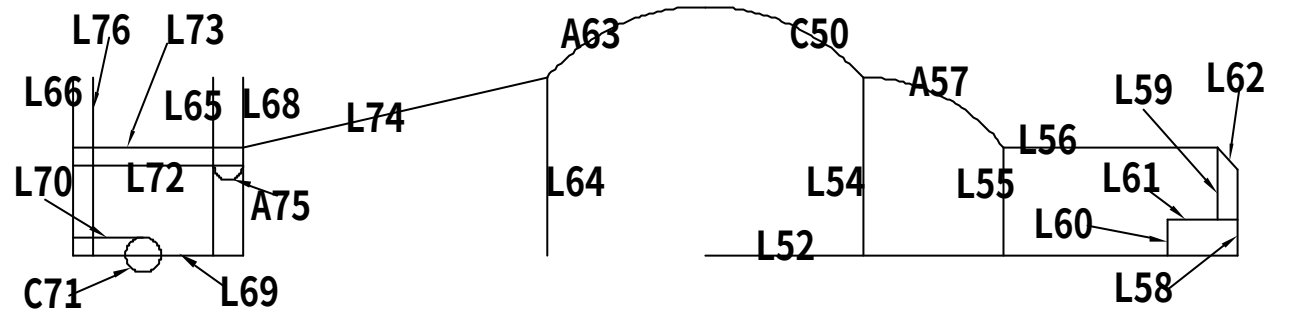

06、 TRim,
07、 Erase,
08、 Offset, 20, L54, **向右點**, →L55
09、 Offset, 15, L52, **向上點**, →L56
10、 Arc, C, int, L52 & L54, int, L55 & L56, int, C50 & L54, →A57
11、 Offset, 33, L55, **向右點**, →L58
12、 Offset, 3, L58, **向左點**, →L59
13、 Offset, 10, L58, **向左點**, →L60
14、 Offset, 5, L52, **向上點**, →L61



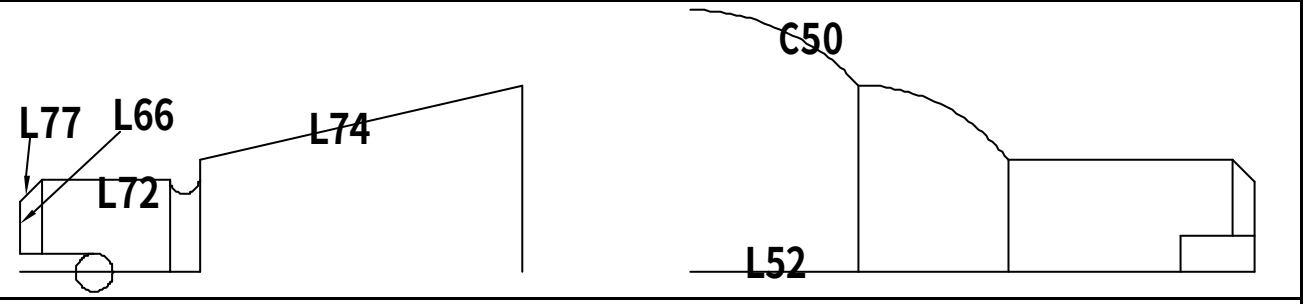

15、 CHAmfer, D, 3, 3, L56, L58, →L62
16、 TRim,
17、 Erase,



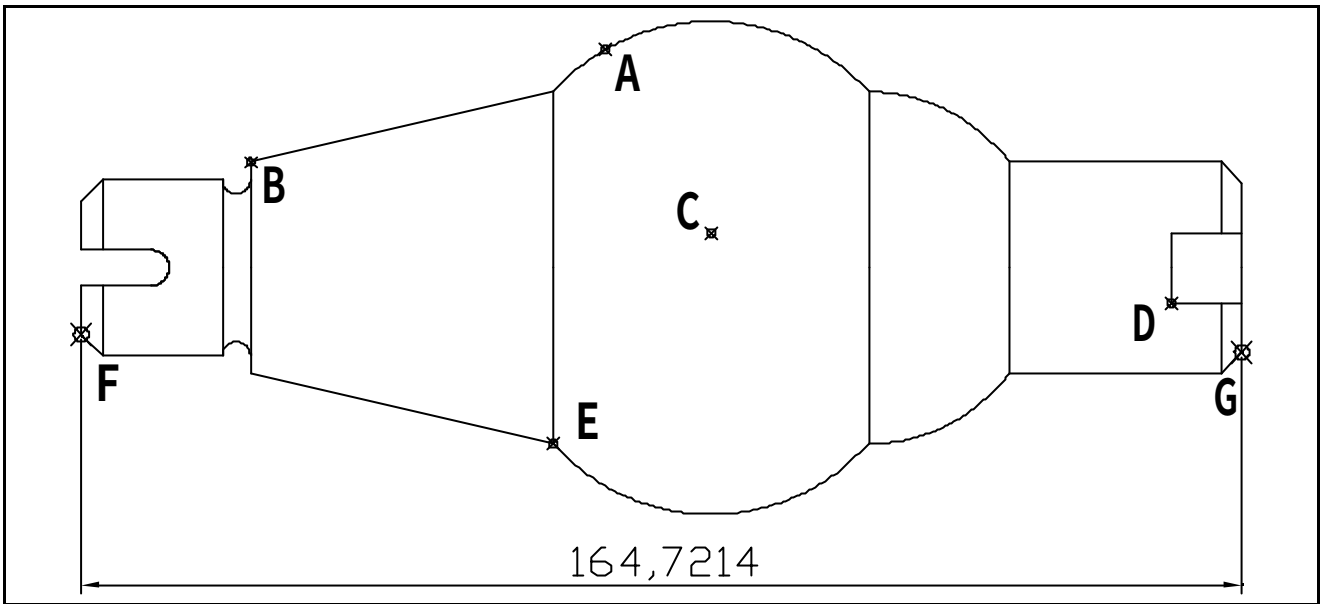

18、 MIrror, C50;L54, end, C50 **左**, end, L52 **左**, →A63;L64
19、 Offset, 47, L64, **向左點**, →L65
20、 Offset, 20, L65, **向左點**, →L66
21、 Offset, 10, L66, **向右點**, →L67
22、 Offset, 4, L65, **向右點**, →L68
23、 Line, end, L66 **下**, end, L68 **下**, →L69
24、 Offset, 2.5, L69, **向上點**, →L70
25、 Circle, int, L67 & L69, 2.5, →C71



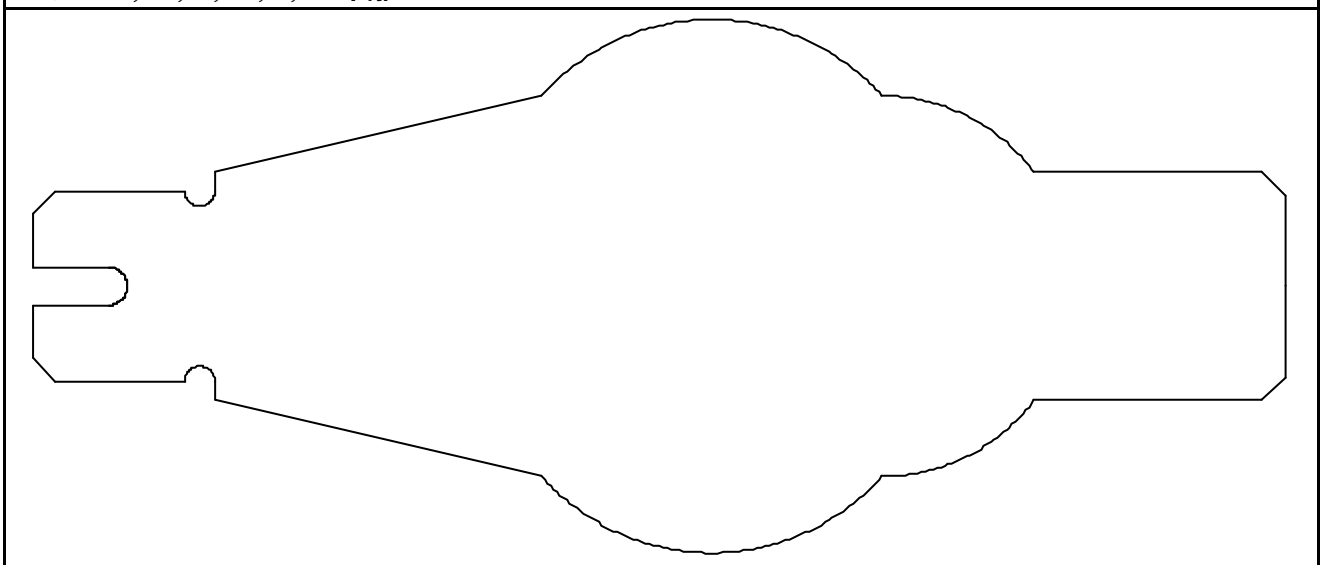

26、 Offset, 12.5, L69, **向上點**, →L72
27、 Offset, 15, L69, **向上點**, →L73
28、 Line, int, A63 & L64, int, L68 & L73, →L74
29、 Arc, int, L65 & L72, E, int, L68 & L72, R, 2, →A75
30、 Offset, 3, L66, **向右點**, →L76




31、 TRim,
32、 Erase, A63, →**刪除** A63
33、 CHAmfer, D, 3, 3, L66, L72, →L77

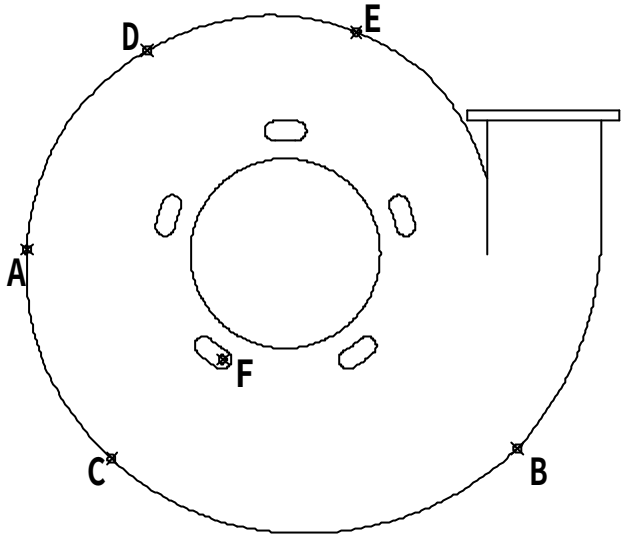



34、 EXtend, L74, C50, →C50 **與** L74 **相交**
35、 MIrror, **上半部物件**, end, L52 **左**, end, L52 **右**, →**下半部物件**
36、 TRim,
37、 Erase,
38、 DimLInear, int, F, int, G, →164.7214
39、 **點弧** A, CTRL+1, →**查幾何總角度值** =96.3794
40、 DIst, int, B, cen, C, →**距離**=66.1212
41、 DIst, int, D, int, E, →**距離**=89.9724



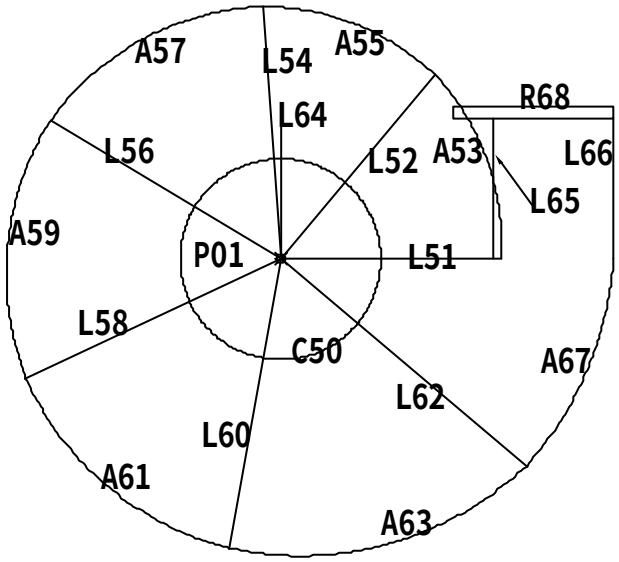
42、 COpy, TRim, Erase, →**上方圖形**
43、 BOundary, **點中間區域**
44、 **點邊緣**, CTRL+1, →**查幾何面積值** =6954.7401

## 2.5.12 AutoCAD2D2008-512

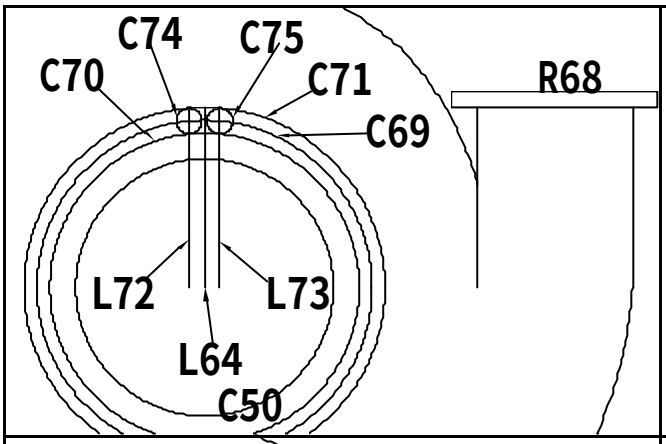
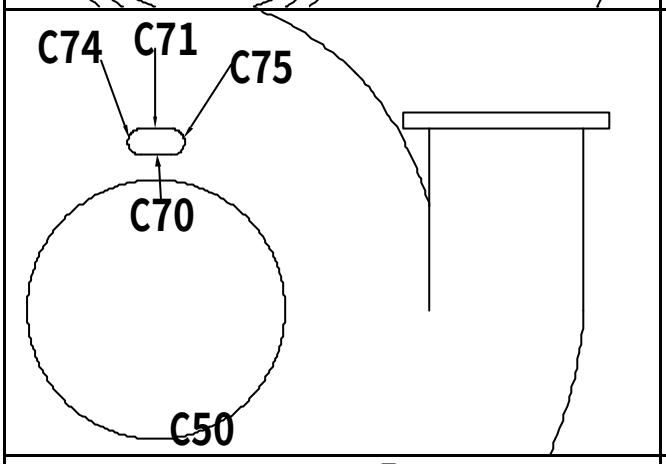
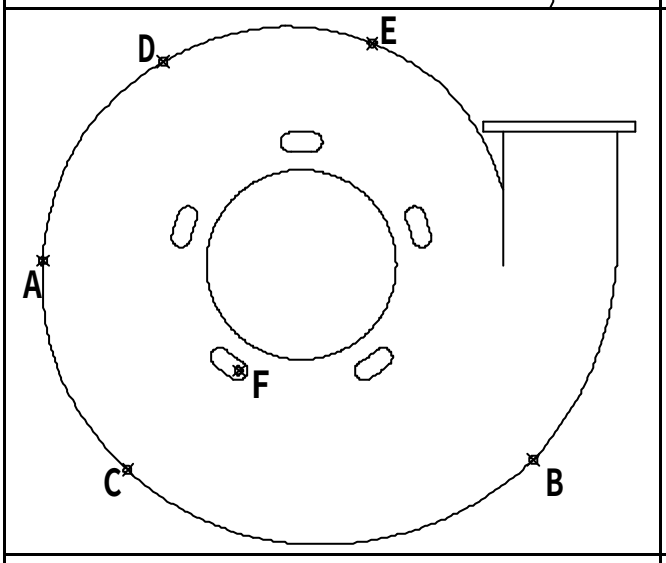
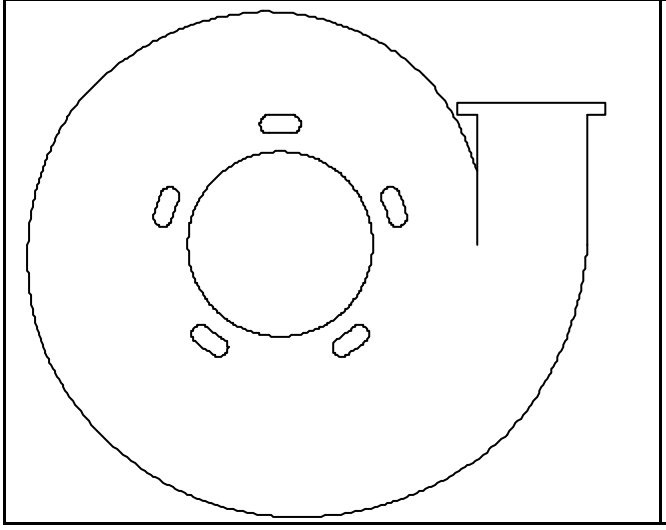



1、 **弧** A **中心點至交點** B **距離為何？** 74.8922
2、 **弧** C **長度為何？** 68.4330
3、 **弧** D **包含角為何？** 51.6647
4、 **中點** E **至中心點** F **距離為何？** 93.2765
5、 **圖形最外圍扣除內孔的面積為何？** 13851.0926

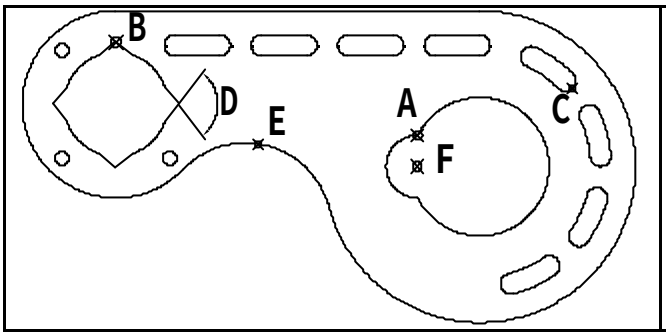
**解題步驟：**




01、 Circle, **任一點** P01, D, 50, →C50
02、 Line, cen, C50, (**正交模式**), **向右** 55, →L51
03、 Line, cen, C50, @60<50, →L52
04、 Arc, end, L51 **右**, E, end, L52 **上**, R, 62, →A53
05、 Line, cen, C50, @63<94, →L54
06、 Arc, end, L52 **上**, E, end, L54 **上**, R, 64, →A55
07、 Line, cen, C50, @67<149, →L56
08、 Arc, end, L54 **上**, E, end, L56 **上**, R, 69, →A57
09、 Line, cen, C50, @70<205, →L58
10、 Arc, end, L56 **上**, E, end, L58 **下**, R, 72, →A59
11、 Line, cen, C50, @73<260, →L60
12、 Arc, end, L58 **下**, E, end, L60 **下**, R, 75, →A61
13、 Line, cen, C50, @80<320, →L62
14、 Arc, end, L60 **下**, E, end, L62 **下**, R, 82, →A63
15、 Line, cen, C50, (**正交模式**), **向上** 35, →L64
16、 Offset, 53(=68-(30/2)), L64, **向右點**, →L65
17、 Offset, 30, L65, **向右點**, →L66,
18、 Arc, end, L62 **下**, E, end, L66 **下**, R, 85, →A67
19、 RECtangle, F, 0, end, L66 **上**, D, 40, 3, **向左上方點**, →R68

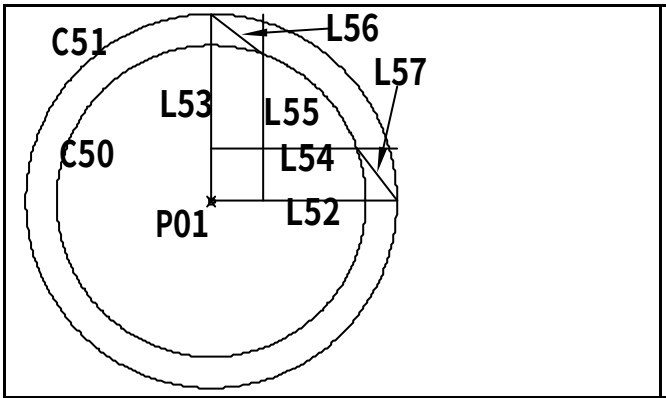
| | |
|---|---|
|  | 20、 Move, R68, end, L66 **上**, (**正交模式**), **向右** 5(=10/2), →R68 **向右移動**<br>21、 TRim,<br>22、 Erase,<br>23、 Circle, cen, C50, D, 65, →C69<br>24、 Offset, 2.5(=5/2), C69, **向下點**, →C70, C69, **向上點**, →C71<br>25、 Offset, 3(=6/2), L64, **向左點**, →L72, L64, **向右點**, →L73<br>26、 Circle, int, C69 & L72, D, 5, →C74<br>27、 Circle, int, C69 & L73, D, 5, →C75 |
|  | 28、 Erase,<br>29、 TRim, |
|  | 30、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 5, **佈滿角度**, 360, **物件**, C70;C71;C74;C75, →**全圓佈滿** 5 **組相同物件**<br>31、 DIst, cen, A, int, B, →**距離**=74.8922<br>32、 **點弧** C, CTRL+1, →**查幾何弧長值**=68.4330<br>33、 **點弧** D, CTRL+1, →**查幾何總角度值** =51.6647<br>34、 DIst, mid, E, cen, F, →**距離**=93.2765 |
|  | 35、 COpy, TRim, Erase, →**左側圖形**<br>36、 Hatch, **點中間區域**, →**斜線區域**<br>37、 **點斜線區域**, CTRL+1, →**查幾何面積值** =13851.0926 |

## 2.5.13 AutoCAD2D2008-513

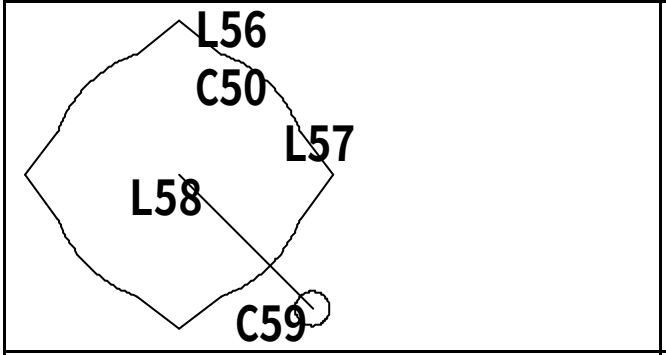



1、 **交點** A **至交點** B **距離為何？** 91.2263
2、 **交點** A **至交點** C **角度為何？** 16.8734
3、 **尺寸** D **角度為何？** 104.6944
4、 **弧** E **中心點至中心點** F **距離為何？** 52.8211
5、 **圖形扣除內孔面積為何？** 7633.9682

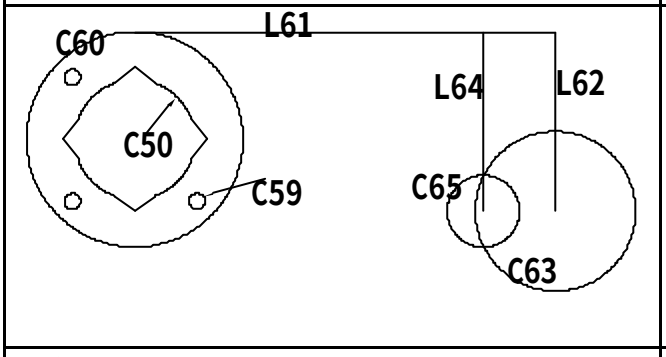
**解題步驟：**



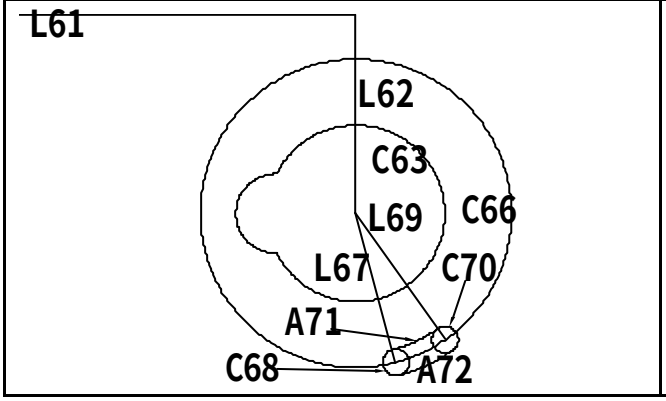

01、 Circle, **任一點** P01, D, 30, →C50
02、 Circle, @(**鎖定上次點**), 18, →C51
03、 Line, qua, C51 **右**, cen, C51, →L52, qua, C51 **上**, →L53
04、 Offset, 5(=10/2), L52, **向上點**, →L54, L53, **向右點**, →L55
05、 Line, int, C51 & L53, int, C50 & L55, →L56
06、 Line, int, C50 & L54, int, C51 & L52, →L57



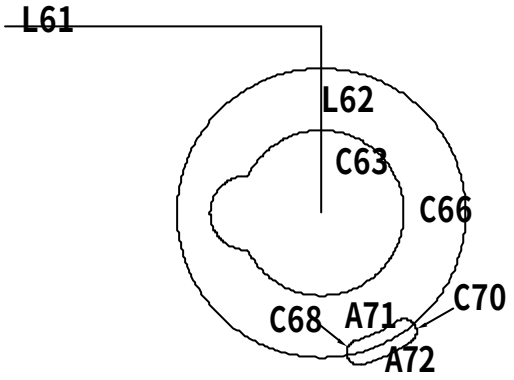
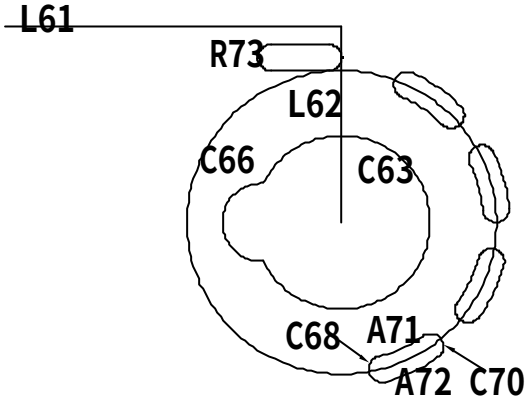
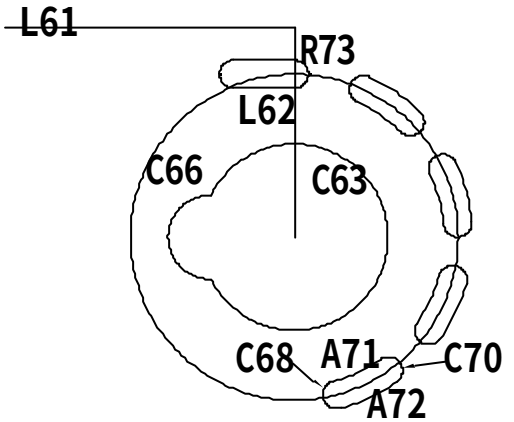
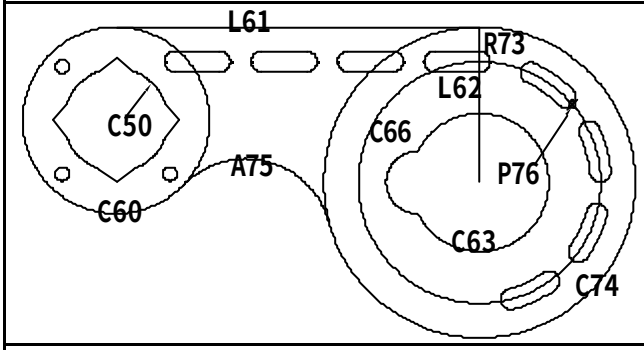
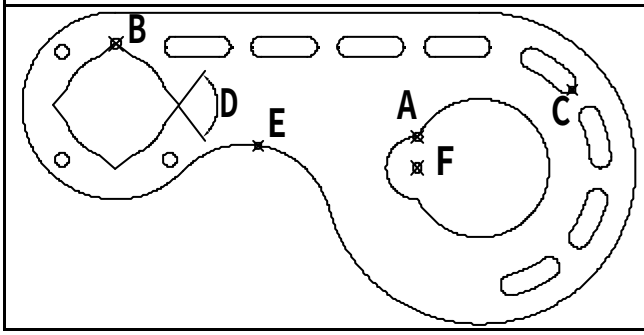

07、 TRim,
08、 Erase,
09、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 4, **佈滿角度**, 360, **物件**, C50;L56;L57, **→全圓佈滿** 4 **組相同物件**
10、 Line, cen, C50, @22<-45, →L58
11、 Circle, end, L58 **下**, D, 4, →C59



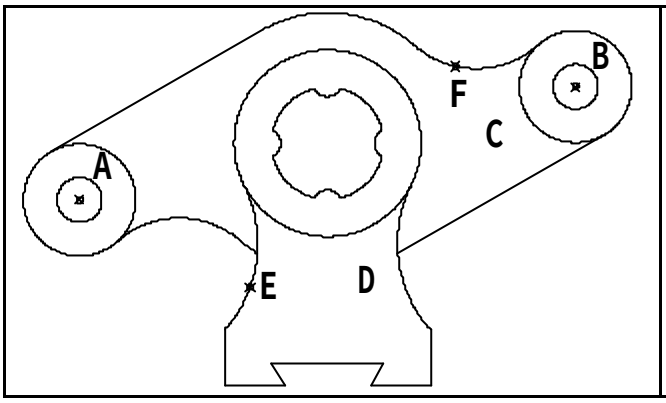

12、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 3, **佈滿角度**, -180, **物件**, C59, **→半圓佈滿** 3 **個相同物件**
13、 Circle, cen, C50, 27, →C60
14、 Line, qua, C60 **上**, (**正交模式**), **向右** 105, →L61, **向下** 45(=90/2), →L62
15、 Circle, end, L62 **下**, D, 40, →C63
16、 Offset, 18(=27-9), L62, **向左點**, →L64
17、 Circle, end, L64 **下**, 9, →C65




18、 Erase,
19、 TRim,
20、 Circle, cen, C63, 35, →C66
21、 Line, cen, C63, @35<-75, →L67
22、 Circle, end, L67 **下**, 3, →C68
23、 ROtate, L67;C68, cen, C63, C, 20, →L69;C70
24、 Arc, C, cen, C63, int, L67 & C68, int, L69 & C70, →A71
25、 Offset, 6(=3x2), A71, **向下點**, →A72

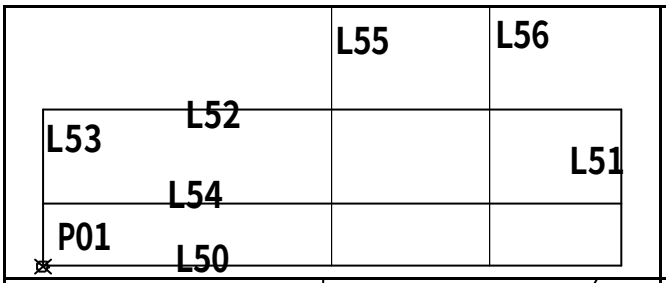
| | |
|---|---|
|  | 26、 TRim,<br>27、 Erase, |
|  | 28、 ARray, **環形陣列**, **中心點**, cen, C63, **項目總數**, 4, **佈滿角度**, 120, **物件**, C68;C70;A71;A72, →**半圓佈滿** 4**個相同物件**<br>29、 RECtangle, F, 3, int, L62 & C66, D, 19, 6, **向左上方點**, →R73 |
|  | 30、 Move, R73, cen, R73 **右**, int, L62 & C66, →R73 **向右下移動** |
|  | 31、 ARray, **矩形陣列**, **列**, 1, **行**, 4, **列偏移**, 0, **行偏移**, -25, **物件**, R73, →**向左佈滿** 4**個相同物件**<br>32、 Circle, cen, C63, D, 90, →C74<br>33、 Fillet, T, T, R, 25, C60 **右下**, C74 **左下**, →A75<br>34、 Point, int, P76(**以利回答第** 2 **題**) |
|  | 35、 DIst, int, A, int, B, →**距離**=91.2263<br>36、 DIst, int, A, nod, C, →XY **平面內角度**= 16.8734<br>37、 DimANgular, D, →104.6944<br>38、 DIst, cen, E, cen, F, →**距離**=52.8211<br>39、 Hatch, **點圖形內區域**, →**斜線區域**<br>40、 **點斜線區域**, CTRL+1, →**查幾何面積值** =7633.9682 |

## 2.5.14 AutoCAD2D2008-514

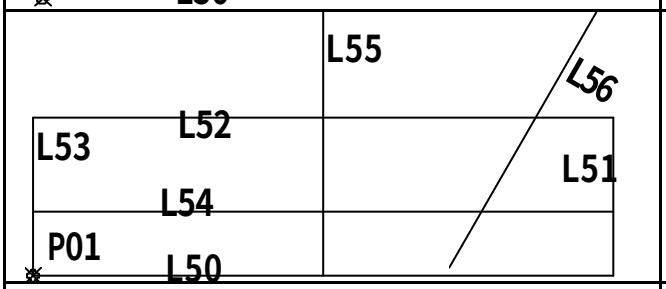



1、 **中心點** A **至中心點** B **水平距離為何？** 132.5833
2、 C **區域面積為何？** 3206.9652
3、 D **區域周長為何？** 244.3491
4、 **弧** E **中心點至中心點** B **距離為何？** 127.1086
5、 **弧** F **包含角為何？** 83.1199

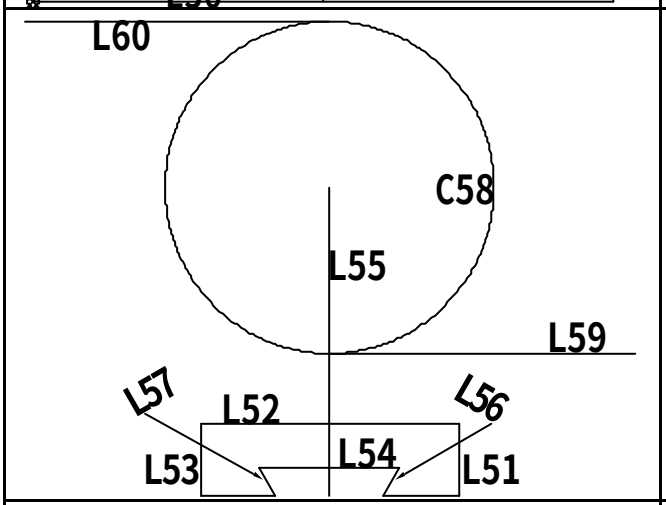
**解題步驟：**



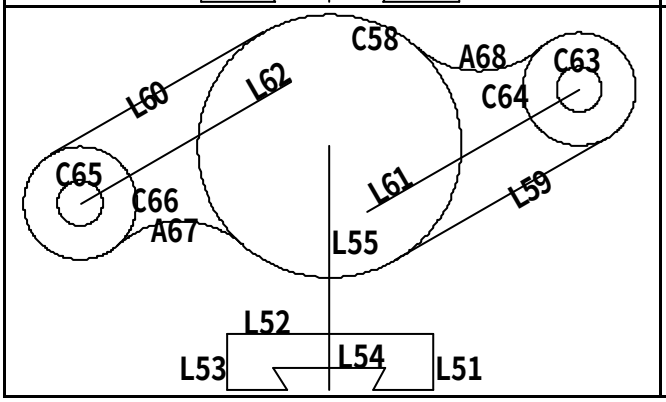

01、 RECtangle, F, 0, **任一點** P01, D, 55, 15, **向右上方點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 6, L50, **向上點**, →L54
04、 Line, mid, L50, (**正交模式**), **向上** 65, →L55
05、 Offset, 15(=30/2), L55, **向右點**, →L56



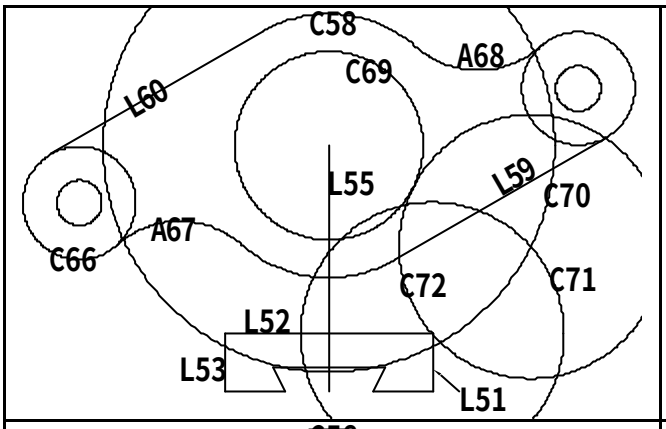
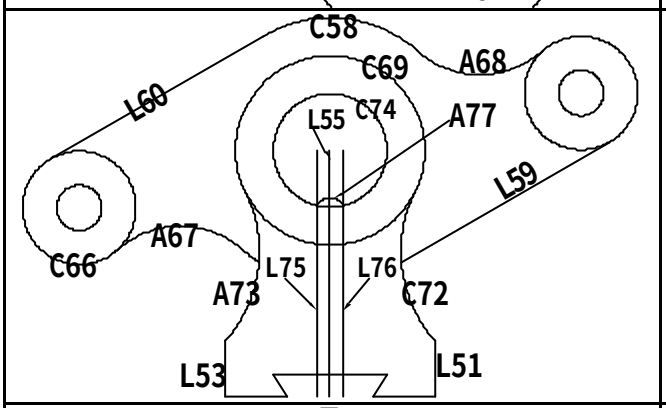
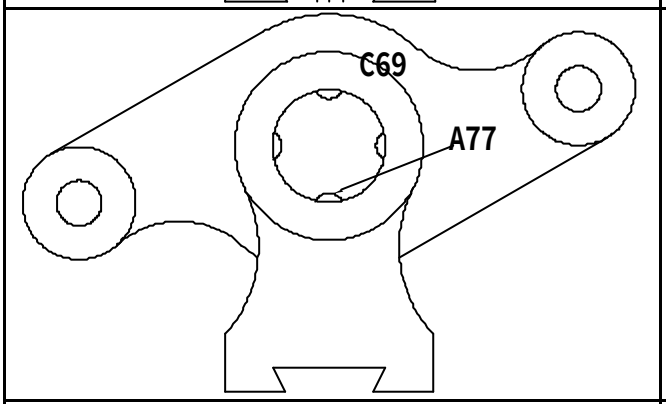
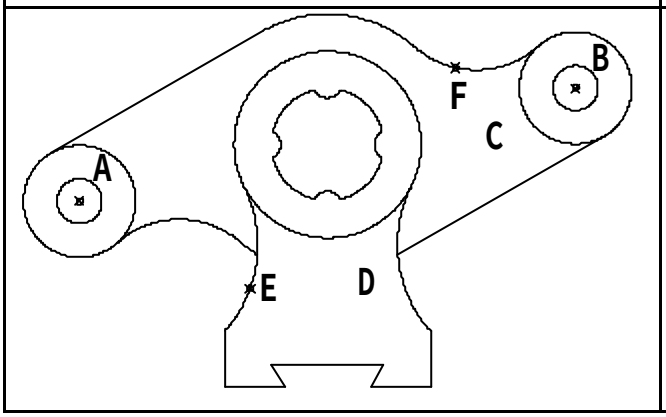

06、 ROtate, L56, int, L54 & L56, -30, →L56 **轉動**



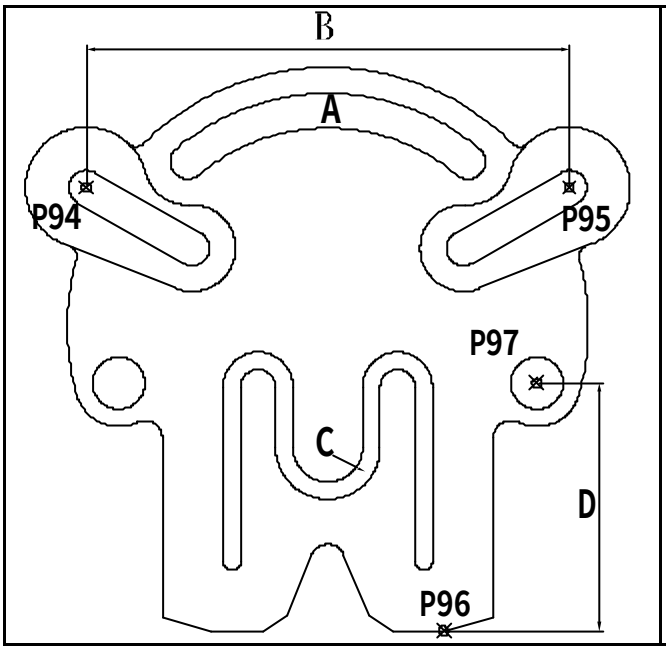

07、 EXtend, L50, L56, →L56 **與** L50 **相交**
08、 TRim,
09、 Erase,
10、 MIrror, L56, end, L55 **上**, end, L55 **下**, →L57
11、 TRim,
12、 Circle, end, L55 **上**, D, 70, →C58
13、 Line, qua, C58 **下**, (**正交模式**), **向右** 65, →L59
14、 Line, qua, C58 **上**, (**正交模式**), **向左** 65, →L60




15、 ROtate, L59;L60, cen, C58, 30, →L59;L60 **轉動**
16、 Offset, 15(=30/2), L59, **向上點**, →L61, L60, **向下點**, →L62
17、 Circle, end, L61 **右**, D, 12, →C63
18、 Circle, @(**鎖定上次點**), D, 30, →C64
19、 COpy, C63;C4, cen, C64, end, L62 **下**, →C65;C66
20、 Fillet, T, T, R, 25, C58 **左下**, C66 **右下**, →A67
21、 Fillet, T, T, R, 25, C58 **右上**, C64 **左上**, →A68

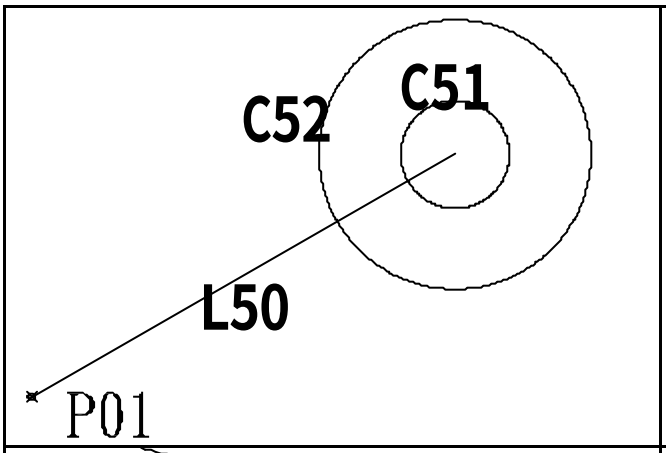
| | |
|---|---|
|  | 22、 Erase,<br>23、 TRim,<br>24、 Circle, cen, C58, D, 50, →C69<br>25、 Offset, 35, C69, **向外點**, →C70<br>26、 Circle, int, L51 & L52, 35, →C71<br>27、 Circle, int, C70 & C71, 35, →C72 |
|  | 28、 Erase,<br>29、 TRim,<br>30、 MIrror, C72, end, L55 **上**, end, L55 **下**, →A73<br>31、 Circle, cen, C69, D, 30, →C74<br>32、 Offset, 3.5(=7/2), L55, **向左點**, →L75, L55, **向右點**, →L76<br>33、 Arc, int, C74 & L76, E, int, C74 & L75, R, 4, →A77 |
|  | 34、 Erase,<br>35、 TRim,<br>36、 ARray, **環形陣列**, **中心點**, cen, C69, **項目總數**, 4, **佈滿角度**, 360, **物件**, A77, **→全圓佈滿** 4 **個相同物件** |
|  | 37、 Erase,<br>38、 TRim,<br>39、 DIst, cen, A, cen, B, →X **差值**=132.5833<br>40、 Hatch, **點** C **區域**, **→斜線區域**<br>41、 **點** C **區域斜線**, CTRL+1, **→查幾何面積值**=3206.9652<br>42、 BOundary, **點** D **區域**<br>43、 **點** D **區邊緣**, CTRL+1, **→查幾何長度值**=244.391<br>44、 DIst, cen, E, cen, B, **→距離**=127.1086<br>45、 **點弧** F, CTRL+1, **→查幾何總角度值**=83.119 |

## 2.5.15 AutoCAD2D2008-515

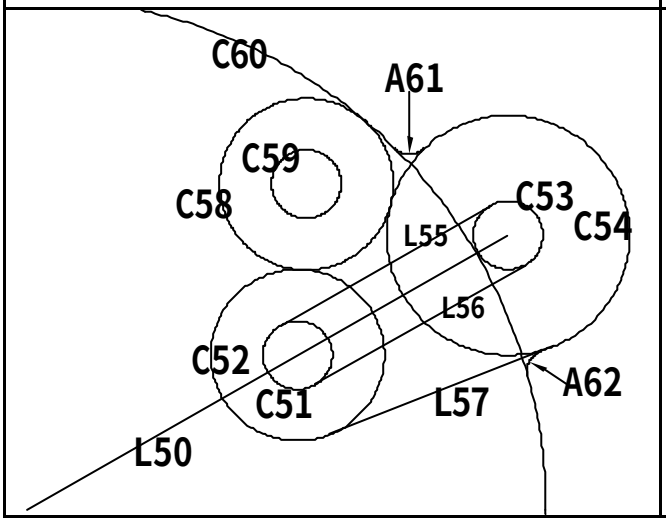



1、 A 區域周長為何？ 206.3819
2、 尺寸 B 長度為何？ 138.5641
3、 C 區域周長為何？ 460.4956
4、 尺寸 D 長度為何？ 71.5249
5、 斜線區域面積為何？ 13617.6592

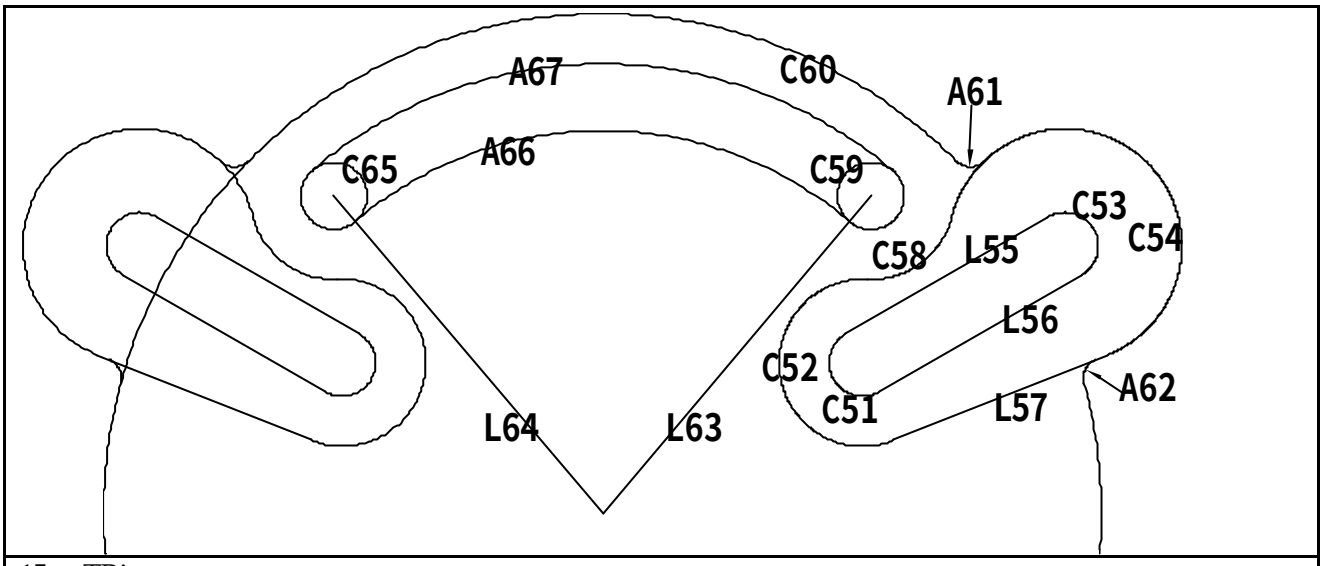
**解題步驟：**



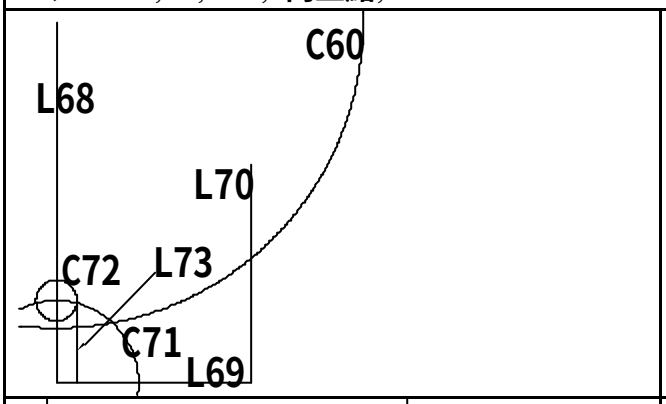

01、 Line, 任一點 P01, @45<30, →L50
02、 Circle, end, L50 上, D, 10, →C51
03、 Circle, @(鎖定上次點), 12.5, →C52



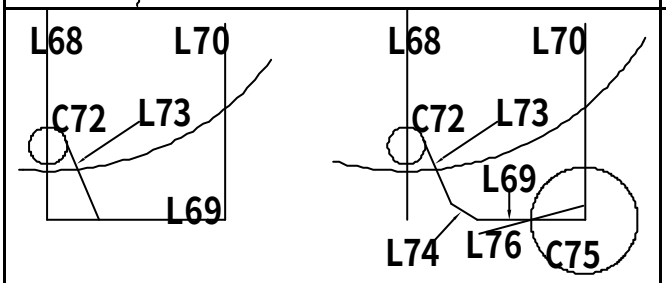

04、 LENgthen, T, 80, L50 上, →L50 向上延伸
05、 Circle, end, L50 上, D, 10, →C53
06、 Circle, @(鎖定上次點), 17.5, →C54
07、 Line, tan, C51 上, tan, C53 上,→L55
08、 Line, tan, C51 下, tan, C53 下,→L56
09、 Line, tan, C52 下, tan, C54 下,→L57
10、 Circle, T, C52 上, C54 左, 12.5, →C58
11、 Circle, cen, C58, 5, →C59
12、 Circle, end, L50 下, tan, C58 右上, →C60
13、 Fillet, T, N, R, 3, C60 右上, C54 左上, →A61
14、 Fillet, T, N, R, 3, C60 右, L57 上, →A62




15、 TRim,
16、 Erase,
17、 Line, cen, C59, cen, C60, →L63
18、 MIrror, **右側物件**, qua, C60 **上**, cen, C60, →L64;C65 **等左側相同物件**
19、 Arc, C, cen, C60, int, C59 & L63, int, C65 & L64, →A66
20、 Offset, 10, A66, **向上點**, →A67



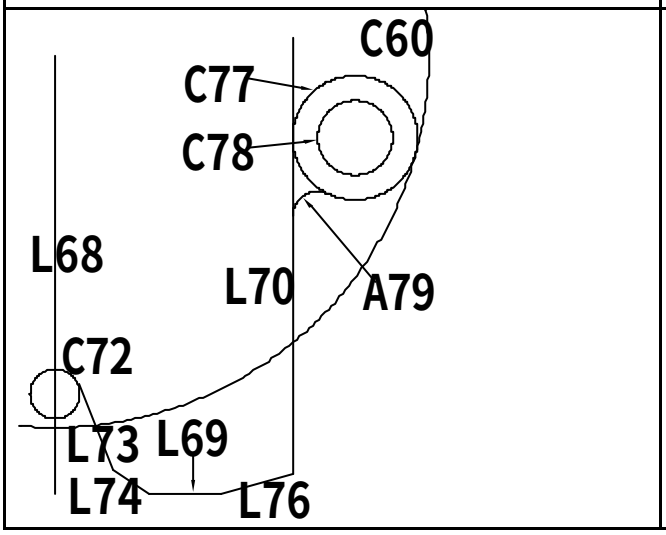

21、 Line, cen, C60, (**正交模式**), **向下** 88, →L68, **向右** 47.5, →L69, **向上** 88, →L70
22、 Circle, int, L68 & L69, 20, →C71
23、 Circle, int, L68 & C71, 5, →C72
24、 Line, qua, C72 **右**, (**正交模式**), **向下** 20, →L73



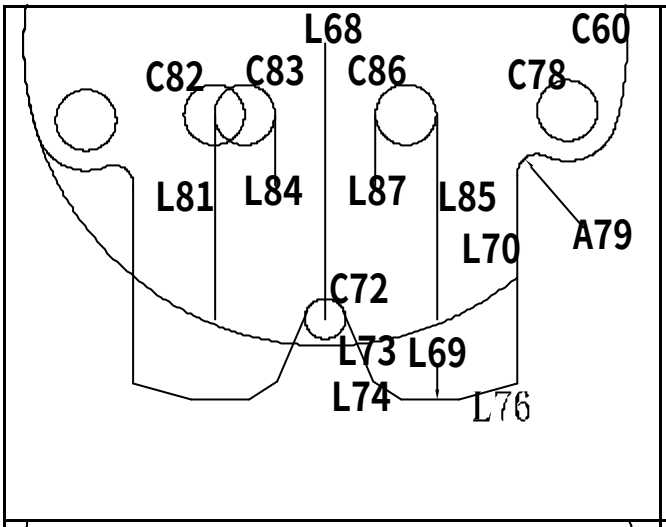

25、 ROtate, L73, cen, C72, 22.5, →L73 **轉動**

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

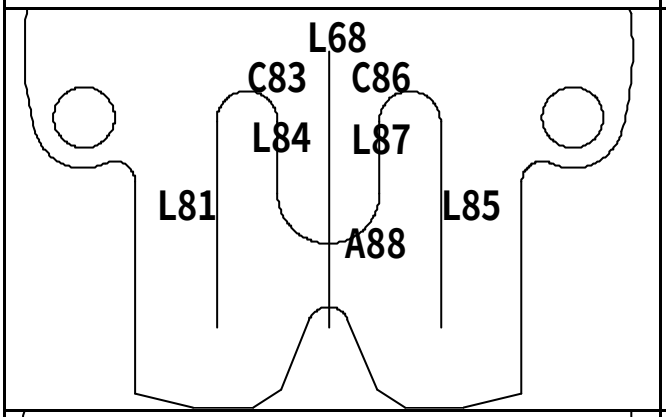
26、 CHAmfer, T, T, D, 5, 5, L69, L73, →L74
27、 Circle, int, L69 & L70, 14.25, →C75
28、 ROtate, L69, int, L69 & C75, C, 15, →L76



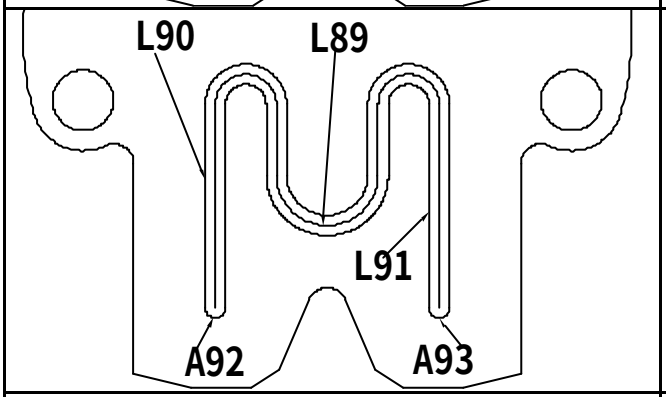

29、 Fillet, T, T, R, 0, L70, L76, →L70 **與** L76**相交**
30、 Fillet, T, T, R, 0, L69, L76, →L69 **與** L76**相交**
31、 Circle, T, L70, C60, 12.5, →C77
32、 Circle, cen, C77, D, 15,→C78
33、 Fillet, T, T, R, 5, L70, C77 **左下**→A79



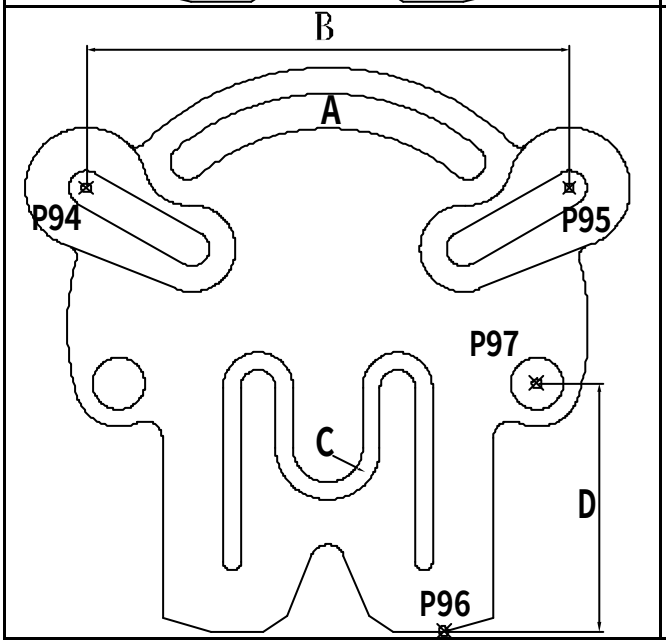

34、 TRim,
35、 Erase,
36、 LENgthen, DE, -20, L68 **下**, →L68 **縮短**
37、 Offset, 27.5, L68, **向左點**, →L81
38、 LENgthen, T, 50.5(=33+17.5), L81 **上**, →L81 **縮短**
39、 Circle, end, L81 **上**, 7.5, →C82
40、 Circle, qua, C82 **右**, 7.5, →C83
41、 Line, qua, C83 **右**, (**正交模式**), **向下** 17.5, →L84
42、 MIrror, C60;C78;C77;A79;L70;L76;L69;L74;L73;L81;C83;L84, end, L68 **上**, end, L68 **下**, →L85~L87 **等左右右對稱物件**




43、 TRim,
44、 Erase,
45、 Arc, C, m2p, end, L84 **下**, end, L87 **下**, end, L84 **下**, end, L87 **下**, →A88



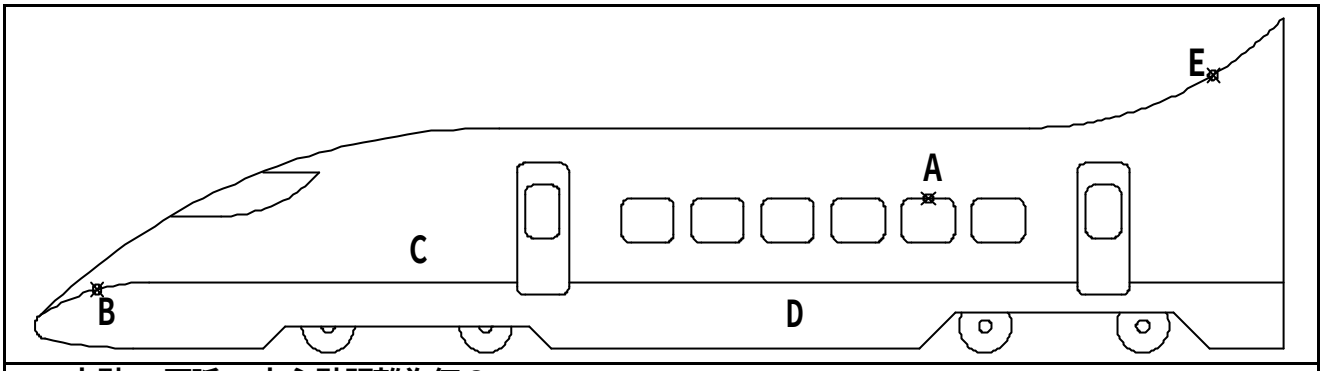

46、 PEdit, L85, Y, J, C86;L87;A88;L84;C83;L81, →L89
47、 Offset, 2.5, L89, **向上點**, →L90
48、 Offset, 2.5, L89, **向下點**, →L91
49、 Arc, C, end, L89 **左下**, end, L90 **左下**, end, L91 **左下**, →A92
50、 Arc, C, end, L89 **右下**, end, L90 **右下**, end, L91 **右下**, →A93




51、 TRim,
52、 Erase,
53、 BOundary, **點** A **區域**
54、 **點** A **區邊緣**, CTRL+1, →**查幾何長度值** =206.3819
55、 DIst, cen, P94, cen, P95, →X **差值**=138.5641
56、 BOundary, **點** C **區域**
57、 **點** C **區邊緣**, CTRL+1, →**查幾何長度值** =460.4956
58、 DIst, int, P96, cen, P97, →Y **差值**=71.5249
59、 Hatch, **點中間區域**, →**斜線區域**
60、 **點斜線區域**, CTRL+1, →**查幾何面積值** =13617.6592

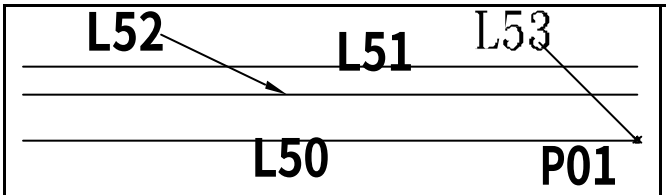
## 2.6 第六類：建築&室內設計應用

### 2.6.1 AutoCAD2D2008-601

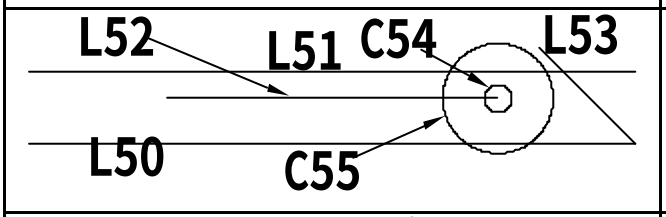



1、 **中點** A **至弧** B **中心點距離為何？** 188.9524
2、 C **區域扣除內孔面積為何？** 7537.1539
3、 D **區域周長為何？** 609.2789
4、 **弧** E **角度為何？** 46.5675
5、 **圖形最外圍所圍成的面積為何？** 12763.5388

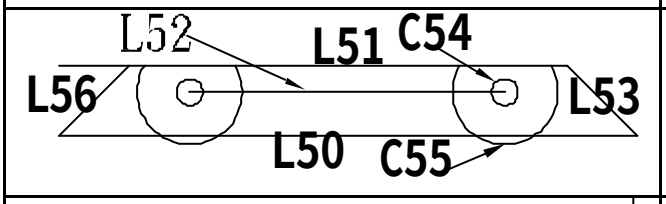
**解題步驟：**



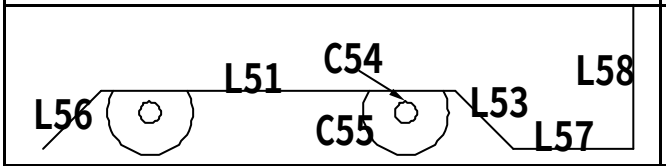

01、 Line, **任一點** P01, (**正交模式**), **向左** 66, →L50
02、 Offset, 8, L50, **向上點**, →L51
03、 Offset, 3, L51, **向下點**, →L52
04、 Line, end, L50 **右**, @15<135, →L53



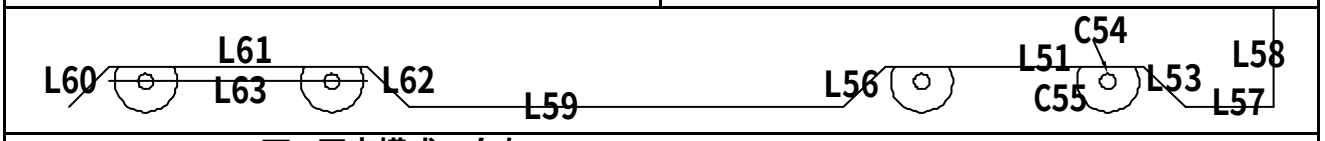

05、 LENgthen, DE, -15, L52 **右**, →L52 **縮短**, L52 **左**, →L52 **再縮短**
06、 Circle, end, L52 **右**, D, 3, →C54
07、 Circle, @(**鎖定上次點**), 6, →C55



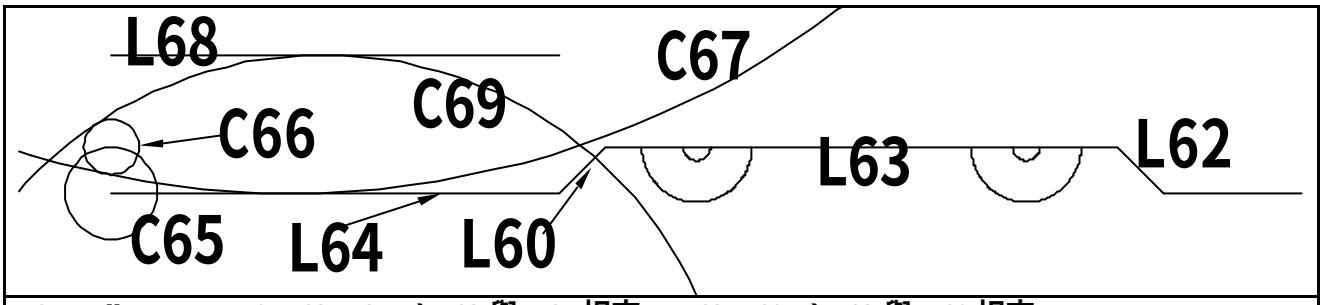

08、 TRim,
09、 MIrror, L53~C55, mid, L50, mid, L52, →L56 **等左側相同物件**



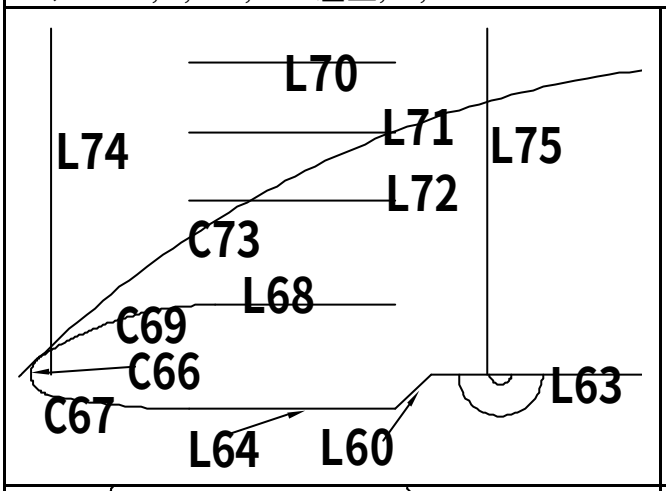

10、 TRim,
11、 Erase,
12、 Line, end, L53 **下**, (**正交模式**), **向右** 17, →L56, **向上** 75, →L58



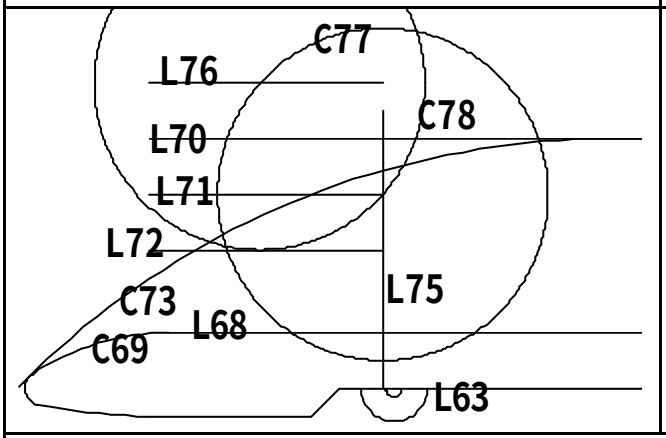

13、 Line, end, L56 **下**, (**正交模式**), **向左** 84, →L59
14、 COpy, L51;L53;C55 **等輪組物件**, end, L53 **下**, end, L59 **左**, →L60~L62 **等左側輪組物件**
15、 Offset, 3, L61, **向下點**, →L63



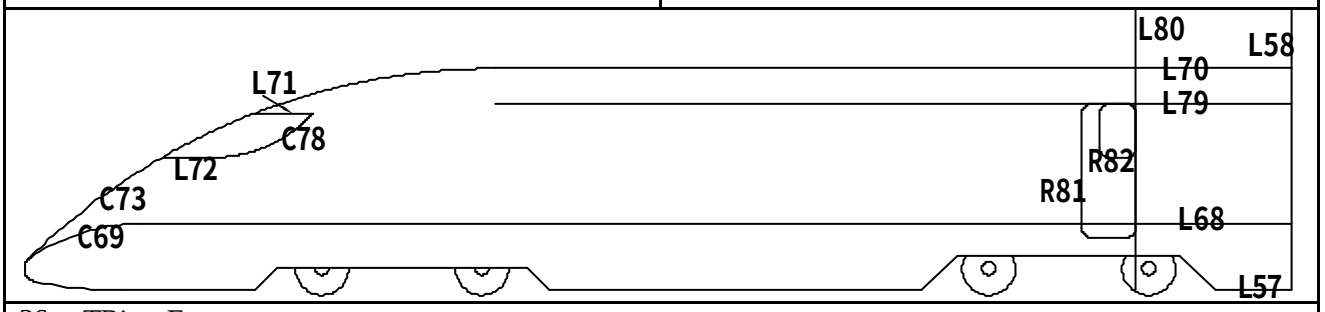

16、 Fillet, T, T, R, 0, L63, L62, →L63 **與** L62 **相交**, L63, L60, →L63 **與** L60 **相交**
17、 Line, end, L60 **下**, (**正交模式**), **向左** 49, →L64
18、 Circle, end, L64 **左**, 5, →C65
19、 Circle, qua, C65 **上**, D, 6, →C66
20、 Circle, T, L64, C66 **下**, 100, →C67
21、 Offset, 15, L64, **向上點**, →L68
22、 Circle, T, L68, C66 **左上**, 45, →C69



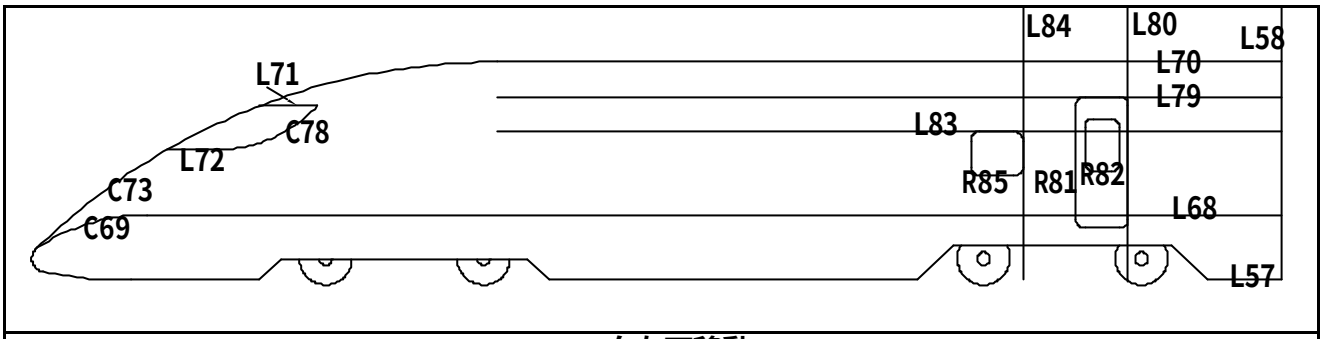

23、 Erase,
24、 TRim,
25、 Offset, 50, L64, **向上點**, →L70
26、 Offset, 10, L70, **向下點**, →L71
27、 Offset, 10, L71, **向下點**, →L72
28、 Circle, T, C66 **左**, L70, 150, →C73
29、 Line, cen, C66, (**正交模式**), **向上** 50, →L74
30、 Offset, 62, L74, **向右點**, →L75



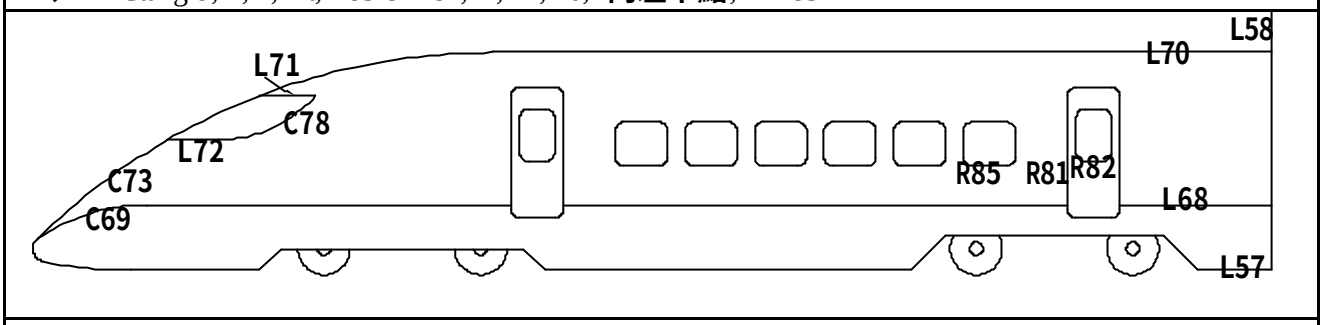

31、 EXtend, L75, L71;L72, →L71 **及** L72 **分別與** L75 **相交**
32、 EXtend, L58, L68;L70, →L68 **及** L70 **分別與** L58 **相交**
33、 Offset, 30, L72, **向上點**, →L76
34、 Circle, int, L71 & L75, 30, →C77
35、 Circle, int, L76 & C77, 30, →C78



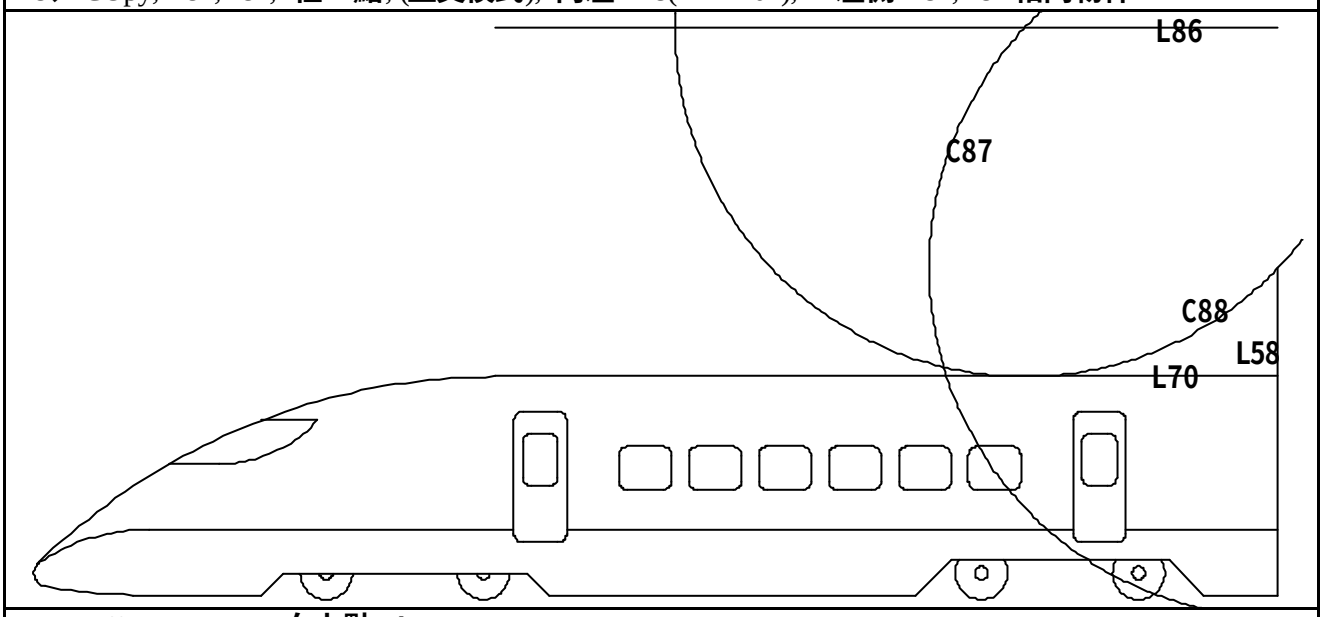

36、 TRim, Erase,
37、 Offset, 8, L70, **向下點**, →L79
38、 Offset, 35, L58, **向左點**, →L80
39、 RECtangle, F, 2, int, L79 & L80, D, 12, 30, **向左下方點**, →R81
40、 RECtangle, F, 2, int, L79 & L80, D, 8, 12, **向左下方點**, →R82



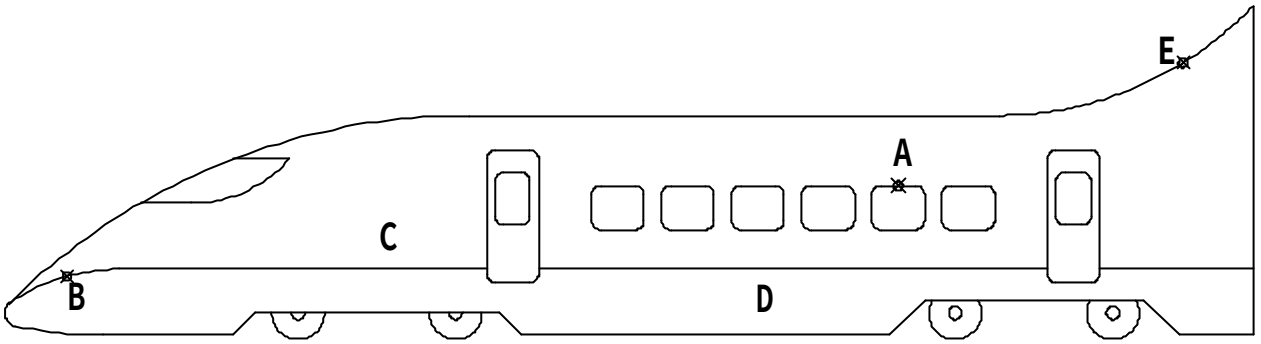

41、 Move, R82, int, L79 & L80, @-2,-5, →R82 **向左下移動**
42、 Offset, 16, L70, **向下點**, →L83
43、 Offset, 24, L80, **向左點**, →L84
44、 RECtangle, F, 2, int, L83 & L84, D, 12, 10, **向左下點**, →R85



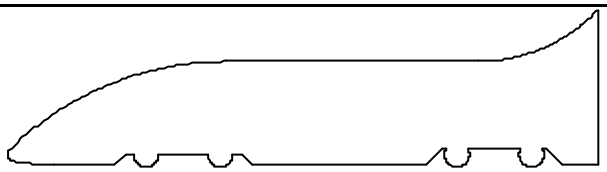

45、 Erase,
46、 TRim,
47、 ARray, **矩形陣列**, **列**, 1, **行**, 6, **列偏移**, 0, **行偏移**, -16, **物件**, R85, →6 **個** R85 **相同物件**
48、 COpy, R81;R82, **任一點**, (**正交模式**), **向左** 128(=24+104), →**左側** R81;R82 **相同物件**




49、 Offset, 80, L70, **向上點**, →L86
50、 Circle, end, L58, 80, →C87
51、 Circle, int, L86 & C87, 80, →C88

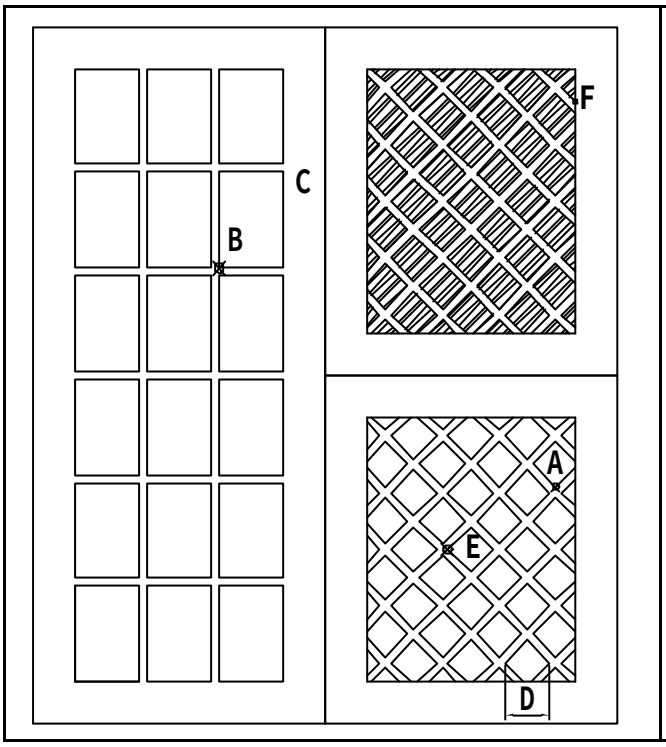



52、 Erase,
53、 TRim,
54、 DIst, mid, A, cen, B, →**距離**=188.9524
55、 Hatch, **點** C **區域**, →**斜線區域**
56、 **點選** C **斜線區域**, CTRL+1, →**查幾何面積值**=7537.1539
57、 BOundary, **點** D **區域**,
58、 **點選** D **區域邊緣**, CTRL+1, →**查幾何長度值**=609.2789
59、 **點選弧** E, CTRL+1, →**查幾何總角度值**=46.5675



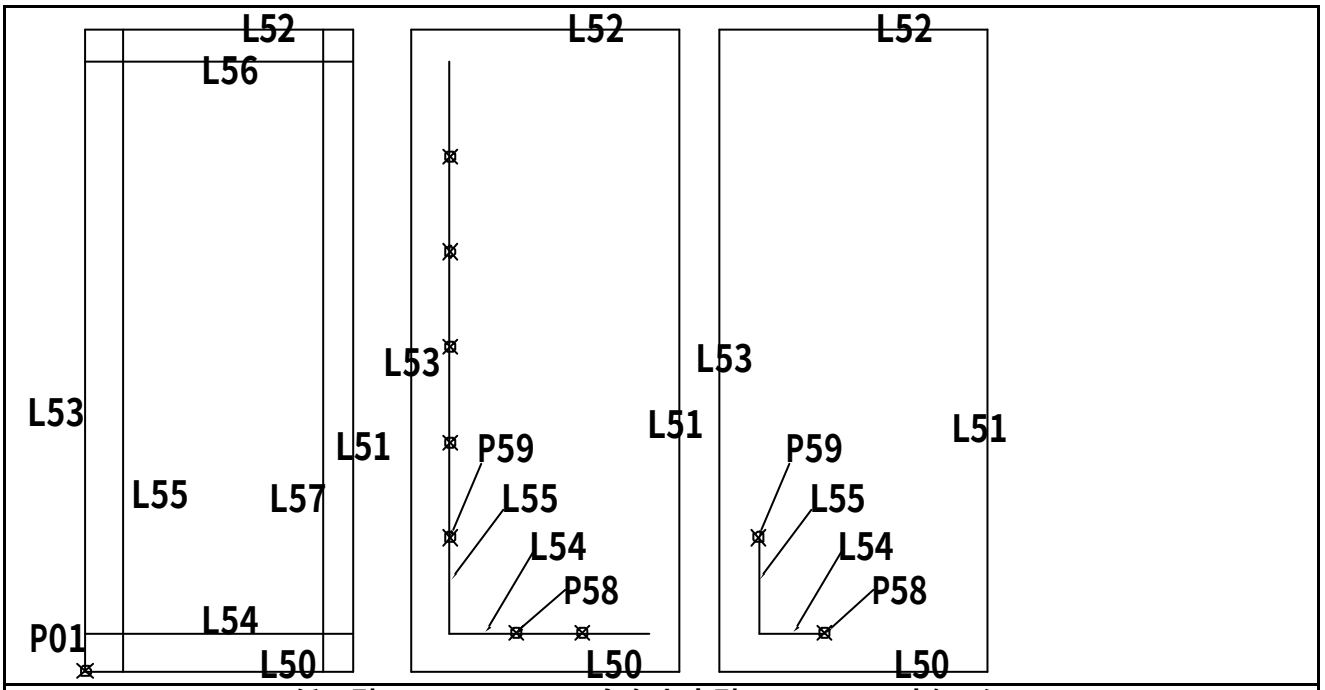
60、 COpy, TRim, Erase, →**左側圖形**
61、 BOundary, **點中間區域**
62、 **點邊緣**, CTRL+1, →**查幾何面積值** =12763.5388

## 2.6.2 AutoCAD2D2008-602




1、 **交點** A **至交點** B **距離為何？**123.9767
2、 **斜線區域面積為何？** 3322.5201
3、 C **區域扣除內孔面積為何？** 8842.1400
4、 **尺寸** D **值為何？** 13.9645
5、 **交點** E **至交點** F **距離為何？** 143.7604

**解題步驟：**




01、 RECtangle, F, 0, **任一點** P01, D, 90, 215, **向右上方點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 12.8, L50, **向上點**, →L54, L53, **向右點**, →L55
04、 Offset, 10.3(=12.8-2.5), L52, **向下點**, →L56, L51, **向左點**, →L57

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05、 TRim,
06、 Erase,
07、 DIVide, L54, 3, →P58 **等** 2 **個點**
08、 DIVide, L55, 6, →P59 **等** 5 **個點**

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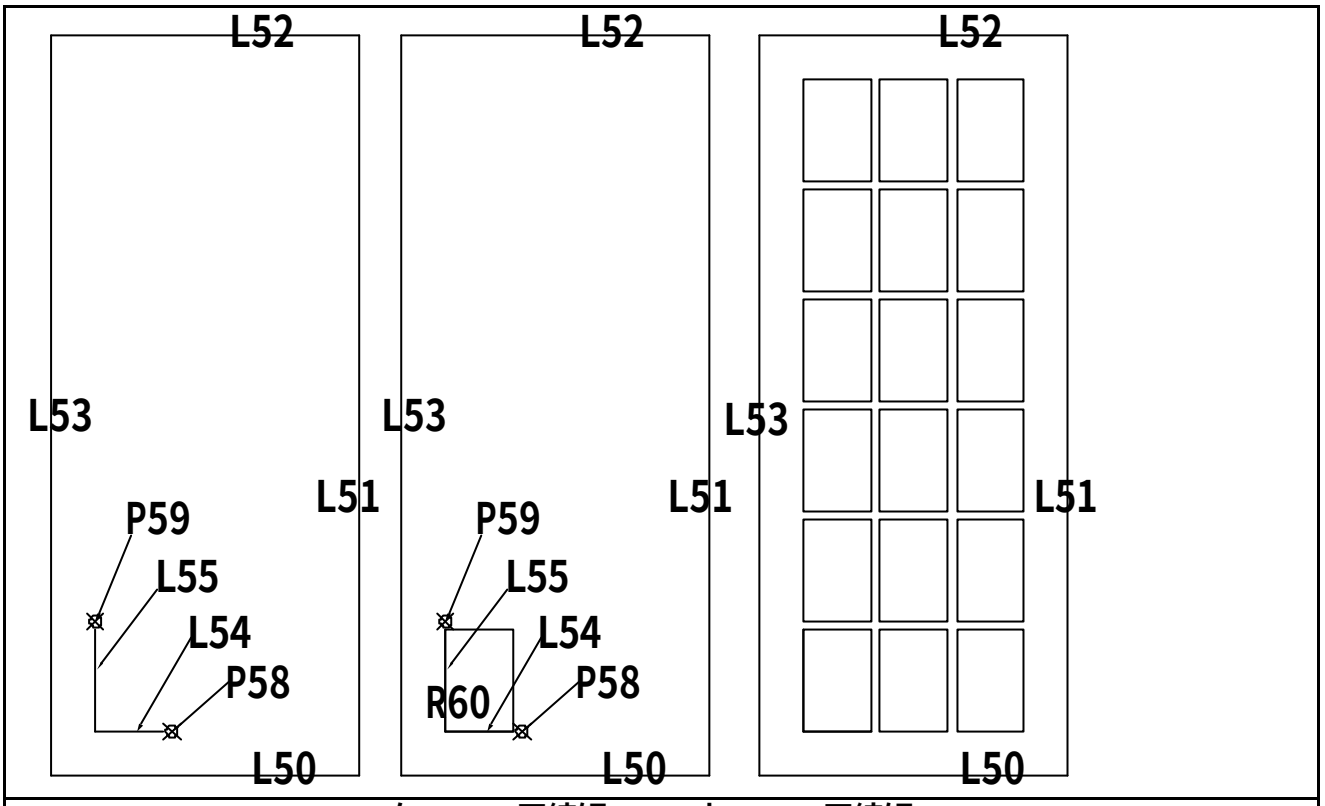
09、 LENgthen, DY, L54 **右**, nod, P58, →L54 **縮短**, L55 **上**, nod, P59, →L55 **縮短**




10、 LENgthen, DE, -2.5, L54 **右**, →L54 **再縮短**, L55 **上**, →L55 **再縮短**

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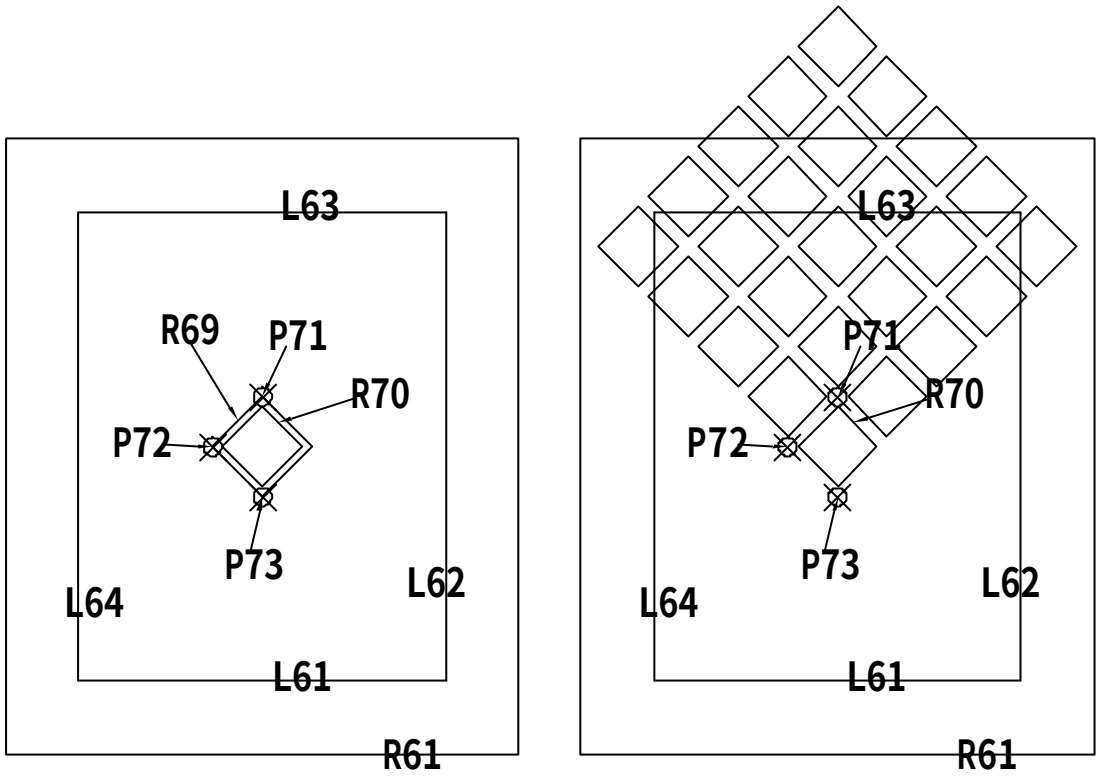


21、 PEdit, L65, Y, J, L66;L67;L68, →R69 之矩形
22、 Offset, 1.25, R69 之矩形, 向內點, →R70 之矩形
23、 POint, int, R69 上, →P71, int, R69 左, →P72, int, R69 下, →P73

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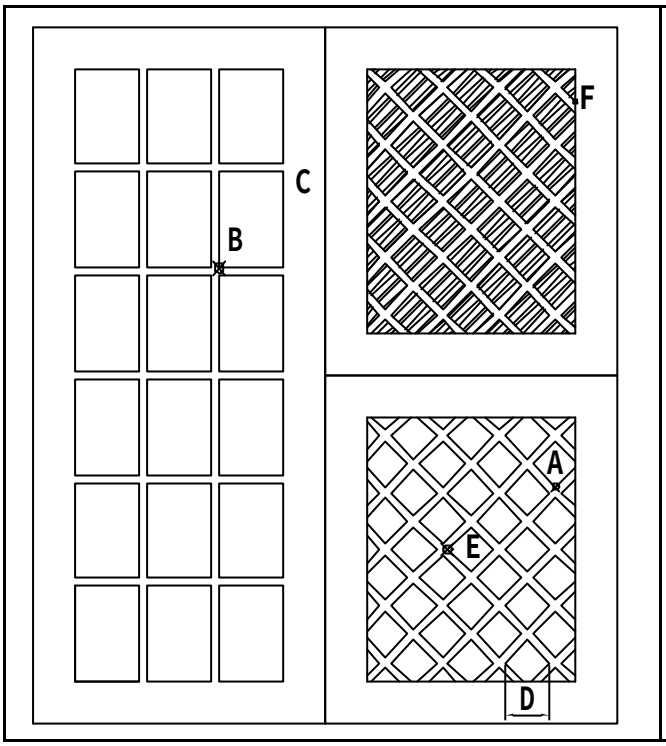



31、 Move, R61;R74, int, R61 **左上**, int, L51 & L52, →**左側圖形**

32、 DIst, int, A, int, B, →**距離**=123.9767

33、 **點斜線區域**, CTRL+1, →**查幾何面積值** =3322.5201

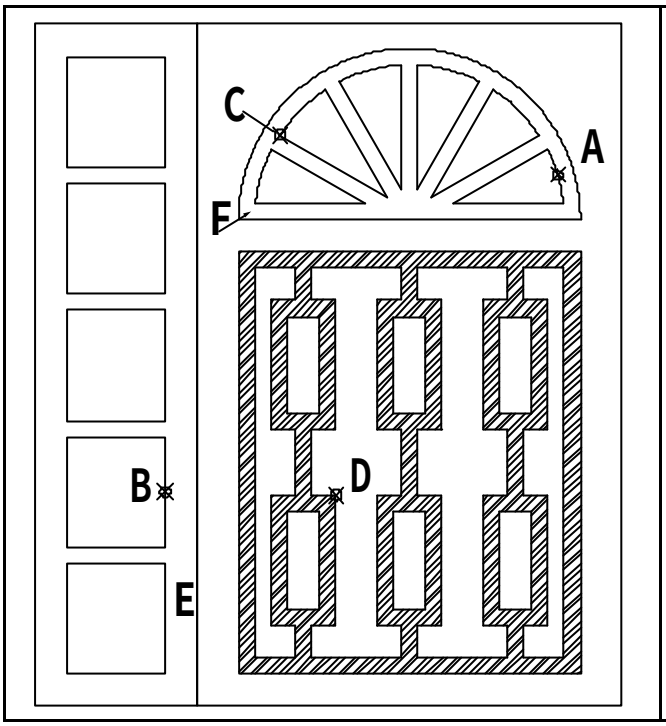
34、 Hatch, **點** C **區域**, →**斜線區域**

35、 **點選** C **斜線區域**, CTRL+1, →**查幾何面積值** =8842.1400

36、 DimLInear, int, D **左**, int, D **右**, →13.9645

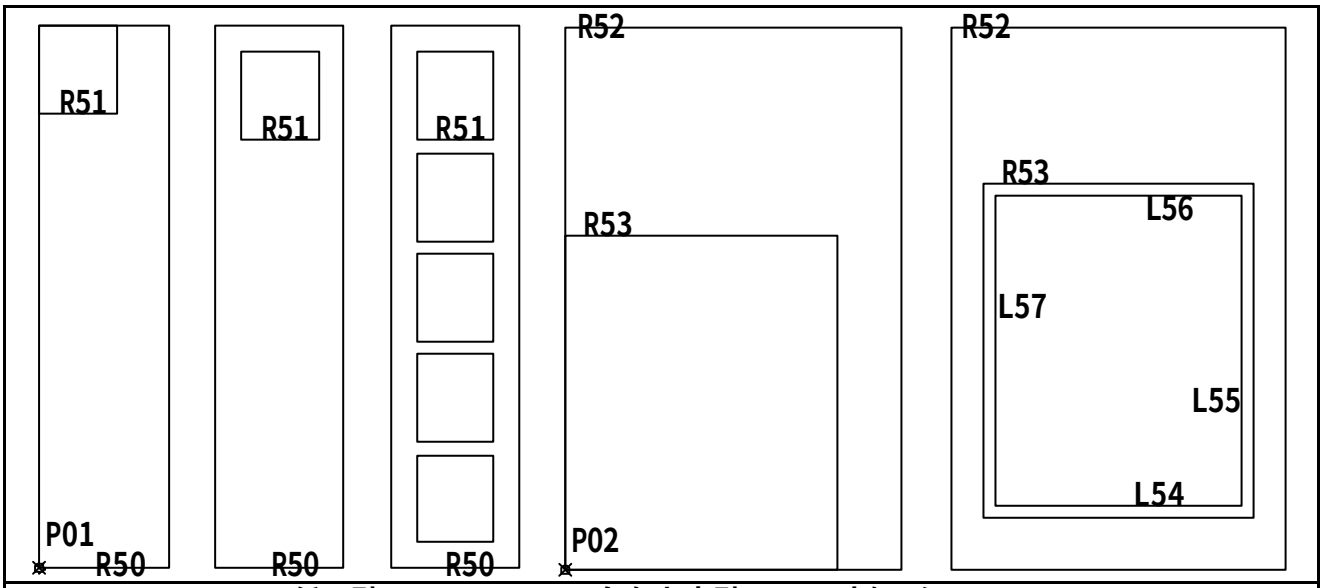
37、 DIst, int, E, int, F, →**距離**=143.7604

## 2.6.3 AutoCAD2D2008-603




1、 **中點** A **至中點** B **距離為何？** 155.0372
2、 **斜線區域面積為何？** 5850.0000
3、 **交點** C **至交點** D **高度為何？** 110.8821
4、 E **區域扣除內孔面積為何？** 5400.0000
5、 F **區域扣除內孔面積為何？** 2249.0432

**解題步驟：**




01、 RECtangle, F, 0, **任一點** P01, D, 50, 210, **向右上方點**, →R50 **之矩形**
02、 RECtangle, F, 0, int, R50 **左上**, D, 30, 34, **向右下方點**, →R51 **之矩形**

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03、 Move, R51 **之矩形**, int, R50 **左上**, @10,-10, →R51 **向右下移動**

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04、 ARray, **矩形陣列**, **列**, 5, **行**, 1, **列偏移**, -39, **行偏移**, 0, **物件**, R51, →5 **個與** R51 **相同之矩形**

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05、 RECtangle, F, 0, **任一點** P02, D, 130, 210, **向右上方點**, →R52 **之矩形**
06、 RECtangle, F, 0, int, R52 **左下**, D, 105, 130, **向右上方點**, →R53 **之矩形**

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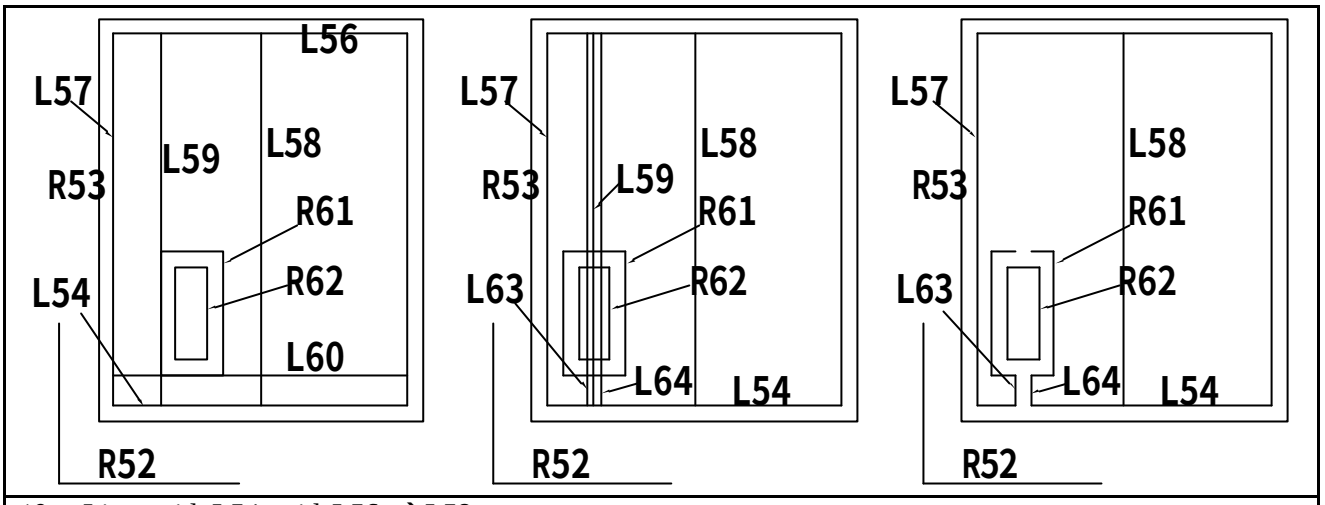
07、 Move, R53 **之矩形**, int, R52 **左下**, @12.5,10, →R53 **向右上移動**
08、 Offset, 5, R53, **向內點**, →L54~L57 **之矩形**
09、 eXplode, L54~L57 **之矩形**, →L54~L57




10、 Line, mid, L54, mid, L56, →L58
11、 Offset, 32.5, L58, **向左點**, →L59
12、 Offset, 10(=15-5), L54, **向上點**, →L60
13、 RECtangle, F, 0, int, L59 & L60, D, 20(=10+5+5), 40, **向右上方點**, →R61 **之矩形**
14、 Offset, 5, R61, **向內點**, →R62 **之矩形**

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15、 Move, R61;R62, mid, R61 **下**, int, L59 & L60, →R61;R62 **向左移動**
16、 Offset, 2.5(=5/2), L59, **向左點**, →L63, L59, **向右點**, →L64

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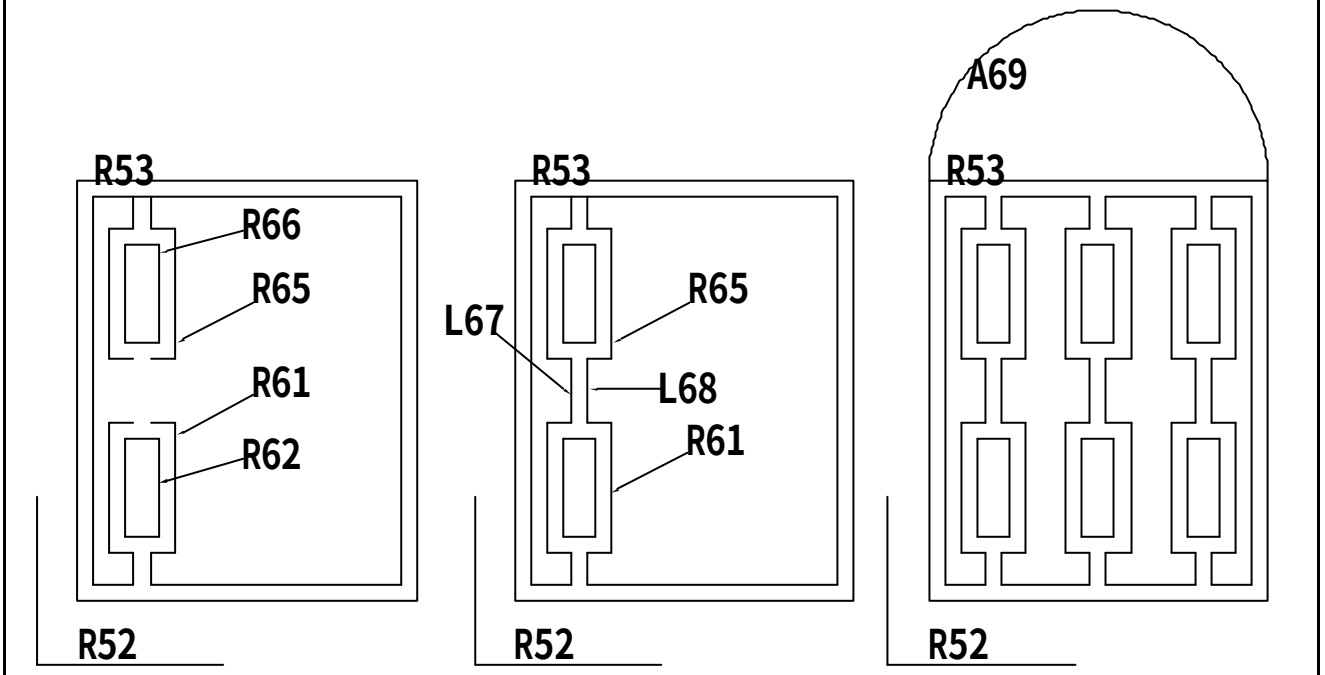
17、 TRim,
18、 Erase,




19、 MIrror, R61;R62, mid, R53 **左**, mid, R53 **右**, →R65;R66

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20、 Line, end, R65 **左下**, end, R61 **左上**, →L67
21、 Line, end, R65 **右下**, end, R61 **右上**, →L68

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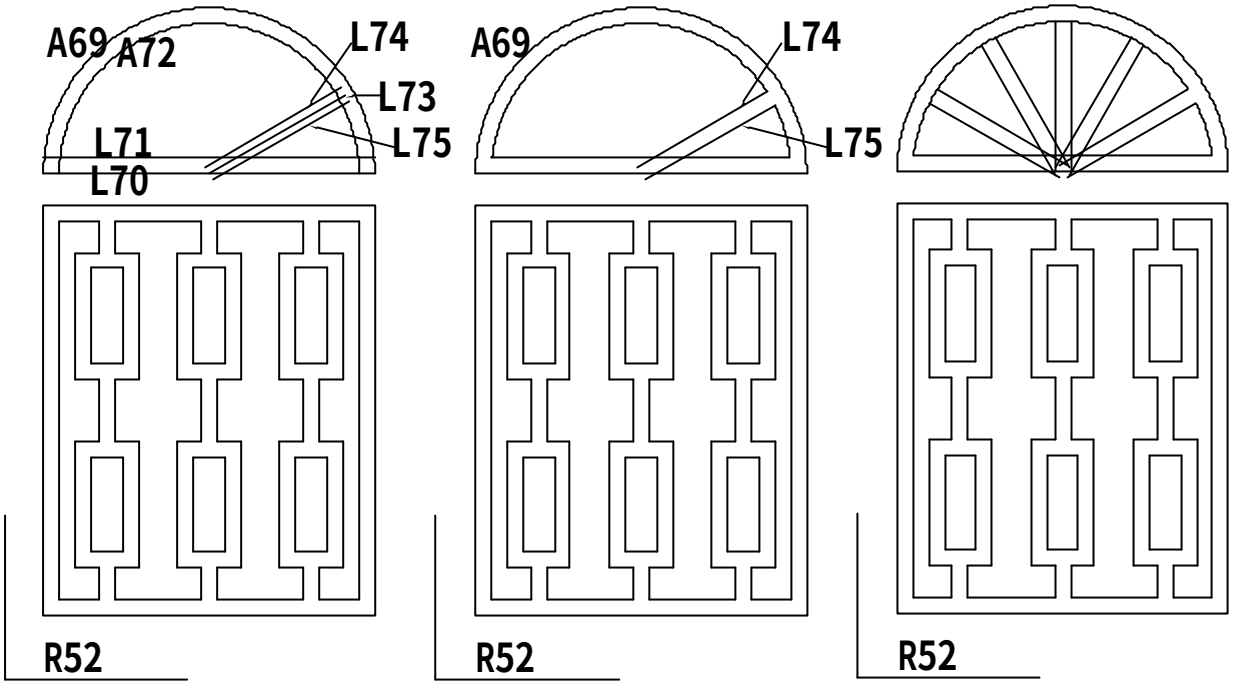
22、 ARray, **矩形陣列**, **列**, 1, **行**, 3, **列偏移**, 0, **行偏移**, 32.5, **物件**, R61 **等物件**, →3 **組物件**
23、 TRim,
24、 Arc, C, mid, R53 **上**, end, R53 **右上**, end, R53 **左上**, →A69




25、 Move, A69, int, R53 **左上**, (**正交模式**), **向上** 10, →A69 **向上移動**
26、 Line, end, A69 **左**, end, A69 **右**, →L70
27、 Offset, 5, L70, **向上點**, →L71
28、 Offset, 5, A69, **向下點**, →A72
29、 Line, cen, A69, @50<30, →L73
30、 Offset, 2.5(=5/2), L73, **向上點**, →L74, L73, **向下點**, →L75

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31、 TRim,
32、 Erase,

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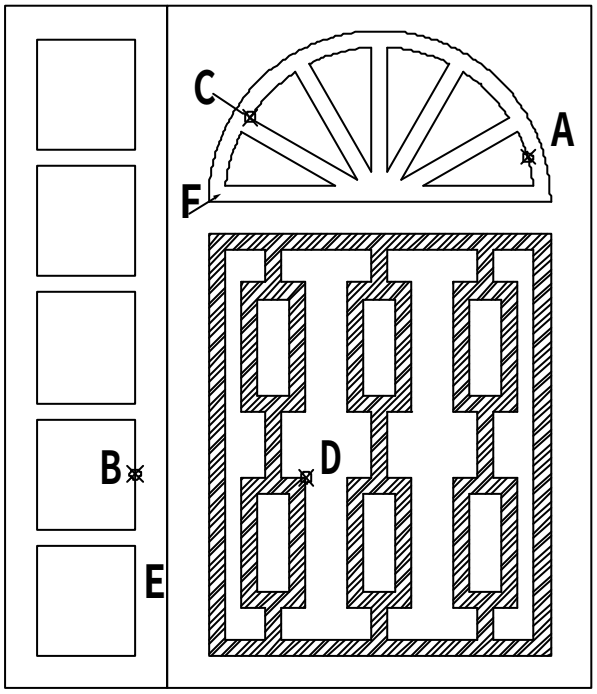
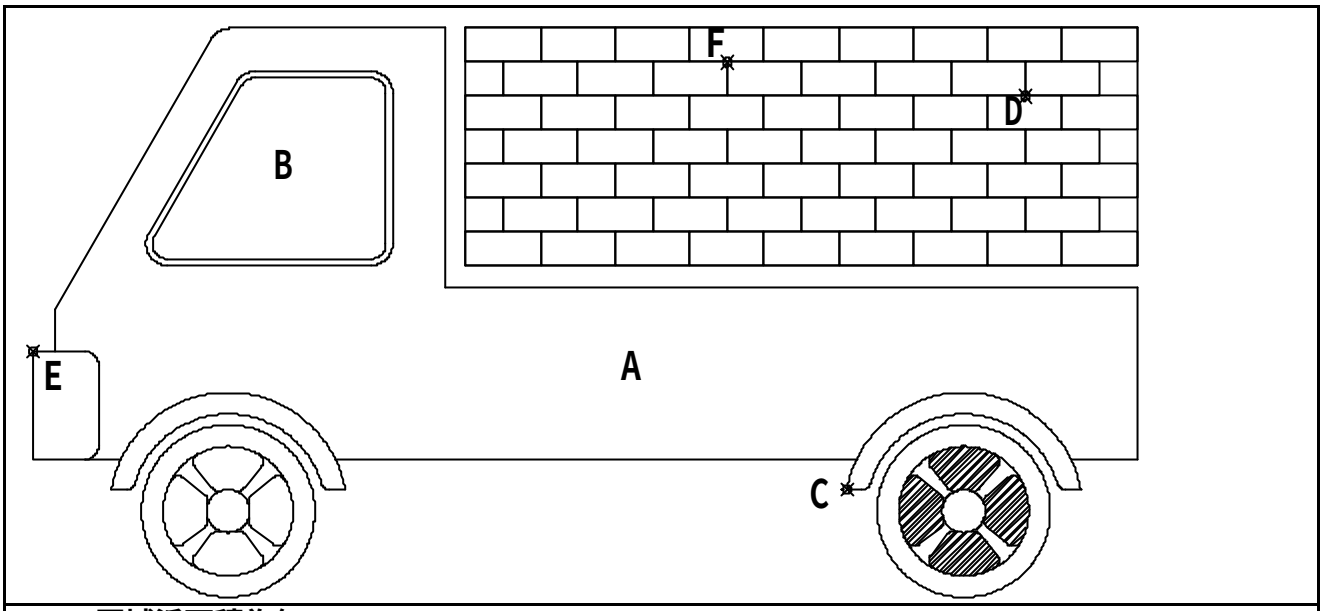
33、 ARray, **環形陣列**, **中心點**, cen, A69, **項目總數**, 5, **佈滿角度**, 120, **物件**, L74;L75, →5 **組與** L74;L75 **相同物件**



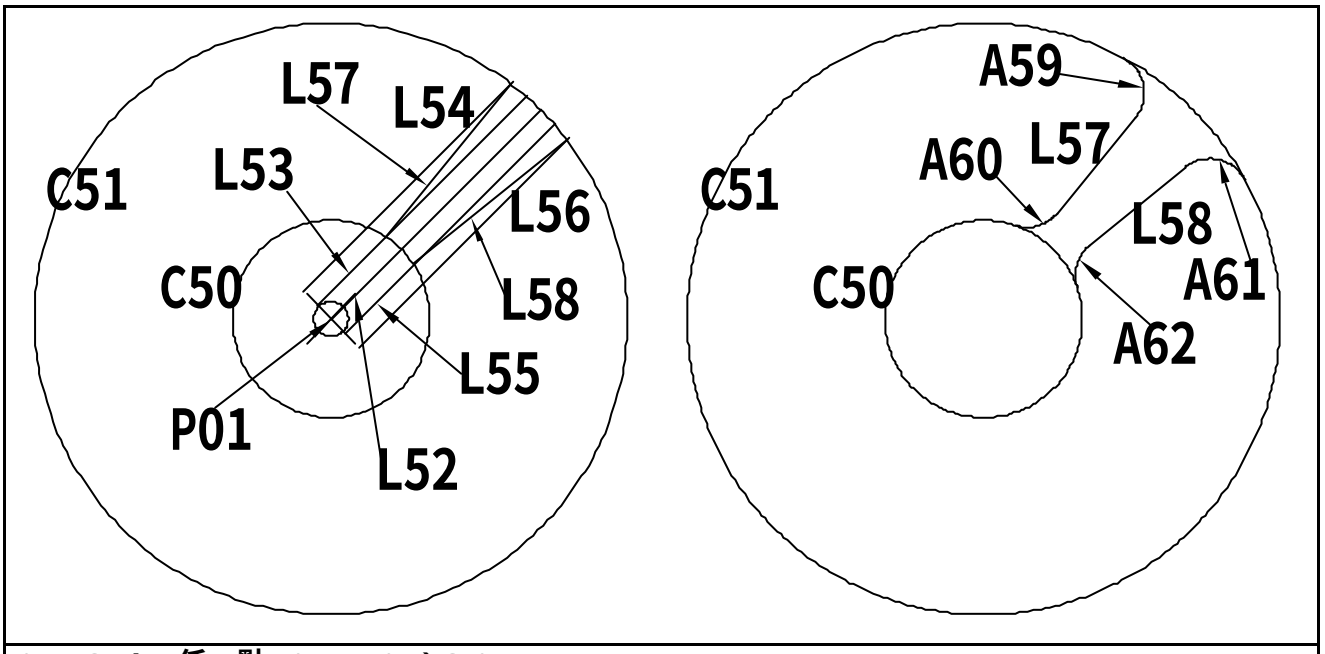

34、 TRim,
35、 Erase,
36、 **將物件以** R50 **之右下角點與** R52 **之左下角點為同點位組合之**
37、 DIst, mid, A, mid, B, →**距離**=155.0372
38、 **點斜線區域**, CTRL+1, →**查幾何面積值** =5850.0000
39、 DIst, int, C, int, D, →Y **差值**=110.8821
40、 Hatch, **點** E **區域**, →**斜線區域**
41、 **點選** E **斜線區域**, CTRL+1, →**查幾何面積值** =5400.0000
42、 Hatch, **點** F **區域**, →**斜線區域**
43、 **點選** F **斜線區域**, CTRL+1, →**查幾何面積值** =2249.0432

## 2.6.4 AutoCAD2D2008-604




1、 A **區域淨面積為何？** 42868.1806
2、 B **區域周長為何？** 342.4738
3、 **交點** C **至交點** D **距離為何？** 200.3118
4、 **交點** E **至交點** F **距離為何？** 347.5464
5、 **斜線區域面積為何？** 1980.1461

**解題步驟：**




01、 Circle, **任一點** P01, D, 20, →C50
02、 Circle, @(**鎖定上次點**), D,60, →C51
03、 Line, @(**鎖定上次點**), @30<45, →L52
04、 Offset, 2, L52, **向上點**, →L53, L53, **向上點**, →L54, L52, **向下點**, →L55, L55, **向下點**, →L56
05、 Line, int, C51 & L54, int, C50 & L53, →L57
06、 Line, int, C51 & L56, int, C50 & L55, →L58

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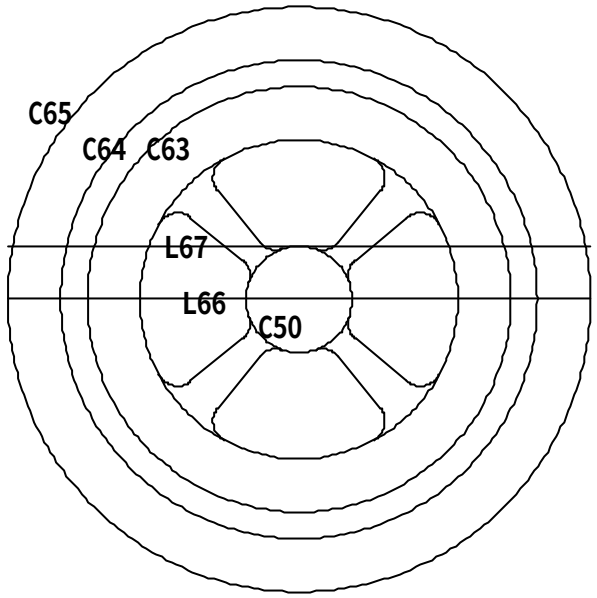
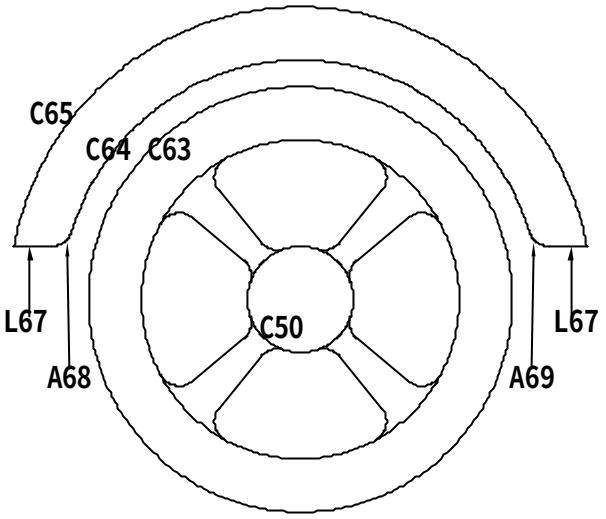



09、 ARray, **環形陣列**, **中心點**, cen, C50, **項目總數**, 4, **佈滿角度**, 360, **物件**, L57~A62, →**全圓佈滿** 4 **組相同物件**

10、 Circle, cen, C50, D, 80, →C63

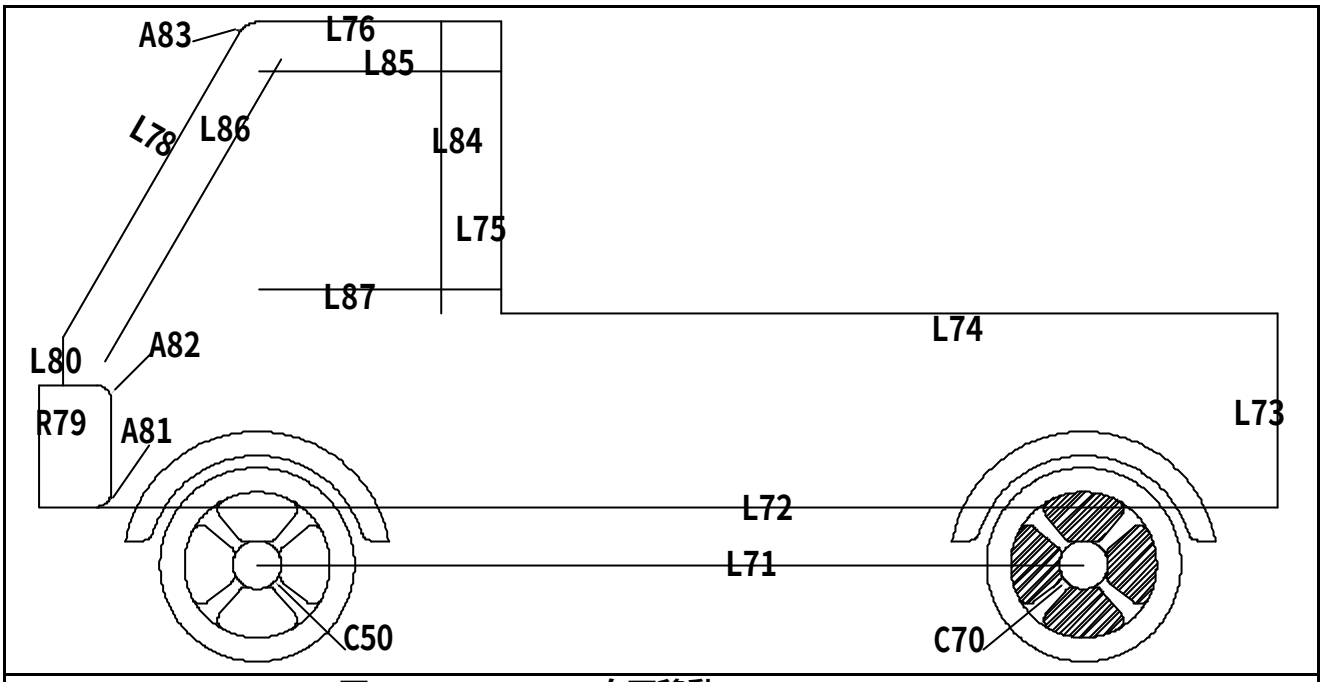
11、 Circle, @(**鎖定上次點**), 45, →C64

12、 Circle, @(**鎖定上次點**), 55, →C65

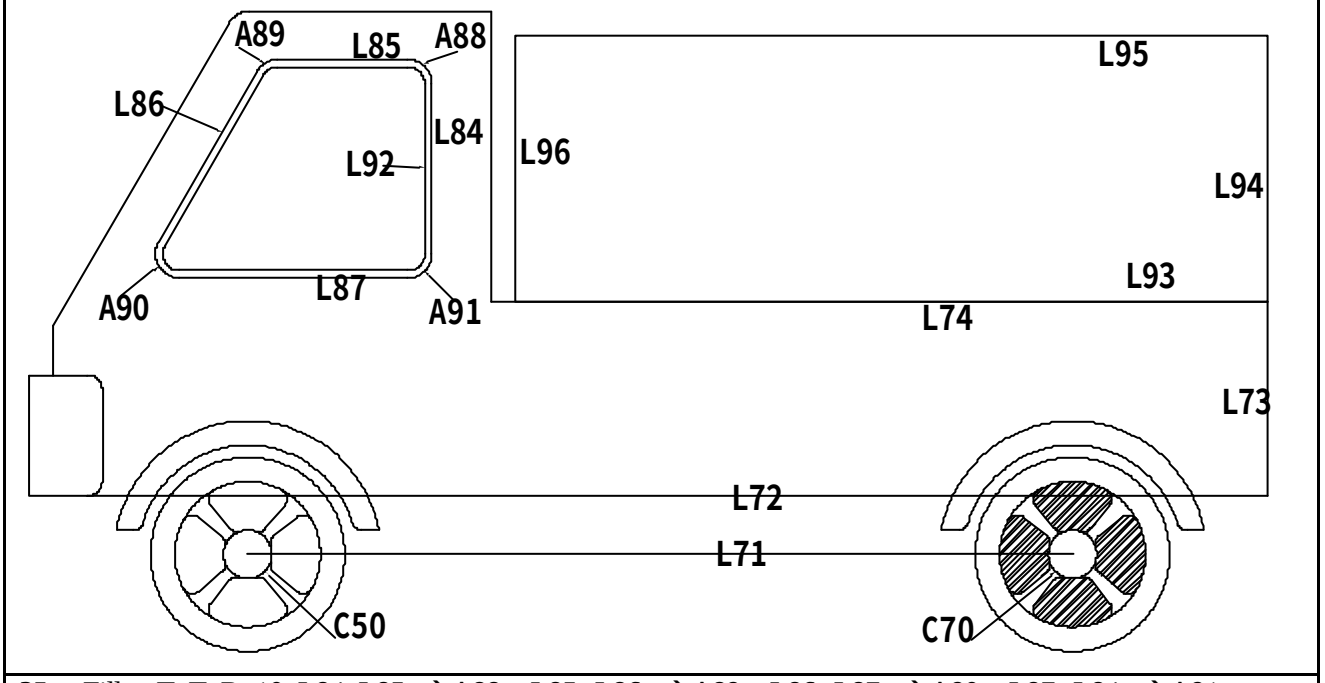
13、 Line, qua, C65 **左**, qua, C65 **右**, →L66

14、 Offset, 10, L66, **向上點**, →L67

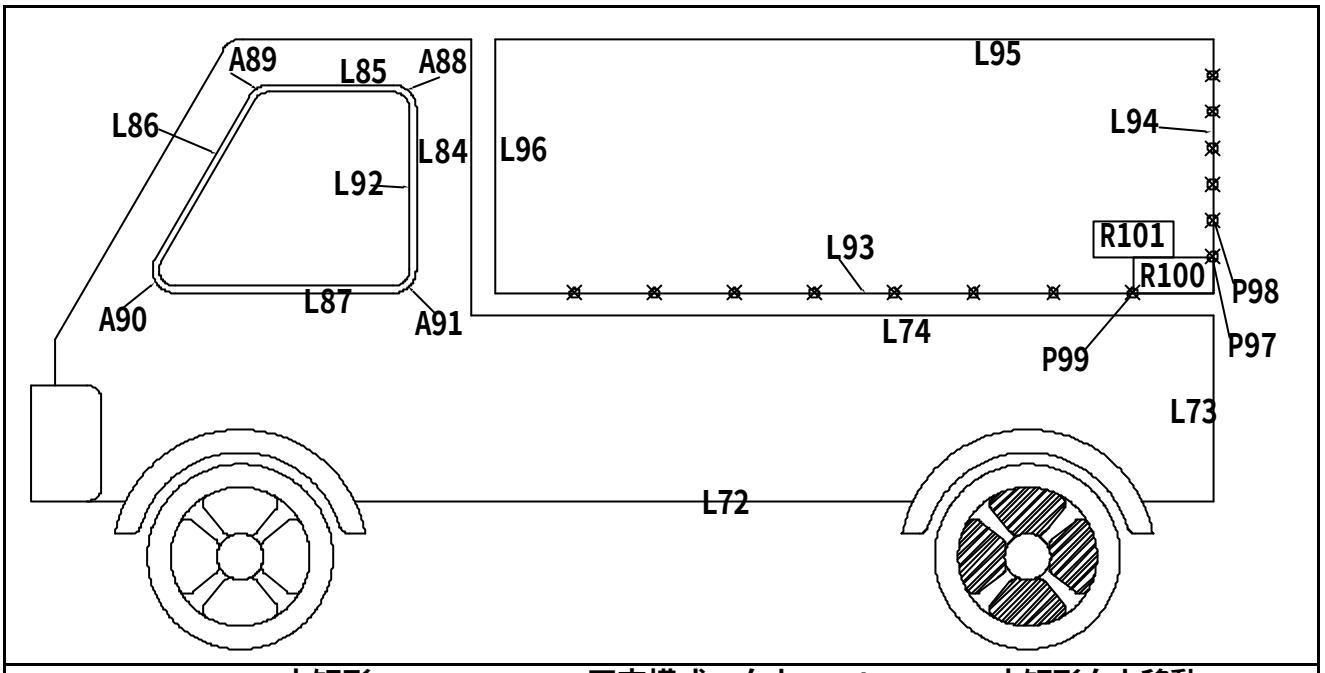
- - - -



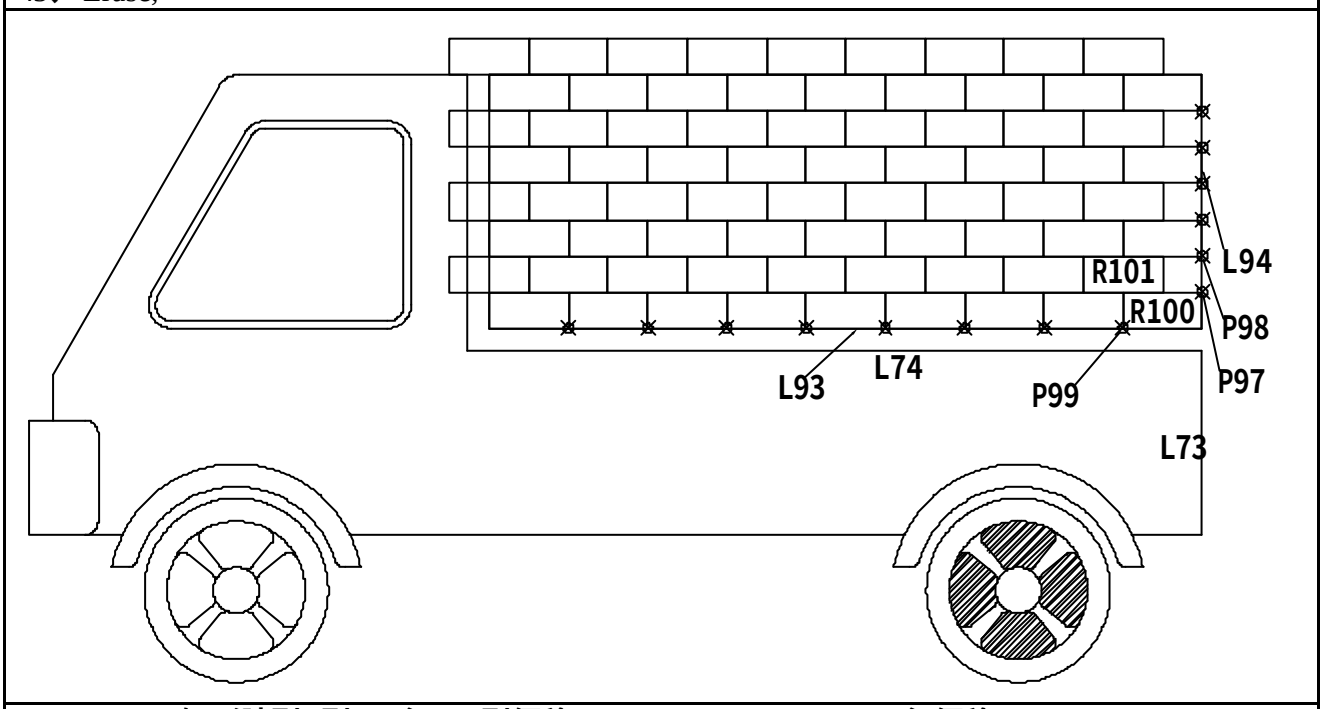

26、 Move, R79, end, L78 **下**, @-10,-20, →R79 **向下移動**
27、 Line, end, L78 **下**, (**正交模式**), **向下** 20, →L80
28、 Fillet, T, T, R, 6, R79 **右**, R79 **下**, →A81, R79 **右**, R79 **上**→A82
29、 Fillet, T, T, R, 10, L76, L78, →A83
30、 Offset, 24, L75, **向左點**, →L84
31、 Offset, 20, L76, **向下點**, →L85, L78, **向右下點**, →L86
32、 Offset, 90, L85, **向下點**, →L87
33、 TRim,
34、 Erase,



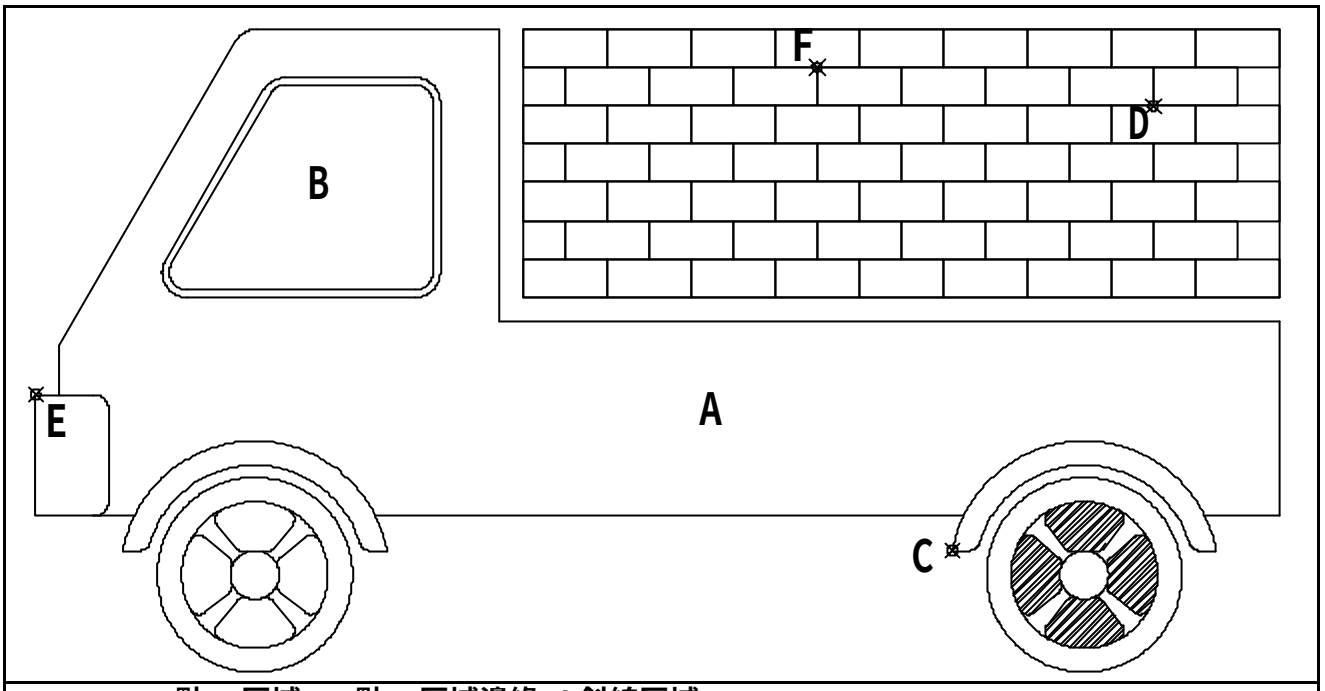

35、 Fillet, T, T, R, 10, L84, L85, →A88, L85, L86, →A89, L86, L87, →A90, L87, L84, →A91
36、 Offset, 3, L84~A91 **等物件**, **向內點**, →L92 **等縮小物件**
37、 RECtangle, F, 0, int, L73 & L74, D, 310, 110, **向左上方點**→L93~L96 **之矩形**




38、 Move, L93~L96 **之矩形**, int, L73 & L74, (**正交模式**), **向上** 10, →L93~L96 **之矩形向上移動**
39、 eXplode, L93~L96 **之矩形**, →L93~L96
40、 DIVide, L94, 7, →P97;P98 **等** 6 **個點**
41、 DIVide, L93, 9, →P99 **等** 8 **個點**
42、 RECtangle, F, 0, int, L93 & L94, D, int, L93 & L94, nod, P99, int, L93 & L94, nod, P97, **向左上方點** →R100 **之矩形**
43、 COpy, R100 **之矩形**, int, L93 & L94, mid, R100 **上中**, →R101 **之矩形**
44、 TRim,
45、 Erase,

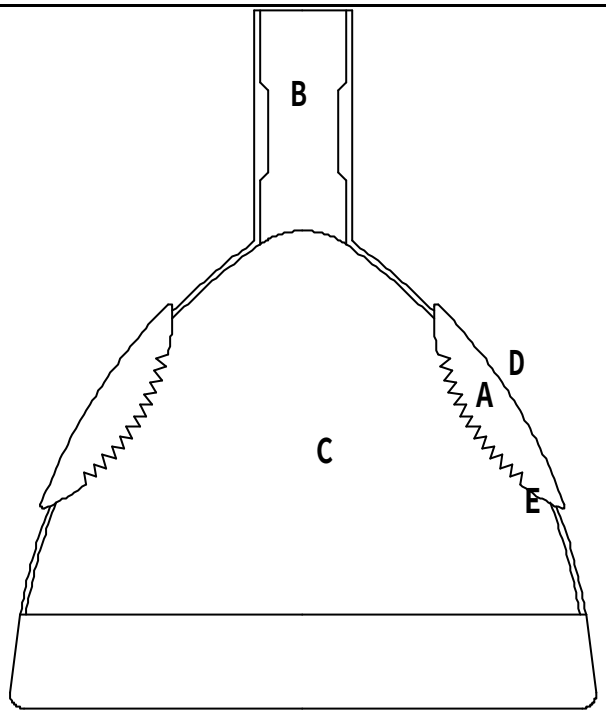



46、 ARray, **矩形陣列**, **列**, 4, **行**, 9, **列偏移**, int, L93 & L94, nod, P98, **行偏移**, int, L93 & L94, nod, P99, **物件**, R100;R101, →36 **組相同之矩形**
47、 TRim,
48、 Erase,



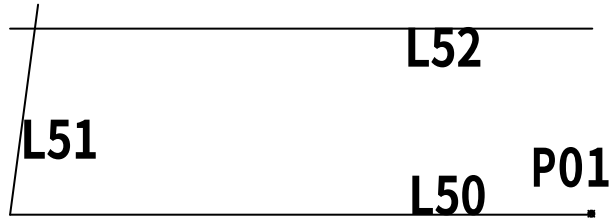

49、 Hatch, **點** A **區域**, B, **點** B **區域邊緣**, →**斜線區域**
50、 **點選** A **斜線區域**, CTRL+1, →**查幾何面積值**=42868.1806
51、 BOundary, **點** B **區域**,
52、 **點選** B **區域邊緣**, CTRL+1, →**查幾何長度值**=342.4738
53、 DIst, int, C, int, D, →**距離**=200.3118
54、 DIst, int, E, int, F, →**距離**=347.5464
55、 **點選斜線區域**, CTRL+1, →**查幾何面積值**=1980.1461

## 2.6.5 AutoCAD2D2008-605

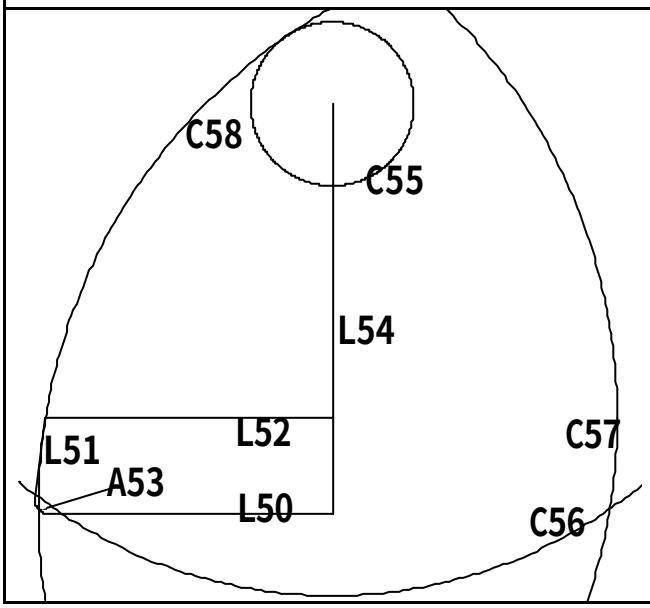



1、 A **區域周長為何？** 241.2301
2、 B **區域面積為何？** 2477.4834
3、 C **區域周長為何？** 689.2216
4、 **弧** D **中心點至弧** E **中心點距離為何？** 272.9962
5、 **圖形最外圍圍成的面積為何？** 32001.0348(32001.0347)

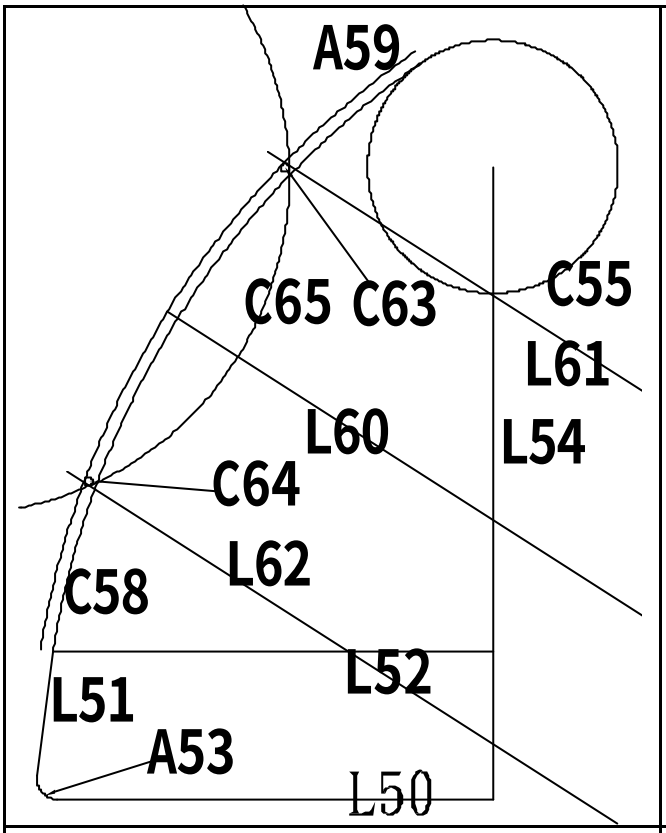
**解題步驟：**



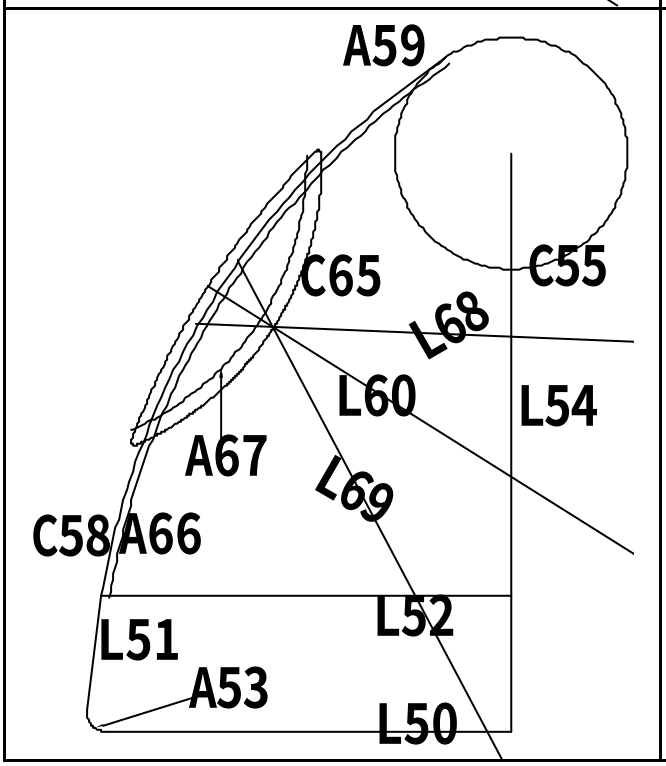

01、 Line, **任一點** P01, (**正交模式**), **向左** 110, →L50, @40<82.5, →L51
02、 Offset, 35, L50, **向上點**, →L52



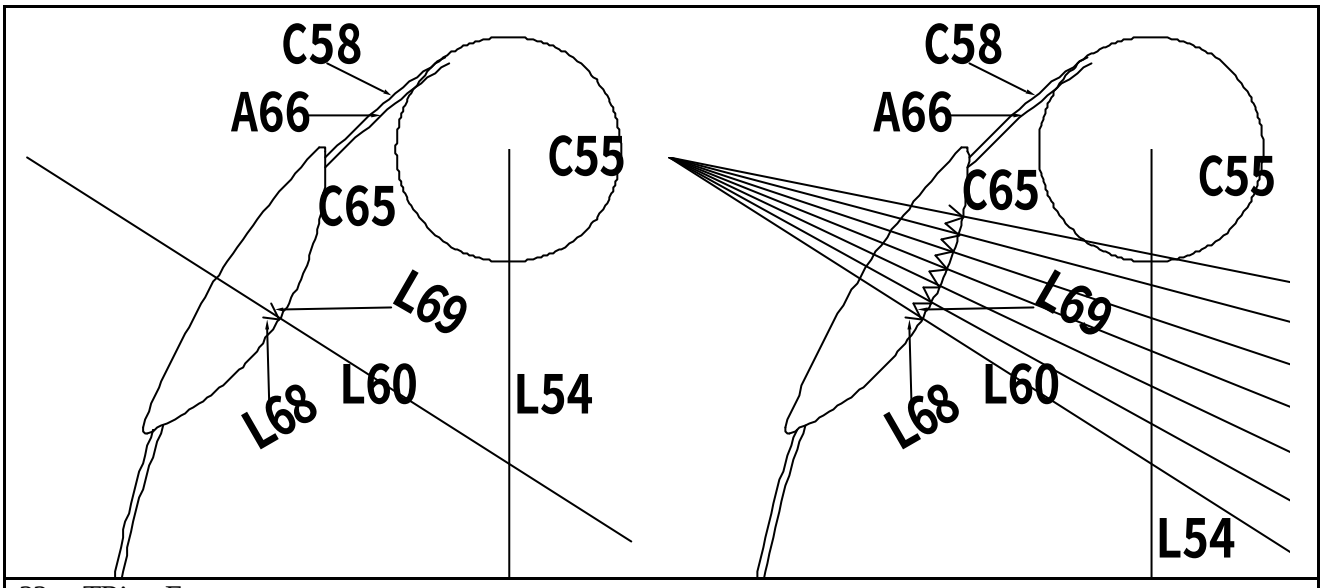

03、 Fillet, T, T, R, 5, L50, L51, →A53
04、 TRim,
05、 Line, end, L50 **右**, (**正交模式**), **向上** 150, →L54
06、 Circle, end, L54 **上**, 30, →C55
07、 Circle, @(**鎖定上次點**), 180(=210-30), →C56
08、 Circle, int, L51 & L52, 210, →C57
09、 Circle, int, C56 & C57, 210, →C58




10、 Offset, 3(=213-210), C58, **向左上點**, →A59
11、 Line, cen, A59, mid, A59, →L60
12、 Offset, 45, L60, **向上點**, →L61, L60, **向下點**, →L62
13、 Circle, T, A59, L61, 1, →C63
14、 Circle, T, A59, L62, 1, →C64
15、 Circle, T, C63 **右**, C64 **下**, 80, →C65

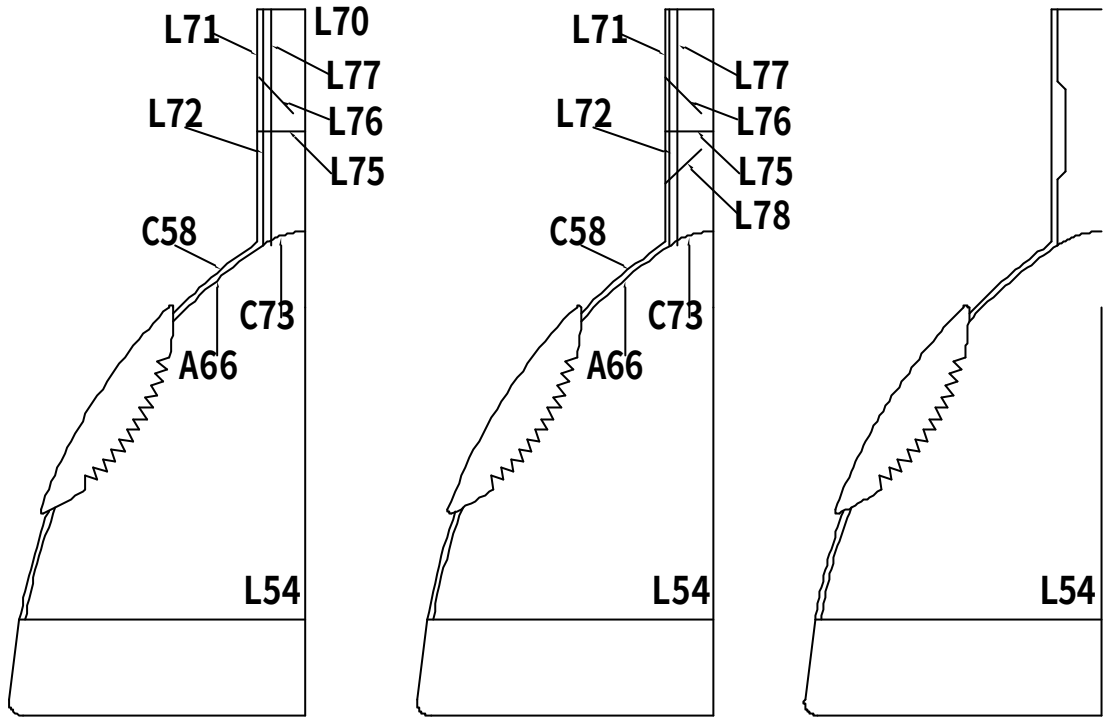



16、 TRim,
17、 Erase,
18、 Offset, 2, C58, **向右點**, →A66
19、 Offset, 4(=80-76), C65, **向左點**, →A67
20、 ROtate, L60, int, L60 & C65, C, 30, →L68
21、 ROtate, L60, int, L60 & C65, C, -30, →L69




22、 TRim, Erase,
23、 LENgthen, DY, L60 **左上**, cen, C65, →L60 **左上端延伸至** C65 **圓心**

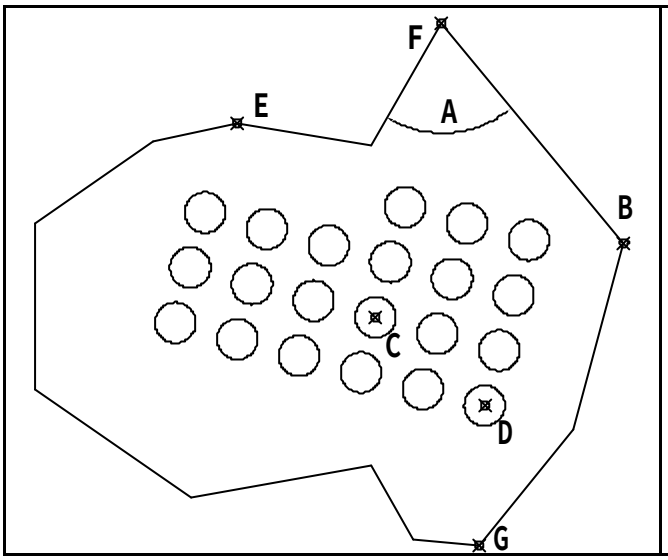
- - - -




36、 ROtate, L76, int, L76 & L72, -45, →L76 **轉動**

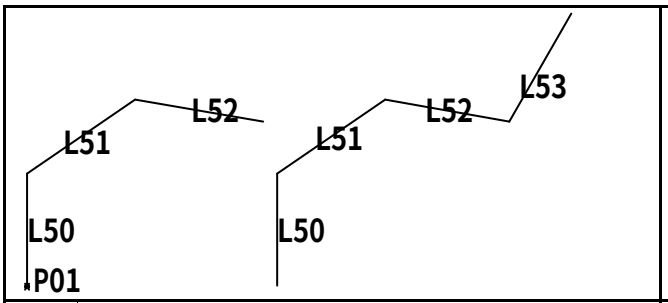
- - - -

## 2.6.6 AutoCAD2D2008-606

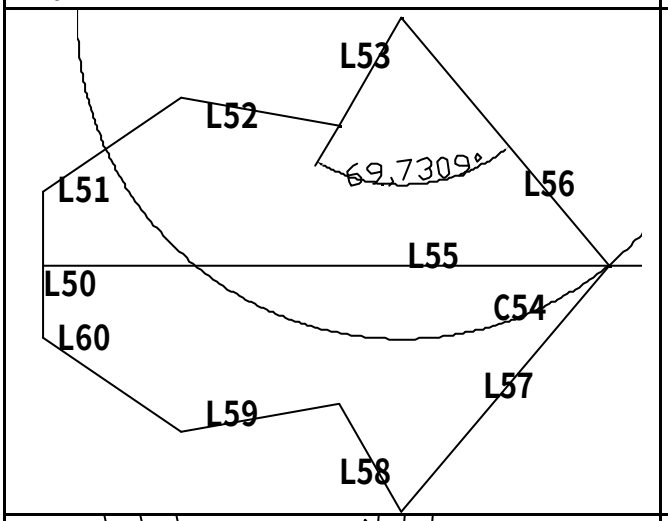



1、 **尺寸** A **角度值為何？** 69.7309
2、 **交點** B **至中心點** C **距離為何？** 102.3973
3、 **中心點** D **至交點** E **距離為何？** 148.3248
4、 **交點** F **至交點** G **高度為何？** 205.6040
5、 **圖形扣除內孔面積為何？** 26090.5460

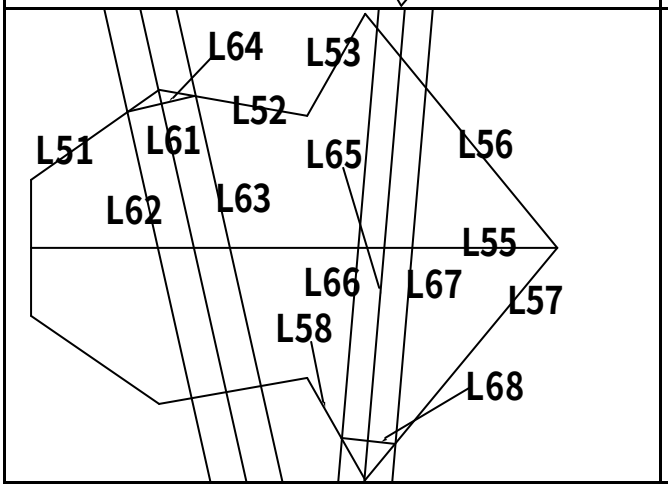
**解題步驟：**



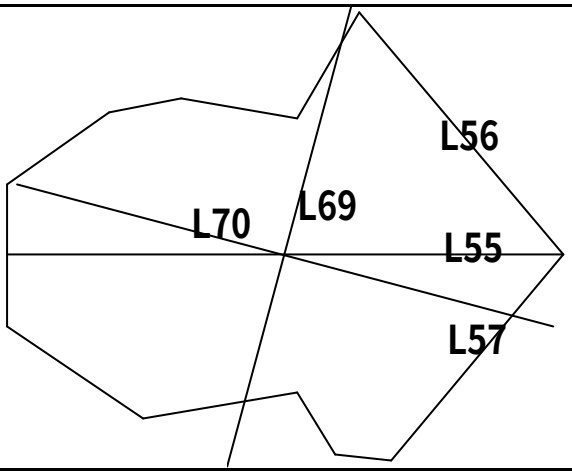
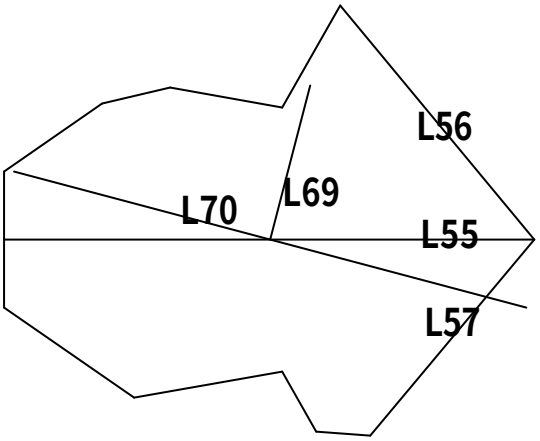
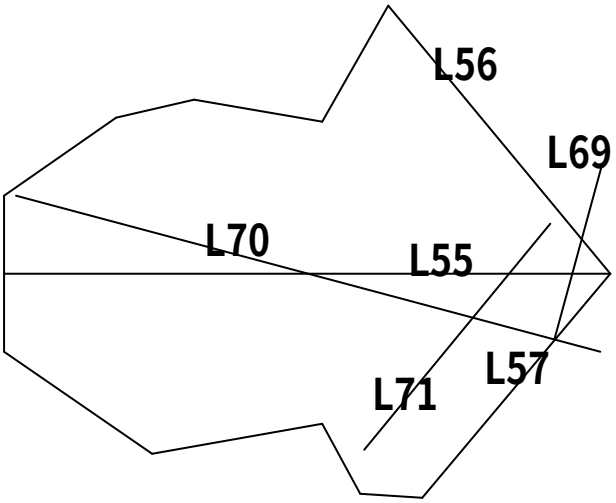
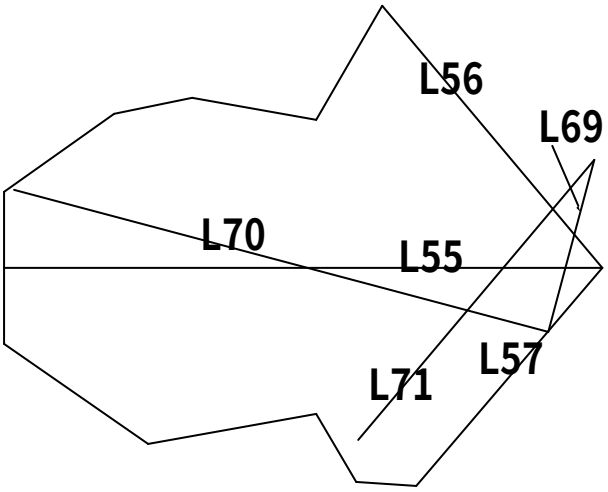

01、 Line, **任一點** P01, (**正交模式**), **向上** 65, →L50, @75<35, →L51
02、 ROtate, L51, end, L51 **上**, C, 135, →L52
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
03、 LENgthen, T, 72, L52 **右**, →L52 **縮短**
04、 ROtate, L52, end, L52 **右**, C, -110, →L53

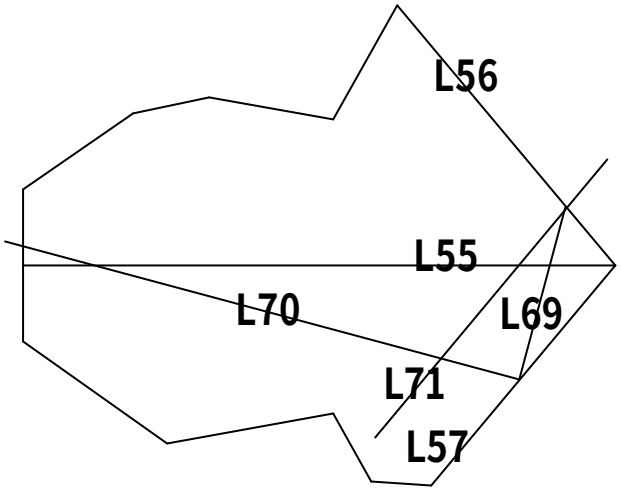
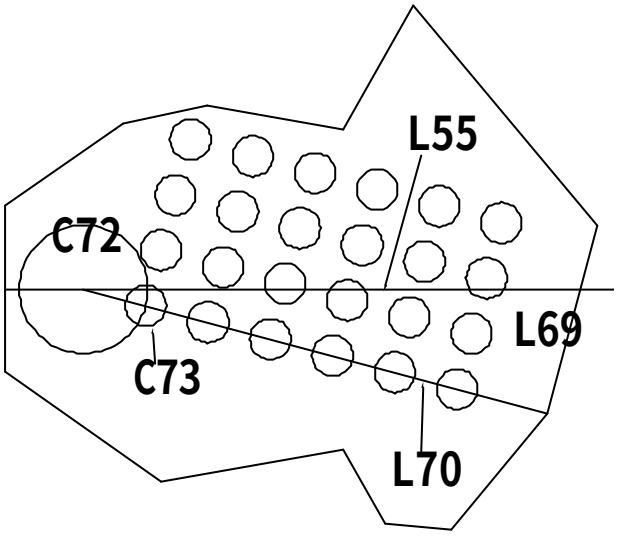
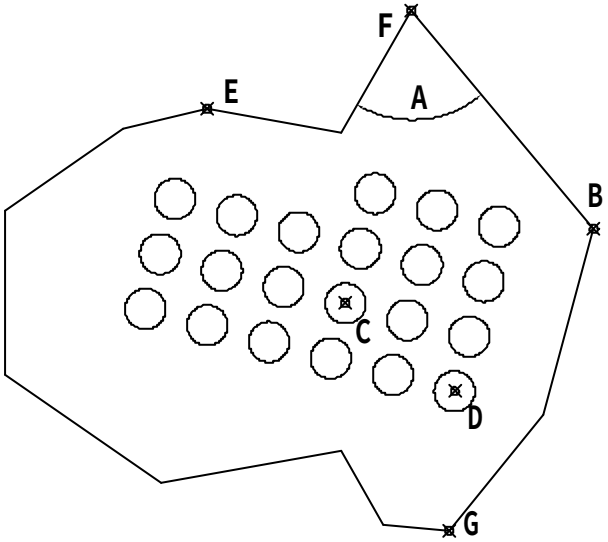



05、 LENgthen, T, 56, L53 **上**, →L53 **縮短**
06、 Circle, end, L53 **上**, 145, →C54
07、 Line, mid, L50, (**正交模式**), **向右交於** C54, →L55
08、 Line, end, L53 **上**, int, C54 & L55, →L56
09、 DimANgular, L53, L56, →**檢查是否為** 69.7309(**即第** 1 **題**)
10、 MIrror, L51~L53;L56, int, L50 & L55, int, C54 & L55, →L57~L60

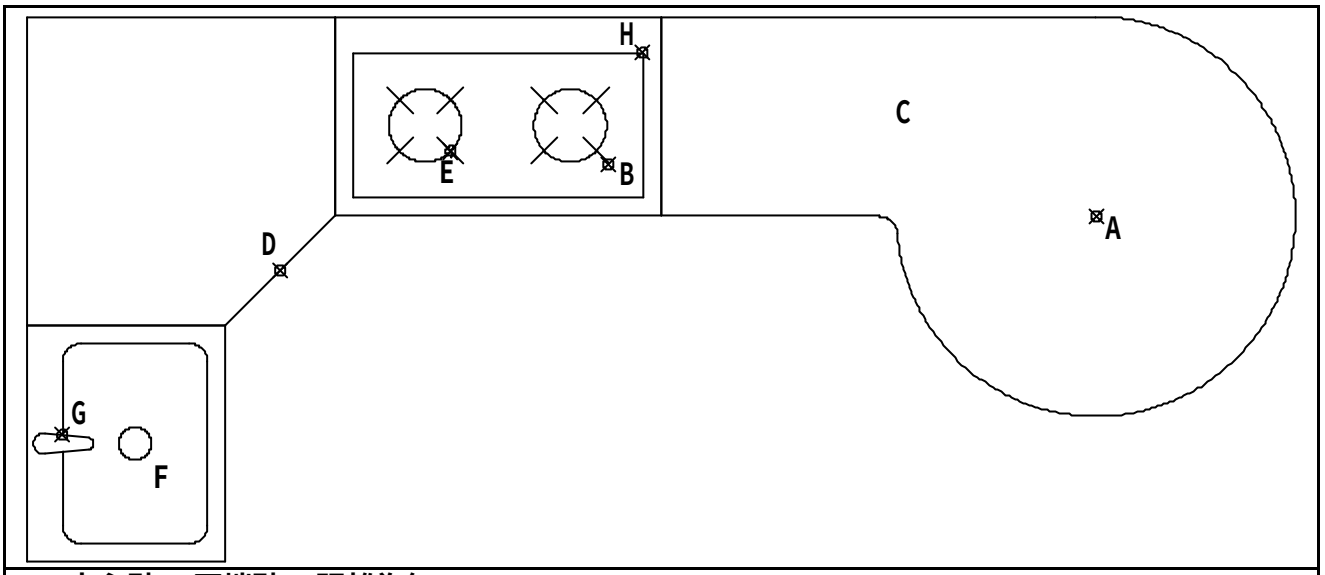



11、 XLine, B, int, L51 & L52, mid, L51, mid, L52, →L61
12、 Offset, 16.85(=33.7/2), L61, **向左點**, →L62, L61, **向右點**, →L63
13、 Line, int, L51 & L62, int, L52 & L63, →L64
14、 XLine, B, int, L57 & L58, mid, L57, mid, L58, →L65
15、 Offset, 12.95(=25.9/2), L65, **向左點**, →L66, L65, **向右點**, →L67
16、 Line, int, L57 & L67, int, L58 & L66, →L68

| | |
|---|---|
|  | 17、 TRim,<br>18、 Erase,<br>19、 ROtate, L55, mid, L55, C, -105, →L69<br>20、 ROtate, L69, int, L55 & L69, C, 90, →L70 |
|  | 21、 TRim, (**刪去** L69 **下端**)<br>22、 LENgthen, T, 76.25, L69 **上**, →L69 **總長** 76.25 |
|  | 23、 Move, L69, int, L55 & L69, int, L57 & L70, →L69 **順** L70 **向右下移動**<br>24、 Offset. T, L57, end, L69 **上**, →L71 |
|  | 25、 EXtend, L69, L71, →L71 **與** L69 **相交** |

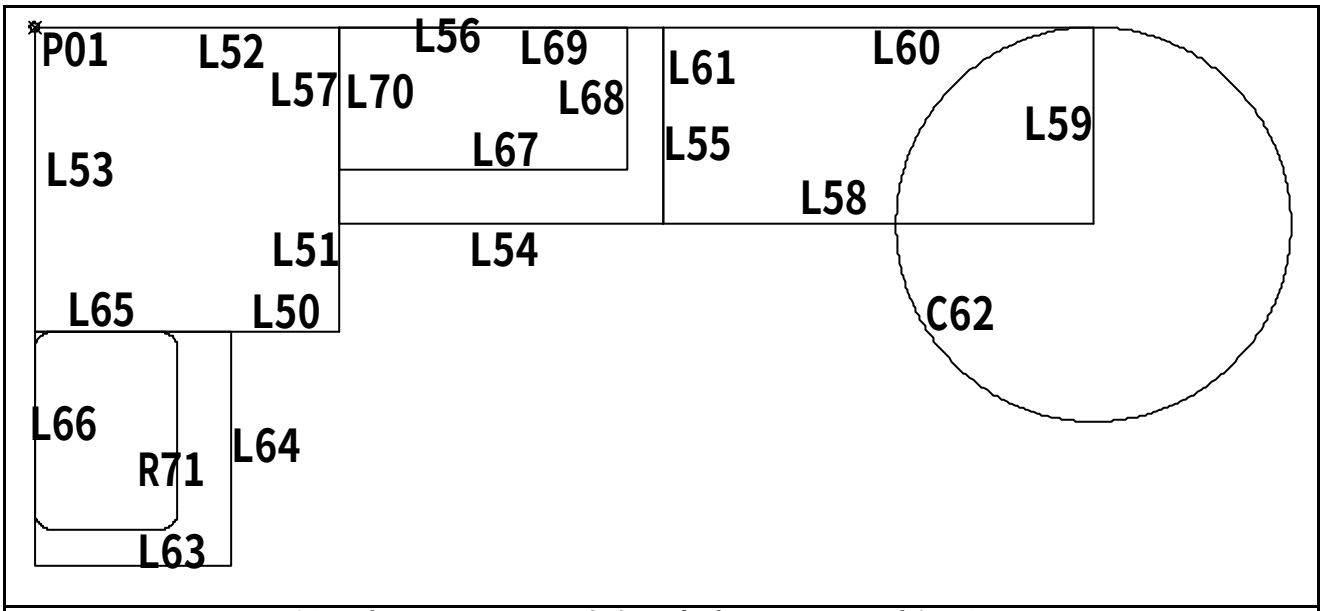
| | |
|---|---|
|  | 26、 Move, L69;L70, int, L69 & L71, int, L56 & L71, →L69;L70 **順** L71 **向左下移動** |
|  | 27、 TRim,<br>28、 Erase,<br>29、 Circle, int, L55 & L70, 25.25, →C72<br>30、 Circle, int, L70 & C72, D, 16, →C73<br>31、 ARray, **矩形陣列**, **列**, 4, **行**, 6, **列偏移**, 22.5, **行偏移**, 25.25, **物件**, C73, **陣列角度**, cen, C72, cen, C73, →24 **個相同之** C73 **物件** |
|  | 32、 Erase,<br>33、 DimANgular, L53, L56, →69.7309<br>34、 DIst, int, B, cen, C, →**距離**=102.3973<br>35、 DIst, cen, D, int, E, →**距離**=148.3248<br>36、 DIst, int, F, int, G, →Y **差值**=205.6040<br>37、 Hatch, **點圖形內區域**, →**斜線區域**<br>38、 **點斜線區域**, CTRL+1, →**查幾何面積值** =26090.5460 |

## 2.6.7 AutoCAD2D2008-607

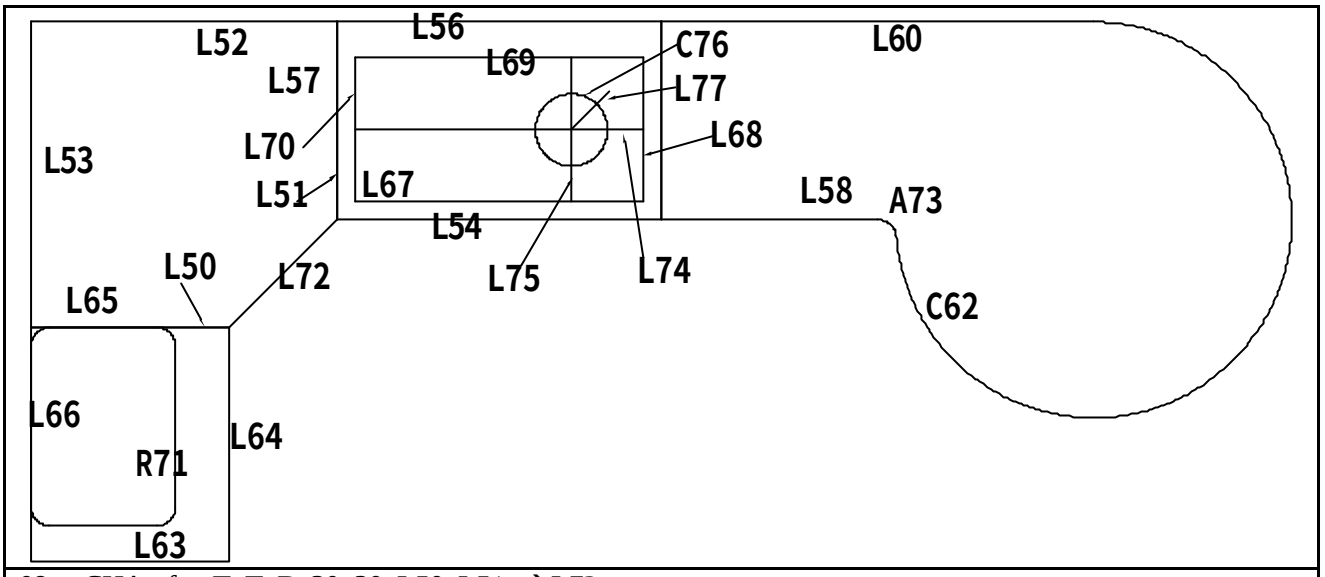



1、 **中心點** A **至端點** B **距離為何？** 135.1620
2、 C **周長為何？** 497.2383
3、 **中點** D **至交點** E **角度為何？** 34.9750
4、 F **區域淨面積為何？** 2084.0474
5、 **交點** G **至交點** H **距離為何？** 191.4349

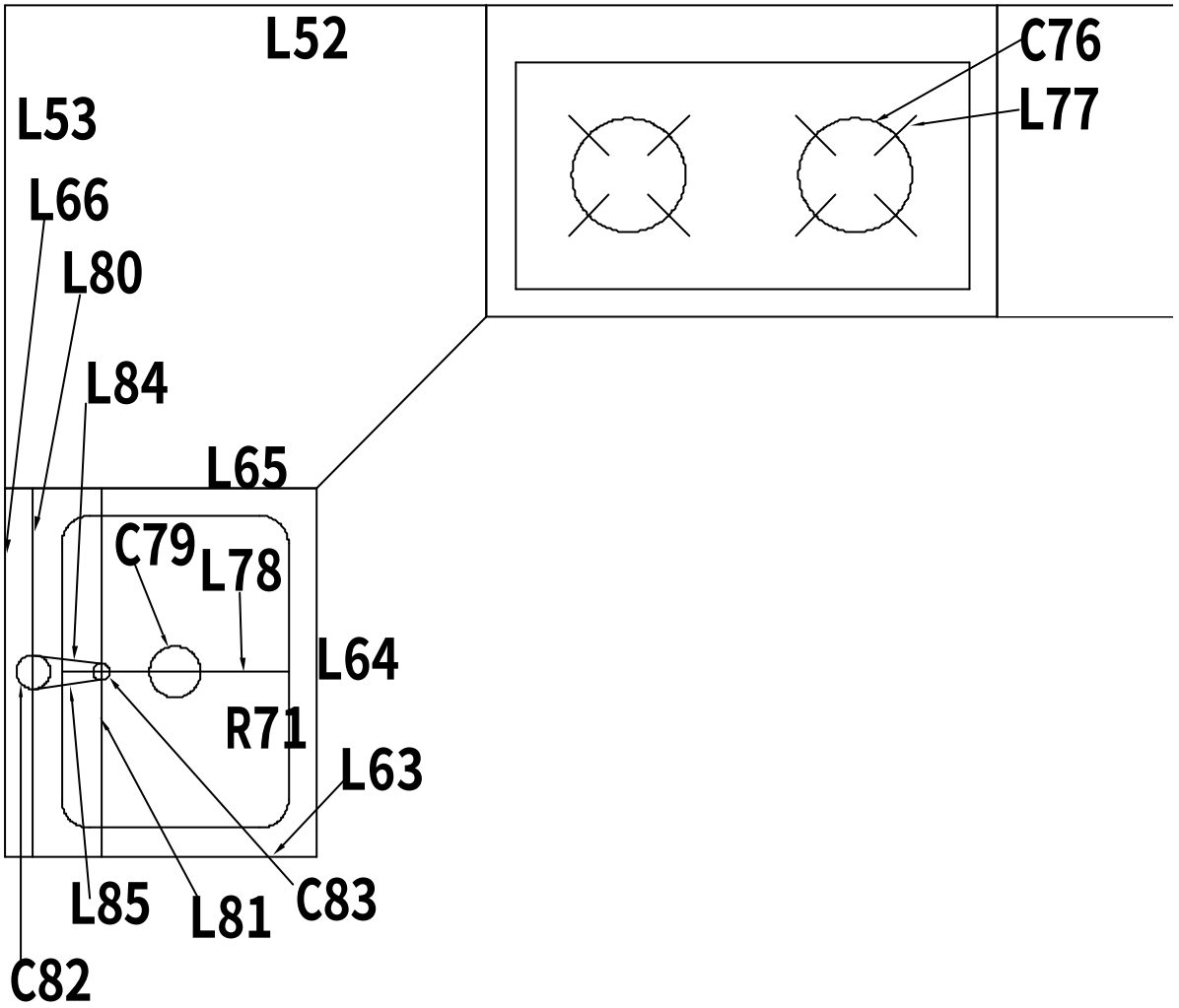
**解題步驟：**



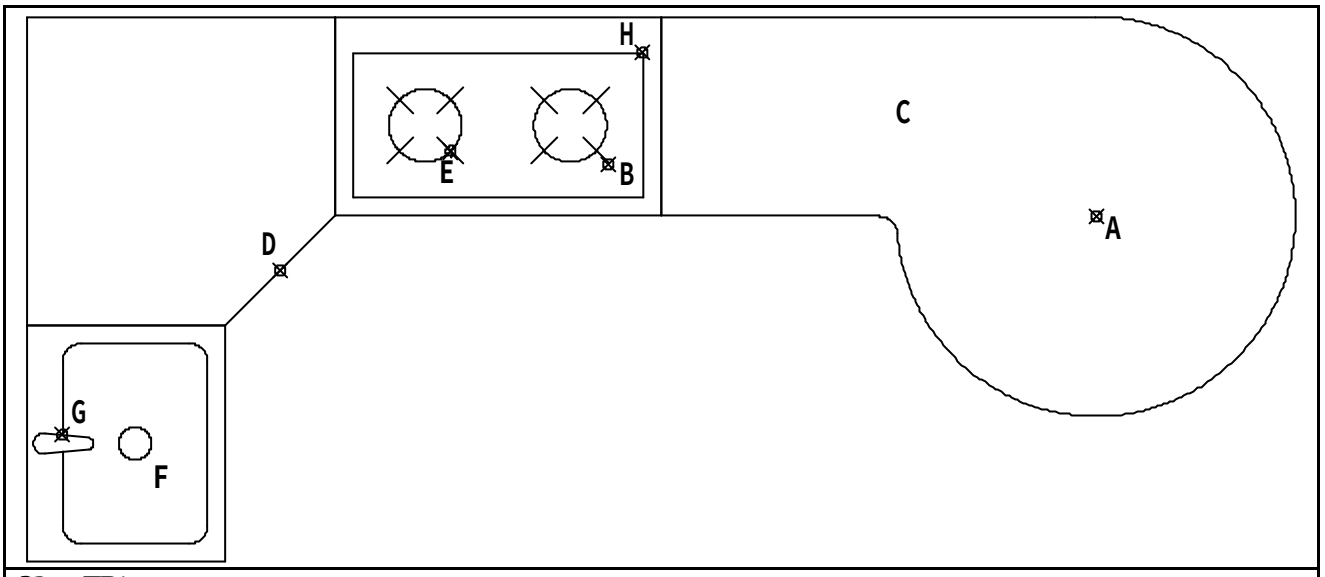

01、 RECtangle, F, 0, **任一點** P01, D, 85, 85, **向右下方點**, →L50~L53 **之矩形**
02、 RECtangle, F, 0, int, L51 & L52, D, 90, 55, **向右下方點**, →L54~L57 **之矩形**
03、 RECtangle, F, 0, int, L55 & L56, D, 120, 55, **向右下方點**, →L58~L61 **之矩形**
04、 Circle, int, L58 & L59, D, 110, →C62
05、 RECtangle, F, 0, int, L50 & L53, D, 55, 65, **向右下方點**, →L63~L66 **之矩形**
06、 RECtangle, F, 0, int, L51 & L52, D, 80, 40, **向右下方點**, →L67~L70 **之矩形**
07、 RECtangle, F, 5, int, L50 & L53, D, 40, 55, **向右下方點**, →R71 **之圓角矩形**




08、 CHAmfer, T, T, D, 30, 30, L50, L51, →L72
09、 eXplode, L58~L61 **之矩形**, →L58~L61
10、 Fillet, T, T, R, 5, L58, C62, →A73
11、 TRim,
12、 Erase,
13、 Move, L67~L70 **之矩形**, int, L51 & L52, @5,-10, →L67~L70 **之矩形向右下移動**
14、 eXplode, L67~L70 **之矩形**, →L67~L70
15、 Line, mid, L68, mid, L70, →L74
16、 Offset, 20, L68, **向左點**, →L75
17、 Circle , int, L74 & L75, D, 20, →C76
18、 Line, @(**鎖定上次點**), @15<45, →L77

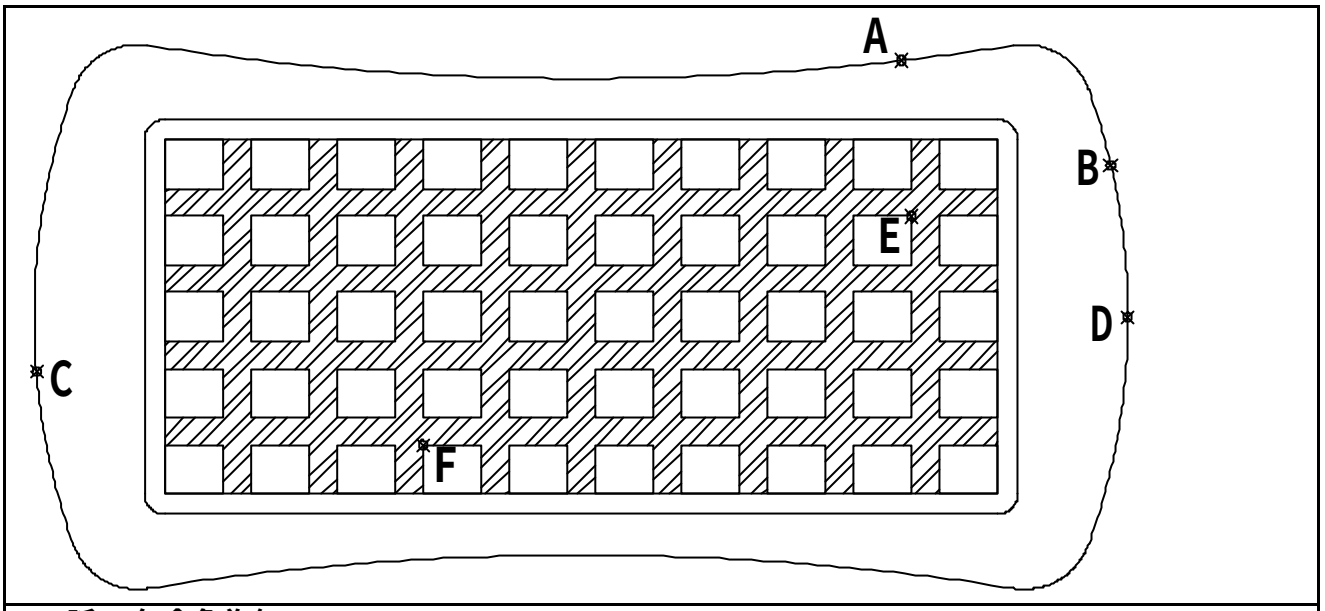



19、 LENgthen, T, 10, L77 下, →L77 向上縮短
20、 ARray, 環形陣列, 中心點, cen, C76, 項目總數, 4, 佈滿角度, 360, 物件, L77, →全圓佈滿 4 個相同物件
21、 COpy, C76;L77 之所有物件, cen, C76, (正交模式), 向左 40, →左側相同物件組
22、 Move, R71 之矩形, int, L65 & L66, @10,-5, →R71 之矩形向右下移動
23、 Line, mid, R71 左, mid, R71 右, →L78
24、 Circle, mid, L78, D, 9, →C79
25、 eXplode, L63~L66 之矩形, →L63~L66
26、 Offset, 5, L66, 向右點, →L80
27、 Offset, 12, L80, 向右點, →L81
28、 Circle, mid, L80, 3, →C82
29、 Circle, int, L78 & L81, 1.5, →C83
30、 Line, tan, C82 上, tan, C83 上, →L84
31、 Line, tan, C82 下, tan, C83 下, →L85



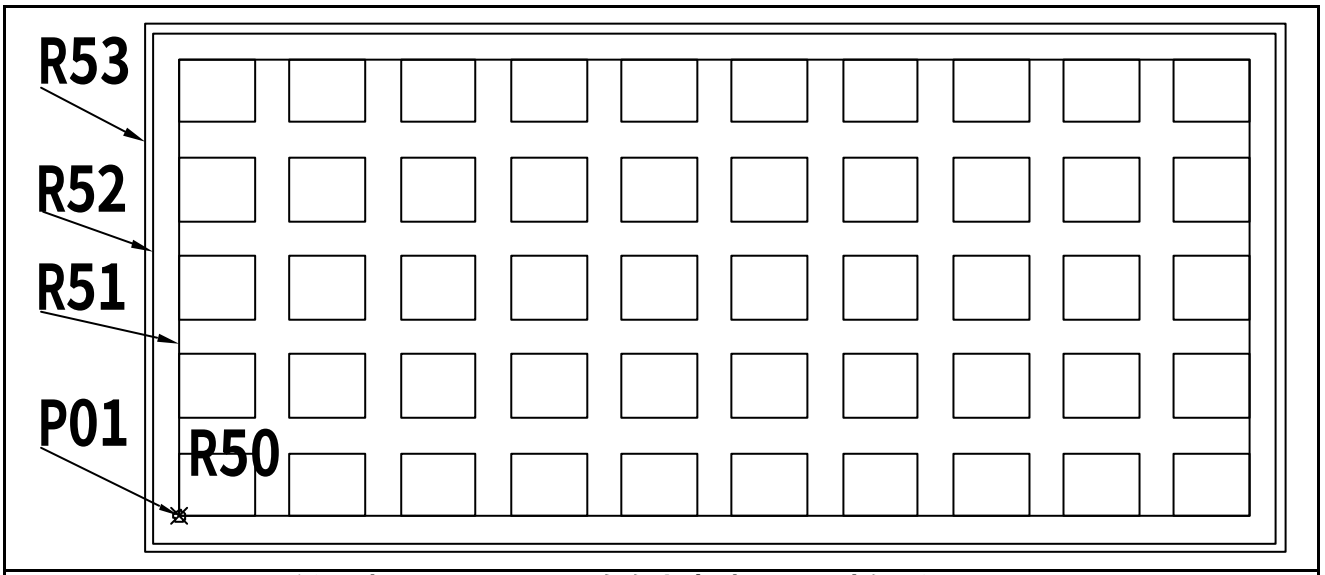

32、 TRim,
33、 Erase,
34、 DIst, cen, A, end, B, →**距離**=135.1620
35、 BOundary, **點** C **區域**,
36、 **點選** C **區域邊緣**, CTRL+1, →**查幾何長度值**=497.2383
37、 DIst, mid, D, int, E, →XY **平面內角度**=34.9750
38、 Hatch, **點** F **區域**, →**斜線區域**
39、 **點選** F **斜線區域**, CTRL+1, →**查幾何面積值**=2084.0474
40、 DIst, int, G, int, H, →**距離**=191.4349

## 2.6.8 AutoCAD2D2008-608

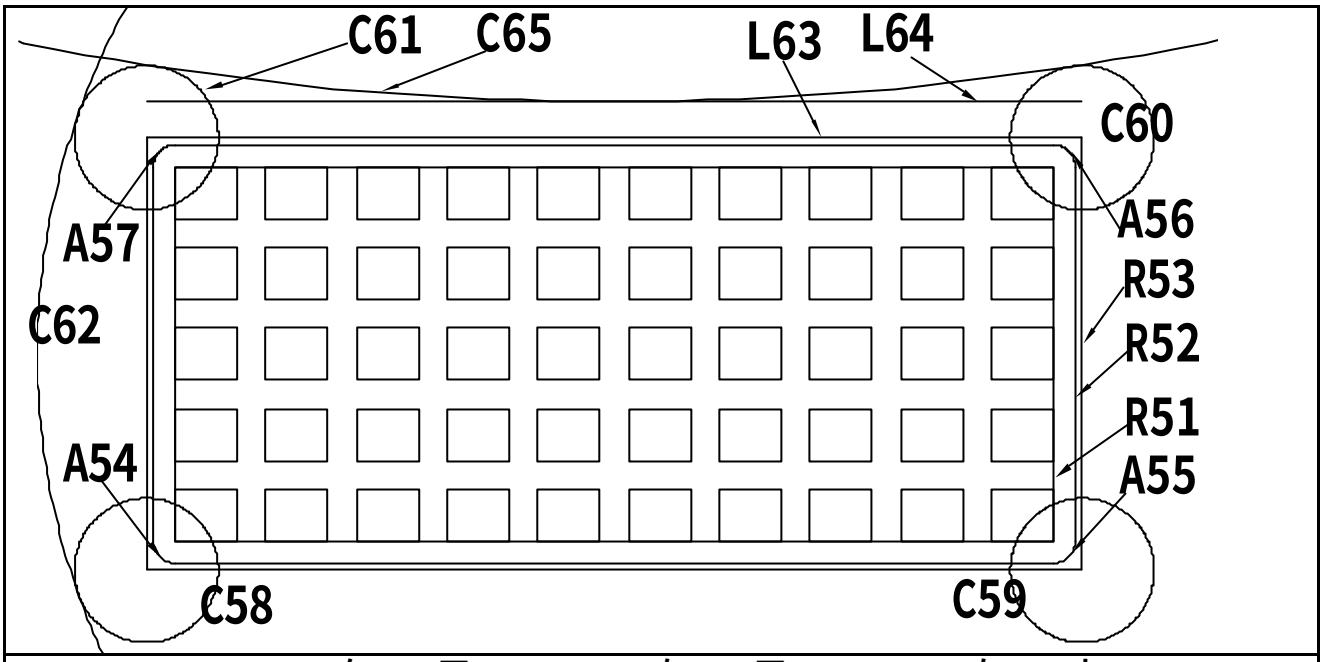



1、 **弧** A **包含角為何？** 17.5948
2、 **弧** B **長度為何？** 67.9674
3、 **中點** C **至中點** D **水平距離為何？** 160.2944
4、 **交點** E **至交點** F **距離為何？** 79.0918
5、 **斜線區域面積為何？** 3248.0000

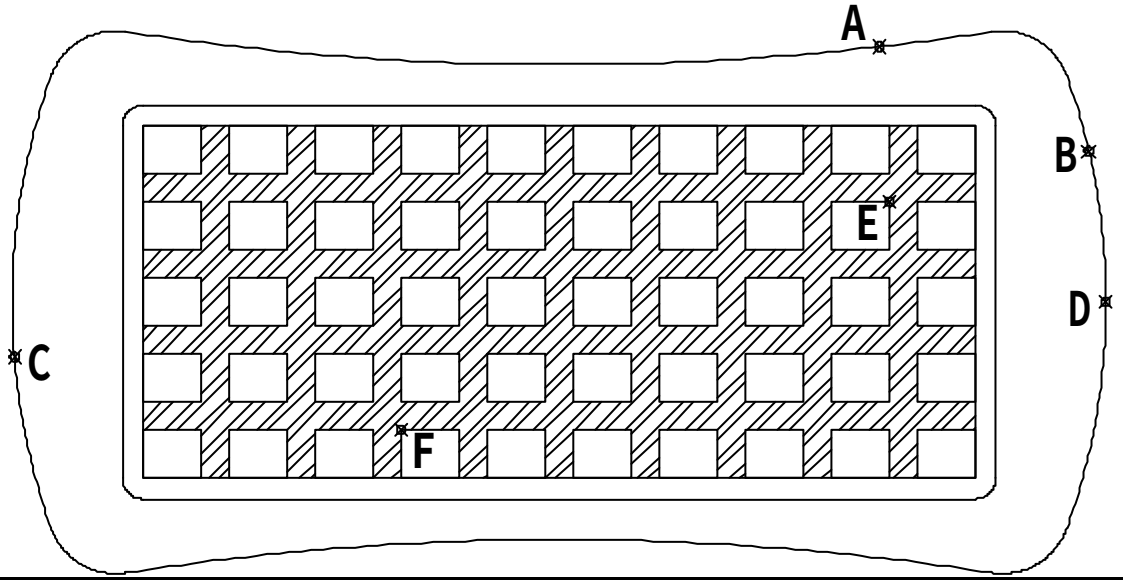
**解題步驟：**




01、 RECtangle, F, 0, **任一點** P01, D, 8.6, 7.2, **向右上方點**, →R50 **之矩形**
02、 ARray, **矩形陣列**, **列**, 5, **行**, 10, **列偏移**, 11.2(=7.2+4), **行偏移**, 12.6(=8.6+4), **物件**, R50, →50 **個相同之矩形**
03、 RECtangle, F, 0, int, R50 **左下**, D, 122, 52, **向右上方點**, →R51 **之矩形**
04、 Offset, 3, R51, **向外點**, →R52 **之矩形**
05、 Offset, 1, R52, **向外點**, →R53 **之矩形**

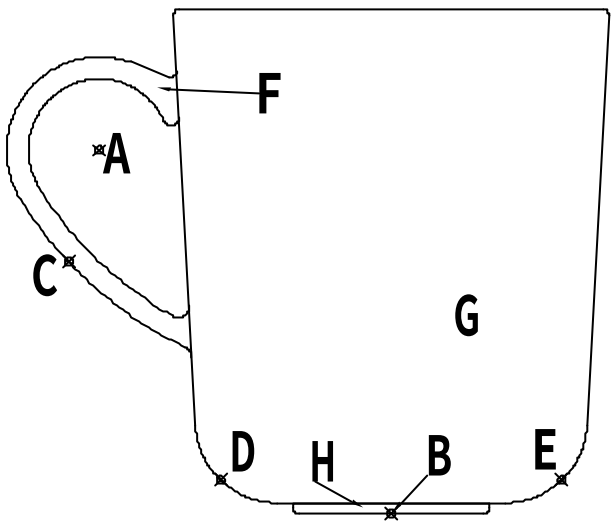



06、 Fillet, T, T, R, 3, R52 **右**, R52 **下**, →A54, R52 **左**, R52 **下**, →A55, R52 **左**, R52 **上**, →A56, R52 **右**, R52 **上**, →A57
07、 Circle, int, R53 **右**, R53 **下**, 10, →C58
08、 COpy, C58, int, R53**右**, R53**下**, int, R53**左**, R53**下**, 10, →C59, int, R53**左**, R53**上**, 10, →C60, int, R53 **右**, R53 **上**, 10, →C61
09、 Circle, T, C61 **左上**, C58 **左下**, 100, →C62
10、 eXplode, R53 **之矩形**, →L63 **等物件**
11、 Offset, 5, L63, **向上點**, →L64
12、 Circle, 3P, tan, , C61 **右上**, tan, L64, tan, C60 **左上**, →C65
13、 TRim, Erase,
14、 MIrror, C62, R52 **上**, R52 **下**, →**右側對稱物件**
15、 MIrror, C65, R52 **左**, R52 **右**, →**下方對稱物件**



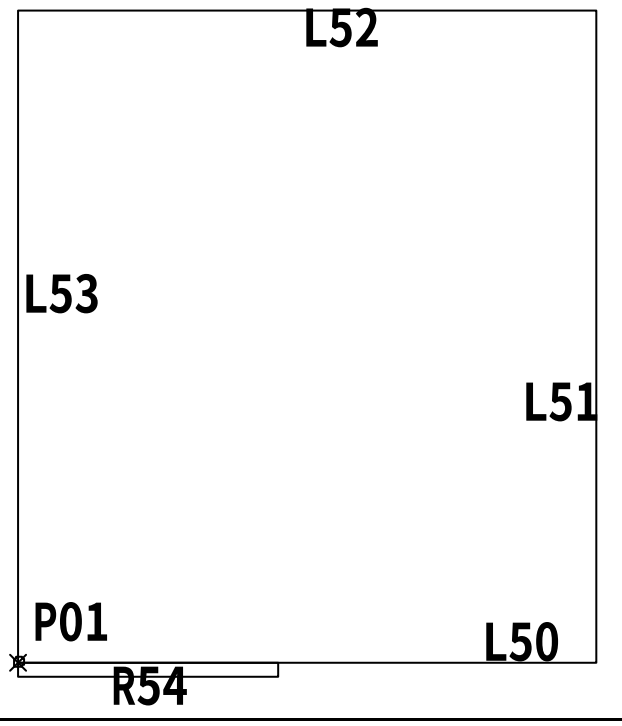

16、 TRim, Erase,
17、 **點選弧** A, CTRL+1, →**查幾何總角度值**=17.5948
18、 **點選弧** B, CTRL+1, →**查幾何弧長值**=67.9674
19、 DIst, mid, C, mid, D, →X **差值**=160.2944
20、 DIst, int, E, int, F, →**距離**=79.0918
21、 Hatch, **點＋字區域**, →**斜線區域**
22、 **點選斜線區域**, CTRL+1, →**查幾何面積值** =3248.0000

## 2.6.9 AutoCAD2D2008-609

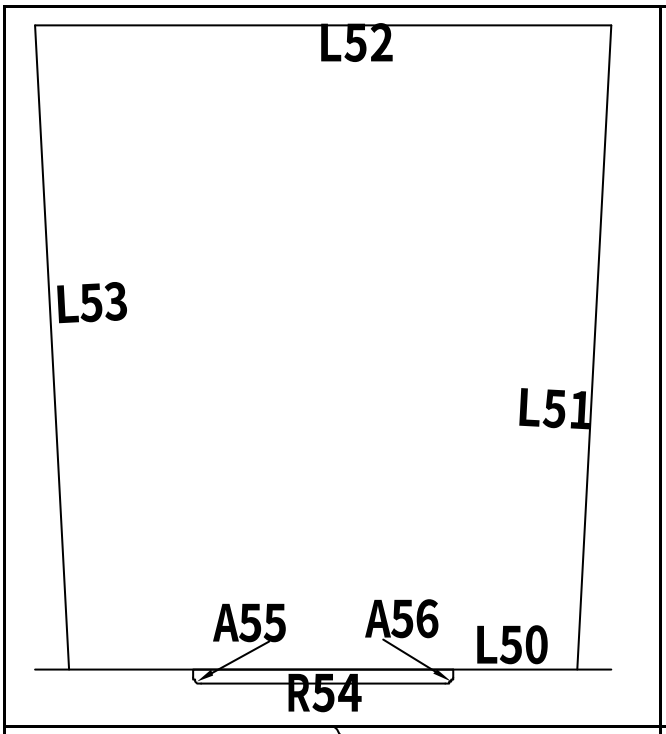



1、 中心點 A 至中點 B 水平距離為何？53.6710
2、 弧 C 包含角為何？ 32.6680
3、 弧 D 中心點至弧 E 中心距為何？ 42.0977
4、 F 區域周長為何？ 197.7225
5、 G、H 區域面積總和為何？ 6761.1975

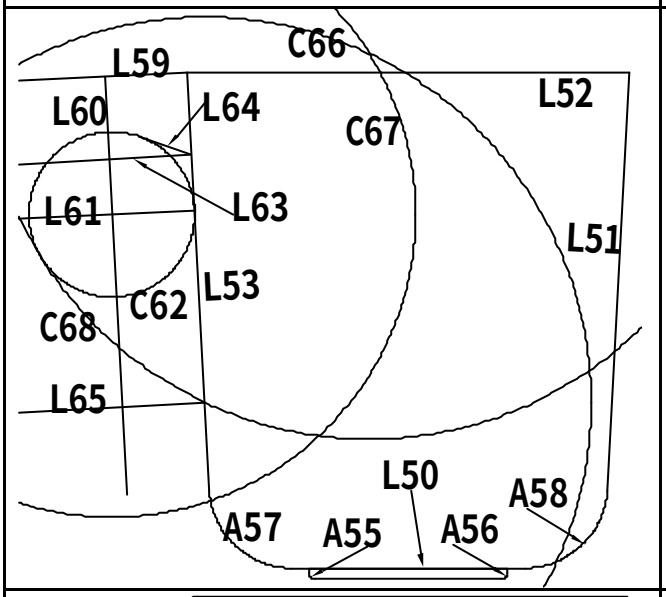
**解題步驟：**



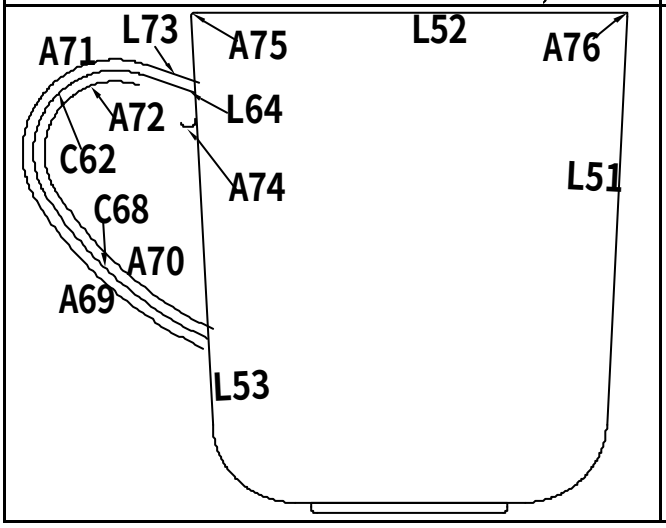

01、 RECtangle, F, 0, 任一點 P01, D, 80, 90, 向右上方點, →L50~L53 之矩形
02、 eXplode, L50~L53 之矩形, →L50~L53
03、 RECtangle, F, 0, int, L50 & L53, D, 36, 2, 向右下方點, →R54 之矩形



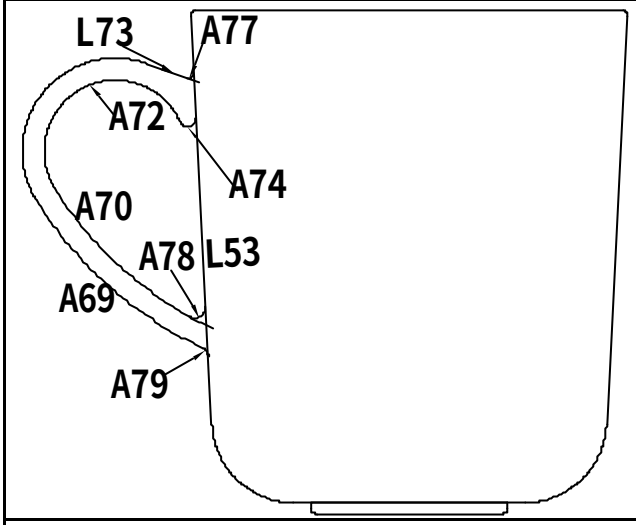
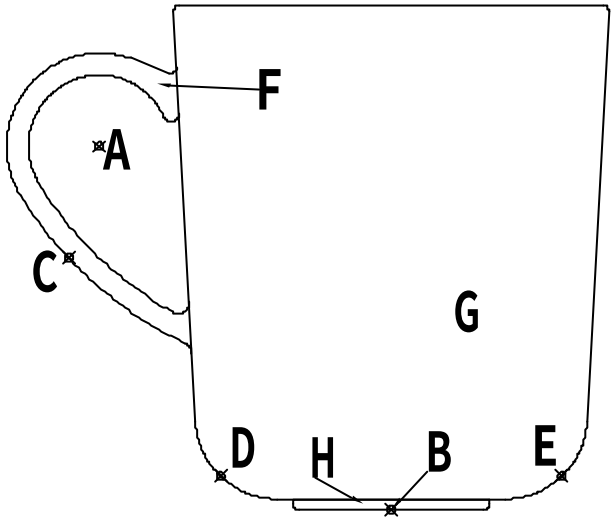

04、 Move, R54, mid, R54 上, mid, L50, →R54 之矩形向右移動
05、 Fillet, T, T, R, 1, R54 左, R54 下, →A55, R54 右, R54 下, →A56
06、 ROtate, L53, int, L52 & L53, 3(=6/2), →L53 轉動
07、 ROtate, L51, int, L52 & L51, -3(=6/2), →L51 轉動



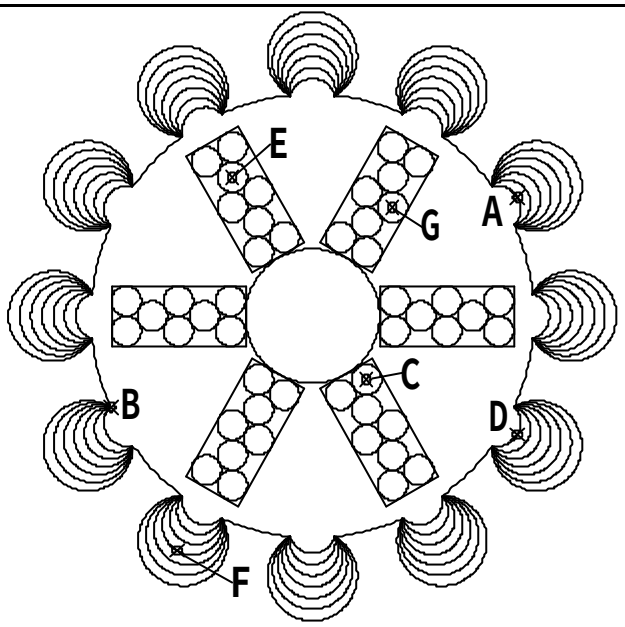

08、 Fillet, T, T, R, 15, L53, L50, →A57, L51, L50, →A58
09、 ROtate, L53, int, L52 & L53, C, -90, →L59
10、 Offset, 15, L53, 向左點, →L60
11、 Offset, 25, L59, 向下點, →L61
12、 Circle, int, L60 & L61, 15, →C62
13、 Offset, 15, L59, 向下點, →L63
14、 Line, int L53 & L63, tan, C62 右上, →L64
15、 Offset, 45, L63, 向下點, →L65
16、 Circle, int, L53 & L65, 70, →C66
17、 Offset, 40(=70-15X2), C62, 向外點, →C67
18、 Circle, int, C66 & C67, 70, →C68




19、 TRim,
20、 Erase,
21、 Offset, 2, C68, 向下點, →A69, C68, 向上點, →A70, C62, 向左點, →A71, C62, 向右, →A72, L64,向上點, →L73
22、 Fillet, T, N, R, 1.5, A72, L53, →A74
23、 Fillet, T, T, R, 1, L53, L52, →A75, L52, L51, →A76

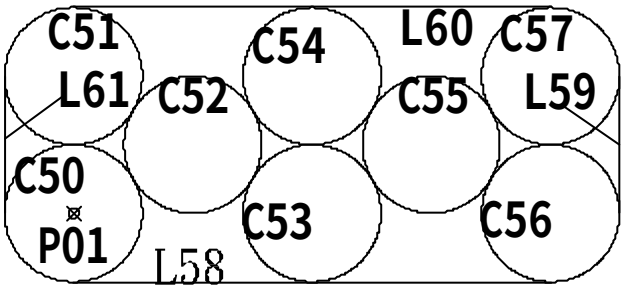
| | |
|---|---|
|  | 24、 EXtend, A74, A72, →A72 與 A74 相交<br>25、 Fillet, T, N, R, 1, L73, L53, →A77<br>26、 Fillet, T, N, R, 2, L53, A70, →A78, L53, A69, →A79 |
|  | 27、 TRim,<br>28、 Erase,<br>29、 DIst, cen, A, mid, B, →X 差值=53.6710<br>30、 點弧 C, CTRL+1, →查幾何總角度值=32.6680<br>31、 DIst, cen, D, cen, E, →距離=42.0977<br>32、 BOundary, 點 F 區域<br>33、 點 F 區域邊緣, CTRL+1, →查幾何長度值=197.7225<br>34、 Hatch, 點 G 及 H 區域, →斜線區域<br>35、 點 G 及 H 斜線區域, CTRL+1, →查幾何面積值=6761.1975 |

## 2.6.10 AutoCAD2D2008-610

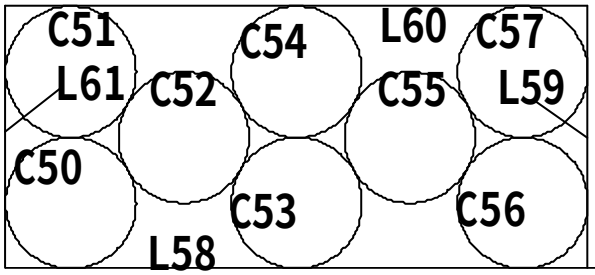



1、 **弧** A **半徑為何？** 10.4142
2、 **交點** B **至中心點** C **距離為何？** 115.1206
3、 **弧** D **中心點至中心點** E **距離為何？** 161.6464
4、 **弧** F **中心點至中心點** G **距離為何？** 167.6600
5、 **圖形最外圍的周長為何？** 1644.8782

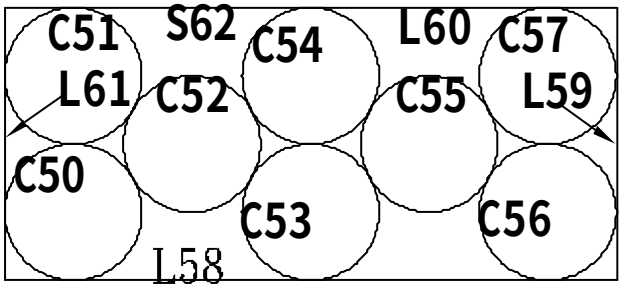
**解題步驟：**



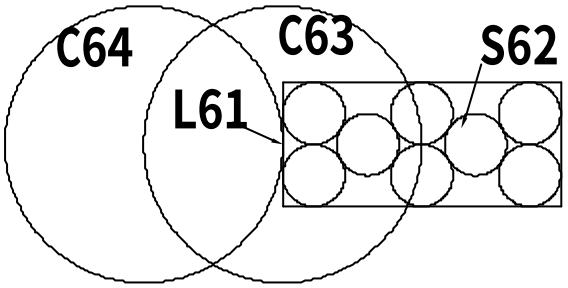

01、 Circle, **任一點** P01, 2, →C50
02、 COpy, C50, qua, C50 **下**, qua, C50 **上**, →C51
03、 Circle, T, C50 **右**, C51 **右**, 2, →C52
04、 MIrror, C50;C51, qua, C52 **下**, qua, C52 **上**, →C53;C54
05、 MIrror, C50~C52, qua, C53 **下**, qua, C54 **上**, →C55~C57
06、 Line, qua, C50 **下**, qua, C56 **下**, →L58
07、 Line, qua, C56 **右**, qua, C57 **右**, →L59
08、 Line, qua, C57 **上**, qua, C51 **上**, →L60
09、 Line, qua, C51 **左**, qua, C50 **左**, →L61



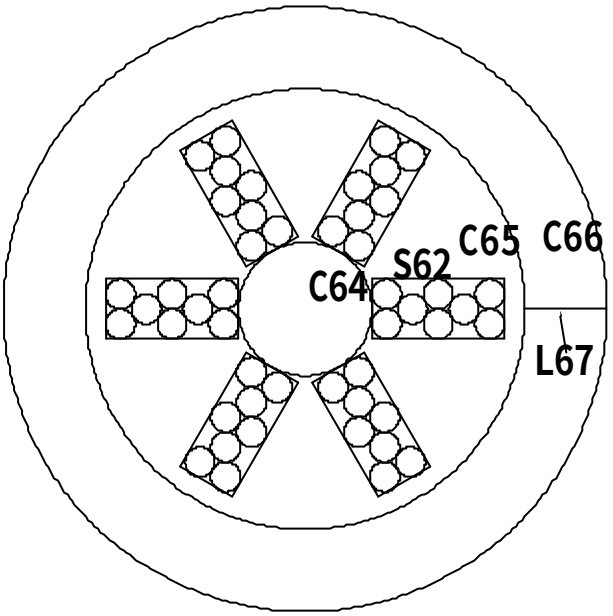

10、 Fillet, T, T, R, 0, L58, L61, →L58 **與** L61 **相交**, L58, L59, →L58 **與** L59 **相交**, L59, L60, →L59 **與** L60 **相交**, L60, L61, →L60 **與** L61 **相交**
11、 LENgthen, T, 60, L58 **右**, →L58 **向右延長為** 60



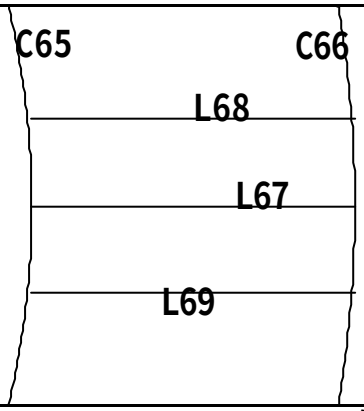

12、 SCale, **所有物件**(L58 **除外**), int, L58 & L61, R, int, L58 & L61, int, L58 & L59, end, L58 **右**, →**所有物件**(L58 **除外**)**等比例放大**(**稱為** S62 **物件組**)

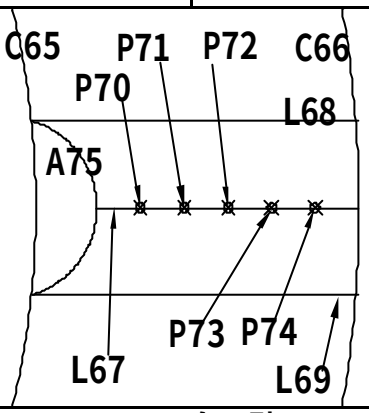


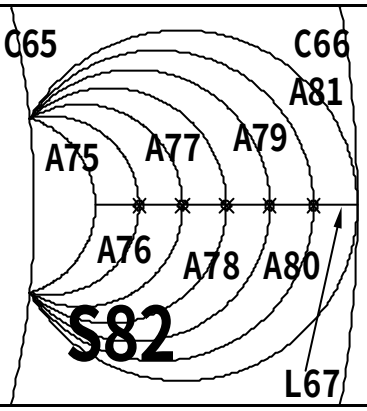

13、 Circle, mid, L61, D, 60, →C63
14、 Circle, qua, C63 **左**, D, 60, →C64




15、 Erase,
16、 ARray, **環形陣列**, **中心點**, cen, C64, **項目總數**, 6, **佈滿角度**, 360, **物件**, S62 **物件組**, →6 **組與** S62 **相同物件組**
17、 Circle, cen, C64, D, 198, →C65
18、 Circle, @(**鎖定上次點**), D, 272, →C66
19、 Line, qua, C65 **右**, qua, C66 **右**, →L67

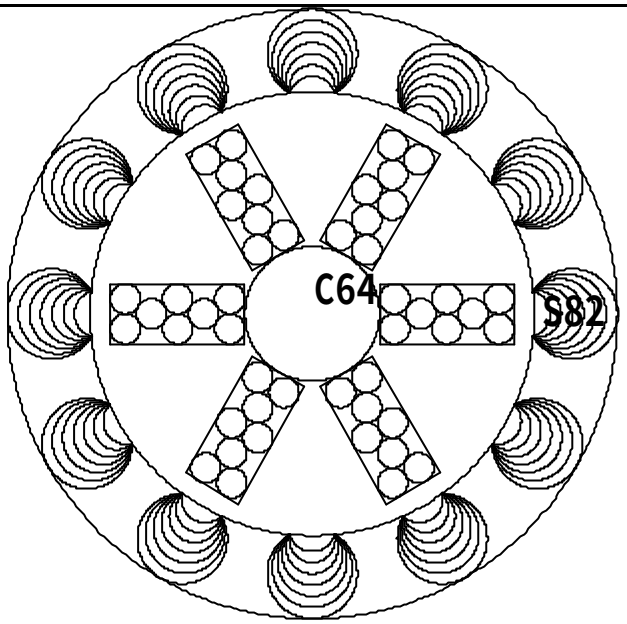
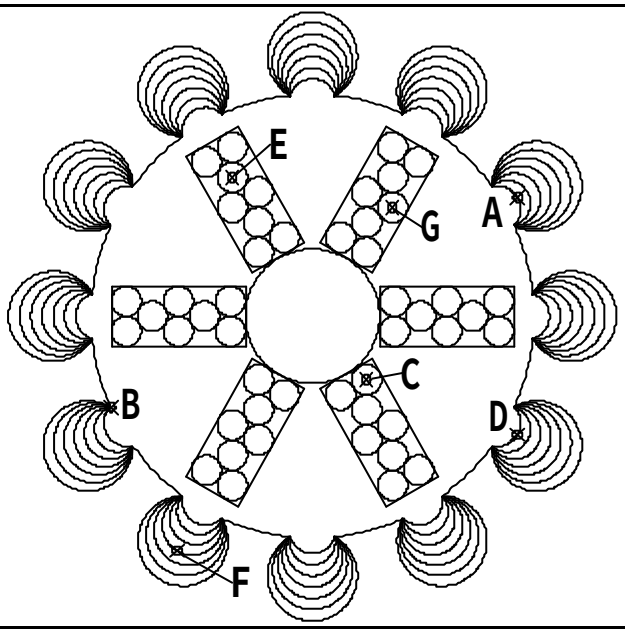
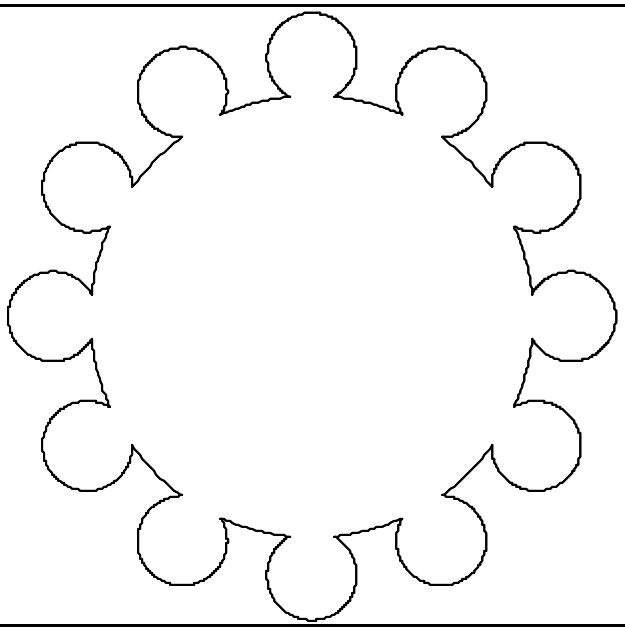


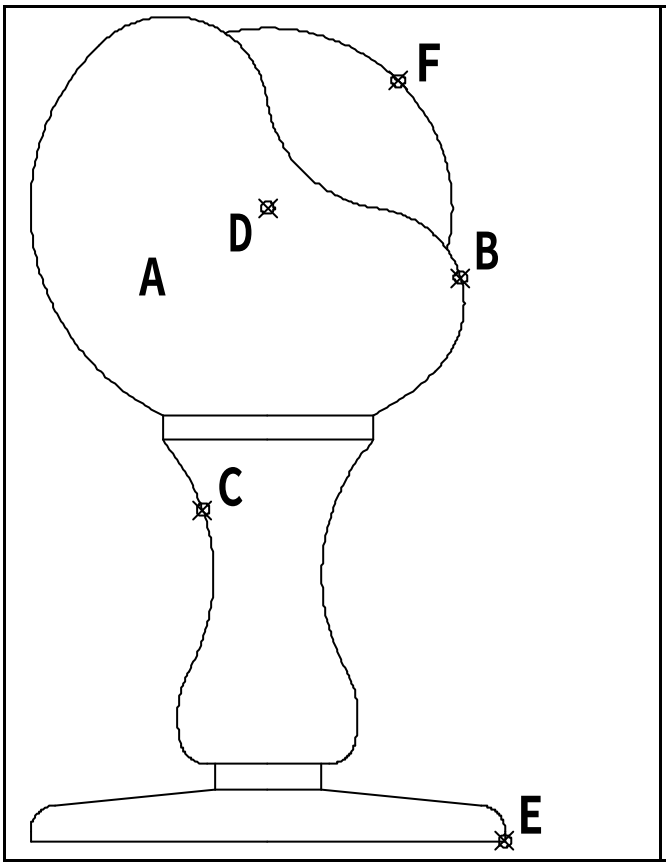




20、 Offset, 10(=20/2), L67, **向上點**, →L68, L67, **向下點**, →L69

- - - -

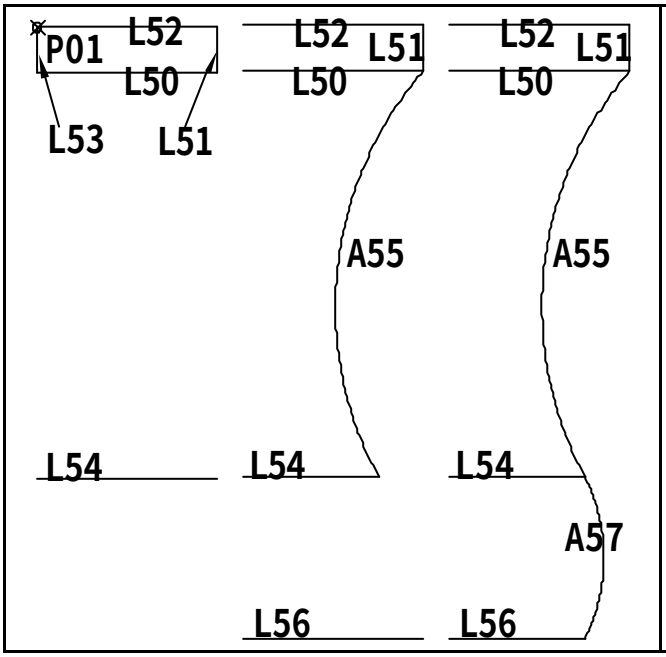
| | |
|---|---|
|  | 26、 TRim,<br>27、 Erase,<br>28、 ARray, **環形陣列**, **中心點**, cen, C64, **項目總數**, 12, **佈滿角度**, 360, **物件**, S82 **物件組**, →12 **組與** S82 **相同物件組** |
|  | 29、 TRim,<br>30、 **點弧** A, CTRL+1, →**查幾何半徑值**=10.4142<br>31、 DIst, int, B, cen, C, →**距離**=115.1206<br>32、 DIst, cen, D, cen, E, →**距離**=161.6464<br>33、 DIst, cen, F, cen, G, →**距離**=167.6600 |
|  | 34、 COpy, TRim, Erase, →**上方圖形**<br>35、 BOundary, **點中間區域**<br>36、 **點邊緣**, CTRL+1, →**查幾何長度值** =1644.8782 |

## 2.6.11 AutoCAD2D2008-611

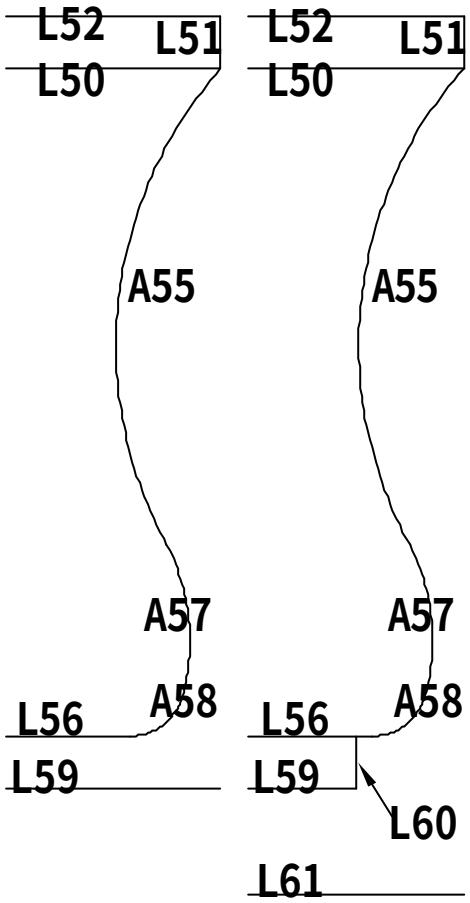



1、 A **區域周長為何？** 255.0865
2、 **弧** B **至弧** C **中心點距離為何？** 86.8249
3、 **中心點** D **至交點** E **高度為何？** 123.3113
4、 **弧** F **包含角為何？** 116.8955
5、 **圖形所圍成最外圍面積為何？** 8067.0387

**解題步驟：**



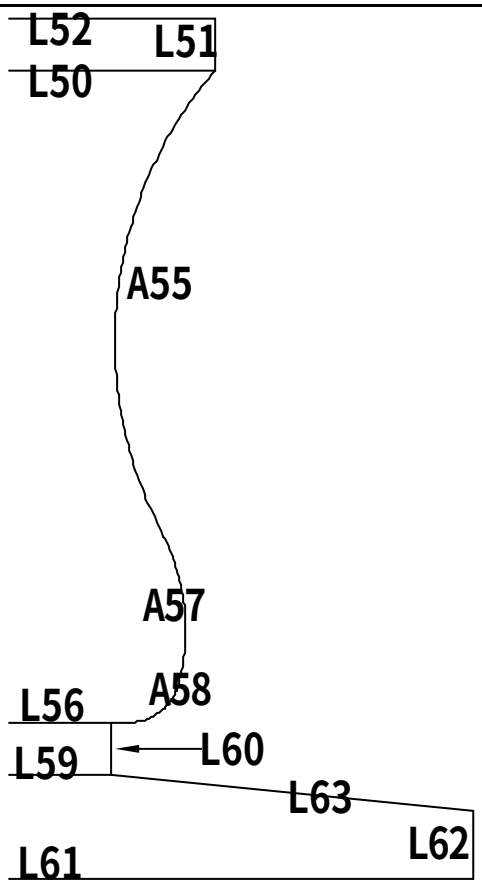

01、 RECtangle, F, 0, **任一點** P01, D, 20, 5, **向右下方點**, →L50~L53 **之矩形**
02、 eXplode, L50~L53 **之矩形**, →L50~L53
03、 Offset, 45, L50, **向下點**, →L54
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
04、 LENgthen, T, 15(=30/2), L54 **右**, →L54 **向左縮短為** 15
05、 Arc, int, L50 & L51, E, end, L54 **右**, R, 40, →A55
06、 Offset, 68, L52, **向下點**, →L56
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
07、 LENgthen, T, 15(=30/2), L56 **右**, →L56 **向左縮短為** 15
08、 Arc, end, L56 **右**, E, int, L54 & A55, R, 20, →A57

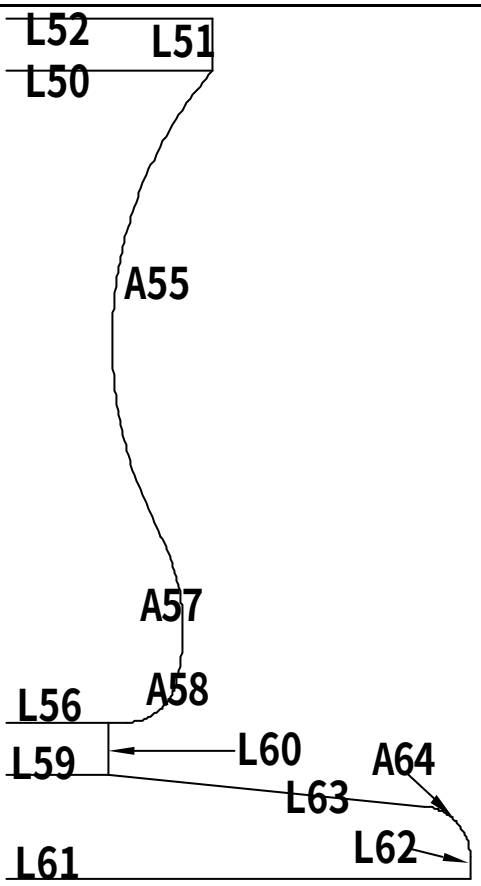



09、 Fillet, T, T, R, 5, L56, A57, →A58
10、 Offset, 5(=73-68), L56, **向下點**, →L59

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

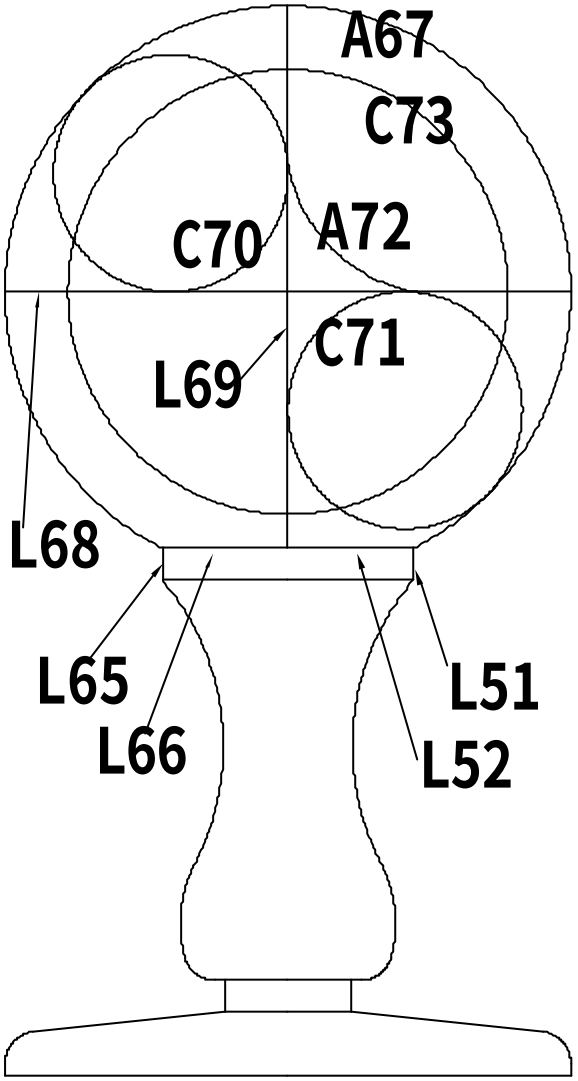
11、 LENgthen, T, 10(=20/2), L59 **右**, →L59 **向左縮短為** 10
12、 Line, end, L59 **右**, (**正交模式**), **向上** 5(=73-68), →L60
13、 Offset, 83, L52, **向下點**, →L61



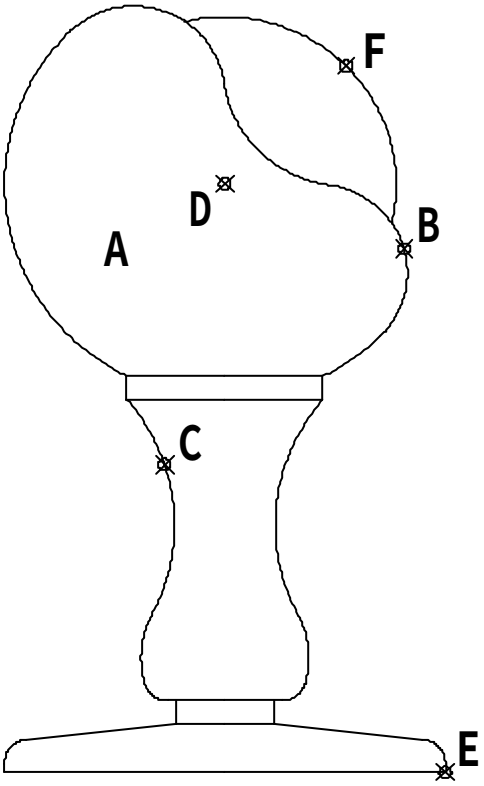




14、 LENgthen, T, 45(=90/2), L61 **右**, →L61 **向右延伸為** 45
15、 Line, end, L61 **右**, (**正交模式**), **向上** 6.5(=83-76.5), →L62, int, L59 & L60, →L63

- - - - - - - - - - - - - - - - - - - -

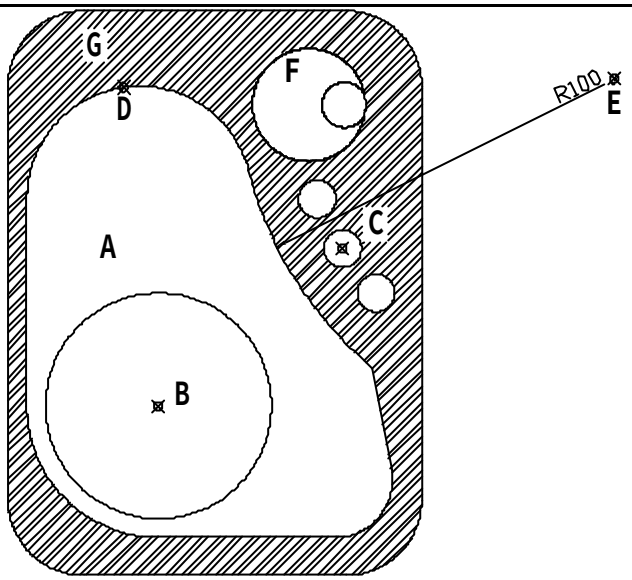



17、 Erase,
18、 MIrror, **右側所有物件**, end, L52 **左**, end, L61 **左**, **→左側** L65;L66 **等相同物件**
19、 Arc, int, L51 & L52, E, int, L65 & L66, R, -45(=90/2), →A67
20、 Line, qua, A67 **左**, qua, A67 **右**, →L68
21、 Line, qua, A67 **上**, end, L52 **左**, →L69
22、 Circle, 3P, tan, A67 **左上**, tan, L68 **左**, tan, L69 **上**, →C70
23、 Circle, 3P, tan, A67 **右下**, tan, L68 **右**, tan, L69 **下**, →C71
24、 Fillet, T, T, R, 25, C70 **右**, C71 **上**, →A72
25、 Circle, cen, C67, 35, →C73



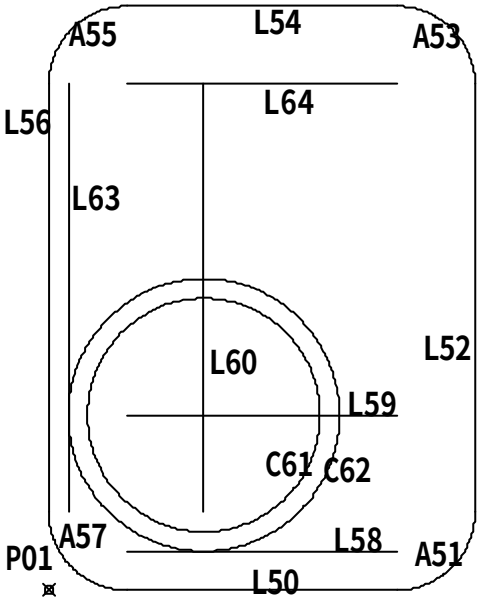

26、 TRim,
27、 Erase,
28、 BOundary, **點** A **區域**
29、 **點** A **區域邊緣**, CTRL+1, **→查幾何長度值** =255.0865
30、 DIst, cen, B, cen, C, **→距離**=86.8249
31、 DIst, cen, D, int, E, →Y **差值**=123.3113
32、 **點弧** F, CTRL+1, **→查幾何總角度值** =116.8955
33、 Hatch, **點所有閉合區域**, **→斜線區域**
34、 **點斜線區域**, CTRL+1, **→查幾何面積值** =8067.0387

## 2.6.12 AutoCAD2D2008-612

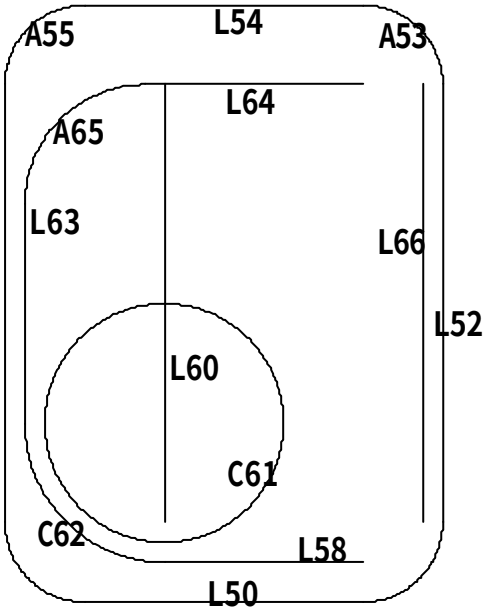



1、 A **區域扣除內孔面積為何？** 5887.3145
2、 **中心點** B **至中心點** C **距離為何？** 64.3821
3、 **弧** D **中心點至中心點** E **水平距離？** 126.0347
4、 F **區域周長為何？** 119.6230
5、 **斜線** G **區域面積為何？** 6499.4114

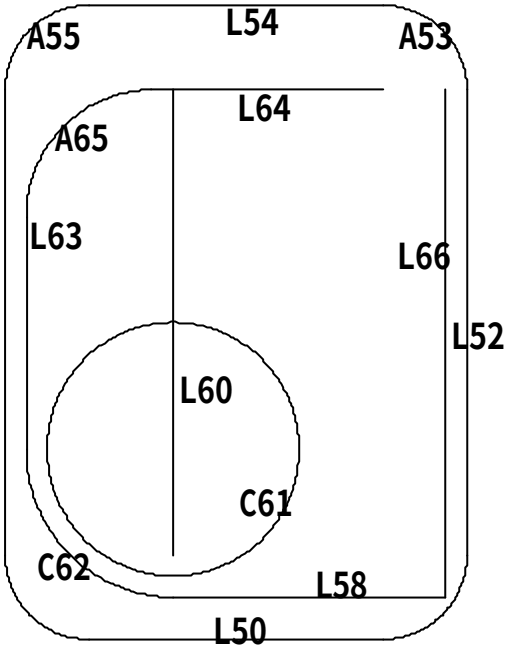
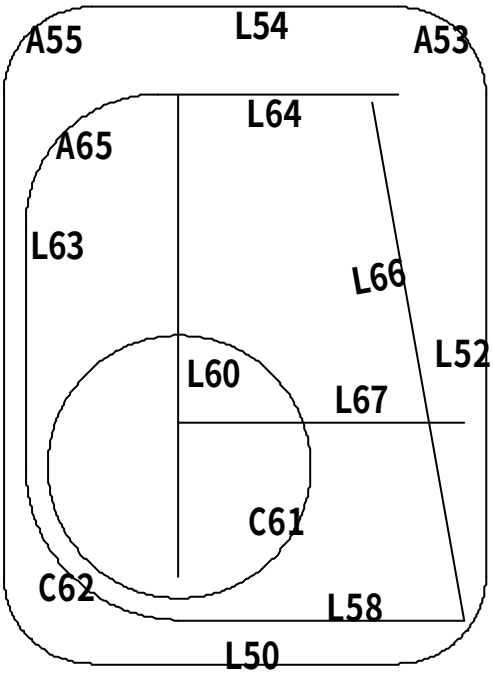
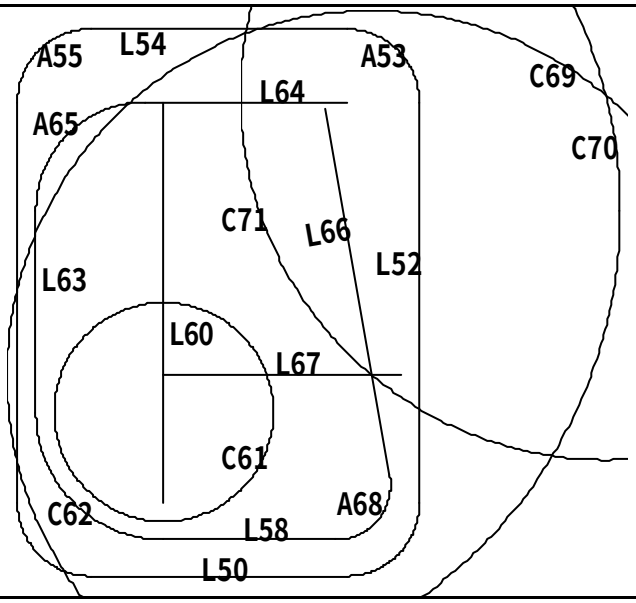
**解題步驟：**

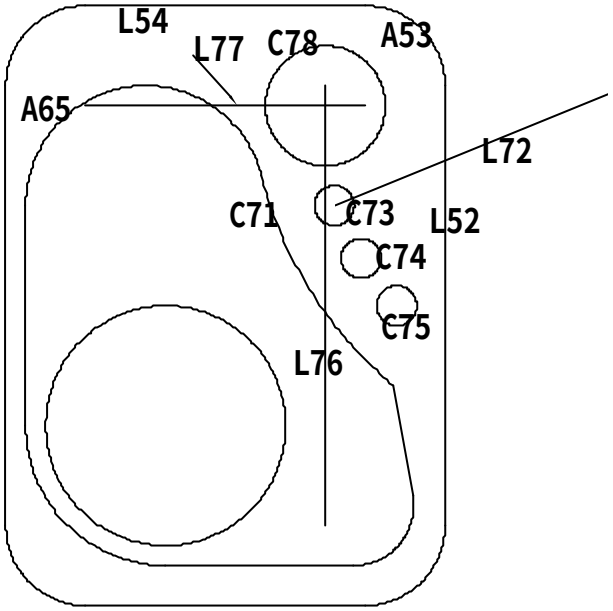



01、 RECtangle, F, 20, **任一點** P01, D, 110, 150, **向右上方點**, →L50~A57 **之圓角矩形**
02、 eXplode, L50~A57 **之圓角矩形**, →L50~A57
03、 Offset, 10, L50, **向上點**, →L58
04、 Offset, 35, L58, **向上點**, →L59
05、 Offset, 40, L56, **向右點**, →L60
06、 Circle, int, L59 & L60, D, 60, →C61
07、 Circle, @(**鎖定上次點**), 35, →C62
08、 Offset, 5, L56, **向右點**, →L63
09、 Offset, 20, L54, **向下點**, →L64

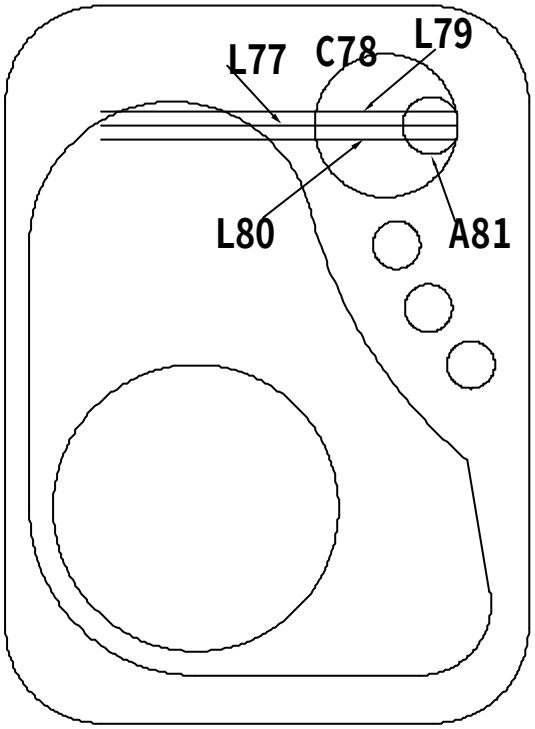



10、 Fillet, T, T, R, 30, L63, L64, →A65
11、 Offset, 5, L52, **向左點**, →L66
12、 TRim,
13、 Erase,

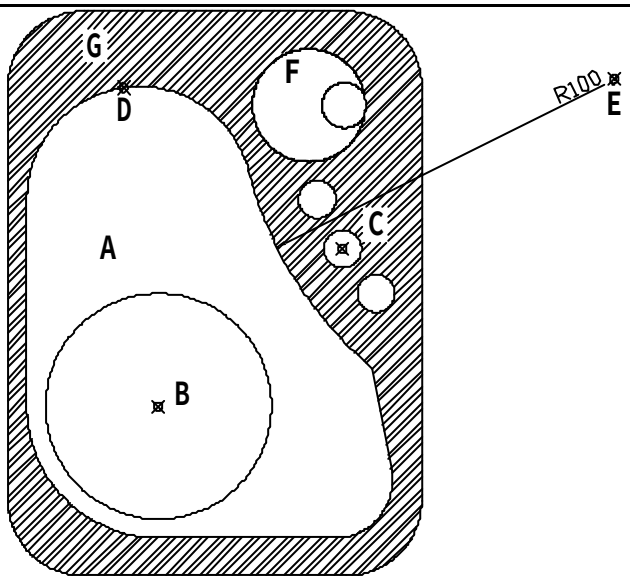
| | |
|---|---|
|  | 14、 Fillet, T, T, R, 0, L58, L66, →L58 **與** L66**相交** |
|  | 15、 ROtate, L66, int, L58 & L66, 10, →L66 **轉動**<br>16、 Offset, 45, L58, **向上點**, →L67 |
|  | 17、 Fillet, T, T, R, 15, L58, L66, →A68<br>18、 Circle, int, L66 & L67, 100, →C69<br>19、 Circle,cen, A65, 130(=30+100), →C70<br>20、 Circle,int, C69 & C70, 100, →C71 |




21、 EXtend, C71, A65, →A65 與 C71 相交
22、 TRim,
23、 Erase,
24、 Line, cen, C71, @85<202, →L72
25、 Circle, end, L72 下, D, 10, →C73
26、 ARray, 環形陣列, 中心點, cen, C71, 項目總數, 3, 佈滿角度, 20, 物件, C73, →C74;C75
27、 Offset, 30, L52, 向左點, →L76
28、 Offset, 25, L54, 向下點, →L77
29、 Circle, int, L76 & L77, D, 30, →C78

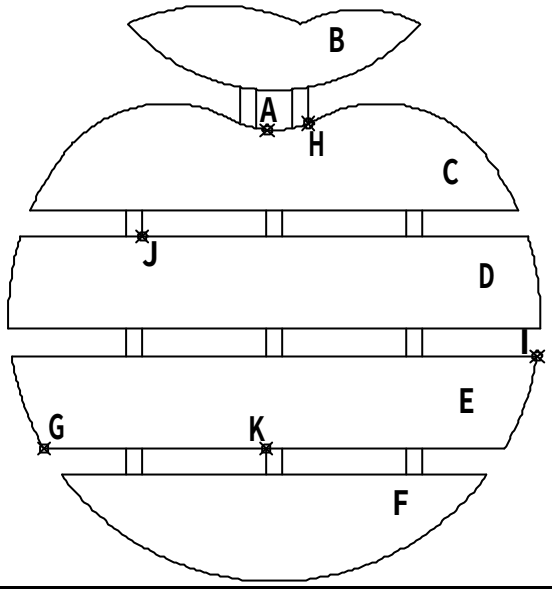



30、 Erase,
31、 EXtend, C78, L77 右, →L77 與 C78 相交
32、 Offset, 3(=6/2), L77, 向上點, →L79
33、 Offset, 3(=6/2), L77, 向下點, →L80
34、 Arc, int, C78 & L79, E, int, C78 & L80, R, 6, →A81



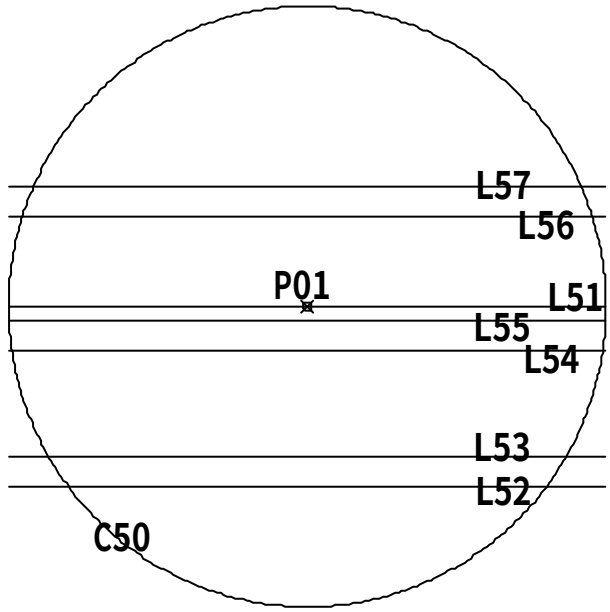

35、 TRim,
36、 Erase,
37、 Hatch, 點 A 區域, →A 區域斜線
38、 點 A 區域斜線, CTRL+1, →查幾何面積值 =5887.3145
39、 DIst, cen, B, cen, C, →距離=64.3821
40、 DIst, cen, D, cen, E, →X 差值=126.0347
41、 BOundary, 點 F 區域,
42、 點 F 區域邊緣, CTRL+1, →查幾何長度值 =119.6230
43、 Hatch, 點 G 區域, B, 點 B 區域邊緣, →G 區域斜線
44、 點 G 區域斜線, CTRL+1, →查幾何面積值 =6499.4114

## 2.6.13 AutoCAD2D2008-613

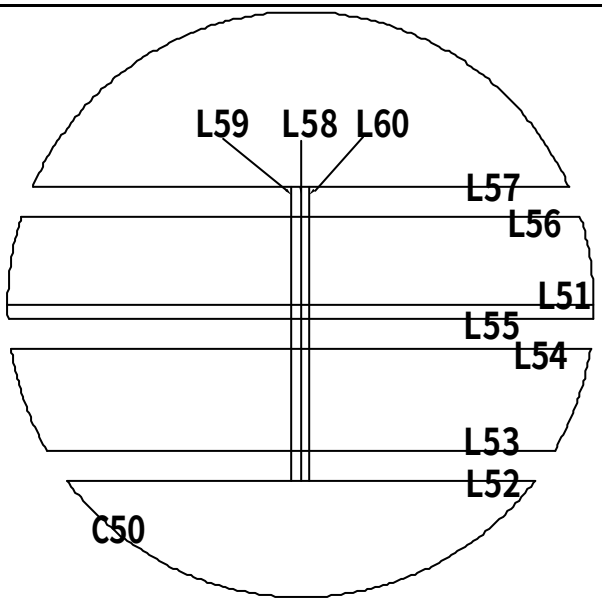



1、 **弧** A **長度為何？** 34.2970
2、 B **區域周長為何？** 238.1872
3、 C+D+E+F **區域面積總和為何？** 23755.3983
4、 **交點** G **至交點** H **距離為何？** 157.8856
5、 ∠IJK **包含角為何？** 42.9602

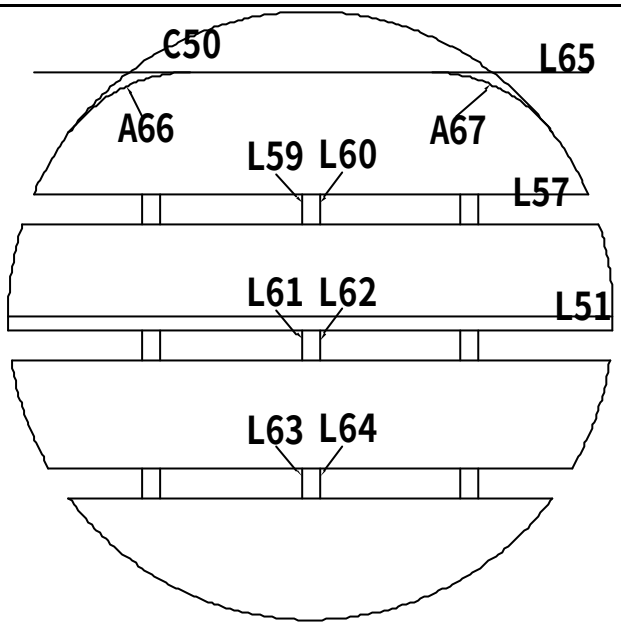
**解題步驟：**



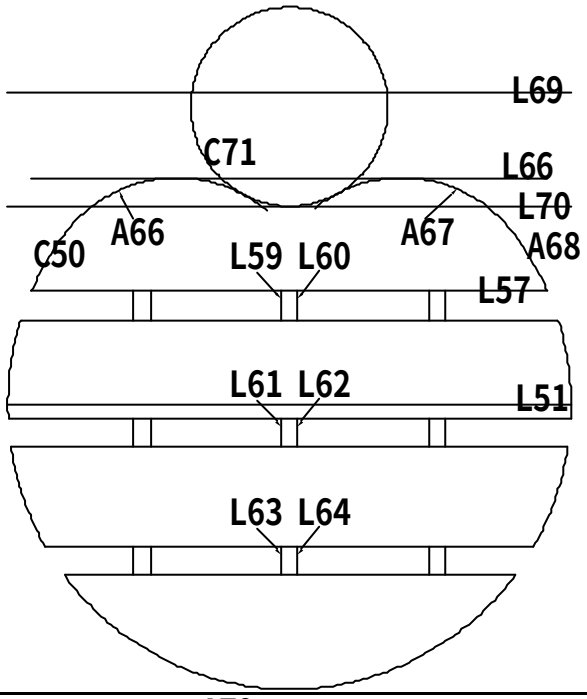

01、 Circle, **任一點** P01, D, 200, →C50
02、 Line, qua, C50 **左**, qua, C50 **右**, →L51
03、 Offset, 60(=(200/2)-40), L51, **向下點**, →L52
04、 Offset, 10, L52, **向上點**, →L53
05、 ARray, **矩形陣列**, **列**, 3, **行**, 1, **列偏移**, 45(=10+35), **行偏移**, 0, **物件**, L52;L53, →L54~L57



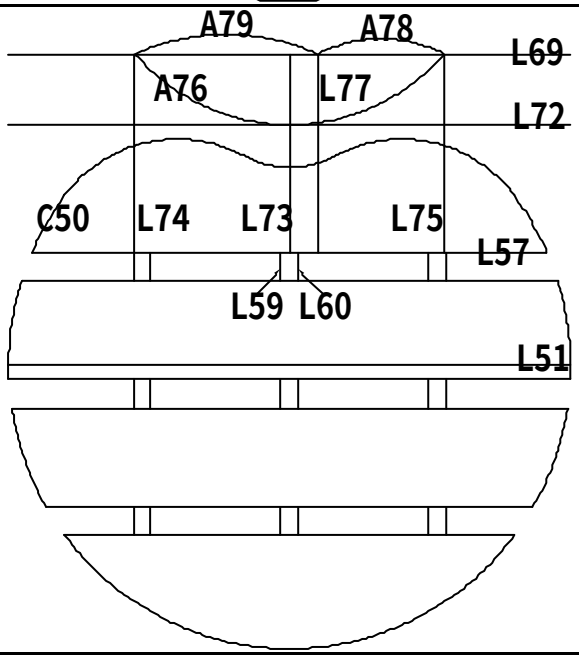

06、 TRim, (**注意** L54 **及** L55 **左右兩端**)
07、 Line, mid, L52, mid, L57, →L58
08、 Offset, 3(=6/2), L58, **向左點**, →L59, L58, **向右點**, →L60



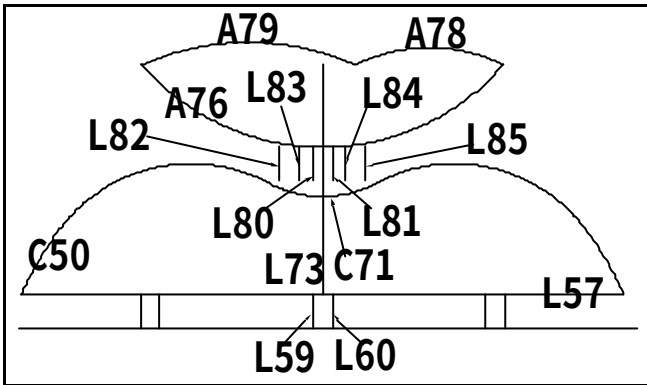
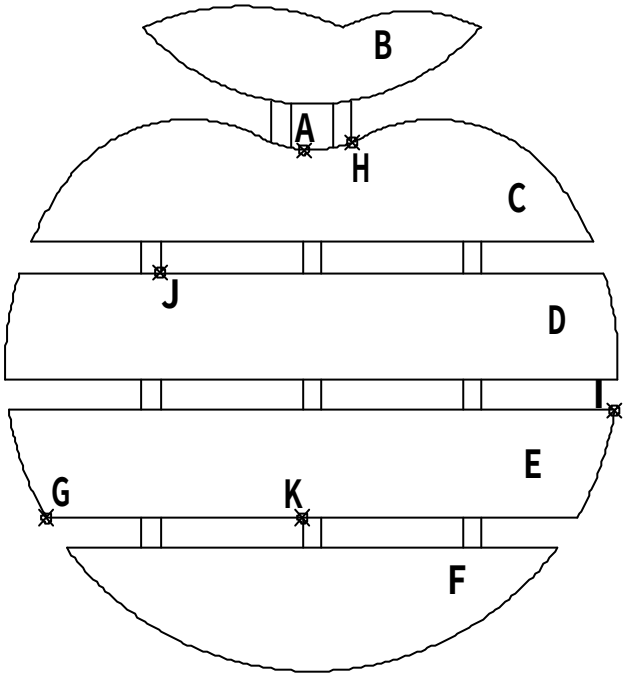
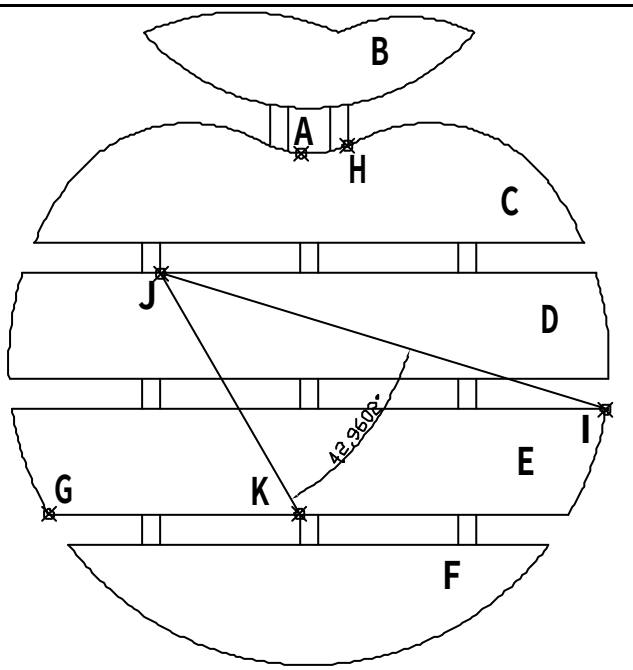

09、 TRim, →L61~L64
10、 COpy, L61~L64, **任一點**, (**正交模式**), **向右** 52.5, →**右側相同物件**, **向左** 52.5, →**左側相同物件**
11、 Offset, 40, L57, **向上點**, →L65
12、 Fillet, T, N, R, 50, C50 **左上**, L65 **左**, →A66, C50 **右上**, L65 **右**, →A67



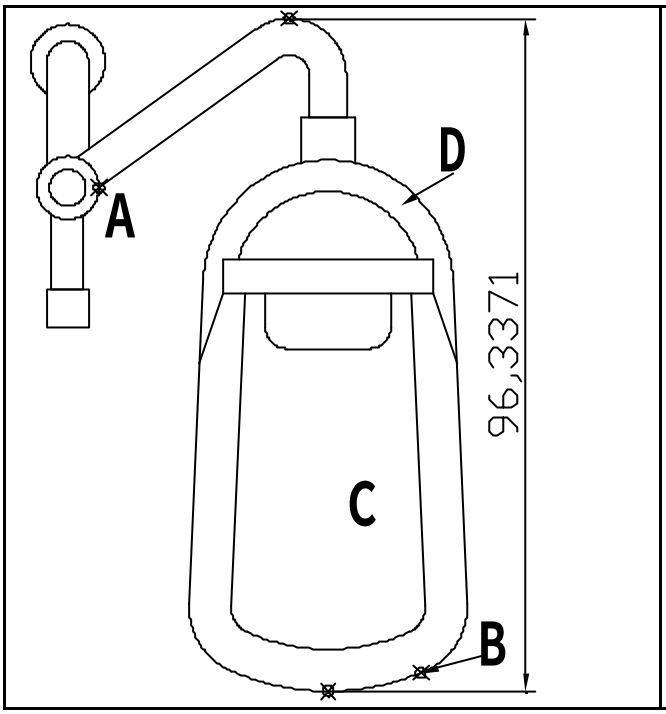

13、 TRim, →A68
14、 Offset, 110, L51, **向上點**, →L69
15、 Offset, 40, L69, **向下點**, →L70
16、 LENgthen, DY, **點** A66 **右上往右下拉**, →A66 **弧段延伸**
17、 LENgthen, DY, **點** A67 **左上往左下拉**, →A67 **弧段延伸**
18、 Circle, 3P, tan, A66 **右上**, tan, L70, tan, A67 **左上**, →C71




19、 TRim,
20、 Erase,
21、 Offset, 25, L69, **向下點**, →L72
22、 Line, mid, L57, mid, L69, →L73
23、 Offset, 55(=110/2), L73, **向左點**, →L74, L73, **向右點**, →L75
24、 Arc, int, L69 & L74, int, L72 & L73, int, L69 & L75, →A76
25、 Offset, 65, L74, **向左點**, →L77
26、 Arc, int, L69 & L75, E, int, L69 & L77, R, 50, →A78
27、 Arc, int, L69 & L77, E, int, L69 & L74, R, 80, →A79

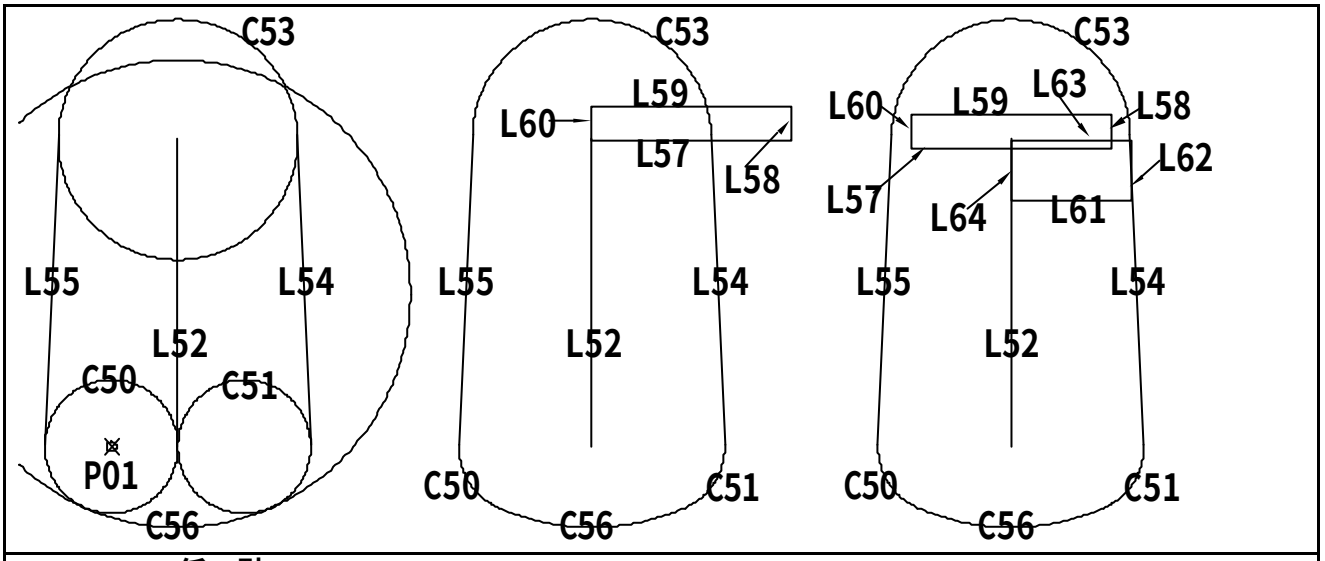
| | |
|---|---|
|  | 28、 Erase,<br>29、 COpy, L59;L60, int, L57 & L73, int, L73 & A76, →L80;L81<br>30、 COpy, L80;L81, int, L73 & A76, (**正交模式**), **向左** 10(=20/2), →L82;L83, **向右** 10(=20/2), →L84;L85 |
|  | 31、 EXtend, C71;A76, **分別點** L82~L85 **之上及下端**, →L82~L85 **之上及下端皆分別與** C71 **及** A76 **相交**<br>32、 Erase, L80;L81<br>33、 **點弧** A, CTRL+1, →**查幾何弧長值**=34.2970<br>34、 BOundary, **點** B **區域**,<br>35、 **點選** B **區域邊緣**, CTRL+1, →**查幾何長度值**=238.1872<br>36、 Hatch, **點** C、D、 E **及** F **區域**, →**區域斜線**<br>37、 **點區域斜線**, CTRL+1, →**查幾何面積值**=23755.3983<br>38、 DIst, int, G, int, H, →**距離**=157.8856 |
|  | 39、 COpy, Line, int, I, int, J, int, K, →**左側圖形**<br>40、 DimANgular, **線** IJ, **線** JK, →42.9602 |

## 2.6.14 AutoCAD2D2008-614




1、 **圖形最大高度為何？** 96.3371
2、 **交點** A **至交點** B **距離為何？** 83.6423
3、 C **區域周長為何？** 159.9772
4、 D **區域周長為何？** 148.9607
5、 **圖形所圍成最外圍面積為何？** 3239.1103

**解題步驟：**




01、 Circle, **任一點** P01, 10, →C50
02、 COpy, C50, cen, C50, (**正交模式**), **向右** 20, →C51
03、 Line, int, C50 & C51, (**正交模式**), **向上** 46, →L52
04、 Circle, end, L52 **上**, 18, →C53
05、 Line, tan, C51 **右**, tan, C53 **右**, →L54
06、 Line, tan, C50 **左**, tan, C53 **左**, →L55
07、 Circle, T, C50 **左下**, C51 **右上**, 35, →C56

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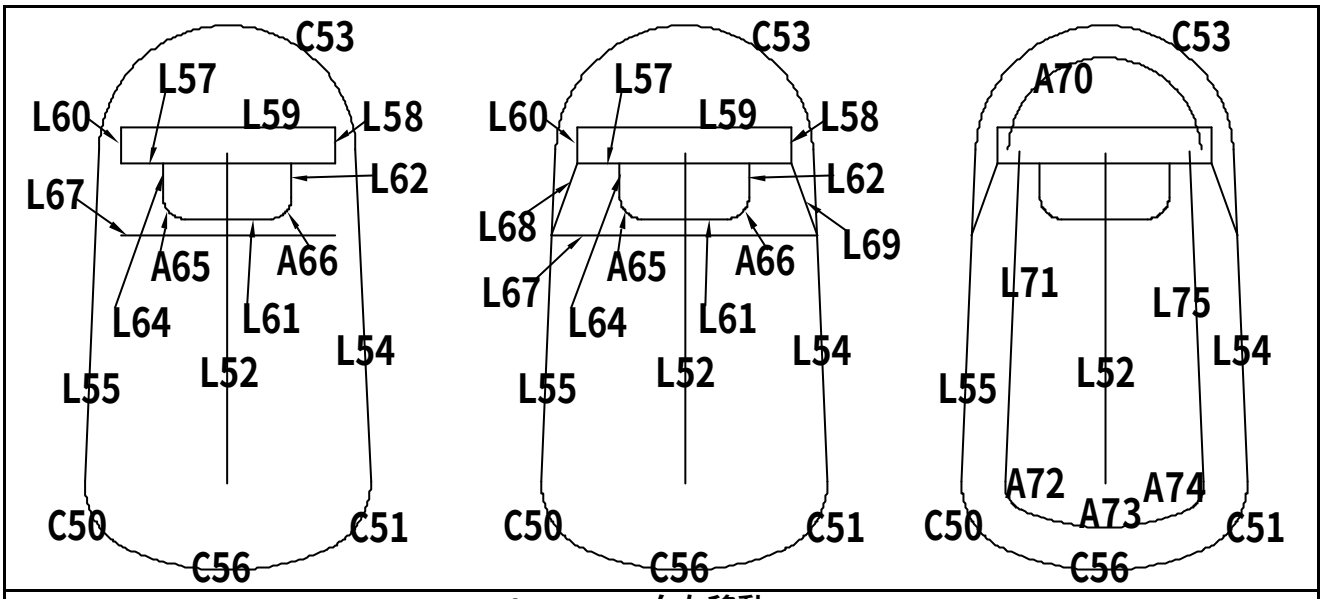


13、 Move, L61~L64, mid, L62, @-9,0, →L61~L64 **向左移動**
14、 eXplode, L61~L64 **之矩形**, →L61~L64
15、 Fillet, T, T, R, 3, L61, L64, →A65, L61, L62 →A66
16、 Offset, 10, L57, **向下點**, →L67

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22、 TRim, Erase,
23、 LENgthen, DE, 24, L52 上, →L52 向上延伸
24、 RECtangle, F, 0, end, L52 上, D, 7.5, 10, 向右下方點, →L76~L79 之矩形

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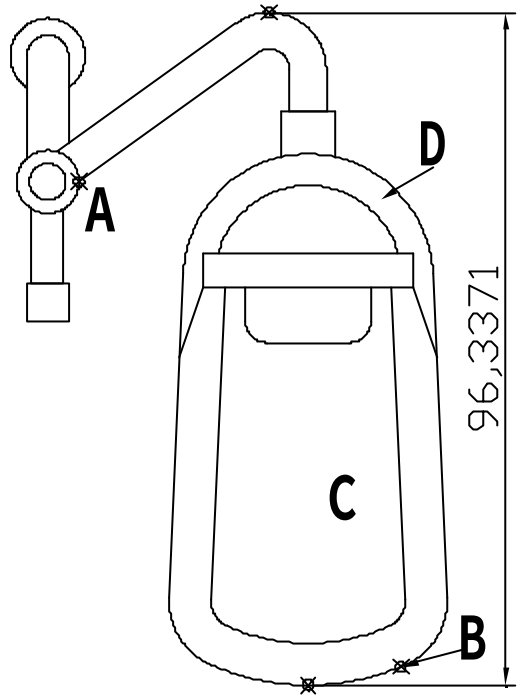
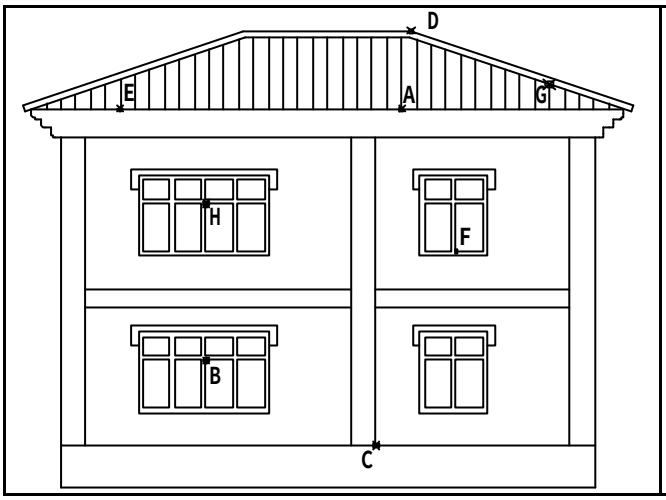


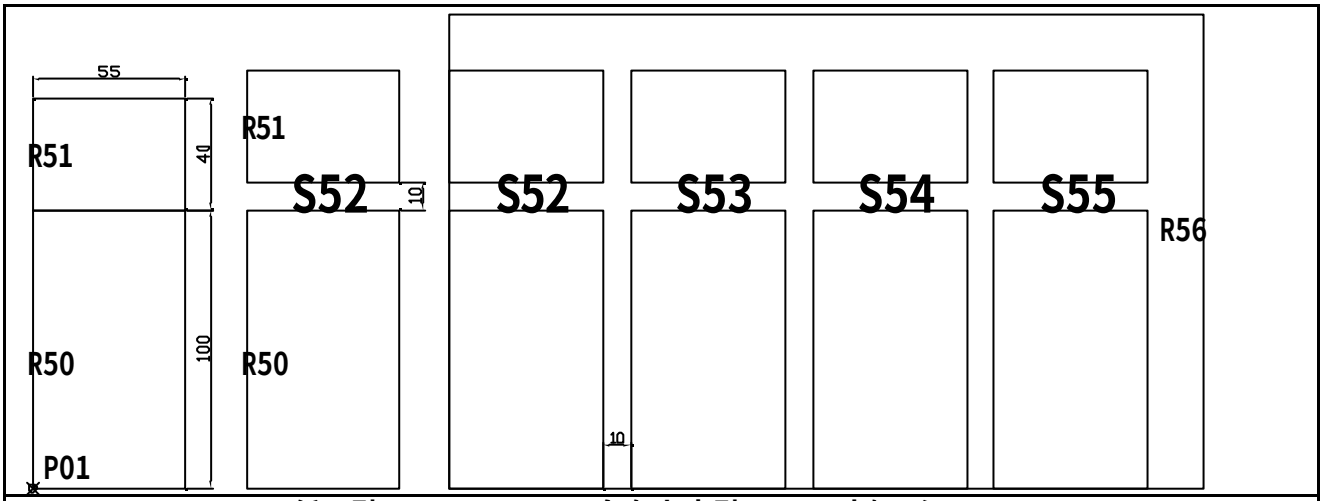

52、 Move, R98, mid, R98 下, mid, R97 上, →R98 向左移動
53、 TRim,
54、 Erase,
55、 DimLInear, qua, A85 上, qua, C56 下, →96.3371
56、 DIst, int, A, int, B, →距離=83.6423
57、 BOundary, 點 C 區域,
58、 點選 C 區域邊緣, CTRL+1, →查幾何長度值=159.9772
59、 BOundary, 點 D 區域,
60、 點選 D 區域邊緣, CTRL+1, →查幾何長度值=148.96072
61、 Hatch, 點所有閉合區域, →區域斜線
62、 點區域斜線, CTRL+1, →查幾何面積值=3239.1103

## 2.6.15 AutoCAD2D2008-615




1、 **交點** A **至交點** B **距離為何？** 665.5120
2、 **當交點** C **座標** 0,0 **交點** D **的** X **座標為何？** 72.4866
3、 **交點** E **至交點** F **距離為何？** 754.2307
4、 **交點** G **至交點** H **距離為何？** 751.0356
5、 **斜線區域總面積為何？** 422400

**解題步驟：**




01、 RECtangle, F, 0, **任一點** P01, D, 55, 100, **向右上方點**, →R50 **之矩形**
02、 RECtangle, F, 0, int, R50 **左上**, D, 55, 40, **向右上方點**, →R51 **之矩形**

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03、 Move, R51, int, R50 **左上**, @0,10, →R51 **向上移動**, R50 & R51 **稱為** S52

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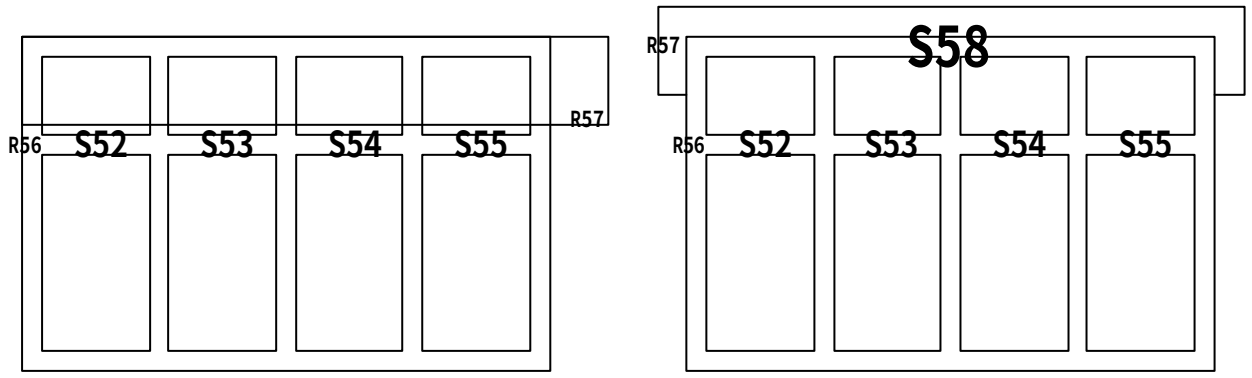
04、 ARray, **矩形陣列**, **列**, 1, **行**, 4, **列偏移**, 0, **行偏移**, 65, **物件**, S52, →S53~S55
05、 RECtangle, F, 0, int, S52 **左下**, D, 270, 170, **向右上方點**, →R56 **之矩形**




06、 Move, R56, int, S52 **左下**, @-10,-10, →R56 **向左下移動**
07、 RECtangle, F, 0, int, R56 **左上**, D, 300, 45, **向右下方點**, →R57 **之矩形**

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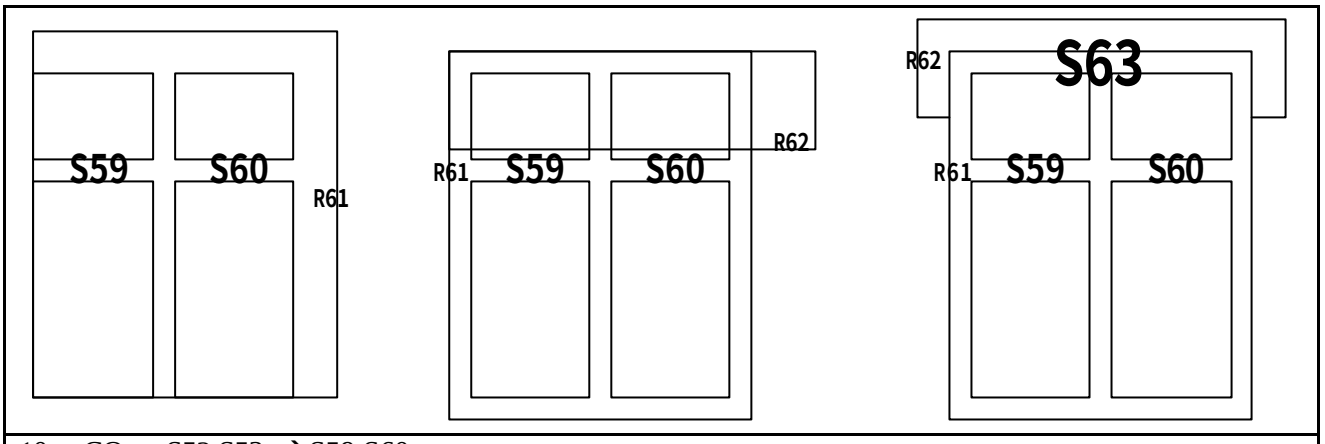
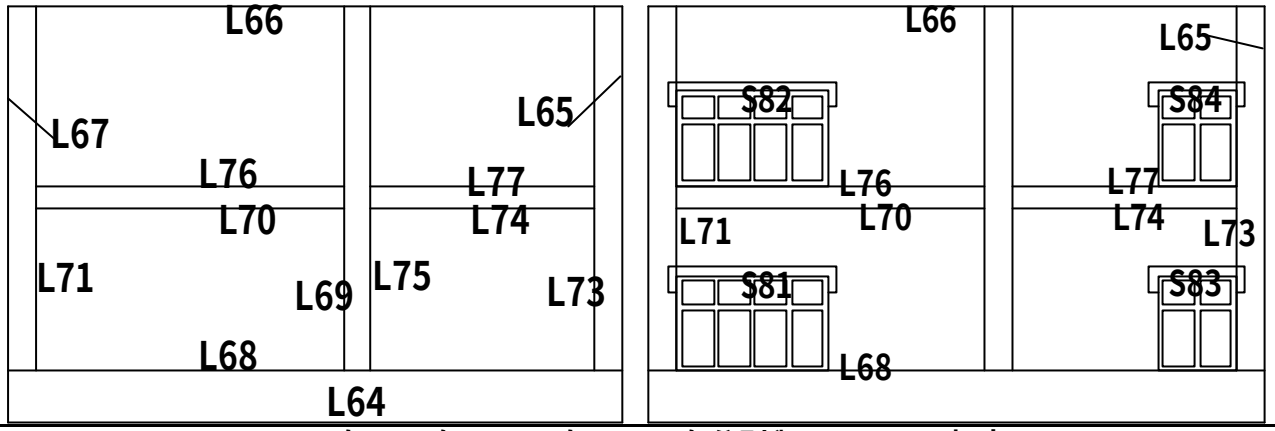
08、 Move, R57, int, R56 **左上**, @-15,15, →R57 **向左上移動**
09、 TRim, Erase,




10、 COpy, S52;S53, →S59;S60
11、 RECtangle, F, 0, int, S59 **左下**, D, 140, 170, **向右上方點**, →R61 **之矩形**

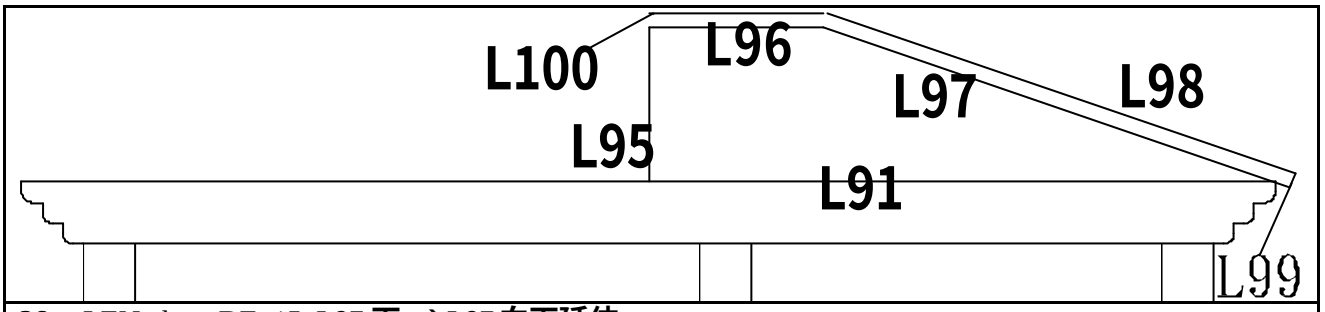
- - - -



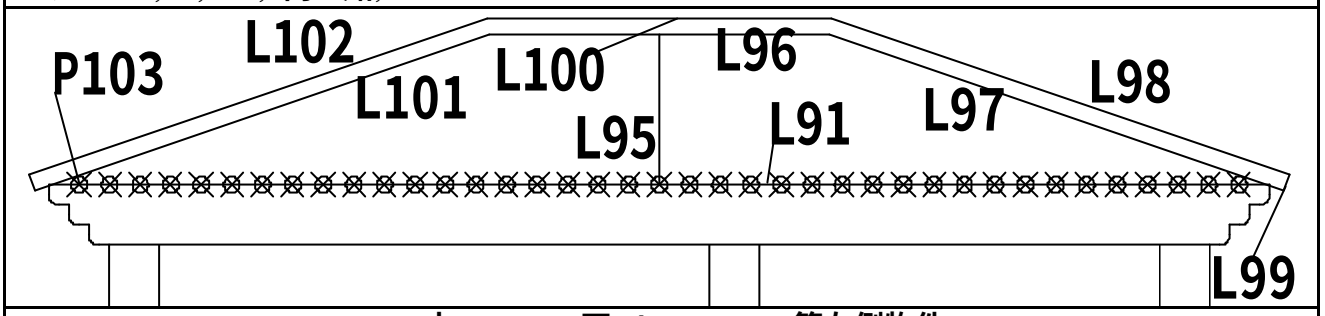

26、 EXtend, L67;L65, L68 **左**;L68 **右**, →L68 **左及** L68 **右分別與** L67 **及** L65 **相交**

27、 EXtend, L66;L68, L71 **上**;L71 **下**; L69 **上**;L69 **下**; L75 **上**;L75 **下**; L73 **上**;L73 **下**, →L71 **上**;L71 **下**; L69 **上**;L69 **下**; L75 **上**;L75 **下**; L73 **上及** L73 **下分別與** L66 **及** L68 **相交**

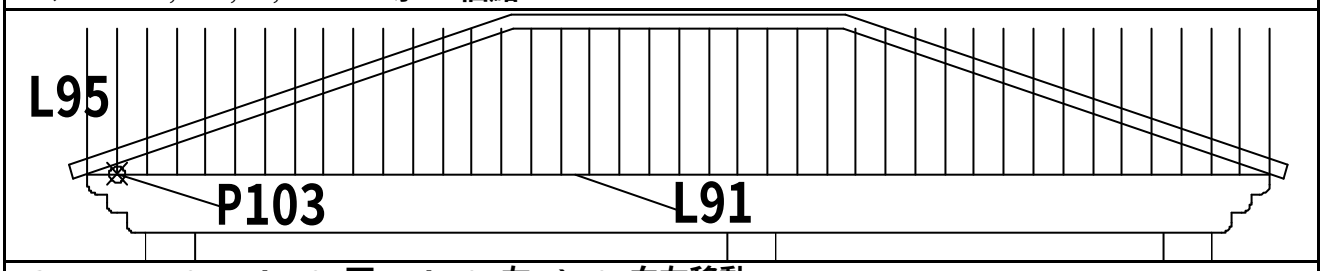
- - - -



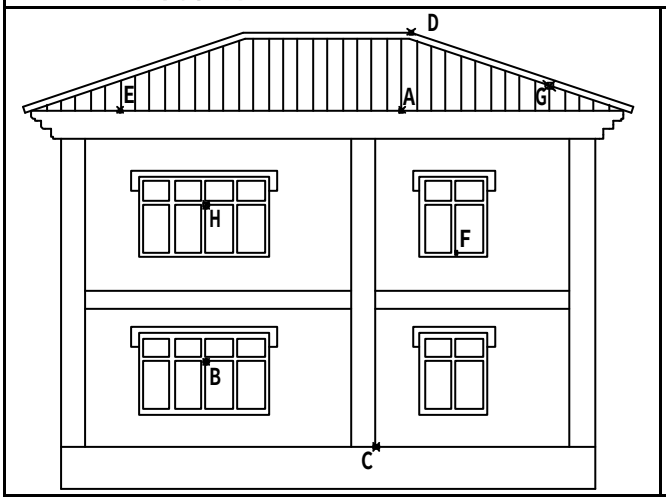

36、 LENgthen, DE, 15, L97 下, →L97 向下延伸
37、 Offset, 15, L97, 向上點, →L98
38、 Line, end, L97 下, end, L98 下, →L99
39、 Offset, 15, L96, 向上點, →L100




40、 MIrror, L96~L100, end, L95 上, end, L95 下, →L101;L102 等左側物件
41、 Fillet, T, T, R, 0, L96, L97, →L96 與 L97 相交, L100, L98, →L100 與 L98 相交, L96, L101, →L96 與 L101 相交, L100, L102,→L100 與 L102 相交,
42、 DIVide, L91, 40, →P103 等 39 個點




43、 Move, L95, end., L95 下, end, L91 左, →L95 向左移動
44、 Erase, 除 P103 之外的所有點
45、 ARray, 矩形陣列, 列, 1, 行, 40, 列偏移, 0, 行偏移, end, L91 左, nod, P103, 物件, L95, →40 個與 L95 相同物件




46、 TRim,
47、 Erase,
48、 DIst, int, A, int, B, →距離=665.5120
49、 Move, all, int, C, 0,0, →所有物件皆移動
50、 ID, int, C, →X=0.0000, Y=0.0000, 用以檢查是否搬移至正確位置
51、 ID, int, D, →X=72.4866, Y=875.0000
52、 DIst, int, E, int, F, →距離=754.2307
53、 DIst, int, G, int, H, →距離=751.0356
54、 Hatch, 分別點每個窗框旁邊之閉合區域, B, 點 24 個大小玻璃框, →區域斜線
55、 點區域斜線, CTRL+1, →查幾何面積值 =422400

# 參考文獻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